Wan Zhexian

Geometry of Classical Groups Over Finite Fields Second Edition

(有限域上典型群的几何学(第二版))

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Graduate Texts in Mathematics

Preface to the Second Edition

As applications of the Anzahl theorems in the geometry of classical groups over finite fields some critical problems of finite vector spaces, Moor-Penrose generalized inverses of matrices over a finite field, and the representation of a form (bilinear, alternate, hermitian, or quadratic, etc.) by another form of the same kind over a finite field are added in this edition.

Wan Zhexian

Beijing, 2002

Preface

This monograph is a comprehensive survey of the results obtained on the geometry of classical groups over finite fields mainly in the 1960s and early 1990s.

For the convenience of the readers I start with the affine geometry and projective geometry over finite fields in Chapters 1 and 2, respectively. Among other things, the affine classification of quadrics is included in Chapter 1, and conics and ovals are studied in detail in Chapter 2. From Chapter 3 and onwards the geometries of symplectic, pseudo-symplectic, unitary, and orthogonal groups are studied in succession. The book ends with two appendices, on the axiomatic projective geometry, and on polar spaces and finite generalized quadrangles, respectively.

Now I shall say a few words about the problems we are going to study in Chapters 3–7, and in addition give some historical remarks.

Let $\mathbb{F}_q$ be the finite field with q elements, where q is a power of a prime, $\mathbb{F}_q^{(n)}$ be the n-dimensional row vector space over $\mathbb{F}_q$, and $GL_n(\mathbb{F}_q)$ be the general linear group of degree n over $\mathbb{F}_q$. It is well-known that $GL_n(\mathbb{F}_q)$ has an action on $\mathbb{F}_q^{(n)}$ and an induced action on the subspaces of $\mathbb{F}_q^{(n)}$, which are thus subdivided into orbits under the action of $GL_n(\mathbb{F}_q)$. It is natural to ask:

(i) How should the orbits be described?

(ii) How many orbits are there?

(iii) What are the lengths of the orbits?

(iv) What is the number of subspaces in any orbit contained in a given subspace?

The answers to these questions are classical and well-known for $GL_n(\mathbb{F}_q)$,

but a natural question arises: If $GL_n(\mathbb{F}_q)$ is replaced by any one of the other classical groups: the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$ (where $n = 2\nu$), the pseudo-symplectic group $Ps_{2\nu+\delta}(\mathbb{F}_q)$ (where q is even, $n = 2\nu + \delta$, and $\delta = 1$ or 2), the unitary group $U_n(\mathbb{F}_q)$ (where q is a square), and the orthogonal group $O_{2\nu+\delta}(\mathbb{F}_q)$ (where $n = 2\nu + \delta$ and $\delta = 0, 1$, or 2), then what will the answers of the four problems mentioned above be? This is what we shall analyse in Chapters 3–7.

In 1937 E. Witt studied problem (i) for the orthogonal group $O_n(F, S)$ defined by an $n \times n$ nonsingular symmetric matrix S over any field F of characterstic $\neq 2$. His famous theorem asserts that two subspaces P_1 and P_2 of $F^{(n)}$ belong to the same orbit under the action of $O_n(F, S)$ if and only if they have the same dimension and $P_1 S\, {}^tP_1$ and $P_2 S\, {}^tP_2$ are cogredient, where P_1 and P_2 also denote matrix representations of the subspaces P_1 and P_2, respectively. Later C. Arf, J. Dieudonné, L. K. Hua, et al. extended Witt's theorem to other classical groups.

In 1958 B. Segre studied problem (iii) for orthogonal groups over any finite field $\mathbb{F}_q$ but he restricted himself to consider only totally isotropic and totally singular subspaces where q is odd and even, respectively. He used the geometrical method and geometrical language. In his address at the International Congress of Mathematicians, Edinburgh, 1958, he announced his formula for the number of subspaces of a given dimension lying on a non-degenerate quadratic in the projective space over any finite field without any restriction on its characteristic.

In the mid-1960s problem (iii) was attacked by three of my students at that time, Z. Dai, X. Feng, and B. Yang, and myself. We obtained the closed formulas for the lengths of all the orbits of subspaces under the action of the symplectic, unitary, and orthogonal groups over finite fields. Our method is algebraic. Closed formulas for problem (iv), i.e., for the number of subspaces in any orbit contained in a given subspace were also obtained by myself in 1966.

In the early 1990s I studied problems (i) and (ii). Conditions satisfied by the numerical invariants characterizing the orbits of subspaces under the symplectic, unitary, and orthogonal groups over finite fields were obtained and then the number of orbits was computed. Problems (i) – (iii) for the pseudo-symplectic groups over finite fields of characteristic 2 were studied together with Y. Liu. Moreover, I also studied problem (iv) for the pseudo-symplectic group and problems (i) – (iv) for the singular symplectic, pseudo-symplectic, unitary, and orthogonal groups.

My interest in the geometry of classical groups over finite fields was arisen by block designs in the mid-1960s and by authentication codes in the early 1990s. By using it, many interesting block designs and authentication codes can be constructed. However, due to the limitation of the thickness of the book only some simple authentication codes constructed from the geometry of classical groups over finite fields are included in this monograph as examples. Using the geometry of classical groups over finite fields one can also construct projective codes with a few distinct weights and study the weight hierarchies of them; they are, however, not included in this monograph.

A large portion of the book is adopted from the lecture notes of a course entitled "Finite Geometry" which I gave at the Department of Information Theory, Lund University. I would like to express my sincere gratitude to Professor Rolf Johannesson who invited me to visit Lund and to give the course and encouraged me to write the present book. My visit in Lund is most fruitful and many of the results on the geometry of classical groups over finite fields that I obtained in Lund are compiled in the present book. The author is also deeply indebted to Mrs. Lena Månsson for her beautiful typesetting, careful and hard work, and most of all her patience and cooperation.

Lund, 1993 Wan Zhexian (Zhe-Xian Wan)

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Chapter 1
Affine Geometry over Finite Fields

1.1 Vector Spaces and Matrices over Finite Fields

Let $\mathbb{F}_q$ be the finite field with q elements, where q is a power of a prime, and n a positive integer. We use

$$\mathbb{F}_q^{(n)} = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{F}_q,\ i = 1, 2, \ldots, n\}$$

to denote the n-dimensional *row vector space* over $\mathbb{F}_q$ formed by the set of all n-tuples (or n-dimensional row vectors)

$$(x_1, x_2, \ldots, x_n), \quad x_i \in \mathbb{F}_q,\ i = 1, 2, \ldots, n,$$

over $\mathbb{F}_q$ with addition and scalar multiplication defined by

$$\begin{aligned}(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) &= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n),\\ x(x_1, x_2, \ldots, x_n) &= (xx_1, xx_2, \ldots, xx_n),\end{aligned}$$

where $x, x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_n \in \mathbb{F}_q$. Let

$$e_1 = (1, 0, 0, \ldots, 0),\ e_2 = (0, 1, 0, \ldots, 0), \ldots,\ e_n = (0, 0, \ldots, 0, 1),$$

then $e_1, e_2, \ldots, e_n$ form a *basis* of $\mathbb{F}_q^{(n)}$, i.e., any vector $(x_1, x_2, \ldots, x_n)$ of $\mathbb{F}_q^{(n)}$ can be written as a linear combination of $e_1, e_2, \ldots, e_n$ with coefficients in $\mathbb{F}_q$:

$$(x_1, x_2, \ldots, x_n) = x_1e_1 + x_2e_2 + \cdots + x_ne_n,$$

and the expression is unique. Of course, any n linearly independent vectors in $\mathbb{F}_q^{(n)}$ form a basis of $\mathbb{F}_q^{(n)}$. Clearly, $\mathbb{F}_q^{(n)}$ has altogether q^n vectors.

Now let P be an m-dimensional vector subspace of $\mathbb{F}_q^{(n)}$, then we write $\dim P = m$. Let $v_1, v_2, \ldots, v_m$ be a basis of P. We notice that $v_1, v_2, \ldots, v_m$ are vectors in $\mathbb{F}_q^{(n)}$. We usually use the $m \times n$ matrix

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix}$$

to represent the vector subspace P, write

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix},$$

i.e., we use the same letter P to denote a matrix which represents the vector subspace P, and call the matrix P a *matrix representation* of the vector subspace P. It should be noted that a matrix representing an m-dimensional vector subspace is an $m \times n$ matrix of rank m. Of course, two $m \times n$ matrices P and Q both of rank m represent the same m-dimensional vector subspace, if and only if there is an $m \times m$ nonsingular matrix A such that

$$P = AQ.$$

Obviously, any m-dimensional vector subspace contains exactly q^m vectors.

Let P_1 and P_2 be two vector subspaces of $\mathbb{F}_q^{(n)}$. The set of vectors belonging to both P_1 and P_2 is a vector subspace of $\mathbb{F}_q^{(n)}$, called the *intersection* of P_1 and P_2 and denoted by $P_1 \cap P_2$. The set of vectors which can be written as sums of a vector of P_1 and a vector of P_2 is also a vector subspace of $\mathbb{F}_q^{(n)}$, called the *join* of P_1 and P_2 and denoted by $P_1 + P_2$. We have the *dimension formula*

$$\dim P_1 + \dim P_2 = \dim (P_1 \cap P_2) + \dim (P_1 + P_2).$$

The set of $n \times n$ nonsingular matrices over $\mathbb{F}_q$ forms a group under matrix multiplication, called the *general linear group* of degree n over $\mathbb{F}_q$ and denoted by $GL_n(\mathbb{F}_q)$. Elements of $GL_n(\mathbb{F}_q)$ can be regarded as linear transformations of $\mathbb{F}_q^{(n)}$. In fact, let $T \in GL_n(\mathbb{F}_q)$, then T carries (or

transforms) the vector $(x_1, x_2, \ldots, x_n)$ of $\mathbb{F}_q^{(n)}$ into $(x_1, x_2, \ldots, x_n)T$. That is, we have a map

$$\mathbb{F}_q^{(n)} \times GL_n(\mathbb{F}_q) \to \mathbb{F}_q^{(n)}$$
$$((x_1, x_2, \ldots, x_n), T) \mapsto (x_1, x_2, \ldots, x_n)T.$$

We also say that there is an *action* of $GL_n(\mathbb{F}_q)$ on $\mathbb{F}_q^{(n)}$, or $GL_n(\mathbb{F}_q)$ *acts* on $\mathbb{F}_q^{(n)}$. This action induces an action of $GL_n(\mathbb{F}_q)$ on the set of $m \times n$ matrices over $\mathbb{F}_q$ and also on the set of m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$: if $T \in GL_n(\mathbb{F}_q)$ and P is an $m \times n$ matrix over $\mathbb{F}_q$ or an m-dimensional vector subspace of $\mathbb{F}_q^{(n)}$, then T carries P into PT.

Theorem 1.1. If $T \in GL_n(\mathbb{F}_q)$ and P is an $m \times n$ matrix of rank m over $\mathbb{F}_q$, then PT is also an $m \times n$ matrix of rank m. Furthermore, if P and Q are both $m \times n$ matrices of rank m over $\mathbb{F}_q$, then there is an element $T \in GL_n(\mathbb{F}_q)$ such that $P = QT$.

Proof. The first statement is clear from linear algebra. We prove the second statement only. Write

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix} \quad \text{and} \quad Q = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{pmatrix},$$

where $v_1, v_2, \ldots, v_m$ are the m rows of P and $u_1, u_2, \ldots, u_m$ are the m rows of Q. Since P is of rank m, $v_1, v_2, \ldots, v_m$ are linearly independent. Thus there exist $n - m$ vectors $v_{m+1}, v_{m+2}, \ldots, v_n$ such that the n vectors $v_1, v_2, \ldots, v_m, v_{m+1}, v_{m+2}, \ldots, v_n$ are linearly independent. Then

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

is an $n \times n$ nonsingular matrix. Similarly, since Q is also of rank m, there exist $n - m$ vectors $u_{m+1}, u_{m+2}, \ldots, u_n$ such that

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

is also an $n \times n$ nonsingular matrix. Let

$$T = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}^{-1} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix},$$

then $T \in GL_n(\mathbb{F}_q)$ and $P = QT$. □

Corollary 1.2. The set of all non-zero vectors of $\mathbb{F}_q^{(n)}$ forms an *orbit*, i.e., a *transitive set*, under $GL_n(\mathbb{F}_q)$. The zero vector $(0, 0, \ldots, 0)$ is left fixed by every element of $GL_n(\mathbb{F}_q)$. □

Theorem 1.3. The set of all m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ forms an orbit under $GL_n(\mathbb{F}_q)$.

Proof. Use matrix representations of vector subspaces of $\mathbb{F}_q^{(n)}$. □

Theorem 1.4. The number of orbits of subspaces of $\mathbb{F}_q^{(n)}$ is $n + 1$. □

Now let us prove some Anzahl theorems in $\mathbb{F}_q^{(n)}$.

Let $N(m, n)$ be the number of m-dimensional vector subspaces in $\mathbb{F}_q^{(n)}$. Our goal is to compute $N(m, n)$. Let $n(m, n)$ be the number of $m \times n$ matrices of rank m over $\mathbb{F}_q$. We know that two $m \times n$ matrices P and Q both of rank m represent the same m-dimensional subspace if and only if there is an $m \times m$ nonsingular matrix A such that $P = AQ$, thus

$$n(m, n) = \mid GL_m(\mathbb{F}_q) \mid N(m, n),$$

where $\mid GL_m(\mathbb{F}_q) \mid$ is the order of $GL_m(\mathbb{F}_q)$. Clearly

$$\mid GL_m(\mathbb{F}_q) \mid = n(m, m).$$

Thus to compute $N(m, n)$ it is sufficient to compute $n(m, n)$.

Lemma 1.5. Let $m \leq n$. Then the number of $m \times n$ matrices of rank m over $\mathbb{F}_q$ is

$$n(m, n) = q^{m(m-1)/2} \prod_{i=n-m+1}^{n} (q^i - 1).$$

Proof. Let P be an $m \times n$ matrix of rank m. Write

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix},$$

where $v_1, v_2, \ldots, v_m$ are the m rows of P. If we want to enumerate how many possible P's there are, it is sufficient to enumerate how many possible v_1's there are, and once a v_1 is chosen to enumerate how many possible v_2's there are, etc. Now, v_1 can be any non-zero vector of $\mathbb{F}_q^{(n)}$, thus there are $q^n - 1$ possible choices for v_1. Once v_1 is chosen, v_2 can be any vector of $\mathbb{F}_q^{(n)}$ which is linearly independent with v_1, thus there are $q^n - q$ possible choices of v_2. After v_1 and v_2 are chosen, v_3 can be any vector of $\mathbb{F}_q^{(n)}$ which is linearly independent with v_1 and v_2; since v_1 and v_2 are linearly independent, there are $q^n - q^2$ possible choices of v_3. Proceeding in this way, finally after $v_1, v_2, \ldots, v_{m-1}$ are chosen, v_m can be any vector of $\mathbb{F}_q^{(n)}$ which is linearly independent with $v_1, v_2, \ldots, v_{m-1}$; but since $v_1, v_2, \ldots, v_{m-1}$ are linearly independent, there are $q^n - q^{m-1}$ possible choices of v_m. Therefore

$$\begin{aligned} n(m,n) &= (q^n - 1)(q^n - q)(q^n - q^2)\ldots(q^n - q^{m-1}) \\ &= q^{m(m-1)/2} \prod_{i=n-m+1}^{n} (q^i - 1). \end{aligned}$$

□

From Lemma 1.5 follows immediately

Theorem 1.6.

$$| \, GL_n(\mathbb{F}_q) \, | = q^{n(n-1)/2} \prod_{i=1}^{n} (q^i - 1).$$

Proof. In fact,

$$| \, GL_n(\mathbb{F}_q) \, | = n(n,n).$$

□

Theorem 1.7. Let $0 \le m \le n$. Then the number of m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ is[1]

$$N(m,n) = \frac{\prod_{i=n-m+1}^{n} (q^i - 1)}{\prod_{i=1}^{m} (q^i - 1)}. \tag{1.1}$$

Proof. In fact, for $m \ge 1$ we have

$$N(m,n) = \frac{n(m,n)}{\mid GL_m(\mathbb{F}_q) \mid}.$$

For $m = 0$, clearly $N(0,n) = 1$ and the right-hand side of (1.1) also equals 1 by our convention. □

For brevity sometimes we use the *Gaussian coefficient*

$$\begin{bmatrix} n \\ m \end{bmatrix}_q = \frac{\prod_{i=n-m+1}^{n} (q^i - 1)}{\prod_{i=1}^{m} (q^i - 1)},$$

where m, n are integers, $m > 0$, and q is a power of a prime. By convention $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = 1$ for all integer n. Then we have

$$N(m,n) = \begin{bmatrix} n \\ m \end{bmatrix}_q.$$

Corollary 1.8. Let $0 \le k \le m \le n$ and $N(k,m,n)$ be the number of k-dimensional vector subspaces contained in a given m-dimensional vector subspace of $\mathbb{F}_q^{(n)}$. Then

$$N(k,m,n) = N(k,m) = \begin{bmatrix} m \\ k \end{bmatrix}_q.$$

□

[1] We adopt the convention that

$$\prod_{i \in \phi} f(i) = 1,$$

where ϕ denotes the empty set.

Corollary 1.9. Let $0 \le k \le m \le n$. Then the number $N'(k,m,n)$ of m-dimensional vector subspaces containing a given k-dimensional vector subspace of $\mathbb{F}_q^{(n)}$ is equal to

$$\begin{bmatrix} n-k \\ m-k \end{bmatrix}_q.$$

Proof. By Theorem 1.3 $N'(k,m,n)$ is independent of the particular k-dimensional vector subspace chosen. Define a $(0,1)$-matrix, whose rows are indexed by the k-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ (they are $N(k,n)$ in number), whose columns are indexed by the m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ (they are $N(m,n)$ in number), and with a 1 or 0 in the (i,j) position of the matrix, if the i-th k-dimensional vector subspace is or is not contained in the j-th m-dimensional vector subspace, respectively. If we count the number of 1's in the matrix by rows, we get $N'(k,m,n)\,N(k,n)$, where $N'(k,m,n)$ is the number of 1's in each row and $N(k,n)$ is the number of rows. If we count the number of 1's in the matrix by columns, we get $N(k,m)\,N(m,n)$, where $N(k,m)$ is the number of 1's in each column and $N(m,n)$ is the number of columns. Thus we must have

$$N'(k,m,n)\,N(k,n) = N(k,m)\,N(m,n),$$

from which we deduce

$$N'(k,m,n) = \begin{bmatrix} n-k \\ m-k \end{bmatrix}_q.$$

□

Let $0 \le r \le \min\{m,n\}$. Denote the number of $m \times n$ matrices of rank r over $\mathbb{F}_q$ by $n(r, m \times n)$. Then as an application of Theorem 1.6 and a generalization of Lemma 1.5 we have

Theorem 1.10. Let $0 \le r \le \min\{m,n\}$. Then the number of $m \times n$ matrices of rank r over $\mathbb{F}_q$ is

$$n(r, m \times n) = q^{r(r-1)/2} \begin{bmatrix} m \\ r \end{bmatrix}_q \prod_{i=n-r+1}^{n} (q^i - 1).$$

□

The proof is left as an exercise (Exercise 1.4).

1.2 Affine Spaces and Affine Groups over Finite Fields

Let $\mathbb{F}_q^{(n)}$ be the n-dimensional row vector space over $\mathbb{F}_q$. The vectors in $\mathbb{F}_q^{(n)}$, both the non-zero vectors and the zero vector, will now be called *points*; the cosets of $\mathbb{F}_q^{(n)}$ relative to any 1-dimensional vector subspace, *lines*; the cosets of $\mathbb{F}_q^{(n)}$ relative to any 2-dimensional vector subspace, *planes*; and the cosets of $\mathbb{F}_q^{(n)}$ relative to any $(n-1)$-dimensional vector subspace, *hyperplanes*. More generally, the cosets of $\mathbb{F}_q^{(n)}$ relative to any r-dimensional vector subspace $(0 \le r \le n)$ will now be called *affine r-flats*, or simply *r-flats*. In particular, 0-flats are points, 1-flats are lines, 2-flats are planes, and $(n-1)$-flats are hyperplanes. An r-flat is said to be *incident* with an s-flat, if the r-flat contains or is contained in the s-flat. Then the point set $\mathbb{F}_q^{(n)}$ with the affine r-flats $(0 \le r \le n)$ and the incidence relation among them defined above is called the n-dimensional *affine space* over $\mathbb{F}_q$ and denoted by $AG(n, \mathbb{F}_q)$. Moreover, the point $(x_1, x_2, \ldots, x_n)$ of $AG(n, \mathbb{F}_q)$ is said to have coordinates $x_1, x_2, \ldots, x_n$. The *dimension* of an r-flat is defined to be r.

Theorem 1.11. A hyperplane of $AG(n, \mathbb{F}_q)$ can be regarded as the solution set of a linear non-homogeneous equation, and conversely.

Proof. Let $U+v$ be a hyperplane, where U is an $(n-1)$-dimensional vector subspace of $\mathbb{F}_q^{(n)}$ and v a vector of $\mathbb{F}_q^{(n)}$. Choose a matrix representation of U, call it U again, then U is an $(n-1) \times n$ matrix of rank $n-1$. From the theory of linear homogeneous equations we know that the solutions of the system of linear homogeneous equations

$$U \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = 0$$

form a 1-dimensional subspace. Let $(a_1, a_2, \ldots, a_n)$ be a non-zero vector in the solution space, then the vector subspace U is the solution space of the linear homogeneous equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0.$$

Let $v = (v_1, v_2, \ldots, v_n)$ and put

$$b = a_1v_1 + a_2v_2 + \cdots + a_nv_n.$$

Then $U + v$ is exactly the set of solutions of the linear non-homogeneous equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

Conversely, let

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \tag{1.2}$$

be a linear non-homogeneous equation. Then the set of solutions of the corresponding linear homogeneous equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$$

is an $(n-1)$-dimensional vector subspace U of $\mathbb{F}_q^{(n)}$. Let $(v_1, v_2, \ldots, v_n)$ be a particular solution of (1.2), then the set of solutions of (1.2) is the hyperplane $U + v$. □

Recall that a system of r linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\cdots \\ a_{r1}x_1 + a_{r2}x_2 + \cdots + a_{rn}x_n &= b_r \end{aligned}$$

is said to be *independent*, if the vectors $(a_{11}, a_{12}, \ldots, a_{1n})$, $(a_{21}, a_{22}, \ldots, a_{2n})$, $\ldots$, $(a_{r1}, a_{r2}, \ldots, a_{rn})$ are linearly independent, and is said to be *consistent*, if the system has a common solution.

More generally, we have

Theorem 1.12. Any r-flat $(0 \le r \le n)$ in $AG(n, \mathbb{F}_q)$ can be regarded as the set of solutions of a system of $n - r$ independent and consistent linear non-homogeneous equations, and conversely. □

The proof of Theorem 1.12 follows also from the theory of linear equations in the same way as Theorem 1.11 and will be left as an exercise.

The 1-dimensional affine space over $\mathbb{F}_q$ consists only of q points and is less interesting. So let us start our discussion with the 2-dimensional affine space $AG(2, \mathbb{F}_q)$, which is usually called the *affine plane* over $\mathbb{F}_q$. There are two kinds of geometric objects in $AG(2, \mathbb{F}_q)$: points and lines. A point is an ordered pair (x, y), where $x, y \in \mathbb{F}_q$ are the coordinates of the point, and a line is a set of points whose coordinates satisfy a linear equation

$$ax + by = c,$$

where $a, b, c \in \mathbb{F}_q$, and a and b are not both zero, called the equation of the line. The equation of a line is uniquely determined up to a non-zero constant multiple of $\mathbb{F}_q$. Two distinct lines of $AG(2, \mathbb{F}_q)$ are said to be *parallel* if they have no point in common. Parallelism is a very important concept in affine geometry.

Theorem 1.13. In $AG(2, \mathbb{F}_q)$,

(i) Through any two distinct points there passes exactly one line. More precisely, the equation of the line passing through two distinct points (x_1, y_1) and (x_2, y_2) is

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0. \tag{1.3}$$

(ii) Either two distinct lines are parallel or they intersect in exactly one point. More precisely, let

$$l_1 : a_1 x + b_1 y = c_1 \tag{1.4}$$

and

$$l_2 : a_2 x + b_2 y = c_2 \tag{1.5}$$

be two distinct lines, then they are parallel or intersect if and only if the matrix

$$\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$$

is of rank 1 or 2, respectively.

(iii) From any point P there passes exactly one line parallel to a given line not containing P.

(iv) If two lines l_1 and l_2 are each parallel to the line l, then l_1 and l_2 are parallel to each other.

Proof. (i) Clearly both (x_1, y_1) and (x_2, y_2) lie on the line with equation (1.3). This proves the existence of a line passing through (x_1, y_1) and (x_2, y_2). Let

$$ax + by = c \tag{1.6}$$

be the equation of a line passing through the points (x_1, y_1) and (x_2, y_2). Then

$$ax_1 + by_1 = c,$$
$$ax_2 + by_2 = c.$$

Solving this system of linear homogeneous equations in a, b, c we get

$$a : b : c = \begin{vmatrix} y_1 & 1 \\ y_2 & 1 \end{vmatrix} : \begin{vmatrix} 1 & x_1 \\ 1 & x_2 \end{vmatrix} : - \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}.$$

Therefore line (1.6) coincides with line (1.3). This proves the uniqueness.

(ii) If

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0,$$

then Cramer's rule gives the unique point of intersection of l_1 and l_2. However, if

$$\operatorname{rank} \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = 1,$$

since l_1 and l_2 are distinct,

$$\operatorname{rank} \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} = 2,$$

and the linear equations (1.4) and (1.5) have no common solution, i.e., the lines l_1 and l_2 are parallel.

(iii) Let P be a point with coordinates (x_1, y_1) and l be a line with equation

$$l : ax + by = c.$$

Let $ax_1 + by_1 = d$, then

$$ax + by = d$$

is the equation of the unique line passing through P and parallel to l.

(iv) Let l_1, l_2, and l have equations

$$l_1 : a_1x + b_1y = c_1,$$
$$l_2 : a_2x + b_2y = c_2,$$

and

$$l : ax + by = c,$$

respectively. l_1 is parallel to l if and only if (a_1, b_1) and (a, b) are linear dependent, and l_2 is parallel to l if and only if (a_2, b_2) and (a, b) are linearly dependent. Since $(a_1, b_1), (a_2, b_2)$ and (a, b) are three non-zero vectors, from the linear dependence of (a_1, b_1) and (a, b), and of (a_2, b_2) and (a, b) we deduce the linear dependence of (a_1, b_1) and (a_2, b_2), thus l_1 and l_2 are parallel. □

Theorem 1.14. (i) There are q^2 points and $q^2 + q$ lines in $AG(2, \mathbb{F}_q)$.

(ii) There are exactly q points on each line.

(iii) The $q^2 + q$ lines of $AG(2, \mathbb{F}_q)$ can be subdivided into $q+1$ subsets, each consisting of q lines, such that any two lines of the same subset are parallel to each other, and any two lines from different sets intersect. Each set is called a *pencil* of parallel lines. Through any point there passes exactly $q + 1$ lines, each froms a parallel pencil.

Proof. (i) A point of $AG(2, \mathbb{F}_q)$ is an ordered pair of elements of $\mathbb{F}_q$, therefore there are q^2 points in $AG(2, \mathbb{F}_q)$. A line of $AG(2, \mathbb{F}_q)$ is the set of points whose coordinates satisfy a linear equation and two linear equations have the same solution set if and only if they are proportional. There are $(q^2 - 1)q$ linear equations in two indeterminates over $\mathbb{F}_q$, and hence there are $(q^2 - 1)q/(q - 1) = q^2 + q$ lines in $AG(2, \mathbb{F}_q)$.

(ii) Since a linear equation in two indeterminates over $\mathbb{F}_q$ has q solutions, a line has q points.

(iii) There are $(q^2 - 1)/(q - 1) = q + 1$ mutually non-proportional non-zero ordered pairs of elements of $\mathbb{F}_q$. For each non-zero ordered pair (a, b), by Theorem 1.13 (ii) the q lines

$$ax + by = c,$$

where c runs through $\mathbb{F}_q$, form a set of parallel lines. For any two lines $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ from different sets, i.e., (a_1, b_1) and (a_2, b_2) are not proportional, by Theorem 1.13 (ii) they must intersect. Therefore the first statement of (iii) is proved. Let (x_0, y_0) be a point. Suppose that $ax + by = c$ is a line passing through (x_0, y_0), i.e.,

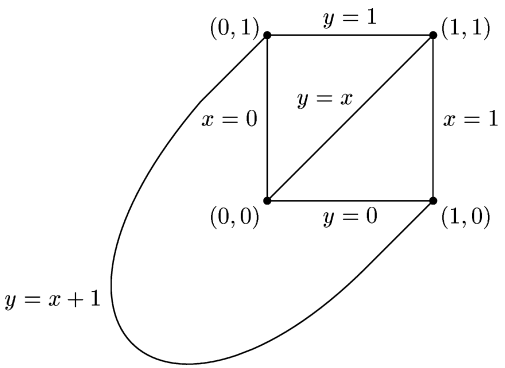
$$ax_0 + by_0 = c.$$

Regard the above equation as a linear homogeneous equation in three indeterminates a, b, and c, it has $q^2 - 1$ non-zero solutions. Two non-zero

solutions give equations of the same line if and only if they are proportional. There are $(q^2-1)/(q-1) = q+1$ mutually non-proportional non-zero solutions. Hence there are $q+1$ lines passing through (x_0, y_0) and they are from different pencils of parallel lines. □

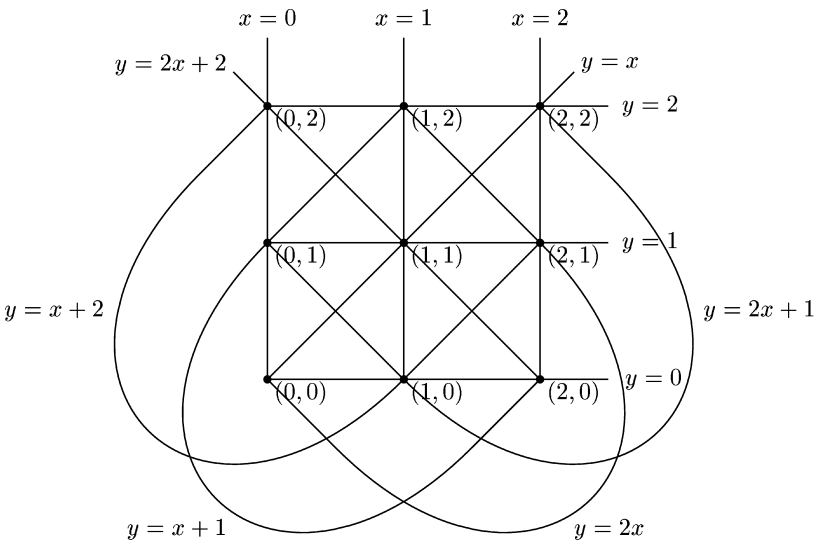
We illustrate Theorem 1.14 with the following examples.

Example 1.1. $AG(2, \mathbb{F}_2)$



There are $2^2 = 4$ points and $2^2 + 2 = 6$ lines in $AG(2, \mathbb{F}_2)$. The 6 lines are divided into 3 different pencils of parallel lines, each consisting of 2 parallel lines. □

Example 1.2. $AG(2, \mathbb{F}_3)$



There are $3^2 = 9$ points and $3^2 + 3 = 12$ lines in $AG(2, \mathbb{F}_3)$. The 12 lines are divided into 4 different pencils of parallel lines, each consisting of 3 lines. □

Now let us give an application of $AG(2, \mathbb{F}_q)$ to authentication codes.

Example 1.3. At first we fix a line l in $AG(2, \mathbb{F}_q)$. The points on l are regarded as *source states*, the points not on l are regarded as *encoding rules*, and the lines not parallel to l and equal to l are regarded as *messages*. Assume that the encoding rules work as follows. Given a source state s (which is a point on l) and an encoding rule e (which is a point not on l), there is a unique line m passing through s and e, then we define $m = e(s)$, i.e., the source state s is encoded by the encoding rule e into the message m. Thus we obtain an *authentication code*.

Let us compute the parameters of the code. We adopt the usual notation that $\mathcal{S}$ represents the set of source states, $\mathcal{M}$ represents the set of messages, and $\mathcal{E}$ represents the set of encoding rules. $AG(2, \mathbb{F}_q)$ has q^2 points and any line of it has q points. Therefore there are q source states and $q^2 - q$ encoding rule, i.e., $| \mathcal{S} | = q$ and $| \mathcal{E} | = q^2 - q$. $AG(2, \mathbb{F}_q)$ has $q^2 + q$ lines and a pencil of parallel lines consists of q lines. Hence there are $(q^2 + q) - q = q^2$ messages, i.e., $| \mathcal{M} | = q^2$.

Now assume that an equally likely probability distribution is given on the set $\mathcal{E}$ of encoding rules, i.e., each encoding rule has a probability $1/(q^2 - q)$ to be chosen by the transmitter/receiver. Suppose that an opponent plays an *impersonation game*. Let us compute the probability P_I that a fraudulent message m sent by the opponent is accepted by the receiver as authentic. A message m is a line which intersects with the set $\mathcal{S} = l$ of source states, hence it contains $q - 1$ points not on l, i.e., $q - 1$ encoding rules. If the encoding rule e used by the transmitter/receiver falls on the line m, then m will be accepted by the receiver as authentic. There are altogether $q^2 - q$ encoding rules, and there are $q - 1$ possibilities that the encoding rule e used by the transmitter/receiver falls on the line m. Hence

$$P_I = \frac{q - 1}{q^2 - q} = \frac{1}{q}.$$

Then suppose that the opponent plays a *substitution game*. After he observed a message m sent by the transmitter to the receiver he sends a fraudulent message m' to the receiver. Of course, he chooses m' not parallel to m and equal to m. Let us compute the probability P_S that the receiver will accept m' as authentic. Once the opponent observed a

message m, the encoding rule e used by the transmitter/receiver must lie on the line m; since e can be any point on m except the point of intersection of m and $\mathcal{S} = l$, there are $q-1$ possibilities. That he chooses m' not parallel to m and equal to m to be sent to the receiver as a fraudulent message means that he guesses the encoding rule to be the point of intersection of m and m', and there is a unique possibility. Hence

$$P_S = \frac{1}{q-1}.$$

□

Now let us come to the study of $AG(n, \mathbb{F}_q)$.

Theorem 1.15. In $AG(n, \mathbb{F}_q)$,

(i) Any two distinct points are joined by exactly one line.

(ii) Any three non-collinear points lie on exactly one plane.

(iii) Any $r+1$ points, not lying on any $(r-1)$-flat, lie on exactly one r-flat $(0 \leq r \leq n)$.

Proof. (i) Let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ be two distinct points in $AG(n, \mathbb{F}_q)$. Regarding $x - y = (x_1 - y_1, x_2 - y_2, \ldots, x_n - y_n)$ as a non-zero vector of $\mathbb{F}_q^{(n)}$, it spans a 1-dimensional vector subspace $\langle x - y \rangle$. Then $\langle x - y \rangle + y$ is a line of $AG(n, \mathbb{F}_q)$ which contains both x and y. This proves the existence of a line passing through x and y.

Now, let $U + u$ be a line in $AG(n, \mathbb{F}_q)$, where U is a 1-dimensional vector subspace of $\mathbb{F}_q^{(n)}$ and $u \in \mathbb{F}_q^{(n)}$. Assume that $x, y \in U+u$. Then $x = u_1+u$, $y = u_2+u$, where $u_1, u_2 \in U$. Hence $U+u = U+y$ and $x-y = u_1-u_2 \in U$. Therefore $U = \langle x-y \rangle$ and $U+u = \langle x-y \rangle + y$. This proves the uniqueness of the line passing through x and y.

Since (ii) is a special case of (iii), it is enough to prove (iii).

(iii) Let $x^{(i)} = (x_1^{(i)}, x_2^{(i)}, \ldots, x_n^{(i)})$, $i = 1, 2, \ldots, r+1$, be $r+1$ points in $AG(n, \mathbb{F}_q)$ and assume that they do not lie on any $(r-1)$-flat. Let

$$v^{(i)} = x^{(i)} - x^{(r+1)} = (x_1^{(i)} - x_1^{(r+1)}, x_2^{(i)} - x_2^{(r+1)}, \ldots, x_n^{(i)} - x_n^{(r+1)}),$$
$$i = 1, 2, \ldots, r,$$

and V be the vector subspace spanned by $v^{(1)}, v^{(2)}, \ldots, v^{(r)}$. Clearly $\dim V \leq r$ and $V + x^{(r+1)}$ is an affine flat containing $x^{(1)}, x^{(2)}, \ldots, x^{(r)}, x^{(r+1)}$. Since $x^{(1)}, x^{(2)}, \ldots, x^{(r+1)}$ do not lie on any $(r-1)$-flat, $\dim V = r$ and $V + x^{(r+1)}$ is an r-flat.

Now let $U + u$ be an r-flat containing $x^{(1)}, x^{(2)}, \ldots, x^{(r+1)}$, where U is an r-dimensional vector subspace and $u \in \mathbb{F}_q^{(n)}$. Then $x^{(i)} = u^{(i)} + u$, where $u^{(i)} \in U,\ i = 1, 2, \ldots, r+1$. Thus $U + u = U + x^{(r+1)}$ and $x^{(i)} - x^{(r+1)} = u^{(i)} - u^{(r+1)} \in U,\ i = 1, 2, \ldots, r$. Therefore U contains the vector subspace V spanned by $x^{(1)} - x^{(r+1)}, x^{(2)} - x^{(r+1)}, \ldots, x^{(r)} - x^{(r+1)}$. Since $\dim U = \dim V = r$, $U = V$ and $U + u = U + x^{(r+1)} = V + x^{(r+1)}$. □

According to Theorem 1.12 we can use a system of $n - 1$ independent and consistent linear non-homogeneous equations to describe a line of $AG(n, \mathbb{F}_q)$ as its solution set. However, another simple way to describe a line is its parametric representation. Let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ be two distinct points in $AG(n, \mathbb{F}_q)$. By the proof of Theorem 1.15, the line l passing through x and y is the set of points in the coset $\langle x - y \rangle + y$, where $\langle x - y \rangle$ is the 1-dimensional vector subspace spanned by the non-zero vector $x - y$. Hence the line passing through x and y is the point set

$$\{\lambda(x_1 - y_1, x_2 - y_2, \ldots, x_n - y_n) + (y_1, y_2, \ldots, y_n) \mid \lambda \in \mathbb{F}_q\},$$

or

$$\{\lambda(x_1, x_2, \ldots, x_n) + (1 - \lambda)(y_1, y_2, \ldots, y_n) \mid \lambda \in \mathbb{F}_q\},$$

or

$$\{\lambda(x_1, x_2, \ldots, x_n) + \mu(y_1, y_2, \ldots, y_n) \mid \lambda, \mu \in \mathbb{F}_q, \lambda + \mu = 1\}.$$

These are called the *parametric representations* of the line passing through x and y.

In a similar way, the plane passing through three non-collinear points $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, and $z = (z_1, z_2, \ldots, z_n)$ has the parametric representation

$$\{\lambda(x_1, x_2, \ldots, x_n) + \mu(y_1, y_2, \ldots, y_n) + \nu(z_1, z_2, \ldots, z_n) \mid \lambda, \mu, \nu \in \mathbb{F}_q, \\ \lambda + \mu + \nu = 1\}.$$

More generally, the r-flat containing $r + 1$ points $x^{(1)}, x^{(2)}, \ldots, x^{(r+1)}$, which do not lie in any $(r-1)$-flat, has the parametric representation

$$\{\lambda_1 x^{(1)} + \lambda_2 x^{(2)} + \ldots + \lambda_{r+1} x^{(r+1)} \mid \lambda_1, \lambda_2, \ldots, \lambda_{r+1} \in \mathbb{F}_q, \\ \lambda_1 + \lambda_2 + \ldots + \lambda_{r+1} = 1\}.$$

We call two flats $U + u$ and $V + v$, where U and V are vector subspaces of $\mathbb{F}_q^{(n)}$ and $u, v \in \mathbb{F}_q^{(n)}$, *parallel*, if $U \subset V$ or $V \subset U$.

Theorem 1.16.

(i) If $n \geq 3$, there exist two skew lines in $AG(n, \mathbb{F}_q)$, i.e., two lines which are neither parallel nor intersecting.

(ii) Through any point of $AG(n, \mathbb{F}_q)$ there passes exactly one r-flat parallel to a given r-flat.

(iii) Two r-flats both parallel to another r-flat are parallel to each other.

Proof. (i) Let $V_i = \langle e_i \rangle$ be the 1-dimensional vector subspace spanned by $e_i (1 \leq i \leq n)$. Then V_1 and $V_2 + (1, 1, \ldots, 1)$ are two skew lines.

(ii) Let x be a point of $AG(n, \mathbb{F}_q)$ and $U + u$ be an r-flat. Clearly $U + x$ is an r-flat passing through x and parallel to $U + u$. Furthermore, let $V + v$ be an r-flat passing through x and parallel to $U + u$. Since $x \in V + v$, $V + v = V + x$. Since $U + u$ and $V + v$ are parallel, $U \subset V$ or $V \subset U$, but both U and V are of dimension r, thus we have necessarily $U = V$. Hence $V + v = V + x = U + x$.

(iii) Let $U_1 + u_1$ and $U_2 + u_2$ be two r-flats and assume that both of them are parallel to another r-flat $V + v$. Then $\dim U_i = \dim V = r$, which together with the parallelism of $U_i + u_i$ and $V + v$ implies $U_i = V (i = 1, 2)$. Hence $U_1 = U_2$. Therefore $U_1 + u_1$ and $U_2 + u_2$ are parallel. □

From the Anzahl theorems in $\mathbb{F}_q^{(n)}$ we deduce immediately the following Anzahl theorems in $AG(n, \mathbb{F}_q)$.

Theorem 1.17. The number of m-flats in $AG(n, \mathbb{F}_q)$ for $0 \leq m \leq n$, is equal to

$$q^{n-m} \begin{bmatrix} n \\ m \end{bmatrix}_q .$$

In particular, there are q^n points and $q^{n-1}(q^n - 1)/(q - 1)$ lines in $AG(n, \mathbb{F}_q)$.

Proof. There are

$$\begin{bmatrix} n \\ m \end{bmatrix}_q$$

m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ and $\mathbb{F}_q^{(n)}$ has q^{n-m} cosets relative to each m-dimensional vector subspace. □

Theorem 1.18. The number of k-flats in $AG(n, \mathbb{F}_q)$ contained in a given m-flat, where $0 \leq k \leq m \leq n$, is equal to

$$q^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q .$$

Proof. Let $U + x$ be a given m-flat. Assume that $V + y$ is a k-flat such that $V + y \subset U + x$. Then $y \in U + x$ and $U + y = U + x$. Thus $V + y \subset U + y$ and $V \subset U$. Hence V is a k-dimensional vector subspace of the m-dimensional vector subspace U, and therefore there are

$$\begin{bmatrix} m \\ k \end{bmatrix}_q$$

possible choices of V. Once V is chosen as a k-dimensional vector subspace of U, any vector $u + x \in U + x$, where $u \in U$, gives a k-flat $V + (u + x) \subset U + (u + x) = U + x$. However, two choices $u_1 + x$ and $u_2 + x \in U + x$ produce the same k-flat $V + (u_1 + x) = V + (u_2 + x)$ if and only if $u_1 - u_2 \in V$, i.e., $V + u_1 = V + u_2$ and, i.e., u_1 and u_2 belong to the same coset of U relative to V. There are q^{m-k} cosets of U relative to V. Hence the theorem follows. □

Theorem 1.19. The number of m-flats in $AG(n, \mathbb{F}_q)$ containing a given k-flat, where $0 \leq k \leq m \leq n$, is equal to

$$\begin{bmatrix} n-k \\ m-k \end{bmatrix}_q .$$

Proof. Let $V + y$ be a given k-flat. Suppose $U + x$ is an m-flat such that $V + y \subset U + x$. Then by the proof of the above theorem, $U + x = U + y$ and $V \subset U$. Thus the number of m-flats containing the k-flat $V + y$ is the number of m-dimensional vector subspaces containing the k-dimensional vector subspace V, which is given by Corollary 1.9. □

Let $U + u$ be an r-flat and $V + v$ be an s-flat, i.e., U and V are vector subspaces of dimensions r and s, respectively, and $u, v \in \mathbb{F}_q^{(n)}$. The set of points belonging to both $U + u$ and $V + v$ is called the *intersection* of $U + u$ and $V + v$, which is denoted by $(U + u) \cap (V + v)$ and will be shown below in Theorem 1.20 to be a flat, if it is non-empty. It follows that the intersection of all flats containing two given flats $U + u$ and $V + v$ is a flat and is the minimum flat containing both $U + u$ and $V + v$, which is called the *join* of $U + u$ and $V + v$ and is denoted by $(U + v) \cup (V + u)$. Denote the dimension of a flat $U + u$ by $\dim(U + u)$.

Theorem 1.20. Let $U + u$ and $V + v$ be flats. Suppose that $(U + u) \cap (V + v) \neq \phi$. Then $(U + u) \cap (V + v)$ is also a flat and we have the *dimension formula*

$$\begin{aligned}&\dim (U + u) + \dim (V + v)\\ &\qquad = \dim ((U + u) \cap (V + v)) + \dim ((U + u) \cup (V + v)).\end{aligned}$$

Proof. Let $x \in (U + u) \cap (V + v)$. Then $U + u = U + x$, $V + v = V + x$. Consequently

$$(U + u) \cap (V + v) = (U + x) \cap (V + x) = (U \cap V) + x,$$

which is clearly a flat.

Denote the join of $U + u$ and $V + v$ by $W + w$, where W is a vector subspace of $\mathbb{F}_q^{(n)}$ and $w \in \mathbb{F}_q^{(n)}$. Let $x \in (U + u) \cap (V + v)$ as above. From $x \in U + u \subset W + w$ we deduce $U + u = U + x$ and $W + w = W + x$. Similarly, $V + v = V + x$. Then from $U + x \subset W + x$ and $V + x \subset W + x$ we deduce $U \subset W$ and $V \subset W$, respectively. Therefore $U \cup V \subset W$ and $(U \cup V) + x \subset W + x$. On the other hand, $U + x \subset (U \cup V) + x, V + x \subset (U \cup V) + x$; but by definition $W + x$ is the minimum flat which contains both $U + x$ and $V + x$, hence $W + x \subset (U \cup V) + x$. Consequently $(U \cup V) + x = W + x = (U + u) \cup (V + v)$.

Then by the dimension formula of vector subspaces of $\mathbb{F}_q^{(n)}$ (cf. Exercise 1.2), we have

$$\begin{aligned}&\dim (U + u) + \dim (V + v)\\ &\qquad = \dim U + \dim V\\ &\qquad = \dim (U \cap V) + \dim (U \cup V)\\ &\qquad = \dim ((U \cap V) + x) + \dim ((U \cup V) + x)\\ &\qquad = \dim ((U + u) \cap (V + v)) + \dim ((U + u) \cup (V + v)).\end{aligned}$$

□

Now we shall introduce the affine group. It is easy to verify that the set of matrices of the form

$$\begin{array}{c} \begin{matrix} n & 1 \end{matrix} \\ \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} \begin{matrix} n \\ 1 \end{matrix} \end{array}, \tag{1.7}$$

where $T \in GL_n(\mathbb{F}_q)$ and $v = (a_1, a_2, \ldots, a_n) \in \mathbb{F}_q^{(n)}$, forms a group under matrix multiplication. This group is called the *affine group* of the n-dimensional affine space over $\mathbb{F}_q$ and will be denoted by $AGL_n(\mathbb{F}_q)$. We

define an *action* of $AGL_n(\mathbb{F}_q)$ on the n-dimensional affine space $AG(n, \mathbb{F}_q)$ as follows:

$$AG(n, \mathbb{F}_q) \times AGL_n(\mathbb{F}_q) \to AG(n, \mathbb{F}_q)$$
$$\left((x_1, x_2, \ldots, x_n), \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}\right) \mapsto (x_1, x_2, \ldots, x_n)T + (a_1, a_2, \ldots, a_n).$$

We say that the element (1.7) determines the *affine transformation*

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_n)T + (a_1, a_2, \ldots, a_n). \tag{1.8}$$

We usually do not distinguish the matrix (1.7) and the affine transformation (1.8) determined by (1.7). Then the elements of $AGL_n(\mathbb{F}_q)$ are also called affine transformations. Notice that the affine transformation (1.8) is a nonsingular linear non-homogeneous transformation, by nonsingularity of (1.8), we mean that $\det T \neq 0$. If $T = (t_{ij})_{1 \leq i,j \leq n}$, then (1.8) can also be written explicitly as

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1 t_{11} + \cdots + x_n t_{n1} + a_1, x_1 t_{12} + \cdots + x_n t_{n2} + a_2,$$
$$\ldots, x_1 t_{1n} + \cdots + x_n t_{nn} + a_n).$$

Two geometric figures in affine space are said to be *affinely equivalent*, if one of them can be carried into the other by an affine transformation.

According to Felix Klein's *Erlangen Program* (cf. Section 1.5), the n-dimensional *affine geometry* is the study of properties of geometric figures in $AG(n, \mathbb{F}_q)$ which are invariant under the n-dimensional affine group.

Let us now study the transitivity properties of the affine group $AGL_n(\mathbb{F}_q)$ on the flats of $AG(n, \mathbb{F}_q)$.

Theorem 1.21. (i) The affine group $AGL_n(\mathbb{F}_q)$ carries r-flats in $AG(n, \mathbb{F}_q)$ into r-flats and preserves parallelism. Moreover, $AGL_n(\mathbb{F}_q)$ is transitive on the set of r-flats in $AG(n, \mathbb{F}_q)$. In particular, $AGL_n(\mathbb{F}_q)$ is transitive on the set of points of $AG(n, \mathbb{F}_q)$.

(ii) $AGL_n(\mathbb{F}_q)$ is doubly transitive on the set of points of $AG(n, \mathbb{F}_q)$.

Proof. (i) Let $U + u$ be an r-flat in $AG(n, \mathbb{F}_q)$, that is, U is an r-dimensional vector subspace of $\mathbb{F}_q^{(n)}$ and $u \in \mathbb{F}_q^{(n)}$. Under the affine transformation (1.8), $U + u$ is carried into $UT + (uT + v)$ which is also an r-flat.

Let $U_1 + u_1$ be an r-flat and $U_2 + u_2$ an s-flat in $AG(n, \mathbb{F}_q)$. Assume that they are parallel, and for definiteness we assume that $U_1 \supset U_2$. Under the

affine transformation (1.8), $U_1 + u_1$ and $U_2 + u_2$ are carried into $U_1T + (u_1T + v)$ and $U_2T + (u_2T + v)$, respectively. From $U_1 \supset U_2$ we deduce $U_1T \supset U_2T$, hence $U_1T + (u_1T + v)$ and $U_2T + (u_2T + v)$ are parallel.

Now let $U_1 + u_1$ and $U_2 + u_2$ be two r-flats in $AG(n,\mathbb{F}_q)$. By Theorem 1.3, there is an element $T \in GL_n(\mathbb{F}_q)$ such that $U_1T = U_2$. Then the affine transformation

$$\begin{pmatrix} T & 0 \\ u_2 - u_1T & 1 \end{pmatrix}$$

carries $U_1 + u_1$ into $U_2 + u_2$. This proves the transitivity of $AGL_n(\mathbb{F}_q)$ on the set of r-flats.

(ii) To prove the double transitivity of $AGL_n(\mathbb{F}_q)$ on the set of points of $AG(n,\mathbb{F}_q)$ it is sufficient to prove that any ordered pair of points v_1 and v_2 in $AG(n,\mathbb{F}_q)$ can be carried under the affine group $AGL_n(\mathbb{F}_q)$ to a particular ordered pair of points, say $(0,0,\ldots,0)$ and e_1. Clearly the affine transformation

$$\begin{pmatrix} I & 0 \\ -v_1 & 1 \end{pmatrix}$$

carries v_1 and v_2 into $(0,0,\ldots,0)$ and $v_2 - v_1$, respectively. By Corollary 1.2 there is an element $T \in GL_n(\mathbb{F}_q)$ such that $(v_2 - v_1)T = e_1$, then the product of affine transformations

$$\begin{pmatrix} I & 0 \\ -v_1 & 1 \end{pmatrix} \begin{pmatrix} T & 0 \\ 0 & 1 \end{pmatrix}$$

carries v_1 and v_2 into $(0,0,\ldots,0)$ and e_1, respectively. □

From Theorem 1.21 it follows that the property a geometric figure being an r-flat is invariant under the affine group, any two r-flats are affinely equivalent, and the set of r-flats is an orbit under the affine group.

1.3 Quadrics in $AG(n,\mathbb{F}_q)$ for q Odd

The set of points $(x_1, x_2, \ldots, x_n)$ in $AG(n,\mathbb{F}_q)$ which satisfy a quadratic equation

$$\sum_{1 \le i \le k \le n} a_{ik} x_i x_k + \sum_{j=1}^{n} b_j x_j + c = 0, \tag{1.9}$$

where a_{ik} $(1 \le i \le k \le n)$, b_j $(1 \le j \le n)$, and c are elements of $\mathbb{F}_q$ and not all a_{ik} are zero, is called a *quadric* in $AG(n,\mathbb{F}_q)$ and the quadratic

equation (1.9) is called its equation. We shall discuss the following two problems:

(i) How can the equation of a quadric be transformed into the simplest form under the affine group? This leads to the classification of quadrics under the affine group.

(ii) Determining the number of points on a quadric.

In this section we shall restrict ourselves to the case when $\mathbb{F}_q$ is of characteristic $\neq 2$, that is, q is a power of an odd prime and leave the case when $\mathbb{F}_q$ is of characteristic 2 to the next section.

From now on we assume that q is a power of an odd prime. Then we may write $a_{ik} = a'_{ik} + a'_{ki}$ with $a'_{ik} = a'_{ki}$ $(1 \leq i < k \leq n)$ in (1.9). If we drop the primes, we get

$$\sum_{i,k=1}^{n} a_{ik}x_ix_k + \sum_{j=1}^{n} b_jx_j + c = 0, \tag{1.10}$$

where

$$a_{ik} = a_{ki} \ \ (1 \leq i, \ k \leq n).$$

Let $b = (b_1, b_2, \ldots, b_n)$ be the row vector whose components are the coefficients of the first degree terms of (1.10), and introduce the matrices

$$A = (a_{ik})_{1 \leq i,k \leq n} \tag{1.11}$$

and

$$\begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2}b & c \end{pmatrix}, \tag{1.12}$$

where ${}^t b$ denotes the transpose of b. These matrices are called the coefficient matrix of the second degree terms and the coefficient matrix of (1.10), respectively. We call a square matrix S *symmetric*, if ${}^t S = S$. Then both matrices (1.11) and (1.12) are symmetric and (1.10) can be written as

$$(x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2}b & c \end{pmatrix} {}^t(x_1, x_2, \ldots, x_n, 1) = 0. \tag{1.13}$$

The affine transformation (1.8) can also be written as

$$(x_1, x_2, \ldots, x_n, 1) \mapsto (x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}. \tag{1.14}$$

Under the affine transformation (1.14) the equation (1.13) is carried into

$$(x_1,\ldots,x_n,1)\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}\begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2}b & c \end{pmatrix}{}^t\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}{}^t(x_1,\ldots,x_n,1)=0.$$

Thus the coefficient matrix A of the second degree terms is subjected to a cogredient transformation

$$A \mapsto TA\,{}^tT,\ T\in GL_n(\mathbb{F}_q)$$

and the coefficient matrix (1.12) is subjected to a special kind of cogredient transformation

$$\begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2}b & c \end{pmatrix}\mapsto\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}\begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2}b & c \end{pmatrix}{}^t\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}.$$

Therefore it is natural to study the normal forms of symmetric matrices relative to cogredience first.

Let S and S_1 be two $n\times n$ matrices over $\mathbb{F}_q$. If there is an $n\times n$ nonsingular matrix Q over $\mathbb{F}_q$ such that $QS\,{}^tQ=S_1$, we say that S is *cogredient* to S_1. If for any vector $x\in\mathbb{F}_q^{(n)}$ such that $xS\,{}^tx=0$ we have necessarily $x=0$, then S is said to be *definite*.

Clearly, matrices cogredient to a symmetric matrix are also symmetric and matrices cogredient to a definite matrix are also definite. Cogredient matrices have the same rank and the determinants of two cogredient nonsingular matrices differ by a non-zero square element of $\mathbb{F}_q$.

Denote the set of non-zero elements of $\mathbb{F}_q$ by $\mathbb{F}_q^*$. We know that $\mathbb{F}_q^*$ is a cyclic group of order $q-1$ with respect to field multiplication, called the *multiplicative group* of $\mathbb{F}_q$. We have assumed that q is odd, hence $\mathbb{F}_q^*$ is a group of even order and the set of square elements of $\mathbb{F}_q^*$, denoted by $\mathbb{F}_q^{*2}$, forms a subgroup of $\mathbb{F}_q^*$ of index 2. In the following we choose a fixed non-square element z in $\mathbb{F}_q^*$, then $\mathbb{F}_q^*=\mathbb{F}_q^{*2}\cup z\mathbb{F}_q^{*2}$ is the decomposition of the group $\mathbb{F}_q^*$ into disjoint cosets relative to the subgroup $\mathbb{F}_q^{*2}$.

Lemma 1.22. Let z be a non-square element of $\mathbb{F}_q^*$. Then the following two 2×2 diagonal matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\qquad\text{and}\qquad\begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}$$

are cogredient.

Proof. We distinguish the cases when $-1 \notin \mathbb{F}_q^{*2}$ and $-1 \in \mathbb{F}_q^{*2}$. Consider first the case when $-1 \notin \mathbb{F}_q^{*2}$. As x runs through $\mathbb{F}_q$, $1+x^2$ runs through $\frac{1}{2}(q+1)$ elements of $\mathbb{F}_q^*$. Thus there is an element $x \in \mathbb{F}_q$ such that $1+x^2$ is a non-square element of $\mathbb{F}_q^*$. Let $(1+x^2)y^2 = z$, where $y \in \mathbb{F}_q^*$. Then

$$\begin{pmatrix} xy & y \\ -y & xy \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} {}^t\!\begin{pmatrix} xy & y \\ -y & xy \end{pmatrix} = \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}.$$

Then consider the case when $-1 \in \mathbb{F}_q^{*2}$. Let $-1 = x^2$, where $x \in \mathbb{F}_q^*$. Then

$$\begin{pmatrix} \frac{1}{2}(1+z) & \frac{1}{2x}(1-z) \\ \frac{x}{2}(1-z) & \frac{1}{2}(1+z) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} {}^t\!\begin{pmatrix} \frac{1}{2}(1+z) & \frac{1}{2x}(1-z) \\ \frac{x}{2}(1-z) & \frac{1}{2}(1+z) \end{pmatrix} = \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}.$$

□

Theorem 1.23. Any $n \times n$ symmetric matrix S of rank r $(0 \le r \le n)$ over $\mathbb{F}_q$ is necessarily cogredient to either[2]

$$\underbrace{\begin{pmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & \ddots & & & & \\ & & & 1 & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix}}_{r \qquad n-r} \quad \text{or} \quad \underbrace{\begin{pmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & \ddots & & & & \\ & & & 1 & & & \\ & & & & z & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix}}_{r-1 \quad 1 \quad n-r}. \tag{1.15}$$

Moreover, these two matrices are not cogredient. Thus the normal forms (1.15) of $n \times n$ symmetric matrices relative to cogredience are uniquely determined.

Proof. At first we assert that any $n \times n$ symmetric matrix S of rank r is cogredient to a diagonal matrix

$$\begin{pmatrix} s_1 & & & & & & \\ & s_2 & & & & 0 & \\ & & \ddots & & & & \\ & & & s_r & & & \\ & & & & 0 & & \\ & 0 & & & & \ddots & \\ & & & & & & 0 \end{pmatrix},$$

[2]We adopt the convention that zeros in matrices are sometimes omitted, i.e., the blanks replace omitted zeros, if it is clear from the context.

where $s_1, s_2, \ldots, s_r \in \mathbb{F}_q^*$. If $S = 0$, the assertion holds automatically. Now suppose that $S \neq 0$. We may assume that there is a non-zero diagonal element in S. In fact, if all the diagonal elements of S are zero, there must be a non-zero non-diagonal element in S; without loss of generality we may assume that the element s_{12} in the $(1,2)$ position of S is non-zero. Then S is cogredient to

$$\begin{pmatrix} 1 & 1 & & \\ 0 & 1 & & \\ & & I & \end{pmatrix} \begin{pmatrix} 0 & s_{12} & * \\ s_{12} & 0 & \\ * & & * \end{pmatrix} {}^t\begin{pmatrix} 1 & 1 & & \\ 0 & 1 & & \\ & & I & \end{pmatrix} = \begin{pmatrix} 2s_{12} & s_{12} & * \\ s_{12} & 0 & \\ * & & * \end{pmatrix},$$

whose element in the $(1,1)$ position is non-zero. Now suppose that S has a non-zero element in (i,i) position. Then $e_i S\, {}^t e_i \neq 0$. The set of vectors in $\mathbb{F}_q^{(n)}$ which satisfy the linear equation

$$(x_1, x_2, \ldots, x_n) S\, {}^t e_i = 0$$

forms an $(n-1)$-dimensional vector subspace, denoted by P. Since $e_i \notin P$, the $n \times n$ matrix

$$\begin{pmatrix} e_i \\ P \end{pmatrix}$$

is nonsingular. We have

$$\begin{pmatrix} e_i \\ P \end{pmatrix} S\, {}^t\begin{pmatrix} e_i \\ P \end{pmatrix} = \begin{pmatrix} e_i S\, {}^t e_i & 0 \\ 0 & P S\, {}^t P \end{pmatrix},$$

where $e_i S\, {}^t e_i \neq 0$ and $P S\, {}^t P$ is an $(n-1) \times (n-1)$ symmetric matrix of rank $r-1$. Then our assertion follows from mathematical induction.

After rearranging $s_1, s_2, \ldots, s_r$ we can assume that $s_1, s_2, \ldots, s_t \in \mathbb{F}_q^{*2}$ and $s_{t+1}, s_{t+2}, \ldots, s_r \notin \mathbb{F}_q^{*2}$. Let $s \in \mathbb{F}_q^{*2}$, then there is at least one element in $\mathbb{F}_q^*$ whose square equals s, denote such an element by $s^{\frac{1}{2}}$ and its inverse by $s^{-\frac{1}{2}}$. Let

$$Q = \begin{pmatrix} s_1^{-\frac{1}{2}} & & & & & & & \\ & \ddots & & & & & & \\ & & s_t^{-\frac{1}{2}} & & & & & \\ & & & (s_{t+1}z^{-1})^{-\frac{1}{2}} & & & & \\ & & & & \ddots & & & \\ & & & & & (s_r z^{-1})^{-\frac{1}{2}} & & \\ & & & & & & 1 & \\ & & & & & & & \ddots \\ & & & & & & & & 1 \end{pmatrix}.$$

Then

$$Q\begin{pmatrix} s_1 & & & & & & \\ & s_2 & & & & & \\ & & \ddots & & & & \\ & & & s_r & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix} {}^tQ = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & z & & & \\ & & & & \ddots & & \\ & & & & & z & \\ & & & & & & 0 \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}.$$

$$\underbrace{\qquad}_{t}\underbrace{\qquad}_{r-t}\underbrace{\qquad}_{n-r}$$

Hence from Lemma 1.22 follows the first statement of Theorem 1.23.

The second statement of Theorem 1.23 follows immediately from the following lemma. □

Lemma 1.24. Let S be an $n \times n$ symmetric matrix of rank r $(1 \le r \le n)$. Suppose that S is cogredient to

$$\begin{matrix} r & n-r \end{matrix}$$
$$\begin{pmatrix} S_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix}.$$

Then $\det S_1 \neq 0$ and the coset $(\det S_1)\mathbb{F}_q^{*2}$ is uniquely determined by S.

Proof. Suppose that S is cogredient to both

$$\begin{matrix} r & n-r \end{matrix} \qquad \begin{matrix} r & n-r \end{matrix}$$
$$\begin{pmatrix} S_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix} \quad \text{and} \quad \begin{pmatrix} S_2 & 0 \\ 0 & 0 \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix}.$$

Since the rank of a matrix is invariant relative to cogredience, $\det S_1 \neq 0$ and $\det S_2 \neq 0$. There is an $n \times n$ nonsingular matrix

$$\begin{matrix} r & n-r \end{matrix}$$
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix}$$

such that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} S_1 & 0 \\ 0 & 0 \end{pmatrix} {}^t\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} S_2 & 0 \\ 0 & 0 \end{pmatrix},$$

from which follows

$$AS_1\,{}^tA = S_2.$$

Therefore A is nonsingular and $\det S_1(\det A)^2 = \det S_2$. Hence $(\det S_1)\mathbb{F}_q^{*2} = (\det S_2)\mathbb{F}_q^{*2}$. □

Now let us come to the following classification theorem of quadrics in $AG(n, \mathbb{F}_q)$.

Theorem 1.25. Any quadric in $AG(n, \mathbb{F}_q)$, where q is odd, is affinely equivalent to a quadric having one of the following quadratic equations as its equation

$$\sum_{i=1}^{r} x_i^2 = 0\,(1 \le r \le n), \tag{1.16}$$

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 = 0\,(r \text{ being even and } 2 \le r \le n), \tag{1.17}$$

$$\sum_{i=1}^{r} x_i^2 + 1 = 0\,(1 \le r \le n), \tag{1.18}$$

$$\sum_{i=1}^{r} x_i^2 + z = 0\,(r \text{ being odd and } 1 \le r \le n), \tag{1.19}$$

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + 1 = 0\,(r \text{ being even and } 2 \le r \le n), \tag{1.20}$$

$$\sum_{i=1}^{r} x_i^2 + x_{r+1} = 0\,(1 \le r \le n-1), \tag{1.21}$$

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + x_{r+1} = 0\,(r \text{ being even and } 2 \le r \le n-1). \tag{1.22}$$

Moreover, any two of these equations are not affinely equivalent, i.e., they cannot be carried into each other by an affine transformation or an affine transformation followed by a multiplication by an element of $\mathbb{F}_q^*$.

Proof. Suppose that a quadric in $AG(n, \mathbb{F}_q)$ is given, whose equation is (1.10)

$$\sum_{i,k=1}^{n} a_{ik}x_ix_k + \sum_{j=1}^{n} b_jx_j + c = 0,$$

where

$$a_{ik} = a_{ki} \ \ (1 \le i, \ k \le n).$$

We have introduced the vector

$$b = (b_1, b_2, \ldots, b_n)$$

and the matrices (1.11)

$$A = (a_{ik})_{1 \le i,k \le n}$$

and (1.12)

$$\begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2} b & c \end{pmatrix}.$$

The equation (1.10) has been written in the form (1.13)

$$(x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} A & \frac{1}{2}{}^t b \\ \frac{1}{2} b & c \end{pmatrix} {}^t(x_1, x_2, \ldots, x_n, 1) = 0.$$

By Theorem 1.23, there is an $n \times n$ nonsingular matrix Q such that

$$QA\,{}^tQ = \begin{pmatrix} \underbrace{\begin{matrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{matrix}}_{r-1} & & & \\ & \underbrace{d}_{1} & & \\ & & \underbrace{\begin{matrix} 0 & & \\ & \ddots & \\ & & 0 \end{matrix}}_{n-r} \end{pmatrix},$$

where $d = 1$ or z and r is the rank of A. Thus under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_n)Q$$

(1.10) is carried into

$$\sum_{i=1}^{r-1} x_i^2 + dx_r^2 + \sum_{j=1}^{n} b'_j x_j + c = 0. \tag{1.23}$$

Then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto \left(x_1 - \frac{b'_1}{2}, \ldots, x_{r-1} - \frac{b'_{r-1}}{2}, x_r - \frac{b'_r}{2d}, x_{r+1}, \ldots, x_n\right)$$

(1.23) is carried into

$$\sum_{i=1}^{r-1} x_i^2 + dx_r^2 + \sum_{j=r+1}^{n} b'_j x_j + c' = 0. \tag{1.24}$$

If all b'_j $(j = r+1, \ldots, n)$ are zero, then (1.24) becomes

$$\sum_{i=1}^{r-1} x_i^2 + dx_r^2 + c' = 0.$$

If $c' = 0$, then we have either

$$\sum_{i=1}^{r} x_i^2 = 0,$$

or

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 = 0.$$

The former case is (1.16). For the latter case, if r is odd, then $r-1$ is even, and by Lemma 1.22, $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 = 0$ can be transformed under an affine transformation into $z\sum_{i=1}^{r} x_i^2 = 0$. After canceling the non-square element z, we have $\sum_{i=1}^{r} x_i^2 = 0$ again. Therefore $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 = 0$ needs only to be listed for r being even; thus we have (1.17).

If $c' \neq 0$, then we have either

$$\sum_{i=1}^{r} x_i^2 + c' = 0, \tag{1.25}$$

or

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + c' = 0. \tag{1.26}$$

In the former case, if $c' \in \mathbb{F}_q^{*2}$, say $c' = t^2$ where $t \in \mathbb{F}_q^*$, then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (tx_1, \ldots, tx_r, x_{r+1}, \ldots, x_n)$$

(1.25) is carried into (1.18) multiplied by c'; thus after canceling the nonzero element c', we obtain (1.18). However, if $c' \notin \mathbb{F}_q^{*2}$, say $c' = t^2 z$, but r is even, then by Lemma 1.22, (1.25) can be carried under an affine transformation into (1.18) multiplied by $t^2 z$; thus after canceling $t^2 z$, we obtain (1.18) again. But if $c' \notin \mathbb{F}_q^{*2}$ and r is odd, (1.25) can be carried under an affine transformation into (1.19) subject to canceling some nonzero constant factor from the equation.

For (1.26), we can proceed in a similar way. If $c' \in \mathbb{F}_q^{*2}$, say $c' = t^2$, where $t \in \mathbb{F}_q^*$, then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (tx_1, \ldots, tx_r, x_{r+1}, \ldots, x_n)$$

(1.26) is carried into $c'(\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + 1) = 0$. After canceling c', we obtain $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + 1 = 0$. However, if r is odd, then $r-1$ is even and by Lemma 1.22 $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + 1 = 0$ is affinely equivalent to $z \sum_{i=1}^{r} x_i^2 + 1 = 0$. Multiplying $z \sum_{i=1}^{r} x_i^2 + 1 = 0$ by z, we obtain $z^2 \sum_{i=1}^{r} x_i^2 + z = 0$. Under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (z^{-1}x_1, z^{-1}x_2, \ldots, z^{-1}x_n)$$

$z^2 \sum_{i=1}^{r} x_i^2 + z = 0$ is carried into (1.19). Hence $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + 1 = 0$ needs to be listed only when r is even, this is (1.20). Now consider the case when $c' \notin \mathbb{F}_q^{*2}$, say $c' = t^2 z$. Under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (tx_1, \ldots, tx_r, x_{r+1}, \ldots, x_n)$$

(1.26) is carried into $t^2(\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + z) = 0$. Canceling t^2, we obtain

$$\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + z = 0. \tag{1.27}$$

If r is odd, then $r-1$ is even, and by Lemma 1.22, (1.27) is affinely equivalent to $z(\sum_{i=1}^{r} x_i^2 + 1) = 0$. Canceling z, we obtain (1.18) again. If r is even, then by Lemma 1.22, (1.27) is affinely equivalent to $x_1^2 + z(\sum_{i=2}^{r} x_i^2 + 1) = 0$. Multiplying by z^{-1}, we obtain $z^{-1}x_1^2 + \sum_{i=2}^{r} x_i^2 + 1 = 0$. Under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (zx_r, x_1, \ldots, x_{r-1}, x_{r+1}, \ldots, x_n)$$

$z^{-1}x_1^2 + \sum_{i=2}^{r} x_i^2 + 1 = 0$ is carried into (1.20).

Now let us consider the case when not all b'_j $(j = r+1, \ldots, n)$ in (1.24) are zero. Without loss of generality we can assume that $b'_{r+1} \neq 0$. Then

under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, \ldots, x_r, b_{r+1}^{'-1}(x_{r+1} - \sum_{j=r+2}^{n} b_j' x_j - c'), x_{r+2}, \ldots, x_n)$$

(1.24) is carried into

$$\sum_{i=1}^{r-1} x_i^2 + dx_r^2 + x_{r+1} = 0, \tag{1.28}$$

where $d = 1$ or z. If $d = 1$, (1.28) is (1.21). If $d = z$ and r is odd, by Lemma 1.22, (1.28) can be carried by an affine transformation into $z\sum_{i=1}^{r} x_i^2 + x_{r+1} = 0$. Under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, \ldots, x_r, zx_{r+1}, x_{r+2}, \ldots, x_n)$$

$z\sum_{i=1}^{r} x_i^2 + x_{r+1} = 0$ is carried into $z(\sum_{i=1}^{r} x_i^2 + x_{r+1}) = 0$. Canceling z, we obtain (1.21) again. Hence $\sum_{i=1}^{r-1} x_i^2 + zx_r^2 + x_{r+1} = 0$ needs to be listed only when r is even.

The first statement of Theorem 1.25 is now completely proved.

Let us come to the proof of the second statement. At first, it is clear that under affine transformations or multiplications by elements of $\mathbb{F}_q^*$, the ranks of the matrices (1.11) and (1.12) are invariant. From the invariance of the rank of A it follows that the equations in (1.16) – (1.22) with different r are not affinely equivalent. It is also clear that the coefficient matrices of the equations (1.16) and (1.17) are of rank r, of (1.18), (1.19), and (1.20) are of rank $r + 1$, and of (1.21) and (1.22) are of rank $r + 2$; hence if we regard (1.16) and (1.17) as the first group, (1.18), (1.19), and (1.20) as the second group, and (1.21) and (1.22) as the third group, then equations with the same r but from different groups are not affinely equivalent. Finally, it follows from Lemma 1.24 that equations (1.16) and (1.17) with the same even r, (1.18) and (1.19) with the same odd r, (1.18) and (1.20) with the same even r, and (1.21) and (1.22) with the same even r are not affinely equivalent. □

Now let us come to the study of the second problem proposed at the beginning of this section, i.e., to determine the number of points on a quadric. It is clear that the number of points on a quadric is invariant under affine transformations. Then it is sufficient to determine the number of points on each of the equations (1.16) – (1.22) listed in Theorem 1.25. We need one more definition.

The quadratic equation (1.9) in n variables and the quadric defined by it are said to be *non-degenerate*[3], if the equation cannot be transformed into an equation with less than n variables under affine transformations.

Thus the equations (1.16) – (1.20) are non-degenerate if $r = n$, and the equations (1.21) and (1.22) are non-degenerate if $r = n - 1$. It is also clear that to determine the number of points on a quadric defined by each of the equations (1.16) – (1.22) it is sufficient to determine the number of points on a non-degenerate quadric defined by each of the equations (1.16) – (1.22) which are non-degenerate. In order to facilitate the computation we shall use another normal form of symmetric matrices relative to cogredience.

Theorem 1.26. Let S be an $n \times n$ symmetric matrix of rank r. If r is odd, we write $r = 2\nu + 1$, then S is cogredient to either

$$\begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 1 & \\ & & & 0^{(n-r)} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & z & \\ & & & 0^{(n-r)} \end{pmatrix}$$

and these two matrices are not cogredient. If r is even, write $r = 2\nu$, then S is cogredient to either

$$\begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 0^{(n-r)} & \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & I^{(\nu-1)} & & & \\ I^{(\nu-1)} & 0 & & & \\ & & 1 & & \\ & & & -z & \\ & & & & 0^{(n-r)} \end{pmatrix}$$

and these two matrices are not cogredient.

Proof. It follows immediately from Theorem 1.23 and Lemma 1.24. For example, when $r = 2\nu + 1$ is odd, the determinants of

$$\begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 1 & \\ & & & 0^{(n-r)} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & z & \\ & & & 0^{(n-r)} \end{pmatrix}$$

are, respectively, $(-1)^\nu$ and $(-1)^\nu z$. If $(-1)^\nu \in \mathbb{F}_q^{*2}$, the first matrix is cogredient to the first of (1.15) and the second to the second of (1.15); and if $(-1)^\nu \notin \mathbb{F}_q^{*2}$, then ν must be odd, $(-1)^\nu z \in \mathbb{F}_q^{*2}$, and the first matrix is cogredient to the second of (1.15) and the second to the first of (1.15). □

[3]This definition also holds for q being even.

If an $n \times n$ symmetric matrix is cogredient to any one of the first three normal forms in Theorem 1.26, then ν is called its *index*; if it is cogredient to the last one, then $\nu - 1$ is called its *index*. It follows from Theorem 1.26 that the index of a symmetric matrix is invariant relative to cogredience.

Theorem 1.27. The numbers of points in $AG(n, \mathbb{F}_q)$ which lie on the non-degenerate quadrics (1.16), (1.17), (1.18), (1.19), (1.20), where $r = n$, and (1.21) and (1.22), where $r = n - 1$, are listed in Table 1.1.

Table 1.1

Equations	n odd	n even
$\sum_{i=1}^{n} x_i^2 = 0$	q^{n-1}	$q^{n-1} + q^{\frac{n}{2}} - q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ $q^{n-1} - q^{\frac{n}{2}} + q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$
$\sum_{i=1}^{n-1} x_i^2 + z x_n^2 = 0$ (n even)		$q^{n-1} - q^{\frac{n}{2}} + q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ $q^{n-1} + q^{\frac{n}{2}} - q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$
$\sum_{i=1}^{n} x_i^2 + 1 = 0$	$q^{n-1} - q^{\frac{n-1}{2}}$, if $-1 \notin \mathbb{F}_q^{*2}$ and $n \equiv 1 \pmod 4$ $q^{n-1} + q^{\frac{n-1}{2}}$, if $-1 \notin \mathbb{F}_q^{*2}$ and $n \equiv 3 \pmod 4$ or $-1 \in \mathbb{F}_q^{*2}$	$q^{n-1} - q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ $q^{n-1} + q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$
$\sum_{i=1}^{n} x_i^2 + z = 0$ (n odd)	$q^{n-1} + q^{\frac{n-1}{2}}$, if $-1 \notin \mathbb{F}_q^{*2}$ and $n \equiv 1 \pmod 4$ $q^{n-1} - q^{\frac{n-1}{2}}$, if $-1 \notin \mathbb{F}_q^{*2}$ and $n \equiv 3 \pmod 4$ or $-1 \in \mathbb{F}_q^{*2}$	
$\sum_{i=1}^{n-1} x_i^2 + z x_n^2 + 1 = 0$ (n even)		$q^{n-1} + q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ $q^{n-1} - q^{\frac{n}{2}-1}$, if $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$
$\sum_{i=1}^{n-1} x_i^2 + x_n = 0$	q^{n-1}	q^{n-1}
$\sum_{i=1}^{n-2} x_i^2 + z x_{n-1}^2 + x_n = 0$ (n odd)	q^{n-1}	

In proving Theorem 1.27, the following lemma will be used.

Lemma 1.28. The number of solutions of the equation

$$x^2 + \alpha y^2 = \beta, \qquad \alpha, \beta \in \mathbb{F}_q^*,$$

is equal to $q-1$ or $q+1$ if the 2×2 diagonal matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}$$

is indefinite or definite, respectively.

Proof. It is sufficient to consider the following two equations

$$x^2 - y^2 = \beta \text{ and } x^2 - zy^2 = \beta, \qquad \beta \in \mathbb{F}_q^*,$$

corresponding to the indefinite and definite cases, respectively.

Consider first $x^2 - y^2 = \beta$. Put $\xi = x + y$ and $\eta = x - y$, then the equation $x^2 - y^2 = \beta$ becomes $\xi\eta = \beta$. For any $\xi \neq 0$, there is a uniquely determined η such that $\xi\eta = \beta$. Thus all together there are $q-1$ solutions (ξ, η) for $\xi\eta = \beta$. Consequently, there are $q-1$ solutions (x, y) for $x^2 - y^2 = \beta$.

Consider next the case $x^2 - zy^2 = \beta$. Since z is a non-square element of $\mathbb{F}_q$, the quadratic polynomial $X^2 - z$ in the indeterminant X is irreducible over $\mathbb{F}_q$. Adjoining a root ζ of the polynomial $X^2 - z$ to $\mathbb{F}_q$, we obtain the finite field $\mathbb{F}_{q^2} = \mathbb{F}_q[\zeta]$, where $\zeta^2 = z$. The equation $x^2 - zy^2 = \beta$ can be written as $x^2 - \zeta^2 y^2 = \beta$ in $\mathbb{F}_{q^2}$, and also as $(x + \zeta y)(x - \zeta y) = \beta$, and the number of solutions of $x^2 - zy^2 = \beta$ is equal to the number of elements $x + \zeta y \in \mathbb{F}_q[\zeta]$ such that $(x + \zeta y)(x - \zeta y) = \beta$.

We assert that $z^{\frac{q-1}{2}} = -1$. Consider the map

$$\begin{aligned} \phi : \mathbb{F}_q^* &\to \{\pm 1\} \\ \gamma &\mapsto \gamma^{\frac{q-1}{2}}. \end{aligned}$$

It is evident that this is a group homomorphism. Under ϕ, the generator of the cyclic group $\mathbb{F}_q^*$ is mapped into -1, hence ϕ is surjective and the kernel of the homomorphism, Ker ϕ, is a subgroup of order $(q-1)/2$ of $\mathbb{F}_q^*$. Clearly $\mathbb{F}_q^{*2} \subseteq \text{Ker } \phi$ and $\left|\mathbb{F}_q^{*2}\right| = (q-1)/2$. Consequently, $\mathbb{F}_q^{*2} = \text{Ker } \phi$. Since $z \notin \mathbb{F}_q^{*2}$, $\phi(z) = z^{\frac{q-1}{2}} = -1$.

From $z^{\frac{q-1}{2}} = -1$ we deduce $\zeta^{q-1} = -1$ and $\zeta^q = -\zeta$. Then the equation $(x + \zeta y)(x - \zeta y) = \beta$ can be written as

$$\begin{aligned}\beta &= (x+\zeta y)(x-\zeta y) = (x+\zeta y)(x+\zeta^q y)\\ &= (x+\zeta y)(x+\zeta y)^q = (x+\zeta y)^{q+1},\end{aligned}$$

and the number of solutions of $x^2 - zy^2 = \beta$ is equal to the number of elements $\eta \in \mathbb{F}_q[\zeta]$ such that $\eta^{q+1} = \beta$.

Consider the map

$$\begin{aligned}\psi : \mathbb{F}_{q^2}^* &\to \mathbb{F}_q^*\\ \xi &\mapsto \xi^{q+1}.\end{aligned}$$

Clearly it is a group homomorphism, $|\ \mathrm{Ker}\ \psi\ | \leq q+1$, and $|\ \mathrm{Im}\ \psi\ | \leq q-1$. But $|\ \mathrm{Ker}\ \psi\ ||\ \mathrm{Im}\ \psi\ | = |\ \mathbb{F}_{q^2}^*\ | = q^2-1$. Hence ψ is necessarily surjective and $\mathrm{Ker}\ \psi = q+1$. The elements in the same coset of $\mathbb{F}_{q^2}^*$ with respect to $\mathrm{Ker}\ \psi$ are mapped into the same element of $\mathbb{F}_q^*$, and elements in different cosets are mapped into different elements of $\mathbb{F}_q^*$. Since each coset contains the same number of elements as $|\ \mathrm{Ker}\ \psi\ |$, viz., $q+1$, the equation $\eta^{q+1} = \beta$ has $q+1$ solutions in $\mathbb{F}_{q^2}^*$. Consequently, the number of solutions of $x^2 - zy^2 = \beta$ is equal to $q+1$. □

Proof of Theorem 1.27. Consider first the equation

$$\sum_{i=1}^{n} x_i^2 = 0. \tag{1.29}$$

We distinguish the cases when n is odd or even. Suppose that n is odd and write $n = 2\nu + 1$. By Theorem 1.26, (1.29) is affinely equivalent to

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + d x_{2\nu+1}^2 = 0, \tag{1.30}$$

where $d = 1$ or z. Since (1.29) and (1.30) are affinely equivalent, they have the same number of points on them. Thus it is sufficient to determine the number of points on (1.30). If we choose $(x_1, x_2, \ldots, x_\nu)$ to be any non-zero vector of $\mathbb{F}_q^{(\nu)}$ and $x_{2\nu+1}$ to be an arbitrary element of $\mathbb{F}_q$, then we obtain a linear equation in $x_{\nu+1}, x_{\nu+2}, \ldots, x_{2\nu}$, which has exactly $q^{\nu-1}$ solutions. Since there are $q^\nu - 1$ possible choices of $(x_1, x_2, \ldots, x_\nu)$ to be a non-zero vector and there are q possible choices of $x_{2\nu+1}$, we obtain

$$(q^\nu - 1) q q^{\nu-1} = (q^\nu - 1) q^\nu$$

solutions of (1.30) with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$. On the other hand, if we choose $(x_1, x_2, \ldots, x_\nu)$ to be the zero vector $(0, 0, \ldots, 0)$, then $x_{2\nu+1}$

must be zero and $x_{\nu+1}, \ldots, x_{2\nu}$ can be arbitrary elements of $\mathbb{F}_q$; and thus we obtain

$$q^\nu$$

solutions of (1.30) with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. Therefore the number of solutions of (1.30) is

$$(q^\nu - 1)q^\nu + q^\nu = q^{2\nu} = q^{n-1}.$$

Suppose now that n is even. By Theorem 1.26, (1.29) is affinely equivalent to either

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu, \tag{1.31}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - z x_{2\nu+2}^2 = 0, \text{ where } n = 2\nu + 2. \tag{1.32}$$

They correspond to the cases when $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ or not, respectively. For (1.31), the number of solutions with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$ is

$$(q^\nu - 1)q^{\nu-1}$$

and the number of solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$ is

$$q^\nu.$$

Therefore the total number of solutions of (1.31) is

$$(q^\nu - 1)q^{\nu-1} + q^\nu = q^{2\nu-1} + q^\nu - q^{\nu-1} = q^{n-1} + q^{\frac{n}{2}} - q^{\frac{n}{2}-1}.$$

For (1.32), if we choose $(x_1, x_2, \ldots, x_\nu)$ to be any non-zero vector of $\mathbb{F}_q^{(\nu)}$, $x_{2\nu+1}$ and $x_{2\nu+2}$ to be arbitrary elements of $\mathbb{F}_q$, then we obtain a linear equation in $x_{\nu+1}, x_{\nu+2}, \ldots, x_{2\nu}$, which has exactly $q^{\nu-1}$ solutions. Since there are $q^\nu - 1$ possible choices of $(x_1, x_2, \ldots, x_\nu)$ to be a non-zero vector and there are q possible choices of each of $x_{2\nu+1}$ and $x_{2\nu+2}$, we obtain

$$(q^\nu - 1)q^2 q^{\nu-1} = (q^\nu - 1)q^{\nu+1}$$

solutions with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$. On the other hand, if we choose $(x_1, x_2, \ldots, x_\nu)$ to be the zero vector $(0, 0, \ldots, 0)$, since $x_{2\nu+1}^2 -$

$zx_{2\nu+2}^2$ is definite, $x_{2\nu+1}$ and $x_{2\nu+2}$ must be zero, and $x_{\nu+1}, x_{\nu+2}, \ldots, x_{2\nu}$ can be arbitrary elements of $\mathbb{F}_q$. Thus we obtain

$$q^\nu$$

solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. Therefore the total number of solutions of (1.32) is

$$(q^\nu - 1)q^{\nu+1} + q^\nu = q^{n-1} - q^{\frac{n}{2}} + q^{\frac{n}{2}-1}.$$

This completes the computation of the number of points on (1.29).

Next, let us consider the equation

$$\sum_{i=1}^{n-1} x_i^2 + zx_n^2 = 0, \tag{1.33}$$

where n is even. By Theorem 1.26, (1.33) is affinely equivalent to

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu, \tag{1.34}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - zx_{2\nu+2}^2 = 0, \text{ where } n = 2\nu + 2. \tag{1.35}$$

They correspond to the cases when $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$ or $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$, respectively. By the computation of the previous paragraph we know that the number of solutions of (1.34) and (1.35) are

$$q^{n-1} + q^{\frac{n}{2}} - q^{\frac{n}{2}-1} \text{ and } q^{n-1} - q^{\frac{n}{2}} + q^{\frac{n}{2}-1},$$

respectively.

Then let us consider the equation

$$\sum_{i=1}^{n} x_i^2 + 1 = 0. \tag{1.36}$$

We distinguish also the cases when n is odd or even. Consider first the case when n is odd. Let $n = 2\nu + 1$. Then by Theorem 1.26, (1.36) is affinely equivalent to

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + 1 = 0 \tag{1.37}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + z x_{2\nu+1}^2 + 1 = 0, \tag{1.38}$$

which corresponds to the cases when $(-1)^{\frac{n-1}{2}} \in \mathbb{F}_q^{*2}$ or not, respectively. By a similar argument we obtain that the number of solutions of (1.37) and also of (1.38) with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$ is

$$(q^\nu - 1)qq^{\nu-1} = (q^\nu - 1)q^\nu.$$

We have also to enumerate the number of solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. For $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$, (1.37) and (1.38) become

$$x_{2\nu+1}^2 + 1 = 0 \tag{1.39}$$

and

$$z x_{2\nu+1}^2 + 1 = 0, \tag{1.40}$$

respectively. If $-1 \in \mathbb{F}_q^{*2}$, we have necessarily the case $(-1)^{\frac{n-1}{2}} \in \mathbb{F}_q^{*2}$ and equation (1.39), which obviously has 2 solutions in $\mathbb{F}_q$. But now $x_{\nu+1}, \ldots, x_{2\nu}$ can be any elements of $\mathbb{F}_q$, hence (1.37) has

$$2q^\nu$$

solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. Now suppose that $-1 \notin \mathbb{F}_q^{*2}$. Then $(-1)^{\frac{n-1}{2}} \in \mathbb{F}_q^{*2}$ if and only if $n \equiv 1 \pmod 4$. For the case $n \equiv 1 \pmod 4$, we have (1.39), which has no solutions in $\mathbb{F}_q$, and hence (1.37) has no solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. For the case $n \equiv 3 \pmod 4$, we have (1.40), which has 2 solutions in $\mathbb{F}_q$, and hence (1.38) has

$$2q^\nu$$

solutions with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$. We conclude that the number of solutions of (1.36) for odd n is

$$(q^\nu - 1)q^\nu = q^{n-1} - q^{\frac{n-1}{2}},$$
$$\text{if } -1 \notin \mathbb{F}_q^{*2} \text{ and } n \equiv 1 \pmod 4,$$

or

$$(q^\nu - 1)q^\nu + 2q^\nu = q^{n-1} + q^{\frac{n-1}{2}},$$
$$\text{if } -1 \in \mathbb{F}_q^{*2}, \text{or } -1 \notin \mathbb{F}_q^{*2} \text{ and } n \equiv 3 \pmod 4.$$

Suppose now that n is even. By Theorem 1.26, (1.36) is affinely equivalent to either

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + 1 = 0, \text{ where } n = 2\nu, \tag{1.41}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - z x_{2\nu+2}^2 + 1 = 0, \text{ where } n = 2\nu + 2. \tag{1.42}$$

They correspond to the cases when $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ or not. By similar arguments we obtain that the number of solutions of (1.41) is

$$(q^\nu - 1)q^{\nu-1} = q^{n-1} - q^{\frac{n}{2}-1}$$

and that the number of solutions of (1.42) is

$$(q^\nu - 1)q^{\nu+1} + q^\nu(q+1) = q^{n-1} + q^{\frac{n}{2}-1},$$

where $q+1$, by Lemma 1.28, is the number of solutions of the equation $x_{2\nu+1}^2 - z x_{2\nu+2}^2 + 1 = 0$. Therefore the total number of solutions of (1.36) for n even is

$$q^{n-1} - q^{\frac{n}{2}-1}, \text{ if } (-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2},$$

or

$$q^{n-1} + q^{\frac{n}{2}-1}, \text{ if } (-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}.$$

Now let us come to the equation

$$\sum_{i=1}^{n} x_i^2 + z = 0, \tag{1.43}$$

where n is odd. Let us write $n = 2\nu + 1$. By Theorem 1.26, (1.43) is affinely equivalent to

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + z = 0 \tag{1.44}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + z x_{2\nu+1}^2 + z = 0, \tag{1.45}$$

which correspond to the cases when $(-1)^\nu \in \mathbb{F}_q^{*2}$ or not. Using a similar argument we obtain that the number of solutions both of (1.44) and of (1.45) for $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$ is

$$(q^\nu - 1)qq^{\nu-1} = q^{2\nu} - q^\nu = q^{n-1} - q^{\frac{n-1}{2}}.$$

However, for $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$, (1.44) and (1.45) reduces to

$$x_{2\nu+1}^2 + z = 0$$

and

$$x_{2\nu+1}^2 + 1 = 0,$$

respectively. If $(-1)^\nu \notin \mathbb{F}_q^{*2}$, then $-1 \notin \mathbb{F}_q^{*2}$, hence $x_{2\nu+1}^2 + 1 = 0$ has no solutions; therefore in case $(-1)^\nu \notin \mathbb{F}_q^{*2}$, the number of solutions of (1.45) for $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$ is 0. If $(-1)^\nu \in \mathbb{F}_q^{*2}$, we have $x_{2\nu+1}^2 + z = 0$; if $-1 \in \mathbb{F}_q^{*2}$, then $x_{2\nu+1}^2 + z = 0$ has no solutions, and if $-1 \notin \mathbb{F}_q^{*2}$, then $x_{2\nu+1}^2 + z = 0$ has 2 solutions. Hence, the number of solutions of (1.44) for $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$ is either 0 or $2q^\nu$ corresponding to the case when both -1 and $(-1)^\nu \in \mathbb{F}_q^{*2}$ and the case when $-1 \notin \mathbb{F}_q^{*2}$ but $(-1)^\nu \in \mathbb{F}_q^{*2}$, respectively. We conclude that the total number of solutions of (1.43) is either

$$q^{n-1} - q^{\frac{n-1}{2}}, \text{ if } -1 \in \mathbb{F}_q^{*2}, \text{ or } -1 \notin \mathbb{F}_q^{*2} \text{ and } n \equiv 3 \pmod 4,$$

or

$$q^{n-1} + q^{\frac{n-1}{2}}, \text{ if } -1 \notin \mathbb{F}_q^{*2} \text{ and } n \equiv 1 \pmod 4.$$

Then we consider the equation

$$\sum_{i=1}^{n-1} x_i^2 + zx_n^2 + 1 = 0, \tag{1.46}$$

where n is even. By Theorem 1.26, (1.46) is affinely equivalent to

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + 1 = 0, \text{ where } n = 2\nu, \tag{1.47}$$

or

$$2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - zx_{2\nu+2}^2 + 1 = 0, \text{ where } n = 2\nu + 2, \tag{1.48}$$

which correspond to the cases when $(-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2}$ or $(-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}$ respectively. Notice that (1.47) is (1.41) and (1.48) is (1.42). Therefore the number of solutions of (1.47) and (1.48) have already been obtained before. They are

$$q^{n-1} - q^{\frac{n}{2}-1} \text{ or } q^{n-1} + q^{\frac{n}{2}-1},$$

respectively. Hence, the number of solutions of (1.46) is

$$q^{n-1} - q^{\frac{n}{2}-1}, \text{ if } (-1)^{\frac{n}{2}} \notin \mathbb{F}_q^{*2},$$

or

$$q^{n-1} + q^{\frac{n}{2}+1}, \text{ if } (-1)^{\frac{n}{2}} \in \mathbb{F}_q^{*2}.$$

Finally, let us consider the equation

$$\sum_{i=1}^{n-2} x_i^2 + dx_{n-1}^2 + x_n = 0, \tag{1.49}$$

where $d = 1$ or z. For any choice of $x_1, x_2, \ldots, x_{n-1}$ in $\mathbb{F}_q$, there is a uniquely determined x_n which satisfies (1.49). Therefore the number of solutions of (1.49) is q^{n-1}.

Now Theorem 1.27 is completely proved. □

1.4 Quadrics in $AG(n,\mathbb{F}_q)$ for q Even

In this section we assume that $\mathbb{F}_q$ is of characteristic 2, that is, q is a power of 2. Then we have $\mathbb{F}_q^2 = \mathbb{F}_q$.

Recall that the set of points $(x_1, x_2, \ldots, x_n)$ in $AG(n,\mathbb{F}_q)$ which satisfy a quadratic equation (1.9)

$$\sum_{1\le i\le k\le n} a_{ik}x_ix_k + \sum_{j=1}^{n} b_jx_j + c = 0,$$

where a_{ik} $(1 \le i \le k \le n)$, b_j $(1 \le j \le n)$, and c are elements of $\mathbb{F}_q$ and not all a_{ik} are zero, is called a *quadric* in $AG(n,\mathbb{F}_q)$ and the quadratic equation (1.9) is called its equation. We can also introduce the matrix of coefficients of the second degree terms

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ 0 & a_{22} & \ldots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix} \tag{1.50}$$

and the coefficient matrix

$$A_1 = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ 0 & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & a_{nn} & b_n \\ 0 & \dots & \dots & 0 & c \end{pmatrix} = \begin{pmatrix} A & {}^t b \\ 0 & c \end{pmatrix}, \tag{1.51}$$

where $b = (b_1, b_2, \dots, b_n)$. Both of them are upper triangular matrices. Then equation (1.9) can be written in matrix form

$$(x_1, x_2, \dots, x_n, 1) \begin{pmatrix} A & {}^t b \\ 0 & c \end{pmatrix} {}^t(x_1, x_2, \dots, x_n, 1) = 0. \tag{1.52}$$

However under the affine transformation

$$(x_1, x_2, \dots, x_n, 1) \mapsto (x_1, x_2, \dots, x_n, 1) \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix},$$

where $T \in GL_n(\mathbb{F}_q)$ and $v \in \mathbb{F}_q^{(n)}$, (1.52) is carried into

$$(x_1, x_2, \dots, x_n, 1) \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} \begin{pmatrix} A & {}^t b \\ 0 & c \end{pmatrix} {}^t \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} {}^t(x_1, x_2, \dots, x_n, 1) = 0,$$

whose matrix of coefficients of the second degree terms and coefficient matrix are

$$T A \, {}^t T$$

and

$$\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} \begin{pmatrix} A & {}^t b \\ 0 & c \end{pmatrix} {}^t \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix},$$

respectively. They are in general no longer upper triangular. So the first problem we encounter is to find a suitable matrix representation of the quadratic equation (1.9). It is helpful to introduce alternate matrices.

Let $K = (k_{ij})_{1 \le i,j \le n}$ be an $n \times n$ matrix. K is said to be *alternate*, if $k_{ij} = k_{ji}$ for all $i \neq j$, $1 \le i, j \le n$, and $k_{ii} = 0$ for all $i = 1, 2, \dots, n$. If K is an alternate matrix, then it is easy to see that $xK\,{}^t x = 0$ for all $x \in \mathbb{F}_q^{(n)}$. Conversely, suppose that $xK\,{}^t x = 0$ for all $x \in \mathbb{F}_q^{(n)}$. From $e_i K\,{}^t e_i = 0$ we deduce $k_{ii} = 0$ $(1 \le i \le n)$, and from $(e_i + e_j)K\,{}^t(e_i + e_j) = 0$ we deduce $k_{ij} = k_{ji}$ for all $i \neq j$. Hence, K is alternate if and only if $xK\,{}^t x = 0$ for all $x \in \mathbb{F}_q^{(n)}$.

Theorem 1.29. Let K be an $n \times n$ alternate matrix over $\mathbb{F}_q$. Then the rank of K is necessarily even. Further, if K is of rank $2\nu(\leq n)$, then K is cogredient to

$$\begin{pmatrix} \underbrace{\begin{matrix} 0 & 1 & & & & & \\ 1 & 0 & & & & & \\ & & 0 & 1 & & & \\ & & 1 & 0 & & & \\ & & & & \ddots & & \\ & & & & & 0 & 1 \\ & & & & & 1 & 0 \end{matrix}}_{2\nu} & & \\ & \underbrace{\begin{matrix} 0 & & \\ & \ddots & \\ & & 0 \end{matrix}}_{n-2\nu} \end{pmatrix},$$

and is also cogredient to

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 0^{(n-2\nu)} \end{pmatrix}.$$

Proof. We apply induction on n. For $n = 1$, $K = (0)$, our theorem holds automatically. For $n = 2$, K has the form

$$K = \begin{pmatrix} 0 & k_{12} \\ k_{12} & 0 \end{pmatrix}.$$

If $k_{12} = 0$, then our theorem also holds automatically; if $k_{12} \neq 0$, then K is cogredient to

$$\begin{pmatrix} 1 & 0 \\ 0 & k_{12}^{-1} \end{pmatrix} K \, {}^t\!\begin{pmatrix} 1 & 0 \\ 0 & k_{12}^{-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now suppose that $n > 2$ and that our theorem holds for $m < n$. Let

$$K = \begin{pmatrix} 0 & k_{12} & k_{13} & \dots & k_{1n} \\ k_{12} & 0 & k_{23} & \dots & k_{2n} \\ k_{13} & k_{23} & 0 & \dots & k_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ k_{1n} & k_{2n} & k_{3n} & \dots & 0 \end{pmatrix}.$$

If $K = 0$, then our theorem is true. Suppose that $K \neq 0$, then there is a $k_{ij} \neq 0$, $i < j$. Interchanging the first row and the i-th row of K and the first column and the i-th column of K simultaneously, we obtain an alternate matrix K_1 which is cogredient to K and whose element at $(1, j)$

position is non-zero. Then interchanging the second row and the j-th row of K_1 and the second column and the j-th column of K_1 simultaneously, we obtain an alternate matrix K_2 which is cogredient to K_1 and hence also to K and whose element at $(1,2)$ position is non-zero. Thus we can assume that $k_{12} \neq 0$. We can assume further that $k_{12} = 1$. In fact, if $k_{12} \neq 0$, K is cogredient to

$$\begin{pmatrix} 1 & & \\ & k_{12}^{-1} & \\ & & I^{(n-2)} \end{pmatrix} K \, {}^t\!\begin{pmatrix} 1 & & \\ & k_{12}^{-1} & \\ & & I^{(n-2)} \end{pmatrix},$$

whose element at $(1,2)$ position is 1. Then K is cogredient to

$$\begin{pmatrix} 1 & & & & & & \\ 0 & 1 & & & & & \\ k_{23} & k_{13} & 1 & & & & \\ \vdots & \vdots & 0 & \ddots & & & \\ \vdots & \vdots & \vdots & \ddots & \ddots & & \\ k_{2n} & k_{1n} & 0 & \dots & 0 & 1 \end{pmatrix} K \, {}^t\!\begin{pmatrix} 1 & & & & & & \\ 0 & 1 & & & & & \\ k_{23} & k_{13} & 1 & & & & \\ \vdots & \vdots & 0 & \ddots & & & \\ \vdots & \vdots & \vdots & \ddots & \ddots & & \\ k_{2n} & k_{1n} & 0 & \dots & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & & & \\ \vdots & \vdots & & K_0 & \\ 0 & 0 & & & \end{pmatrix},$$

where K_0 is an $(n-2) \times (n-2)$ alternate matrix. Therefore our theorem follows from induction hypothesis. □

Now let $K = (k_{ij})_{1 \leq i,j \leq n+1}$ be an $(n+1) \times (n+1)$ alternate matrix. Then clearly

$$\begin{aligned} &(x_1, x_2, \dots, x_n, 1)\ (A_1 + K)\ {}^t(x_1, x_2, \dots, x_n, 1) \\ &\quad = (x_1, x_2, \dots, x_n, 1)\ A_1\ {}^t(x_1, x_2, \dots, x_n, 1) \end{aligned}$$

and hence the quadratic equation

$$(x_1, x_2, \dots, x_n, 1)\ (A_1 + K)\ {}^t(x_1, x_2, \dots, x_n, 1) = 0$$

defines the same quadric as (1.52). Therefore it is natural not to distinguish square matrices differing by an alternate matrix in studying quadrics over $\mathbb{F}_q$ for q even. This leads to the following definition.

Denote the set of all $n \times n$ alternate matrices over $\mathbb{F}_q$ by $\mathcal{K}_n$. Two $n \times n$ matrices A and B over $\mathbb{F}_q$ are said to be *congruent* $\bmod \mathcal{K}_n$, if $A + B \in \mathcal{K}_n$, which is usually denoted by

$$A \equiv B \pmod{\mathcal{K}_n}$$

or simply, $A \equiv B$.

Two $n \times n$ matrices A and B over $\mathbb{F}_q$ are said to be *'cogredient'*, if there is a nonsingular $n \times n$ matrix Q over $\mathbb{F}_q$ such that $QA\,{}^tQ \equiv B$. It can be easily proved that this is an equivalence relation. Therefore the set of $n \times n$ matrices over $\mathbb{F}_q$ is partitioned into mutually disjoint classes of 'cogredient' matrices. It can also be proved that matrices 'cogredient' to an alternate matrix are also alternate and matrices 'cogredient' to a definite matrix are also definite, while a matrix A is said to be *definite* if $xA\,{}^tx = 0$, where $x \in \mathbb{F}_q^{(n)}$, implies $x = 0$.

At first, we shall study the normal form of $n \times n$ matrices relative to 'cogredience'. We start with a preliminary result.

Theorem 1.30. Any $n \times n$ matrix over $\mathbb{F}_q$ is 'cogredient' to a matrix of the form

$$\begin{pmatrix} A & I^{(p)} & & \\ & B & & \\ & & C & \\ & & & 0 \end{pmatrix}, \tag{1.53}$$

where A and B are $p \times p$ diagonal matrices, and C is a $d \times d$ definite matrix. Moreover, p and d are uniquely determined and $d = 0$ or 1; in case $d = 1$, we can have $C = (1)$.

Proof. Let G be an $n \times n$ matrix over $\mathbb{F}_q$. Then $G + {}^tG \in \mathcal{K}_n$. By Theorem 1.29 we can assume that rank $(G + {}^tG) = 2p$ $(0 \le p \le [\frac{n}{2}])$ and there is a nonsingular $n \times n$ matrix Q such that

$$Q(G + {}^tG)\,{}^tQ = \begin{pmatrix} 0 & I^{(p)} & \\ I^{(p)} & 0 & \\ & & 0 \end{pmatrix}. \tag{1.54}$$

If we write $QG\,{}^tQ$ in block form

$$QG\,{}^tQ = \begin{array}{c} \begin{matrix} p & p & n-2p \end{matrix} \\ \begin{pmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{pmatrix} \end{array} \begin{matrix} p \\ p \\ n-2p \end{matrix},$$

then from (1.54) we deduce ${}^tA_1 = A_1$, ${}^tB_2 = B_2$, ${}^tC_3 = C_3$, ${}^tB_1 + A_2 = I$, ${}^tC_1 + A_3 = 0$, and ${}^tC_2 + B_3 = 0$. Thus we have

$$QG\,{}^tQ \equiv \begin{array}{c} \begin{matrix} p & p & n-2p \end{matrix} \\ \begin{pmatrix} A & I & \\ & B & \\ & & C_4 \end{pmatrix} \end{array} \begin{matrix} p \\ p \\ n-2p \end{matrix},$$

where A, B, and C_4 are diagonal matrices. Write $C_4 = [\gamma_{2p+1}, \gamma_{2p+2}, \ldots, \gamma_n]$. If there is a non-zero row vector $x = (x_{2p+1}, x_{2p+2}, \ldots, x_n)$ such that $xC_4\,{}^t x = \gamma_{2p+1}x_{2p+1}^2 + \gamma_{2p+2}x_{2p+2}^2 + \cdots + \gamma_n x_n^2 = 0$, we can assume without loss of generality that $x_n \neq 0$. Let

$$Q_1 = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ x_{2p+1} & x_{2p+2} & \ldots & x_{n-1} & x_n \end{pmatrix},$$

then

$$Q_1 C_4\,{}^t Q_1 = [\gamma_{2p+1}, \gamma_{2p+2}, \ldots, \gamma_{n-1}, 0].$$

Proceeding in this way, we finally reduce G to the form (1.53) relative to 'cogredience', where C is a $d \times d$ definite diagonal matrix. We have to prove that $d \leq 1$. Suppose that $d \geq 2$ and write $C = [\gamma_1, \gamma_2 \ldots, \gamma_d]$. Of course all γ_i $(1 \leq i \leq d)$ are non-zero. Since every element of $\mathbb{F}_q$ has a unique square root, for any $x \in \mathbb{F}_q^*$ the symbols $x^{\frac{1}{2}}$ and $x^{-\frac{1}{2}}$ are meaningful. Then we have

$$(\gamma_1^{-\frac{1}{2}}, \gamma_2^{-\frac{1}{2}}, 0, \ldots, 0) \begin{pmatrix} \gamma_1 & & & 0 \\ & \gamma_2 & & \\ & & \ddots & \\ 0 & & & \gamma_d \end{pmatrix} {}^t(\gamma_1^{-\frac{1}{2}}, \gamma_2^{-\frac{1}{2}}, 0, \ldots, 0) = 0,$$

which contradicts the assumption that C is definite. Furthermore, if $d = 1$, then clearly $C = (\gamma)$ is 'cogredient' to $(\gamma^{-\frac{1}{2}})(\gamma)\,{}^t(\gamma^{-\frac{1}{2}}) = (1)$.

Now, let us prove the uniqueness of p and d in (1.53). Let

$$G = \begin{pmatrix} A & I^{(p)} & & \\ & B & & \\ & & C & \\ & & & 0 \end{pmatrix} \quad \text{and} \quad G_1 = \begin{pmatrix} A_1 & I^{(p_1)} & & \\ & B_1 & & \\ & & C_1 & \\ & & & 0 \end{pmatrix}$$

be two $n \times n$ matrices of the form (1.53) and assume that they are 'cogredient', i.e., there is a nonsingular matrix Q such that $QG\,{}^t Q \equiv G_1$. Then $Q(G + {}^t G)\,{}^t Q = G_1 + {}^t G_1$, from which we deduce $2p = \text{rank}\,(G + {}^t G) = \text{rank}\,(G_1 + {}^t G_1) = 2p_1$ and $p = p_1$. Let $r = n - 2p - d$ and $r_1 = n - 2p - d_1$. Write Q in block form

$$Q = \begin{array}{c} \begin{matrix} 2p & n-2p \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix} \begin{matrix} 2p \\ n-2p \end{matrix} \end{array}.$$

Then from $Q(G + {}^tG){}^tQ = G_1 + {}^tG_1$, we deduce $Q_{21} = 0$, hence Q_{22} is nonsingular and

$$Q_{22} \begin{pmatrix} \overset{d}{C} & \overset{r}{} \\ & 0 \end{pmatrix} {}^tQ_{22} \equiv \begin{pmatrix} \overset{d_1}{C_1} & \overset{r_1}{} \\ & 0 \end{pmatrix}.$$

Since Q_{22} is nonsingular, $d = 0$ if and only if $d_1 = 0$. Therefore d is uniquely determined. □

It follows from Theorem 1.30 that both p and d are uniquely determined by G, i.e., they are invariants relative to 'cogredience'. Hence the number $2p + d$ is also an invariant of G relative to 'cogredience' and is called the *'rank'* of G. Moreover, d takes the value 0 or 1 and is called the *defect* of G. If $d = 0$, G is said to be *without defect*. If the 'rank' of an $n \times n$ matrix G is equal to n, i.e., $n = 2p + d$, then G is said to be *regular*. Clearly, an $n \times n$ matrix G is regular if and only if it is 'cogredient' to a matrix of the form

$$\begin{pmatrix} A & I & \\ & B & \\ & & C \end{pmatrix},$$

where A and B are diagonal matrices, C disappears or C is the 1×1 matrix (1). Matrices 'cogredient' to a regular matrix are also regular.

Now we introduce a subset of $\mathbb{F}_q$ as follows

$$N = \{x^2 + x \mid x \in \mathbb{F}_q\}.$$

Then we have

Lemma 1.31. N is an additive subgroup of $\mathbb{F}_q$ of index 2 and $N^2 = N$.

Proof. The map $x \mapsto x^2 + x$ from the additive group of $\mathbb{F}_q$ into itself is a homomorphism of groups. The kernel of this map is $\{0, 1\}$ and its image is N. Hence N is an additive subgroup of $\mathbb{F}_q$ of index 2. Furthermore, from $\mathbb{F}_q^2 = \mathbb{F}_q$ we deduce $N^2 = N$. □

From now on we choose a fixed element $\alpha \in \mathbb{F}_q$ but $\alpha \notin N$.

We need more lemmas.

Lemma 1.32. The 2×2 matrix over $\mathbb{F}_q$

$$\begin{pmatrix} a & 1 \\ & b \end{pmatrix}$$

is 'cogredient' to either

$$\begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix} \tag{1.55}$$

or

$$\begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix} \tag{1.56}$$

according to $ab \in N$ or $ab \notin N$, respectively. Moreover, the first one of the above two matrices is not definite and the second one is definite.

Proof. If $a = b = 0$, there is nothing to prove. If $a = 0$ and $b \neq 0$, then

$$\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ & b \end{pmatrix} {}^t\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix}.$$

We can treat the case $a \neq 0$ and $b = 0$ in a similar way.

Now suppose that $ab \neq 0$. Consider first the case that $ab \in N$. There is an $x \in \mathbb{F}_q^*$ such that $ab = (ax)^2 + ax$. Thus

$$\begin{pmatrix} x & 1 \\ & x^{-1} \end{pmatrix} \begin{pmatrix} a & 1 \\ & b \end{pmatrix} {}^t\begin{pmatrix} x & 1 \\ & x^{-1} \end{pmatrix} \equiv \begin{pmatrix} 0 & 1 \\ & x^{-2}b \end{pmatrix},$$

which reduces to one of the cases we considered before.

Then consider the case $ab \notin N$. Since $\mathbb{F}_q^2 = \mathbb{F}_q$, there is a $\delta \in \mathbb{F}_q^*$ such that $\delta^2 a = b\delta^{-2}$. Let $\lambda = \delta^2 a$, then

$$\begin{pmatrix} \delta & \\ & \delta^{-1} \end{pmatrix} \begin{pmatrix} a & 1 \\ & b \end{pmatrix} {}^t\begin{pmatrix} \delta & \\ & \delta^{-1} \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ & \lambda \end{pmatrix}$$

and $\lambda^2 = ab \notin N$. By Lemma 1.31, from $\alpha \notin N$ we deduce $\alpha^2 \notin N$. Again by the lemma, from $\lambda^2 \notin N$ and $\alpha^2 \notin N$ we deduce $\alpha^2 + \lambda^2 \in N$, i.e., there is an element $t \in \mathbb{F}_q$ such that $\alpha^2 + \lambda^2 = t^2 + t$. Consequently,

$$\begin{pmatrix} (\alpha\lambda^{-1})^{\frac{1}{2}} & \\ (\alpha\lambda)^{-\frac{1}{2}}t & (\alpha^{-1}\lambda)^{\frac{1}{2}} \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ & \lambda \end{pmatrix} {}^t\begin{pmatrix} (\alpha\lambda^{-1})^{\frac{1}{2}} & \\ (\alpha\lambda)^{-\frac{1}{2}}t & (\alpha^{-1}\lambda)^{\frac{1}{2}} \end{pmatrix} \equiv \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}.$$

Clearly (1.55) is not definite. It remains to show that (1.56) is definite. Suppose that there is a non-zero vector (x, y) such that $\alpha x^2 + xy + \alpha y^2 = 0$. Without loss of generality we can assume that $x \neq 0$, then $\alpha^2 = (\alpha x^{-1}y)^2 + \alpha x^{-1}y \in N$ and hence $\alpha \in N$, contrary to the choice of $\alpha \notin N$. □

Lemma 1.33. The 4×4 matrix

$$\begin{pmatrix} \alpha & & 1 & \\ & \alpha & & 1 \\ & & \alpha & \\ & & & \alpha \end{pmatrix}$$

is 'cogredient' to

$$\begin{pmatrix} 0 & & 1 & \\ & 0 & & 1 \\ & & 0 & \\ & & & 0 \end{pmatrix}.$$

Proof. We compute

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \alpha & \alpha & 1 & 0 \\ 0 & 1 & \alpha & \alpha \end{pmatrix} \begin{pmatrix} \alpha & & 1 & \\ & \alpha & & 1 \\ & & \alpha & \\ & & & \alpha \end{pmatrix} {}^t\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \alpha & \alpha & 1 & 0 \\ 0 & 1 & \alpha & \alpha \end{pmatrix} \equiv \begin{pmatrix} 0 & & 1 & \\ & 0 & & 1 \\ & & 0 & \\ & & & 0 \end{pmatrix}. \square$$

Lemma 1.34. The 3×3 matrix

$$\begin{pmatrix} \alpha & 1 & \\ & \alpha & \\ & & 1 \end{pmatrix}$$

is 'cogredient' to

$$\begin{pmatrix} 0 & 1 & \\ & 0 & \\ & & 1 \end{pmatrix}.$$

Proof. We compute

$$\begin{pmatrix} 1 & 0 & \alpha^{\frac{1}{2}} \\ 0 & 1 & \alpha^{\frac{1}{2}} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha & 1 & \\ & \alpha & \\ & & 1 \end{pmatrix} {}^t\begin{pmatrix} 1 & 0 & \alpha^{\frac{1}{2}} \\ 0 & 1 & \alpha^{\frac{1}{2}} \\ 0 & 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} 0 & 1 & \\ & 0 & \\ & & 1 \end{pmatrix}. \qquad \square$$

From Theorem 1.30 and the above lemmas, we deduce immediately

Theorem 1.35. Any $n\times n$ matrix over $\mathbb{F}_q$ is 'cogredient' to a matrix of one and only one of the following forms

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ & 0 & \\ & & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & & \\ & 0 & & \\ & & 1 & \\ & & & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}, \quad (1.57)$$

where α is a fixed element of $\mathbb{F}_q$ not belonging to N.

Proof. It is enough to show that the three forms of matrices listed in the theorem are not 'cogredient' to each other. Since the 'rank' of the second matrix is odd and the 'ranks' of the other two are even, the second one can not be 'cogredient' to the other two. Suppose that a matrix of the first form is 'cogredient' to a matrix of the third form, then they must have the same 'rank' and we can assume that they are

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ & 0 & \\ & & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & I^{(\nu-1)} & & & \\ & 0 & & & \\ & & \alpha & 1 & \\ & & & \alpha & \\ & & & & 0 \end{pmatrix}.$$

Then there is an $n \times n$ nonsingular matrix

$$Q = \begin{matrix} \\ 2\nu \\ n-2\nu \end{matrix} \begin{matrix} \begin{matrix} 2\nu & n-2\nu \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix} \end{matrix}$$

such that

$$\begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix} \begin{pmatrix} 0 & I^{(\nu)} & \\ & 0 & \\ & & 0 \end{pmatrix} {}^t\!\begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix} \equiv \begin{pmatrix} 0 & I^{(\nu-1)} & & & \\ & 0 & & & \\ & & \alpha & 1 & \\ & & & \alpha & \\ & & & & 0 \end{pmatrix}.$$

It follows that

$$Q_{11} \begin{pmatrix} 0 & I^{(\nu)} \\ & 0 \end{pmatrix} {}^t\!Q_{11} \equiv \begin{pmatrix} 0 & I^{(\nu-1)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix},$$

and hence

$$Q_{11} \begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix} {}^t\!Q_{11} = \begin{pmatrix} 0 & I^{(\nu-1)} & & \\ I^{(\nu-1)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 0 \end{pmatrix}.$$

Therefore Q_{11} is nonsingular. Then the number of solutions of the equation

$$(x_1, x_2, \ldots, x_{2\nu}) \begin{pmatrix} 0 & I^{(\nu)} \\ & 0 \end{pmatrix} {}^t(x_1, x_2, \ldots, x_{2\nu}) = 0,$$

or of the equation

$$x_1 x_{\nu+1} + x_2 x_{\nu+2} + \cdots + x_\nu x_{2\nu} = 0 \tag{1.58}$$

should be equal to the number of solutions of the equation

$$(x_1, x_2, \ldots, x_{2\nu}) \begin{pmatrix} 0 & I^{(\nu-1)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix} {}^t(x_1, x_2, \ldots, x_{2\nu}) = 0$$

or of the equation

$$x_1x_\nu + x_2x_{\nu+1} + \cdots + x_{\nu-1}x_{2\nu-2} + \alpha x_{2\nu-1}^2 + x_{2\nu-1}x_{2\nu} + \alpha x_{2\nu}^2 = 0. \quad (1.59)$$

It is easy to compute that the number of solutions of (1.58) is $q^{2\nu-1} + q^\nu - q^{\nu-1}$ and that of (1.59) is $q^{2\nu-1} - q^\nu + q^{\nu-1}$ (Exercise 1.11); they are not equal and we obtain a contradiction. □

If an $n \times n$ matrix is 'cogredient' to any one of the matrices in (1.57), then ν is called its *index*. It follows from Theorem 1.35 that the index of a matrix is invariant relative to 'cogredience'.

Now we state the classification theorem of quadrics in $AG(n, \mathbb{F}_q)$ for q even as follows.

Theorem 1.36. Under affine transformations any quadric in $AG(n, \mathbb{F}_q)$, where q is even, can be carried into a quadric with one of the following quadratic equations as its equation

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} = 0 \, (1 \le \nu \le \frac{n}{2}), \quad (1.60)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + 1 = 0 \, (1 \le \nu \le \frac{n}{2}), \quad (1.61)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1} = 0 \, (1 \le \nu \le \frac{n-1}{2}), \quad (1.62)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 = 0 \, (0 \le \nu \le \frac{n-1}{2}), \quad (1.63)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+2} = 0 \, (0 \le \nu \le \frac{n-2}{2}), \quad (1.64)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1} = 0 \, (0 \le \nu \le \frac{n-1}{2}), \quad (1.65)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1} + \alpha = 0 \, (0 \le \nu \le \frac{n-1}{2}), \quad (1.66)$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 = 0 \tag{1.67}$$

$$(0 \le \nu \le \frac{n-2}{2}),$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 + 1 = 0 \tag{1.68}$$

$$(0 \le \nu \le \frac{n-2}{2}),$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 + x_{2\nu+3} = 0 \tag{1.69}$$

$$(0 \le \nu \le \frac{n-3}{2}),$$

where α is a fixed element of $\mathbb{F}_q$ not belonging to N. Moreover, any two of these equations cannot be transformed into each other by an affine transformation or by an affine transformation followed by multiplying by an element of $\mathbb{F}_q^*$.

Proof. Suppose that a quadric in $AG(n, \mathbb{F}_q)$ is given, whose equation is (1.9)

$$\sum_{1 \le i \le k \le n} a_{ik} x_i x_k + \sum_{j=1}^{n} b_j x_j + c = 0.$$

After introducing the vector $b = (b_1, b_2, \ldots, b_n)$ and the matrices

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ & a_{22} & \ldots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{pmatrix}$$

and

$$A_1 = \begin{pmatrix} A & {}^t b \\ & c \end{pmatrix},$$

we can write (1.9) in the form (1.52)

$$(x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} A & {}^t b \\ & c \end{pmatrix} {}^t(x_1, x_2, \ldots, x_n, 1) = 0.$$

Under the affine transformation

$$(x_1, x_2, \ldots, x_n, 1) \mapsto (x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix},$$

where $T \in GL_n(\mathbb{F}_q)$ and $v \in \mathbb{F}_q^{(n)}$, the equation (1.52) is carried into

$$(x_1, x_2, \ldots, x_n, 1) \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} \begin{pmatrix} A & {}^t b \\ & c \end{pmatrix} {}^t\!\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} {}^t(x_1, x_2, \ldots, x_n, 1) = 0.$$

Thus A is subjected to a 'cogredient' transformation

$$A \mapsto TA\,{}^tT$$

and A_1 is subjected to a special kind of 'cogredient' transformation

$$\begin{pmatrix} A & {}^t b \\ & c \end{pmatrix} \mapsto \begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix} \begin{pmatrix} A & {}^t b \\ & c \end{pmatrix} {}^t\!\begin{pmatrix} T & 0 \\ v & 1 \end{pmatrix}.$$

By Theorem 1.35, there is an $n \times n$ nonsingular matrix Q such that

$$QA\,{}^tQ \equiv \begin{pmatrix} 0 & I^{(\nu)} & \\ & 0 & \\ & & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & & \\ & 0 & & \\ & & 1 & \\ & & & 0 \end{pmatrix}, \text{ or } \begin{pmatrix} 0 & I^{(\nu)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \\ & & & & 0 \end{pmatrix},$$

where ν is the index of A. Then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_n)Q$$

(1.9) is transformed into one of the following three forms

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \sum_{j=1}^{n} b'_j x_j + c = 0, \tag{1.70}$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + \sum_{j=1}^{n} b'_j x_j + c = 0, \tag{1.71}$$

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1} x_{2\nu+2} + \alpha x_{2\nu+2}^2 + \sum_{j=1}^{n} b'_j x_j + c = 0. \tag{1.72}$$

Consider the equation (1.70) first. We have necessarily that $\nu \geq 1$. Under the affine transformation

$$\begin{aligned}(x_1, x_2, \ldots, x_n) \mapsto (&x_1 + b'_{\nu+1}, \ldots, x_\nu + b'_{2\nu}, x_{\nu+1} + b'_1, \ldots, \\ &x_{2\nu} + b'_\nu, x_{2\nu+1}, \ldots, x_n)\end{aligned}$$

(1.70) is carried into the following form

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \sum_{j=2\nu+1}^{n} b'_j x_j + c' = 0. \tag{1.73}$$

If all b'_j $(j = 2\nu+1, \ldots, n)$ and c' are zero, we obtain (1.60). If all b'_j $(j = 2\nu+1, \ldots, n)$ are zero but $c' \neq 0$, then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (c'x_1, c'x_2, \ldots, c'x_\nu, x_{\nu+1}, \ldots, x_{2\nu}, \ldots, x_n)$$

(1.73) is carried into (1.61) multiplied by c'; after canceling c', we obtain (1.61). Now if not all b'_j $(j = 2\nu+1, \ldots, n)$ are zero, without loss of generality we assume that $b'_{2\nu+1} \neq 0$, then the inverse of the following affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto \left(x_1, x_2, \ldots, x_{2\nu}, \sum_{j=2\nu+1}^{n} b'_j x_j + c', x_{2\nu+2}, \ldots, x_n\right)$$

carries (1.73) into (1.62).

Next, consider the equation (1.71). In this case we have $\nu \geq 0$. As in the above case, (1.71) can be carried under an affine transformation into the following form

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + \sum_{j=2\nu+1}^{n} b'_j x_j + c' = 0. \tag{1.74}$$

If all b'_j $(j = 2\nu+1, \ldots, n)$ are zero, then the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto \left(x_1, x_2, \ldots, x_{2\nu}, x_{2\nu+1} + c'^{\frac{1}{2}}, x_{2\nu+2}, \ldots, x_n\right)$$

carries (1.74) into (1.63). If not all b'_j $(j = 2\nu+1, \ldots, n)$ are zero, we have to distinguish the case when not all b'_j $(j = 2\nu+2, \ldots, n)$ are zero and the case when all b'_j $(j = 2\nu+2, \ldots, n)$ are zero. If not all b'_j $(j = 2\nu+2, \ldots, n)$ are zero, without loss of generality we assume that $b'_{2\nu+2} \neq 0$, then the inverse of the following affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto \left(x_1, x_2, \ldots, x_{2\nu}, x_{2\nu+1}, \sum_{j=2\nu+1}^{n} b'_j x_j + c', x_{2\nu+3}, \ldots, x_n\right)$$

carries (1.74) into (1.64). If all b'_j $(j = 2\nu+2, \ldots, n)$ are zero, but $b'_{2\nu+1} \neq 0$, then the affine transformation

$$\begin{aligned}(x_1, x_2, \ldots, x_n) \mapsto (&b'_{2\nu+1}x_1, b'_{2\nu+1}x_2, \ldots, b'_{2\nu+1}x_{2\nu}, \\ &b'_{2\nu+1}x_{2\nu+1}, x_{2\nu+2}, \ldots, x_n)\end{aligned}$$

carries (1.74) into

$$b_{2\nu+1}^{\prime 2}\left(\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1}\right) + c' = 0. \tag{1.75}$$

Multiplying (1.75) by $b_{2\nu+1}^{\prime -2}$, we obtain

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1} + c'' = 0, \tag{1.76}$$

where $c'' = b_{2\nu+1}^{\prime -2} c'$. If $c'' \in N$, i.e., there is a $\beta \in \mathbb{F}_q$ such that $c'' = \beta^2 + \beta$, then the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_{2\nu}, x_{2\nu+1} + \beta, x_{2\nu+2}, \ldots, x_n)$$

carries (1.76) into (1.65). If $c'' \notin N$, then $c'' + \alpha \in N$ and there is $\beta_1 \in \mathbb{F}_q$ such that $c'' + \alpha = \beta_1^2 + \beta_1$, thus the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_{2\nu}, x_{2\nu+1} + \beta_1, x_{2\nu+2}, \ldots, x_n)$$

carries (1.76) into (1.66).

Finally we have to consider the equation (1.72). In this case we also have $\nu \geq 0$. As in the above cases, (1.72) can be carried by an affine transformation into the following form

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1} x_{2\nu+2} + \alpha x_{2\nu+2}^2 + \sum_{j=2\nu+1}^{n} b_j' x_j + c' = 0,$$

which can then be carried by the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_{2\nu}, x_{2\nu+1} + b'_{2\nu+2}, x_{2\nu+2} + b'_{2\nu+1}, x_{2\nu+3}, \ldots, x_n)$$

into the following form

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1} x_{2\nu+2} + \alpha x_{2\nu+2}^2 + \sum_{j=2\nu+3}^{n} b_j' x_j + c'' = 0. \tag{1.77}$$

If all b_j' $(j = 2\nu + 3, \ldots, n)$ and c'' are zero, then (1.77) is (1.67). If all b_j' $(j = 2\nu + 3, \ldots, n)$ are zero but $c'' \neq 0$, then under the affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (c''^{\frac{1}{2}} x_1, c''^{\frac{1}{2}} x_2, \ldots, c''^{\frac{1}{2}} x_{2\nu+2}, x_{2\nu+3}, \ldots, x_n)$$

(1.77) is carried into (1.68) multiplied by c''; after canceling c'', we obtain (1.68). Now suppose that not all b'_j $(j = 2\nu+3, \ldots, n)$ are zero; without loss of generality we assume that $b'_{2\nu+3} \neq 0$. Then the inverse of the following affine transformation

$$(x_1, x_2, \ldots, x_n) \mapsto (x_1, x_2, \ldots, x_{2\nu+2}, \sum_{j=2\nu+3}^{n} b'_j x_j + c'', x_{2\nu+4}, \ldots, x_n)$$

carries (1.77) into (1.69).

Hence the first statement of Theorem 1.36 is completely proved.

The second statement of the theorem follows from the invariance of the indices, 'ranks', and defects of the matrix of coefficients of the second degree terms A and the coefficient matrix A_1 under affine transformations and multiplications by elements of $\mathbb{F}_q^*$. □

The following lemma will be used for computing the number of points in $AG(n, \mathbb{F}_q)$ lying on a non-degenerate quadric.

Lemma 1.37. The number of solutions of the equation

$$\alpha x^2 + xy + \alpha y^2 = \beta,$$

where $\alpha, \beta \in \mathbb{F}_q^*$ and $\alpha \notin N$, is $q+1$.

Proof. Consider the set of all 2-dimensional non-zero vectors over $\mathbb{F}_q$, which is denoted by $\mathbb{F}_q^{(2)*}$. It is known that $|\mathbb{F}_q^{(2)*}| = q^2 - 1$. Two non-zero vectors (x, y) and (x_1, y_1) are said to be equivalent, if there is a $\lambda \in \mathbb{F}_q^*$ such that $(x, y) = \lambda(x_1, y_1)$. This is an equivalence relation and the set $\mathbb{F}_q^{(2)*}$ is partitioned into $q+1$ equivalence classes, each consisting of $q-1$ non-zero vectors. By Lemma 1.32, (1.56) is definite. Thus for any non-zero vector (x, y) in $\mathbb{F}_q^{(2)*}$, $\alpha x^2 + xy + \alpha y^2 \neq 0$. Since $\mathbb{F}_q^2 = \mathbb{F}_q$, as (x, y) runs through an equivalence class, $\alpha x^2 + xy + \alpha y^2$ runs through $\mathbb{F}_q^*$. Therefore, for any fixed $\beta \in \mathbb{F}_q^*$, the equation $\alpha x^2 + xy + \alpha y^2 = \beta$ has one and only one solution in each equivalence class. Consequently, the equation $\alpha x^2 + xy + \alpha y^2 = \beta$ has $q+1$ solutions. □

Finally we have

Theorem 1.38. The numbers of points in $AG(n, \mathbb{F}_q)$, which lie on the non-degenerate quadrics (1.60) – (1.69), where $n = 2\nu$ in (1.60) and (1.61), $n = 2\nu+1$ in (1.62), (1.63), (1.65), and (1.66), $n = 2\nu+2$ in (1.64), (1.67), and (1.68), and $n = 2\nu+3$ in (1.69) are listed in Table 1.2.

The proof is similar to the proof of Theorem 1.27 and will be omitted.

Table 1.2

equations	number of points
$\sum_{i=1}^{\nu} x_i x_{\nu+i} = 0$ $(n = 2\nu)$	$q^{n-1} + q^{\frac{n}{2}} - q^{\frac{n}{2}-1}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + 1 = 0$ $(n = 2\nu)$	$q^{n-1} - q^{\frac{n}{2}-1}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1} = 0$ $(n = 2\nu+1)$	q^{n-1}
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 = 0$ $(n = 2\nu+1)$	q^{n-1}
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+2} = 0$ $(n = 2\nu+2)$	q^{n-1}
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1} = 0$ $(n = 2\nu+1)$	$q^{n-1} + q^{\frac{n-1}{2}}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 + x_{2\nu+1} + \alpha = 0$ $(n = 2\nu+1)$	$q^{n-1} - q^{\frac{n-1}{2}}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 = 0$ $(n = 2\nu+2)$	$q^{n-1} - q^{\frac{n}{2}} + q^{\frac{n}{2}-1}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 + 1 = 0$ $(n = 2\nu+2)$	$q^{n-1} + q^{\frac{n}{2}-1}$
$\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 + x_{2\nu+3} = 0$ $(n = 2\nu+3)$	q^{n-1}

1.5 Comments

The results of Section 1.1 are classical. However, Theorems 1.3 (the transitivity theorem), 1.4, and 1.7 (the Anzahl theorem) and Corollaries 1.8, 1.9 are the germ of our study in later chapters.

The presentation of affine spaces over $\mathbb{F}_q$ in Section 1.2 is standard. The Erlangen Program is the famous principle proposed by Felix Klein in his inaugural public address, when he accepted a Chair of Mathematics at the University of Erlangen, Germany in 1872. It says that

> "A geometry is the set of properties of figures which are invariant under the nonsingular linear transformations of some group."

cf. Klein 1921. In the Erlangen Program, Klein pointed out the intimate relationship between geometries, groups, and invariants. The Erlangen Program has been one of the guiding principles for geometric studies since then and it is also the guiding principle for the present book.

In Section 1.3, Theorems 1.23 and 1.26, and Lemma 1.28 are due to L.E. Dickson, cf. Dickson 1900. Theorem 1.25 may be known to geometers, but the author could not find a reference for it, so he published a letter in Chinese Science Bulletin, cf. Wan 1991a. Theorem 1.27 was also announced in Wan 1991a.

Theorem 1.35 and Lemma 1.37 in Section 1.4 are also due to L.E. Dickson, but the derivation of Theorem 1.35 given in this book seems to be new. For Theorems 1.36 and 1.38, cf. Wan 1991b.

1.6 Exercises

1.1 Determine the number of generator matrices of a linear (n, k) code over $\mathbb{F}_q$ and also the number of its parity check matrices.

1.2 Let P_1 and P_2 be vector subspaces of $\mathbb{F}_q^{(n)}$. Show that $P_1 \cap P_2$ and $P_1 + P_2$ are vector subspaces and then prove the dimension formula

$$\dim P_1 + \dim P_2 = \dim (P_1 \cap P_2) + \dim (P_1 + P_2).$$

1.3 Prove that the set of $n \times n$ nonsingular matrices over $\mathbb{F}_q$ forms a group under matrix multiplication.

1.4 Prove Proposition 1.10.

(Hint: As a first step prove that the set of all $m \times n$ matrices of rank r over $\mathbb{F}_q$ is an orbit under the action of $GL_m(\mathbb{F}_q) \times GL_n(\mathbb{F}_q)$ on the set of all $m \times n$ matrices defined by

$$(X, (A, B)) \mapsto A^{-1} X B,$$

where $A \in GL_m(\mathbb{F}_q)$, $B \in GL_n(\mathbb{F}_q)$, and X is any $m \times n$ matrix over $\mathbb{F}_q$. Then determine the stabilizer H of

$$\begin{array}{c} \begin{matrix} r & n-r \end{matrix} \\ \begin{pmatrix} I^{(r)} & 0 \\ 0 & 0 \end{pmatrix} \begin{matrix} r \\ m-r \end{matrix} \end{array},$$

in the group $GL_m(\mathbb{F}_q) \times GL_n(\mathbb{F}_q)$. The index $GL_m(\mathbb{F}_q) \times GL_n(\mathbb{F}_q) : H$ will give the result.)

1.5 Let $0 \le k \le m \le n$ and P be a given k-dimensional vector subspace of $\mathbb{F}_q^{(n)}$. Prove that

a) The quotient space $\mathbb{F}_q^{(n)}/P$ is isomorphic to the $(n-k)$-dimensional row vector space $\mathbb{F}_q^{(n-k)}$ over $\mathbb{F}_q$.

b) There is a bijection from the set of m-dimensional vector subspaces of $\mathbb{F}_q^{(n)}$ containing P to the set of $(m-k)$-dimensional vector subspaces of $\mathbb{F}_q^{(n-k)}$. Hence deduce that $N'(k,m,n) = N(m-k, n-k)$.

1.6 In $AG(2, \mathbb{F}_5)$:

a) Find the equation of the line joining the points (3,4) and (2,1).

b) Find the intersection of the lines $3x + 4y = 2$ and $2x + 3y = 4$.

c) Find the equation of the line parallel to $3x + 4y = 1$ and passing through the point (2,3).

1.7 How many points and lines are there in $AG(2, \mathbb{F}_{2^2})$? Write down the equations of all lines in $AG(2, \mathbb{F}_{2^2})$, find the points contained in each line, find the lines passing through each point, and divide the lines into pencils of parallel lines.

1.8 Let t be a primitive element of $\mathbb{F}_{2^3}$. In $AG(2, \mathbb{F}_{2^3})$:

a) Find all points on the line $x + t^3 y = t^5$.

b) Find the equation of the line passing through the point $(1, t^6)$ and parallel to the line $y = t^5 x + t^3$.

c) Find the point of intersection of the lines $t^5 x + t^3 y + 1 = 0$ and $t^2 x + t^4 y + t^6 = 0$.

d) Find the equation of the line joining $(1, t^5)$ and $(t^6, 1)$.
(Hint: In doing arithmetic in $\mathbb{F}_{2^3}$, use the minimal polynomial of t.)

1.9 Prove Theorem 1.12.

1.10 In $AG(n, \mathbb{F}_q)$ fix an m-flat F $(1 \le m \le n-1)$. The points on F are regarded as source states, the points not on F are regarded as encoding rules, and the lines intersecting F and not lying on F are regarded as messages. Assume that the encoding rules work as follows: Given a message s (which is a point in F) and an encoding rule e (which is a point not in F), there is a unique line m passing through s and e, then we define $m = e(s)$, i.e., the source state s is encoded by the encoding rule e into the message m. Then we obtain an authentication code. Compute the parameters of the code. Assume that there is an equally distributed probability distribution on the set of encoding rules. Determine the probability that an opponent can deceive the transmitter/receiver when he plays an impersonation or a substitution game.

1.11 Prove that the number of solutions of equation (1.58) is $q^{2\nu-1} + q^\nu - q^{\nu-1}$ and that of equation (1.59) is $q^{2\nu-1} - q^\nu + q^{\nu-1}$.

1.12 Give the normal forms of the equations of quadrics in $AG(2, \mathbb{F}_q)$ under affine equivalence, where q may be odd or even. Draw the figures of the normal forms for the cases $q = 2$ and 3.

1.13 Prove Theorem 1.38.

1.14 Let A be a nonempty subset of points in $AG(n, \mathbb{F}_q)$. Prove that A is an affine flat if and only if for any x, $y \in A$ and $\lambda \in \mathbb{F}_q$, $\lambda x + (1 - \lambda)y \in A$.

Chapter 2
Projective Geometry over Finite Fields

2.1 Projective Spaces and Projective Groups over Finite Fields

Let n be an integer ≥ 1 and $\mathbb{F}_q^{(n+1)}$ be the $(n+1)$-dimensional row vector space over $\mathbb{F}_q$. The 1-dimensional vector subspaces of $\mathbb{F}_q^{(n+1)}$ will now be called *points*, the 2-dimensional, 3-dimensional, and n-dimensional vector subspaces of $\mathbb{F}_q^{(n+1)}$ will be called *lines*, *planes*, and *hyperplanes*, respectively. More generally, the $(r+1)$-dimensional vector subspaces of $\mathbb{F}_q^{(n+1)}$ will be called *projective r-flats*, or simply r-flats $(0 \leq r \leq n)$ in this chapter. Thus 0-flats, 1-flats, 2-flats, and $(n-1)$-flats are points, lines, planes, and hyperplanes, respectively. An r-flat is said to be *incident* with an s-flat, if the r-flat as a vector subspace contains or is contained in the s-flat as a vector subspace. Then the set of points, i.e., the set of 1-dimensional vector subspaces of $\mathbb{F}_q^{(n+1)}$, together with the r-flats $(0 \leq r \leq n)$ and the incidence relation among them defined above is called the n-dimensional *projective space* over $\mathbb{F}_q$ and is denoted by $PG(n, \mathbb{F}_q)$.

To be concrete, let us introduce the coordinate description of $PG(n, \mathbb{F}_q)$. Let P be a point of $PG(n, \mathbb{F}_q)$, that is, P is a 1-dimensional vector subspace of $\mathbb{F}_q^{(n+1)}$. Let $(x_0, x_1, \ldots, x_n)$ be a non-zero vector in P, then

$$P = \{\lambda(x_0, x_1, \ldots, x_n) \mid \lambda \in \mathbb{F}_q\}.$$

For any $\lambda \in \mathbb{F}_q^*$, we shall call the non-zero vector $(\lambda x_0, \lambda x_1, \ldots, \lambda x_n)$ a *system of coordinates* or simply the *coordinates* of the point P. We also say that $(\lambda x_0, \lambda x_1, \ldots, \lambda x_n)$ is the point P. Clearly, a system of coordinates of a point P is uniquely determined up to a non-zero constant multiple of $\mathbb{F}_q$.

$PG(n, \mathbb{F}_q)$ can also be regarded as being obtained from $AG(n, \mathbb{F}_q)$ by

adding a set of points at infinity, which constitute the hyperplane at infinity. We shall illustrate this point of view with the *projective plane* $PG(2, \mathbb{F}_q)$.

Let (x_1, x_2) be a point in $AG(2, \mathbb{F}_q)$ and call it P. A 3-tuple (y_0, y_1, y_2) over $\mathbb{F}_q$ with $y_0 \neq 0$ is called a *system of homogeneous coordinates* or simply the *homogeneous coordinates* of the point P if

$$x_1 = y_0^{-1} y_1, \ x_2 = y_0^{-1} y_2.$$

In particular, $(1, x_1, x_2)$ is a system of homogeneous coordinates of P. The original coordinates x_1, x_2 of P are also called the *non-homogeneous coordinates*. Of course, two 3-tuples (y_0, y_1, y_2) and (z_0, z_1, z_2) with $y_0 \neq 0$ and $z_0 \neq 0$ are both homogeneous coordinates of the same point of $AG(2, \mathbb{F}_q)$ if and only if there is a $\lambda \in \mathbb{F}_q^*$ such that

$$(z_0, z_1, z_2) = \lambda(y_0, y_1, y_2).$$

Call the points of $AG(2, \mathbb{F}_q)$ the *finite points*.

Let

$$a_0 + a_1 x_1 + a_2 x_2 = 0 \tag{2.1}$$

be a line in $AG(2, \mathbb{F}_q)$. Under the substitution $x_1 = y_0^{-1} y_1, x_2 = y_0^{-1} y_2$, the line (2.1) is carried into a linear homogeneous equation

$$a_0 y_0 + a_1 y_1 + a_2 y_2 = 0. \tag{2.2}$$

Of course, the homogeneous coordinates of the finite points on (2.1) satisfy equation (2.2). Besides these points, there are also solutions $\lambda(0, a_2, -a_1)$ of (2.2) where $\lambda \in \mathbb{F}_q^*$; and these solutions are not homogeneous coordinates of the finite points lying on (2.1). We regard $\lambda(0, a_2, -a_1)$ with $\lambda \in \mathbb{F}_q^*$ as a system of homogeneous coordinates of a *point at infinity*. That is, besides the finite points on (2.2), we add a point at infinity on it. By Theorem 1.14, (2.2) has q finite points, now including the point at infinity on it, it has $q + 1$ points altogether. Now let

$$a_0 + a_1 x_1 + a_2 x_2 = 0$$

and

$$b_0 + b_1 x_1 + b_2 x_2 = 0$$

be a pair of parallel lines in $AG(2, \mathbb{F}_q)$, i.e., $a_1 : a_2 = b_1 : b_2$. Under the substitution $x_1 = y_0^{-1} y_1, x_2 = y_0^{-1} y_1$, they are carried into

$$a_0 y_0 + a_1 y_1 + a_2 y_2 = 0$$

and

$$b_0 y_0 + b_1 y_1 + b_2 y_2 = 0,$$

respectively. The point at infinity on these two lines are $(0, a_2, -a_1)$ and $(0, b_2, -b_1)$, respectively. Since $a_1 : a_2 = b_1 : b_2$, there is a $\lambda \in \mathbb{F}_q^*$ such that $(b_1, b_2) = \lambda(a_1, a_2)$. Thus $(0, b_2, -b_1) = \lambda(0, a_2, -a_1)$, and $(0, a_2, -a_1)$ and $(0, b_2, -b_1)$ are two systems of homogeneous coordinates of the same point at infinity. We have proved that to each pencil of parallel lines in $AG(2, \mathbb{F}_q)$ a common point at infinity is added. Furthermore, it is easy to show that the homogeneous coordinates of the points at infinity added to two intersecting lines are not proportional, which means that the points at infinity added to two intersecting lines are distinct. We know from Theorem 1.14 that there are $q + 1$ pencils of parallel lines in $AG(2, \mathbb{F}_q)$, hence altogether $q + 1$ points at infinity are added to $AG(2, \mathbb{F}_q)$. On the other hand, there are $q^2 - 1$ non-zero 3-tuples of the form $(0, y_1, y_2)$; they are divided into subsets, each of which consists of $q - 1$ pairwise proportional 3-tuples, for instance $\{\lambda(0, y_1, y_2) \mid \lambda \in \mathbb{F}_q^*\}$; the 3-tuples in the same subset are systems of homogeneous coordinates of the same point at infinity; hence, altogether we have $(q^2 - 1)/(q - 1) = q + 1$ points at infinity. All these $q+1$ points at infinity satisfy the linear homogeneous equation

$$y_0 = 0,$$

which will be regarded as a line, called the *line at infinity*. We know from Theorem 1.14 that $AG(2, \mathbb{F}_q)$ has q^2 finite points and $q^2 + q$ lines, thus with $q + 1$ points at infinity and a line at infinity added to it we obtain an extended plane with $q^2 + q + 1$ points and $q^2 + q + 1$ lines.

Finally, we notice that the set of all systems of homogeneous coordinates of a finite point with the zero vector $(0, 0, 0)$ added forms a 1-dimensional vector subspace of the 3-dimensional row vector space $\mathbb{F}_q^{(3)}$, whose non-zero vectors have non-zero first components. The set of all systems of homogeneous coordinates of a point at infinity with the zero vector added forms a 1-dimensional vector subspace of $\mathbb{F}_q^{(3)}$, whose non-zero vectors have zero first components. Altogether we obtain $q^2 + q + 1$ 1-dimensional vector subspaces of $\mathbb{F}_q^{(3)}$, and they exhaust all the 1-dimensional vector subspaces. Therefore the set of points of the extended plane is in one-to-one correspondence with the set of 1-dimensional vector subspaces of $\mathbb{F}_q^{(3)}$.

Moreover, we see easily that the systems of homogeneous coordinates of all the points on a line in our extended plane with the zero vector added constitute a 2-dimensional vector subspace of $\mathbb{F}_q^{(3)}$. There is a one-to-one correspondence between the set of lines of our extended plane and the set of 2-dimensional vector subspaces of $\mathbb{F}_q^{(3)}$. Hence our extended plane is actually $PG(2, \mathbb{F}_q)$.

The above construction of $PG(2, \mathbb{F}_q)$ from $AG(2, \mathbb{F}_q)$ implies also that a line in $PG(2, \mathbb{F}_q)$ consists of the $q+1$ points whose coordinates are solutions of a linear homogeneous equation

$$a_0 y_0 + a_1 y_1 + a_2 y_2 = 0.$$

Example 2.1. We know from Example 1.1 that $AG(2, \mathbb{F}_2)$ has 4 points

$$(0,0), (1,0), (0,1), (1,1);$$

they have homogeneous coordinates

$$(1,0,0), (1,1,0), (1,0,1), (1,1,1),$$

respectively. $AG(2, \mathbb{F}_2)$ has 3 pencils of parallel lines

$$\begin{matrix} x_1 = 0 \\ x_1 = 1 \end{matrix}, \quad \begin{matrix} x_2 = 0 \\ x_2 = 1 \end{matrix}, \quad \begin{matrix} x_1 + x_2 = 0 \\ x_1 + x_2 = 1 \end{matrix}.$$

In homogeneous coordinates, they have the equations

$$\begin{matrix} x_1 = 0 \\ x_0 + x_1 = 0 \end{matrix}, \quad \begin{matrix} x_2 = 0 \\ x_0 + x_2 = 0 \end{matrix}, \quad \begin{matrix} x_1 + x_2 = 0 \\ x_0 + x_1 + x_2 = 0 \end{matrix},$$

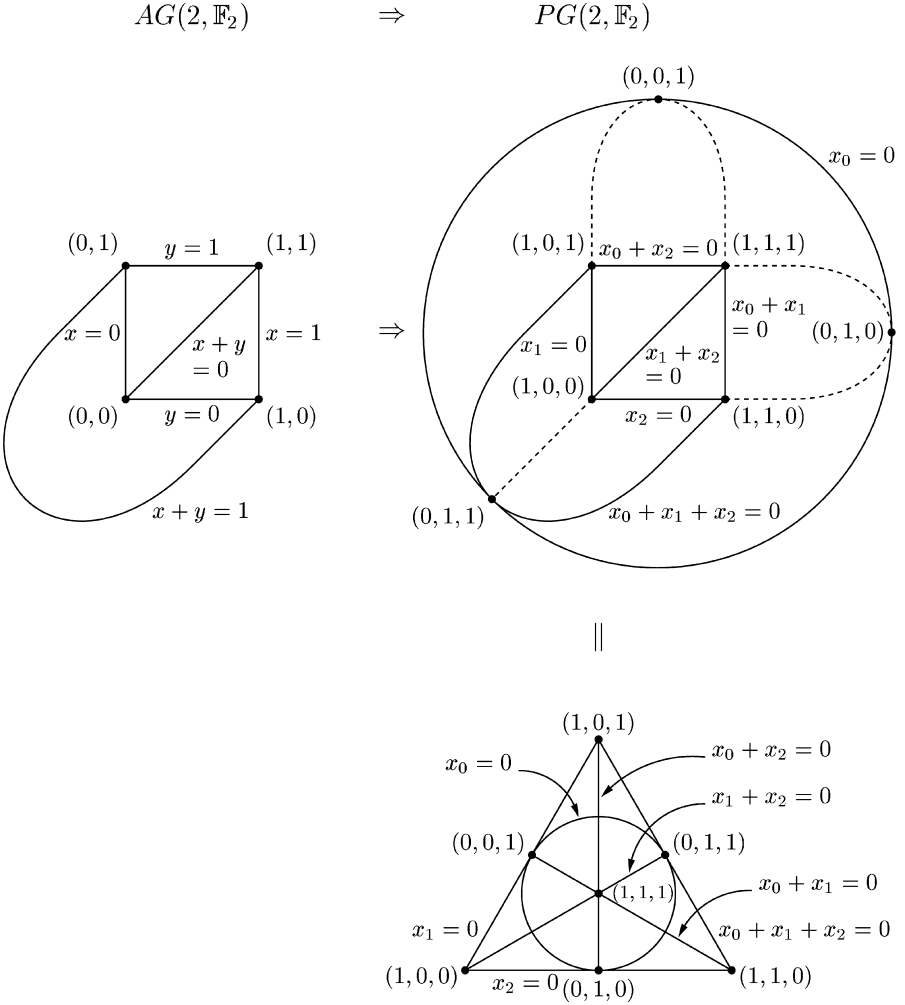
respectively, and the following 3 points at infinity

$$(0,0,1), (0,1,0), (0,1,1)$$

are added to them, respectively. Moreover, these points at infinity lie on the line at infinity

$$x_0 = 0.$$

If we add the 3 points at infinity and the line at infinity to $AG(2, \mathbb{F}_2)$, we obtain the $PG(2, \mathbb{F}_2)$, which is sometimes called the *Fano plane*. The following figures illustrate the process to obtain $PG(2, \mathbb{F}_2)$ from $AG(2, \mathbb{F}_2)$:



$\square$

The process to obtain $PG(2,\mathbb{F}_q)$ from $AG(2,\mathbb{F}_q)$ by adding a set of points at infinity shows that any two parallel lines in $AG(2,\mathbb{F}_q)$ intersect at a point at infinity. Hence any two lines in $PG(2,\mathbb{F}_q)$ has a point of intersection. We collect the main properties of $PG(2,\mathbb{F}_q)$ in the following theorem:

Theorem 2.1. In $PG(2,\mathbb{F}_q)$

(i) There are q^2+q+1 points and q^2+q+1 lines; every line contains $q+1$ points and through every point passes $q+1$ lines.

(ii) Any two distinct points are joined by exactly one line; any two distinct lines intersect in exactly one point.

(iii) Three distinct points (a_0, a_1, a_2), (b_0, b_1, b_2), and (c_0, c_1, c_2) are *collinear*, i.e., lie on the same line, if and only if

$$\begin{vmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{vmatrix} = 0;$$

three distinct lines

$$\begin{aligned} a_0x_0 + a_1x_1 + a_2x_2 &= 0, \\ b_0x_0 + b_1x_1 + b_2x_2 &= 0, \\ c_0x_0 + c_1x_1 + c_2x_2 &= 0 \end{aligned}$$

are *concurrent*, i.e., intersect in a common point, if and only if

$$\begin{vmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{vmatrix} = 0.$$

Proof. (i) is clear from the construction of $PG(2, \mathbb{F}_q)$ from $AG(2, \mathbb{F}_q)$.

(ii) Let P_1 and P_2 be two distinct points with coordinates (a_0, a_1, a_2) and (b_0, b_1, b_2), respectively. Then (a_0, a_1, a_2) and (b_0, b_1, b_2) are not proportional, i.e., the matrix

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \end{pmatrix}$$

is of rank 2. Therefore the solutions of the system of linear homogeneous equations

$$\left.\begin{aligned} a_0x_0 + a_1x_1 + a_2x_2 &= 0, \\ b_0x_0 + b_1x_1 + b_2x_2 &= 0 \end{aligned}\right\} \tag{2.3}$$

form a 1-dimensional vector subspace of $\mathbb{F}_q^{(3)}$. Let (f_0, f_1, f_2) be a non-zero vector in the solution set. Then both (a_0, a_1, a_2) and (b_0, b_1, b_2) lie on the line

$$f_0x_0 + f_1x_1 + f_2x_2 = 0. \tag{2.4}$$

This proves that any two distinct points are joined by a line. Moreover, if both (a_0, a_1, a_2) and (b_0, b_1, b_2) lie on the line

$$g_0x_0 + g_1x_1 + g_2x_2 = 0, \tag{2.5}$$

then (g_0, g_1, g_2) is also a solution of (2.3), and hence is proportional to (f_0, f_1, f_2). Consequently, (2.4) and (2.5) define the same line. This proves that any two distinct points are joined by exactly one line.

That any two distinct lines intersect in exactly one point can be proved in a similar way.

(iii) Three distinct points (a_0, a_1, a_2), (b_0, b_1, b_2), and (c_0, c_1, c_2) are collinear if and only if there is a line, i.e., a linear homogeneous equation

$$ax_0 + bx_1 + cx_2 = 0,$$

where a, b, c are not all zero, such that

$$\begin{aligned} aa_0 + ba_1 + ca_2 &= 0, \\ ab_0 + bb_1 + cb_2 &= 0, \\ ac_0 + bc_1 + cc_2 &= 0. \end{aligned}$$

That is, the above system of linear homogeneous equations have a non-zero solution (a, b, c), and this is equivalent to

$$\begin{vmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{vmatrix} = 0.$$

Similarly, three distinct lines

$$\begin{aligned} a_0x_0 + a_1x_1 + a_2x_2 &= 0, \\ b_0x_0 + b_1x_1 + b_2x_2 &= 0, \\ c_0x_0 + c_1x_1 + c_2x_2 &= 0 \end{aligned}$$

are concurrent if and only if the above system of linear homogeneous equations have a non-zero solution. The latter is equivalent to

$$\begin{vmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{vmatrix} = 0. \qquad \square$$

Now we shall give two examples showing how $PG(2, \mathbb{F}_q)$ is used to construct error-correcting codes and authentication codes, respectively.

Example 2.2. Let us use the projective plane $PG(2, \mathbb{F}_q)$ to construct an *error-correcting code*. $PG(2, \mathbb{F}_q)$ has $q^2 + q + 1$ points, and we call these

points $1, 2, \ldots, q^2+q+1$. The components of the vectors of the (q^2+q+1)-dimensional row vector space $\mathbb{F}_q^{(q^2+q+1)}$ are indexed by $1, 2, \ldots, q^2+q+1$. We define a binary code C of code length q^2+q+1, which consists of the following four kinds of code words:

(i) The zero vector $(0, 0, \ldots, 0)$.

(ii) To each line l of $PG(2, \mathbb{F}_q)$, there corresponds a code word whose i-th component is 1 if and only if the point i lies on l. There are altogether q^2+q+1 code words of this type, each of which is of weight $q+1$.

(iii) To each line l of $PG(2, \mathbb{F}_q)$, there corresponds a code word whose i-th component is 1 if and only if the point i does not lie on l. There are also q^2+q+1 code words of this type, but each of which is of weight q^2.

(iv) The all one vector $(1, 1, \ldots, 1)$.

It is clear that $\mid C \mid = 2(q^2+q+2)$. It can be easily verified that the minimum distance of C is $q+1$ and hence C is a $[\frac{q}{2}]$-error-correcting code.

In particular, we take $q = 2$, the code defined above is of code length 7, contains 16 code words and is single-error correcting. Since

$$\mid C \mid \left(\binom{7}{0} + \binom{7}{1}(2-1) \right) = 16 \cdot 8 = 2^7, \qquad \square$$

the code is *perfect*.

Example 2.3. We can also use the projective plane $PG(2, \mathbb{F}_q)$ to construct authentication codes as we did in the affine case. At first we fix a line l in $PG(2, \mathbb{F}_q)$. The points on l are regarded as source states, the points not on l are regarded as encoding rules, and the lines distinct from l are regarded as messages. Assume that the encoding rules work as follows. Given a source state s (which is a point on l) and an encoding rule e (which is a point not on l), there is a unique line m passing through s and e. Then we define $m = e(s)$, i.e., the source state s is encoded by the encoding rule e into the message m. Thus we obtain an authentication code.

Let us compute the parameters of the code. $PG(2, \mathbb{F}_q)$ has q^2+q+1 points and any line in it has $q+1$ points. Therefore there are $q+1$ source states and $(q^2+q+1)-(q+1) = q^2$ encoding rules. Moreover, $PG(2, \mathbb{F}_q)$ has q^2+q+1 lines, hence there are $(q^2+q+1)-1 = q^2+q$ messages.

Assuming that an equally likely probability distribution is given on the set $\mathcal{E}$ of encoding rules, we can compute the probability P_I that a fraudulent message m' sent by the opponent is accepted by the receiver as authentic

when he plays an impersonation game and the probability P_S that after the opponent observed a message m sent by the transmitter to the receiver, a fraudent message m' sent by the opponent is accepted by the receiver as authentic when he plays a substitution game. Proceeding as in the affine case, we obtain

$$P_I = \frac{q}{q^2} = \frac{1}{q},$$

$$P_S = \frac{1}{q}.$$

□

Now let us return to the study of $PG(n, \mathbb{F}_q)$. According to the definition of $PG(n, \mathbb{F}_q)$ given at the beginning of this section, the set of r-flats of $PG(n, \mathbb{F}_q)$ and the set of $(r+1)$-dimensional vector subspaces of $\mathbb{F}_q^{(n+1)}$ are in one-to-one correspondence. The r-flat F corresponding to an $(r+1)$-dimensional vector subspace U can be regarded as the set of points whose coordinates are the non-zero vectors of U. After adopting this point of view, we have

Theorem 2.2. A hyperplane in $PG(n, \mathbb{F}_q)$ is the set of points whose coordinates are non-zero solutions of a linear homogeneous equation in $n+1$ unknowns, and conversely.

Proof. Let H be a hyperplane of $PG(n, \mathbb{F}_q)$, then H corresponds uniquely to an n-dimensional vector subspace U of $\mathbb{F}_q^{(n+1)}$ such that H is the set of 1-dimensional vector subspaces contained in U. Let $v_1, v_2, \ldots, v_n$ be a basis of U. Then

$$U = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

is a matrix representation of the n-dimensional vector subspace U. Consider the system of linear homogeneous equations in $n+1$ unknowns

$$U \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

Since rank $U = n$, the solutions of this system of equations form a 1-dimensional vector subspace P of $\mathbb{F}_q^{(n+1)}$. Let $(a_0, a_1, \ldots, a_n)$ be a non-zero vector of P, then the n-dimensional vector subspace U is the solution space of the following linear homogeneous equation in $n+1$ unknowns

$$a_0x_0 + a_1x_1 + \cdots + a_nx_n = 0. \tag{2.6}$$

Hence the hyperplane H is the set of points whose coordinates are non-zero solutions of (2.6).

Conversely, let (2.6) be given. Then the set of solutions of (2.6) form an n-dimensional vector subspace U of $\mathbb{F}_q^{(n+1)}$, and hence the hyperplane corresponding to U is the set of points whose coordinates are non-zero solutions of (2.6). □

The linear homogeneous equation (2.6), whose non-zero solutions are all the points of a hyperplane H, is called the *equation of the hyperplane* H. Clearly, it is uniquely determined up to a non-zero constant multiple of $\mathbb{F}_q$.

In the same way we can prove also

Theorem 2.3. Any r-flat $(0 \leq r \leq n)$ of $PG(n, \mathbb{F}_q)$ is the set of points whose coordinates are non-zero solutions of a system of $n-r$ independent linear homogeneous equations in $n+1$ unknowns, and conversely.

We define the *dimension* of an r-flat F in $PG(n, \mathbb{F}_q)$ to be r and denote $\dim F = r$; however the corresponding $(r+1)$-dimensional vector subspace U is of dimension $r+1$. Moreover, we define the empty set ϕ of points in $PG(n, \mathbb{F}_q)$ to be of dimension -1 and denote $\dim \phi = -1$.

Let R_1 and R_2 be flats of $PG(n, \mathbb{F}_q)$. The set of points contained in both R_1 and R_2 is called the *intersection* of R_1 and R_2, and denoted by $R_1 \cap R_2$. It can be easily seen that $R_1 \cap R_2$ is a flat of $PG(n, \mathbb{F}_q)$. The minimal flat containing both R_1 and R_2 is called the *join* of R_1 and R_2, and denoted by $R_1 \cup R_2$. From the dimension formula of $\mathbb{F}_q^{(n+1)}$, we deduce immediately the dimension formula of $PG(n, \mathbb{F}_q)$.

Theorem 2.4. Let R_1 and R_2 be two flats of $PG(n, \mathbb{F}_q)$. Then

$$\dim R_1 + \dim R_2 = \dim (R_1 \cap R_2) + \dim (R_1 \cup R_2). \qquad \square$$

Also, from the Anzahl theorems in $\mathbb{F}_q^{(n)}$, i.e., Theorem 1.7, Corollary 1.8, and Corollary 1.9, we deduce immediately the following Anzahl theorems in $PG(n, \mathbb{F}_q)$:

Theorem 2.5. In $PG(n, \mathbb{F}_q)$,

(i) The number of m-flats $(0 \leq m \leq n)$ is

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix}_q .$$

In particular, $PG(n, \mathbb{F}_q)$ has $q^n + q^{n-1} + \cdots + q + 1$ points and $q^n + q^{n-1} + \cdots + q + 1$ hyperplanes.

(ii) The number of k-flats contained in a given m-flat $(0 \leq k \leq m \leq n)$ is

$$\begin{bmatrix} m+1 \\ k+1 \end{bmatrix}_q .$$

(iii) The number of m-flats containing a given k-flat $(0 \leq k \leq m \leq n)$ is

$$\begin{bmatrix} n-k \\ m-k \end{bmatrix}_q . \qquad \square$$

It follows from Theorem 2.1 (ii) that there is no parallelism in the projective plane $PG(2, \mathbb{F}_q)$. However, if $n \geq 3$, there exists a pair of *skew lines* in $PG(n, \mathbb{F}_q)$, that is, a pair of lines which do not intersect.

Theorem 2.6. For $n \geq 3$, there exists a pair of skew lines in $PG(n, \mathbb{F}_q)$.

Proof. For simplicity let us assume that $n = 3$. It is clear that both

$$V_1 = \{(x_0, x_1, 0, 0) \mid x_0, x_1 \in \mathbb{F}_q\}$$

and

$$V_2 = \{(0, 0, x_2, x_3) \mid x_2, x_3 \in \mathbb{F}_q\}$$

are 2-dimensional vector subspaces of $\mathbb{F}_q^{(4)}$ and that $V_1 \cap V_2 = (0)$. Hence the lines corresponding to them are two skew lines. $\square$

Theorem 2.7. In $PG(n, \mathbb{F}_q)$,

(i) Any two distinct points are joined by a unique line.

(ii) Any three points, which are not collinear, lie on exactly one plane.

(iii) Any $r+1$ points, which do not lie on any $(r-1)$-flat, lie on exactly one r-flat $(1 \leq r \leq n)$. Moreover, on any r-flat $(1 \leq r \leq n)$ there exist $r+1$ points, which do not lie on any $(r-1)$-flat.

Proof. We only need to prove (iii). Let $P_0, P_1, \ldots, P_r$ be $r+1$ points which do not lie on any $(r-1)$-flat. Let $(a_0^{(i)}, a_1^{(i)}, \ldots, a_n^{(i)})$ be the coordinates of P_i $(0 \leq i \leq r)$. Denote the vector subspace spanned by the $r+1$ vectors $(a_0^{(i)}, a_1^{(i)}, \ldots, a_n^{(i)})$, $i = 0, 1, \ldots, r$, by U. Then $\dim U \leq r+1$. Denote the flat corresponding to U by F. By hypothesis, $\dim F \geq r$, and hence $\dim U = \dim F + 1 \geq r+1$. Consequently, $\dim U = r+1$ and hence $\dim F = r$. Of course, F contains $P_0, P_1, \ldots, P_r$. This proves the existence part of the first assertion in (iii). Now let F' be another r-flat containing $P_0, P_1, \ldots, P_r$. Then there is a vector subspace U' of dimension $r+1$ corresponding to F'. Thus $P_0, P_1, \ldots, P_r$ and hence U are contained in U'. But $\dim U = \dim U' = r+1$, hence $U = U'$ and therefore $F = F'$.

Now let F be an r-flat and U be the corresponding vector subspace of dimension $r+1$. Let $v_0, v_1, \ldots, v_r$ be a basis of U. Then $v_0, v_1, \ldots, v_r$ are the coordinates of $r+1$ points $P_0, P_1, \ldots, P_r$, which do not lie on any $(r-1)$-flat. □

Any $r+1$ points which do not lie on any $(r-1)$-flat are said to be *independent* and any set of $r+1$ independent points is simply called an *independent set of $r+1$ points*. From the proof of Theorem 2.7 follows also the following lemma:

Lemma 2.8. Let $P_0, P_1, \ldots, P_r$ be $r+1$ points in $PG(n, \mathbb{F}_q)$ and P_i have coordinates $(a_0^{(i)}, a_1^{(i)}, \ldots, a_n^{(i)})$ $(0 \leq i \leq r)$. Then $P_0, P_1, \ldots, P_r$ form an independent set of $r+1$ points if and only if the $r+1$ vectors $(a_0^{(0)}, a_1^{(0)}, \ldots, a_n^{(0)}), (a_0^{(1)}, a_1^{(1)}, \ldots, a_n^{(1)}), \ldots, (a_0^{(r)}, a_1^{(r)}, \ldots, a_n^{(r)})$ are linearly independent. □

According to Theorem 2.3 a line can be regarded as the set of points whose coordinates are non-zero solutions of $n-1$ independent linear homogeneous equations in $n+1$ variables. However as in the affine case a line in $PG(n, \mathbb{F}_q)$ has also a parametric representation. Let P and Q be two distinct points on a line l with coordinates $(x_0, x_1, \ldots, x_n)$ and $(y_0, y_1, \ldots, y_n)$, respectively. Then

$$l = \{\lambda(x_0, x_1, \ldots, x_n) + \mu(y_0, y_1, \ldots, y_n) \mid \lambda, \mu \in \mathbb{F}_q \text{ and are not both zero}\}.$$

More generally, let $P_0, P_1, \ldots, P_r$ be $r+1$ points not lying on any $(r-1)$-flat and let the coordinates of P_i be $(x_0^{(i)}, x_1^{(i)}, \ldots, x_n^{(i)})$ $(0 \leq i \leq r)$. Then the unique r-flat passing through $P_0, P_1, \ldots, P_r$ is

$$\{\lambda_0(x_0^{(0)}, x_1^{(0)}, \ldots, x_n^{(0)}) + \lambda_1(x_0^{(1)}, x_1^{(1)}, \ldots, x_n^{(1)}) + \cdots + \lambda_r(x_0^{(r)}, x_1^{(r)}, \ldots, x_n^{(r)}) \mid \lambda_0, \lambda_1, \ldots, \lambda_r \in \mathbb{F}_q \text{ and are not all zero}\}.$$

Now let us introduce the projective group.

Any $T \in GL_{n+1}(\mathbb{F}_q)$ defines a point to point transformation of $PG(n, \mathbb{F}_q)$ in the following way:

$$\begin{aligned} PG(n, \mathbb{F}_q) &\to PG(n, \mathbb{F}_q) \\ (x_0, x_1, \ldots, x_n) &\to (x_0, x_1, \ldots, x_n)T. \end{aligned} \tag{2.7}$$

This is well-defined. In fact, if $(y_0, y_1, \ldots, y_n)$ is another system of coordinates of the point $(x_0, x_1, \ldots, x_n)$, that is, there is a non-zero element $\lambda \in \mathbb{F}_q$ such that $(y_0, y_1, \ldots, y_n) = \lambda(x_0, x_1, \ldots, x_n)$, then $(y_0, y_1, \ldots, y_n)T = \lambda(x_0, x_1, \ldots, x_n)T$. The transformation (2.7) is called a *projective transformation* of $PG(n, \mathbb{F}_q)$, and denoted by $\overline{T}$.

Lemma 2.9. Two elements T and T_1 of $GL_{n+1}(\mathbb{F}_q)$ define the same projective transformation if and only if there is an element $\lambda \in \mathbb{F}_q^*$ such that $T_1 = \lambda T$.

Proof. Suppose that there is an element $\lambda \in \mathbb{F}_q^*$ such that $T_1 = \lambda T$. Then under the projective transformation $\overline{T}_1$, defined by T_1, we have

$$\begin{aligned} (x_0, x_1, \ldots, x_n) \to (x_0, x_1, \ldots, x_n)T_1 &= (x_0, x_1, \ldots, x_n)\lambda T \\ &= \lambda(x_0, x_1, \ldots, x_n)T. \end{aligned}$$

Hence $\overline{T}_1 = \overline{T}$.

Conversely, suppose that $\overline{T}_1 = \overline{T}$. Let

$$T_1 = (s_{ij})_{0 \le i,j \le n} \qquad \text{and} \qquad T = (t_{ij})_{0 \le i,j \le n}.$$

Then for each i $(0 \le i \le n)$ $e_i T_1$ and $e_i T$ are proportional, thus there is an element $\lambda_i \in \mathbb{F}_q^*$ such that

$$(s_{i0}, s_{i1}, \ldots, s_{in}) = \lambda_i(t_{i0}, t_{i1}, \ldots, t_{in}).$$

But for all $i \ne j$ $(e_i + e_j)T_1$ and $(e_i + e_j)T$ are also proportional, it follows that $\lambda_0 = \lambda_1 = \cdots = \lambda_n = \lambda$, say. Therefore $T_1 = \lambda T$. □

In virtue of Lemma 2.9, it is natural to introduce the factor group

$$PGL_{n+1}(\mathbb{F}_q) = GL_{n+1}(\mathbb{F}_q)/Z_{n+1},$$

where

$$Z_{n+1} = \{\lambda I^{(n+1)} \mid \lambda \in \mathbb{F}_q^*\}.$$

We leave as an exercise to prove that Z_{n+1} is the *center* of $GL_{n+1}(\mathbb{F}_q)$. The image of $T \in GL_{n+1}(\mathbb{F}_q)$ in $PGL_{n+1}(\mathbb{F}_q)$ is also denoted by $\overline{T}$ and called a *projective transformation*. The group $PGL_{n+1}(\mathbb{F}_q)$ is called the *projective general linear group* of degree $n+1$ over $\mathbb{F}_q$. Notice that there is an *action* of $PGL_{n+1}(\mathbb{F}_q)$ on $PG(n, \mathbb{F}_q)$ defined as follows

$$\begin{aligned} PG(n, \mathbb{F}_q) \times PGL_{n+1}(\mathbb{F}_q) &\to PG(n, \mathbb{F}_q) \\ ((x_0, x_1, \ldots, x_n), \overline{T}) &\to (x_0, x_1, \ldots, x_n)T. \end{aligned}$$

Two geometric figures in the projective space are said to be *projectively equivalent*, if one of them can be carried into the other by a projective transformation.

According to the Erlangen Program, *projective geometry* is the study of properties of geometric figures in projective space which are invariant under the projective general linear group.

Clearly, r-flats are carried into r-flats by any projective transformation. Thus the properties that a geometric figure being a flat and the dimension of a flat are invariant under the projective general linear group.

Now let us study the transitivity properties of $PGL_{n+1}(\mathbb{F}_q)$.

Theorem 2.10. (i) The group $PGL_{n+1}(\mathbb{F}_q)$ is doubly transitive on the set of points of $PG(n, \mathbb{F}_q)$.

(ii) The group $PGL_{n+1}(\mathbb{F}_q)$ is transitive on the set of independent triples of points of $PG(n, \mathbb{F}_q)$.

(iii) The group $PGL_{n+1}(\mathbb{F}_q)$ is transitive on the set of independent sets of $r+1$ points $(0 \le r \le n)$.

(iv) The group $PGL_{n+1}(\mathbb{F}_q)$ is transitive on the set of r-flats $(0 \le r \le n)$.

(v) The group $PGL_{n+1}(\mathbb{F}_q)$ is transitive on the set of $(n+2)$-subsets of points such that any $n+1$ of them are independent.

Proof. It is sufficient to prove (iii) and (v), since (i) and (ii) are special cases of (iii), and (iv) is a consequence of (iii) by Theorem 2.7.

Let $P_0, P_1, \ldots, P_r$ be $r+1$ independent points and the coordinates of P_i be $(a_{i0}, a_{i1}, \ldots, a_{in})$ $(0 \le i \le r)$. It is easy to see that the following $r+1$ points

$$E_0 = (1, 0, \ldots, 0), E_1 = (0, 1, 0, \ldots, 0), \ldots, E_r = (0, 0, \ldots, 0, 1, 0, \ldots, 0)$$

are independent. If we can find a projective transformation which carries $P_0, P_1, \ldots, P_r$ into $E_0, E_1, \ldots, E_r$, respectively, then (iii) will follow. Form the $(r+1) \times (n+1)$ matrix

$$\begin{pmatrix} a_{00} & a_{01} & \ldots & a_{0n} \\ a_{10} & a_{11} & \ldots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r0} & a_{r1} & \ldots & a_{rn} \end{pmatrix}.$$

It follows from Lemma 2.8 that the above matrix is of rank $r+1$. We can supplement $n-r$ more rows to the above matrix such that the obtained matrix

$$T = \begin{pmatrix} a_{00} & a_{01} & \ldots & a_{0n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r0} & a_{r1} & \ldots & a_{rn} \\ a_{r+1\,0} & a_{r+1\,1} & \ldots & a_{r+1\,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n0} & a_{n1} & \ldots & a_{nn} \end{pmatrix}$$

is nonsingular. Then $(a_{i0}, a_{i1}, \ldots, a_{in})T^{-1} = e_i,\ i = 0, 1, \ldots, r$, and hence $P_i\overline{T}^{-1} = E_i,\ i = 0, 1, \ldots, r$. This proves (iii).

Now let $P_0, P_1, \ldots, P_{n+1}$ be $n+2$ points such that any $n+1$ of them are independent. It is clear that any $n+1$ of the following $n+2$ points

$$E_0 = (1, 0, 0, \ldots, 0), E_1 = (0, 1, 0, \ldots, 0), \ldots,$$
$$E_n = (0, 0, \ldots, 0, 1), E\ = (1, 1, \ldots, 1)$$

are independent. If we can find a projective transformation which carries $P_0, P_1, \ldots, P_{n+1}$ into $E_0, E_1, \ldots, E_n, E$, respectively, then (v) will follow. By (iii), there is a projective transformation $\overline{T}$ such that $P_i\overline{T} = E_i,\ i = 0, 1, \ldots, n$. Assume that

$$P_{n+1}\overline{T} = (x_0, x_1, \ldots, x_n).$$

Then any $n+1$ of the $n+2$ points $E_0, E_1, \ldots, E_n$, and $(x_0, x_1, \ldots, x_n)$ are independent. It follows from Lemma 2.8 that $x_0 \neq 0, x_1 \neq 0, \ldots, x_n \neq 0$. Thus the projective transformation

$$\overline{\begin{pmatrix} x_0^{-1} & & & \\ & x_1^{-1} & & \\ & & \ddots & \\ & & & x_n^{-1} \end{pmatrix}}$$

will leave each of the $n+1$ points $E_0, E_1, \ldots, E_n$ invariant and carry the point $(x_0, x_1, \ldots, x_n)$ into E. This proves (v). □

Notice that in contrast to Theorem 2.10 (v) for $r < n$ the group $PGL_{n+1}(\mathbb{F}_q)$ is not transitive on the set of $(r+2)$-subsets of points such that any $r+1$ of them are independent. For example, when $n = 2$ and $r = 1$ consider a set of 3 collinear points and a set of 3 non-collinear points. Both of them are 3-subsets of points such that any 2 of them are independent, but there does not exist an element in $PG_3(\mathbb{F}_q)$ carrying one of them to the other.

Projective transformations are useful in proving theorems in $PG(n, \mathbb{F}_q)$. The following two theorems are interesting examples.

Theorem 2.11. Given a pair of skew lines in $PG(3, \mathbb{F}_q)$ and a point on neither of the lines, there is a unique line which intersects each of the two lines and passes through the point, or simply, *lies on all the three.*

Proof. Let L_1 and L_2 be the given pair of skew lines and P be the given point on neither of them. Let the equations of L_1 be

$$\begin{cases} a_0x_0 + a_1x_1 + a_2x_2 + a_3x_3 = 0, \\ b_0x_0 + b_1x_1 + b_2x_2 + b_3x_3 = 0. \end{cases}$$

Let the equations of L_2 be

$$\begin{cases} c_0x_0 + c_1x_1 + c_2x_2 + c_3x_3 = 0, \\ d_0x_0 + d_1x_1 + d_2x_2 + d_3x_3 = 0. \end{cases}$$

Then

$$\operatorname{rank} \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \end{pmatrix} = \operatorname{rank} \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ d_0 & d_1 & d_2 & d_3 \end{pmatrix} = 2.$$

Since L_1 and L_2 are skew,

$$\operatorname{rank} \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \\ c_0 & c_1 & c_2 & c_3 \\ d_0 & d_1 & d_2 & d_3 \end{pmatrix} = 4.$$

Let

$$T = \begin{pmatrix} a_0 & b_0 & c_0 & d_0 \\ a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{pmatrix}^{-1}.$$

Then T defines a projective transformation $\overline{T}$ of $PG(3, \mathbb{F}_q)$

$$(x_0, x_1, x_2, x_3) \mapsto (x_0, x_1, x_2, x_3)T,$$

which carries L_1 and L_2 into

$$L_1' : \begin{cases} x_0 = 0, \\ x_1 = 0 \end{cases} \qquad \text{and} \qquad L_2' : \begin{cases} x_2 = 0, \\ x_3 = 0, \end{cases}$$

respectively. Let the image P' of P under T have coordinates (p_0, p_1, p_2, p_3). Since P lies on neither L_1 nor L_2, P' lies on neither L_1' nor L_2'. Then at least one of p_0 and p_1 is non-zero and at least one of p_2 and p_3 is non-zero. Thus

$$L' : \begin{cases} p_1 x_0 - p_0 x_1 = 0, \\ p_3 x_2 - p_2 x_3 = 0 \end{cases}$$

is a line lying on L_1', L_2', and P'. Therefore the preimage L of L' under the projective transformation $\overline{T}$ is a line lying on L_1, L_2, and P.

Now assume that L_0 is another line lying on L_1, L_2, and P. Then L and L_0 intersect at P and they determine a plane π. Consequently, both the point of intersection of L and L_1 and the point of intersection of L_0 and L_1 lie in π, and hence L_1 lie in π. Similarly, L_2 also lie in π. Being coplanar lines, L_1 and L_2 must intersect, this contradicts the assumption that they are skew. □

The following theorem gives a remarkable distinction between a $PG(2, \mathbb{F}_q)$ with q even and a $PG(2, \mathbb{F}_q)$ with q odd. At first we state the following definition.

A *complete quadrangle* in the projective plane is a configuration consisting of (i) four points, no three of them are collinear, and (ii) six lines, each of them joining a pair of the points. The four points of a complete quadrangle are called its *vertices*, and the six lines are called its *diagonals*. Let A, B, C, D be the four vertices of a complete quadrangle; the points of intersection of AB and CD, of AC and BD, and of AD and BC are called the *diagonal points* of the complete quadrangle.

Theorem 2.12. The three diagonal points of a complete quadrangle in $PG(2, \mathbb{F}_q)$ are collinear if and only if q is even.

Proof. Let A, B, C, and D be the four vertices of a complete quadrangle. Then no three of them are on a line. By Theorem 2.10 (v), there is a

projective transformation $\overline{T}$ such that

$$A\overline{T} = E_0 = (1,0,0),\ B\overline{T} = E_1 = (0,1,0),$$
$$C\overline{T} = E_2 = (0,0,1), \text{ and } D\overline{T} = E = (1,1,1).$$

It is clear that under a projective transformation a complete quadrangle is carried into a complete quadrangle, thus it is sufficient to prove our theorem for the transformed complete quadrangle under $\overline{T}$.

The vertices of the transformed complete quadrangle are E_0, E_1, E_2, and E. The line joining E_0 and E_1 is $x_2 = 0$, the line joining E_2 and E is $x_0 - x_1 = 0$, and the point of intersection of these two lines is $(1,1,0)$. Similarly, the point of intersection of the lines joining E_0 and E_2 and the line joining E_1 and E is $(1,0,1)$; the point of intersection of the line joining E_0 and E and the line joining E_1 and E_2 is $(0,1,1)$. Now

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -2.$$

Therefore by Theorem 2.1 (iii) the three diagonal points $(1,1,0)$, $(1,0,1)$, and $(0,1,1)$ are collinear if and only if q is even. □

2.2 Quadrics in $PG(n, \mathbb{F}_q)$

The set of points $(x_0, x_1, \ldots, x_n)$ of $PG(n, \mathbb{F}_q)$ which satisfy a quadratic homogeneous equation

$$\sum_{0 \le i \le k \le n} a_{ik} x_i x_k = 0, \tag{2.8}$$

where a_{ik} $(0 \le i \le k \le n)$ are elements of $\mathbb{F}_q$ and not all a_{ik} are zero, is called a *quadric* in $PG(n, \mathbb{F}_q)$ and the quadratic equation (2.8) is called its equation.

At first we would like to study how the equation of a quadric can be transformed into the simplest form under the projective group. The solution of this problem is much simpler than the corresponding problem in the affine case we discussed before. As in the affine case we have to distinguish the cases when q is odd or even.

For q being odd, we have

Theorem 2.13. Assume that q is odd. Then under projective transformations a quadric in $PG(n, \mathbb{F}_q)$ can be carried into a quadric with one of

the following quadratic homogeneous equations as its equation

$$\sum_{i=0}^{r} x_i^2 = 0 \, (0 \le r \le n) \tag{2.9}$$

$$\sum_{i=0}^{r-1} x_i^2 + z x_r^2 = 0 \, (r \text{ being odd and } 0 < r \le n), \tag{2.10}$$

where z is a fixed non-square element of $\mathbb{F}_q$. Moreover, any two of these equations are not projectively equivalent, i.e., they cannot be carried from one into another by a projective transformation or a projective transformation followed by a multiplication by an element of $\mathbb{F}_q^*$.

Proof. As in the affine case, we can assume that in equation (2.8) of a quadric the summation range is $0 \le i, k \le n$ and $a_{ik} = a_{ki}$ $(0 \le i, k \le n)$. Introduce the coefficient matrix

$$A = (a_{ik})_{0 \le i,k \le n},$$

which is an $(n+1) \times (n+1)$ symmetric matrix, then (2.8) can be written in the following matrix form

$$(x_0, x_1, \ldots, x_n) A \, {}^t(x_0, x_1, \ldots, x_n) = 0. \tag{2.11}$$

Under a projective transformation

$$(x_0, x_1, \ldots, x_n) \mapsto (x_0, x_1, \ldots, x_n) T, \tag{2.12}$$

where $T \in GL_{n+1}(\mathbb{F}_q)$, (2.11) is carried into

$$(x_0, x_1, \ldots, x_n) T A \, {}^t T \, {}^t(x_0, x_1, \ldots, x_n) = 0.$$

Hence the coefficient matrix A is subjected to the cogredient transformation

$$A \mapsto T A \, {}^t T.$$

Assume that A is of rank $r+1$ $(0 \le r \le n)$; clearly the rank is invariant under $PGL_{n+1}(\mathbb{F}_q)$. By Theorem 1.23 there is an $(n+1) \times (n+1)$

nonsingular matrix T such that

$$TA\,{}^tT = \begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & \ddots & & & & & \\ & & & 1 & & & & \\ & & & & d & & & \\ & & & & & 0 & & \\ & & & & & & \ddots & \\ & & & & & & & 0 \end{pmatrix}, \qquad \begin{matrix} \underbrace{\qquad\qquad} & \underbrace{\quad} & \underbrace{\qquad\qquad} \\ r & 1 & n-r \end{matrix}$$

where $d = 1$ or z. Therefore under a projective transformation (2.12), (2.11) can be carried into either (2.9) or

$$\sum_{i=0}^{r-1} x_i^2 + zx_r^2 = 0.$$

If the latter case occurs and r is even, then by Lemma 1.22 it can be carried under a projective transformation into

$$z\sum_{i=0}^{r} x_i^2 = 0.$$

Canceling z, we obtain (2.9) again. But, if r is odd, then by Lemma 1.24 (2.9) and (2.10) are not projectively equivalent. □

From Theorem 1.26 we deduce

Theorem 2.14. Assume that q is odd. Then under projective transformations a quadric in $PG(n, \mathbb{F}_q)$ can be carried into a quadric with one of the following quadratic homogeneous equations as its equation

$$x_0^2 + \sum_{i=1}^{\nu} x_i x_{\nu+i} = 0 \ (0 \le 2\nu \le n), \tag{2.13}$$

$$\sum_{i=0}^{\nu-1} x_i x_{\nu+i} = 0 \ (2 \le 2\nu \le n+1), \tag{2.14}$$

$$x_0^2 - zx_1^2 + \sum_{i=2}^{\nu+1} x_i x_{\nu+i} = 0 \ (0 \le 2\nu \le n-1). \tag{2.15}$$

Moreover, any two of these equations are not projectively equivalent.

Proof. As in the proof of Theorem 2.13, we can assume that the equation of the quadric is written in the matrix form (2.11)

$$(x_0, x_1, \ldots, x_n) A\, {}^t(x_0, x_1, \ldots, x_n) = 0,$$

where A is an $(n+1)\times(n+1)$ symmetric matrix. Suppose that A is of rank r. If r is odd, we write $r = 2\nu + 1$, then $0 \le 2\nu \le n$. By Theorem 1.26, there is an $(n+1) \times (n+1)$ nonsingular matrix T such that

$$TA\,{}^tT = \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & d & \\ & & & 0^{(n+1-r)} \end{pmatrix},$$

where $d = 1$ or z. Therefore under a projective transformation (2.12), (2.11) can be carried into

$$2\sum_{i=0}^{\nu-1} x_i x_{\nu+i} + dx_{2\nu}^2 = 0. \tag{2.16}$$

Then under the projective transformation

$$(x_0, x_1, \ldots, x_n) \mapsto ((d/2)x_1, \ldots, (d/2)x_\nu, x_{\nu+1}, \ldots, x_{2\nu}, x_0, x_{2\nu+1}, \ldots, x_n)$$

(2.16) is carried into (2.13) multiplied by d. Canceling d, we obtain (2.13). If r is even, we write $r = 2\nu$, then $2 \le 2\nu \le n+1$. By Theorem 1.26, there is an $(n+1) \times (n+1)$ nonsingular matrix T such that

$$TA\,{}^tT = \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 0^{(n+1-2\nu)} & \end{pmatrix} \text{ or } \begin{pmatrix} 0 & I^{(\nu-1)} & & & \\ I^{(\nu-1)} & 0 & & & \\ & & 1 & & \\ & & & -z & \\ & & & & 0^{(n+1-2\nu)} \end{pmatrix}.$$

Therefore under a projective transformation (2.12), (2.11) can be carried into

$$2\sum_{i=0}^{\nu-1} x_i x_{\nu+i} = 0 \tag{2.17}$$

or

$$2\sum_{i=0}^{\nu-2} x_i x_{\nu-1+i} + x_{2\nu-2}^2 - z x_{2\nu-1}^2 = 0. \tag{2.18}$$

Then the projective transformation

$$(x_0, x_1, \ldots, x_n) \mapsto (\frac{1}{2}x_0, \ldots, \frac{1}{2}x_{\nu-1}, x_\nu, \ldots, x_n)$$

carries (2.17) into (2.14), and the projective transformation

$$(x_0, x_1, \ldots, x_n) \mapsto (\frac{1}{2}x_2, \ldots, \frac{1}{2}x_\nu, x_{\nu+1}, \ldots, x_{2\nu-1}, x_0, x_1, x_{2\nu}, \ldots, x_n)$$

carries (2.18) into

$$x_0^2 - zx_1^2 + \sum_{i=2}^{\nu} x_i x_{\nu-1+i} = 0. \tag{2.19}$$

If we write $\nu + 1$ for ν in (2.19), then (2.15) is obtained.

Since the rank of A is invariant under projective transformations, (2.13) cannot be projectively equivalent to (2.14) and (2.15). By Lemma 1.24, (2.14) and (2.15) are also not projectively equivalent. □

The quadratic homogeneous equation (2.8) in $n + 1$ variables and the quadric defined by it are said to be *non-degenerate*, if it cannot be transformed into an equation with less than $n + 1$ variables under projective transformations.

Corollary 2.15. Assume that q is odd. If n is even, then under projective transformations a non-degenerate quadric in $PG(n, \mathbb{F}_q)$ can be carried into

$$x_0^2 + \sum_{i=1}^{\nu} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu. \tag{2.20}$$

If n is odd, then it can be carried into either

$$\sum_{i=0}^{\nu-1} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu - 1, \tag{2.21}$$

or

$$x_0^2 - zx_1^2 + \sum_{i=2}^{\nu+1} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu + 1, \tag{2.22}$$

and these two equations are not projectively equivalent. □

Now let us study the case when q is even. From Theorem 1.35 we deduce immediately

Theorem 2.16. Assume that q is even. Then under projective transformations any quadric in $PG(n, \mathbb{F}_q)$ can be carried into a quadric with one of the following quadratic homogeneous equations as its equation

$$x_0^2 + \sum_{i=1}^{\nu} x_i x_{\nu+i} = 0 \ (0 \le 2\nu \le n), \tag{2.23}$$

$$\sum_{i=0}^{\nu-1} x_i x_{\nu+i} = 0 \ (2 \le 2\nu \le n+1), \tag{2.24}$$

$$\alpha x_0^2 + x_0 x_1 + \alpha x_1^2 + \sum_{i=2}^{\nu+1} x_i x_{\nu+i} = 0 \ (0 \le 2\nu \le n-1), \tag{2.25}$$

where α is a fixed element of $\mathbb{F}_q$ not belonging to $N = \{x^2 + x \mid x \in \mathbb{F}_q\}$. Moreover, any two of these equations are not projectively equivalent. □

Corollary 2.17. Assume that q is even. If n is even, then under projective transformations a non-degenerate quadric in $PG(n, \mathbb{F}_q)$ can be carried into

$$x_0^2 + \sum_{i=1}^{\nu} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu. \tag{2.26}$$

If n is odd, then it can be carried into either

$$\sum_{i=0}^{\nu-1} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu - 1, \tag{2.27}$$

or

$$\alpha x_0^2 + x_0 x_1 + \alpha x_1^2 + \sum_{i=2}^{\nu+1} x_i x_{\nu+i} = 0, \text{ where } n = 2\nu + 1, \tag{2.28}$$

and α is a fixed element of $\mathbb{F}_q$ not belonging to N, and these two equations are not projectively equivalent. □

For n being odd, a non-degenerate quadric whose equation is projectively equivalent to (2.21) or (2.27) (corresponding to the cases when q is odd

or even, respectively) is said to be of the *hyperbolic type*, and a non-degenerate quadric whose equation is projectively equivalent to (2.22) or (2.28) is said to be of the *elliptic type*.

Now let us determine the number of points on a quadric in $PG(n, \mathbb{F}_q)$. As in the affine case it is sufficient to determine the number of points on a non-degenerate quadric. We have

Theorem 2.18. For a given non-degenerate quadric in $PG(n, \mathbb{F}_q)$ write $n+1 = 2\nu+\delta$, where $\delta = 1$ if n is even and $\delta = 0$ or 2 if the non-degenerate quadric is of the hyperbolic type or the elliptic type, respectively. Then the number of points on the non-degenerate quadric is equal to

$$\frac{(q^\nu - 1)(q^{\nu+\delta-1} + 1)}{q-1}. \qquad \square$$

This theorem can be proved by the same technique we used in computing the number of points on a non-degenerate quadric in $AG(n, \mathbb{F}_q)$ and its proof will be left as an exercise. Notice that in the formula in Theorem 2.18 we need not to distinguish the cases when q is even or odd.

Finally we remark that it follows from Theorem 2.18 that Theorems 2.13, 2.14, 2.16 give us not only the projective classification of quadratic homogeneous equations but also the projective classification of quadrics in $PG(n, \mathbb{F}_q)$. That is, quadrics defined by different equations listed in Theorem 2.13 or in Theorem 2.14, or in Theorem 2.16 are not projectively equivalent.

2.3 Conics in $PG(2, \mathbb{F}_q)$

By a *conic* in $PG(2, \mathbb{F}_q)$ we mean a non-degenerate quadric in $PG(2, \mathbb{F}_q)$. It follows from Corollaries 2.15 and 2.17 that the equation of a conic can be carried by a projective transformation into the normal form $x_0^2 + x_1x_2 = 0$. It follows from Theorem 2.18 that there are $q+1$ distinct points on it. We shall include these results in the following theorem for later reference and give a simple direct proof of the latter.

Theorem 2.19. The equation of any conic can be carried by a projective transformation into the following form

$$x_0^2 + x_1x_2 = 0. \tag{2.29}$$

The number of points on any conic is $q+1$. Furthermore, any three distinct points on a conic are not collinear.

Proof. Without loss of generality we can assume that the equation of the conic is (2.29). Obviously, there is only one point on (2.29) with $x_1 = 0$, that is $(0, 0, 1)$. For the points on (2.29) with $x_1 \neq 0$, we can assume that $x_1 = 1$ and get q points $(t, 1, -t^2)$, $t \in \mathbb{F}_q$. Thus, altogether there are $q+1$ points on (2.29).

It remains to prove that any three distinct points of (2.29) are not collinear. There are two cases to distinguish. The first case is that $(0, 0, 1)$ is one of the three points. Let $(t_1, 1, -t_1^2)$ and $(t_2, 1, -t_2^2)$, where $t_1, t_2 \in \mathbb{F}_q$ and $t_1 \neq t_2$, be the other two points. Since

$$\begin{vmatrix} 0 & 0 & 1 \\ t_1 & 1 & -t_1^2 \\ t_2 & 1 & -t_2^2 \end{vmatrix} = t_1 - t_2 \neq 0,$$

the three points $(0, 0, 1)$, $(t_1, 1, -t_1^2)$, and $(t_2, 1, -t_2^2)$ are not collinear. The second case is that $(0, 0, 1)$ is not among the three points. Let $(t_i, 1, -t_i^2)$, $i = 1, 2, 3$, where t_1, t_2, t_3 are three distinct elements of $\mathbb{F}_q$, be the three distinct points. Since

$$\begin{vmatrix} t_1 & 1 & -t_1^2 \\ t_2 & 1 & -t_2^2 \\ t_3 & 1 & -t_3^2 \end{vmatrix} = (t_1 - t_2)(t_2 - t_3)(t_3 - t_1) \neq 0,$$

the three points $(t_i, 1, -t_i^2)$, $i = 1, 2, 3$, are also not collinear. □

Suppose that a conic is given. A line which meets the conic in exactly one point is called a *tangent*. A line which intersects the conic in two distinct points is called a *secant*.

Theorem 2.20. (i) Through each point on a conic in $PG(2, \mathbb{F}_q)$ there passes one and only one tangent of the conic.

(ii) In $PG(2, \mathbb{F}_q)$, where q is even, all the tangents of a conic meet in a single point.

(iii) In $PG(2, \mathbb{F}_q)$, where q is odd, any three distinct tangents of a conic are not concurrent. Moreover, through a point not on the conic there passes two or no tangents of the conic.

Proof. Without loss of generality we assume that the equation of the conic is (2.29).

(i) Let (a_0, a_1, a_2) be a point on the conic, i.e., $a_0^2 + a_1 a_2 = 0$. At first we prove that there is a tangent of the conic at the point (a_0, a_1, a_2). Consider

first the case when q is odd. We assert that

$$(a_0, a_1, a_2) \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t(x_0, x_1, x_2) = 0 \tag{2.30}$$

is the equation of a tangent of the conic at the point (a_0, a_1, a_2). (2.30) is a linear homogeneous equation in x_0, x_1, x_2; hence it defines a line in $PG(2, \mathbb{F}_q)$. Clearly (a_0, a_1, a_2) lies on the line. Suppose that there is another point (b_0, b_1, b_2) of the conic lying also on the line, i.e., $b_0^2 + b_1 b_2 = 0$ and (b_0, b_1, b_2) satisfies (2.30). Choose any point (c_0, c_1, c_2) in $PG(2, \mathbb{F}_q)$ such that (a_0, a_1, a_2), (b_0, b_1, b_2), and (c_0, c_1, c_2) are linearly independent. Then

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t \begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & * \\ 0 & 0 & * \\ * & * & * \end{pmatrix}$$

is singular, which is a contradiction. Therefore the line (2.30) meets the conic (2.29) only at (a_0, a_1, a_2), in other words, (2.30) is a tangent of the conic at (a_0, a_1, a_2). Now suppose that q is even. We can prove in the same way that

$$(a_0, a_1, a_2) \begin{pmatrix} 0 & & \\ & 0 & 1 \\ & 1 & 0 \end{pmatrix} {}^t(x_0, x_1, x_2) = 0 \tag{2.31}$$

is the equation of a tangent of the conic at the point (a_0, a_1, a_2). In fact, let (b_0, b_1, b_2) be another point of the conic lying also on the line, i.e., $b_0^2 + b_1 b_2 = 0$ and (b_0, b_1, b_2) satisfies (2.31). Choose any point (c_0, c_1, c_2) in $PG(2, \mathbb{F}_q)$ such that (a_0, a_1, a_2), (b_0, b_1, b_2), and (c_0, c_1, c_2) are linearly independent. Then

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & 1 \\ & 0 & 0 \end{pmatrix} {}^t \begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} \equiv \begin{pmatrix} 0 & 0 & * \\ 0 & 0 & * \\ 0 & 0 & * \end{pmatrix}$$

is not regular, which is a contradiction.

Next we are going to prove that there is only one tangent of the conic at the point (a_0, a_1, a_2) of the conic. Suppose that there are m tangents of the conic at the point (a_0, a_1, a_2), where $m \geq 2$. Through each point of $PG(2, \mathbb{F}_q)$ there pass exactly $q + 1$ distinct lines. Besides the m tangents passing through (a_0, a_1, a_2), there are $q + 1 - m$ lines through (a_0, a_1, a_2).

By Theorem 2.19 each of these $q+1-m$ lines contains at most one point of the conic distinct from (a_0, a_1, a_2). Then we obtain at most $1+(q+1-m) \leq q$ points of the conic. However, the points on the $q+1$ lines through (a_0, a_1, a_2) exhaust all the points of $PG(2, \mathbb{F}_q)$; in particular every point of the conic should lie on one of these lines. By Theorem 2.19 there are $q+1$ points on the conic, and thus we have a contradiction. (i) is proved completely.

(ii) Suppose that q is even. We know from the proof of Theorem 2.19 that $(0,0,1)$ and $(t, 1, -t^2), t \in \mathbb{F}_q$, are all the points on the conic (2.29). By the proof of (i), the tangent of the conic at $(0,0,1)$ is

$$(0,0,1)\begin{pmatrix} 0 & & \\ & 0 & 1 \\ & 1 & 0 \end{pmatrix} {}^t(x_0, x_1, x_2) = 0,$$

i.e.,

$$x_1 = 0,$$

and the tangent of the conic at $(t, 1, -t^2)$ is

$$(t, 1, -t^2)\begin{pmatrix} 0 & & \\ & 0 & 1 \\ & 1 & 0 \end{pmatrix} {}^t(x_0, x_1, x_2) = 0,$$

i.e.,

$$-t^2 x_1 + x_2 = 0.$$

Clearly all these $q+1$ tangents of the conic pass through $(1,0,0)$.

(iii) Now suppose that q is odd. By the proof of (i), the tangent of the conic at $(0,0,1)$ is

$$x_1 = 0,$$

and the tangent of the conic at $(t, 1, -t^2)$ is

$$tx_0 - \frac{1}{2}t^2 x_1 + \frac{1}{2}x_2 = 0.$$

Choose any three tangents of the conic. If $x_1 = 0$ is one of them, then let the other two be $t_1 x_0 - \frac{1}{2}t_1^2 x_1 + \frac{1}{2}x_2 = 0$ and $t_2 x_0 - \frac{1}{2}t_2^2 x_1 + \frac{1}{2}x_2 = 0$, where $t_1 \neq t_2$. Since

$$\begin{vmatrix} 0 & 1 & 0 \\ t_1 & -\frac{1}{2}t_1^2 & \frac{1}{2} \\ t_2 & -\frac{1}{2}t_2^2 & \frac{1}{2} \end{vmatrix} = -\frac{1}{2}(t_1 - t_2) \neq 0,$$

these three tangents are not concurrent. If $x_1 = 0$ is not among the three tangents, then let the three tangents be $t_i x_0 - \frac{1}{2}t_i^2 x_1 + \frac{1}{2}x_2 = 0,\ i = 1, 2, 3$, where t_1, t_2, t_3 are pairwise distinct. Since

$$\begin{vmatrix} t_1 & -\frac{1}{2}t_1^2 & \frac{1}{2} \\ t_2 & -\frac{1}{2}t_2^2 & \frac{1}{2} \\ t_3 & -\frac{1}{2}t_3^2 & \frac{1}{2} \end{vmatrix} = -\frac{1}{4}(t_1 - t_2)(t_2 - t_3)(t_3 - t_1) \neq 0,$$

these three tangents are also not concurrent.

It follows immediately that through a point not on the conic there passes $0, 1$, or 2 tangents of the conic. Suppose that there is a point P not on the conic such that only one tangent passes through it. Besides the tangent through P there are q lines through P, which are not tangents of the conic. By Theorem 2.19 on each of these q lines there are 0 or 2 points of the conic. Then the number of points on the conic should be odd. However the number of points on the conic is $q + 1$, which is even, if q is odd. We get a contradiction. Therefore through a point not on the conic there passes two or no tangents of the conic. □

For the case when q is odd, we can distinguish the points through which there pass two or no tangents. First we need more definitions. Let (a_0, a_1, a_2) be a point not on the conic (2.29), i.e., $a_0^2 + a_1 a_2 \neq 0$. We say that the point (a_0, a_1, a_2) not on (2.29) is of the *square type* or the *non-square type* with respect to the conic (2.29), if $a_0^2 + a_1 a_2 \in \mathbb{F}_q^{*2}$ or not, respectively. Clearly, this definition is independent of the particular choice of the coordinates of the point.

Theorem 2.21. Suppose that q is odd. Let $O_3(\mathbb{F}_q)$ be the set of linear transformations which leaves the quadratic form $x_0^2 + x_1 x_2$ invariant. Then $O_3(\mathbb{F}_q)$ is a subgroup of $GL_3(\mathbb{F}_q)$. Denote the image of $O_3(\mathbb{F}_q)$ in the projective general linear group $PGL_3(\mathbb{F}_q)$ by $PO_3(\mathbb{F}_q)$. Then the set of points on the conic $x_0^2 + x_1 x_2 = 0$ in $PG(2, \mathbb{F}_q)$ forms an orbit under $PO_3(\mathbb{F}_q)$, so does the set of points of the square type with respect to the conic and the set of points of the non-square type.

Proof. Clearly, $T \in O_3(\mathbb{F}_q)$ if and only if

$$T \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t T = \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix}.$$

We verify immediately that $O_3(\mathbb{F}_q)$ is a subgroup of $GL_3(\mathbb{F}_q)$.

For the second assertion of the theorem, we prove only that the set of points of the square type forms an orbit under $PO_3(\mathbb{F}_q)$ as an example. The proofs of the other two cases run similarly.

Clearly, a point of the square type cannot be carried by an element of $PO_3(\mathbb{F}_q)$ to a point on the conic or a point of the non-square type. Now let (a_0, a_1, a_2) be a point of the square type, i.e., $a_0^2 + a_1 a_2 = \lambda^2$ for some $\lambda \in \mathbb{F}_q^*$. Replacing (a_0, a_1, a_2) by $(\lambda^{-1}a_0, \lambda^{-1}a_1, \lambda^{-1}a_2)$ we can assume that $a_0^2 + a_1 a_2 = 1$. The set of points (x_0, x_1, x_2) in $PG(2, \mathbb{F}_q)$ which satisfy the linear homogeneous equation

$$(a_0, a_1, a_2) \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t(x_0, x_1, x_2) = 0$$

is a line of $PG(2, \mathbb{F}_q)$, call it l. Let (b_0, b_1, b_2) and (c_0, c_1, c_2) be two distinct points on l. Clearly, $(a_0, a_1, a_2) \notin l$. Hence the 3×3 matrix

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix}$$

is nonsingular and

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t \begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & S_2 \end{pmatrix},$$

where S_2 is a 2×2 nonsingular symmetric matrix. By Theorem 1.23, there is a 2×2 nonsingular matrix Q_2 such that

$$Q_2 S_2 {}^t Q_2 = \begin{pmatrix} 1 & 0 \\ 0 & d \end{pmatrix},$$

where $d = 1$ or z if $-1 \in \mathbb{F}_q^{*2}$ or not, respectively, and z is a fixed non-square element of $\mathbb{F}_q^*$. Write

$$Q_2 \begin{pmatrix} b_0 & b_1 & b_2 \\ c_0 & c_1 & c_2 \end{pmatrix} = \begin{pmatrix} d_0 & d_1 & d_2 \\ f_0 & f_1 & f_2 \end{pmatrix},$$

then

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ d_0 & d_1 & d_2 \\ f_0 & f_1 & f_2 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t \begin{pmatrix} a_0 & a_1 & a_2 \\ d_0 & d_1 & d_2 \\ f_0 & f_1 & f_2 \end{pmatrix} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & d \end{pmatrix}.$$

Let (a_0', a_1', a_2') be another point of the square type and we can also assume that $a_0'^2 + a_1'a_2' = 1$. By the above proof, there is a 3×3 nonsingular matrix

$$\begin{pmatrix} a_0' & a_1' & a_2' \\ d_0' & d_1' & d_2' \\ f_0' & f_1' & f_2' \end{pmatrix}$$

such that

$$\begin{pmatrix} a_0' & a_1' & a_2' \\ d_0' & d_1' & d_2' \\ f_0' & f_1' & f_2' \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & \frac{1}{2} \\ & \frac{1}{2} & 0 \end{pmatrix} {}^t\begin{pmatrix} a_0' & a_1' & a_2' \\ d_0' & d_1' & d_2' \\ f_0' & f_1' & f_2' \end{pmatrix} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & d \end{pmatrix}.$$

Let

$$T = \begin{pmatrix} a_0 & a_1 & a_2 \\ d_0 & d_1 & d_2 \\ f_0 & f_1 & f_2 \end{pmatrix}^{-1} \begin{pmatrix} a_0' & a_1' & a_2' \\ d_0' & d_1' & d_2' \\ f_0' & f_1' & f_2' \end{pmatrix},$$

then $T \in O_3(\mathbb{F}_q)$ and

$$(a_0, a_1, a_2)\, T = (a_0', a_1', a_2').$$

□

Let us now prove the following theorem which is more precise than Theorem 2.20 (iii).

Theorem 2.22. Suppose that q is odd. Then through a point of the square type with respect to the conic $x_0^2 + x_1x_2 = 0$ there pass two tangents of the conic, and through a point of the non-square type with respect to the conic there passes no tangent.

Proof. By Theorem 2.21, it is sufficient to prove this theorem for a particular point of the square type and also for a particular point of the non-square type.

At first we take the point $(1, 0, 0)$, which is of the square type. By the proof of Theorem 2.20, $x_1 = 0$ is a tangent of the conic at $(0, 0, 1)$. It is clear that the point $(1, 0, 0)$ lies on the line $x_1 = 0$. Therefore by Theorem 2.20 (iii), there pass two tangents of the conic through the point $(1, 0, 0)$.

Next, let z be a fixed non-square element of $\mathbb{F}_q^*$ and take the point $(0, 1, z)$, which is a point of the non-square type. The general equation of a line through $(0, 1, z)$ is

$$\lambda x_0 + \mu(zx_1 - x_2) = 0,$$

where $\lambda, \mu \in \mathbb{F}_q$ and are not all zero. If $\lambda = 0$, then the line becomes $zx_1 - x_2 = 0$. If $-1 \in \mathbb{F}_q^{*2}$, then the line $zx_1 - x_2 = 0$ does not intersect $x_0^2 + x_1x_2 = 0$ and hence is not a tangent; if $-1 \notin \mathbb{F}_q^{*2}$, then $-z \in \mathbb{F}_q^{*2}$, say $-z = \eta^2$, and $(\pm\eta, 1, z)$ are two points of intersection of the line $zx_1 - x_2 = 0$ and the conic $x_0^2 + x_1x_2 = 0$, and hence $zx_1 - x_2 = 0$ is also not a tangent. If $\lambda \neq 0$, then we can assume that $\lambda = 1$. Since $z \notin \mathbb{F}_q^{*2}$, $1 - 4z\mu^2 \neq 0$. By simple calculation we can prove that if $1 - 4z\mu^2 \notin \mathbb{F}_q^{*2}$, the line $x_0 + \mu(zx_1 - x_2) = 0$ does not intersect the conic $x_0^2 + x_1x_2 = 0$, and if $1 - 4z\mu^2 \in \mathbb{F}_q^{*2}$, there are two points of intersection. Hence $x_0 + \mu(zx_1 - x_2) = 0$ is not a tangent in either case. Therefore through the point $(0, 1, z)$ there passes no tangent of the conic. □

In $PG(2, \mathbb{F}_2)$, a conic has exactly three points; they are not collinear and hence they are the three vertices of a triangle. In $PG(2, \mathbb{F}_3)$, a conic has exactly four points, no three of them are collinear, and hence they are the four vertices of a complete quadrangle. In $PG(2, \mathbb{F}_q)$ with $q \geq 4$, the number of points on a conic is ≥ 5. Moreover, we have

Theorem 2.23. In $PG(2, \mathbb{F}_q)$ with $q \geq 4$, through any five points, no three of which are collinear, passes a unique conic.

Proof. At first we prove that there exists at least one quadric in $PG(2, \mathbb{F}_q)$ passing through any five given points, no three of which are collinear. Let

$$\sum_{0 \leq i \leq j \leq 2} a_{ij}x_ix_j = 0 \tag{2.32}$$

be the equation of a quadric passing through five given points, no three of which are collinear. Substituting the coordinates of the five points into this equation one by one, we obtain five linear homogeneous equations in the six coefficients $a_{00}, a_{01}, a_{02}, a_{11}, a_{12}$, and a_{22}. There exists at least one non-zero solution of this system of equations, which gives a quadric passing through the five given points.

Then we prove that any quadric in $PG(2, \mathbb{F}_q)$ passing through five given points, no three of which are collinear, must be non-degenerate. By Theorems 2.14 and 2.16 degenerate quadrics in $PG(2, \mathbb{F}_q)$ can be carried under projective transformations into one of the following forms

$x_0^2 = 0$,
$x_0x_1 = 0$,
$x_0^2 - zx_1^2 = 0$, if q is odd and z is a fixed non-square element of $\mathbb{F}_q^*$,

$$\alpha x_0^2 + x_0x_1 + \alpha x_1^2 = 0, \text{ if } q \text{ is even and } \alpha \text{ is a fixed element of } \mathbb{F}_q \text{ not belonging to } N.$$

Clearly, $x_0^2 = 0$ is the equation of a repeated line, $x_0x_1 = 0$ is the equation of a pair of distinct lines, and $x_0^2 - zx_1^2 = 0$ and $\alpha x_0^2 + x_0x_1 + \alpha x_1^2 = 0$ are equations of the single point $(0, 0, 1)$. Thus on any one of them there do not exist five points such that no three of them are collinear. Hence any quadric in $PG(2, \mathbb{F}_q)$ passing through five given points, no three of which are collinear, must be non-degenerate and hence must be a conic.

Finally we prove that there is only one conic passing through any five given points, no three of which are collinear. Suppose that there are two conics passing through five given points, no three of which are collinear. Under projective transformations we can assume that one of the conics is (2.29) and the other is (2.32). By the proof of Theorem 2.19 we know that the points on $x_0^2 + x_1x_2 = 0$ are $(0, 0, 1)$ and $(t, 1, -t^2)$, $t \in \mathbb{F}_q$. Consider first the case that the coordinates of the five given points are $(t_i, 1, -t_i^2)$, $i = 1, 2, 3, 4, 5$. Substituting $(t_i, 1, -t_i^2)$, $i = 1, 2, 3, 4, 5$, into (2.32), we know that t_i $(i = 1, 2, 3, 4, 5)$ are roots of an equation of degree ≤ 4, so the equation must be an identity, from which it follows that $a_{01} = a_{02} = a_{11} = a_{22} = 0$ and $a_{00} = a_{12}$. Thus (2.32) becomes $a_{00}x_0^2 + a_{00}x_1x_2 = 0$. Canceling a_{00}, we obtain (2.29). Consider then the case that the coordinates of the five given points are $(0, 0, 1)$ and $(t_i, 1, -t_i^2)$, $i = 1, 2, 3, 4$. Substituting $(0, 0, 1)$ in (2.32), we obtain that $a_{22} = 0$. Then substituting $(t_i, 1, -t_i^2)$, $i = 1, 2, 3, 4$ into (2.32), we know that t_i $(i = 1, 2, 3, 4)$ are roots of an equation of degree ≤ 3, by the same reasoning as the above case we have the same conclusion. Therefore there is only one conic passing through any five given points, no three of which are collinear. □

2.4 Ovals in $PG(2, \mathbb{F}_q)$

Let $m(2, q)$ denote the maximal number of points in $PG(2, \mathbb{F}_q)$ such that no three of them are collinear. Then an *oval* in $PG(2, \mathbb{F}_q)$ is defined to be a set of $m(2, q)$ points in $PG(2, \mathbb{F}_q)$ such that no three of them are collinear. To study ovals, the first problem is to determine the number $m(2, q)$. It follows from Theorem 2.19 that no three of the $q+1$ points of a conic are collinear. Thus $m(2, q) \geq q+1$. Before we determine the exact value of $m(2, q)$ let us show how error-correcting codes can be constructed using ovals.

Example 2.4. For an (n, k) linear code over $\mathbb{F}_q$, the parity check matrix is an $n \times (n - k)$ matrix. Write $r = n - k$. If the code is a single-

error-correcting code, then its minimum weight is ≥ 3. Hence any two rows of its parity check matrix are linearly independent. Therefore the all zero vector cannot appear as a row of the parity check matrix and any two rows of it cannot differ by a non-zero element of $\mathbb{F}_q$. Thus we may regard the rows of the parity check matrix as distinct points of the projective space $PG(r-1,\mathbb{F}_q)$. Assume that r is fixed. In order to obtain a linear single-error-correcting code with maximum information rate $k/n = (n-r)/n = 1 - r/n$, n should be as large as possible. Thus if all the points in $PG(r-1,\mathbb{F}_q)$ appear as rows in the parity check matrix, a linear code with maximum rate will be obtained. The number of points in $PG(r-1,\mathbb{F}_q)$ is $(q^r-1)/(q-1)$. Therefore, for fixed r we have a linear single-error-correcting code with code length $n = (q^r-1)/(q-1)$, which attains the maximum information rate. This code is actually the Hamming code over $\mathbb{F}_q$.

We are also interested in linear error-correcting codes over $\mathbb{F}_q$ with minimum distance ≥ 4, i.e., linear single-error-correcting and double-error-detecting codes. For such an $(n, n-r)$ linear code, any three rows of its parity check matrix must be linearly independent, i.e., the set of points in $PG(r-1,\mathbb{F}_q)$ corresponding to the rows of the parity check matrix has the property that any three points in the set must be non-collinear. In particular, for $r = 3$, a maximal set of points in $PG(2,\mathbb{F}_q)$ with such a property is an oval. Hence, for $r = 3$ there exists a linear single-error-correcting and double-error-detecting code over $\mathbb{F}_q$ with code length $m(2,q)$ which attains maximum information rate $1 - 3/m(2,q)$. □

Now let us return to the determination of the exact value of $m(2,q)$. For later applications we shall start more generally with k-arcs in $PG(2,\mathbb{F}_q)$. A *k-arc* in $PG(2,\mathbb{F}_q)$ is a set of k points in $PG(2,\mathbb{F}_q)$ such that no three of them are collinear. Clearly, $k \leq m(2,q)$ and an oval is a $m(2,q)$-arc in $PG(2,\mathbb{F}_q)$. Let $\mathcal{K}$ be a k-arc and l a line in $PG(2,\mathbb{F}_q)$, l is called a *secant*, a *tangent*, or an *external line* of $\mathcal{K}$, if it intersects $\mathcal{K}$ in two distinct points, in exactly one point, or does not intersect $\mathcal{K}$, respectively. For a point $P \in \mathcal{K}$, denote by $t(P)$ the number of tangents of $\mathcal{K}$ through P. For a point $Q \notin \mathcal{K}$, denote by $\sigma_1(Q)$ and $\sigma_2(Q)$ the number of tangents and the number of secants of $\mathcal{K}$ through Q, respectively. First we prove the following two lemmas.

Lemma 2.24. For any point P on a k-arc $\mathcal{K}$ in $PG(2,\mathbb{F}_q)$,

$$t(P) = q + 2 - k,$$

which is independent of P. In particular, for any point P on an oval $\mathcal{O}$,

$$t(P) = q + 2 - m(2, q).$$

Proof. Through P, there pass $q+1$ lines of $PG(2, \mathbb{F}_q)$ altogether. To each point P' of $\mathcal{K}$ distinct from P, there corresponds a secant of $\mathcal{K}$ joining P and P'. Altogether, there are $k-1$ secants through P. All the other lines passing through P are tangents of $\mathcal{K}$. They are $(q+1)-(k-1) = q+2-k$ in number. Hence $t(P) = q + 2 - k$. □

Lemma 2.25. For any point Q not on a k-arc $\mathcal{K}$ in $PG(2, \mathbb{F}_q)$,

$$\sigma_1(Q) + 2\sigma_2(Q) = k.$$

Proof. Each tangent of $\mathcal{K}$ passing through Q meets $\mathcal{K}$ in exactly one point, and each secant of $\mathcal{K}$ passing through Q meets $\mathcal{K}$ in exactly two distinct points. Hence $\sigma_1(Q) + 2\sigma_2(Q) = k$. □

Then we have

Theorem 2.26.

$$m(2, q) = \begin{cases} q + 2 & \text{for } q \text{ even}, \\ q + 1 & \text{for } q \text{ odd}. \end{cases}$$

Proof. For any point P on an oval $\mathcal{O}$, it is clear that $t(P) \geq 0$. Thus it follows from Lemma 2.24 that $m(2, q) \leq q + 2$. On the other hand, by Theorem 2.19 the $q + 1$ points on any conic is a $(q + 1)$-arc in $PG(2, \mathbb{F}_q)$, hence $q + 1 \leq m(2, q)$. Therefore $q + 1 \leq m(2, q) \leq q + 2$.

Let q be even. By Theorem 2.20 (ii), the $q + 1$ tangents of a conic $\mathcal{C}$ are concurrent. Let their common point of intersection be P. Then $\mathcal{C} \cup \{P\}$ is a set of $q + 2$ points, no three of which are collinear. Thus $m(2, q) = q + 2$ for q even.

Now let q be odd. Suppose that $m(2, q) = q + 2$, i.e., there is an oval $\mathcal{O}$ in $PG(2, \mathbb{F}_q)$ consisting of $q + 2$ points. Then by Lemma 2.24 we must have $t(P) = 0$ for any $P \in \mathcal{O}$, and hence $\sigma_1(Q) = 0$ for any $Q \notin \mathcal{O}$. But then by Lemma 2.25 we should have $2\sigma_2(Q) = q + 2$, which is impossible, because q is odd. Therefore $m(2, q) = q + 1$ for q odd. □

In the following we want to study how the ovals are. We have to distinguish the cases when q is odd and even.

Let us consider first the case when q is odd. In this case an oval contains $q+1$ points and, by Lemma 2.24, through any point on the oval there is one and only one tangent to the oval. Moreover, we have

Lemma 2.27. Let q be odd and $\mathcal{O}$ an oval in $PG(2,\mathbb{F}_q)$. Then every point not on $\mathcal{O}$ lies on exactly two or no tangents of $\mathcal{O}$.

Proof. Let Q be a point not on $\mathcal{O}$. It is sufficient to prove that if there is a tangent of $\mathcal{O}$ passing through Q, then there are exactly two tangents of $\mathcal{O}$ passing through Q. Now let l be a tangent of $\mathcal{O}$ through Q and let P be the point of intersection of l and $\mathcal{O}$. By Lemma 2.25 $\sigma_1(Q) + 2\sigma_2(Q) = q+1$. Since $q+1$ is even and l is a tangent of $\mathcal{O}$ through Q, $\sigma_1(Q) \geq 2$, i.e., there is a tangent of $\mathcal{O}$ through Q, which is not l. Similarly, for any point $Q' \in l \setminus \{P\}$, there is a tangent of $\mathcal{O}$ through Q', which is not l. These q tangents, none of which is l, together with l are $q+1$ in number, and hence they constitute the whole set of tangents of $\mathcal{O}$. Consequently, through every point of $l \setminus \{P\}$, in particular Q, there pass exactly two tangents of $\mathcal{O}$. □

Lemma 2.28. Let q be odd and $\mathcal{O}$ an oval in $PG(2,\mathbb{F}_q)$. Assume that $E_0 = (1,0,0)$, $E_1 = (0,1,0)$, and $E_2 = (0,0,1)$ lie on $\mathcal{O}$. Suppose that the three tangents of $\mathcal{O}$ at E_0, E_1, and E_2 have equations

$$x_1 - ax_2 = 0,\ x_2 - bx_0 = 0,\ \text{and}\ x_0 - cx_1 = 0,$$

respectively, where $a, b, c \in \mathbb{F}_q$. Then $abc = -1$.

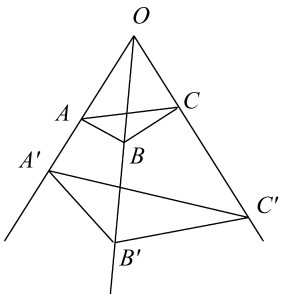
Proof. Let $P_j = (d_0^{(j)}, d_1^{(j)}, d_2^{(j)})$, $j = 1, 2, \ldots, q-2$, be points of $\mathcal{O}$ distinct from E_0, E_1, and E_2. Since $\mathcal{O}$ is an oval and $E_0, E_1, E_2 \in \mathcal{O}$, $d_0^{(j)} \neq 0, d_1^{(j)} \neq 0$, and $d_2^{(j)} \neq 0$. Clearly the lines P_jE_0, P_jE_1, and P_jE_2 have equations

$$x_1 - e_jx_2 = 0,\ x_2 - f_jx_0 = 0,\ \text{and}\ x_0 - g_jx_1 = 0,$$

respectively, where $e_j = d_1^{(j)}/d_2^{(j)}$, $f_j = d_2^{(j)}/d_0^{(j)}$, and $g_j = d_0^{(j)}/d_1^{(j)}$. Hence $e_jf_jg_j = 1$ for $j = 1, 2, \ldots, q-2$. Through E_0 there are altogether $q+1$ lines, two of them are E_0E_1 and E_0E_2 which have equations $x_2 = 0$ and $x_1 = 0$, respectively. Among the other $q-1$ lines through E_0 we have $q-2$ secants P_jE_0: $x_1 - e_jx_2 = 0$, $j = 1, 2, \ldots, q-2$ and one tangent. We have assumed that the equations of the tangent through E_0 is $x_1 - ax_2 = 0$. Therefore $e_1, e_2, \ldots, e_{q-2}$, and a are the $q-1$ non-zero elements of $\mathbb{F}_q$. We know that the product of all non-zero elements of a finite field $\mathbb{F}_q$ is

$(-1)^q$, which is now equal to -1, because q is odd. Thus $a\prod_{j=1}^{q-2} e_j = -1$. Similarly, $b\prod_{j=1}^{q-2} f_j = -1$ and $c\prod_{j=1}^{q-2} g_j = -1$. Since $e_j f_j g_j = 1$ for $j = 1, 2, \ldots, q-2$, $abc = -1$. □

Two triangles ABC and $A'B'C'$ are said to be in *perspective*, if the lines AA', BB', and CC' have a common point of intersection O, and O is called the *center of perspectivity* of the two triangles.



Lemma 2.29. Let q be odd and $\mathcal{O}$ an oval in $PG(2, \mathbb{F}_q)$. Then the triangle formed by any three points of $\mathcal{O}$ and the triangle formed by the three tangents at these three points are in perspective.

Proof. The properties that a geometric figure is an oval, a line a tangent of an oval, and two triangles in perspective, are invariant under projective transformations. Thus, by Theorem 2.10 (ii) we can assume that the three points of $\mathcal{O}$ are $E_0 = (1, 0, 0)$, $E_1 = (0, 1, 0)$, and $E_2 = (0, 0, 1)$. Then we can assume that the tangents of $\mathcal{O}$ at E_0, E_1, and E_2 are

$$l_0 : x_1 - ax_2 = 0,\ l_1 : x_2 - bx_0 = 0,\ \text{and}\ l_2 : x_0 - cx_1 = 0,$$

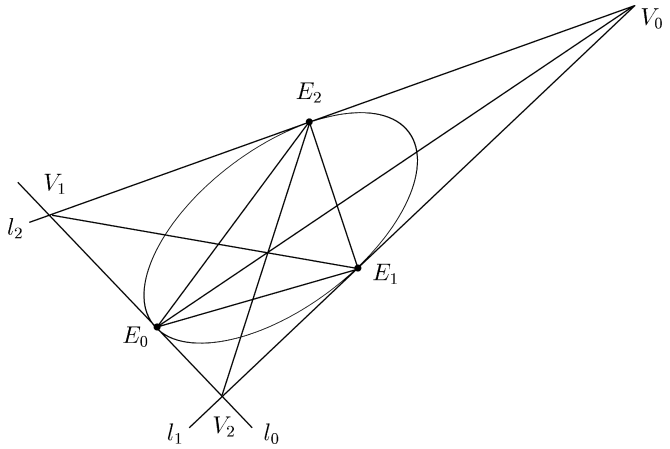
respectively. By Lemma 2.28, $abc = -1$. It is easy to verify that l_0 and l_1 intersect at $V_2 = (c, -1, bc)$, l_1 and l_2 intersect at $V_0 = (ca, a, -1)$, and l_2 and l_0 intersect at $V_1 = (-1, ab, b)$. Then the lines E_0V_0, E_1V_1, and E_2V_2 have the equations

$$x_1 + ax_2 = 0,\ x_2 + bx_0 = 0,\ \text{and}\ x_0 + cx_1 = 0,$$

respectively. We have

$$\begin{vmatrix} 0 & 1 & a \\ b & 0 & 1 \\ 1 & c & 0 \end{vmatrix} = 1 + abc = 0,$$

hence the lines E_0V_0, E_1V_1, and E_2V_2 are concurrent, i.e., the triangles $E_0E_1E_2$ and $V_0V_1V_2$ are in perspective. □



Theorem 2.30. In $PG(2, \mathbb{F}_q)$ for q odd, every oval is a conic.

Proof. Let $\mathcal{O}$ be an oval. We can assume that the points $E_0 = (1, 0, 0)$, $E_1 = (0, 1, 0)$, and $E_2 = (0, 0, 1)$ are on $\mathcal{O}$, and that $E = (1, 1, 1)$ is the center of perspectivity of the triangle $E_0E_1E_2$ and the triangle $V_0V_1V_2$ formed by the three tangents of $\mathcal{O}$ at E_0, E_1, and E_2. By the proof of the above lemma, we can assume that E_0V_0, E_1V_1, and E_2V_2 have the equations

$$x_1 + ax_2 = 0,\ x_2 + bx_0 = 0,\ \text{and}\ x_0 + cx_1 = 0,$$

respectively. Since $E = (1, 1, 1)$ is the common point of intersection of E_0V_0, E_1V_1, and E_2V_2, $a = b = c = -1$. Thus $V_0 = (1, -1, -1)$, $V_1 = (-1, 1, -1)$, and $V_2 = (-1, -1, 1)$, and the tangents l_0, l_1, and l_2 of $\mathcal{O}$ at E_0, E_1, and E_2 have the equations

$$l_0 : x_1 + x_2 = 0,\ l_1 : x_2 + x_0 = 0,\ \text{and}\ l_2 : x_0 + x_1 = 0,$$

respectively.

Let $P = (y_0, y_1, y_2)$ be any point of $\mathcal{O}$ different from E_0, E_1, and E_2 and let the tangent of $\mathcal{O}$ at P be

$$l : k_0x_0 + k_1x_1 + k_2x_2 = 0.$$

Then l does not pass through E_0, E_1, and E_2, from which we deduce $k_0 \neq 0$, $k_1 \neq 0$, and $k_2 \neq 0$. By Lemma 2.27, l cannot pass through V_0, V_1, and V_2, from which we deduce that $k_0 - k_1 - k_2 \neq 0$, $-k_0 + k_1 - k_2 \neq 0$, and $-k_0 - k_1 + k_2 \neq 0$. Now let us apply Lemma 2.29 to the triangle

PE_1E_2 and the triangle formed by the three tangents of $\mathcal{O}$ at P, E_1, and E_2. The point of intersection of the tangents l_1 and l_2 is $V_0 = (1, -1, -1)$. Denote the point of intersection of the tangents l_2 and l of $\mathcal{O}$ by R_1, then $R_1 = (k_2, -k_2, k_1 - k_0)$. Similarly, denote the point of intersection of the tangents l and l_1 of $\mathcal{O}$ by R_2, then $R_2 = (-k_1, k_0 - k_2, k_1)$. The lines passing through P and V_0, R_1 and E_1, and R_2 and E_2 have the equations

$$\begin{array}{llllll} (y_1 - y_2)x_0 & - & (y_0 + y_2)x_1 & + & (y_0 + y_1)x_2 & = 0, \\ (k_0 - k_1)x_0 & & & + & k_2 x_2 & = 0, \\ (k_0 - k_2)x_0 & + & k_1 x_1 & & & = 0, \end{array}$$

respectively. By Lemma 2.29, these three lines are concurrent, i.e.,

$$\begin{vmatrix} y_1 - y_2 & -(y_0 + y_2) & y_0 + y_1 \\ k_0 - k_1 & 0 & k_2 \\ k_0 - k_2 & k_1 & 0 \end{vmatrix} = 0.$$

Expanding, we obtain

$$(k_0 - k_1 - k_2)[k_1(y_0 + y_1) - k_2(y_0 + y_2)] = 0.$$

Since $k_0 - k_1 - k_2 \neq 0$,

$$k_1(y_0 + y_1) = k_2(y_0 + y_2).$$

Similarly, from the triangles PE_2E_0 and PE_0E_1, we deduce that

$$\begin{aligned} k_2(y_1 + y_2) &= k_0(y_1 + y_0), \\ k_0(y_2 + y_0) &= k_1(y_2 + y_1). \end{aligned}$$

Hence,

$$k_0 : k_1 : k_2 = (y_1 + y_2) : (y_2 + y_0) : (y_0 + y_1).$$

But P is on l, i.e., $k_0y_0 + k_1y_1 + k_2y_2 = 0$. Hence $y_1y_2 + y_2y_0 + y_0y_1 = 0$. Therefore every point of $\mathcal{O}$ is on the conic

$$x_1x_2 + x_2x_0 + x_0x_1 = 0.$$

Both $\mathcal{O}$ and the conic have the same number of points, viz., $q + 1$. Hence they must coincide. □

We recall that a conic is defined algebraically, viz., by a non-degenerate quadratic homogeneous equation. Thus Theorem 2.30 can be regarded as a characterization of the conic by geometric properties when q is odd.

Now let us assume that q is even and study the ovals in $PG(2, \mathbb{F}_q)$. Call the common point of intersection of all the tangents of a conic the *nucleus* of the conic. In the proof of Theorem 2.26 we have already proved that a conic with its nucleus added is an oval. Such an oval is called a *regular oval*. An oval which is not regular is called an *irregular oval*. The following lemma is immediate.

Lemma 2.31. In $PG(2, \mathbb{F}_q)$ for q even, an oval is regular if and only if it contains a conic.

Proof. If an oval contains a conic, then the line joining the remaining point of the oval not lying on the conic and any point of the conic must be a tangent of the conic. Therefore the remaining point of the oval not lying on the conic must be the nucleus of the conic. □

Theorem 2.32. Every oval in $PG(2, \mathbb{F}_2)$ or $PG(2, \mathbb{F}_4)$ is regular.

Proof. In $PG(2, \mathbb{F}_2)$ an oval consists of four points, any three of which form a conic. By Lemma 2.31, it is regular.

In $PG(2, \mathbb{F}_4)$ an oval consists of six points and by Theorems 2.23 and 2.19 any five of them form a conic. Again by Lemma 2.31, it is regular. □

However, for $q > 4$, there are ovals in $PG(2, \mathbb{F}_q)$ which are not regular. At first, we shall obtain a normal form for ovals in $PG(2, \mathbb{F}_q)$. We need the following definition.

A polynomial $f(x)$ over $\mathbb{F}_q$ is called a *permutation polynomial* of $\mathbb{F}_q$, if $f(x)$ runs through $\mathbb{F}_q$ as x runs through $\mathbb{F}_q$.

Theorem 2.33. In $PG(2, \mathbb{F}_q)$ for q even and $q > 2$, any oval can be transformed under projective transformations to the following form:

$$\mathcal{O}(f) = \{(1, t, f(t)) \mid t \in \mathbb{F}_q\} \cup \{(0, 1, 0), (0, 0, 1)\},$$

where $f(x) \in \mathbb{F}_q[x]$ is such that

(i) $f(x)$ is a permutation polynomial of $\mathbb{F}_q$ with degree $\partial^0 f < q$ and $f(0) = 0, f(1) = 1$;

(ii) for each $a \in \mathbb{F}_q$, $g_a(x) = (f(x+a) + f(a))/x$ is a permutation polynomial of $\mathbb{F}_q$ with $g_a(0) = 0$.

Conversely, every such set $\mathcal{O}(f)$ is an oval.

Proof. Let $\mathcal{O}$ be an oval in $PG(2, \mathbb{F}_q)$. After applying a projective transformation, we can assume that $\mathcal{O}$ contains the following four points

$$E_0 = (1,0,0),\ E_1 = (0,1,0),\ E_2 = (0,0,1),\ E = (1,1,1).$$

The line through E_1 and E_2 cannot contain any other point of $\mathcal{O}$. Hence the other q points of $\mathcal{O}$ distinct from E_1 and E_2 have the coordinates

$$(1, a_i, b_i),\ a_i, b_i \in \mathbb{F}_q,\ i = 1, 2, \ldots, q.$$

If there is a pair of distinct indices i and j $(1 \leq i, j \leq q)$ such that $a_i = a_j$ but $b_i \neq b_j$, then $E_2 = (0,0,1), (1, a_i, b_i)$, and $(1, a_i, b_j)$ are three collinear points of $\mathcal{O}$; that is impossible. Similarly, there is no pair of distinct indices i and j $(1 \leq i, j \leq q)$ such that $a_i \neq a_j$ but $b_i = b_j$. Hence

$$\{a_1, a_2, \ldots, a_q\} = \{b_1, b_2, \ldots, b_q\} = \mathbb{F}_q$$

and by Lagrange's interpolation formula there exists a unique polynomial $f(x) \in \mathbb{F}_q[x]$ such that $\partial^0 f < q$ and

$$f(a_i) = b_i\ (1 \leq i \leq q).$$

Such an $f(x)$ is a permutation polynomial of $\mathbb{F}_q$ by definition. Therefore $\mathcal{O} = \mathcal{O}(f)$. Since E_0 and E are contained in $\mathcal{O}$, we have

$$f(0) = 0,\ f(1) = 1.$$

It remains to show that (ii) is equivalent to the condition that no three points of $\mathcal{O}(f) \setminus \{E_1, E_2\}$ are collinear. The latter condition is equivalent to

$$\begin{vmatrix} 1 & b & f(b) \\ 1 & c & f(c) \\ 1 & d & f(d) \end{vmatrix} \neq 0$$

for any three distinct elements $b, c, d \in \mathbb{F}_q$, and is then equivalent to

$$(f(b) + f(c))(b + c)^{-1} \neq (f(b) + f(d))(b + d)^{-1}$$

for any three distinct elements $b, c, d \in \mathbb{F}_q$, and hence is equivalent to the condition that for each $a \in \mathbb{F}_q$

$$(f(t) + f(a))(t + a)^{-1}$$

takes a different value for each $t \in \mathbb{F}_q$ and $t \neq a$. Substituting $x+a$ for t yields that the polynomial

$$g_a(x) = (f(x+a) + f(a))/x$$

defines a permutation of $\mathbb{F}_q^*$. Notice that the polynomials

$$g_a(x) \text{ and } \sum_{c\in\mathbb{F}_q} g_a(c)(1-(x-c)^{q-1})$$

take the same value $g_a(c)$ for all $c \in \mathbb{F}_q$ and are both of degree $\leq q-1$, hence they must be equal, i.e.,

$$g_a(x) = \sum_{c\in\mathbb{F}_q} g_a(c)(1-(x-c)^{q-1}).$$

Comparing coefficients of x^{q-1} on both sides, we get

$$0 = \sum_{c\in\mathbb{F}_q} g_a(c) = g_a(0) + \sum_{c\in\mathbb{F}_q^*} g_a(c) = g_a(0) + \sum_{c\in\mathbb{F}_q^*} c = g_a(0) + \sum_{c\in\mathbb{F}_q} c.$$

Since $q > 2$, we have $\sum_{c\in\mathbb{F}_q} c = 0$ and then $g_a(0) = 0$. Hence $g_a(x)$ is a permutation polynomial with $g_a(0) = 0$. □

Corollary 2.34. If $\mathcal{O}(f)$ with $f(x) = \sum_{i=1}^{q-1} b_i x^i$ is an oval in $PG(2,\mathbb{F}_q)$ with q even and $q > 2$, then $f(x) = \sum_{j=1}^{(q-2)/2} b_{2j} x^{2j}$.

Proof. For $a \in \mathbb{F}_q$, condition (ii) of Theorem 2.33 yields

$$0 = g_a(0) = b_1 + b_3 a^2 + b_5 a^4 + \cdots + b_{q-1} a^{q-2},$$

which implies $b_1 = b_3 = \cdots = b_{q-1} = 0$. □

Corollary 2.35. The set $\mathcal{O}(x^k)$ with $1 \leq k < q$ is an oval in $PG(2,\mathbb{F}_q)$ with q even and $q > 2$, if and only if the following conditions hold:

(i) $(k, q-1) = 1$,

(ii) $(k-1, q-1) = 1$,

(iii) $((x+1)^k + 1)/x$ is a permutation polynomial of $\mathbb{F}_q$.

Proof. x^k is a permutation polynomial if and only if $(k, q-1) = 1$. Hence (i) is equivalent to condition (i) of Theorem 2.33 for $f(x) = x^k$. We have $g_0(x) = x^{k-1}$, similarly (ii) is equivalent to condition (ii) of Theorem 2.33 for $f(x) = x^k$ and $a = 0$. For $a \neq 0$, $g_a(x) = a^{k-1} g_1(a^{-1}x)$, where $g_1(x) = ((x+1)^k + 1)/x$. Hence $g_a(x)$ is a permutation polynomial if and only if $g_1(x)$ is one. In particular, condition (ii) of Theorem 2.33 for $f(x) = x^k$ and $a \neq 0$ implies (iii) of the present corollary. Conversely, if $g_1(x)$ is a permutation polynomial, then from $g_1(0) \in \mathbb{F}_2$ and $g_1(1) = 1$ it follows that $g_1(0) = 0$, and hence $g_a(0) = 0$. Thus condition (ii) of Theorem 2.33 for every $a \neq 0$ is also satisfied. □

Corollary 2.36. The set $\mathcal{O}(x^{2^n})$ with $1 \leq n < h$ is an oval in $PG(2, \mathbb{F}_{2^h})$ if and only if $(n, h) = 1$.

Proof. $(2^n, 2^h - 1) = 1$, so condition (i) of Corollary 2.35 is satisfied. Now $((x+1)^{2^n} + 1)/x = x^{2^n - 1}$, so condition (iii) of Corollary 2.35 holds if and only if $(2^n - 1, 2^h - 1) = 1$, i.e., condition (ii) of Corollary 2.35 holds. But $(2^n - 1, 2^h - 1) = 1$ holds if and only if $(n, h) = 1$. □

Corollary 2.37. In $PG(2, \mathbb{F}_{2^h})$ with $h > 1$, $\mathcal{O}(x^{2^n})$ with $1 \leq n < h$ is a regular oval if and only if $n = 1$ or $n = h - 1$.

Proof. Clearly $\mathcal{O}(x^2)$ contains the conic $x_1^2 + x_0 x_2 = 0$ and $\mathcal{O}(x^{2^{h-1}})$ contains the conic $x_2^2 + x_0 x_1 = 0$. Hence by Lemma 2.31 both $\mathcal{O}(x^2)$ and $\mathcal{O}(x^{2^{h-1}})$ are regular ovals.

Now let $1 < n < h - 1$ and suppose that $\mathcal{O}(x^{2^n})$ is a regular oval. By Lemma 2.31 $\mathcal{O}(x^{2^n})$ contains a conic $\mathcal{C}$. Let the equation of $\mathcal{C}$ be

$$\sum_{0 \leq i \leq j \leq 2} a_{ij} x_i x_j = 0.$$

$\mathcal{C}$ has at least $2^h - 1$ points in common with $\mathcal{O}(x^{2^n}) \setminus \{(0,1,0), (0,0,1)\} = \{(1, c, c^{2^n}) \mid c \in \mathbb{F}_{2^h}\}$. In other words there are at least $2^h - 1$ elements t of $\mathbb{F}_{2^h}$ such that $(1, t, t^{2^n})$ lies on $\mathcal{C}$. That is, the equation

$$a_{00} + a_{01}t + a_{02}t^{2^n} + a_{11}t^2 + a_{12}t^{2^n+1} + a_{22}t^{2^{n+1}} = 0$$

has at least $2^h - 1$ distinct roots in $\mathbb{F}_{2^h}$. But the equation is of degree $\leq 2^{n+1}$ and from $n < h - 1$ we deduce $2^{n+1} < 2^h - 1$; we obtain a contradiction. Therefore $\mathcal{O}(x^{2^n})$ cannot be a regular oval. □

Theorem 2.38. In $PG(2, \mathbb{F}_{2^h})$ with $h = 5$ or $h \geq 7$, there exist irregular ovals. More precisely, for $h = 5$ and $h \geq 7$ there exists at least one n such that $(n, k) = 1$ and $1 < n < h - 1$, so the oval $\mathcal{O}(x^{2^n})$ in $PG(2, \mathbb{F}_{2^h})$ is irregular.

Proof. Denote the number of integers n with $1 \leq n \leq h$ such that $(n, h) = 1$ by $\phi(h)$, which is usually called the *Euler ϕ-function*. Since $(1, h) = 1$ and $(h-1, h) = 1$, it follows from Corollaries 2.36 and 2.37 that irregular ovals in $PG(2, \mathbb{F}_{2^h})$ exist if $\phi(h) > 2$. It is known that $\phi(5) = 4$ and $\phi(h) > 2$ for $h \geq 7$. Hence our theorem follows. □

Theorem 2.20 (ii) tells us that when q is even all the tangents of a conic in $PG(2, \mathbb{F}_q)$ meet in a single point. For a $(q+1)$-arc in $PG(2, \mathbb{F}_q)$, we have the same conclusion.

Theorem 2.39. In $PG(2, \mathbb{F}_q)$, when q is even, all the tangents of a $(q+1)$-arc meet in a single point.

Proof. Let $\mathcal{K}$ be a $(q+1)$-arc in $PG(2, \mathbb{F}_q)$. By Lemma 2.24 through every point of $\mathcal{K}$ there passes exactly one tangent of $\mathcal{K}$. Hence altogether there are $q+1$ tangents of $\mathcal{K}$. Let l_1 and l_2 be two tangents of $\mathcal{K}$. l_1 and l_2 must intersect, call their point of intersection P. Clearly P is not on $\mathcal{K}$. We assert that any line passing through P must be a tangent of $\mathcal{K}$. Let l be a line passing through P but distinct from l_1 and l_2, and suppose that l is not a tangent. Let Q be any point on l, by Lemma 2.25 $\sigma_1(Q) + 2\sigma_2(Q) = q+1$, from which it follows that there must be a tangent of $\mathcal{K}$ passing through Q. Since l has exactly $q+1$ points and $\mathcal{K}$ has exactly $q+1$ tangents, there is exactly one tangent of $\mathcal{K}$ through each point of l. But l_1 and l_2 are two tangents of $\mathcal{K}$ passing through P, which is on l. This contradiction proves our assertion. Therefore all the tangents of $\mathcal{K}$ meet at P. □

As for a conic, the common point of intersection of all the tangents to a $(q+1)$-arc is called its *nucleus*.

Corollary 2.40. For q even, a $(q+1)$-arc in $PG(2, \mathbb{F}_q)$ can be uniquely completed to an oval by adding its nucleus. □

2.5 Comments

The presentation of projective spaces over $\mathbb{F}_q$ in Section 2.1 is standard. For the Fano plane, cf. Fano 1892. The authentication code constructed

from $PG(2, \mathbb{F}_q)$ in Example 2.3 is due to E. N. Gilbert, F. J. MacWilliams, and N. J. A. Sloane, cf. Gilbert et al. 1974.

Theorems 2.13, 2.14, and 2.16 in Section 2.2 are immediate consequences of the corresponding Theorems 1.23, 1.26, and 1.35 of L. E. Dickson. Theorem 2.18 is due to E. J. F. Primrose, cf. Primrose 1952.

Theorems 2.19, 2.20, and 2.23 in Section 2.3 are well known. The main idea in the proof of Theorem 2.20 is the polarization method of obtaining the equation of the tangent of a conic section through a point on it, which we learned in plane analytic geometry, but now the method is put into the matrix form. Theorem 2.22 is due to B. Qvist, cf. Qvist 1962, however its proof given in this book, based on Theorem 2.21, seems to be new.

Most of the main results of Section 2.4 are due to B. Segre. In particular, for Theorem 2.30, cf. Segre 1955 and for Theorem 2.33, its Corollaries and Theorem 2.38, cf. Segre 1957, 1962. Theorem 2.26 is due to R. C. Bose, cf. Bose 1947. Theorem 2.32 is due to M. Sce, cf. Sce 1960.

It was proved by M. Sce in 1960 that every oval in $PG(2, \mathbb{F}_{2^3})$ is regular. In 1958 L. Lunelli and M. Sce found, using a computer, an irregular oval in $PG(2, \mathbb{F}_{2^4})$. In 1975 M. Hall showed, with the aid of a computer, that in $PG(2, \mathbb{F}_{2^4})$ every irregular oval is projectively equivalent to the one found by Lunelli and Sce. In 1991 C. M. O'Keefe and T. Penttila gave a proof of this result without using a computer. In 1994 the classification of irregular ovals in $PG(2, \mathbb{F}_{2^5})$ under projective transformations was achieved by T. Penttila and G. F. Royle.

2.6 Exercises

2.1 Draw a diagram showing the points and lines of $AG(2, \mathbb{F}_3)$. Add a point at infinity to each pencil of parallel lines of $AG(2, \mathbb{F}_3)$ and a line at infinity, then obtain $PG(2, \mathbb{F}_3)$.

2.2 Prove Theorem 2.3.

2.3 Show that $Z_{n+1} = \{\lambda I^{(n+1)} \mid \lambda \in \mathbb{F}_q^*\}$ is the center of $GL_{n+1}(\mathbb{F}_q)$.

2.4 Determine the order of $PGL_{n+1}(\mathbb{F}_q)$.

2.5 In $PG(n, \mathbb{F}_q)$ select a particular m-flat $S (1 \leq m \leq n-1)$. The s-flats $(0 \leq s \leq m-1)$ contained in S are regarded as source states, the points not on S are regarded as encoding rules, and the $(s+1)$-flats not contained in S but intersect S in s-flats are regarded as messages. Given any source state s (which is an s-flat contained in S), and any encoding rule e (which is a point not on S) the minimal flat m containing both s and e

will be regarded as the message into which the source state s is encoded by the encoding rule e. Then we obtain an authentication code. Compute the parameters of this code. Assume that there is an equally distributed probability distribution given on the set of encoding rules. Determine the probabilities that an opponent can deceive the transmitter/receiver when he plays an impersonation or a substitution game.

2.6 In $PG(s+t+1, \mathbb{F}_q)$, where both s and t are ≥ 0, let us fix an s-flat U. The points on U are regarded as source states, the t-flats of $PG(s+t+1, \mathbb{F}_q)$ which do not meet U are regarded as encoding rules, and the $(t+1)$-flats which meet U in a single point are regarded as messages. Given a source state s and an encoding rule e, the minimal flat m containing both s and e will be regarded as the message into which the source state s is encoded by the encoding rule e. Repeat the preceding exercise.

2.7 Prove Theorem 2.18.

2.8 In $PG(2, \mathbb{F}_4)$ consider the complete quadrangle with vertices $A = (1, 1, 1+\beta)$, $B = (0, 1, \beta)$, $C = (1, 1, \beta)$, and $D = (1, 1+\beta, \beta)$, where β is a primitive element of $\mathbb{F}_4$. Find the diagonal points and verify that they are collinear.

2.9 Prove that there is one and only one conic passing through the four points A, B, C, D of the preceding exercise and the point $E = (1, 1+\beta, 1+\beta)$. Determine all its tangents and the point F they meet. Show that any three of the six points A, B, C, D, E, and F are not collinear.

2.10 Show that for any conic in $PG(2, \mathbb{F}_5)$ the tangents do not meet in the same point.

2.11 Prove that $PO_3(\mathbb{F}_q)$ is triply transitively on the set of points lying on the conic $x_0^2 + x_1 x_2 = 0$. (Note: For even q $O_3(\mathbb{F}_q)$ is defined to be the group formed by all 3×3 matrices T such that

$$T \begin{pmatrix} 1 & & \\ & 0 & 1 \\ & 0 & 0 \end{pmatrix} {}^tT = \begin{pmatrix} 1 & & \\ & 0 & 1 \\ & 0 & 0 \end{pmatrix}$$

and $PO_3(\mathbb{F}_q)$ is defined to be the image of $O_3(\mathbb{F}_q)$ in $PGL_3(\mathbb{F}_q)$. We have $PO_3(\mathbb{F}_q) = O_3(\mathbb{F}_q)$ for even q.)

2.12 Let q be a power of 2. Prove that the set of points of $PG(2, \mathbb{F}_q)$ is partitioned into three orbits under $PO_3(\mathbb{F}_q)$: one consisting of all points on the conic, one consisting of the nucleus alone, and one consisting of all the remaining points.

2.13 Prove that the number of tangents of a k-arc in $PG(2, \mathbb{F}_q)$ is $k(q+2-k)$, the number of secants is $k(k-1)/2$, and the number of external lines is $q(q-1)/2 + (q+2-k)(q+2-k-1)/2$.

2.14 Prove that in $PG(2, \mathbb{F}_q)$, where q is even, if two ovals have more than half their points in common, then they must coincide.

Chapter 3
Geometry of Symplectic Groups over Finite Fields

Following Hermann Weyl, we call the general linear groups, the symplectic groups, the orthogonal groups, and the unitary groups together the *classical groups*. The geometry of linear groups over finite fields has been studied in Chapter 1, in particular Section 1.1, which discusses the geometry of $GL_n(\mathbb{F}_q)$ and Chapter 2, which discusses the geometry of $PGL_n(\mathbb{F}_q)$. We devote the present chapter to the study of the geometry of the symplectic groups over finite fields and later chapters to the study of the geometry of the pseudo symplectic groups, the orthogonal groups, and the unitary groups over finite fields. One can adopt either the vector space language as we did in Section 1.1 or the projective geometry language as we did in Chapter 2. However we prefer the vector space language. But sometimes we also adopt the projective geometry language.

3.1 Symplectic Groups and Symplectic Spaces over Finite Fields

Let $\mathbb{F}_q$ be a field with q elements, where q may be odd or even. An $n \times n$ matrix $K = (k_{ij})_{1 \le i,j \le n}$ is said to be *alternate* if $k_{ij} = -k_{ji}$ for $i \neq j$, $1 \le i,\ j \le n$, and $k_{ii} = 0$ for $i = 1, 2, \ldots, n$. An equivalent definition for K to be alternate is that $xK\,{}^t x = 0$ for all $x \in \mathbb{F}_q^{(n)}$. The equivalence of these two definitions can be proved in the same way as we did in Section 1.4 for the case when q is even. Generalizing Theorem 1.29, we have

Theorem 3.1. Let K be an $n \times n$ alternate matrix over $\mathbb{F}_q$. The rank of K is necessarily even. Furthermore, if K is of rank 2ν ($\le n$), then K is cogredient to

$$\underbrace{\begin{pmatrix} 0 & 1 & & & & & & & \\ -1 & 0 & & & & & & & \\ & & 0 & 1 & & & & & \\ & & -1 & 0 & & & & & \\ & & & & \ddots & & & & \\ & & & & & 0 & 1 & & \\ & & & & & -1 & 0 & & \\ & & & & & & & 0 & \\ & & & & & & & & \ddots \\ & & & & & & & & & 0 \end{pmatrix}}_{2\nu \qquad\qquad n-2\nu}$$

and is also cogredient to

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ -I^{(\nu)} & 0 & \\ & & 0^{(n-2\nu)} \end{pmatrix}. \qquad \square$$

The proof of this theorem is the same as the proof of Theorem 1.29 and will be omitted.

If K is an $n \times n$ alternate matrix of rank 2ν, then ν is called the *index* of K.

Corollary 3.2. Two $n \times n$ alternate matrices over $\mathbb{F}_q$ are cogredient, if and only if they have the same rank, and if and only if they have the same index. $\square$

Corollary 3.3. Let K be an $n \times n$ nonsingular alternate matrix, then n is necessarily even. Suppose that $n = 2\nu$, then K is cogredient to

$$\begin{pmatrix} 0 & I^{(\nu)} \\ -I^{(\nu)} & 0 \end{pmatrix}.$$

$\square$

From now on we assume that $n = 2\nu$ is even. Let K be a $2\nu \times 2\nu$ nonsingular alternate matrix over $\mathbb{F}_q$. A $2\nu \times 2\nu$ matrix T over $\mathbb{F}_q$ is called a *symplectic matrix* with respect to K if

$$TK\,{}^tT = K.$$

Clearly, $2\nu \times 2\nu$ symplectic matrices with respect to the nonsingular alternate matrix K are nonsingular and they form a group with respect

to matrix multiplication, called the *symplectic group* of degree 2ν with respect to K over $\mathbb{F}_q$ and denoted by $Sp_{2\nu}(\mathbb{F}_q, K)$.

Let K_1 and K_2 be two $2\nu \times 2\nu$ nonsingular alternate matrices over $\mathbb{F}_q$. By Corollary 3.2 they are cogredient, i.e., there is a $2\nu \times 2\nu$ nonsingular matrix Q over $\mathbb{F}_q$ such that $QK_1{}^tQ = K_2$. Then $T \in Sp_{2\nu}(\mathbb{F}_q, K_1)$ if and only if $QTQ^{-1} \in Sp_{2\nu}(\mathbb{F}_q, K_2)$. In fact, if $T \in Sp_{2\nu}(\mathbb{F}_q, K_1)$, i.e., $TK_1{}^tT = K_1$, then

$$\begin{aligned}(QTQ^{-1})K_2{}^t(QTQ^{-1}) &= QTQ^{-1}K_2{}^t(Q^{-1})^tT^tQ \\ &= QTK_1{}^tT^tQ \\ &= QK_1{}^tQ \\ &= K_2.\end{aligned}$$

Furthermore, it can be readily verified that the map

$$\begin{aligned}Sp_{2\nu}(\mathbb{F}_q, K_1) &\to Sp_{2\nu}(\mathbb{F}_q, K_2) \\ T &\mapsto QTQ^{-1}\end{aligned}$$

is an isomorphism. Thus, in discussing symplectic groups, without loss of generality we can choose any particular $2\nu \times 2\nu$ nonsingular alternate matrix K and study $Sp_{2\nu}(\mathbb{F}_q, K)$.

From now on let us take

$$K = \begin{pmatrix} 0 & I^{(\nu)} \\ -I^{(\nu)} & 0 \end{pmatrix}, \tag{3.1}$$

denote the symplectic group with respect to K over $\mathbb{F}_q$ simply by $Sp_{2\nu}(\mathbb{F}_q)$, and call it the *symplectic group* of degree 2ν over $\mathbb{F}_q$.

Clearly, $Sp_{2\nu}(\mathbb{F}_q)$ is a subgroup of $GL_{2\nu}(\mathbb{F}_q)$ and the action of $GL_{2\nu}(\mathbb{F}_q)$ on $\mathbb{F}_q^{(2\nu)}$ defined in Section 1.1 induces an action of $Sp_{2\nu}(\mathbb{F}_q)$ on $\mathbb{F}_q^{(2\nu)}$ as follows:

$$\begin{aligned}\mathbb{F}_q^{(2\nu)} \times Sp_{2\nu}(\mathbb{F}_q) &\to \mathbb{F}_q^{(2\nu)} \\ ((x_1, x_2, \ldots, x_{2\nu}), T) &\mapsto (x_1, x_2, \ldots, x_{2\nu})T.\end{aligned}$$

The elements of $Sp_{2\nu}(\mathbb{F}_q)$ are also called *symplectic transformations*. The vector space $\mathbb{F}_q^{(2\nu)}$ together with the above group action of the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$, is called the 2ν-dimensional *symplectic space* over $\mathbb{F}_q$. According to the Erlangen Program the *geometry of the symplectic group* $Sp_{2\nu}(\mathbb{F}_q)$ or simply the *symplectic geometry* over $\mathbb{F}_q$ is the study of properties of the symplectic space $\mathbb{F}_q^{(2\nu)}$ which are invariant under the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$.

At first, we want to study how the subspaces of $\mathbb{F}_q^{(2\nu)}$ are partitioned into orbits under the action of $Sp_{2\nu}(\mathbb{F}_q)$. We need more definitions.

A vector $x \in \mathbb{F}_q^{(2\nu)}$ is called an *isotropic vector*, if $xK\,{}^t x = 0$. Clearly

$$\begin{aligned}&(x_1, \ldots, x_\nu, x_{\nu+1}, \ldots, x_{2\nu})K\,{}^t(x_1, \ldots, x_\nu, x_{\nu+1}, \ldots, x_{2\nu})\\ &\quad = (-x_{\nu+1}, \ldots, -x_{2\nu}, x_1, \ldots, x_\nu)\,{}^t(x_1, \ldots, x_\nu, x_{\nu+1}, \ldots, x_{2\nu})\\ &\quad = 0,\end{aligned}$$

hence all vectors of $\mathbb{F}_q^{(2\nu)}$ are isotropic.

Let P be an m-dimensional vector subspace of $\mathbb{F}_q^{(2\nu)}$. As before we use the same letter P to denote a matrix representation of the vector subspace P, i.e., P is an $m \times 2\nu$ matrix of rank m whose rows form a basis of P. It is easy to see that $PK\,{}^tP$ is an alternate matrix. By Theorem 3.1, let the rank of $PK\,{}^tP$ be $2s$, then we call the vector subspace P a subspace of *type* (m, s). Clearly $s \le \nu$ and $2s \le m$. In particular, subspaces of type $(m, 0)$ are called m-dimensional *totally isotropic subspaces*, and subspaces of type $(2s, s)$ are called $2s$-dimensional *non-isotropic subspaces*. It is clear that a subspace P is totally isotropic if and only if $PK\,{}^tP = 0$, and it is non-isotropic if and only if $PK\,{}^tP$ is nonsingular.

For example, since all vectors of $\mathbb{F}_q^{(2\nu)}$ are isotropic, all 1-dimensional subspaces are totally isotropic. If $s \le \nu$, then

$$\begin{array}{cccc} {\scriptstyle s} & {\scriptstyle \nu-s} & {\scriptstyle s} & {\scriptstyle \nu-s} \\ \big(I^{(s)} & 0 & 0 & 0 \big) \end{array} \tag{3.2}$$

is a totally isotropic subspace of dimension s, and

$$\begin{array}{cccc} {\scriptstyle s} & {\scriptstyle \nu-s} & {\scriptstyle s} & {\scriptstyle \nu-s} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & I^{(s)} & 0 \end{pmatrix} \end{array} \tag{3.3}$$

is a non-isotropic subspace of dimension $2s$. More generally, if $i_1, i_2, \ldots, i_s$ are s indices such that $1 \le i_1 < i_2 < \cdots < i_s \le \nu$, then the subspace

$$\langle e_{i_1}, e_{i_2}, \ldots, e_{i_s} \rangle$$

generated by $e_{i_1}, e_{i_2}, \ldots, e_{i_s}$ is an s-dimensional totally isotropic subspace, and the subspace

$$\langle e_{i_1}, e_{i_2}, \ldots, e_{i_s}, e_{\nu+i_1}, e_{\nu+i_2}, \ldots, e_{\nu+i_s} \rangle$$

generated by $e_{i_1}, e_{i_2}, \ldots, e_{i_s}, e_{\nu+i_1}, e_{\nu+i_2}, \ldots, e_{\nu+i_s}$ is a $2s$-dimensional non-isotropic subspace. If $2s \le m \le \nu + s$, then

$$\begin{array}{c} \begin{matrix} s & m-2s & \nu+s-m & s & m-2s & \nu+s-m \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 \\ 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{3.4}$$

is a subspace of type (m, s).

Two vectors x and y of $\mathbb{F}_q^{(2\nu)}$ are said to be *orthogonal* (with respect to K), if $xK\,{}^t y = 0$. Clearly, if $xK\,{}^t y = 0$, then $yK\,{}^t x = 0$. Of course, every vector is orthogonal to itself, i.e., $xK\,{}^t x = 0$ for every vector x of $\mathbb{F}_q^{(2\nu)}$. To say that a vector is orthogonal to itself is equivalent to say that it is an isotropic vector.

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu)}$. Denote by $P^\perp$ the set of vectors which are orthogonal to every vector of P, i.e.,

$$P^\perp = \{y \in \mathbb{F}_q^{(2\nu)} \mid yK\,{}^t x = 0 \text{ for all } x \in P\}.$$

Obviously, $P^\perp$ is a $(2\nu - m)$-dimensional subspace of $\mathbb{F}_q^{(2\nu)}$ and is called the *dual subspace* of P. For example, the dual subspace of (3.2) is

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(\nu-s)} & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s)} \end{pmatrix}, \end{array}$$

and the dual subspace of (3.3) is

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s \end{matrix} \\ \begin{pmatrix} 0 & I^{(\nu-s)} & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s)} \end{pmatrix}. \end{array}$$

More generally, the dual subspace of (3.4) which is of type (m, s) is

$$\begin{array}{c} \begin{matrix} s & m-2s & \nu+s-m & s & m-2s & \nu+s-m \end{matrix} \\ \begin{pmatrix} 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\nu+s-m)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m)} \end{pmatrix} \end{array}$$

and is of type $(2\nu - m, \nu + s - m)$.

Theorem 3.4. A subspace P is totally isotropic if and only if $P \subseteq P^{\perp}$; and P is non-isotropic if and only if $P \cap P^{\perp} = (0)$.

Proof. By definition a subspace P is totally isotropic if $PK\,{}^tP = 0$, and $PK\,{}^tP = 0$ if and only if $P \subseteq P^{\perp}$, whence the first assertion is proved.

Now let us prove the second assertion. Notice that

$$\begin{pmatrix} P \\ P^{\perp} \end{pmatrix} K \,{}^t\!\begin{pmatrix} P \\ P^{\perp} \end{pmatrix} = \begin{pmatrix} PK\,{}^tP & 0 \\ 0 & P^{\perp}K\,{}^t(P^{\perp}) \end{pmatrix}.$$

If $P \cap P^{\perp} = (0)$, then

$$\begin{pmatrix} P \\ P^{\perp} \end{pmatrix}$$

is a $2\nu \times 2\nu$ nonsingular matrix, and hence $PK\,{}^tP$ is nonsingular, i.e., P is non-isotropic. Conversely, suppose that there is a non-zero vector $x \in P \cap P^{\perp}$. Let P be an m-dimensional subspace, then there is a $1 \times m$ vector $\xi \neq 0$ such that $x = \xi P$. Thus $\xi PK\,{}^tP = xK\,{}^tP = 0$, and it follows that P is not non-isotropic. □

In studying the transitivity properties of $Sp_{2\nu}(\mathbb{F}_q)$ on the set of subspaces of $\mathbb{F}_q^{(2\nu)}$, the following lemma is crucial.

Lemma 3.5. Let P be a subspace of type (m, s) in $\mathbb{F}_q^{(2\nu)}$. Then $2s \leq m \leq \nu + s$ and there is an $m \times m$ nonsingular matrix Q and a $(2\nu - m) \times 2\nu$ matrix Z such that

$$\begin{pmatrix} QP \\ Z \end{pmatrix} K \,{}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix} = \begin{pmatrix} 0 & I^{(s)} & & & & \\ -I^{(s)} & 0 & & & & \\ & & 0 & I^{(m-2s)} & & \\ & & -I^{(m-2s)} & 0 & & \\ & & & & 0 & I^{(\sigma)} \\ & & & & -I^{(\sigma)} & 0 \end{pmatrix}, \quad (3.5)$$

where $\sigma = \nu + s - m$.

Proof. P is a subspace of type (m, s) and hence $PK\,{}^tP$ is an $m \times m$ matrix of rank $2s$. By Theorem 3.1, there is an $m \times m$ nonsingular matrix Q such that

$$Q(PK\,{}^tP)\,{}^tQ = \begin{pmatrix} 0 & I^{(s)} & \\ -I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix}.$$

Hence $2s \leq m$. There is a $(2\nu - m) \times 2\nu$ matrix Z_1, such that

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix}$$

is nonsingular. We may write

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix} K \, {}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} = \begin{pmatrix} 0 & I^{(s)} & 0 & K_{14} \\ -I^{(s)} & 0 & 0 & K_{24} \\ 0 & 0 & 0^{(m-2s)} & K_{34} \\ -{}^tK_{14} & -{}^tK_{24} & -{}^tK_{34} & K_{44} \end{pmatrix},$$

where K_{44} is a $(2\nu - m) \times (2\nu - m)$ alternate matrix. Then we have

$$\begin{pmatrix} I^{(s)} & & & \\ & I^{(s)} & & \\ & & I^{(m-2s)} & \\ {}^tK_{24} & -{}^tK_{14} & 0 & I^{(2\nu-m)} \end{pmatrix} \begin{pmatrix} 0 & I^{(s)} & 0 & K_{14} \\ -I^{(s)} & 0 & 0 & K_{24} \\ 0 & 0 & 0^{(m-2s)} & K_{34} \\ -{}^tK_{14} & -{}^tK_{24} & -{}^tK_{34} & K_{44} \end{pmatrix}$$

$$\times \, {}^t\begin{pmatrix} I^{(s)} & & & \\ & I^{(s)} & & \\ & & I^{(m-2s)} & \\ {}^tK_{24} & -{}^tK_{14} & 0 & I^{(2\nu-m)} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & I^{(s)} & 0 & 0 \\ -I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & K_{34} \\ 0 & 0 & -{}^tK_{34} & L_{44} \end{pmatrix},$$

where $L_{44} = K_{44} + {}^tK_{24}K_{14} - {}^tK_{14}K_{24}$. Since K is nonsingular, K_{34} is of rank $m - 2s$. Thus $m - 2s \leq 2\nu - m$, which gives $m \leq \nu + s$, and there is a $(2\nu - m) \times (2\nu - m)$ nonsingular matrix B such that

$$K_{34} \, {}^tB = (I^{(m-2s)} \; 0^{(m-2s,2\sigma)}).$$

Then

$$\begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & B \end{pmatrix} \begin{pmatrix} 0 & I^{(s)} & & \\ -I^{(s)} & 0 & & \\ & & 0 & K_{34} \\ & & -{}^tK_{34} & L_{44} \end{pmatrix} {}^t\begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & B \end{pmatrix}$$

$$= \begin{pmatrix} 0 & I^{(s)} & & & \\ -I^{(s)} & 0 & & & \\ & & 0 & I^{(m-2s)} & 0 \\ & & -I^{(m-2s)} & K_{44} & K_{45} \\ & & 0 & -{}^tK_{45} & K_{55} \end{pmatrix},$$

where K_{44} is an $(m-2s)\times(m-2s)$ alternate matrix and K_{55} is a $2\sigma\times 2\sigma$ alternate matrix. Let T be the matrix obtained from K_{44} by replacing all elements below the main diagonal by zeros. Then

$$K_{44} = T - {}^tT,$$

and

$$\begin{pmatrix} I^{(s)} & & & & \\ & I^{(s)} & & & \\ & & I^{(m-2s)} & & \\ & & -T & I^{(m-2s)} & \\ & & {}^tK_{45} & & I^{(2\sigma)} \end{pmatrix} \begin{pmatrix} 0 & I^{(s)} & & & \\ -I^{(s)} & 0 & & & \\ & & 0 & I^{(m-2s)} & \\ & & -I^{(m-2s)} & K_{44} & K_{45} \\ & & & -{}^tK_{45} & K_{55} \end{pmatrix}$$

$$\times {}^t\begin{pmatrix} I^{(s)} & & & & \\ & I^{(s)} & & & \\ & & I^{(m-2s)} & & \\ & & -T & I^{(m-2s)} & \\ & & {}^tK_{45} & & I^{(2\sigma)} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & I^{(s)} & & & \\ -I^{(s)} & 0 & & & \\ & & 0 & I^{(m-2s)} & \\ & & -I^{(m-2s)} & 0 & 0 \\ & & & 0 & K_{55} \end{pmatrix}.$$

Since K is nonsingular, K_{55} is nonsingular. By Corollary 3.3 there is a $2\sigma\times 2\sigma$ nonsingular matrix R such that

$$RK_{55}\,{}^tR = \begin{pmatrix} 0 & I^{(\sigma)} \\ -I^{(\sigma)} & 0 \end{pmatrix}.$$

Write

$$\begin{pmatrix} I^{(s)} & & & & \\ & I^{(s)} & & & \\ & & I^{(m-2s)} & & \\ & & & I^{(m-2s)} & \\ & & & & R \end{pmatrix} \begin{pmatrix} I^{(s)} & & & & \\ & I^{(s)} & & & \\ & & I^{(m-2s)} & & \\ & & -T & I^{(m-2s)} & \\ & & {}^tK_{45} & & I \end{pmatrix}$$

$$\times \begin{pmatrix} I^{(s)} & & & \\ & I^{(s)} & & \\ & & I^{(m-2s)} & \\ & & & B \end{pmatrix} \begin{pmatrix} I^{(s)} & & & \\ & I^{(s)} & & \\ & & I^{(m-2s)} & \\ {}^tK_{24} & -{}^tK_{14} & & I^{(2\nu-m)} \end{pmatrix}$$

$$= \begin{matrix} \\ m \\ 2\nu - m \end{matrix} \begin{matrix} \begin{matrix} m & 2\nu - m \end{matrix} \\ \begin{pmatrix} I^{(m)} & \\ L_1 & L_2 \end{pmatrix} \end{matrix},$$

and let

$$\begin{pmatrix} I^{(m)} & \\ L_1 & L_2 \end{pmatrix} \begin{pmatrix} QP \\ Z_1 \end{pmatrix} = \begin{pmatrix} QP \\ Z \end{pmatrix}.$$

Then

$$\begin{pmatrix} QP \\ Z \end{pmatrix} K \, {}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix} = \begin{pmatrix} 0 & I^{(s)} & & & & \\ -I^{(s)} & 0 & & & & \\ & & 0 & I^{(m-2s)} & & \\ & & -I^{(m-2s)} & 0 & & \\ & & & & 0 & I^{(\sigma)} \\ & & & & -I^{(\sigma)} & 0 \end{pmatrix}.$$

Therefore Lemma 3.5 is completely proved. □

From Lemma 3.5 we deduce immediately the following necessary and sufficient condition for the existence of subspaces of type (m, s) in the 2ν-dimensional symplectic space.

Theorem 3.6. Subspaces of type (m, s) exist in the 2ν-dimensional symplectic space if and only if $2s \leq m \leq \nu + s$.

Proof. Lemma 3.5 asserts that if there is a subspace of type (m, s) in the 2ν-dimensional symplectic space, then $2s \leq m \leq \nu + s$. Conversely, if $2s \leq m \leq \nu + s$, then (3.4) is a subspace of type (m, s). □

Moreover, the following transitivity theorem of subspaces of the same type under $Sp_{2\nu}(\mathbb{F}_q)$ also follows from Lemma 3.5.

Theorem 3.7. Let P_1 and P_2 be two m-dimensional subspaces of $\mathbb{F}_q^{(2\nu)}$. Then there is a $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = AP_2T$, where A is an $m \times m$ nonsingular matrix, if and only if P_1 and P_2 are of the same type. In other words, $Sp_{2\nu}(\mathbb{F}_q)$ acts transitively on each set of subspaces of the same type.

Proof. Suppose that there is a $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = AP_2T$, where A is an $m \times m$ nonsingular matrix. Then

$$P_1 K \, {}^t\!P_1 = AP_2TK \, {}^t\!T \, {}^t\!P_2 \, {}^t\!A = AP_2K \, {}^t\!P_2 \, {}^t\!A,$$

i.e., $P_1 K\,{}^tP_1$ and $P_2 K\,{}^tP_2$ are cogredient. Hence P_1 and P_2 are of the same type.

Conversely, suppose that P_1 and P_2 are of the same type (m, s). By Lemma 3.5 there are $m \times m$ nonsingular matrices Q_1 and Q_2, and $(2\nu - m) \times 2\nu$ matrices Z_1 and Z_2 such that both

$$\begin{pmatrix} Q_1P_1 \\ Z_1 \end{pmatrix} K \,{}^t\!\begin{pmatrix} Q_1P_1 \\ Z_1 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} Q_2P_2 \\ Z_2 \end{pmatrix} K \,{}^t\!\begin{pmatrix} Q_2P_2 \\ Z_2 \end{pmatrix}$$

are equal to the right-hand side of (3.5). Thus

$$\begin{pmatrix} Q_2P_2 \\ Z_2 \end{pmatrix}^{-1} \begin{pmatrix} Q_1P_1 \\ Z_1 \end{pmatrix} \in Sp_{2\nu}(\mathbb{F}_q).$$

Let

$$T = \begin{pmatrix} Q_2P_2 \\ Z_2 \end{pmatrix}^{-1} \begin{pmatrix} Q_1P_1 \\ Z_1 \end{pmatrix},$$

then

$$P_1 = (Q_1^{-1}Q_2)P_2T.$$

□

Corollary 3.8. Totally isotropic subspaces of $\mathbb{F}_q^{(2\nu)}$ are of dimension $\leq \nu$, and there exist totally isotropic subspaces of dimension ν. Therefore maximal totally isotropic subspaces are of dimension ν. Moreover, any totally isotropic subspace is contained in a maximal totally isotropic subspace.

Proof. By Theorem 3.6, totally isotropic subspaces are of dimension $\leq \nu$. It is easy to verify that

$$(I^{(\nu)}\ 0^{(\nu)})$$

is a totally isotropic subspace of dimension ν. Therefore maximal totally isotropic subspaces are of dimension ν. Clearly, subspaces of a totally isotropic subspace are also totally isotropic. Hence we deduce that $(I^{(\nu)}\ 0^{(\nu)})$ has totally isotropic subspaces of any dimension $\leq \nu$. By Theorem 3.7, $Sp_{2\nu}(\mathbb{F}_q)$ acts transitively on the set of totally isotropic subspaces of the same dimension. Hence any totally isotropic subspace is contained in a totally isotropic subspace of dimension ν. □

Corollary 3.9. The dual subspace of a subspace of type (m, s) is of type $(2\nu - m, \nu + s - m)$.

Proof. Let P be a subspace of type (m, s). Denote the subspace (3.4) by P_0. We know that P_0 is also of type (m, s). By Theorem 3.7, there is a $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P = AP_0T$, where $A \in GL_m(\mathbb{F}_q)$. For $x \in \mathbb{F}_q^{(2\nu)}$, $xK\,{}^tP_0 = 0$ if and only if $(xT)K\,{}^tP = 0$. That is, $x \in P_0^{\perp}$ if and only if $xT \in P^{\perp}$. Therefore $P^{\perp} = P_0^{\perp}T$, where $P^{\perp}$ and $P_0^{\perp}$ are regarded as subspaces. Since $P_0^{\perp}$ is of type $(2\nu - m, \nu + s - m)$, so is $P^{\perp}$. □

By Theorems 3.6 and 3.7 we can enumerate the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu)}$ under $Sp_{2\nu}(\mathbb{F}_q)$. We have

Theorem 3.10. The number of orbits of subspaces of $\mathbb{F}_q^{(2\nu)}$ under $Sp_{2\nu}(\mathbb{F}_q)$ is

$$\frac{1}{2}(\nu + 1)(\nu + 2).$$

Proof. From Theorem 3.7 we know that two subspaces of $\mathbb{F}_q^{(2\nu)}$ can be carried from one into the other under $Sp_{2\nu}(\mathbb{F}_q)$ if and only if they are of the same type. Hence by Theorem 3.6, the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu)}$ under $Sp_{2\nu}(\mathbb{F}_q)$ is equal to the number of pairs (m, s) of non-negative integers m and s which satisfy the condition $2s \le m \le \nu + s$. Clearly $0 \le m \le 2\nu$. From $2s \le m \le \nu + s$, we deduce $m - \nu \le s \le [m/2]$. If $m \le \nu$, then s can take $[m/2] + 1$ values: $0, 1, \ldots, [m/2]$; if $m > \nu$, then s can take $[m/2] - (m - \nu) + 1$ values: $m - \nu, m - \nu + 1, \ldots, [m/2]$. Consequently, the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu)}$ under $Sp_{2\nu}(\mathbb{F}_q)$ is equal to

$$\sum_{m=0}^{\nu}([m/2] + 1) + \sum_{m=\nu+1}^{2\nu}([m/2] - (m - \nu) + 1) = \frac{1}{2}(\nu + 1)(\nu + 2).$$

□

Denote by $\mathcal{M}(m, s; 2\nu)$ the set of subspaces of $\mathbb{F}_q^{(2\nu)}$ of type (m, s) and by $N(m, s; 2\nu)$ the number of subspaces of $\mathbb{F}_q^{(2\nu)}$ of type (m, s), i.e.,

$$N(m, s; 2\nu) = |\ \mathcal{M}(m, s; 2\nu)\ |\ .$$

Our next goal is to compute $N(m, s; 2\nu)$. Since $\mathcal{M}(m, s; 2\nu)$ is an orbit of subspaces under $Sp_{2\nu}(\mathbb{F}_q)$, $N(m, s; 2\nu)$ is equal to the index of the *stabilizer* of any element of $\mathcal{M}(m, s; 2\nu)$ in $Sp_{2\nu}(\mathbb{F}_q)$. However, in the

following we shall compute $N(m, s; 2\nu)$ in the same way as we did in Section 1.1 to compute $N(m, n)$, the number of m-dimensional subspaces in $\mathbb{F}_q^{(n)}$. For this purpose, the following result, which is slightly stronger than Theorem 3.7, is needed.

Lemma 3.11. Let P_1 and P_2 be two $m \times 2\nu$ matrices of rank m. Then there exists an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = P_2T$ if and only if $P_1K\,{}^tP_1 = P_2K\,{}^tP_2$. □

Proof. Suppose that there is a $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = P_2T$, then

$$P_1K\,{}^tP_1 = P_2TK\,{}^tT\,{}^tP_2 = P_2K\,{}^tP_2.$$

Now suppose that $P_1K\,{}^tP_1 = P_2K\,{}^tP_2$. Then in the proof of Lemma 3.5 we can take the same $m \times m$ nonsingular matrix Q and two $(2\nu - m) \times 2\nu$ matrices Z_1 and Z_2 such that both

$$\begin{pmatrix} QP_1 \\ Z_1 \end{pmatrix} K \begin{pmatrix} QP_1 \\ Z_1 \end{pmatrix}^t \qquad \text{and} \qquad \begin{pmatrix} QP_2 \\ Z_2 \end{pmatrix} K \begin{pmatrix} QP_2 \\ Z_2 \end{pmatrix}^t$$

are equal to the right-hand side of (3.5). Then

$$\begin{pmatrix} QP_1 \\ Z_1 \end{pmatrix} K \begin{pmatrix} QP_1 \\ Z_1 \end{pmatrix}^t = \begin{pmatrix} QP_2 \\ Z_2 \end{pmatrix} K \begin{pmatrix} QP_2 \\ Z_2 \end{pmatrix}^t.$$

Thus

$$T = \begin{pmatrix} QP_2 \\ Z_2 \end{pmatrix}^{-1} \begin{pmatrix} QP_1 \\ Z_1 \end{pmatrix} \in Sp_{2\nu}(\mathbb{F}_q)$$

and

$$P_1 = P_2T.$$

□

Finally, let us give some examples of authentication codes constructed from symplectic geometries over finite fields.

Example 3.1. Let us fix a non-zero vector v in the symplectic space $\mathbb{F}_q^{(2\nu)}$, where $\nu \geq 2$, and take the 2-dimensional totally isotropic subspaces containing v as source states, the 1-dimensional subspaces not contained in $\langle v \rangle^\perp$ as encoding rules, and the subspaces of type (3,1) containing v as messages. Given a source state s (which is a 2-dimensional totally isotropic subspace containing v) and an encoding rule e (which is a 1-dimensional

subspace not contained in $\langle v\rangle^{\perp}$), then the subspace m generated by s and e is of type (3,1), and we define $m = e(s)$, i.e., the source state s is encoded by the encoding rule e into the message m. Thus we obtain an authentication code. □

Example 3.2. Let us fix a subspace P of type (3,1) in the symplectic space $\mathbb{F}_q^{(2\nu)}$, where $\nu \geq 2$, and a vector $v \in P$ such that $v \notin P^{\perp}$. The 2-dimensional non-isotropic subspaces contained in P and containing v are regarded as source states, the 2-dimensional non-isotropic subspaces not contained in P but containing v are regarded as encoding rules, and the subspaces of type (3,1), which contains v and intersect P in 2-dimensional non-isotropic subspaces, are regarded as messages. Assume that the encoding rules work as follows. Given a source state s (which is a 2-dimensional non-isotropic subspace contained in P and containing v) and an encoding rule e (which is a 2-dimensional non-isotropic subspace not contained in P but containing v), the subspace m generated by s and e is of type (3,1) and of course $m \cap P = s$ is a 2-dimensional non-isotropic subspace, then we define $m = e(s)$, i.e., the source state s is encoded by the encoding rule e into the message m. Thus we obtain an authentication code. □

3.2 Anzahl Theorems in Symplectic Geometry over Finite Fields

We follow the notation of the previous section. The main goal of the present section is to compute $N(m, s; 2\nu)$. If $s > \nu$, $2s > m$, or $\nu + s - m < 0$, then we will have $N(m, s; 2\nu) = 0$. Hence in the following we always assume that

$$2s \leq m \leq \nu + s.$$

Let

$$M(m,s) = \begin{pmatrix} 0 & I^{(s)} & \\ -I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix}. \tag{3.6}$$

If P is a subspace of type (m, s), then by Theorem 3.1 we can choose a matrix representation P of the subspace P such that

$$PK\,{}^tP = M(m,s). \tag{3.7}$$

Denote by $n(m, s; 2\nu)$ the number of $m \times 2\nu$ matrices P of rank m which satisfy (3.7). Then we have

Lemma 3.12. Let $2s \leq m \leq \nu + s$. Then

$$n(m, s; 2\nu) =\mid Sp_{2s}(\mathbb{F}_q) \mid\mid GL_{m-2s}(\mathbb{F}_q) \mid q^{2s(m-2s)} N(m, s; 2\nu).$$

Proof. Let P_1 and P_2 be two $m \times 2\nu$ matrices of rank m which satisfy (3.7). If P_1 and P_2 are matrix representations of the same subspace, then there is an $m \times m$ nonsingular matrix T such that

$$P_1 = TP_2.$$

From $P_1 K\, {}^tP_1 = P_2 K\, {}^tP_2 = M(m, s)$ we deduce

$$TM(m, s)\, {}^tT = M(m, s). \tag{3.8}$$

The set of all $m \times m$ nonsingular matrices T which satisfy (3.8) forms a group, denoted by $\mathcal{G}(m, s)$[1]. For any $T \in \mathcal{G}(m, s)$, $T \neq I$ implies $TP_2 \neq P_2$. Hence,

$$n(m, s; 2\nu) =\mid \mathcal{G}(m, s) \mid N(m, s; 2\nu). \tag{3.9}$$

Let us compute $\mid \mathcal{G}(m, s) \mid$.

Let

$$K_1 = \begin{pmatrix} 0 & I^{(s)} \\ -I^{(s)} & 0 \end{pmatrix},$$

and write T in block forms

$$T = \begin{array}{c} \begin{matrix} 2s & m-2s \end{matrix} \\ \begin{pmatrix} A & B \\ C & D \end{pmatrix} \end{array} \begin{matrix} 2s \\ m-2s \end{matrix}.$$

From (3.8) we deduce

$$AK_1\, {}^tA = K_1,$$
$$AK_1\, {}^tC = 0,$$

and

$$CK_1\, {}^tC = 0.$$

It follows that $A \in Sp_{2\nu}(K_1)$ and $C = 0$. That is, $\mathcal{G}(m, s)$ consists of all $m \times m$ matrices of the form

$$\begin{pmatrix} A & B \\ 0 & D \end{pmatrix},$$

[1] $\mathcal{G}(m, s)$ will be called the singular symplectic group in the next section and its geometry will be studied there.

where $A \in Sp_{2\nu}(K_1)$, B is any $2s \times (m-2s)$ matrix, and $D \in GL_{m-2s}(\mathbb{F}_q)$. Consequently,

$$| \mathcal{G}(m,s) | = | Sp_{2s}(\mathbb{F}_q) | \, | GL_{m-2s}(\mathbb{F}_q) | \, q^{2s(m-2s)} \tag{3.10}$$

and Lemma 3.12 follows from (3.9) and (3.10). □

Clearly,

$$| Sp_{2s}(\mathbb{F}_q) | = n(2s, s; 2s)$$

and we have already computed $| GL_{m-2s}(\mathbb{F}_q) |$ in Theorem 1.6. Thus to compute $N(m, s; 2\nu)$ it is sufficient to compute $n(m, s; 2\nu)$.

Let us introduce the following notations. Let A be an $m \times n$ matrix and $0 \leq r \leq \min(m, n)$. Denote by A_r the upper left corner $r \times r$ submatrix of A. Now let $2s \leq m \leq \nu + s$ and $0 \leq r_1, r_2, r_1 + r_2 \leq m$. Let $n(r_1)$ be the number of $r_1 \times 2\nu$ matrices P_1 of rank r_1 such that $P_1 K {}^t P_1 = M(m, s)_{r_1}$. For a fixed $r_1 \times 2\nu$ matrix P_1 of rank r_1 such that $P_1 K {}^t P_1 = M(m, s)_{r_1}$, let $n(P_1, r_2)$ be the number of $r_2 \times 2\nu$ matrices P_2 such that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank $r_1 + r_2$ and $PK {}^t P = M(m, s)_{r_1 + r_2}$. Then we have

Lemma 3.13. Let $2s \leq m \leq \nu + s$ and $0 \leq r_1, r_2, r_1 + r_2 \leq m$. Then $n(P_1, r_2)$ is independent of the choice of P_1, and

$$n(r_1 + r_2) = n(r_1) n(P_1, r_2)$$

for any choice of P_1.

Proof. Let Q_1 be another $r_1 \times 2\nu$ matrix of rank r_1 such that $Q_1 K {}^t Q_1 = M(m, s)_{r_1}$. Then $P_1 K {}^t P_1 = Q_1 K {}^t Q_1$. By Lemma 3.11 there is an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $Q_1 = P_1 T$. Thus, if P_2 is an $r_2 \times 2\nu$ matrix such that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank $r_1 + r_2$ and $PK {}^t P = M(m, s)_{r_1 + r_2}$, then $P_2 T$ is also an $r_2 \times 2\nu$ matrix such that

$$Q = \begin{pmatrix} Q_1 \\ P_2 T \end{pmatrix} = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} T$$

is of rank $r_1 + r_2$ and $QK\,{}^tQ = M(m,s)_{r_1+r_2}$. Conversely, if Q_2 is an $r_2 \times 2\nu$ matrix such that

$$Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$$

is of rank $r_1 + r_2$ and $QK\,{}^tQ = M(m,s)_{r_1+r_2}$, then Q_2T^{-1} is an $r_2 \times 2\nu$ matrix such that

$$P = \begin{pmatrix} P_1 \\ Q_2T^{-1} \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} T^{-1}$$

is of rank $r_1 + r_2$ and $PK\,{}^tP = M(m,s)_{r_1+r_2}$. Hence our lemma follows. □

Now we are ready to compute $n(m,s;2\nu)$. Let us start with a special case.

Lemma 3.14. Let $s \le \nu$. Then

$$n(s,0;2\nu) = q^{\frac{s(s-1)}{2}} \prod_{i=\nu-s+1}^{\nu} (q^{2i}-1).$$

Proof. $n(s,0;2\nu)$ is the number of $s \times 2\nu$ matrices P of rank s such that $PK\,{}^tP = 0$. Write

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_s \end{pmatrix},$$

where $v_1, v_2, \ldots, v_s$ are the s row vectors of P. Then

$$v_iK\,{}^tv_j = 0 \ (1 \le i,j \le s),$$

i.e., $v_1, v_2, \ldots, v_s$ are s linearly independent mutually orthogonal isotropic vectors of $\mathbb{F}_q^{(2\nu)}$. Since any vector of $\mathbb{F}_q^{(2\nu)}$ is isotropic, we may choose v_1 to be any non-zero vector of $\mathbb{F}_q^{(2\nu)}$. Thus there are $q^{2\nu}-1$ possible choices of v_1. Once v_1 is chosen, we may choose v_2 to be any vector in the dual subspace $\langle v_1\rangle^\perp$ of the 1-dimensional subspace $\langle v_1\rangle$ spanned by v_1 such that v_1 and v_2 are linearly independent. Since $v_1 \in \langle v_1\rangle^\perp$, there are $q^{2\nu-1}-q$ possible choices of v_2. Once v_1 and v_2 are chosen, they span a 2-dimensional totally isotropic subspace $\langle v_1, v_2\rangle$. We may choose v_3 to

be any vector in the dual subspace $\langle v_1, v_2\rangle^\perp$ of the 2-dimensional totally isotropic subspace $\langle v_1, v_2\rangle$ such that v_1, v_2, and v_3 are linearly independent. Since $\langle v_1, v_2\rangle \subseteq \langle v_1, v_2\rangle^\perp$, there are $q^{2\nu-2} - q^2$ possible choices of v_3. Proceeding in this way, finally when $v_1, v_2, \ldots, v_{s-1}$ are chosen, they span an $(s-1)$-dimensional totally isotropic subspace $\langle v_1, v_2, \ldots, v_{s-1}\rangle$, and we may choose v_s to be any vector of $\langle v_1, v_2, \ldots, v_{s-1}\rangle^\perp$ not contained in $\langle v_1, v_2, \ldots, v_{s-1}\rangle$. Thus there are $q^{2\nu-s+1} - q^{s-1}$ possible choices of v_s. Therefore

$$\begin{aligned} n(s,0;2\nu) &= (q^{2\nu}-1)(q^{2\nu-1}-q)\ldots(q^{2\nu-s+1}-q^{s-1}) \\ &= q^{\frac{s(s-1)}{2}} \prod_{i=\nu-s+1}^{\nu} (q^{2i}-1). \end{aligned}$$

□

Lemma 3.15. Let $s \leq \nu$. Then

$$n(2s,s;2\nu) = q^{(2\nu-s)s} \prod_{i=\nu-s+1}^{\nu} (q^{2i}-1).$$

Proof. $n(2s,s;2\nu)$ is the number of $2s \times 2\nu$ matrices P of rank $2s$ such that

$$PK\,{}^tP = \begin{pmatrix} 0 & I^{(s)} \\ -I^{(s)} & 0 \end{pmatrix}.$$

Write

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{matrix} s \\ s \end{matrix}.$$

Then

$$\begin{aligned} P_1 K\,{}^tP_1 &= 0^{(s)}, \\ P_1 K\,{}^tP_2 &= I^{(s)}, \end{aligned}$$

and

$$P_2 K\,{}^tP_2 = 0^{(s)}.$$

Thus there are $n(s,0,2\nu)$ possible choices of P_1.

By Lemma 3.13, once P_1 has been chosen, the number of possible choices of P_2 is independent of the particular choice of P_1. Hence we may choose

$$P_1 = (I^{(s)}\ 0^{(s,2\nu-s)})$$

and then compute the number of possible choices of P_2. Write

$$P_2 = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_s \end{pmatrix},$$

where $v_1, v_2, \ldots, v_s$ are the s row vectors of P_2. From

$$e_1 K^t v_1 = 1,\ e_2 K^t v_1 = e_3 K^t v_1 = \cdots = e_s K^t v_1 = 0$$

we deduce that v_1 is necessarily of the form

$$(x_1, x_2, \ldots, x_\nu, 1, \underbrace{0, 0, \ldots, 0}_{s-1}, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}),$$

where $x_1, x_2, \ldots, x_\nu, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}$ can be arbitrary elements of $\mathbb{F}_q$. Thus there are $q^{2\nu-s}$ possible choices of v_1.

Again by Lemma 3.13, once P_1 and v_1 have been chosen, the number of possible choices of v_2 is independent of the particular choices of P_1 and v_1. Hence we may choose P_1 as above and choose $v_1 = e_{\nu+1}$, and then compute the number of possible choices of v_2. From

$$e_1 K^t v_2 = 0,\ e_2 K^t v_2 = 1,\ e_3 K^t v_2 = \cdots = e_s K^t v_2 = 0,\ e_{\nu+1} K^t v_2 = 0$$

we deduce that v_2 is necessarily of the form

$$(0, x_2, \ldots, x_\nu, 0, 1, \underbrace{0, \ldots, 0}_{s-2}, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}),$$

where $x_2, \ldots, x_\nu, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}$ can be arbitrary elements of $\mathbb{F}_q$. Thus there are $q^{2\nu-s-1}$ possible choices of v_2.

Proceeding in this way, once P_1 and $v_1, v_2, \ldots, v_{s-1}$ have been chosen, by Lemma 3.13, the number of possible choices of v_s is independent of the particular choices of P_1 and $v_1, v_2, \ldots, v_{s-1}$. Thus we may choose P_1 as above and choose $v_1 = e_{\nu+1}, v_2 = e_{\nu+2}, \ldots, v_{s-1} = e_{\nu+s-1}$, and then compute the number of possible choices of v_s. From

$$\begin{aligned} e_1 K^t v_s = e_2 K^t v_s = \cdots = e_{s-1} K^t v_s = 0,\ e_s K^t v_s = 1, \\ e_{\nu+1} K^t v_s = e_{\nu+2} K^t v_s = \cdots = e_{\nu+s-1} K^t v_s = 0 \end{aligned}$$

we deduce that v_s is necessarily of the form

$$(\underbrace{0, 0, \ldots, 0}_{s-1}, x_s, x_{s+1}, \ldots, x_\nu, \underbrace{0, \ldots, 0}_{s-1}, 1, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}),$$

where $x_s, x_{s+1}, \ldots, x_\nu, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu}$ can be arbitrary elements of $\mathbb{F}_q$. Hence there are $q^{2\nu-2s+1}$ possible choices of v_s.

Thus it follows from Lemmas 3.13 and 3.14 that

$$\begin{aligned} n(2s, s; 2\nu) &= n(s, 0; 2\nu)\, q^{2\nu-s} q^{2\nu-s-1} \ldots q^{2\nu-2s+1} \\ &= n(s, 0; 2\nu)\, q^{(2\nu-s)s-\frac{s(s-1)}{2}} \\ &= q^{(2\nu-s)s} \prod_{i=\nu-s+1}^{\nu} (q^{2i}-1). \end{aligned}$$

□

The following theorem of Dickson on the orders of the symplectic groups over $\mathbb{F}_q$ is a special case of Lemma 3.15.

Theorem 3.16.

$$\mid Sp_{2\nu}(\mathbb{F}_q) \mid = q^{\nu^2} \prod_{i=1}^{\nu} (q^{2i}-1).$$

Proof. In fact, $\mid Sp_{2\nu}(\mathbb{F}_q) \mid = n(2\nu, \nu; 2\nu)$. □

Lemma 3.17. Let $2s \le m \le \nu + s$. Then

$$n(m, s; 2\nu) = q^{(2\nu-s)s+\frac{(m-2s)(m-2s-1)}{2}} \prod_{i=\nu+s-m+1}^{\nu} (q^{2i}-1).$$

Proof. $n(m, s; 2\nu)$ is the number of $m \times 2\nu$ matrices P of rank m such that $PK\,{}^tP = M(m, s)$. Write

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{matrix} 2s \\ m-2s \end{matrix},$$

then

$$\begin{aligned} P_1 K\, {}^tP_1 &= \begin{pmatrix} 0 & I^{(s)} \\ -I^{(s)} & 0 \end{pmatrix}, \\ P_1 K\, {}^tP_2 &= 0, \end{aligned}$$

and

$$P_2 K\, {}^tP_2 = 0^{(m-2s)}.$$

Thus there are $n(2s, s; 2\nu)$ possible choices of P_1. Once P_1 has been chosen, by Lemma 3.13, the number of possible choices of P_2 is independent of the particular choice of P_1. Thus we may choose

$$P_1 = \begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & I^{(s)} & 0 \end{pmatrix} \begin{matrix} s \\ s \end{matrix} \end{array}$$

and then compute the number of possible choices of P_2. From $P_1 K\, {}^tP_2 = 0$ we deduce that P_2 is necessarily of the form

$$P_2 = \begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s \end{matrix} \\ \begin{pmatrix} 0 & P_{21} & 0 & P_{22} \end{pmatrix} \end{array}.$$

Then from $P_2 K\, {}^tP_2 = 0$ we deduce

$$(P_{21} P_{22}) \begin{pmatrix} 0 & I^{(\nu-s)} \\ -I^{(\nu-s)} & 0 \end{pmatrix} {}^t(P_{21} P_{22}) = 0.$$

Consequently, there are $n(m-2s, 0; 2(\nu-s))$ possible choices of P_2. Thus our lemma follows from Lemmas 3.13, 3.14, and 3.15. □

From Lemmas 3.12, 3.17, and Theorems 1.6, 3.16 we deduce the following Anzahl theorem in the symplectic geometry over finite fields.

Theorem 3.18. Let $2s \le m \le \nu + s$. Then the number of subspaces of type (m, s) in the 2ν-dimensional symplectic space over $\mathbb{F}_q$ is given by

$$N(m, s; 2\nu) = q^{2s(\nu+s-m)} \frac{\displaystyle\prod_{i=\nu+s-m+1}^{\nu} (q^{2i}-1)}{\displaystyle\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s} (q^i - 1)}.$$ □

Corollary 3.19. Let $m \le \nu$. Then the number of m-dimensional totally isotropic subspaces in the 2ν-dimensional symplectic space over $\mathbb{F}_q$ is

$$N(m, 0; 2\nu) = \frac{\displaystyle\prod_{i=\nu-m+1}^{\nu} (q^{2i}-1)}{\displaystyle\prod_{i=1}^{m}(q^i - 1)}.$$ □

Corollary 3.20. Let $s \leq \nu$. Then the number of $2s$-dimensional non-isotropic subspaces in the 2ν-dimensional symplectic space over $\mathbb{F}_q$ is

$$N(2s, s; 2\nu) = q^{2s(\nu-s)} \frac{\prod_{i=\nu-s+1}^{\nu} (q^{2i} - 1)}{\prod_{i=1}^{s} (q^{2i} - 1)}.$$ □

3.3 Singular Symplectic Geometry over Finite Fields

We would also like to compute the number of subspaces of type (m_1, s_1) contained in a given subspace of type (m, s) in the 2ν-dimensional symplectic space over $\mathbb{F}_q$. This leads us naturally to introduce the singular symplectic spaces over $\mathbb{F}_q$.

Now let

$$K_l = \begin{pmatrix} K & \\ & 0^{(l)} \end{pmatrix},$$

where K is the $2\nu \times 2\nu$ alternate matrix (3.1) as before. The set of all $(2\nu + l) \times (2\nu + l)$ nonsingular matrices T over $\mathbb{F}_q$ satisfying

$$T K_l {}^t T = K_l$$

forms a group, called the *singular symplectic group* of degree $2\nu + l$ and index ν over $\mathbb{F}_q$ and denoted by $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$. For $l = 0$, it reduces to the ordinary symplectic group $Sp_{2\nu,\nu}(\mathbb{F}_q) = Sp_{2\nu}(\mathbb{F}_q)$. It can be readily verified that $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$ consists of all $(2\nu + l) \times (2\nu + l)$ nonsingular matrices of the form

$$T = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} 2\nu \\ l \end{matrix} \end{array},$$

where $T_{11} K {}^t T_{11} = K$ and T_{22} is nonsingular. At first, let us compute the order of $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$.

Theorem 3.21.

$$| \, Sp_{2\nu+l,\nu}(\mathbb{F}_q) \, | = q^{\nu^2 + 2\nu l + \frac{l(l-1)}{2}} \prod_{i=1}^{\nu} (q^{2i} - 1) \prod_{i=1}^{l} (q^i - 1).$$

Proof. In fact,

$$\mid Sp_{2\nu+l,\nu}(\mathbb{F}_q) \mid = \mid Sp_{2\nu}(\mathbb{F}_q) \mid\mid GL_l(\mathbb{F}_q) \mid q^{2\nu l}.$$

Hence our theorem follows from Theorems 1.6 and 3.16. □

Let $\mathbb{F}_q^{(2\nu+l)}$ be the $(2\nu+l)$-dimensional vector space of row vectors over $\mathbb{F}_q$. We have an action of $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$ on $\mathbb{F}_q^{(2\nu+l)}$ defined as follows:

$$\mathbb{F}_q^{(2\nu+l)} \times Sp_{2\nu+l,\nu}(\mathbb{F}_q) \to \mathbb{F}_q^{(2\nu+l)}$$
$$((x_1, x_2, \ldots, x_{2\nu+l}),\ T) \mapsto (x_1, x_2, \ldots, x_{2\nu+l})T.$$

The vector space $\mathbb{F}_q^{(2\nu+l)}$ together with this action of $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$ is called the *singular symplectic space* over $\mathbb{F}_q$. First we study how the subspaces of $\mathbb{F}_q^{(2\nu+l)}$ are subdivided into orbits under the singular symplectic group $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$.

An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+l)}$ is said to be of *type* (m, s), if $PK_l\,{}^tP$ is cogredient to $M(m, s)$ (cf. (3.6)). Denote the set of all subspaces of type (m, s) in $\mathbb{F}_q^{(2\nu+l)}$ by $\mathcal{M}(m, s; 2\nu+l, \nu)$ and let

$$N(m, s; 2\nu+l, \nu) = \mid \mathcal{M}(m, s; 2\nu+l, \nu) \mid .$$

For the ordinary symplectic geometry, i.e., the case $l = 0$, $\mathcal{M}(m, s; 2\nu + 0, \nu)$ is simply $\mathcal{M}(m, s; 2\nu)$ which was studied previously in the last two sections.

Let e_i $(1 \le i \le 2\nu+l)$ be the row vector in $\mathbb{F}_q^{(2\nu+l)}$ whose i-th coordinate is 1 and all other coordinates are 0. Denote by E the l-dimensional subspace of $\mathbb{F}_q^{(2\nu+l)}$ generated by $e_{2\nu+1}, e_{2\nu+2}, \ldots, e_{2\nu+l}$. An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+l)}$ is called a *subspace of type* (m, s, k) if

(i) $PK_l\,{}^tP$ is cogredient to $M(m, s)$, and

(ii) $\dim(P \cap E) = k$.

Denote the set of all subspaces of type (m, s, k) in $\mathbb{F}_q^{(2\nu+l)}$ by $\mathcal{M}(m, s, k; 2\nu+l, \nu)$, then we prove

Theorem 3.22. $\mathcal{M}(m, s, k; 2\nu+l, \nu)$ is non-empty if and only if

$$k \le l \ \text{ and } \ 2s \le m - k \le \nu + s, \tag{3.11}$$

and if and only if

$$\max\{0, m-\nu-s\} \le k \le \min\{l, m-2s\}. \tag{3.12}$$

Moreover, if $\mathcal{M}(m, s, k; 2\nu+l, \nu)$ is non-empty, then it forms an orbit of subspaces under $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$.

Proof. That conditions (3.11) and (3.12) are equivalent is obvious.

Suppose that condition (3.11) hold. Then it is easy to verify that

$$\begin{array}{c} \begin{array}{cccccccc} s & m-k-2s & \nu+s-m+k & s & m-k-2s & \nu+s-m+k & k & l-k \end{array} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & I^{(m-k-2s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I^{(k)} & 0 \end{pmatrix} \end{array}$$

is a subspace of type (m, s, k) in the $(2\nu + l)$-dimensional singular symplectic space $\mathbb{F}_q^{(2\nu+l)}$. This proves the sufficiency of condition (3.11).

Conversely, suppose that P is a subspace of type (m, s, k) in the $(2\nu + l)$-dimensional singular symplectic space $\mathbb{F}_q^{(2\nu+l)}$. Since $P \cap E \subseteq E$ and $\dim(P \cap E) = k$, we have $k \leq l$. Moreover, we may take

$$P = \begin{array}{c} \begin{array}{cc} 2\nu & l \end{array} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \begin{array}{c} m-k \\ k \end{array} \end{array}$$

to be a matrix representation of the subspace P, where

$$\begin{aligned} \text{rank } P_{11} &= m-k, \\ \text{rank } P_{22} &= k, \end{aligned}$$

and

$$P_{11} K\, {}^t P_{11} = M(m-k, s).$$

By Theorem 3.6, we have

$$2s \leq m-k \leq \nu+s.$$

This proves the necessity of condition (3.11).

Now let Q be another subspace of type (m, s, k). Similarly, we may take

$$Q = \begin{array}{c} \begin{array}{cc} 2\nu & l \end{array} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \begin{array}{c} m-k \\ k \end{array} \end{array}$$

to be a matrix representation of the subspace Q, where

$$\begin{aligned} \text{rank } Q_{11} &= m-k, \\ \text{rank } Q_{22} &= k, \end{aligned}$$

and

$$Q_{11}K\,{}^tQ_{11} = M(m-k,s).$$

By Lemma 3.11 there is an element $T_{11} \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_{11} = Q_{11}T_{11}$. On the other hand, both P_{22} and Q_{22} are subspaces of dimension k in $\mathbb{F}_q^{(l)}$ and hence by Theorem 1.1 there is a nonsingular matrix $T_{22} \in GL_l(\mathbb{F}_q)$ such that $P_{22} = Q_{22}T_{22}$. But

$$T = \begin{pmatrix} T_{11} & 0 \\ 0 & T_{22} \end{pmatrix} \in Sp_{2\nu+l,\nu}(\mathbb{F}_q)$$

and

$$QT = \begin{pmatrix} P_{11} & Q_{12}T_{22} \\ 0 & P_{22} \end{pmatrix}.$$

Since P_{11} is an $(m-k) \times 2\nu$ matrix of rank $m-k$, we can find a $2\nu \times l$ matrix X such that

$$P_{11}X = P_{12} - Q_{12}T_{22}.$$

Let

$$T_1 = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{matrix} 2\nu \\ l \end{matrix} \end{array},$$

then $T_1 \in Sp_{2\nu+l,\nu}(\mathbb{F}_q)$ and $QTT_1 = P$. The proof of Theorem 3.22 is now completed. □

Corollary 3.23. $\mathcal{M}(m,s;2\nu+l,\nu)$ is non-empty if and only if

$$\max\{0, m-\nu-s\} \le \min\{l, m-2s\}.$$

Moreover, $\mathcal{M}(m,s;2\nu+l,\nu)$ is the disjoint union of the orbits $\mathcal{M}(m,s,k;2\nu+l,\nu)$ under $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$, where k satisfies (3.12). □

Now let

$$N(m,s,k;2\nu+l,\nu) = \mid \mathcal{M}(m,s,k;2\nu+l,\nu) \mid .$$

It follows from Theorem 3.22 that $N(m,s,k;2\nu+l,\nu) = 0$ whenever condition (3.11) does not hold. However, if condition (3.11) holds, then we have

Theorem 3.24. Let

$$\max\{0, m-\nu-s\} \le k \le \min\{l, m-2s\}.$$

Then

$$N(m,s,k;2\nu+l,\nu) = q^{2s(\nu+s-m+k)+(m-k)(l-k)} \frac{\prod_{i=\nu+s-m+k+1}^{\nu}(q^{2i}-1)\prod_{i=l-k+1}^{l}(q^i-1)}{\prod_{i=1}^{s}(q^{2i}-1)\prod_{i=1}^{m-k-2s}(q^i-1)\prod_{i=1}^{k}(q^i-1)}. \tag{3.13}$$

Proof. By the proof of Theorem 3.22 a subspace P of type (m,s,k) in the $(2\nu+l)$-dimensional singular symplectic space over $\mathbb{F}_q$ can be represented by an $m \times (2\nu+l)$ matrix P of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \end{array} \begin{matrix} m-k \\ k \end{matrix},$$

where

$$\begin{aligned} &\text{rank } P_{11} = m-k, \\ &\text{rank } P_{22} = k, \end{aligned}$$

and

$$P_{11} K\, {}^tP_{11} = M(m-k,s).$$

Denote by $n(m,s,k;2\nu+l,\nu)$ the number of such matrices. Then

$$n(m,s,k;2\nu+l,\nu) = n(m-k,s;2\nu)q^{(m-k)l}n(k,l), \tag{3.14}$$

where $n(m-k,s;2\nu)$ denotes the number of $(m-k)\times 2\nu$ matrices P_{11} of rank $m-k$, which satisfy $P_{11}K\,{}^tP_{11} = M(m-k,s)$, and $n(k,l)$ denotes the number of $k \times l$ matrices of rank k.

Now suppose that two $m \times (2\nu+l)$ matrices

$$P = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \end{array} \begin{matrix} m-k \\ k \end{matrix} \quad \text{and} \quad \mathrm{Q} = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \end{array} \begin{matrix} m-k \\ k \end{matrix},$$

where

$$\begin{aligned} &\text{rank } P_{11} = \text{ rank } Q_{11} = m-k, \\ &\text{rank } P_{22} = \text{ rank } Q_{22} = k, \end{aligned}$$

and

$$P_{11}K\,{}^tP_{11} = Q_{11}K\,{}^tQ_{11} = M(m-k,s),$$

represent the same subspace. Then there is an $m \times m$ nonsingular matrix T such that $P = TQ$. Thus T is necessarily of the form

$$T = \begin{array}{c} \begin{matrix} m-k & \quad k \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array}, \tag{3.15}$$

where both T_{11} and T_{22} are nonsingular. From $P_{11}K\,{}^tP_{11} = Q_{11}K\,{}^tQ_{11} = M(m-k,s)$ and $P_{11} = T_{11}Q_{11}$, it follows that

$$T_{11}M(m-k,s)\,{}^tT_{11} = M(m-k,s). \tag{3.16}$$

The set of $(m-k)\times(m-k)$ nonsingular matrices T_{11} satisfying (3.16) is the singular symplectic group $Sp_{m-k,s}(\mathbb{F}_q)$. All $m \times m$ nonsingular matrices T of the form (3.15) in which T_{11} satisfies (3.16) form a group, denoted by $\mathcal{G}$. Then

$$\mid \mathcal{G} \mid = \mid Sp_{m-k,s}(\mathbb{F}_q) \mid\mid GL_k(\mathbb{F}_q) \mid q^{(m-k)k}. \tag{3.17}$$

Moreover, if $TQ = Q$ for $T \in \mathcal{G}$, then we have necessarily $T = I$. Consequently,

$$n(m,s,k;2\nu+l,\nu) = \mid \mathcal{G} \mid N(m,s,k;2\nu+l,\nu). \tag{3.18}$$

From (3.14), (3.17), and (3.18), we deduce

$$N(m,s,k;2\nu+l,\nu) = \frac{n(m-k,s;2\nu)q^{(m-k)(l-k)}n(k,l)}{\mid Sp_{m-k,s}(\mathbb{F}_q) \mid\mid GL_k(\mathbb{F}_q) \mid}.$$

By the proof of Lemma 3.12

$$n(m-k,s;2\nu) = \mid Sp_{m-k,s}(\mathbb{F}_q) \mid N(m-k,s;2\nu),$$

where $N(m-k,s;2\nu)$ denotes the number of subspaces of $\mathbb{F}_q^{(2\nu)}$ which are of type $(m-k,s)$. Thus

$$N(m,s,k;2\nu+l,\nu) = \frac{N(m-k,s;2\nu)}{\mid GL_k(\mathbb{F}_q) \mid} q^{(m-k)(l-k)}n(k,l).$$

The explicit formulas of $N(m-k,s;2\nu)$, $\mid GL_k(\mathbb{F}_q) \mid$, and $n(k,l)$ can be obtained from Theorem 3.18, Theorem 1.6, and Lemma 1.5, respectively. Substituting these formulas into the above formula, we obtain (3.13). □

Corollary 3.25. Let $\max\{0, m-\nu-s\} \leq \min\{l, m-2s\}$. Then

$$N(m, s; 2\nu+l, \nu) = \sum_{k=\max\{0,m-\nu-s\}}^{\min\{l,m-2s\}} N(m, s, k; 2\nu+l, \nu),$$

where $N(m, s, k; 2\nu+l, \nu)$ is given by (3.13). □

Now we shall apply the Anzahl theorem given above to obtain an Anzahl theorem in the ordinary symplectic geometry.

Let $\mathbb{F}_q^{(2\nu)}$ be the 2ν-dimensional row space on which the ordinary symplectic group $Sp_{2\nu}(\mathbb{F}_q)$ acts. Let P be a fixed subspace of type (m, s) in $\mathbb{F}_q^{(2\nu)}$. Denote by $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ the set of subspaces of type (m_1, s_1) contained in P. Let

$$N(m_1, s_1; m, s; 2\nu) = \mid \mathcal{M}(m_1, s_1; m, s; 2\nu) \mid .$$

By Theorem 3.7 $N(m_1, s_1; m, s; 2\nu)$ is independent of the particular choice of the subspace P of type (m, s). We have

Theorem 3.26. $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ is non-empty if and only if

$$\left.\begin{array}{r} 2s \leq m \leq \nu+s, \\ 2s_1 \leq m_1 \leq \nu+s_1, \\ \text{and} \\ 0 \leq s-s_1 \leq m-m_1, \end{array}\right\} \tag{3.19}$$

and these three conditions are equivalent to

$$\left.\begin{array}{r} 2s \leq m \leq \nu+s \\ \text{and} \\ \max\{0, m_1-s-s_1\} \leq \min\{m-2s, m_1-2s_1\}. \end{array}\right\} \tag{3.20}$$

□

Proof. The equivalence of (3.19) and (3.20) can be proved easily and will be left as an exercise.

Now suppose that (3.19) and (3.20) hold. Then subspaces of type (m, s) and of type (m_1, s_1) exist in $\mathbb{F}_q^{(2\nu)}$. Let P be a subspace of type (m, s) and assume that P is represented by an $m \times 2\nu$ matrix P of rank m such that

$$PK\,{}^tP = M(m, s).$$

It follows from Corollary 3.23 that the second condition in (3.20) implies that $\mathcal{M}(m_1, s_1; 2s + (m - 2s), s)$ is non-empty. Let P_1 be an element of $\mathcal{M}(m_1, s_1; 2s+(m-2s), s)$ and assume that P_1 is represented by an $m_1 \times m$ matrix P_1 of rank m_1 such that

$$P_1 M(m, s)\, {}^t P_1 = M(m_1, s_1).$$

Then $P_1 P$ is an $m_1 \times 2\nu$ matrix of rank m_1 such that

$$(P_1 P) K\, {}^t(P_1 P) = M(m_1, s_1),$$

i.e., $P_1 P$ is a subspace of type (m_1, s_1) of $\mathbb{F}_q^{(2\nu)}$. Clearly, $P_1 P$ as a subspace of $\mathbb{F}_q^{(2\nu)}$ is contained in the subspace P. Hence $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ is non-empty.

Conversely, suppose that $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ is non-empty. Then both subspaces of type (m, s) and of type (m_1, s_1) exist in $\mathbb{F}_q^{(2\nu)}$. Therefore the first two conditions in (3.19) hold. We choose a fixed subspace P of type (m, s) such that

$$P K\, {}^t P = M(m, s).$$

Let P_1 be the subspace of type (m_1, s_1) contained in P. P_1, being a subspace of P, can be represented by an $m_1 \times m$ matrix P_1 of rank m_1, and then P_1, being a subspace of $\mathbb{F}_q^{(2\nu)}$, is represented by the $m_1 \times 2\nu$ matrix $P_1 P$. We may assume

$$P_1 P K\, {}^t P\, {}^t P_1 = M(m_1, s_1),$$

i.e.,

$$P_1 M(m, s)\, {}^t P_1 = M(m_1, s_1).$$

Therefore the second condition in (3.20) follows from Corollary 3.23. □

Theorem 3.27. Let

$$2s \leq m \leq \nu + s$$

and

$$\max\{0, m_1 - s - s_1\} \leq \min\{m - 2s, m_1 - 2s_1\}.$$

Then

$$N(m_1, s_1; m, s; 2\nu) = \sum_{k=\max\{0, m_1-s-s_1\}}^{\min\{m-2s, m_1-2s_1\}} q^{2s_1(s+s_1-m_1+k)+(m_1-k)(m-2s-k)} \times \frac{\prod_{i=s+s_1-m_1+k+1}^{s}(q^{2i}-1)\prod_{i=m-2s-k+1}^{m-2s}(q^i-1)}{\prod_{i=1}^{s_1}(q^{2i}-1)\prod_{i=1}^{m_1-2s_1-k}(q^i-1)\prod_{i=1}^{k}(q^i-1)}.$$

Proof. From the proof of Theorem 3.26 it follows that

$$N(m_1, s_1; m, s; 2\nu) = N(m_1, s_1; 2s + (m-2s), s).$$

Hence Theorem 3.27 follows immediately from Corollary 3.25 and Theorem 3.24. □

We remark that knowing $N(m_1, s_1; m, s; 2\nu)$ we can compute easily the number $N'(m_1, s_1; m, s; 2\nu)$ of subspaces of type (m, s) containing a given subspace of type (m_1, s_1) as we did in Corollary 1.9.

Now let us study the similar problem in the singular symplectic geometry. We consider again the $(2\nu + l)$-dimensional row space $\mathbb{F}_q^{(2\nu+l)}$ over $\mathbb{F}_q$ on which the singular symplectic group $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$ acts. Let P be a subspace of type (m, s, k) of $\mathbb{F}_q^{(2\nu+l)}$. Denote by $\mathcal{M}(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu)$ the set of subspaces of type (m_1, s_1, k_1) contained in P. Let

$$N(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu) = \mid \mathcal{M}(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu) \mid,$$

which by Theorem 3.22 is independent of the particular choice of the subspace P of type (m, s, k). At first we prove

Theorem 3.28. $\mathcal{M}(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu)$ is non-empty if and only if

$$\left.\begin{aligned} k_1 \le k \le l, \\ 2s \le m - k \le \nu + s, \\ 2s_1 \le m_1 - k_1 \le \nu + s_1, \\ 0 \le s - s_1 \le (m-k) - (m_1 - k_1), \end{aligned}\right\} \quad (3.21)$$

and

and these four conditions are equivalent to

$$\left.\begin{array}{c} k_1 \le k \le l, \\ 2s \le m-k \le \nu+s, \\ \text{and} \\ \max\{0, m_1-k_1-s-s_1\} \le \min\{m-k-2s, m_1-k_1-2s_1\}. \end{array}\right\} \tag{3.22}$$

Proof. The equivalence of (3.21) and (3.22) can be proved easily and the details are omitted.

Now suppose that (3.21) holds. By Theorem 3.22 subspaces of types (m,s,k) and (m_1,s_1,k_1) exist in $\mathbb{F}_q^{(2\nu+l)}$. Let P be a subspace of type (m,s,k). We can assume that P has a matrix representation, call it P again, of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & \quad l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array}, \tag{3.23}$$

where

$$\left.\begin{array}{l} \operatorname{rank} P_{11} = m-k, \\ \operatorname{rank} P_{22} = k, \\ \text{and} \\ P_{11} K\, {}^tP_{11} = M(m-k,s). \end{array}\right\} \tag{3.24}$$

It follows from Corollary 3.23 that the last condition in (3.22) implies that $\mathcal{M}(m_1-k_1, s_1; 2s+(m-k-2s), s)$ is non-empty. Let Q_{11} be a subspace contained in this set and assume that Q_{11} is represented by an $(m_1-k_1)\times(m-k)$ matrix of rank m_1-k_1 such that

$$Q_{11} M(m-k,s)\, {}^tQ_{11} = M(m_1-k_1, s_1).$$

Let Q_{22} be a $k_1 \times k$ matrix of rank k_1. Then it is easy to verify that

$$\begin{pmatrix} Q_{11} & \\ & Q_{22} \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ & P_{22} \end{pmatrix}$$

is a subspace of type (m_1,s_1,k_1) in $\mathbb{F}_q^{(2\nu+l)}$ and contained in P. Hence $\mathcal{M}(m_1,s_1,k_1;m,s,k;2\nu+l,\nu)$ is non-empty.

Conversely, suppose that $\mathcal{M}(m_1,s_1,k_1;m,s,k;2\nu+l,\nu)$ is non-empty. Then there is subspace Q of type (m_1,s_1,k_1) contained in a subspace P of type (m,s,k). $Q \subset P$ implies $Q\cap E \subset P\cap E$, and it follows that

$$k_1 = \dim(Q\cap E) \le \dim(P\cap E) = k.$$

Thus the first condition in (3.22) is obtained. From the existence of subspaces of type (m, s, k) the second condition in (3.22) follows. As in the above paragraph we can assume that P is of the form (3.23) and that (3.24) holds. Clearly, P_{22} is a matrix representation of $P \cap E$. From $Q \cap E \subset P \cap E$ it follows that there is a $k_1 \times k$ matrix Q_{22} of rank k_1 such that $Q \cap E$, being a subspace of E, is represented by the $k_1 \times l$ matrix $Q_{22}P_{22}$. Then Q can be represented by an $m_1 \times (2\nu + l)$ matrix of the form

$$Q = \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix}, \tag{3.25}$$

where Q_{11} is an $(m_1 - k_1) \times (m - k)$ matrix of rank $m_1 - k_1$. We have

$$QK_l\,{}^tQ = \begin{pmatrix} Q_{11}M(m-k,s)\,{}^tQ_{11} & \\ & 0^{(k_1)} \end{pmatrix}.$$

Since Q is of type (m_1, s_1, k_1), we can assume that

$$Q_{11}M(m-k,s)\,{}^tQ_{11} = M(m_1 - k_1, s_1). \tag{3.26}$$

Then the last condition in (3.22) follows from Corollary 3.23. □

Theorem 3.29. Assume that (3.22) holds. Then

$$\begin{aligned} &N(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu) \\ &\quad = N(m_1 - k_1, s_1; 2s + (m - k - 2s), s)N(k_1, k)q^{(m_1-k_1)(k-k_1)}, \end{aligned} \tag{3.27}$$

where $N(m_1 - k_1, s_1; 2s + (m - k - 2s), s)$ and $N(k_1, k)$ are given in Corollary 3.25 and Theorem 1.7, respectively. □

Proof. Let P be a subspace of type (m, s, k) of $\mathbb{F}_q^{(2\nu+l)}$. We can assume that P has a matrix representation of the form (3.23)

$$P = \begin{array}{c} \begin{matrix} 2\nu & l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array}$$

and that (3.24) holds. Let Q be a subspace of type (m_1, s_1, k_1) and contained in P. By the proof of Theorem 3.28 we can assume that Q has a matrix representation of the form (3.25)

$$Q = \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix},$$

where Q_{11} is an $(m_1-k_1)\times(m-k)$ matrix of rank m_1-k_1 over $\mathbb{F}_q$ such that (3.26)

$$Q_{11}M(m-k,s)\,{}^tQ_{11}=M(m_1-k_1,s_1)$$

holds and Q_{22} is a $k_1\times k$ matrix of rank k_1 over $\mathbb{F}_q$. Denote the number of $(m_1-k_1)\times(m-k)$ matrix Q_{11} satisfying (3.26) by $n(m_1-k_1,s_1;m-k,s)$, then

$$\begin{aligned}&n(m_1-k_1,s_1;m-k,s)\\&\quad=\mid Sp_{2s_1+(m_1-k_1-2s_1),s_1}(\mathbb{F}_q)\mid N(m_1-k_1,s_1;2s+(m-k-2s),s).\end{aligned}\tag{3.28}$$

Moreover, we have

$$n(k_1,k)=GL_{k_1}(\mathbb{F}_q)N(k_1,k),\tag{3.29}$$

where $n(k_1,k)$ is the number of $k_1\times k$ matrices of rank k_1 over $\mathbb{F}_q$, and $N(k_1,k)$ is the number of k_1-dimensional subspace contained in a k-dimensional vector space over $\mathbb{F}_q$. Denote by $n(m_1,s_1,k_1;m,s,k;2\nu+l,\nu)$ the number of $m_1\times m$ matrices of the form

$$\begin{array}{c}\begin{matrix}m-k & k\end{matrix}\\ \begin{pmatrix}Q_{11} & Q_{12}\\ 0 & Q_{22}\end{pmatrix}\begin{matrix}m_1-k_1\\ k_1\end{matrix}\end{array},$$

where Q_{11} is an $(m_1-k_1)\times(m-k)$ matrix of rank m_1-k_1 and satisfying (3.26) and Q_{22} is a $k_1\times k$ matrix of rank k_1. Then

$$\begin{aligned}&n(m_1,s_1,k_1;m,s,k;2\nu+l,\nu)\\&\quad=n(m_1-k_1,s_1;m-k,s)q^{(m_1-k_1)k}n(k_1,k).\end{aligned}\tag{3.30}$$

Now suppose that two $m_1\times m$ matrices

$$Q=\begin{pmatrix}Q_{11} & Q_{12}\\ 0 & Q_{22}\end{pmatrix}\qquad\text{and}\qquad R=\begin{pmatrix}R_{11} & R_{12}\\ 0 & R_{22}\end{pmatrix},$$

where Q_{11} and R_{11} are $(m_1-k_1)\times(m-k)$ matrices of rank m_1-k_1 such that

$$Q_{11}M(m-k,s)\,{}^tQ_{11}=R_{11}M(m-k,s)\,{}^tR_{11}=M(m_1-k_1,s_1),\tag{3.31}$$

and Q_{22} and R_{22} are $k_1\times k$ matrices of rank k_1, represent the same subspace. Then there is an $m_1\times m_1$ nonsingular matrix T such that $Q=TR$. Thus T is necessarily of the form

$$T=\begin{array}{c}\begin{matrix}m_1-k_1 & k_1\end{matrix}\\ \begin{pmatrix}T_{11} & T_{12}\\ 0 & T_{22}\end{pmatrix}\begin{matrix}m_1-k_1\\ k_1\end{matrix}\end{array},\tag{3.32}$$

where both T_{11} and T_{22} are nonsingular. $Q = TR$ implies, in particular,

$$Q_{11} = T_{11}R_{11}. \tag{3.33}$$

From (3.31) and (3.33) we deduce

$$T_{11}M(m_1 - k_1, s_1)\,{}^tT_{11} = M(m_1 - k_1, s_1). \tag{3.34}$$

The set of $(m_1 - k_1) \times (m_1 - k_1)$ nonsingular matrices T_{11} satisfying (3.34) is the singular symplectic group $Sp_{2s_1+(m_1-k_1-2s_1),s_1}(\mathbb{F}_q)$. All $m_1 \times m_1$ nonsingular matrices T of the form (3.32) in which T_{11} satisfies (3.34) form a group, denoted by $\mathcal{G}$. Clearly,

$$|\ \mathcal{G}\ | = |\ Sp_{2s_1+(m_1-k_1-2s_1),s_1}(\mathbb{F}_q)\ ||\ GL_{k_1}(\mathbb{F}_q)\ |\ q^{(m_1-k_1)k_1}. \tag{3.35}$$

Moreover, if $TQ = Q$ for $T \in \mathcal{G}$, then we have necessarily $T = I$. Consequently,

$$n(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu) = |\ \mathcal{G}\ |\ N(m_1, s_1, k_1; m, s, k; 2\nu + l, \nu). \tag{3.36}$$

Equating the right-hand sides of (3.30) and (3.36), substituting (3.28), (3.29) and (3.35) into the equation thus obtained, and then cancelling $|\ Sp_{2s_1+(m_1-k_1-2s_1),s_1}(\mathbb{F}_q)\ |$ and $|\ GL_{k_1}(\mathbb{F}_q)\ |$, we obtain (3.27). □

We remark also that knowing $N(m_1, s_1, k_1; m, s, k; 2\nu)$ we can compute the number $N'(m_1, s_1, k_1; m, s, k; 2\nu)$ of subspaces of type (m, s, k) containing a given subspace of type (m_1, s_1, k_1) as we did in Corollary 1.9.

Finally as another application of Theorem 3.21 let us compute the number of $n \times n$ alternate matrices of a given rank over a finite field.

Denote by $\mathcal{K}(n)$ the set of all $n \times n$ alternate matrices over $\mathbb{F}_q$. Define an action of $GL_n(\mathbb{F}_q)$ on $\mathcal{K}(n)$ as follows:

$$\begin{aligned} \mathcal{K}(n) \times GL_n(\mathbb{F}_q) &\to \mathcal{K}(n) \\ (K, \quad T) &\mapsto {}^tTKT. \end{aligned}$$

Then $\mathcal{K}(n)$ is subdivided into orbits under the action of $GL_n(\mathbb{F}_q)$. By Theorem 3.1 each orbit is characterized by the rank of the matrices in it, and the rank of an alternate matrix is necessarily even. The orbit of all $n \times n$ alternate matrices of rank 2ν $(0 \le 2\nu \le n)$ will be denoted by $\mathcal{K}(n, \nu)$. Then we have

Theorem 3.30. Let $0 \leq 2\nu \leq n$. Then

$$| \mathcal{K}(n,\nu) | = q^{\nu(\nu-1)} \frac{\prod_{i=n-2\nu+1}^{n} (q^i - 1)}{\prod_{i=1}^{\nu} (q^{2i} - 1)}.$$

Proof. Let

$$K_{n-2\nu} = \begin{pmatrix} K & \\ & 0^{(n-2\nu)} \end{pmatrix},$$

where K is the $2\nu \times 2\nu$ nonsingular alternate matrix (3.1). Clearly the stabilizer of $K_{n-2\nu}$ in $GL_n(\mathbb{F}_q)$ is the singular symplectic group $Sp_{n,\nu}(\mathbb{F}_q)$, and

$$| \mathcal{K}(n,\nu) | = GL_n(\mathbb{F}_q) : Sp_{n,\nu}(\mathbb{F}_q).$$

Hence our theorem follows from Theorems 1.6 and 3.21. □

Corollary 3.31. The number of $2\nu \times 2\nu$ nonsingular alternate matrices over $\mathbb{F}_q$ is

$$q^{\nu(\nu-1)} \prod_{i=1}^{\nu} (q^{2i-1} - 1). \qquad \square$$

3.4 Symplectic Polarities and Polar Spaces Arising from Them

From now on we change our language to the projective geometry language. For any $n > 0$ the n-dimensional projective space $PG(n, \mathbb{F}_q)$ arises from $\mathbb{F}_q^{(n+1)}$ by regarding the 1-dimensional vector subspaces as points and the $(r+1)$-dimensional vector subspaces as (projective) r-flats $(0 \leq r \leq n)$ as we did in Section 2.1. In order to acquire a better geometric background we begin with the study of collineations and correlations in $PG(n, \mathbb{F}_q)$.

A bijective map from the set of flats in $PG(n, \mathbb{F}_q)$ to itself

$$\tau : P \mapsto \tau(P)$$

is called a *collineation* of $PG(n, \mathbb{F}_q)$, if for any two flats P and Q

$$P \subset Q \Rightarrow \tau(P) \subset \tau(Q)$$

and for any flat P

$$\dim \tau(P) = \dim P.$$

Clearly, projective transformations of $PG(n, \mathbb{F}_q)$ introduced in Section 2.1 induce collineations of $PG(n, \mathbb{F}_q)$. More generally, let $x \mapsto \tilde{x}$ be an automorphism of $\mathbb{F}_q$ and let $T \in GL_{n+1}(\mathbb{F}_q)$, then the map

$$\tau : (x_0, x_1, \ldots, x_n) \mapsto (\tilde{x}_0, \tilde{x}_1, \ldots, \tilde{x}_n)T$$

is a well-defined bijective map from the set of points of $PG(n, \mathbb{F}_q)$ to itself and induces a collineation of $PG(n, \mathbb{F}_q)$ in a natural way: an r-flat P is mapped into the r-flat $\tilde{P}T$, where P also denotes a matrix representation of the $(r+1)$-dimensional vector subspace of $\mathbb{F}_q^{(n+1)}$ corresponding to the r-flat P and $\tilde{P}$ denotes the matrix obtained from P by applying the automorphism $x \mapsto \tilde{x}$ of $\mathbb{F}_q$ to all the elements of P.

Similarly, a bijective map from the set of flats in $PG(n, \mathbb{F}_q)$ to itself

$$\sigma : P \mapsto \sigma(P)$$

is called a *correlation* of $PG(n, \mathbb{F}_q)$, if for any two flats P and Q,

$$P \subset Q \Rightarrow \sigma(P) \supset \sigma(Q)$$

and for any flat P,

$$\dim P + \dim \sigma(P) = n - 1.$$

Clearly, a correlation of $PG(n, \mathbb{F}_q)$ carries points into hyperplanes and hyperplanes into points.

Let us agree on representing the hyperplane in $PG(n, \mathbb{F}_q)$ with equation

$$a_0x_0 + a_1x_1 + \cdots + a_nx_n = 0,$$

where $a_0, a_1, \ldots, a_n$ are elements of $\mathbb{F}_q$ and are not all zero, by the non-zero column vector

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix}$$

(in contrast with the non-zero row vectors representing the points). Then two non-zero column vectors

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{pmatrix}$$

represent the same hyperplane if and only if there is a $\lambda \in \mathbb{F}_q^*$ such that $a_i = \lambda b_i$ for all $i = 0, 1, \ldots, n$. Let $x \mapsto \tilde{x}$ be an automorphism of $\mathbb{F}_q$ and $T \in GL_{n+1}(\mathbb{F}_q)$, then

$$\sigma : (x_0, x_1, \ldots, x_n) \mapsto T \begin{pmatrix} \tilde{x}_0 \\ \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix}$$

is a well-defined bijective map from the set of points of $PG(n, \mathbb{F}_q)$ to the set of its hyperplanes. Let P be any r-flat ($0 \leq r \leq n$) and we use the same letter P to denote a matrix representation of the $(r+1)$-dimensional vector subspace of $\mathbb{F}_q^{(n+1)}$ corresponding to the flat P. We write

$$P = \begin{pmatrix} a_{00} & a_{01} & \ldots & a_{0n} \\ a_{10} & a_{11} & \ldots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r0} & a_{r1} & \ldots & a_{rn} \end{pmatrix}$$

and then define

$$\sigma(P) = \bigcap_{i=0}^{r} T \begin{pmatrix} \tilde{a}_{i0} \\ \tilde{a}_{i1} \\ \vdots \\ \tilde{a}_{in} \end{pmatrix},$$

where the right-hand side denotes the intersection of the hyperplanes

$$\sigma(a_{i0}, a_{i1}, \ldots, a_{in}) = T \begin{pmatrix} \tilde{a}_{i0} \\ \tilde{a}_{i1} \\ \vdots \\ \tilde{a}_{in} \end{pmatrix}, \ i = 0, 1, \ldots, r.$$

We have to prove that σ is well-defined on every r-flat. In fact, let

$$W = \bigcap_{i=0}^{r} T \begin{pmatrix} \tilde{a}_{i0} \\ \tilde{a}_{i1} \\ \vdots \\ \tilde{a}_{in} \end{pmatrix},$$

then a point $(x_0, x_1, \ldots, x_n) \in W$ if and only if

$$(x_0, x_1, \ldots, x_n)T \begin{pmatrix} \tilde{a}_{i0} \\ \tilde{a}_{i1} \\ \vdots \\ \tilde{a}_{in} \end{pmatrix} = 0, \ i = 0, 1, \ldots, r,$$

and, hence, if and only if

$$(x_0, x_1, \ldots, x_n)T\,{}^t\tilde{P} = 0. \tag{3.37}$$

Consequently, $\sigma(P)$ is independent of the particular matrix representation of the vector subspace corresponding to the flat P, i.e., $\sigma(P)$ is well-defined. Furthermore, if $\sigma(P) = \sigma(Q)$, i.e., for any point $(x_0, x_1, \ldots, x_n)$ (3.37) holds if and only if

$$(x_0, x_1, \ldots, x_n)T\,{}^t\tilde{Q} = 0$$

holds. By the theory of linear homogeneous equations, $\tilde{P}\,{}^tT$ and $\tilde{Q}\,{}^tT$ are different matrix representations of the same vector subspace, so are P and Q. Thus $P = Q$ as flats. This proves that σ is injective. Hence σ is a bijective map from the set of flats of $PG(n, \mathbb{F}_q)$ to itself. It is clear from the definition of σ that for any flat P, $\dim P + \dim \sigma(P) = n - 1$ and for any two flats P and Q, $P \subset Q$ implies $\sigma(P) \supset \sigma(Q)$. Therefore σ is a correlation of $PG(n, \mathbb{F}_q)$.

Now let us express the image of a hyperplane

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix}$$

under the above defined correlation σ in terms of the hyperplane. We know that $\sigma({}^t(a_0, a_1, \ldots, a_n))$ is a point. Let

$$\sigma \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = (b_0, b_1, \ldots, b_n).$$

For any point $(x_0, x_1, \ldots, x_n)$ lying on the hyperplane ${}^t(a_0, a_1, \ldots, a_n)$, i.e.,

$$(x_0, x_1, \ldots, x_n) \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = 0, \tag{3.38}$$

the hyperplane $\sigma(x_0, x_1, \ldots, x_n) = T\,{}^t(\tilde{x}_0, \tilde{x}_1, \ldots, \tilde{x}_n)$ should contain the point $\sigma({}^t(a_0, a_1, \ldots, a_n)) = (b_0, b_1, \ldots, b_n)$, i.e.,

$$(b_0, b_1, \ldots, b_n)T \begin{pmatrix} \tilde{x}_0 \\ \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix} = 0.$$

But (3.38) is equivalent to

$$(\tilde{a}_0, \tilde{a}_1, \ldots, \tilde{a}_n) \begin{pmatrix} \tilde{x}_0 \\ \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{pmatrix} = 0.$$

Consequently, there is a $\lambda \in \mathbb{F}_q^*$ such that

$$(b_0, b_1, \ldots, b_n)T = \lambda(\tilde{a}_0, \tilde{a}_1, \ldots, \tilde{a}_n),$$

which implies

$$(b_0, b_1, \ldots, b_n) = \lambda(\tilde{a}_0, \tilde{a}_1, \ldots, \tilde{a}_n)T^{-1}.$$

Hence,

$$\sigma : \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto (\tilde{a}_0, \tilde{a}_1, \ldots, \tilde{a}_n)T^{-1}.$$

A correlation of order 2 is called a *polarity*. Let σ be a polarity of $PG(n, \mathbb{F}_q)$. For any point $(x_0, x_1, \ldots, x_n)$ we call the hyperplane $\sigma(x_0, x_1, \ldots, x_n)$ the *polar* of the point $(x_0, x_1, \ldots, x_n)$ and for any hyperplane ${}^t(a_0, a_1, \ldots, a_n)$ we call the point $\sigma({}^t(a_0, a_1, \ldots, a_n))$ the *pole* of the hyperplane ${}^t(a_0, a_1, \ldots, a_n)$. It is clear that the hyperplane ${}^t(a_0, a_1, \ldots, a_n)$ is

the polar of the point $(x_0, x_1, \ldots, x_n)$ if and only if the point $(x_0, x_1, \ldots, x_n)$ is the pole of the hyperplane ${}^t(a_0, a_1, \ldots, a_n)$. A flat P of $PG(n, \mathbb{F}_q)$ is called an *absolute* of a polarity σ, if $P \subset \sigma(P)$ or $P \supset \sigma(P)$.

After the foregoing preparation let us now come to the main theme of the present section.

Let us assume that $n = 2\nu - 1$ and consider the $(2\nu - 1)$-dimensional projective space $PG(2\nu - 1, \mathbb{F}_q)$ arising from the vector space $\mathbb{F}_q^{(2\nu)}$. Let K be a $2\nu \times 2\nu$ nonsingular alternate matrix over $\mathbb{F}_q$. Let P be an r-flat of $PG(2\nu-1, \mathbb{F}_q)$ and the $(r+1)$-dimensional subspace of $\mathbb{F}_q^{(2\nu)}$ corresponding to P will also be denoted by P. Then the dual subspace of P,

$$P^\perp = \{x \in \mathbb{F}_q^{(2\nu)} \mid xK\,{}^tP = 0\},$$

is a $(2\nu-r-1)$-dimensional vector subspace of $\mathbb{F}_q^{(2\nu)}$ and the corresponding flat, again denoted by $P^\perp$, is a $(2\nu-r-2)$-flat of $PG(2\nu-1, \mathbb{F}_q)$. We have $\dim P + \dim P^\perp = 2\nu - 2$. Moreover, $P \subset Q$ implies $P^\perp \supset Q^\perp$. Therefore the map

$$P \mapsto P^\perp$$

from the set of flats of $PG(2\nu - 1, \mathbb{F}_q)$ to itself is a correlation of $PG(2\nu - 1, \mathbb{F}_q)$. Since

$$(P^\perp)^\perp = P,$$

this correlation is a polarity, called the *symplectic polarity* of $PG(2\nu-1, \mathbb{F}_q)$ defined by the $2\nu \times 2\nu$ nonsingular alternate matrix K. It is easy to see that this symplectic polarity can also be regarded as the map induced by the bijective map

$$(x_1, x_2, \ldots, x_{2\nu}) \mapsto K \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{2\nu} \end{pmatrix}$$

from the set of points of $PG(2\nu - 1, \mathbb{F}_q)$ to the set of its hyperplanes.

Denote by $\mathcal{P}(K)$ the set of absolutes of dimension $\leq \nu - 1$ of the symplectic polarity of $PG(2\nu - 1, \mathbb{F}_q)$ defined by K and call it the *polar space* arising from the symplectic polarity of $PG(2\nu - 1, \mathbb{F}_q)$ defined by K. The object of our present study is $\mathcal{P}(K)$.

An m-flat P of $PG(2\nu-1,\mathbb{F}_q)$ is called *totally isotropic* with respect to K, if its corresponding vector subspace, denoted also by P, of dimension $m+1$ is totally isotropic with respect to K, i.e., $PK\,{}^tP=0$. We know from Corollary 3.8 that maximal totally isotropic flats of $PG(2\nu-1,\mathbb{F}_q)$ are of dimension $\nu-1$. It follows from Theorem 3.4 that $\mathcal{P}(K)$ is the set of all totally isotropic flats with respect to K in $PG(2\nu-1,\mathbb{F}_q)$.

Let K_1 and K_2 be two $2\nu\times 2\nu$ nonsingular alternate matrices over $\mathbb{F}_q$. By Corollary 3.2 there is a $2\nu\times 2\nu$ nonsingular matrix A over $\mathbb{F}_q$ such that $K_1=AK_2\,{}^tA$. It can be readily verified that the map

$$\begin{aligned}\mathcal{P}(K_1)&\to\mathcal{P}(K_2)\\ P&\mapsto PA\end{aligned}$$

is a bijection such that $\dim PA=\dim P$ for any flat P in $\mathcal{P}(K_1)$ and that for any two flats P and Q in $\mathcal{P}(K_1)$, $P\supset Q$ implies $PA\supset QA$. Therefore in studying properties concerning dimension and inclusion between elements in polar spaces arising from symplectic polarities, without loss of generality we can choose a particular $2\nu\times 2\nu$ nonsingular alternate matrix K and study $\mathcal{P}(K)$. From now on we take K to be the $2\nu\times 2\nu$ nonsingular alternate matrix (3.1).

Theorem 3.32. The set $\mathcal{P}(K)$ of totally isotropic flats of $PG(2\nu-1,\mathbb{F}_q)$ has the following properties:

(i) $\nu-1$ is the maximal dimension of totally isotropic flats. For any $r(-1\le r\le\nu-1)$, every totally isotropic r-flat together with the s-flats $(0\le s\le r)$ it contains forms an r-dimensional projective space isomorphic[2] to $PG(r,\mathbb{F}_q)$.

(ii) The intersection of any two totally isotropic flats is a totally isotropic flat.

(iii) Given a totally isotropic $(\nu-1)$-flat P and a point $p\notin P$, there exists a unique totally isotropic flat Q containing p such that $\dim(P\cap Q)=\nu-2$; Q contains all those points of P, each of which generates a totally isotropic 1-flat with p.

(iv) There exist two disjoint totally isotropic $(\nu-1)$-flats.

Proof. (i) By Corollary 3.8, $\nu-1$ is the maximal dimension of totally isotropic flats. Now let P be any totally isotropic r-flat. If $r=-1$

[2]For the definition of isomorphism of projective spaces, see Section A.6.

or 0, there is nothing to prove[3]. Suppose that $r \geq 1$. Since all flats of $PG(2\nu-1, \mathbb{F}_q)$ contained in P are totally isotropic, it follows from Theorem A.18 and Corollary A.23 that P together with all the totally isotropic s-flats $(0 \leq s \leq r)$ it contains as its flats forms a finite dimensional projective space P. It remains to prove that P is isomorphic to $PG(r, \mathbb{F}_q)$.

Let $v_0, v_1, \ldots, v_r$ be a basis of the $(r+1)$-dimensional vector subspace corresponding to the r-flat P and p be any point of P. Then p corresponds to a 1-dimensional subspace. Let v be any non-zero vector in this 1-dimensional subspace. Then v can be expressed as

$$v = p_0 v_0 + p_1 v_1 + \cdots + p_r v_r.$$

Define a map

$$\begin{aligned} P &\to PG(r, \mathbb{F}_q) \\ p &\mapsto (p_0, p_1, \ldots, p_r). \end{aligned}$$

It can be readily verified that the map is well-defined and is an isomorphism of projective spaces.

(ii) Let P and Q be two totally isotropic flats. By Theorem 3.4, $P \subset P^\perp$ and $Q \subset Q^\perp$. Hence,

$$P \cap Q \subset P \subset P^\perp \subset (P \cap Q)^\perp.$$

Again by Theorem 3.4, $P \cap Q$ is also totally isotropic.

(iii) Let a totally isotropic $(\nu-1)$-flat P and a point $p \notin P$ be given. Then $P \not\subset p^\perp$; in fact, if $P \subset p^\perp$, then $P = P^\perp \supset (p^\perp)^\perp = p$, a contradiction. But $\dim p^\perp = 2\nu - 2$, hence, $P \cup p^\perp = PG(2\nu-1, \mathbb{F}_q)$. It follows from the dimension formula of the projective space, i.e., Theorem 2.4, that $\dim(P \cap p^\perp) = \nu - 2$. Let $Q = (P \cap p^\perp) \cup p$, then Q is a totally isotropic $(\nu-1)$-flat containing p such that $\dim(P \cap Q) = \nu - 2$ and that Q contains all those points of P, each of which generates a totally isotropic line with p.

It remains to prove the uniqueness of Q. Let Q be any totally isotropic flat containing p such that $\dim(P \cap Q) = \nu - 2$. From $\dim(P \cap Q) = \nu - 2$, $p \notin P$, and $p \in Q$, we deduce $\dim Q \geq \nu - 1$. But since Q is

[3]We adopt the convention that by definition a (-1)-dimensional projective space is an empty set, a 0-dimensional projective space consists of a single point, whose flats are the point itself and the empty set, and a 1-dimensional projective space is an arbitrary set of cardinality at least 2, whose flats are the arbitrary set itself (the unique line), its subsets with a single element (the points), and the empty set.

totally isotropic, we must have $\dim Q = \nu - 1$. Since Q contains p, we have necessarily $Q = (P \cap Q) \cup p$ and $P \cap Q = P \cap p^{\perp}$. Consequently, $Q = (P \cap p^{\perp}) \cup p$.

(iv) The following two totally isotropic $(\nu - 1)$-flats

$$\begin{pmatrix} I^{(\nu)} & 0^{(\nu)} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0^{(\nu)} & I^{(\nu)} \end{pmatrix}$$

are clearly disjoint. □

Now we introduce the abstract definition of the polar space as follows:

A *polar space* $\mathcal{P}$ of rank n, where n is an integer ≥ 1, is defined to be a set S of points together with some distinguished subsets called subspaces such that

PS1 A subspace together with the subspaces it contains forms an i-dimensional projective space with $-1 \leq i \leq n - 1$.

PS2 The intersection of any two subspaces is a subspace.

PS3 Given a subspace P of dimension $n - 1$ and a point $p \in S \backslash P$ there exists a unique subspace Q containing p such that $\dim (P \cap Q) = n - 2$; Q contains all those points of P such that each of them is contained in a 1-dimensional subspace together with p.

PS4 There exist two disjoint subspaces of dimension $n - 1$.

PS1, PS2, PS3, and PS4 are called the axioms of polar spaces.

Then Theorem 3.32 can be restated simply as follows:

Theorem 3.33. The set $\mathcal{P}(K)$ of totally isotropic flats of $PG(2\nu - 1, \mathbb{F}_q)$ with respect to the $2\nu \times 2\nu$ nonsingular alternate matrix (3.1) forms a polar space of rank ν. □

We have called $\mathcal{P}(K)$ the polar space arising from a symplectic polarity. We shall give other examples of polar spaces in later chapters.

Polar spaces of rank 2 are also called *generalized quadrangles*. Let us now analyze how the axioms of the polar space look like in the case of generalized quadrangles. PS1 now says that besides the (-1)-dimensional subspace, which is the empty set, there are only 0-dimensional subspaces, which are points, and 1-dimensional subspaces, which will be called lines in the sequel; each line contains at least two points. PS2 says that any

two lines intersect in at most one point; in other words, any two points are contained in at most one line. PS3 says that given a line l and a point p not on l there exists a unique line l' containing p and intersecting l in a point. Finally, PS4 says that there exist two non-intersecting lines. From the above analysis, we have

Theorem 3.34. A set of points in which some proper subsets are distinguished and called lines is a generalized quadrangle if and only if the following properties hold:

GQ1 Any line has at least two distinct points.

GQ2 Any two distinct points are on at most one line.

GQ3 Given a line and a point $p \notin l$, there exists a unique line l' containing p and intersecting l in a point.

GQ4 There exist two non-intersecting lines. □

GQ1, GQ2, GQ3, and GQ4 are usually called the axioms of generalized quadrangles.

As a special case of Theorem 3.33 we have the following example of generalized quadrangles.

Corollary 3.35. The set of points of $PG(3, \mathbb{F}_q)$ together with the set of totally isotropic lines with respect to the 4×4 nonsingular alternate matrix

$$\begin{pmatrix} 0 & I^{(2)} \\ -I^{(2)} & 0 \end{pmatrix}$$

forms a generalized quadrangle. □

The generalized quadrangle given in Corollary 3.35 is usually denoted by $W(q)$. Clearly all lines of $W(q)$ contain the same number of points and it follows from Theorem 3.7 that all points are on the same number of lines. Denote by $s+1$ the number of points lying on any line and by $t+1$ the number of lines passing through any point. Then s and t are called the parameters of the generalized quadrangle.

We have

Corollary 3.36. The generalized quadrangle $W(q)$ has parameters

$$s = q, \quad t = q.$$

Proof. Since every line of $W(q)$ is a line of $PG(3, \mathbb{F}_q)$, it contains $q+1$ points and we have $s = q$. By Theorem 3.7, to determine t we need only to determine the number of lines of $W(q)$ passing through the point $(1,0,0,0)$. We can assume that any line of $PG(3, \mathbb{F}_q)$ containing the point $(1,0,0,0)$ is of the form

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x_2 & x_3 & x_4 \end{pmatrix},$$

where x_2, x_3, and x_4 are not all zero. If it is a line of $W(q)$, then it must be totally isotropic, i.e., we must have $x_3 = 0$. Therefore $t+1 = (q^2-1)/(q-1) = q+1$ and consequently $t = q$. □

Finally we shall use the symplectic polarity to deduce a closed formula for the number of subspaces of type (m,s) in $\mathbb{F}_q^{(2\nu)}$ containing a given subspace of type (m_1, s_1). Here we use the vector space language again. Let P_1 be a given subspace of type (m_1, s_1) in $\mathbb{F}_q^{(2\nu)}$. Denote the set of subspaces of type (m,s) in $\mathbb{F}_q^{(2\nu)}$ containing P_1 by $\mathcal{M}'(m_1, s_1; m, s; 2\nu)$. Let

$$N'(m_1, s_1; m, s; 2\nu) = \mid \mathcal{M}'(m_1, s_1; m, s; 2\nu) \mid .$$

By Theorem 3.7, $N'(m_1, s_1; m, s; 2\nu)$ is independent of the particular choice of the subspace P_1 of type (m_1, s_1) in $\mathbb{F}_q^{(2\nu)}$. At first, we have

Theorem 3.37. $\mathcal{M}'(m_1, s_1; m, s; 2\nu)$ is non-empty if and only if (3.19) holds.

Proof. Suppose that $\mathcal{M}'(m_1, s_1; m, s; 2\nu)$ is non-empty. Then we can assume that for a given subspace P_1 of type (m_1, s_1) in $\mathbb{F}_q^{(2\nu)}$ there is a subspace P of type (m,s) in $\mathbb{F}_q^{(2\nu)}$ such that $P \supset P_1$. From Theorem 3.7 it follows that given a subspace Q of type (m,s) there is a subspace Q_1 of type (m_1, s_1) such that $Q_1 \subset Q$. Hence $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ is non-empty. It follows from Theorem 3.26 that (3.19) holds.

Now, suppose that (3.19) holds. By Theorem 3.26, $\mathcal{M}(m_1, s_1; m, s; 2\nu)$ is non-empty. That is, given any subspace P of type (m,s) in $\mathbb{F}_q^{(2\nu)}$ there is a subspace P_1 of type (m_1, s_1) in $\mathbb{F}_q^{(2\nu)}$ such that $P \supset P_1$. By Theorem 3.7 given a subspace Q_1 of type (m_1, s_1) there is a subspace Q of type (m,s) such that $Q_1 \subset Q$. This proves that $\mathcal{M}'(m_1, s_1; m, s; 2\nu)$ is non-empty. □

Theorem 3.38. Let (3.19) hold. Then

$$N'(m_1, s_1; m, s; 2\nu) = \sum_{k=\max\{0, m_1-s-s_1\}}^{\min\{m-2s, m_1-2s_1\}} q^{2(\nu+s-m)(s+s_1-m_1+k)+(2\nu-m-k)(m_1-2s_1-k)}$$

$$\times \frac{\prod_{i=s+s_1-m_1+k+1}^{\nu+s_1-m_1}(q^{2i}-1)\prod_{i=m_1-2s_1-k+1}^{m_1-2s_1}(q^i-1)}{\prod_{i=1}^{\nu+s-m}(q^{2i}-1)\prod_{i=1}^{m-2s-k}(q^i-1)\prod_{i=1}^{k}(q^i-1)}.$$

Proof. Let P_1 be a subspace of type (m_1, s_1) in $\mathbb{F}_q^{(2\nu)}$. By Corollary 3.9, $P_1^\perp$ is a subspace of type $(2\nu - m_1, \nu + s_1 - m_1)$ in $\mathbb{F}_q^{(2\nu)}$. If P is a subspace of type (m, s) in $\mathbb{F}_q^{(2\nu)}$ containing P_1, then $P^\perp$ is a subspace of type $(2\nu - m, \nu + s - m)$ contained in $P_1^\perp$. Therefore,

$$N'(m_1, s_1; m, s; 2\nu) = N(2\nu - m, \nu + s - m; 2\nu - m_1, \nu + s_1 - m_1; 2\nu).$$

Hence our theorem follows immediately from Theorem 3.27. □

3.5 Comments

Symplectic groups over finite fields were first studied by L. E. Dickson, in particular, Theorem 3.16 is due to him, cf. Dickson 1900. J. Dieudonné then studied symplectic groups over any field, cf. Dieudonné 1948. All the results of Section 3.1 hold not only for finite fields but also for any field and the same proofs apply. Theorem 3.1 is well known. Theorem 3.7, the transitivity theorem in the symplectic geometry, is due to J. Dieudonné, cf. Dieudonné 1948, and is the analogy of the Witt's theorem in the orthogonal geometry over any field of characteristic $\neq 2$, cf. Witt 1937. Lemma 3.5 is due to the author, from which we deduce not only Theorem 3.7 but also Theorems 3.6, 3.10, and Lemma 3.11, which are also due to the author, cf. Wan 1991c. Lemma 3.11 can be regarded as an improvement of Theorem 3.7 and is used in proving the Anzahl theorems in Section 3.2.

All the results of Section 3.2, except Theorem 3.16, are due to the author, cf. Wan 1964, 1965, 1995. Corollary 3.19 was also obtained independently by V. Pless, cf. Pless 1965.

All the results of Section 3.3, except Theorem 3.30, are due to the author, cf. Wan 1992e and 1992f. Theorem 3.30 was obtained by L. Carlitz when the characteristic is greater than 2 and by Y. Wang for the general case, cf. Carlitz 1954b and Wang 1980.

The presentation of polarities in $PG(n, \mathbb{F}_q)$ in Section 3.4 is standard. The definitions of the polar space and the generalized quadrangle are due to J. Tits; for the polar space, cf. Tits 1974, and for the generalized quadrangle, cf. Tits 1959. All the theorems and corollaries of this section, except

Theorems 3.37 and 3.38, are due to Tits. Of course, we can define the polar space arising from the symplectic polarity in $PG(2\nu-1,\mathbb{F}_q)$ directly as the set of all totally isotropic flats with respect to K in $PG(2\nu-1,\mathbb{F}_q)$ as we shall do in Section 7.4 for the orthogonal geometry over any finite field of characteristic 2, but the present approach reveals some geometric background.

3.6 Exercises

3.1 Check that all the results of Section 3.1 hold for any commutative field.

3.2 Prove that $Sp_2(\mathbb{F}_q)$ consists of all 2×2 matrices of determinant 1 over $\mathbb{F}_q$.

3.3 Determine the center of $Sp_{2\nu}(\mathbb{F}_q)$.

3.4 Determine the stabilizer of the following totally isotropic subspace of dimension s

$$\begin{array}{cccccc} s & \nu-s & s & \nu-s & & \\ (\ I^{(s)} & 0 & 0 & 0 &) & , \end{array}$$

where $s\le\nu$, and compute the index of the stabilizer in $Sp_{2\nu}(\mathbb{F}_q)$, which gives us again $N(s,0;2\nu)$.

3.5 Compute the parameters of the authentication codes constructed in Examples 3.1 and 3.2, and also P_I and P_S, assuming that there is an equally distributed probability distribution on the set of encoding rules.

3.6 Construct more authentication codes from geometry of symplectic groups over finite fields.

3.7 Prove the equivalence of (3.19) and (3.20).

3.8 Compute the number of orbits of subspaces of the $(2\nu+l)$-dimensional singular symplectic space over $\mathbb{F}_q$ under $Sp_{2\nu+l,\nu}(\mathbb{F}_q)$.

3.9 Let $2s\le m\le\nu+s$, $2s_1\le m_1\le\nu+s_1$, and P and Q be subspaces of types (m,s) and (m_1,s_1), respectively, in the 2ν-dimensional symplectic space over $\mathbb{F}_q$. Assume that $P\supseteq Q$. Prove that there is a matrix representation P of the subspace P such that the first m_1 rows of the matrix P span Q and

$$PK\,{}^tP=\begin{pmatrix} 0 & I^{(s_1)} & & & & & & & \\ -I^{(s_1)} & 0 & & & & & & & \\ & & 0 & & I^{(s_2)} & & & & \\ & & & 0^{(\sigma_1)} & & & & & \\ & & -I^{(s_2)} & & 0 & & & & \\ & & & & & 0 & I^{(s_3)} & & \\ & & & & & -I^{(s_3)} & 0 & & \\ & & & & & & & 0^{(\sigma_2)} & \end{pmatrix}$$

where s_2 and s_3 are nonnegative integers, $\sigma_1 = m_1 - 2s_1 - s_2 \geq 0$, $\sigma_2 = m - m_1 - s_2 - 2s_3 \geq 0$, $s = s_1 + s_2 + s_3$, and $m - m_1 \geq s - s_1 \geq 0$.

Chapter 4
Geometry of Pseudo-symplectic Groups over Finite Fields of Characteristic 2

4.1 Pseudo-symplectic Groups over Finite Fields of Characteristic 2

In this chapter we assume that $\mathbb{F}_q$ is a finite field of characteristic 2.

Let S be an $n \times n$ nonsingular non-alternate symmetric matrix over $\mathbb{F}_q$. An $n \times n$ matrix T over $\mathbb{F}_q$ is said to be *pseudo-symplectic* with respect to S, if

$$TS\,{}^tT = S.$$

Clearly, $n \times n$ pseudo-symplectic matrices with respect to S are nonsingular and they form a group with respect to matrix multiplication, called the *pseudo-symplectic group* with respect to S over $\mathbb{F}_q$ and denoted by $Ps_n(\mathbb{F}_q, S)$.

Two $n \times n$ non-alternate symmetric matrices S_1 and S_2 over $\mathbb{F}_q$ are said to be *cogredient*, if there is an $n \times n$ nonsingular matrix P such that $S_1 = PS_2\,{}^tP$. Let S_1 and S_2 be two nonsingular non-alternate symmetric matrices and suppose that they are cogredient, i.e., $S_1 = PS_2\,{}^tP$ for some $n \times n$ nonsingular matrix T. Then we have an isomorphism

$$\begin{aligned} Ps_n(\mathbb{F}_q, S_1) &\to Ps_n(\mathbb{F}_q, S_2) \\ T &\mapsto P^{-1}TP. \end{aligned}$$

Therefore, it is sufficient to study the pseudo-symplectic groups with respect to the normal forms of nonsingular non-alternate symmetric matrices under cogredience.

Theorem 4.1. Let S be an $n \times n$ symmetric matrix over $\mathbb{F}_q$. Then S is cogredient to one and only one of the following three normal forms:

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 0^{(n-2\nu)} \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 1 & \\ & & & 0^{(n-2\nu-1)} \end{pmatrix},$$

and

$$\begin{pmatrix} 0 & I^{(\nu)} & & & \\ I^{(\nu)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 1 & \\ & & & & 0^{(n-2\nu-2)} \end{pmatrix},$$

which are of ranks 2ν, $2\nu+1$, and $2\nu+2$, respectively.

Proof. Suppose that S is of rank r. Then there is an $n \times n$ nonsingular matrix P such that

$$PS = \begin{pmatrix} S_1 \\ 0 \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix},$$

where S_1 is an $r \times n$ matrix of rank r. Then

$$PS^tP = \begin{matrix} \\ \end{matrix} \begin{matrix} r & n-r \end{matrix} \\ \begin{pmatrix} S_{11} & S_{12} \\ 0 & 0 \end{pmatrix} \begin{matrix} r \\ n-r \end{matrix}.$$

Since PS^tP is symmetric, $S_{12} = 0$ and S_{11} is an $r \times r$ nonsingular symmetric matrix. Thus it is sufficient to consider the cogredience of nonsingular symmetric matrices.

From now on suppose that S is an $n \times n$ nonsingular symmetric matrix. A vector $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}_q^{(n)}$ is said to be *isotropic* with respect to S, if $xS^tx = 0$. We use P to denote the set of all isotropic vectors in $\mathbb{F}_q^{(n)}$ with respect to S. If $x, y \in P$, i.e., $xS^tx = 0$, $yS^ty = 0$, then $(x+y)S^t(x+y) = 0$ and $(\lambda x)S^t(\lambda x) = 0$ for any $\lambda \in \mathbb{F}_q$. Hence, P is a subspace of $\mathbb{F}_q^{(n)}$. Suppose that $\dim P = m$. We may choose an $(n-m) \times n$ matrix Z such that

$$\begin{pmatrix} P \\ Z \end{pmatrix}$$

is an $n \times n$ nonsingular matrix. Then S is cogredient to

$$\begin{pmatrix} P \\ Z \end{pmatrix} S \, {}^t\!\begin{pmatrix} P \\ Z \end{pmatrix} = \begin{pmatrix} PS\,{}^tP & PS\,{}^tZ \\ ZS\,{}^tP & ZS\,{}^tZ \end{pmatrix}.$$

It is easy to see that all vectors in $\mathbb{F}_q^{(m)}$ are isotropic with respect to $PS\,{}^tP$, hence $PS\,{}^tP$ is alternate. Assume that rank $PS\,{}^tP = 2\nu$, then $PS\,{}^tP$ is cogredient to

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 0^{(m-2\nu)} \end{pmatrix}.$$

Thus we may assume that S is cogredient to a matrix of the form

$$\begin{array}{c} \begin{matrix} \nu & \nu & m-2\nu & n-m \end{matrix} \\ \begin{pmatrix} 0 & I & & A_1 \\ I & 0 & & A_2 \\ & & 0 & A_3 \\ {}^tA_1 & {}^tA_2 & {}^tA_3 & A_4 \end{pmatrix} \end{array} \begin{matrix} \nu \\ \nu \\ m-2\nu \\ n-m \end{matrix}.$$

Clearly, S is then cogredient to

$$\begin{array}{c} \begin{matrix} \nu & \nu & m-2\nu & n-m \end{matrix} \\ \begin{pmatrix} 0 & I & & 0 \\ I & 0 & & 0 \\ & & 0 & A_3 \\ 0 & 0 & {}^tA_3 & B \end{pmatrix} \end{array} \begin{matrix} \nu \\ \nu \\ m-2\nu \\ n-m \end{matrix}.$$

Since S is nonsingular, rank $A_3 = m - 2\nu$. Thus S is cogredient to

$$\begin{array}{c} \begin{matrix} \nu & \nu & m-2\nu & m-2\nu & n-2m+2\nu \end{matrix} \\ \begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & B_1 & B_2 \\ & & 0 & {}^tB_2 & B_3 \end{pmatrix} \end{array} \begin{matrix} \nu \\ \nu \\ m-2\nu \\ m-2\nu \\ n-2m+2\nu \end{matrix},$$

and then cogredient to

$$\begin{array}{c} \begin{matrix} \nu & \nu & m-2\nu & m-2\nu & n-2m+2\nu \end{matrix} \\ \begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & B_1 & 0 \\ & & 0 & 0 & B_3 \end{pmatrix} \end{array} \begin{matrix} \nu \\ \nu \\ m-2\nu \\ m-2\nu \\ n-2m+2\nu \end{matrix}.$$

Notice that both B_1 and B_3 are symmetric matrices. Let T be the $(m-2\nu)\times(m-2\nu)$ matrix obtained from B_1 by replacing all elements on and below the main diagonal by zeros. If we put

$$Q=\begin{array}{c}\begin{array}{ccccc}\nu & \nu & m-2\nu & m-2\nu & n-2m+2\nu\end{array}\\ \begin{pmatrix} I & & & & \\ & I & & & \\ & & I & & \\ & & T & I & \\ & & & & I \end{pmatrix}\end{array}\begin{array}{c}\nu\\ \nu\\ m-2\nu\\ m-2\nu\\ n-2m+2\nu\end{array},$$

then S is cogredient to

$$Q\begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & B_1 & 0 \\ & & 0 & 0 & B_3 \end{pmatrix}{}^tQ=\begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & C & 0 \\ & & 0 & 0 & B_3 \end{pmatrix},$$

where $C=[b_{11},b_{22},\ldots,b_{m-2\nu,m-2\nu}]$ is a diagonal matrix whose diagonal elements are $b_{11},b_{22},\ldots,b_{m-2\nu,m-2\nu}$ in succession. We assumed that the set of all isotropic vectors with respect to S forms an m-dimensional subspace, thus the set of all isotropic vectors with respect to any matrix cogredient to S also forms an m-dimensional subspace. Consequently, all the $b_{11},b_{22},\ldots,b_{m-2\nu,m-2\nu}$ are nonzero. Since $\mathbb{F}_q$ is a finite field of characteristic 2, every nonzero element of $\mathbb{F}_q$ is a square element of $\mathbb{F}_q$. Thus we may assume that S is cogredient to

$$\begin{array}{c}\begin{array}{ccccc}\nu & \nu & m-2\nu & m-2\nu & n-2m+2\nu\end{array}\\ \begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & I & 0 \\ & & 0 & 0 & B_3 \end{pmatrix}\end{array}\begin{array}{c}\nu\\ \nu\\ m-2\nu\\ m-2\nu\\ n-2m+2\nu\end{array}.$$

Clearly, there is no nonzero isotropic vectors with respect to B_3. It can be proved that B_3 is cogredient to the $(n-2m+2\nu)\times(n-2m+2\nu)$ identity matrix (Exercise 4.1). Consequently, S is cogredient to

$$\begin{array}{c}\begin{array}{ccccc}\nu & \nu & m-2\nu & m-2\nu & n-2m+2\nu\end{array}\\ \begin{pmatrix} 0 & I & & & \\ I & 0 & & & \\ & & 0 & I & 0 \\ & & I & I & 0 \\ & & 0 & 0 & I \end{pmatrix}\end{array}\begin{array}{c}\nu\\ \nu\\ m-2\nu\\ m-2\nu\\ n-2m+2\nu\end{array}.$$

Since all isotropic vectors of S form an m-dimensional subspace, there do not exist nonzero isotropic vectors with respect to the lower right corner $(n-m)\times(n-m)$ submatrix

$$\begin{array}{c} \begin{matrix} m-2\nu & n-2m+2\nu \end{matrix} \\ \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \begin{matrix} m-2\nu \\ n-2m+2\nu \end{matrix} \end{array}$$

of the above matrix. But it is clear that there exist nonzero isotropic vectors with respect to the $r\times r$ identity matrix $I^{(r)}$, unless $r=1$. Consequently, S is cogredient to one of the following three matrices:

$$\begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 1 \end{pmatrix}.$$

It remains to show that any two of the three normal forms of Theorem 4.1 are not cogredient to each other. Since the second one is of odd rank and the other two are of even rank, the second one cannot be cogredient to the other two. Though

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 0^{(n-2\nu)} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & I^{(\nu-1)} & & & \\ I^{(\nu-1)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 1 & \\ & & & & 0^{(n-2\nu-2)} \end{pmatrix}$$

are both of rank 2ν, but the first one is alternate and the second one is not, so they are not cogredient to each other either. □

Corollary 4.2. Let S be an $n\times n$ nonsingular non-alternate matrix. If $n=2\nu+1$ is odd, then S is cogredient to

$$S_1 = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix}, \tag{4.1}$$

and if $n=2\nu+2$ is even, then S is cogredient to

$$S_2 = \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 1 \end{pmatrix}. \tag{4.2}$$

□

In the following we shall study the pseudo-symplectic groups $Ps_{2\nu+1}(\mathbb{F}_q, S_1)$ and $Ps_{2\nu+2}(\mathbb{F}_q, S_2)$. We write simply $Ps_{2\nu+\delta}(\mathbb{F}_q)$ for $Ps_{2\nu+\delta}(\mathbb{F}_q, S_\delta)$, where $\delta = 1$ or 2. The group $Ps_{2\nu+\delta}(\mathbb{F}_q)$ acts on $\mathbb{F}_q^{(2\nu+\delta)}$ in the usual way:

$$\begin{aligned} \mathbb{F}_q^{(2\nu+\delta)} \times Ps_{2\nu+\delta}(\mathbb{F}_q) &\to \mathbb{F}_q^{(2\nu+\delta)} \\ ((x_1, x_2, \ldots, x_{2\nu+\delta}), T) &\mapsto (x_1, x_2, \ldots, x_{2\nu+\delta})T. \end{aligned}$$

The vector space $\mathbb{F}_q^{(2\nu+\delta)}$ together with this group action is called the *pseudo-symplectic space* over the finite field $\mathbb{F}_q$ of characteristic 2, and the *pseudo-symplectic geometry* aims at the study of properties of the pseudo-symplectic space which are invariant under the pseudo-symplectic group.

Let us first study the pseudo-symplectic group $Ps_{2\nu+1}(\mathbb{F}_q)$. Write

$$K = \begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix}.$$

Then S_1 can be written as

$$S_1 = \begin{pmatrix} K & \\ & 1 \end{pmatrix}.$$

Let $T_1 \in Ps_{2\nu+1}(\mathbb{F}_q, S_1)$ and write T_1 in the same block form as S_1:

$$T_1 = \begin{pmatrix} T & {}^t b \\ c & d \end{pmatrix},$$

where b and c are 2ν-dimensional vectors. Let $x = (x_1, x_2, \ldots, x_{2\nu+1}) \in \mathbb{F}_q^{(2\nu+1)}$. Then $xS_1\,{}^t x = x_{2\nu+1}^2$ and, hence, x is isotropic with respect to S_1 if and only if $x_{2\nu+1} = 0$. The image of an isotropic vector under any element of $Ps_{2\nu+1}(\mathbb{F}_q)$ is also isotropic, from which it follows that $(x_1, x_2, \ldots, x_{2\nu}, 0)T_1$ is isotropic for any $x_1, x_2, \ldots, x_{2\nu} \in \mathbb{F}_q$ and, hence, $b = 0$. Clearly the image of $e_{2\nu+1} = (0, 0, \ldots, 0, 1)$ under T_1 is (c, d). Thus $1 = e_{2\nu+1}S_1\,{}^t e_{2\nu+1} = (c, d)S_1\,{}^t(c, d) = d^2$ and, hence, $d = 1$. Since $(0, 0, \ldots, 0, x_{2\nu+1})$ for all $x_{2\nu+1} \in \mathbb{F}_q$ are all the vectors which are orthogonal to all the isotropic vectors of $\mathbb{F}_q^{(2\nu+1)}$, for any $x_{2\nu+1} \in \mathbb{F}_q$ there must be a $y_{2\nu+1} \in \mathbb{F}_q$ such that

$$(0, 0, \ldots, 0, x_{2\nu+1})T_1 = (0, 0, \ldots, 0, y_{2\nu+1}).$$

Consequently, $c = 0$. Therefore, T_1 is of the form

$$T_1 = \begin{pmatrix} T & 0 \\ 0 & 1 \end{pmatrix},$$

where $T \in Sp_{2\nu}(\mathbb{F}_q)$. Hence, $Ps_{2\nu+1}(\mathbb{F}_q) \simeq Sp_{2\nu}(\mathbb{F}_q)$. We have proved

Theorem 4.3. $Ps_{2\nu+1}(\mathbb{F}_q)$ consists of all matrices of the form

$$\begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} T & 0 \\ 0 & 1 \end{pmatrix} \begin{matrix} 2\nu \\ 1 \end{matrix} \end{array},$$

where $T \in Sp_{2\nu}(\mathbb{F}_q)$. Hence, $Ps_{2\nu+1}(\mathbb{F}_q) \simeq Sp_{2\nu}(\mathbb{F}_q)$ and

$$\mid Ps_{2\nu+1}(\mathbb{F}_q) \mid = q^{\nu^2} \prod_{i=1}^{\nu} (q^{2i} - 1).$$

□

Corollary 4.4. With respect to S_1, the vector $(x_1, x_2, \ldots, x_{2\nu+1})$ is non-isotropic if and only if $x_{2\nu+1} \neq 0$; in particular, $e_{2\nu+1}$ is non-isotropic. The dual subspace $\langle e_{2\nu+1} \rangle^{\perp}$ of $\langle e_{2\nu+1} \rangle$ with respect to S_1, i.e.,

$$\langle e_{2\nu+1} \rangle^{\perp} = \{x \in \mathbb{F}_q^{(2\nu+1)} \mid x S_1 \, {}^t e_{2\nu+1} = 0\},$$

is the 2ν-dimensional subspace $\langle e_1, e_2, \ldots, e_{2\nu} \rangle$ on which $Ps_{2\nu+1}(\mathbb{F}_q)$ acts as the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$ does. Moreover, only $e_{2\nu+1}$ and its multiples are left fixed by every element of $Ps_{2\nu+1}(\mathbb{F}_q)$. □

Now let us study the group $Ps_{2\nu+2}(\mathbb{F}_q)$. Write

$$S_2 = \begin{pmatrix} K & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}.$$

Let $T_2 \in Ps_{2\nu+2}(\mathbb{F}_q)$ and write T_2 in the same block form as S_2:

$$T_2 = \begin{pmatrix} T & {}^t b_1 & {}^t b_2 \\ c_1 & d_{11} & d_{12} \\ c_2 & d_{21} & d_{22} \end{pmatrix},$$

where b_1, b_2, c_1, c_2 are 2ν-dimensional row vectors and $d_{11}, d_{12}, d_{21}, d_{22} \in \mathbb{F}_q$. Let $x = (x_1, x_2, \ldots, x_{2\nu+2}) \in \mathbb{F}_q^{(2\nu+2)}$. Then $x S_2 \, {}^t x = x_{2\nu+2}^2$. Hence, x is isotropic with respect to S_2 if and only if $x_{2\nu+2} = 0$. Since the image of an isotropic vector under T_2 is also isotropic and $(x_1, x_2, \ldots, x_{2\nu+1}, 0)$ is isotropic for any $x_1, x_2, \ldots, x_{2\nu+1} \in \mathbb{F}_q$, it follows that $b_2 = 0$ and $d_{12} = 0$. Notice that the image of $e_{2\nu+2}$ under T_2 is (c_2, d_{21}, d_{22}), thus, $1 = e_{2\nu+2} S_2 \, {}^t e_{2\nu+2} = (c_2, d_{21}, d_{22}) S_2 \, {}^t (c_2, d_{21}, d_{22}) = d_{22}^2$ and, hence, $d_{22} =$

1. From $T_2 S_2 {}^t T_2 = S_2$ we deduce that $T \in Sp_{2\nu}(\mathbb{F}_q)$, $c_1 = 0$, $d_{11} = 1$, and $TK {}^t c_2 + {}^t b_1 = 0$. Therefore,

$$T_2 = \begin{pmatrix} T & {}^t b_1 & 0 \\ 0 & 1 & 0 \\ c_2 & d_{21} & 1 \end{pmatrix},$$

where $T \in Sp_{2\nu}(\mathbb{F}_q)$, $TK {}^t c_2 + {}^t b_1 = 0$, and d_{21} is an arbitrary element of $\mathbb{F}_q$. It is easy to check that the map

$$T_2 = \begin{pmatrix} T & {}^t b_1 & 0 \\ 0 & 1 & 0 \\ c_2 & d_{21} & 1 \end{pmatrix} \mapsto T$$

is a homomorphism from $Ps_{2\nu+2}(\mathbb{F}_q)$ onto $Sp_{2\nu}(\mathbb{F}_q)$ and the kernel $\mathcal{K}_1$ of this homomorphism consists of all matrices of the form

$$\begin{matrix} 2\nu & 1 & 1 \\ \end{matrix}$$
$$\begin{pmatrix} I & {}^t b_1 & 0 \\ 0 & 1 & 0 \\ c_2 & d_{21} & 1 \end{pmatrix} \begin{matrix} 2\nu \\ 1 \\ 1 \end{matrix},$$

where $K {}^t c_2 + {}^t b_1 = 0$. Moreover, the map

$$\begin{pmatrix} I & {}^t b_1 & 0 \\ 0 & 1 & 0 \\ c_2 & d_{21} & 1 \end{pmatrix} \mapsto c_2$$

is a homomorphism from $\mathcal{K}_1$ onto the abelian group of $\mathbb{F}_q^{(2\nu)}$ and the kernel $\mathcal{K}_2$ of this homomorphism consists of all matrices of the form

$$\begin{pmatrix} I & 0 & 0 \\ 0 & 1 & 0 \\ 0 & d_{21} & 1 \end{pmatrix}.$$

Clearly, $| \mathbb{F}_q^{(2\nu)} | = q^{2\nu}$, $| \mathcal{K}_2 | = q$ and, hence, $| \mathcal{K}_1 | = q^{2\nu+1}$. We have proved

Theorem 4.5. *$Ps_{2\nu+2}(\mathbb{F}_q)$ consists of all matrices of the form*

$$\begin{matrix} 2\nu & 1 & 1 \\ \end{matrix}$$
$$\begin{pmatrix} T & {}^t b & 0 \\ 0 & 1 & 0 \\ c & d & 1 \end{pmatrix} \begin{matrix} 2\nu \\ 1 \\ 1 \end{matrix},$$

where $T \in Sp_{2\nu}(\mathbb{F}_q)$, $TK\,{}^tc + {}^tb = 0$, and d is an arbitrary element of $\mathbb{F}_q$. Furthermore, $Ps_{2\nu+2}(\mathbb{F}_q)$ has a normal series

$$Ps_{2\nu+2}(\mathbb{F}_q) \supset \mathcal{K}_1 \supset \mathcal{K}_2 \supset \{I\},$$

where $\mathcal{K}_1$ and $\mathcal{K}_2$ consist of elements of $Ps_{2\nu+2}(\mathbb{F}_q)$ which are of the form

$$\begin{pmatrix} I & {}^tb & 0 \\ 0 & 1 & 0 \\ c & d & 1 \end{pmatrix},$$

where $K\,{}^tc + {}^tb = 0$, and of the form

$$\begin{pmatrix} I & 0 & 0 \\ 0 & 1 & 0 \\ 0 & d & 1 \end{pmatrix},$$

respectively. Moreover, $Ps_{2\nu+2}(\mathbb{F}_q)/\mathcal{K}_1 \simeq Sp_{2\nu}(\mathbb{F}_q), \mathcal{K}_1/\mathcal{K}_2$ is abelian of order $q^{2\nu}$ and $\mathcal{K}_2$ is abelian of order q. Hence

$$\mid Ps_{2\nu+2}(\mathbb{F}_q) \mid = q^{(\nu+1)^2} \prod_{i=1}^{\nu} (q^{2i} - 1).$$

□

Corollary 4.6. With respect to S_2, the vector $(x_1, x_2 \ldots, x_{2\nu+2})$ is non-isotropic if and only if $x_{2\nu+2} \neq 0$; in particular, $e_{2\nu+1}$ is isotropic and $e_{2\nu+2}$ is non-isotropic. The 2-dimensional subspace $\langle e_{2\nu+1}, e_{2\nu+2}\rangle$ is non-degenerate, i.e., $\langle e_{2\nu+1}, e_{2\nu+2}\rangle S_2\,{}^t\langle e_{2\nu+1}, e_{2\nu+2}\rangle$ is nonsingular, and the dual subspace of $\langle e_{2\nu+1}, e_{2\nu+2}\rangle$ with respect to S_2, i.e.,

$$\langle e_{2\nu+1}, e_{2\nu+2}\rangle^{\perp} = \{x \in \mathbb{F}_q^{(2\nu+2)} \mid xS_2\,{}^te_{2\nu+1} = 0 \text{ and } xS_2\,{}^te_{2\nu+2} = 0\},$$

is the 2ν-dimensional subspace $\langle e_1, e_2, \ldots, e_{2\nu}\rangle$ on which $Ps_{2\nu+2}(\mathbb{F}_q)$ acts as the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$ does. Moreover, only $e_{2\nu+1}$ and its multiples are left fixed by every element of $Ps_{2\nu+2}(\mathbb{F}_q)$. □

We can combine the orders of the pseudo-symplectic groups obtained in Theorems 4.3 and 4.5 into a single theorem:

Theorem 4.7.

$$\mid Ps_{2\nu+\delta}(\mathbb{F}_q) \mid = q^{(\nu+\delta-1)^2} \prod_{i=1}^{\nu} (q^{2i} - 1).$$

□

Now let us introduce the singular pseudo-symplectic group. Let

$$S_{\delta,l} = \begin{pmatrix} S_\delta & \\ & 0^{(l)} \end{pmatrix},$$

where S_δ ($\delta = 1$ or 2) is the $(2\nu+\delta) \times (2\nu+\delta)$ non-alternate symmetric matrix (4.1) or (4.2), respectively, as before. The set of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices T satisfying

$$TS_{\delta,l}\,{}^tT = S_{\delta,l}$$

forms a group with respect to matrix multiplication, called the *singular pseudo-symplectic group* of degree $2\nu+\delta+l$ and rank $2\nu+\delta$ over $\mathbb{F}_q$ and denoted by $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$. For $l = 0$, it reduces to the ordinary pseudo-symplectic group $Ps_{2\nu+\delta}(\mathbb{F}_q)$. It can be verified that $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$ consists of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices of the form

$$T = \begin{array}{c} \begin{matrix} 2\nu+\delta & \quad l \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} 2\nu+\delta \\ l \end{matrix} \end{array}, \tag{4.3}$$

where $T_{11}S_\delta\,{}^tT_{11} = S_\delta$ and T_{22} is nonsingular. We have

Theorem 4.8.

$$|\,Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)\,| = q^{(\nu+\delta-1)^2+(2\nu+\delta)l+\frac{l(l-1)}{2}} \prod_{i=1}^{\nu}(q^{2i}-1) \prod_{i=1}^{l}(q^i-1).$$

Proof. In fact

$$|\,Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)\,| = |\,Ps_{2\nu+\delta}(\mathbb{F}_q)\,|\,|\,GL_l(\mathbb{F}_q)\,|\,q^{(2\nu+\delta)l}.$$

Hence our theorem follows from Theorems 1.6 and 4.7. □

As an application of Theorem 4.8, let us now compute the number of $n \times n$ non-alternate symmetric matrices of a given rank r ($0 \le r \le n$) over $\mathbb{F}_q$.

Theorem 4.9. The number of $n \times n$ non-alternate symmetric matrices of rank $r = 2\nu+\delta$, where $0 \le r \le n$ and $\delta = 1$ or 2, is given by

$$q^{\nu(\nu+1)} \frac{\prod_{i=n-2\nu-\delta+1}^{n}(q^i-1)}{\prod_{i=1}^{\nu}(q^{2i}-1)}.$$

Proof. Define an action of $GL_n(\mathbb{F}_q)$ on the set $\mathcal{S}(n)$ of all $n \times n$ non-alternate symmetric matrices as follows:

$$\begin{array}{ccc} \mathcal{S}(n) \times GL_n(\mathbb{F}_q) & \to & \mathcal{S}(n) \\ (S, \quad T) & \mapsto & TS\,{}^tT. \end{array}$$

By Theorem 4.1, those matrices of $\mathcal{S}(n)$ of the same rank r form an orbit under $GL_n(\mathbb{F}_q)$. If r is odd, write $r = 2\nu + 1$, and if r is even, write $r = 2\nu + 2$. Then we have $r = 2\nu + \delta$, where $\delta = 1$ or 2. Let

$$S_{\delta,n-r} = \begin{pmatrix} S_\delta & \\ & 0^{(n-r)} \end{pmatrix},$$

where S_δ ($\delta = 1$ or 2) is given by (4.1) or (4.2), respectively. Then $S_{\delta,n-r}$ is a non-alternate symmetric matrix of rank $r = 2\nu + \delta$ and the stabilizer of $S_{\delta,n-r}$ in $GL_n(\mathbb{F}_q)$ is $Ps_{2\nu+\delta+(n-r),2\nu+\delta}(\mathbb{F}_q)$. Therefore the number of $n \times n$ non-alternate symmetric matrices of rank $r = 2\nu + \delta$ is given by the index

$$GL_n(\mathbb{F}_q) : Ps_{2\nu+\delta+(n-r),2\nu+\delta}(\mathbb{F}_q).$$

Hence our theorem follows from Theorems 1.6 and 4.8. □

Corollary 4.10. The number of $(2\nu + \delta) \times (2\nu + \delta)$ nonsingular non-alternate symmetric matrices, where $\delta = 1$ or 2, is given by

$$q^{\nu(\nu+1)} \frac{\prod_{i=1}^{2\nu+\delta} (q^i - 1)}{\prod_{i=1}^{\nu} (q^{2i} - 1)}.$$

□

4.2 Pseudo-symplectic Geometry over Finite Fields of Characteristic 2

Now we would like to study how the subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ are subdivided into orbits under the pseudo-symplectic group $Ps_{2\nu+\delta}(\mathbb{F}_q)$ and to compute the number of orbits and the length of each orbit.

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu+\delta)}$. By Theorem 4.1 $PS_\delta\,{}^tP$ is cogredient to one of the following three normal forms

$$M(m, 2s, s) = \begin{pmatrix} 0 & I^{(s)} & \\ I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix}, \tag{4.4}$$

$$M(m,2s+1,s)=\begin{pmatrix} 0 & I^{(s)} & & \\ I^{(s)} & 0 & & \\ & & 1 & \\ & & & 0^{(m-2s-1)} \end{pmatrix}, \tag{4.5}$$

and

$$M(m,2s+2,s)=\begin{pmatrix} 0 & I^{(s)} & & & \\ I^{(s)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 1 & \\ & & & & 0^{(m-2s-2)} \end{pmatrix} \tag{4.6}$$

for some s such that $0 \le s \le [m/2]$. We say that P is a *subspace of type* $(m, 2s+\tau, s, \varepsilon)$, where $\tau = 0, 1,$ or 2 and $\varepsilon = 0$ or 1, if

(i) $PS_\delta\,{}^tP$ is cogredient to $M(m, 2s+\tau, s)$, and

(ii) $e_{2\nu+1} \notin P$ or $e_{2\nu+1} \in P$ according to $\varepsilon = 0$ or $\varepsilon = 1$, respectively.

In particular, subspaces of type $(m,0,0,0)$ and $(m,0,0,1)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ are called *totally isotropic subspaces* of dimension m.

At first, we have

Theorem 4.11. In the $(2\nu+\delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$, subspaces of type $(m, 2s+\tau, s, \varepsilon)$ exist if and only if

$$(\tau,\varepsilon)=\begin{cases} (0,0),(1,0),(1,1), \text{ or } (2,0), & \text{when } \delta=1, \\ (0,0),(0,1),(1,0),(2,0), \text{ or } (2,1), & \text{when } \delta=2, \end{cases} \tag{4.7}$$

and

$$2s+\max\{\tau,\varepsilon\} \le m \le \nu+s+[(\tau+\delta-1)/2]+\varepsilon. \tag{4.8}$$

Proof. Consider first the case $\delta = 1$. Let $\sigma = \nu + s - m + [\tau/2] + \varepsilon$. It is easy to verify that if both (4.7) and (4.8) hold, then

$$\begin{array}{c} \begin{matrix} s & m-2s & \sigma & s & m-2s & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.9}$$

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.10}$$

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.11}$$

and

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-2 & \sigma & s & 1 & m-2s-2 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{4.12}$$

are subspaces of types $(m, 2s, s, 0)$, $(m, 2s+1, s, 0)$, $(m, 2s+1, s, 1)$, and $(m, 2s+2, s, 0)$, respectively. Hence the sufficiency of conditions (4.7) and (4.8) is proved.

To prove the necessity of (4.7) it is enough to prove that if $\tau = 0$ or 2, then we have necessarily $\varepsilon = 0$. Suppose first that $\tau = 0$ and let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu+1)}$ such that $PS_1\,{}^tP = M(m, 2s, s)$ (see (4.4)). Then every vector of P is isotropic with respect to S_1, but $e_{2\nu+1}$ is not. Therefore $e_{2\nu+1} \notin P$, and hence $\varepsilon = 0$. Next suppose that $\tau = 2$ and let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu+1)}$ such that $PS_1\,{}^tP = M(m, 2s+2, s)$ (see (4.6)). If $e_{2\nu+1} \in P$, then P would contain an isotropic vector not contained in $\langle e_{2\nu+1}\rangle^{\perp} = \{e_1, e_2, \ldots, e_{2\nu}\}$, which is impossible. Hence $e_{2\nu+1} \notin P$, that is, $\varepsilon = 0$.

To prove the necessity of (4.8) we have to distinguish the four possible cases of (τ, ε).

(i) $(\tau, \varepsilon) = (0, 0)$. Let P be a subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. We can assume that $PS_1\,{}^tP = M(m, 2s, s)$. Clearly all vectors of P are isotropic. By Corollary 4.4, $P \subset \langle e_1, e_2, \ldots, e_{2\nu}\rangle$ and hence P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \end{pmatrix} \end{array},$$

where P_1 is an $m \times 2\nu$ matrix of rank m. From $PS_1\,{}^tP = M(m, 2s, s)$, we deduce $P_1 K\,{}^tP_1 = M(m, 2s, s)$, that is, P_1 is a subspace of type (m, s) in the 2ν-dimensional symplectic space. By Theorem 3.6 we have $2s \le m \le \nu + s$, which proves the necessity of (4.8) for the case $(\tau, \varepsilon) = (0, 0)$.

(ii) $(\tau, \varepsilon) = (1, 0)$. Let P be a subspace of type $(m, 2s+1, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. Thus $e_{2\nu+1} \notin P$. We can assume that $PS_1\,{}^tP = M(m, 2s+1, s)$, then $2s+1 \leq m$ and P is necessarily of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \\ v & 1 \\ P_2 & 0 \end{pmatrix} \end{array} \begin{array}{l} 2s \\ 1 \\ m-2s-1 \end{array}.$$

From rank $P = m$, we deduce

$$\operatorname{rank} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m - 1.$$

Since $e_{2\nu+1} \notin P$, v is non-zero and

$$\operatorname{rank} \begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = m.$$

From $PS_1\,{}^tP = M(m, 2s+1, s)$ we deduce

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} K \,{}^t\!\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = M(m, 2s, s),$$

that is,

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix}$$

is a subspace of type (m, s) in the 2ν-dimensional symplectic space. By Theorem 3.6 we have $m \leq \nu + s$. Combining the inequalities $2s+1 \leq m$ and $m \leq \nu + s$ we obtain $2s+1 \leq m \leq \nu + s$.

(iii) $(\tau, \varepsilon) = (1, 1)$. Let P be a subspace of type $(m, 2s+1, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. Thus $e_{2\nu+1} \in P$. We can choose a matrix representation of P of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \\ 0 & 1 \end{pmatrix} \end{array} \begin{array}{l} m-1 \\ 1 \end{array},$$

where P_1 is an $(m-1) \times 2\nu$ matrix of rank $m-1$. We have

$$PS_1\,{}^tP = \begin{pmatrix} P_1 K\,{}^tP_1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $PS_1\,{}^tP$ is cogredient to $M(m, 2s+1, s)$, $P_1K\,{}^tP_1$ is cogredient to $M(m-1, 2s, s)$, that is, P_1 is a subspace of type $(m-1, s)$ in the 2ν-dimensional symplectic space. Therefore $2s \le m-1 \le \nu+s$, and hence $2s+1 \le m \le \nu+s+1$.

(iv) $(\tau, \varepsilon) = (2, 0)$. Let P be a subspace of type $(m, 2s+2, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. We can assume that $PS_1\,{}^tP = M(m, 2s+2, s)$. Then P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \\ v & 1 \\ P_2 & 0 \end{pmatrix} \end{array} \begin{array}{c} \\ \begin{matrix} 2s+1 \\ 1 \\ m-2s-2 \end{matrix} \end{array},$$

where v is a non-zero 2ν-dimensional row vector. From rank $P = m$ we deduce

$$\operatorname{rank} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-1.$$

Since $e_{2\nu+1} \notin P$, we have also

$$\operatorname{rank} \begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = m.$$

From $PS_1\,{}^tP = M(m, 2s+2, s)$ we deduce

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} K \,{}^t\!\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = \begin{pmatrix} 0 & I^{(s)} & & & \\ I^{(s)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 0 & \\ & & & & 0^{(m-2s-2)} \end{pmatrix},$$

that is,

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix}$$

is a subspace of type $(m, s+1)$ in the 2ν-dimensional symplectic space. Therefore we have $2s+2 \le m \le \nu+s+1$.

Thus we have completed the proof of Theorem 4.11 for the case $\delta = 1$.

Now let us consider the case $\delta = 2$. Let $\sigma = \nu + s - m + [(\tau+1)/2] + \varepsilon$. It is easy to verify that if both (4.7) and (4.8) hold, then

$$\begin{array}{c} \begin{matrix} s & m-2s & \sigma & s & m-2s & \sigma & 2 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.13}$$

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.14}$$

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.15}$$

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-2 & \sigma & s & 1 & m-2s-2 & \sigma & 1 & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{4.16}$$

and

$$\begin{array}{c} \begin{matrix} s & m-2s-2 & \sigma & s & m-2s-2 & \sigma & 1 & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{4.17}$$

are subspaces of types $(m, 2s, s, 0)$, $(m, 2s, s, 1)$, $(m, 2s+1, s, 0)$, $(m, 2s+2, s, 0)$, and $(m, 2s+2, s, 1)$, respectively. This proves the sufficiency of conditions (4.7) and (4.8).

To prove the necessity of condition (4.7) it is sufficient to prove that if $\tau = 1$, then $\varepsilon = 0$. Suppose that $\tau = 1$ and let P be a subspace of type $(m, 2s+1, s, \varepsilon)$. Then $PS_2\,{}^tP$ is cogredient to $M(m, 2s+1, s)$, thus P must contain a vector whose $(2\nu+2)$-th component is 1. If $e_{2\nu+1} \in P$, then we can assume that P has a matrix representation of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} 0 & 1 & 0 \\ v & 0 & 1 \\ P_1 & 0 & 0 \end{pmatrix} \begin{matrix} 1 \\ 1 \\ m-2 \end{matrix} \end{array}.$$

Thus

$$PS_2\,{}^tP = \begin{array}{c} \begin{matrix} 1 & 1 & m-2 \end{matrix} \\ \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & vK\,{}^tP_1 \\ 0 & P_1K\,{}^tv & P_1K\,{}^tP_1 \end{pmatrix} \end{array} \begin{matrix} 1 \\ 1 \\ m-2 \end{matrix}.$$

Let

$$A = \begin{pmatrix} 0 & P_1K\,{}^tv \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

then $PS_2\,{}^tP$ is cogredient to

$$\begin{pmatrix} I^{(2)} & \\ A & I^{(m-2)} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & vK\,{}^tP_1 \\ 0 & P_1K\,{}^tv & P_1K\,{}^tP_1 \end{pmatrix} {}^t\begin{pmatrix} I^{(2)} & \\ A & I^{(m-2)} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & \\ 1 & 1 & \\ & & P_1K\,{}^tP_1 \end{pmatrix}.$$

But $P_1K\,{}^tP_1$ is alternate and hence is of even rank. It follows that $PS_2\,{}^tP$ is of even rank. We have assumed that P is of type $(m, 2s+1, s, \varepsilon)$, hence $PS_2\,{}^tP$ is of rank $2s+1$. We obtain a contradiction. Therefore $e_{2\nu+1} \notin P$, that is, $\varepsilon = 0$.

To prove the necessity of condition (4.8) we distinguish the five possible cases of (τ, ε).

(i) $(\tau, \varepsilon) = (0, 0)$. Let P be a subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+2)}$. We can assume that $PS_2\,{}^tP = M(m, 2s, s)$. Then every vector of P is isotropic with respect to S_2. By Corollary 4.6, P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & {}^tu & 0 \end{pmatrix} \end{array}.$$

Since $e_{2\nu+1} \notin P$, rank $P_1 = m$. From $PS_2\,{}^tP = M(m, 2s, s)$ we deduce $P_1K\,{}^tP_1 = M(m, 2s, s)$. Thus P_1 is a subspace of type (m, s) in the 2ν-dimensional symplectic space. By Theorem 3.6 $2s \le m \le \nu + s$.

(ii) $(\tau, \varepsilon) = (0, 1)$. Let P be a subspace of type $(m, 2s, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. Then $e_{2\nu+1} \in P$. We can assume that $PS_2\,{}^tP = M(m, 2s, s)$ and that P has a matrix representation of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \end{array} \begin{matrix} m-1 \\ 1 \end{matrix},$$

where P_1 is an $(m-1)\times 2\nu$ matrix of rank $m-1$. From $PS_2\,{}^tP = M(m,2s,s)$ we deduce $P_1K\,{}^tP_1 = M(m-1,2s,s)$. Thus, P_1 is a subspace of type $(m-1,s)$ in the 2ν-dimensional symplectic space. Therefore $2s \le m-1 \le \nu+s$ and hence $2s+1 \le m \le \nu+s+1$.

(iii) $(\tau,\varepsilon)=(1,0)$. Let P be a subspace of type $(m,2s+1,s,0)$ in $\mathbb{F}_q^{(2\nu+2)}$. We can assume that $PS_2\,{}^tP = M(m,2s+1,s)$. It follows that the last coordinate of every row of P except the $(2s+1)$-th row is 0 and the $(2s+1)$-th row of P is of the form $(x,y,1)$, where $x\in\mathbb{F}_q^{(2\nu)}$ and $y\in\mathbb{F}_q$. Let

$$T_1 = \begin{pmatrix} I & K\,{}^tx & 0 \\ 0 & 1 & 0 \\ x & y & 1 \end{pmatrix}. \tag{4.18}$$

Then $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$,

$$(x,y,1)T_1 = (0,0,1) = e_{2\nu+2},$$

and the last coordinate of every row of PT_1 different from the $(2s+1)$-th row is 0. Since $(x,y,1)S_2\,{}^tv = 0$ for every row v of P different from the $(2s+1)$-th row, $e_{2\nu+2}S_2\,{}^t(vT_1) = 0$ for every row vT_1 of PT_1 different from the $(2s+1)$-th row. Hence the $(2\nu+1)$-th coordinate of every row of PT_1 different from the $(2s+1)$-th row is also 0. Therefore PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \end{array} \begin{matrix} 2s \\ 1 \\ m-2s-1 \end{matrix}.$$

Thus

$$\operatorname{rank}\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-1.$$

From $(PT_1)S_2\,{}^t(PT_1) = M(m,2s+1,s)$ we deduce also

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K\,{}^t\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-1,2s,s).$$

That is,

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is a subspace of type $(m-1,s)$ in the 2ν-dimensional symplectic space. Therefore $2s \le m-1 \le \nu+s$ and hence $2s+1 \le m \le \nu+s+1$.

(iv) $(\tau,\varepsilon)=(2,0)$. Let P be a subspace of type $(m,2s+2,s,0)$ in $\mathbb{F}_q^{(2\nu+2)}$. We can assume that $PS_2\,{}^tP = M(m,2s+2,s)$. Clearly $2s+2 \le m$. As in the above case there is a $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that the $(2s+2)$-th row of PT_1 is $e_{2\nu+2}$ and the last coordinate of every row of PT_1 different from the $(2s+2)$-th row is 0. From

$$(PT_1)S_2\,{}^t(PT_1) = PS_2\,{}^tP = M(m,2s+2,s),$$

we deduce that the $(2\nu+1)$-th coordinate of every row of PT_1 different from the $(2s+1)$-th row is also 0, while the $(2\nu+1)$-th coordinate of the $(2s+1)$-th row is 1. Therefore PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ v & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \begin{matrix} 2s \\ 1 \\ 1 \\ m-2s-2 \end{matrix} \end{array}.$$

Clearly

$$\operatorname{rank}\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-2.$$

Since $e_{2\nu+1} \notin P$ and $e_{2\nu+1} = e_{2\nu+1}T \notin PT_1$, we must have

$$\operatorname{rank}\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = m-1.$$

From $(PT_1)S_2\,{}^t(PT_1) = M(m,2s+2,s)$, we deduce

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} K \,{}^t\!\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = M(m-1,2s,s).$$

Thus

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix}$$

is a subspace of type $(m-1,s)$ in the 2ν-dimensional symplectic space. Therefore $2s \le m-1 \le \nu+s$ and hence $2s+1 \le m \le \nu+s+1$.

Combining this inequality with the previous one, $2s+2 \leq m$, we have $2s+2 \leq m \leq \nu+s+1$.

(v) $(\tau,\varepsilon)=(2,1)$. Let P be a subspace of type $(m,2s+2,s,1)$ in $\mathbb{F}_q^{(2\nu+2)}$. We can assume that $PS_2\,{}^tP = M(m,2s+2,s)$. As in case (iv), there is a $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ v & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \end{array} \begin{matrix} 2s \\ 1 \\ 1 \\ m-2s-2 \end{matrix} .$$

Clearly

$$\operatorname{rank}\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-2.$$

From $(PT_1)S_2\,{}^t(PT_1) = M(m,2s+2,s)$ we deduce

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K\,{}^t\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-2,2s,s).$$

Thus

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is a subspace of type $(m-2,s)$ in the 2ν-dimensional symplectic space. Consequently, $2s \leq m-2 \leq \nu+s$ and hence $2s+2 \leq m \leq \nu+s+2$.

The proof of Theorem 4.11 for the case $\delta=2$ is also completed. □

Let us denote the set of subspaces of type $(m,2s+\tau,s,\varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ by $\mathcal{M}(m,2s+\tau,s,\varepsilon;2\nu+\delta)$, where $\delta=1$ or 2, $\tau=0,1$, or 2, and $\varepsilon=0$ or 1. If either (4.7) or (4.8) fails to hold, then by Theorem 4.11 $\mathcal{M}(m,2s+\tau,s,\varepsilon;2\nu+\delta)=\phi$. However, if both (4.7) and (4.8) hold, then we have

Theorem 4.12. Assume that both (4.7) and (4.8) are satisfied. Then $\mathcal{M}(m,2s+\tau,s,\varepsilon;2\nu+\delta)$ is an orbit of subspaces under $Ps_{2\nu+\delta}(\mathbb{F}_q)$.

Proof. Consider first the case $\delta=1$. The four possible cases of (τ,ε) can be treated in a similar way, so we discuss only the case $(\tau,\varepsilon)=(0,0)$ as an example.

Let P be a subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. By the proof of Theorem 4.11 we can assume that P is of the form

$$\begin{matrix} & 2\nu & 1 \\ P = & (P_1 & 0) \end{matrix},$$

where P_1 is an $m \times 2\nu$ matrix of rank m and that $P_1 K\, {}^t P_1 = M(m, 2s, s)$. Now let Q be another subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. We can also assume that Q is of the form

$$\begin{matrix} & 2\nu & 1 \\ Q = & (Q_1 & 0) \end{matrix},$$

where Q_1 is an $m \times 2\nu$ matrix of rank m and that $Q_1 K\, {}^t Q_1 = M(m, 2s, s)$. By Theorem 3.11, there is an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = Q_1 T$. Consequently, $P = QT_1$, where

$$T_1 = \begin{pmatrix} T & \\ & 1 \end{pmatrix} \in Ps_{2\nu+1}(\mathbb{F}_q).$$

Hence $\mathcal{M}(m, 2s, s, 0; 2\nu + 1)$ is an orbit of $Ps_{2\nu+1}(\mathbb{F}_q)$.

Then let us consider the case $\delta = 2$. We distinguish the five possible cases of (τ, ε) as we did in Theorem 4.11.

(i) $(\tau, \varepsilon) = (0, 0)$. Let P be a subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+2)}$. By the proof of Theorem 4.11 we can assume that P has a matrix representation of the form

$$\begin{matrix} & 2\nu & 1 & 1 \\ P = & (P_1 & {}^t u & 0) \end{matrix},$$

where P_1 is an $m \times 2\nu$ matrix of rank m and $P_1 K\, {}^t P_1 = M(m, 2s, s)$. Similarly, if Q is another subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+2)}$, then Q has a matrix representation of the form

$$\begin{matrix} & 2\nu & 1 & 1 \\ Q = & (Q_1 & {}^t v & 0) \end{matrix},$$

where Q_1 is an $m \times 2\nu$ matrix of rank m and $Q_1 K\, {}^t Q_1 = M(m, 2s, s)$. By Theorem 3.11 there is an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that $P_1 = Q_1 T$. Since both u and v are m-dimensional row vectors and P_1 is of rank m, we can choose a 2ν-dimensional row vector x such that $P_1 K\, {}^t x + {}^t v = {}^t u$. Then

$$(P_1 \quad {}^t u \quad 0) = (Q_1 \quad {}^t v \quad 0) \begin{pmatrix} T & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} I & K\, {}^t x & \\ & 1 & \\ x & 0 & 1 \end{pmatrix},$$

where

$$\begin{pmatrix} T & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} I & K\,{}^t x & \\ & 1 & \\ x & 0 & 1 \end{pmatrix} \in Ps_{2\nu+2}(\mathbb{F}_q).$$

This proves that $\mathcal{M}(m, 2s, 0; 2\nu+2)$ is an orbit.

(ii) $(\tau, \varepsilon) = (0, 1)$. This case can be treated in the same way as the case $\delta = 1$, and the details will be omitted.

(iii) $(\tau, \varepsilon) = (1, 0)$. Let P be a subspace of type $(m, 2s+1, s, 0)$ in $\mathbb{F}_q^{(2\nu+2)}$. By the proof of Theorem 4.11 we can assume that there is a matrix representation P of the subspace P and an element $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \end{array} \begin{array}{l} 2s \\ 1 \\ m-2s-1 \end{array},$$

where

$$\operatorname{rank} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-1$$

and

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K \, {}^t \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-1, 2s, s).$$

Similarly, if $Q \in \mathcal{M}(m, 2s+1, s, 0; 2\nu+2)$, then we can assume that there is a matrix representation Q of the subspace Q and an element $T_2 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that

$$QT_2 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 & 0 \\ 0 & 0 & 1 \\ Q_2 & 0 & 0 \end{pmatrix} \end{array} \begin{array}{l} 2s \\ 1 \\ m-2s-1 \end{array},$$

where

$$\operatorname{rank} \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = m-1$$

and

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} K \, {}^t \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = M(m-1, 2s, s).$$

By Theorem 3.11 there is an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} T.$$

Let

$$T_3 = \begin{pmatrix} T & \\ & I^{(2)} \end{pmatrix},$$

then $T_3 \in Ps_{2\nu+2}(\mathbb{F}_q)$ and $P = QT_2T_3T_1^{-1}$.

(iv) $(\tau, \varepsilon) = (2, 0)$. This case can be treated in the same way as case (iii) and the details will be omitted.

(v) $(\tau, \varepsilon) = (2, 1)$. Let P be a subspace of type $(m, 2s+2, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By the proof of Theorem 4.11 we can assume that there is a matrix representation P of the subspace P and an element $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ v & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \begin{matrix} 2s \\ 1 \\ 1 \\ m-2s-2 \end{matrix} \end{array},$$

where

$$\operatorname{rank} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m - 2$$

and

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-2, 2s, s).$$

Since $\varepsilon = 1$, $e_{2\nu+1} \in P$. Thus we must have

$$v = u_1P_1 + u_2P_2,$$

where $u_1 \in \mathbb{F}_q^{(2s)}$ and $u_2 \in \mathbb{F}_q^{(m-2s-2)}$. $(PT_1)S_2 \, {}^t(PT_1) = M(m, 2s+2, s)$ implies $(v \ 1 \ 0)S_2 \, {}^t(P_1 \ 0 \ 0) = 0$ and $P_2K \, {}^tP_1 = 0$, which together imply $u_1 = 0$ and $v = u_2P_2$. Let

$$A = \begin{pmatrix} I^{(2s)} & & & \\ & 1 & & u_2 \\ & & 1 & \\ & & & I^{(m-2s-2)} \end{pmatrix},$$

then

$$APT_1 = \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix}.$$

Similarly, let Q be another subspace of type $(m, 2s+2, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$, then we can assume that there is a matrix representation Q of the subspace Q and an element $T_2 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) and an $m \times m$ matrix B which has the same form as A such that

$$BQT_2 = \begin{pmatrix} Q_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ Q_2 & 0 & 0 \end{pmatrix},$$

where

$$\operatorname{rank} \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = m - 2$$

and

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = M(m-2, 2s, s).$$

By Theorem 3.1 there is an element $T \in Sp_{2\nu}(\mathbb{F}_q)$ such that

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} T.$$

Let

$$T_3 = \begin{pmatrix} T & \\ & I^{(2)} \end{pmatrix},$$

then $T_3 \in Ps_{2\nu+2}(\mathbb{F}_q)$ and $AP = BQT_2T_3T_1^{-1}$. Therefore P and Q belong to the same orbit under $Ps_{2\nu+2}(\mathbb{F}_q)$.

Theorem 4.12 is now completely proved. □

From Theorems 4.11 and 4.12 we deduce

Theorem 4.13. In the $(2\nu+\delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$, the number of orbits of subspaces is equal to

$$\frac{1}{2}(\nu+1)[(\nu+4)\delta + 3\nu],$$

where $\delta = 1$ or 2.

Proof. By Theorems 4.11 and 4.12, the number of orbits of subspaces is equal to the number of possible 4-tuples $(m, s, \tau, \varepsilon)$ satisfying (4.7) and (4.8).

Consider first the case $\delta = 1$. There are four possible values for (τ, ε), i.e., $(0,0), (1,0), (1,1)$ and $(2,0)$.

(a) $(\tau, \varepsilon) = (0,0)$. Subspaces of type $(m, 2s, s, 0)$ exist if and only if

$$2s \le m \le \nu + s.$$

Clearly the case $m = 2\nu + 1$ is impossible. Therefore

$$0 \le m \le 2\nu \ \text{ and } \ m - \nu \le s \le \frac{m}{2}.$$

By the proof of Theorem 3.10, the number of subspace types of the form $(m, 2s, s, 0)$ is equal to

$$\frac{1}{2}(\nu + 1)(\nu + 2).$$

(b) $(\tau, \varepsilon) = (1,0)$. Subspaces of type $(m, 2s+1, s, 0)$ exist if and only if

$$2s + 1 \le m \le \nu + s.$$

Clearly the cases $m = 0$ and $m = 2\nu + 1$ are impossible. Therefore

$$1 \le m \le 2\nu \ \text{ and } \ m - \nu \le s \le \frac{m-1}{2}.$$

For $m \le \nu$, s can take $[\frac{m-1}{2}] + 1$ values: $0, 1, \ldots, [\frac{m-1}{2}]$; and for $m > \nu$, s can take $[\frac{m-1}{2}] - (m - \nu) + 1$ values: $m - \nu, m - \nu + 1, \ldots, [\frac{m-1}{2}]$. Hence the number of subspace types of the form $(m, 2s+1, s, 0)$ is equal to

$$\sum_{m=1}^{\nu} ([\frac{m-1}{2}] + 1) + \sum_{m=\nu+1}^{2\nu} ([\frac{m-1}{2}] - (m - \nu) + 1) = \frac{1}{2}\nu(\nu + 1).$$

(c) $(\tau, \varepsilon) = (1,1)$. Subspaces of type $(m, 2s+1, s, 1)$ exist if and only if

$$2s + 1 \le m \le \nu + s + 1.$$

Clearly the case $m = 0$ is impossible. Therefore

$$1 \le m \le 2\nu + 1 \ \text{ and } \ m - \nu - 1 \le s \le \frac{m-1}{2}.$$

For $m \le \nu+1$, s can take $[\frac{m-1}{2}]+1$ values; and for $m > \nu+1$, s can take $[\frac{m-1}{2}]-(m-\nu-1)+1$ values. Hence the number of subspace types of the form $(m, 2s+1, s, 1)$ is equal to

$$\sum_{m=1}^{\nu+1}([\frac{m-1}{2}]+1)+\sum_{m=\nu+2}^{2\nu+1}([\frac{m-1}{2}]-(m-\nu-1)+1)=\frac{1}{2}(\nu+1)(\nu+2).$$

(d) $(\tau,\varepsilon)=(2,0)$. Subspaces of type $(m, 2s+2, s, 0)$ exist if and only if

$$2s+2 \le m \le \nu+s+1.$$

Clearly the cases $m=0,1$, and $2\nu+1$ are impossible. Therefore

$$2 \le m \le 2\nu \text{ and } m-\nu-1 \le s \le \frac{m-2}{2}.$$

For $m \le \nu+1$, s can take $[\frac{m-2}{2}]+1$ values; and for $m > \nu+1$, s can take $[\frac{m-2}{2}]-(m-\nu-1)+1$ values. Thus the number of subspace types of the form $(m, 2s+2, s, 0)$ is equal to

$$\sum_{m=2}^{\nu+1}([\frac{m-2}{2}]+1)+\sum_{m=\nu+2}^{2\nu}([\frac{m-2}{2}]-(m-\nu-1)+1)=\frac{1}{2}\nu(\nu+1).$$

Summing up, we obtain the number of orbits of subspaces in $\mathbb{F}_q^{(2\nu+1)}$, which is equal to

$$2(\nu+1)^2.$$

Then consider the case $\delta=2$. There are five possibilities for (τ,ε), i.e., $(0,0),(0,1),(1,0),(2,0)$, and $(2,1)$.

(a) $(\tau,\varepsilon)=(0,0)$. Subspaces of type $(m, 2s, s, 0)$ exist if and only if

$$2s \le m \le \nu+s.$$

Clearly the cases $m=2\nu+1$ and $2\nu+2$ are impossible. Therefore

$$0 \le m \le 2\nu \text{ and } m-\nu \le s \le \frac{m}{2}.$$

By the proof of Theorem 3.10, the number of subspace types of the form $(m, 2s, s, 0)$ is equal to

$$\frac{1}{2}(\nu+1)(\nu+2).$$

(b) $(\tau, \varepsilon) = (0, 1)$. Subspaces of type $(m, 2s, s, 1)$ exist if and only if

$$2s + 1 \leq m \leq \nu + s + 1.$$

Clearly the cases $m = 0$ and $m = 2\nu + 2$ are impossible. Therefore

$$1 \leq m \leq 2\nu + 1 \text{ and } m - \nu - 1 \leq s \leq \frac{m-1}{2}.$$

By the calculation of case (c) for the case $\delta = 1$, the number of subspace types of the form $(m, 2s, s, 1)$ is equal to

$$\frac{1}{2}(\nu + 1)(\nu + 2).$$

(c) $(\tau, \varepsilon) = (1, 0)$. Subspaces of type $(m, 2s + 1, s, 0)$ exist if and only if

$$2s + 1 \leq m \leq \nu + s + 1.$$

The cases $m = 0$ and $m = 2\nu + 2$ are impossible. Therefore

$$1 \leq m \leq 2\nu + 1 \text{ and } m - \nu - 1 \leq s \leq \frac{m-1}{2}.$$

As the above case (b), the number of subspace types of the form $(m, 2s + 1, s, 0)$ is equal to

$$\frac{1}{2}(\nu + 1)(\nu + 2).$$

(d) $(\tau, \varepsilon) = (2, 0)$. Subspaces of type $(m, 2s + 2, s, 0)$ exist if and only if

$$2s + 2 \leq m \leq \nu + s + 1.$$

Clearly the cases $m = 0, 1,\ 2\nu + 1$, and $2\nu + 2$ are impossible. Therefore

$$2 \leq m \leq 2\nu \text{ and } m - \nu - 1 \leq s \leq \frac{m-2}{2}.$$

The same calculation of case (d) for the case $\delta = 1$ gives the number of subspace types of the form $(m, 2s + 2, s, 0)$, which is

$$\frac{1}{2}\nu(\nu + 1).$$

(e) $(\tau, \varepsilon) = (2, 1)$. Subspaces of type $(m, 2s + 2, s, 1)$ exist if and only if

$$2s + 2 \leq m \leq \nu + s + 2.$$

Clearly the cases $m = 0$ and $m = 1$ are impossible. Therefore

$$2 \le m \le 2\nu + 2 \text{ and } m - \nu - 2 \le s \le \frac{m-2}{2}.$$

If $m \le \nu + 2$, s can take $[\frac{m-2}{2}] + 1$ values. If $m > \nu + 2$, s can take $[\frac{m-2}{2}] - (m - \nu - 2) + 1$ values. Thus the number of subspace types of the form $(m, 2s + 2, s, 1)$ is equal to

$$\sum_{m=2}^{\nu+2}([\frac{m-2}{2}]+1) + \sum_{m=\nu+3}^{2\nu+2}([\frac{m-2}{2}] - (m-\nu-2)+1) = \frac{1}{2}(\nu+1)(\nu+2).$$

Summing up, we get the number of orbits of subspaces in $\mathbb{F}_q^{(2\nu+2)}$, which is equal to

$$4 \cdot \frac{1}{2}(\nu+1)(\nu+2) + \frac{1}{2}\nu(\nu+1) = \frac{1}{2}(\nu+1)(5\nu+8).$$

□

Let

$$N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta) = \mid \mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta) \mid .$$

Our next goal is to compute $N(m, 2s + \tau, s, \varepsilon; 2\nu + \delta)$. It follows from Theorem 4.11 that $N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta) = 0$ if either (4.7) or (4.8) fails to hold. However, if both (4.7) and (4.8) are satisfied, then we have the following Anzahl theorem in the $(2\nu + \delta)$-dimensional pseudo-symplectic geometry over finite fields of characteristic 2.

Theorem 4.14. Assume that both (4.7) and (4.8) are satisfied. Then

$$\begin{aligned} &N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta) \\ &= q^{n_0 + 2(s+(2-\delta)[\tau/2])(\nu+s-m+\delta[(\tau+1)/2]+(\delta-1)(\tau-1)(\tau-2)\varepsilon/2)} \\ &\quad \times \frac{\prod_{i=\nu+s-m+[(\tau+\delta-1)/2]+\varepsilon+1}^{\nu} (q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s-\max(\tau,\varepsilon)} (q^i-1)}, \end{aligned} \tag{4.19}$$

where $n_0 = 0$ if $\delta = 1$, and $n_0 = m, 0, 2(\nu+1) - m, 2(\nu+1) - m$, or $2(\nu+1) - m$ corresponding to the cases $(\tau, \varepsilon) = (0,0), (0,1), (1,0), (2,0)$, or $(2,1)$, respectively, if $\delta = 2$.

Proof. Consider first the case $\delta = 1$. There are four possible cases for (τ, ε), i.e., $(0,0), (1,0), (1,1)$, and $(2,0)$.

(i) $(\tau, \varepsilon) = (0,0)$. Let $P \in \mathcal{M}(m, 2s, s, 0; 2\nu+1)$ and assume that $PS_1\,{}^tP = M(m, 2s, s)$. By the proof of Theorem 4.11 P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \end{pmatrix} \end{array},$$

where P_1 is an $m \times 2\nu$ matrix of rank m such that $P_1 K\,{}^tP_1 = M(m, 2s, s)$. The number of $m \times 2\nu$ matrices P_1 of rank m satisfying $P_1 K\,{}^tP_1 = M(m, 2s, s)$ was denoted by $n(m, s; 2\nu)$ in Section 3.2. Let

$$Q = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 \end{pmatrix} \end{array}$$

be an $m \times (2\nu+1)$ matrix of rank m satisfying $Q_1 K\,{}^tQ_1 = M(m, 2s, s)$. Suppose that P and Q represent the same subspace of type $(m, 2s, s, 0)$. Then $Q = TP$, where $T \in GL_m(\mathbb{F}_q)$. From $P_1 K\,{}^tP_1 = Q_1 K\,{}^tQ_1 = M(m, 2s, s)$ we deduce $TM(m, 2s, s)\,{}^tT = M(m, 2s, s)$, i.e., $T \in Sp_{m,s}(\mathbb{F}_q)$. Since P is of rank m, $TP = P$ implies $T = I$. Therefore

$$N(m, 2s, s, 0; 2\nu+1) = \frac{n(m, s; 2\nu)}{|\, Sp_{m,s}(\mathbb{F}_q)\,|}.$$

By Lemma 3.12

$$n(m, s; 2\nu) = |\, Sp_{m,s}(\mathbb{F}_q)\,|\, N(m, 2; 2\nu).$$

Hence

$$\begin{aligned} N(m, 2s, s, 0; 2\nu+1) &= N(m, s; 2\nu) \\ &= q^{2s(\nu+s-m)} \frac{\displaystyle\prod_{i=\nu+s-m+1}^{\nu} (q^{2i}-1)}{\displaystyle\prod_{i=1}^{s}(q^{2i}-1)\prod_{i=1}^{m-2s}(q^i-1)}. \end{aligned}$$

(ii) $(\tau, \varepsilon) = (1,0)$. Let $P \in \mathcal{M}(m, 2s+1, s, 0; 2\nu+1)$ and assume that $PS_1\,{}^tP = M(m, 2s+1, s)$. By the proof of Theorem 4.11 P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \\ v & 1 \\ P_2 & 0 \end{pmatrix} \end{array} \begin{array}{l} 2s \\ 1 \\ m-2s-1 \end{array},$$

where

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} \tag{4.20}$$

is of rank m and

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = M(m, 2s, s). \tag{4.21}$$

By Lemma 3.17 the number of $m \times 2\nu$ matrices (4.20) of rank m satisfying (4.21) is known to be

$$n(m, s; 2\nu) = q^{(2\nu-s)s+\frac{(m-2s)(m-2s-1)}{2}} \prod_{i=\nu+s-m+1}^{\nu} (q^{2i} - 1).$$

Let

$$Q = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 \\ u & 1 \\ Q_2 & 0 \end{pmatrix} \end{array} \begin{array}{c} 2s \\ 1 \\ m-2s-1 \end{array}$$

be such that

$$\operatorname{rank} \begin{pmatrix} Q_1 \\ u \\ Q_2 \end{pmatrix} = m$$

and

$$QS_1 \, {}^t\!Q = M(m, 2s+1, s).$$

Suppose that P and Q represent the same subspace of type $(m, 2s+1, s, 0)$. Then $Q = TP$, where $T \in GL_m(\mathbb{F}_q)$. From $PS_1 \, {}^t\!P = QS_1 \, {}^t\!Q = M(m, 2s+1, s)$ we deduce $TM(m, 2s+1, s) \, {}^t\!T = M(m, 2s+1, s)$, i.e., $T \in Ps_{m,2s+1}(\mathbb{F}_q)$. Since P is an $m \times (2\nu+1)$ matrix of rank m, $TP = P$ implies $T = I$. Consequently

$$N(m, 2s+1, s, 0; 2\nu+1) = \frac{n(m, s; 2\nu)}{| \, Ps_{m,2s+1}(\mathbb{F}_q) \, |}$$

$$= q^{2s(\nu+s-m+1)} \frac{\displaystyle\prod_{i=\nu+s-m+1}^{\nu} (q^{2i} - 1)}{\displaystyle\prod_{i=1}^{s} (q^{2i} - 1) \prod_{i=1}^{m-2s-1} (q^i - 1)}.$$

(iii) $(\tau, \varepsilon) = (1, 1)$. This case can be treated in the same way as case (i), but the details will be omitted. We have

$$N(m, 2s+1, s, 1; 2\nu+1) = N(m-1, s; 2\nu)$$
$$= q^{2s(\nu+s-m+1)} \frac{\prod_{i=\nu+s-m+2}^{\nu} (q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s-1} (q^i-1)}.$$

(iv) $(\tau, \varepsilon) = (2, 0)$. Let $P \in \mathcal{M}(m, 2s+2, s, 0; 2\nu+1)$ and assume that $PS_1\,{}^tP = M(m, 2s+2, s)$. By the proof of Theorem 4.11 P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 \\ v & 1 \\ P_2 & 0 \end{pmatrix} \end{array} \begin{array}{c} 2s+1 \\ 1 \\ m-2s-2 \end{array},$$

where

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} \tag{4.22}$$

is of rank m and

$$\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ v \\ P_2 \end{pmatrix} = \begin{pmatrix} 0 & I^{(s)} & & & \\ I^{(s)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 0 & \\ & & & & 0^{(m-2s-2)} \end{pmatrix}. \tag{4.23}$$

By Lemma 3.17 the number of $m \times 2\nu$ matrices (4.22) of rank m satisfying (4.23) is known to be

$$n(m, s+1; 2\nu) = q^{(2\nu-s-1)(s+1)+\frac{(m-2s-2)(m-2s-3)}{2}} \prod_{i=\nu+s-m+2}^{\nu} (q^{2i}-1).$$

Let

$$Q = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 \\ u & 1 \\ Q_2 & 0 \end{pmatrix} \end{array} \begin{array}{c} 2s+1 \\ 1 \\ m-2s-2 \end{array}$$

be such that

$$\operatorname{rank}\begin{pmatrix} Q_1 \\ u \\ Q_2 \end{pmatrix} = m$$

and

$$QS_1\,{}^tQ = M(m, 2s+2, s).$$

Suppose that P and Q represent the same subspace of type $(m, 2s+2, s, 0)$. Then $Q = TP$, where $T \in GL_m(\mathbb{F}_q)$. From $PS_1\,{}^tP = QS_1\,{}^tQ = M(m, 2s+2, s)$ we deduce $TM(m, 2s+2, s)\,{}^tT = M(m, 2s+2, s)$, i.e., $T \in Ps_{m,2s+2}(\mathbb{F}_q)$. Furthermore, $TP = P$ implies $T = I$. Consequently

$$N(m, 2s+2, s, 0; 2\nu+1) = \frac{n(m, s+1; 2\nu)}{|\,Ps_{m,2s+2}(\mathbb{F}_q)\,|}$$

$$= q^{2(s+1)(\nu+s-m+1)} \frac{\displaystyle\prod_{i=\nu+s-m+2}^{\nu} (q^{2i}-1)}{\displaystyle\prod_{i=1}^{s}(q^{2i}-1)\prod_{i=1}^{m-2s-2}(q^i-1)}.$$

These assemble to give Theorem 4.14 for the case $\delta = 1$.

Now let us consider the case $\delta = 2$. We compute $N(m, 2s+\tau, s, \varepsilon; 2\nu+2)$ by considering the five possible cases of (τ, ε) one by one.

(i) $(\tau, \varepsilon) = (0, 0)$. Let $P \in \mathcal{M}(m, 2s, s, 0; 2\nu+2)$ and assume that $PS_2\,{}^tP = M(m, 2s, s)$. By the proof of Theorem 4.11 P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & {}^tu & 0 \end{pmatrix} \end{array},$$

where P_1 is an $m \times 2\nu$ matrix of rank m such that $P_1K\,{}^tP_1 = M(m, 2s, s)$ and u is an m-dimensional row vector. The number of such matrices P is $q^m n(m, s; 2\nu)$.

Let

$$Q = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & {}^tv & 0 \end{pmatrix} \end{array},$$

where Q_1 is an $m \times 2\nu$ matrix of rank m satisfying $Q_1K\,{}^tQ_1 = M(m, 2s, s)$ and v is an m-dimensional row vector. Suppose that P and Q represent the

same subspace of type $(m, 2s, s, 0)$, then there is an element $T \in GL_m(\mathbb{F}_q)$ such that $Q = TP$. Thus $Q_1 = TP_1$ and ${}^t v = T\,{}^t u$. From $P_1 K\,{}^t P_1 = Q_1 K\,{}^t Q_1 = M(m, 2s, s)$ we deduce $T \in Sp_{m,s}(\mathbb{F}_q)$. Since P is of rank m, $TP = P$ implies $T = I$. Therefore

$$N(m, 2s, s, 0; 2\nu + 2) = \frac{q^m n(m, s; 2\nu)}{\mid Sp_{m,s}(\mathbb{F}_q) \mid}.$$

By Lemma 3.12

$$n(m, s; 2\nu) = \mid Sp_{m,s}(\mathbb{F}_q) \mid N(m, s; 2\nu).$$

Therefore

$$\begin{aligned} N(m, 2s, s, 0; 2\nu + 2) &= q^m N(m, s; 2\nu) \\ &= q^{2s(\nu+s-m)+m} \frac{\prod_{i=\nu+s-m+1}^{\nu} (q^{2i} - 1)}{\prod_{i=1}^{s}(q^{2i} - 1) \prod_{i=1}^{m-2s} (q^i - 1)}. \end{aligned}$$

(ii) $(\tau, \varepsilon) = (0, 1)$. Similar to cases (i) and (iii) when $\delta = 1$ we can prove

$$\begin{aligned} N(m, 2s, s, 1; 2\nu + 2) &= N(m - 1, s; 2\nu) \\ &= q^{2s(\nu+s-m+1)} \frac{\prod_{i=\nu+s-m+2}^{\nu} (q^{2i} - 1)}{\prod_{i=1}^{s}(q^{2i} - 1) \prod_{i=1}^{m-2s-1} (q^i - 1)}. \end{aligned}$$

(iii) $(\tau, \varepsilon) = (1, 0)$. Let $P \in \mathcal{M}(m, 2s + 1, s, 0; 2\nu + 2)$ and assume that $PS_2\,{}^t P = M(m, 2s + 1, s)$. By the proof of Theorem 4.11 there is an element $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that PT_1 has a matrix representation of the form

$$PT_1 = \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{c} \begin{array}{ccc} 2\nu & 1 & 1 \end{array} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \end{array} \begin{array}{l} 2s \\ 1 \\ m - 2s - 1 \end{array},$$

where

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \tag{4.24}$$

is of rank $m-1$ and

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-1, 2s, s). \tag{4.25}$$

Thus

$$P = \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_1^{-1}.$$

By Lemma 3.17 the number of $(m-1)\times 2\nu$ matrices (4.24) of rank $m-1$ which satisfy (4.25) is known to be $n(m-1, s; 2\nu)$. Denote by $\mathcal{K}_1$ the subgroup of $Ps_{2\nu+2}(\mathbb{F}_q)$ formed by all the elements of the form (4.18). Then given any $(m-1)\times 2\nu$ matrix (4.24) of rank $m-1$ which satisfy (4.25) and any $T_2 \in \mathcal{K}_1$ we have an $m \times (2\nu+2)$ matrix

$$\begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_2$$

of rank m such that

$$\left(\begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_2\right) S_2 \, {}^t\!\left(\begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_2\right) = M(m, 2s+1, s).$$

In this way obtain $n(m-1, s; 2\nu) \mid \mathcal{K}_1 \mid$ different matrices. Two such matrices

$$P = \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_2 \qquad \text{and} \qquad Q = \begin{pmatrix} Q_1 & 0 & 0 \\ 0 & 0 & 1 \\ Q_2 & 0 & 0 \end{pmatrix} T_3,$$

where both

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$$

are of rank $m-1$ and

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = M(m-1, 2s, s),$$

represent the same subspace of type $(m, 2s+1, s, 0)$ if and only if $P = TQ$ where $T \in GL_m(\mathbb{F}_q)$. We have $PS_2 \, {}^tP = QS_2 \, {}^tQ = M(m, 2s+1, s)$ and hence $T \in Ps_{m,2s+1}(\mathbb{F}_q)$. Furthermore, $TP = P$ implies $T = I$. Therefore

$$N(m, 2s+1, s, 0; 2\nu+2) = \frac{n(m-1, s; 2\nu) \mid \mathcal{K}_1 \mid}{\mid Ps_{m,2s+1}(\mathbb{F}_q) \mid}.$$

It follows from Lemma 3.17, Theorem 4.8, and $\mid \mathcal{K}_1 \mid = q^{2\nu+1}$ that

$$N(m, 2s+1, s, 0; 2\nu+2)$$
$$= q^{2s(\nu+s-m+2)+2(\nu+1)-m} \frac{\prod_{i=\nu+s-m+2}^{\nu} (q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s-1} (q^i-1)}.$$

(iv) $(\tau, \varepsilon) = (2, 0)$. This case can be treated in the same way as case (iii) above, but the details will be omitted. We have

$$N(m, 2s+2, s, 0; 2\nu+2) = \frac{n(m-1, s; 2\nu) \mid \mathcal{K}_1 \mid}{\mid Ps_{m,2s+2}(\mathbb{F}_q) \mid}$$
$$= q^{2s(\nu+s-m+2)+2(\nu+1)-m} \frac{\prod_{i=\nu+s-m+2}^{\nu} (q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s-2} (q^i-1)}.$$

(v) $(\tau, \varepsilon) = (2, 1)$. Let $P \in \mathcal{M}(m, 2s+2, s, 1; 2\nu+2)$ and assume that $PS_2\,{}^tP = M(m, 2s+2, s)$. By the proofs of Theorems 4.11 and 4.12 there is an element $T_1 \in Ps_{2\nu+2}(\mathbb{F}_q)$ of the form (4.18) such that PT_1 is of the form

$$PT_1 = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ u_2P_2 & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} \end{array} \begin{matrix} 2s \\ 1 \\ 1 \\ m-2s-2 \end{matrix},$$

where

$$\operatorname{rank} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = m-2,$$

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} K \, {}^t\!\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = M(m-2, 2s, s),$$

and

$$u_2 \in \mathbb{F}_q^{(m-2s-2)}.$$

Therefore

$$P = \begin{pmatrix} P_1 & 0 & 0 \\ u_2 P_2 & 1 & 0 \\ 0 & 0 & 1 \\ P_2 & 0 & 0 \end{pmatrix} T_1^{-1}.$$

Proceeding in the same way as case (iii) above, we obtain

$$N(m, 2s+2, s, 1; 2\nu+2) = \frac{n(m-2, s; 2\nu) q^{m-2s-2} \mid \mathcal{K}_1 \mid}{\mid Ps_{m,2s+2}(\mathbb{F}_q) \mid}$$

$$= q^{2s(\nu+s-m+2)+2(\nu+1)-m} \frac{\prod_{i=\nu+s-m+3}^{\nu} (q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-2s-2} (q^i-1)}.$$

The details of case (v) will be omitted. □

Corollary 4.15. Let $m \leq \nu+\delta-1$. Then the number of totally isotropic subspaces of dimension m in the $(2\nu+\delta)$-dimensional pseudo-symplectic space is given by

$$N(m, 0, 0, 0; 2\nu+1) = \frac{\prod_{i=0}^{m-1} (q^{2(\nu-i)}-1)}{\prod_{i=1}^{m}(q^i-1)}, \quad \text{when } \delta = 1,$$

and

$$N(m, 0, 0, 0; 2\nu+2) + N(m, 0, 0, 1; 2\nu+2)$$

$$= \frac{(q^{2\nu+2-m}-1) \prod_{i=1}^{m-1} (q^{2\nu+2-2i}-1)}{\prod_{i=1}^{m}(q^i-1)}, \quad \text{when } \delta = 2,$$

respectively. □

4.3 Singular Pseudo-symplectic Geometry over Finite Fields of Characteristic 2

The singular pseudo-symplectic group $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$ defined in Section 4.1 has an action on the vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ defined as follows:

$$\begin{aligned}\mathbb{F}_q^{(2\nu+\delta+l)} \times Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q) &\to \mathbb{F}_q^{(2\nu+\delta+l)}\\ ((x_1, x_2, \ldots, x_{2\nu+\delta+l}), T) &\mapsto (x_1, x_2, \ldots, x_{2\nu+\delta+l})T.\end{aligned}$$

The vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ together with this action is called the *singular pseudo-symplectic space* of dimension $2\nu+\delta+l$ over $\mathbb{F}_q$. This action induces an action on the set of subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$ such that an element $T \in Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$ carries a subspace P into PT. Now we are going to study how the subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$ are divided into orbits under the action of $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$ and to compute the length of each orbit.

An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is said to be of type $(m, 2s+\tau, s, \varepsilon)$, where $\tau = 0, 1$, or 2 and $\varepsilon = 0$ or 1, if $PS_{\delta,l}\,{}^tP$ is cogredient to $M(m, 2s+\tau, s)$ and P does not or does contain a vector of the form

$$\begin{cases} (\underbrace{0, 0, \ldots, 0}_{2\nu}, 1, x_{2\nu+2}, \ldots, x_{2\nu+1+l}), & \text{when } \delta = 1,\\ (\underbrace{0, 0, \ldots, 0}_{2\nu}, 1, 0, x_{2\nu+3}, \ldots, x_{2\nu+2+l}), & \text{when } \delta = 2, \end{cases} \tag{4.26}$$

corresponding to the cases $\varepsilon = 0$ or 1, respectively. Denote the set of subspaces of type $(m, 2s+\tau, s, \varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$ by $\mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta)$ and let

$$\begin{aligned}&N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta)\\ &\quad = \mid \mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta) \mid .\end{aligned}$$

Let E be the subspace of $\mathbb{F}_q^{(2\nu+\delta+l)}$ generated by $e_{2\nu+\delta+1}, \ldots, e_{2\nu+\delta+l}$. Then $\dim E = l$. An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is called a *subspace of type* $(m, 2s+\tau, s, \varepsilon, k)$ if

(i) P is a subspace of type $(m, 2s+\tau, s, \varepsilon)$, and

(ii) $\dim(P \cap E) = k$.

Denote by $\mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ the set of subspaces of type $(m, 2s+\tau, s, \varepsilon, k)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$ and let

$$\begin{aligned}&N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)\\ &\quad = \mid \mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \mid .\end{aligned}$$

At first, we have

Theorem 4.16. $\mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ is non-empty if and only if

$$\left.\begin{array}{l} (\tau,\varepsilon) = \left\{\begin{array}{ll} (0,0),(1,0),(1,1), \text{ or } (2,0), & \text{when } \delta=1,\\ (0,0),(0,1),(1,0),(2,0), \text{ or } (2,1), & \text{when } \delta=2, \end{array}\right.\\ \quad k \le l, \\ \text{and} \\ \quad 2s+\max\{\tau,\varepsilon\} \le m-k \le \nu+s+[(\tau+\delta-1)/2]+\varepsilon \end{array}\right\} \tag{4.27}$$

hold simultaneously, and if and only if

$$\left.\begin{array}{l} (\tau,\varepsilon) = \left\{\begin{array}{ll} (0,0),(1,0),(1,1), \text{ or } (2,0), & \text{when } \delta=1,\\ (0,0),(0,1),(1,0),(2,0), \text{ or } (2,1), & \text{when } \delta=2, \end{array}\right.\\ \text{and} \\ \quad \max\{0, m-\nu-s-[(\tau+\delta-1)/2]-\varepsilon\} \le k \\ \qquad \le \min\{l, m-2s-\max\{\tau,\varepsilon\}\} \end{array}\right\} \tag{4.28}$$

hold simultaneously.

Proof. That (4.27) and (4.28) are equivalent is obvious.

Now suppose that (4.27) holds. From the first and third condition of (4.27), by Theorem 4.11 we deduce that there is a subspace P of type $(m-k, 2s+\tau, s, \varepsilon)$ in the $(2\nu+\delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$. The matrix representation P of the subspace P is an $(m-k)\times(2\nu+\delta)$ matrix. Since $k \le l$,

$$\begin{array}{c} \begin{array}{ccc} 2\nu+\delta & k & l-k \end{array} \\ \left(\begin{array}{ccc} P & 0 & 0 \\ 0 & I^{(k)} & 0 \end{array}\right) \begin{array}{c} m-k \\ k \end{array} \end{array}$$

is a subspace of type $(m, 2s+\tau, s, \varepsilon, k)$ in the $(2\nu+\delta+l)$-dimensional singular pseudo-symplectic space over $\mathbb{F}_q$. The sufficiency of (4.27) is thus proved.

Conversely, suppose that P is a subspace of type $(m, 2s+\tau, s, \varepsilon, k)$ in the $(2\nu+\delta+l)$-dimensional singular pseudo-symplectic space $\mathbb{F}_q^{(2\nu+\delta+l)}$. Since $P \cap E \subseteq E$ and $\dim(P \cap E) = k$, we have $k \le l$, which is the second condition of (4.27). Moreover, we can assume that P has a matrix representation of the form

$$P = \begin{array}{c} \begin{array}{cc} 2\nu+\delta & l \end{array} \\ \left(\begin{array}{cc} P_{11} & P_{12} \\ 0 & P_{22} \end{array}\right) \begin{array}{c} m-k \\ k \end{array} \end{array},$$

such that

$$\begin{aligned} \text{rank } P_{11} &= m-k, \\ \text{rank } P_{22} &= k, \\ P_{11}S_\delta\, {}^tP_{11} &= M(m-k, 2s+\tau, s), \end{aligned}$$

and $e_{2\nu+1} \notin P_{11}$ or $e_{2\nu+1} \in P_{11}$ corresponding to the cases $\varepsilon = 0$ or 1, respectively. Then P_{11} is a subspace of type $(m-k, 2s+\tau, s, \varepsilon)$ in the $(2\nu+\delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$. By Theorem 4.11 we have the first and third condition of (4.27). □

Corollary 4.17. $\mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta)$ is non-empty if and only if

$$\left.\begin{aligned} (\tau,\varepsilon) &= \begin{cases} (0,0),(1,0),(1,1), \text{ or } (2,0), & \text{when } \delta=1, \\ (0,0),(0,1),(1,0),(2,0), \text{ or } (2,1), & \text{when } \delta=2, \end{cases} \\ &\text{and} \\ &\max\{0, m-\nu-s-[(\tau+\delta-1)/2]-\varepsilon\} \\ &\quad \le \min\{l, m-2s-\max\{\tau,\varepsilon\}\} \end{aligned}\right\} \quad (4.29)$$

hold simultaneously. □

Theorem 4.18. Let (4.28) be satisfied. Then $\mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ is an orbit of subspaces under $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$.

Proof. Let $P \in \mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ and we can assume that $PS_{\delta,l}\, {}^tP = M(m, 2s+\tau, s)$. Let $T \in Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$, then $TS_{\delta,l}\, {}^tT = S_{\delta,l}$ and, hence, $(PT)S_{\delta,l}\, {}^t(PT) = M(m, 2s+\tau, s)$. Since T is of the form (4.3), by Corollaries 4.4 and 4.6 a vector of the form (4.26) is carried by T into a vector of the same form and $ET = E$. Hence $\dim(PT \cap E) = \dim(PT \cap ET) = \dim(P \cap E)T = \dim P \cap E$. Therefore $PT \in \mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$.

On the other hand, let $P, Q \in \mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$. By the proof of Theorem 4.16 we can assume that P has a matrix representation of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu+\delta & l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array},$$

such that

$$\begin{aligned} \text{rank } P_{11} &= m-k, \\ \text{rank } P_{22} &= k, \\ P_{11} S_\delta {}^t P_{11} &= M(m-k, 2s+\tau, s), \end{aligned}$$

and $e_{2\nu+1} \notin P_{11}$ or $e_{2\nu+1} \in P_{11}$ corresponding to the cases $\varepsilon = 0$ or 1, respectively and that Q has a matrix representation of the form

$$Q = \begin{array}{c} \begin{matrix} 2\nu+\delta & l \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array},$$

such that

$$\begin{aligned} \text{rank } Q_{11} &= m-k, \\ \text{rank } Q_{22} &= k, \\ Q_{11} S_\delta {}^t Q_{11} &= M(m-k, 2s+\tau, s), \end{aligned}$$

and $e_{2\nu+1} \notin Q_{11}$ or $e_{2\nu+1} \in Q_{11}$ corresponding to the cases $\varepsilon = 0$ or 1, respectively. By Theorem 4.12 there is an element $T_{11} \in Ps_{2\nu+\delta}(\mathbb{F}_q)$ and an element $A_{11} \in GL_{m-k}(\mathbb{F}_q)$ such that $P_{11} = A_{11} Q_{11} T_{11}$. Since both P_{22} and Q_{22} are $k \times l$ matrices of rank k, by Theorem 1.1 there is an element $T_{22} \in GL_l(\mathbb{F}_q)$ such that $P_{22} = Q_{22} T_{22}$. Clearly

$$T = \begin{pmatrix} T_{11} & \\ & T_{22} \end{pmatrix} \in Ps_{2\nu+\delta+l, 2\nu+\delta}(\mathbb{F}_q)$$

and

$$AQT = \begin{pmatrix} P_{11} & A_{11} Q_{12} T_{22} \\ & P_{22} \end{pmatrix},$$

where

$$A = \begin{pmatrix} A_{11} & \\ & I^{(k)} \end{pmatrix}.$$

Since P_{11} is an $(m-k) \times (2\nu+\delta)$ matrix of rank $m-k$, we can find a $(2\nu+\delta) \times l$ matrix X such that

$$P_{11} X = P_{12} - A_{11} Q_{12} T_{22}.$$

Let

$$T_1 = \begin{pmatrix} I^{(2\nu+\delta)} & X \\ & I^{(l)} \end{pmatrix},$$

then $T_1 \in Ps_{2\nu+\delta+l, 2\nu+\delta}(\mathbb{F}_q)$ and $AQTT_1 = P$. □

Corollary 4.19. Let (4.29) be satisfied. Then $\mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta)$ is the disjoint union of orbits $\mathcal{M}(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ under $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$, where k runs through all the integers satisfying the second condition of (4.28). □

Theorem 4.20. Let (4.28) be satisfied. Then

$$\begin{aligned} &N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \\ &= q^{n_0+(m-k)(l-k)+2(s+(2-\delta)[\tau/2])(\nu+s-m+k+\delta[(\tau+1)/2]+(\delta-1)(\tau-1)(\tau-2)\varepsilon/2)} \\ &\quad \times \frac{\displaystyle\prod_{i=\nu+s-m+k+[(\tau+\delta-1)/2]+\varepsilon+1}^{\nu} (q^{2i}-1) \prod_{i=l-k+1}^{l} (q^i-1)}{\displaystyle\prod_{i=1}^{s}(q^{2i}-1) \prod_{i=1}^{m-k-2s-\max\{\tau,\varepsilon\}} (q^i-1) \prod_{i=1}^{k}(q^i-1)}, \end{aligned} \tag{4.30}$$

where $n_0 = 0$ if $\delta = 1$, and $n_0 = m-k, 0, 2(\nu+1)-m+k, 2(\nu+1)-m+k$, or $2(\nu+1)-m+k$ corresponding to the cases $(\tau, \varepsilon) = (0,0), (0,1), (1,0), (2,0)$, or $(2,1)$, respectively, if $\delta = 2$.

Proof. Denote by $n(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ the number of $m \times (2\nu+\delta+l)$ matrices P of the form

$$P = \begin{array}{c} \\ m-k \\ k \end{array} \begin{array}{c} \begin{array}{cc} 2\nu+\delta & l \end{array} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \end{array},$$

where

$$\begin{aligned} \operatorname{rank} P_{11} &= m-k, \\ \operatorname{rank} P_{22} &= k, \\ P_{11} S_\delta {}^t P_{11} &= M(m-k, 2s+\tau, s), \end{aligned}$$

and $e_{2\nu+1} \notin P_{11}$ or $e_{2\nu+1} \in P_{11}$ corresponding to the cases $\varepsilon = 0$ or 1, respectively. Denote by $n(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ the number of $(m-k) \times (2\nu+\delta)$ matrices P_{11} of rank $m-k$ such that $P_{11} S_\delta {}^t P_{11} = M(m-k, 2s+\tau, s)$ and $e_{2\nu+1} \notin P_{11}$ or $e_{2\nu+1} \in P_{11}$ corresponding to the cases $\varepsilon = 0$ or 1, respectively. Denote by $n(k,l)$ the number of $k \times l$ matrices of rank k. Then

$$\begin{aligned} &n(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = n(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta) q^{(m-k)l} n(k,l). \end{aligned}$$

Clearly

$$\begin{aligned} &n(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta) \\ &\quad = \mid Ps_{m-k,2s+\tau}(\mathbb{F}_q) \mid N(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta). \end{aligned}$$

By Lemma 1.5 we have

$$n(k,l) = q^{\frac{k(k-1)}{2}} \prod_{i=l-k+1}^{l} (q^i - 1).$$

Hence,

$$\begin{aligned} &n(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \\ &= \mid Ps_{m-k,2s+\tau}(\mathbb{F}_q) \mid N(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta) q^{(m-k)l+\frac{k(k-1)}{2}} \prod_{i=l-k+1}^{l} (q^i - 1). \end{aligned} \tag{4.31}$$

We recall that $\mid Ps_{m-k,2s+\tau}(\mathbb{F}_q) \mid$ is given by Theorem 4.8 and $N(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ is given by Theorem 4.14.

Now suppose that two $m \times (2\nu+\delta+l)$ matrices

$$P = \begin{array}{c} \begin{matrix} 2\nu+\delta & l \end{matrix} \\ \begin{pmatrix} P_{11} & P_{12} \\ 0 & P_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array} \quad \text{and} \quad \mathrm{Q} = \begin{array}{c} \begin{matrix} 2\nu+\delta & l \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ 0 & Q_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array}$$

represent the same subspace, where

$$\left.\begin{array}{rcl} \text{rank } P_{11} &=& \text{rank } Q_{11} = m-k, \\ \text{rank } P_{22} &=& \text{rank } Q_{22} = k, \\ P_{11} S_\delta {}^t P_{11} &=& Q_{11} S_\delta {}^t Q_{11} = M(m-k, 2s+\tau, s), \end{array}\right\} \tag{4.32}$$

and $e_{2\nu+1}$ is not or is contained in both P_{11} and Q_{11} corresponding to cases $\varepsilon = 0$ or 1, respectively. Then there is an $m \times m$ nonsingular matrix T such that $P = TQ$. T is necessarily of the form

$$T = \begin{array}{c} \begin{matrix} m-k & k \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} m-k \\ k \end{matrix} \end{array}, \tag{4.33}$$

where both T_{11} and T_{22} are nonsingular. Thus $P_{11} = T_{11} Q_{11}$. From the third condition in (4.32) it follows that

$$T_{11} M(m-k, 2s+\tau, s) \, {}^t T_{11} = M(m-k, 2s+\tau, s). \tag{4.34}$$

The set of $(m-k)\times(m-k)$ nonsingular matrices T_{11} satisfying (4.34) is the singular pseudo-symplectic group $Ps_{m-k,2s+\tau}(\mathbb{F}_q)$. All $m\times m$ nonsingular matrices T of the form (4.33) in which $T_{11}\in Ps_{m-k,2s+\tau}(\mathbb{F}_q)$ form a group, denoted by $\mathcal{G}$. Clearly

$$\mid \mathcal{G} \mid = \mid Ps_{m-k,2s+\tau}(\mathbb{F}_q) \mid\mid GL_k(\mathbb{F}_q) \mid q^{(m-k)k}. \tag{4.35}$$

Moreover, if $TQ = Q$ for $T\in\mathcal{G}$, then we have necessarily $T = I$. Consequently,

$$\begin{aligned} &n(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = \mid \mathcal{G} \mid N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta). \end{aligned} \tag{4.36}$$

From (4.31), (4.35), and (4.36), we deduce

$$\begin{aligned} &N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta) \\ &= \frac{N(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta)}{\mid GL_k(\mathbb{F}_q) \mid} q^{(m-k)(l-k)+\frac{k(k-1)}{2}} \prod_{i=l-k+1}^{l} (q^i - 1). \end{aligned} \tag{4.37}$$

Substituting the formula of $GL_k(\mathbb{F}_q)$ in Theorem 1.6 and the formula of $N(m-k, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ in Theorem 4.14 into (4.37), we obtain (4.30). □

Corollary 4.21. Let (4.29) be satisfied. Then

$$\begin{aligned} &N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = \sum_k N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta), \end{aligned} \tag{4.38}$$

where k runs from $\max\{0, m-\nu-s-[(\tau+\delta-1)/2]-\varepsilon\}$ to $\min\{l, m-2s-\max\{\tau,\varepsilon\}\}$ and $N(m, 2s+\tau, s, \varepsilon, k; 2\nu+\delta+l, 2\nu+\delta)$ is given by (4.30). □

For latter purpose, let us introduce

$$\begin{aligned} &\mathcal{M}(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = \bigcup_{\varepsilon=0,1} \mathcal{M}(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta) \end{aligned} \tag{4.39}$$

and

$$\begin{aligned} &N(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = \mid \mathcal{M}(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta) \mid . \end{aligned} \tag{4.40}$$

From Corollary 4.17 we deduce

Corollary 4.22. $\mathcal{M}(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta)$ is non-empty if and only if there is an $\varepsilon = 0$ or 1 such that (4.29) holds. □

If (4.29) does not hold for $\varepsilon = 0$ and also for $\varepsilon = 1$, then $N(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta) = 0$. However, if there is an $\varepsilon = 0$ or 1 such that (4.29) holds, then from (4.39) and (4.40) we deduce

Corollary 4.23. Assume that there is an $\varepsilon = 0$ or 1 such that (4.29) is satisfied. Then

$$\begin{aligned} &N(m, 2s+\tau, s; 2\nu+\delta+l, 2\nu+\delta) \\ &\quad = \sum_{\varepsilon=0,1} N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta), \end{aligned} \tag{4.41}$$

where $N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta)$ is given by (4.38) if (4.29) holds, and $N(m, 2s+\tau, s, \varepsilon; 2\nu+\delta+l, 2\nu+\delta) = 0$ if (4.29) does not hold. □

4.4 Pseudo-symplectic Geometry over Finite Fields of Characteristic 2 Again

In the present section let us consider again the $(2\nu+\delta)$-dimensional pseudo-symplectic space $\mathbb{F}_q^{(2\nu+\delta)}$ on which the pseudo-symplectic group $Ps_{2\nu+\delta}(\mathbb{F}_q)$ acts, where $\delta = 1$ or 2 as before.

Let P be a fixed subspace of type $(m, 2s+\tau, s, \varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Denote by $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ the set of subspaces of type $(m_1, 2s_1+\tau_1, s_1, \varepsilon_1)$ contained in P. Let

$$\begin{aligned} &N(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta) \\ &\quad = \mid \mathcal{M}(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta) \mid . \end{aligned}$$

It follows from Theorem 4.12 that $N(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ is independent of the particular choice of the subspace P and depends only on its type $(m, 2s+\tau, s, \varepsilon)$. At first we have

Theorem 4.24. Assume that subspaces of type $(m, 2s+\tau, s, \varepsilon)$ exist in the $(2\nu+\delta)$-dimensional pseudo-symplectic space $\mathbb{F}_q^{(2\nu+\delta)}$. Then

(i) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+\tau, s, 0; 2\nu+\delta) = \phi$.

(ii) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+\tau, s, 0; 2\nu+\delta) \neq \phi$ if and only if

$$\left.\begin{array}{l} \text{(a) both} \\ \qquad \tau_1 = 0 \\ \text{and} \\ \qquad \max\{0, m_1 - s - s_1\} \le \min\{m-2s, m_1-2s_1\} \\ \text{holds, when } \tau = 0, \text{ or} \\ \text{(b) there is an } \varepsilon_1 = 0 \text{ or } 1 \text{ such that} \\ (\tau_1, \varepsilon_1) = \begin{cases} (0,0), (1,0), (1,1), \text{ or } (2,0), & \text{when } \tau = 1, \\ (0,0), (0,1), (1,0), (2,0), \text{ or } (2,1), & \text{when } \tau = 2, \end{cases} \\ \text{and} \\ \qquad \max\{0, m_1 - s - s_1 - [(\tau_1+\tau-1)/2] - \varepsilon_1\} \\ \qquad\quad \le \min\{m-2s-\tau, m_1-2s_1-\max\{\tau_1, \varepsilon_1\}\}. \end{array}\right\} \tag{4.42}$$

(iii) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+\tau, s, 1; 2\nu+1) \neq \phi$ if and only if

$$\left.\begin{array}{l} \tau = 1 \\ \text{and} \\ \max\{0, m_1 - s - (s_1 + [\tau_1/2])\} \\ \quad \le \min\{m-1-2s, m_1 - 2(s_1+[\tau_1/2])\}. \end{array}\right\} \tag{4.43}$$

(iv) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+\tau, s, 1; 2\nu+2) \neq \phi$ if and only if

$$\left.\begin{array}{l} 2s_1 + \tau_1 \le m_1, \\ (\tau, \tau_1) = (0,0), (2,0), (2,1), \text{ or } (2,2), \\ \text{and} \\ \max\{0, m_1 - [(\tau_1+1)/2] - s - s_1\} \\ \le \min\{m - [(\tau+2)/2] - 2s, m_1 - [(\tau_1+1)/2] - 2s_1\}. \end{array}\right\} \tag{4.44}$$

(v) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+\tau, s, 1; 2\nu+1) \neq \phi$ if and only if

$$\left.\begin{array}{l} \tau = \tau_1 = 1 \\ \text{and} \\ \max\{0, m_1 - 1 - s - s_1\} \\ \quad \le \max\{m-1-2s, m_1-1-2s_1\}. \end{array}\right\} \tag{4.45}$$

(vi) $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+\tau, s, 1; 2\nu+2) \neq \phi$ if and only if

$$\left.\begin{array}{l} (\tau, \tau_1) = (0,0), (2,0), \text{ or } (2,2), \\ \text{and} \\ \max\{0, m_1 - (\tau_1+2)/2 - s - s_1\} \\ \quad \le \min\{m - (\tau+2)/2 - 2s, m_1 - (\tau_1+2)/2 - 2s_1\}. \end{array}\right\} \tag{4.46}$$

Proof. (i) is obvious.

(ii) Suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+\tau, s, 0; 2\nu+\delta)$ is non-empty. Then there is a subspace P of type $(m, 2s+\tau, s, 0)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ and a subspace Q of type $(m_1, 2s_1+\tau_1, s_1, 0)$ contained in P. We can assume that

$$PS_\delta\,{}^tP = M(m, 2s+\tau, s).$$

Q, being a subspace of P, can be represented by an $m_1 \times m$ matrix Q of rank m_1. Then Q, being a subspace of $\mathbb{F}_q^{(2\nu+\delta)}$, is represented by the $m_1 \times (2\nu+\delta)$ matrix QP. We can also assume that

$$(QP)S_\delta\,{}^t(QP) = M(m_1, 2s_1+\tau_1, s_1).$$

Thus

$$QM(m, 2s+\tau, s)\,{}^tQ = M(m_1, 2s_1+\tau_1, s_1),$$

i.e., $Q \in \mathcal{M}(m_1, 2s_1+\tau_1, s_1; 2s+\tau+(m-2s-\tau), 2s+\tau)$. By Corollaries 3.23 and 4.22 we have (4.42).

Conversely, suppose that (4.42) holds. Let P be a subspace of type $(m, 2s+\tau, s, 0)$ and assume that P is represented by an $m \times (2\nu+\delta)$ matrix P such that

$$PS_\delta\,{}^tP = M(m, 2s+\tau, s).$$

By Corollaries 3.23 and 4.22, condition (4.42) implies that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1; 2s+\tau+(m-2s-\tau), 2s+\tau)$ is non-empty. Let $Q \in \mathcal{M}(m_1, 2s_1+\tau_1, s_1; 2s+\tau+(m-2s-\tau), 2s+\tau)$. We can assume that Q is represented by an $m_1 \times m$ matrix Q of rank m_1 such that

$$QM(m, 2s+\tau, s)\,{}^tQ = M(m_1, 2s_1+\tau_1, s_1).$$

Thus

$$(QP)S_\delta\,{}^t(QP) = M(m_1, 2s_1+\tau_1, s_1).$$

Since $e_{2\nu+1} \notin P$, $e_{2\nu+1} \notin QP$ either. Therefore QP is a subspace of type $(m_1, 2s_1+\tau_1, s_1, 0)$ and contained in P. Hence $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+\tau, s, 0; 2\nu+\delta)$ is non-empty.

(iii) Let P be a subspace of type $(m, 2s+\tau, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. By Theorem 4.11 we have necessarily $\tau = 1$. Hence P is a subspace of type

$(m, 2s+1, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. By the proof of Theorem 4.11 we can assume that P has a matrix representation of the form

$$P = \begin{array}{c} \\ \\ m-1 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 2\nu & 1 \end{array} \\ \begin{pmatrix} P_1 & 0 \\ 0 & 1 \end{pmatrix} \end{array},$$

where P_1 is an $(m-1) \times 2\nu$ matrix of rank $m-1$ such that

$$P_1 K\, {}^tP_1 = M(m-1, 2s, s).$$

Now suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+1, s, 1; 2\nu+1)$ is non-empty. Then there is a subspace Q of type $(m_1, 2s_1+\tau_1, s_1, 0)$ and contained in P. Q, being an m_1-dimensional subspace of P and not containing $e_{2\nu+1}$, can be represented by an $m_1 \times m$ matrix Q of the form

$$Q = \begin{array}{c} \\ \end{array} \begin{array}{c} \begin{array}{cc} m-1 & 1 \end{array} \\ \begin{pmatrix} Q_1 & {}^tv \end{pmatrix} \end{array},$$

where Q_1 is an $m_1 \times (m-1)$ matrix of rank m_1 and

$$v = (\underbrace{0, 0, \ldots, 0}_{m_1}) \text{ or } (\underbrace{0, 0, \ldots, 0}_{m_1-1}, 1).$$

Then Q, being a subspace of $\mathbb{F}_q^{(2\nu+1)}$, is represented by the $m_1 \times (2\nu+1)$ matrix

$$QP = \begin{pmatrix} Q_1P_1 & {}^tv \end{pmatrix}.$$

Since Q is a subspace of type $(m_1, 2s_1+\tau_1, s_1, 0)$, $(QP)S_1\,{}^t(QP)$ is cogredient to $M(m_1, 2s_1+\tau_1, s_1)$. But

$$(QP)S_1\,{}^t(QP) = Q_1 M(m-1, 2s, s)\,{}^tQ_1 + {}^tvv.$$

Therefore $v = (0, 0, \ldots, 0)$ if and only if $\tau_1 = 0$, and $v = (0, 0, \ldots, 0, 1)$ if and only if $\tau_1 = 1$ or 2. If $\tau_1 = 0$ or 1, then $Q_1 \in \mathcal{M}(m_1, s_1; 2s+(m-1-2s), s)$ and hence by Corollary 3.23 we have

$$\max\{0, m_1 - s - s_1\} \leq \min\{m-1-2s, m_1-2s_1\}. \tag{4.47}$$

If $\tau_1 = 2$, then $Q_1 \in \mathcal{M}(m_1, s_1+1; 2s+(m-1-2s), s)$ and again by Corollary 3.23 we have

$$\max\{0, m_1 - s - (s_1+1)\} \leq \min\{m-1-2s, m_1-2(s_1+1)\}. \tag{4.48}$$

Combining (4.47) and (4.48), we obtain the second condition of (4.43).

Conversely, suppose that (4.43) holds. By Corollary 3.23 this implies that there is a $Q_1 \in \mathcal{M}(m_1, s_1 + [\tau_1/2]; 2s + (m-1-2s), s)$. We can assume that Q_1 is represented by an $m_1 \times (m-1)$ matrix Q_1 of rank m_1 such that

$$Q_1 M(m-1, 2s, s)\,{}^t Q_1 = M(m_1, 2(s_1 + [\tau_1/2]), s_1 + [\tau_1/2]).$$

Thus

$$Q_1 P_1 K\,{}^t(Q_1 P_1) = M(m_1, 2(s_1 + [\tau_1/2]), s_1 + [\tau_1/2]).$$

Let

$$Q = \begin{array}{c} \begin{matrix} m-1 & 1 & \end{matrix} \\ \begin{cases} (Q_1 & 0), & \text{if } \tau_1 = 0 \\ (Q_1 & {}^t e_{2s_1+1}), & \text{if } \tau_1 = 1, \\ (Q_1 & {}^t e_{2s_1+2}), & \text{if } \tau_1 = 2. \end{cases} \end{array}$$

Then $(QP)S_1\,{}^t(QP)$ is equal to or cogredient to $M(m_1, 2s_1 + \tau_1, s_1)$, if $\tau_1 = 0, 1$ or $\tau_1 = 2$, respectively. Clearly $e_{2\nu+1} \notin QP$. Hence QP is a subspace of type $(m_1, 2s_1 + \tau_1, s_1, 0)$ and contained in P. Therefore $\mathcal{M}(m_1, 2s_1 + \tau_1, s_1, 0; m, 2s+1, s, 1; 2\nu+1)$ is non-empty.

(iv) Let P be a subspace of type $(m, 2s + \tau, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By Theorem 4.11 we have $\tau = 0$ or 2. We distinguish these two cases.

(a) $\tau = 0$. Then P is a subspace of type $(m, 2s, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By the proof of Theorem 4.11 we can assume that P has a matrix representation of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{matrix} m-1 \\ 1 \end{matrix} \end{array},$$

where P_1 is an $(m-1) \times 2\nu$ matrix of rank $m-1$ and

$$P_1 K\,{}^t P_1 = M(m-1, 2s, s).$$

Now suppose that $\mathcal{M}(m_1, 2s_1 + \tau_1, s_1, 0; m, 2s, s, 1; 2\nu+2)$ is non-empty. Then there is a subspace Q of type $(m_1, 2s_1 + \tau_1, s_1, 0)$ contained in P. Thus $2s_1 + \tau_1 \le m_1$. Q, being an m_1-dimensional subspace of P and not containing $e_{2\nu+1}$, can be represented by an $m_1 \times m$ matrix Q of the form

$$Q = \begin{array}{c} \begin{matrix} m-1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & {}^t v \end{pmatrix} \end{array},$$

where Q_1 is an $m_1 \times (m-1)$ matrix of rank m_1 and v is an m_1-dimensional row vector. Then, Q, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, is represented by the $m_1 \times (2\nu+2)$ matrix

$$QP = (\ Q_1P_1 \quad {}^tv \quad 0\ ).$$

Since Q is a subspace of type $(m_1, 2s_1+\tau_1, s_1, 0)$, $(QP)S_2\,{}^t(QP)$ is cogredient to $M(m_1, 2s_1+\tau_1, s_1)$. Clearly,

$$(QP)S_2\,{}^t(QP) = Q_1M(m-1, 2s, s)\,{}^tQ_1.$$

Since $Q_1M(m-1, 2s, s)\,{}^tQ_1$ is alternate, $\tau_1 = 0$. Hence Q_1 is a subspace of type (m_1, s_1) in the singular symplectic space $\mathbb{F}_q^{(2s+(m-1-2s))}$. By Corollary 3.23, we have

$$\max\{0, m_1 - s - s_1\} \le \min\{m-1-2s, m_1-2s_1\}. \tag{4.49}$$

(b) $\tau = 2$. Then P is a subspace of type $(m, 2s+2, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By Theorem 4.12 and the proofs of Theorems 4.11 and 4.12, without loss of generality we can assume that P is of the following form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{matrix} m-2 \\ 1 \\ 1 \end{matrix} \end{array},$$

where P_1 is of rank $m-2$ and

$$P_1K\,{}^tP_1 = M(m-2, 2s, s).$$

Then

$$PS_2\,{}^tP = \begin{pmatrix} M(m-2, 2s, s) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}. \tag{4.50}$$

Now suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 0; m, 2s+2, s, 1; 2\nu+2) \neq \phi$. Then there is a subspace Q of type $(m_1, 2s_1+\tau_1, s_1, 0)$ contained in P. Thus $2s_1+\tau_1 \le m_1$. We distinguish the cases $\tau_1 = 0, 1$, and 2.

(b_0) $\tau_1 = 0$. Then Q is a subspace of type $(m_1, 2s_1, s_1, 0)$ contained in P. Thus Q consists of isotropic vectors with respect to S_2 and $e_{2\nu+1} \notin Q$. Therefore, being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q of the form

$$Q = \begin{array}{c} \begin{matrix} m-2 & 1 & 1 \end{matrix} \\ (\ Q_1 \quad {}^tv \quad 0\ ) \end{array},$$

where Q_1 is an $m_1 \times (m-2)$ matrix of rank m_1 and v is an m_1-dimensional row vector. Then, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu+2)$ matrix

$$QP = (\ Q_1P_1 \quad {}^t v \quad 0\ ).$$

Since Q is a subspace of type $(m_1, 2s_1, s_1, 0)$ in $\mathbb{F}_q^{(2\nu+2)}$, we can assume that

$$(QP)S_2\,{}^t(QP) = M(m_1, 2s_1, s_1).$$

Consequently,

$$Q_1 M(m-2, 2s, s)\,{}^t Q_1 = M(m_1, 2s_1, s_1).$$

That is, Q_1 is a subspace of type (m_1, s_1) in the singular symplectic space $\mathbb{F}_q^{(2s+(m-2-2s))}$. By Corollary 3.23, we have

$$\max\{0, m_1 - s - s_1\} \le \min\{m-2-2s, m_1 - 2s_1\}. \tag{4.51}$$

(b_1) $\tau_1 = 1$. Then Q is a subspace of type $(m_1, 2s_1+1, s_1, 0)$ contained in P. Being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q of rank m_1. Thus, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu+2)$ matrix QP. We can assume that

$$(QP)S_2\,{}^t(QP) = \begin{pmatrix} M(m_1-1, 2s_1, s_1) & \\ & 1 \end{pmatrix}. \tag{4.52}$$

Substituting the expression of $PS_2\,{}^tP$ in (4.50) into the above equation, we obtain

$$Q \begin{pmatrix} M(m-2, 2s, s) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix} {}^t Q = \begin{pmatrix} M(m_1-1, 2s_1, s_1) & \\ & 1 \end{pmatrix}.$$

It follows that Q is of the form

$$Q = \begin{array}{c} \begin{matrix} m-2 & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & {}^t u_1 & 0 \\ Q_2 & u_2 & 1 \end{pmatrix} \begin{matrix} m_1 - 1 \\ 1 \end{matrix} \end{array}.$$

Then

$$QP = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1P_1 & {}^t u_1 & 0 \\ Q_2P_1 & u_2 & 1 \end{pmatrix} \begin{matrix} m_1 - 1 \\ 1 \end{matrix} \end{array}.$$

Let

$$T = \begin{pmatrix} I^{(2\nu)} & K\,{}^tP_1\,{}^tQ_2 & \\ & 1 & \\ Q_2P_1 & u_2 & 1 \end{pmatrix},$$

then $T \in Ps_{2\nu+2}(\mathbb{F}_q)$ and

$$QPT = \begin{pmatrix} Q_1P_1 & Q_1P_1K\,{}^tP_1\,{}^tQ_2 + {}^tu_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Write $Q_1P_1K\,{}^tP_1\,{}^tQ_2 + {}^tu_1 = {}^tv_1$ for simplicity, then

$$\begin{aligned}(QPT)S_2\,{}^t(QPT) &= \begin{pmatrix} (Q_1P_1)K\,{}^t(Q_1P_1) & {}^tv_1 \\ v_1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} (Q_1M(m-2,2s,s)\,{}^tQ_1 & {}^tv_1 \\ v_1 & 1 \end{pmatrix}.\end{aligned}$$

On the other hand, from (4.52) we have

$$(QPT)S_2\,{}^t(QPT) = \begin{pmatrix} M(m_1-1,2s_1,s_1) & \\ & 1 \end{pmatrix}.$$

Therefore $v_1 = 0$ and

$$Q_1M(m-2,2s,s)\,{}^tQ_1 = M(m_1-1,2s_1,s_1).$$

Hence,

$$QPT = \begin{pmatrix} Q_1P_1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

from which it follows that rank $Q_1 = m_1 - 1$. Thus Q_1 is a subspace of type (m_1-1, s_1) in the $(2s+(m-2-2s))$-dimensional singular symplectic space. By Corollary 3.23, we have

$$\max\{0, m_1-1-s-s_1\} \le \min\{m-2-2s, m_1-1-2s_1\}. \quad (4.53)$$

(b_2) $\tau_1 = 2$. Then Q is a subspace of type $(m_1, 2s_1+2, s_1, 0)$ contained in P. Being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q of rank m_1. Thus, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu+2)$ matrix QP. We can assume that

$$(QP)S_2\,{}^t(QP) = \begin{pmatrix} M(m_1-2,2s_1,s_1) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}. \quad (4.54)$$

Substituting the expression of $PS_2\,{}^tP$ in (4.50) into the above equation, we obtain

$$Q\begin{pmatrix} M(m-2,2s,s) & & \\ & \begin{matrix}0&1\\1&1\end{matrix} & \\ & & \end{pmatrix}{}^tQ = \begin{pmatrix} M(m_1-2,2s_1,s_1) & & \\ & \begin{matrix}0&1\\1&1\end{matrix} & \\ & & \end{pmatrix}.$$

Thus we can assume that Q is of the form

$$Q = \begin{matrix} \\ Q_1 \\ Q_2 \\ Q_3 \end{matrix}\begin{matrix} \begin{matrix} m-2 \end{matrix} \\ \begin{pmatrix} Q_1 \\ Q_2 \\ Q_3 \end{pmatrix} \end{matrix}\begin{matrix} \begin{matrix} 1 & 1 \end{matrix} \\ \begin{pmatrix} {}^tu_1 & 0 \\ u_2 & 0 \\ u_3 & 1 \end{pmatrix} \end{matrix}\begin{matrix} \\ m_1-2 \\ 1 \\ 1 \end{matrix}.$$

Then

$$QP = \begin{matrix} \\ Q_1P_1 \\ Q_2P_1 \\ Q_3P_1 \end{matrix}\begin{matrix} \begin{matrix} 2\nu \end{matrix} \\ \begin{pmatrix} Q_1P_1 \\ Q_2P_1 \\ Q_3P_1 \end{pmatrix} \end{matrix}\begin{matrix} \begin{matrix} 1 & 1 \end{matrix} \\ \begin{pmatrix} {}^tu_1 & 0 \\ u_2 & 0 \\ u_3 & 1 \end{pmatrix} \end{matrix}\begin{matrix} \\ m_1-2 \\ 1 \\ 1 \end{matrix}.$$

Let

$$T = \begin{pmatrix} I^{(2\nu)} & K\,{}^tP_1\,{}^tQ_3 & \\ & 1 & \\ Q_3P_1 & u_3 & 1 \end{pmatrix},$$

then $T \in Ps_{2\nu+2}(\mathbb{F}_q)$ and

$$QPT = \begin{pmatrix} Q_1P_1 & Q_1P_1K\,{}^tP_1\,{}^tQ_3 + {}^tu_1 & \\ Q_2P_1 & Q_2P_1K\,{}^tP_1\,{}^tQ_3 + u_2 & \\ & & 1 \end{pmatrix}.$$

Write $Q_1P_1K\,{}^tP_1\,{}^tQ_3 + {}^tu_1 = {}^tv_1$ and $Q_2P_1K\,{}^tP_1\,{}^tQ_3 + u_2 = v_2$ for simplicity, then

$$\begin{aligned}(QPT)S_2\,{}^t(QPT) &= \begin{pmatrix} (Q_1P_1)K\,{}^t(Q_1P_1) & (Q_1P_1)K\,{}^t(Q_2P_1) & {}^tv_1 \\ (Q_2P_1)K\,{}^t(Q_1P_1) & 0 & v_2 \\ v_1 & v_2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} Q_1M(m{-}2,2s,s)\,{}^tQ_1 & Q_1M(m{-}2,2s,s)\,{}^tQ_2 & {}^tv_1 \\ Q_2M(m{-}2,2s,s)\,{}^tQ_1 & 0 & v_2 \\ v_1 & v_2 & 1 \end{pmatrix}.\end{aligned}$$

On the other hand, from (4.54) we have

$$(QPT)S_2\,{}^t(QPT) = \begin{pmatrix} M(m_1-2,2s_1,s_1) & & \\ & \begin{matrix}0&1\\1&1\end{matrix} & \\ & & \end{pmatrix}.$$

Therefore $v_1 = 0$, $v_2 = 1$, and

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} M(m-2, 2s, s) \, {}^t\!\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = M(m_1 - 1, 2s_1, s_1). \tag{4.55}$$

Hence,

$$QPT = \begin{pmatrix} Q_1P_1 & 0 & 0 \\ Q_2P_1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{4.56}$$

Since Q is of rank m_1, from (4.56) we deduce rank $Q_1 = m_1 - 2$. Since $e_{2\nu+1} \notin QP$ and $e_{2\nu+1}T = e_{2\nu+1}$, we have $e_{2\nu+1} \notin QPT$. Therefore

$$\operatorname{rank} \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = m_1 - 1.$$

Thus from (4.55) we know that the subspace

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$$

is of type $(m_1 - 1, s_1)$ in the singular symplectic space $\mathbb{F}_q^{(2s+(m-2-2s))}$. By Corollary 3.23, we have

$$\max\{0, m_1 - 1 - s - s_1\} \leq \min\{m - 2 - 2s, m_1 - 1 - 2s_1\}. \tag{4.57}$$

Assembling (4.49), (4.51), (4.53), and (4.57), we obtain the last condition in (4.44).

Conversely, suppose that (4.44) holds.

Consider first the case $\tau = 0$. Then $\tau_1 = 0$ and the last condition in (4.44) reduces to (4.49). By Corollary 3.23, $\mathcal{M}(m_1, s_1; 2s + (m - 1 - 2s), s)$ is non-empty. Let $Q_1 \in \mathcal{M}(m_1, s_1; 2s+(m-1-2s), s)$. We can assume that Q_1 is represented by an $m_1 \times (m-1)$ matrix Q_1 of rank m_1 such that

$$Q_1 M(m-1, 2s, s) \, {}^t Q_1 = M(m_1, 2s_1, s_1).$$

Let

$$\begin{array}{c} \qquad\quad m-1 \quad 1 \\ Q = \begin{pmatrix} Q_1 & 0 \end{pmatrix}, \end{array}$$

then

$$QP = \begin{pmatrix} Q_1P_1 & 0 & 0 \end{pmatrix}$$

and

$$(QP)S_2\,{}^t(QP) = M(m_1, 2s_1, s_1).$$

Hence QP is a subspace of type $(m_1, 2s_1, s_1, 0)$ contained in P.

Next consider the case $\tau = 2$. Now the last condition in (4.44) becomes

$$\begin{aligned}&\max\{0, m_1 - [(\tau_1+1)/2] - s - s_1\}\\ &\quad\le \min\{m-2-2s, m_1 - [(\tau_1+1)/2] - 2s_1\}.\end{aligned}$$

By Corollary 3.23, there is a $Q_1 \in \mathcal{M}(m_1 - [(\tau_1+1)/2], s_1; 2s + (m-2-2s), s)$. We can assume that

$$Q_1 M(m-2, 2s, s)\,{}^tQ_1 = M(m_1 - [(\tau_1+1)/2], 2s_1, s_1).$$

Let

$$Q = \begin{cases} \overset{m-2}{(Q_1} \quad \overset{1}{0} \quad \overset{1}{0}), & \text{if } \tau_1 = 0,\\[2ex] \begin{pmatrix} Q_1 & 0 & 0\\ 0 & 0 & 1\end{pmatrix} \begin{matrix} m_1-1\\ 1\end{matrix}, & \text{if } \tau_1 = 1,\\[2ex] \begin{pmatrix} Q_1 & {}^tv & 0\\ 0 & 0 & 1\end{pmatrix} \begin{matrix} m_1-1\\ 1\end{matrix}, & \text{if } \tau_1 = 2,\end{cases}$$

where

$$v = (\underbrace{0, 0, \ldots, 0}_{m_1-2}, 1).$$

Since $2s_1 + \tau_1 \le m_1$, it is easy to verify that QP is a subspace of type $(m_1, 2s_1 + \tau_1, s_1, 0)$ contained in P.

The proof of case (iv) is now completed.

(v) Let P be a subspace of type $(m, 2s+\tau, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. By Theorem 4.11 we have necessarily $\tau = 1$. Hence P is a subspace of type $(m, 2s+1, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. By the proof of Theorem 4.11 we can assume that P has a matrix representation of the form

$$P = \begin{pmatrix} P_1 & 0\\ 0 & 1\end{pmatrix} \begin{matrix} m-1\\ 1\end{matrix},$$

$$\overset{2\nu}{\quad} \quad \overset{1}{\quad}$$

where P_1 is an $(m-1)\times 2\nu$ matrix of rank $m-1$ such that

$$P_1 K\, {}^tP_1 = M(m-1, 2s, s).$$

Now suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+1, s, 1; 2\nu+1)$ is non-empty. Again by Theorem 4.11, we have necessarily $\tau_1 = 1$ and there is a subspace Q of type $(m_1, 2s_1+1, s_1, 1)$ and contained in P. Being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q. Since $e_{2\nu+1} \in Q$, we can assume that

$$Q = \begin{array}{c} \begin{matrix} m-1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 \\ 0 & 1 \end{pmatrix} \begin{matrix} m_1-1 \\ 1 \end{matrix} \end{array},$$

where Q_1 is an $(m_1-1)\times(m-1)$ matrix of rank m_1-1. Then, being a subspace of $\mathbb{F}_q^{(2\nu+1)}$, Q is represented by the $m_1 \times (2\nu+1)$ matrix

$$QP = \begin{array}{c} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{pmatrix} Q_1P_1 & 0 \\ 0 & 1 \end{pmatrix} \begin{matrix} m-1 \\ 1 \end{matrix} \end{array}.$$

We have

$$\begin{aligned} (QP)S_1\, {}^t(QP) &= \begin{pmatrix} Q_1P_1K\, {}^t(Q_1P_1) & \\ & 1 \end{pmatrix} \\ &= \begin{pmatrix} Q_1M(m-1, 2s, s)\, {}^tQ_1 & \\ & 1 \end{pmatrix}. \end{aligned}$$

Since Q is of type $(m_1, 2s_1+1, s_1, 1)$, $(QP)S_1\, {}^t(QP)$ is cogredient to $M(m_1, 2s_1+1, s_1)$. Hence $Q_1M(m-1, 2s, s)\, {}^tQ_1$ is cogredient to $M(m_1-1, 2s_1, s_1)$. That is, $Q_1 \in \mathcal{M}(m_1-1, s_1; 2s+(m-1-2s), s)$. Therefore the second condition in (4.45) follows from Corollary 3.23.

Conversely, suppose that (4.45) holds. By Corollary 3.23, there is a $Q_1 \in \mathcal{M}(m_1-1, s_1; 2s+(m-1-2s), s)$. We can assume that Q_1 is represented by an $(m_1-1)\times(m-1)$ matrix Q_1 of rank m_1-1 such that

$$Q_1M(m-1, 2s, s)\, {}^tQ_1 = M(m_1-1, 2s_1, s_1).$$

Let

$$Q = \begin{pmatrix} Q_1 & 0 \\ 0 & 1 \end{pmatrix},$$

then QP is an $m_1 \times (2\nu+1)$ matrix of rank m_1 such that $e_{2\nu+1} \in QP$ and

$$(QP)S_1\,{}^t(QP) = \begin{pmatrix} M(m_1-1, 2s_1, s_1) & \\ & 1 \end{pmatrix}.$$

That is, $QP \in \mathcal{M}(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1)$.

(vi) Let P be a subspace of type $(m, 2s+\tau, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By Theorem 4.11, we have $\tau = 0$ or 2. We distinguish these two cases.

(a) $\tau = 0$. Then P is a subspace of type $(m, 2s, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. By the proof of Theorem 4.11 we can assume that P is of the form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{matrix} m-1 \\ 1 \end{matrix} \end{array},$$

where P_1 is an $(m-1) \times 2\nu$ matrix of rank $m-1$ and $P_1 K\,{}^t P_1 = M(m-1, 2s, s)$.

Now suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s, s, 1; 2\nu+2) \neq \phi$. We assert that $\tau_1 = 0$. In fact, by Theorem 4.11 we have $\tau_1 = 0$ or 2. If $\tau_1 = 2$, let Q be a subspace of type $(m_1, 2s_1+2, s_1, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$, then there is a vector in Q whose $(2\nu+2)$-th coordinate is 1, hence $Q \not\subset P$. Therefore we must have $\tau_1 = 0$. Let Q be a subspace of type $(m_1, 2s_1, s_1, 1)$ contained in P. We can assume that being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q of the form

$$Q = \begin{array}{c} \begin{matrix} m-1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 \\ 0 & 1 \end{pmatrix} \begin{matrix} m_1-1 \\ 1 \end{matrix} \end{array},$$

where Q_1 is an $(m_1-1) \times (m-1)$ matrix of rank m_1-1. Then being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu+2)$ matrix

$$QP = \begin{pmatrix} Q_1 P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Thus

$$QPS_2\,{}^t(QP) = \begin{pmatrix} Q_1 M(m-1, 2s, s)\,{}^t Q_1 & \\ & 0 \end{pmatrix}.$$

Since Q is a subspace of type $(m_1, 2s_1, s_1, 1)$, we can assume that

$$Q_1 M(m-1, 2s, s)\,{}^t Q_1 = M(m_1-1, 2s_1, s_1).$$

That is, Q_1 is a subspace of type (m_1-1, s_1) in the $(2s+(m-1-2s))$-dimensional singular symplectic space. By Corollary 3.23,

$$\max\{0, m_1-1-s-s_1\} \le \min\{m-1-2s, m_1-1-2s_1\}. \quad (4.58)$$

(b) $\tau = 2$. Then P is a subspace of type $(m, 2s+2, s, 1)$ in $\mathbb{F}_q^{(2\nu+2)}$. As in case (iv), without loss of generality we can assume that P is of the following form

$$P = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{matrix} m-2 \\ 1 \\ 1 \end{matrix} \end{array},$$

where P_1 is of rank $m-2$ and

$$P_1 K\, {}^tP_1 = M(m-2, 2s, s).$$

Then we have (4.50)

$$PS_2\, {}^tP = \begin{pmatrix} M(m-2, 2s, s) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}.$$

Now suppose that $\mathcal{M}(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+2, s, 1; 2\nu+2) \neq 0$. By Theorem 4.11 we have necessarily $\tau_1 = 0$ or 2. We distinguish these two cases.

(b_0) $\tau_1 = 0$. There is a subspace Q of type $(m_1, 2s_1, s_1, 1)$ contained in P. Then Q consists of isotropic vectors with respect to S_2 and $e_{2\nu+1} \in Q$. Therefore, being a subspace of P, Q is represented by an $m_1 \times m$ matrix of the form

$$Q = \begin{array}{c} \begin{matrix} m-2 & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{matrix} m_1-1 \\ 1 \end{matrix} \end{array},$$

where Q_1 is an $(m_1-1)\times(m-2)$ matrix of rank m_1-1. Then, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu+2)$ matrix

$$QP = \begin{pmatrix} Q_1P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

From (4.50) follows

$$(QP)S_2\, {}^t(QP) = \begin{pmatrix} Q_1 M(m-2, 2s, s)\, {}^tQ_1 & \\ & 0 \end{pmatrix}.$$

Since Q is of type $(m_1, 2s_1, s_1, 1)$, we can assume that

$$Q_1 M(m-2, 2s, s)\,{}^t Q_1 = M(m_1 - 1, 2s_1, s_1).$$

That is, Q_1 is a subspace of type $(m_1 - 1, s_1)$ in the singular symplectic space $\mathbb{F}_q^{(2s+(m-2-2s))}$. Hence by Corollary 3.23, we have

$$\max\{0, m_1 - 1 - s - s_1\} \le \min\{m - 2 - 2s, m_1 - 1 - 2s_1\}. \quad (4.59)$$

(b_2) $\tau_1 = 2$. There is a subspace Q of type $(m_1, 2s_1 + 2, s_1, 1)$ contained in P. Being a subspace of P, Q is represented by an $m_1 \times m$ matrix Q. Thus, being a subspace of $\mathbb{F}_q^{(2\nu+2)}$, Q is represented by the $m_1 \times (2\nu + 2)$ matrix QP. We can assume that

$$(QP) S_2\,{}^t(QP) = \begin{pmatrix} M(m_1 - 2, 2s_1, s_1) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}.$$

From (4.50) we deduce

$$Q \begin{pmatrix} M(m-2, 2s, s) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix} {}^t Q = \begin{pmatrix} M(m_1 - 2, 2s_1, s_1) & & \\ & 0 & 1 \\ & 1 & 1 \end{pmatrix}.$$

Thus we can assume that Q is of the form

$$Q = \begin{array}{c} \begin{matrix} m-2 & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 & {}^t u_1 & 0 \\ Q_2 & u_2 & 0 \\ Q_3 & u_3 & 1 \end{pmatrix} \begin{matrix} m_1 - 2 \\ 1 \\ 1 \end{matrix} \end{array}.$$

Then

$$QP = \begin{array}{c} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} Q_1 P_1 & {}^t u_1 & 0 \\ Q_2 P_1 & u_2 & 0 \\ Q_3 P_1 & u_3 & 1 \end{pmatrix} \begin{matrix} m_1 - 2 \\ 1 \\ 1 \end{matrix} \end{array}.$$

Let

$$T = \begin{pmatrix} I^{(2\nu)} & K\,{}^t P_1\,{}^t Q_3 & \\ & 1 & \\ Q_3 P_1 & u_3 & 1 \end{pmatrix}.$$

Repeating the proof of subcase (b_2) of case $\tau = 2$ in (iv) we have

$$QPT = \begin{pmatrix} Q_1P_1 & 0 & 0 \\ Q_2P_1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$Q_1 M(m-2, 2s, s)^t Q_1 = M(m_1 - 2, 2s_1, s_1).$$

Thus rank $Q_1 = m_1 - 2$ and Q_1 is a subspace of type $(m_1 - 2, s_1)$ in the singular symplectic space $\mathbb{F}_q^{(2s+(m-2-2s))}$. By Corollary 3.23, we have

$$\max\{0, m_1 - 2 - s - s_1\} \le \min\{m - 2 - 2s, m_1 - 2 - 2s_1\}. \quad (4.60)$$

Assembling (4.58), (4.59), and (4.60), we obtain the second condition in (4.46).

Conversely, suppose that (4.46) holds. Similar to case (iv), we can prove that $\mathcal{M}(m_1, 2s_1 + \tau_1, s_1, 1; m, 2s + \tau, s, 1; 2\nu + 2) \neq \phi$; but the details will be omitted. □

From the proof of Theorem 4.24 follows

Theorem 4.25. Assume that subspaces of type $(m, 2s + \tau, s, \varepsilon)$ exist in $(2\nu + \delta)$-dimensional pseudo-symplectic space $\mathbb{F}_q^{(2\nu+\delta)}$. We have

(i) $N(m_1, 2s_1 + \tau_1, s_1, 1; m, 2s + \tau, s, 0; 2\nu + \delta) = 0$.

(ii) If (4.42) holds, then

$$\begin{aligned} &N(m_1, 2s_1 + \tau_1, s_1, 0; m, 2s + \tau, s, 0; 2\nu + \delta) \\ &\quad = N(m_1, 2s_1 + \tau_1, s_1; 2s + \tau + (m - 2s - \tau), 2s + \tau). \end{aligned}$$

(iii) If (4.43) holds, then

$$\begin{aligned} &N(m_1, 2s_1 + \tau_1, s_1, 0; m, 2s + 1, s, 1; 2\nu + 1) \\ &\quad = N(m_1, s_1 + [\tau_1/2]; 2s + (m - 1 - 2s), s). \end{aligned}$$

(iv) If (4.44) holds, then

$$\begin{aligned} &N(m_1, 2s_1 + \tau_1, s_1, 0; m, 2s + \tau, s, 1; 2\nu + 2) \\ &\quad = q^{n_1} N(m_1 - [(\tau_1 + 1)/2], s_1; 2s + (m - [(\tau + 2)/2] - 2s), s), \end{aligned}$$

where $n_1 = m_1, m_1, m - m_1 + 2s_1$, or $2(m - m_1 + s_1)$ corresponding to the cases $(\tau, \tau_1) = (0, 0), (2, 0), (2, 1)$, or $(2, 2)$, respectively.

(v) If (4.45) holds, then

$$N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) = N(m_1-1, s_1; 2s+(m-1-2s), s).$$

(vi) If (4.46) holds, then

$$N(m_1, 2s_1+\tau_1, s_1, 1; m, 2s+\tau, s, 1; 2\nu+2) = q^{n_2} N(m_1-[(\tau_1+2)/2], s_1; 2s+m-[(\tau+2)/2]-2s, s),$$

where $n_2 = 0, 0,$ or $m - m_1 + 2s_1$ corresponding to the cases $(\tau, \tau_1) = (0,0), (2,0),$ or $(2,2)$, respectively. □

4.5 Pseudo-symplectic Polarities

Let $n = 2\nu + \delta$, where $\delta = 1$ or 2 as before, and let S_δ be the matrix (4.1) or (4.2).

Consider the projective space $PG(n-1, \mathbb{F}_q)$ over the finite field $\mathbb{F}_q$ of characteristic 2. Let P be an r-flat in $PG(n-1, \mathbb{F}_q)$. Define

$$P^\perp = \{x \in PG(n-1, \mathbb{F}_q) \mid xS_\delta\, {}^t y = 0 \text{ for all } y \in P\}.$$

Clearly, $P^\perp$ is also a flat and we call it the *dual flat* of P with respect to S_δ. It can be readily verified that for any flats P and Q of $PG(n-1, \mathbb{F}_q)$,

$$P \subset Q \Rightarrow P^\perp \supset Q^\perp,$$
$$\dim P + \dim P^\perp = n-2,$$

and

$$(P^\perp)^\perp = P.$$

Therefore the map

$$P \mapsto P^\perp$$

from the set of flats of $PG(n-1, \mathbb{F}_q)$ to itself is a polarity of $PG(n-1, \mathbb{F}_q)$, called the *pseudo-symplectic polarity* of $PG(n-1, \mathbb{F}_q)$.

A flat of $PG(n-1, \mathbb{F}_q)$ is called *totally isotropic* with respect to S_δ, if the vector subspace of $\mathbb{F}_q^{(n)}$ corresponding to it is totally isotropic. When $\delta = 1$, we define $\mathcal{P}(S_1)$ to be the set of all totally isotropic flats with respect to S_1, and when $\delta = 2$, we define $\mathcal{P}(S_2)$ to be the set of all totally isotropic flats with respect to S_2 and containing the point $\langle e_{2\nu+1} \rangle$. Then we have

Theorem 4.26. Let $n = 2\nu + \delta$, where $\delta = 1$ or 2. Then both $\mathcal{P}(S_1)$ and $\mathcal{P}(S_2)$ are polar spaces of rank ν and isomorphic[1] to the polar space arising from a symplectic polarity of $PG(2\nu - 1, \mathbb{F}_q)$.

Proof. Consider the case $\delta = 1$ first. Let $P \in \mathcal{P}(S_1)$. Then P has a matrix representation of the form

$$P = \begin{matrix} \\ (\,P_1 \end{matrix} \begin{matrix} \begin{matrix} 2\nu & 1 \end{matrix} \\ \begin{matrix} \quad & 0\,) \end{matrix} \end{matrix},$$

where $P_1 K\, {}^t P_1 = 0$. It is easy to verify that the map

$$P \mapsto P_1$$

from $\mathcal{P}(S_1)$ to the polar space $\mathcal{P}(K)$ arising from the symplectic polarity of $PG(2\nu - 1, \mathbb{F}_q)$ is an isomorphism of their incidence structures. Therefore $\mathcal{P}(S_1)$ is a polar space and isomorphic to $\mathcal{P}(K)$.

Then consider the case $\delta = 2$. Let $P \in \mathcal{P}(S_2)$. Then P has a matrix representation of the form

$$P = \begin{matrix} \\ \\ \end{matrix} \begin{matrix} \begin{matrix} 2\nu & 1 & 1 \end{matrix} \\ \begin{pmatrix} P_1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \end{matrix},$$

where $P_1 K\, {}^t P_1 = 0$. As the above case, the map

$$P \to P_1$$

from $\mathcal{P}(S_2)$ to the polar space $\mathcal{P}(K)$ arising from the symplectic polarity of $PG(2\nu - 1, \mathbb{F}_q)$ is an isomorphism of their incidence structures. Therefore $\mathcal{P}(S_2)$ is a polar space and isomorphic to $\mathcal{P}(K)$. □

However, when $\delta = 2$, neither the set of all totally isotropic flats not containing $e_{2\nu+1}$ nor the set of all totally isotropic flats forms a polar space. The proof will be left as an exercise.

If we adopt the vector space language, then the dual $P^\perp$ of an m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta)}$ is a subspace of dimension $2\nu+\delta-m$ and is called the *dual subspace* of P with respect to S_δ. Let P be a subspace of type $(m, 2s+\tau, s, \varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta)}$, and denote the type of $P^\perp$ by $(m, 2s+\tau, s, \varepsilon)^\perp$. By Theorem 4.12, $(m, 2s+\tau, s, \varepsilon)^\perp$ depends only on the type of P and not on the particular choice of P. We have

[1] For the definition of isomorphism of polar spaces, see Section B.1.

Theorem 4.27.

(a) In $\mathbb{F}_q^{(2\nu+1)}$,

(i) $(m, 2s, s, 0)^\perp = (2\nu + 1 - m, 2(\nu + s - m) + 1, \nu + s - m, 1)$,
(ii) $(m, 2s + 1, s, 0)^\perp = (2\nu + 1 - m, 2(\nu + s - m) + 2, \nu + s - m, 0)$,
(iii) $(m, 2s + 1, s, 1)^\perp = (2\nu + 1 - m, 2(\nu + s - m + 1), \nu + s - m + 1, 0)$,
(iv) $(m, 2s+2, s, 0)^\perp = (2\nu+1-m, 2(\nu+s-m+1)+1, \nu+s-m+1, 0)$.

(b) In $\mathbb{F}_q^{(2\nu+2)}$,

(i) $(m, 2s, s, 0)^\perp = (2\nu + 2 - m, 2(\nu + s - m) + 2, \nu + s - m, 1)$,
(ii) $(m, 2s, s, 1)^\perp = (2\nu + 2 - m, 2(\nu + s - m + 1), \nu + s - m + 1, 1)$,
(iii) $(m, 2s+1, s, 0)^\perp = (2\nu+2-m, 2(\nu+s-m+1)+1, \nu+s-m+1, 0)$,
(iv) $(m, 2s+2, s, 0)^\perp = (2\nu+2-m, 2(\nu+s-m+1)+2, \nu+s-m+1, 0)$,
(v) $(m, 2s + 2, s, 1)^\perp = (2\nu + 2 - m, 2(\nu + s - m + 2), \nu + s - m + 2, 0)$.

Proof. (a) Let P be a subspace of type $(m, 2s + \tau, s, \varepsilon)$ in $\mathbb{F}_q^{(2\nu+1)}$. By Theorem 4.11, $(\tau, \varepsilon) = (0, 0), (1, 0), (1, 1)$ or $(2, 0)$. We consider these four cases one by one.

(i) $(\tau, \varepsilon) = (0, 0)$. We know from the proof of Theorem 4.11 that (4.9) is a subspace of type $(m, 2s, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. Clearly, the dual of (4.9) is

$$\begin{array}{c} \begin{array}{ccccccc} s & m-2s & \nu+s-m & s & m-2s & \nu+s-m & 1 \end{array} \\ \begin{pmatrix} 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\nu+s-m)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array},$$

which is a subspace of type $(2\nu + 1 - m,\ 2(\nu + s - m) + 1,\ \nu + s - m, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. Therefore

$$(m, 2s, s, 0)^\perp = (2\nu + 1 - m,\ 2(\nu + s - m) + 1,\ \nu + s - m, 1).$$

(ii) $(\tau, \varepsilon) = (1, 0)$. We know that (4.10) is a subspace of type $(m, 2s + 1, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. Clearly, the dual of (4.10) is

$$\begin{array}{c} \begin{array}{ccccccccc} s & 1 & m-2s-1 & \nu+s-m & s & 1 & m-2s-1 & \nu+s-m & 1 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu+s-m)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m)} & 0 \end{pmatrix} \end{array},$$

which is a subspace of type $(2\nu+1-m,\ 2(\nu+s-m)+2,\ \nu+s-m, 0)$. Therefore

$$(m, 2s+1, s, 0)^{\perp} = (2\nu+1-m,\ 2(\nu+s-m)+2,\ \nu+s-m, 0).$$

(iii) $(\tau, \varepsilon) = (1,1)$. Since $P \mapsto P^{\perp}$ is a polarity, it follows from case (i) that

$$(m, 2s+1, s, 1)^{\perp} = (2\nu+1-m,\ 2(\nu+s-m+1),\ \nu+s-m+1, 0).$$

(iv) $(\tau, \varepsilon) = (2,0)$. Similarly, it follows from case (ii) that

$$(m, 2s+2, s, 0)^{\perp} = (2\nu+1-m,\ 2(\nu+s-m+1)+1,\ \nu+s-m+1, 0).$$

(b) Let P be a subspace of type $(m, 2s+\tau, s, \varepsilon)$ in $\mathbb{F}_q^{(2\nu+2)}$. By Theorem 4.11, $(\tau, \varepsilon) = (0,0), (0,1), (1,0), (2,0)$ or $(2,1)$. We consider these five cases one by one.

(i) $(\tau, \varepsilon) = (0,0)$. From the proof of Theorem 4.11 we know that (4.13) is a subspace of type $(m, 2s, s, 0)$. Clearly, the dual of (4.13) is

$$\begin{array}{c} \begin{array}{ccccccc} s & m-2s & \nu+s-m & s & m-2s & \nu+s-m & 2 \end{array} \\ \begin{pmatrix} 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\nu+s-m)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I^{(2)} \end{pmatrix}, \end{array}$$

which is a subspace of type $(2\nu+2-m,\ 2(\nu+s-m)+2,\ \nu+s-m, 1)$. Therefore

$$(m, 2s, s, 0)^{\perp} = (2\nu+2-m,\ 2(\nu+s-m)+2,\ \nu+s-m, 1).$$

(ii) $(\tau, \varepsilon) = (0,1)$. We know that (4.14) is a subspace of type $(m, 2s, s, 1)$. Clearly, the dual of (4.14) is

$$\begin{array}{c} \begin{array}{ccccccccc} s & m-2s-1 & \nu+s-m+1 & s & m-2s-1 & \nu+s-m+1 & 1 & 1 \end{array} \\ \begin{pmatrix} 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \end{array}$$

which is a subspace of type $(2\nu+2-m,\ 2(\nu+s-m+1),\ \nu+s-m+1, 1)$. Therefore

$$(m, 2s, s, 1)^{\perp} = (2\nu+2-m,\ 2(\nu+s-m+1),\ \nu+s-m+1, 1).$$

(iii) $(\tau,\varepsilon)=(1,0)$. We know that (4.15) is a subspace of type $(m,2s+1,s,0)$. Clearly, the dual of (4.15) is

$$\begin{array}{cccccccc} s & m-2s-1 & \nu+s-m+1 & s & m-2s-1 & \nu+s-m+1 & 1 & 1 \\ \begin{pmatrix} 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array},$$

which is a subspace of type $(2\nu+2-m,\ 2(\nu+s-m+1)+1,\ \nu+s-m+1,0)$. Therefore

$$(m,2s+1,s,0)^{\perp}=(2\nu+2-m,\ 2(\nu+s-m+1)+1,\ \nu+s-m+1,0).$$

(iv) $(\tau,\varepsilon)=(2,0)$. We know that (4.16) is a subspace of type $(m,2s+2,s,0)$. Clearly, the dual of (4.16) is

$$\begin{array}{cccccccccc} s & 1 & m-2s-2 & \nu+s-m+1 & s & 1 & m-2s-2 & \nu+s-m+1 & 1 & 1 \\ \begin{pmatrix} 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\nu+s-m+1)} & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array},$$

which is of type $(2\nu+2-m,\ 2(\nu+s-m+1)+2,\ \nu+s-m+1,0)$. Therefore

$$(m,2s+2,s,0)^{\perp}=(2\nu+2-m,\ 2(\nu+s-m+1)+2,\ \nu+s-m+1,0).$$

(v) $(\tau,\varepsilon)=(2,1)$. Since $P\to P^{\perp}$ is a polarity, it follows from case (i) that

$$(m,2s+2,s,1)^{\perp}=(2\nu+2-m,\ 2(\nu+s-m+2),\ \nu+s-m+2,0).$$

□

Finally, we remark that using the pseudo-symplectic polarity we can deduce a closed formula for the number of subspaces of type $(m,2s+\tau,s,\varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ containing a given subspace of type $(m_1,2s_1+\tau_1,s_1,\varepsilon_1)$.

Let P_1 be a given subspace of type $(m_1,2s_1+\tau_1,s_1,\varepsilon_1)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Denote by $\mathcal{M}'(m_1,2s_1+\tau_1,s_1,\varepsilon_1;m,2s+\tau,s,\varepsilon;2\nu+\delta)$ the set of subspaces of type $(m,2s+\tau,s,\varepsilon)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ containing P_1. Let

$$\begin{aligned} &N'(m_1,2s_1+\tau_1,s_1,\varepsilon_1;m,2s+\tau,s,\varepsilon;2\nu+\delta) \\ &\quad =\mid \mathcal{M}'(m_1,2s_1+\tau_1,s_1,\varepsilon_1;m,2s+\tau,s,\varepsilon;2\nu+\delta)\mid . \end{aligned}$$

By Theorem 4.12, $N'(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ is independent of the particular choice of the subspace P_1 of type $(m_1, 2s_1+\tau_1, s_1, \varepsilon_1)$ in $\mathbb{F}_q^{(2\nu+\delta)}$, and hence the notation $N'(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta)$ is justified.

If we replace all the $\mathcal{M}$ in Theorem 4.24 by $\mathcal{M}'$, then the theorem so obtained also holds. Moreover, we have

Theorem 4.28. Assume that in $\mathbb{F}_q^{(2\nu+\delta)}$ subspaces of types $(m, 2s+\tau, s, \varepsilon)$ and $(m_1, 2s_1+\tau_1, s_1, \varepsilon_1)$ exist. Then

$$\begin{aligned} &N'(m_1, 2s_1+\tau_1, s_1, \varepsilon_1; m, 2s+\tau, s, \varepsilon; 2\nu+\delta) \\ &\quad = N((m, 2s+\tau, s, \varepsilon)^{\perp}; (m_1, 2s_1+\tau_1, s_1, \varepsilon_1)^{\perp}; 2\nu+\delta), \end{aligned}$$

where $(m, 2s+\tau, s, \varepsilon)^{\perp}$ and $(m_1, 2s_1+\tau_1, s_1, \varepsilon_1)^{\perp}$ are given by Theorem 4.27, and $N((m, 2s+\tau, s, \varepsilon)^{\perp}; (m_1, 2s_1+\tau_1, s_1, \varepsilon_1)^{\perp}; 2\nu+\delta)$, is given by Theorem 4.25. □

4.6 Comments

Theorem 4.1 in Section 4.1 is wellknown. The pseudo-symplectic group over any field of characteristic 2 was first studied by J. Dieudonné, cf. Dieudonné 1948. His main results specialized to the finite fields are Theorems or Corollaries 4.3 – 4.7. The definition of the singular pseudo-symplectic group and the results following it in Section 4.1 are contained in Liu and Wan 1991.

All the main results of Section 4.2 are due to Y. Liu and Z. Wan, cf. Liu and Wan 1991. But it should be remarked that Theorem 4.12 can also be deduced from Witt's theorem for non-alternate symmetric bilinear forms over fields of characteristic 2, which is due to V. Pless, cf. Pless 1964; however, the present proof leads also directly to a proof of Theorem 4.14. Moreover, Corollary 4.15 is due to V. Pless, cf. Pless 1965.

All the results from Sections 4.3 to 4.5 are contained in Wan 1992c.

4.7 Exercises

4.1 Prove that if there is no nonzero isotropic vector with respect to a given $n \times n$ symmetric matrix over a finite field of characteristic 2, the matrix is cogredient to the $n \times n$ identity matrix.

4.2 Fill in the missing proofs for those cases not given in the proof of Theorem 4.12.

4.3 Do the same thing for the proof of Theorem 4.14.

4.4 Compute the number of orbits of subspaces of the $(2\nu+\delta+l)$-dimensional singular pseudo-symplectic space over the finite field $\mathbb{F}_q$ of characteristic 2 under $Ps_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$.

4.5 Give the missing part of the proof of case (vi) of Theorem 4.24.

4.6 Prove Theorem 4.25.

4.7 Prove that in the $(2\nu+2)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$ neither the set of all totally isotropic flats not containing $e_{2\nu+1}$ nor the set of all totally isotropic flats forms a polar space.

Chapter 5
Geometry of Unitary Groups over Finite Fields

5.1 Unitary Geometry over Finite Fields

In this chapter let $\mathbb{F}_{q^2}$ be a finite field with q^2 elements, where q is a power of a prime. $\mathbb{F}_{q^2}$ has an *involutive automorphism*, i.e., an automorphism of order 2

$$a \mapsto \bar{a} = a^q,$$

and the fixed field of this automorphism is $\mathbb{F}_q$. Let us start with the following lemma which will be used several times in the sequel.

Lemma 5.1. For any $\lambda \in \mathbb{F}_q^*$, the equation $x\bar{x} = \lambda$ has exactly $q+1$ solutions in $\mathbb{F}_{q^2}^*$. And for any $\lambda \in \mathbb{F}_q$, the equation $x + \bar{x} = \lambda$ has exactly q solutions in $\mathbb{F}_{q^2}$.

Proof. Consider the map

$$\begin{aligned} \phi : \mathbb{F}_{q^2}^* &\to \mathbb{F}_q^* \\ x &\mapsto x\bar{x}. \end{aligned}$$

Clearly it is a homomorphism of groups. The kernel $\mathrm{Ker}\ \phi$ consists of all solutions of the equation $x\bar{x} = x^{q+1} = 1$, hence $\mid \mathrm{Ker}\ \phi \mid \leq q+1$. On the other hand, $\mid \mathrm{Im}\ \phi \mid \leq \mid \mathbb{F}_q^* \mid = q-1$. But $\mid \mathbb{F}_{q^2}^* \mid = \mid \mathrm{Ker}\ \phi \mid\mid \mathrm{Im}\ \phi \mid$ and $\mid \mathbb{F}_{q^2}^* \mid = q^2 - 1$. Therefore ϕ is a surjection and $\mid \mathrm{Ker}\ \phi \mid = q+1$. Consequently, for any $\lambda \in \mathbb{F}_q^*$, the equation $x\bar{x} = \lambda$ has exactly $q+1$ solutions in $\mathbb{F}_{q^2}^*$. This proves the first assertion of the lemma.

Similarly, consider the map

$$\begin{aligned} \psi : \mathbb{F}_{q^2} &\to \mathbb{F}_q \\ x &\mapsto x + \bar{x}. \end{aligned}$$

This is a homomorphism from the additive group of $\mathbb{F}_{q^2}$ into the additive group of $\mathbb{F}_q$. Ker ψ consists of all solutions of the equation $x + \bar{x} = x + x^q = 0$, hence $|\text{ Ker } \psi | \leq q$. On the other hand, $|\text{ Im } \psi | \leq |\mathbb{F}_q| = q$. From $|\mathbb{F}_{q^2}| = |\text{ Ker } \psi || \text{ Im } \psi |$ and $|\mathbb{F}_{q^2}| = q^2$ we deduce that ψ is a surjection and $|\text{ Ker } \psi | = q$. Consequently, for any $\lambda \in \mathbb{F}_q$, the equation $x + \bar{x} = \lambda$ has exactly q solutions, whence the second assertion of the lemma is proved. □

Let

$$A = (a_{ik})_{1 \leq i \leq m, 1 \leq k \leq n}$$

be an $m \times n$ matrix over $\mathbb{F}_{q^2}$. We use $\bar{A}$ to denote the matrix obtained from A by applying the automorphism $a \mapsto \bar{a}$ to all the mn elements of A, i.e.,

$$\bar{A} = (\bar{a}_{ik})_{1 \leq i \leq m, 1 \leq k \leq n}.$$

Clearly, the ranks of A and $\bar{A}$ are equal. Two $n \times n$ matrices A and B over $\mathbb{F}_{q^2}$ are said to be *cogredient*, if there is an $n \times n$ nonsingular matrix Q such that $QA\,{}^t\bar{Q} = B$. It is clear that cogredient matrices over $\mathbb{F}_{q^2}$ have the same rank. An $n \times n$ matrix H over $\mathbb{F}_{q^2}$ is said to be *Hermitian*, if ${}^t\bar{H} = H$. And it is also clear that matrices over $\mathbb{F}_{q^2}$ cogredient to an Hermitian matrix are Hermitian.

Theorem 5.2. Let $r \leq n$. Then an $n \times n$ Hermitian matrix of rank r is necessarily cogredient to

$$\begin{pmatrix} I^{(r)} & \\ & 0^{(n-r)} \end{pmatrix}.$$

Proof. Let $H = (h_{ik})_{1 \leq i,k \leq n}$ be an $n \times n$ Hermitian matrix of rank r. At first we apply induction on n to prove that H is cogredient to a diagonal matrix. When $n = 1$, this is obvious. Now assume that $n \geq 2$ and that the assertion holds for all $m < n$. If $H = 0$, our assertion holds automatically. If $H \neq 0$, we can assume that $h_{11} \neq 0$. In fact, if $h_{11} = 0$, but there is a non-zero diagonal element h_{ii} with $i > 1$, then interchanging the first and the i-th rows of H and the first and the i-th columns of H simultaneously, we obtain an Hermitian matrix which is cogredient to H and whose element at $(1,1)$ position is non-zero. If all the diagonal elements of H are zero, since $H \neq 0$, there is a non-zero non-diagonal element of H. We can assume that $h_{12} \neq 0$. For if $h_{ij} \neq 0$ with

$i < j$, interchanging the first and the i-th rows and the first and the i-th columns of H simultaneously and then interchanging the second and the j-th rows and the second and the j-th columns of the matrix thus obtained simultaneously, we obtain an Hermitian matrix which is cogredient to H and whose element at $(1,2)$ position is non-zero. Therefore we can assume that

$$H = \begin{pmatrix} 0 & h_{12} & H_{12} \\ \bar{h}_{12} & 0 & \\ {}^t\bar{H}_{12} & & H_{22} \end{pmatrix},$$

where $h_{12} \neq 0$. Let

$$Q = \begin{pmatrix} 1 & \lambda & \\ 0 & 1 & \\ & & I^{(n-2)} \end{pmatrix},$$

then the element at $(1,1)$ position of $QH\,{}^t\bar{Q}$ is $\lambda\bar{h}_{12} + h_{12}\bar{\lambda}$. By Lemma 5.1, let x be an element of $\mathbb{F}_{q^2}$ which satisfies the equation $x + \bar{x} = 1$ and put $\lambda = x\bar{h}_{12}^{-1}$, then the element at $(1,1)$ position of $QH\,{}^t\bar{Q}$ is 1. Therefore we can always assume that $h_{11} \neq 0$.

Write H in block form

$$H = \begin{pmatrix} h_{11} & H_{12} \\ {}^t\bar{H}_{12} & H_{22} \end{pmatrix},$$

where $\bar{h}_{11} = h_{11}$ and ${}^t\bar{H}_{22} = H_{22}$. Let

$$Q = \begin{pmatrix} 1 & \\ -h_{11}^{-1}\,{}^t\bar{H}_{12} & I^{(n-1)} \end{pmatrix},$$

then

$$QH\,{}^t\bar{Q} = \begin{pmatrix} h_{11} & 0 \\ 0 & -h_{11}^{-1}\,{}^t\bar{H}_{12}H_{12} + H_{22} \end{pmatrix},$$

where $-h_{11}^{-1}\,{}^t\bar{H}_{12}H_{12} + H_{22}$ is an $(n-1) \times (n-1)$ Hermitian matrix of rank $r-1$. By induction hypothesis, $-h_{11}^{-1}\,{}^t\bar{H}_{12}H_{12} + H_{22}$ is cogredient to a diagonal matrix, so does H.

After permuting the diagonal elements of the diagonal matrix to which H is cogredient, we can assume that H is cogredient to a diagonal matrix

$$\begin{pmatrix} h_{11} & & & & & & \\ & h_{22} & & & & & \\ & & \ddots & & & & \\ & & & h_{rr} & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix},$$

where $h_{11}h_{22}\dots h_{rr} \neq 0$. From ${}^t\bar{H} = H$, we deduce that $\bar{h}_{ii} = h_{ii}$, $i = 1, 2, \dots, r$. By Lemma 5.1 there are $\mu_i \in \mathbb{F}_{q^2}^*$, $i = 1, 2, \dots, r$, such that $\mu_i\bar{\mu}_i = h_{ii}$. Then

$$\begin{pmatrix} \mu_1^{-1} & & & \\ & \ddots & & \\ & & \mu_r^{-1} & \\ & & & I^{(n-r)} \end{pmatrix} \begin{pmatrix} h_{11} & & & \\ & \ddots & & \\ & & h_{rr} & \\ & & & 0^{(n-r)} \end{pmatrix} \overline{{}^t\begin{pmatrix} \mu_1^{-1} & & & \\ & \ddots & & \\ & & \mu_r^{-1} & \\ & & & I^{(n-r)} \end{pmatrix}}$$

$$= \begin{pmatrix} I^{(r)} & \\ & 0^{(n-r)} \end{pmatrix}.$$

□

Corollary 5.3. Two $n \times n$ Hermitian matrices over $\mathbb{F}_{q^2}$ are cogredient if and only if they have the same rank. □

Corollary 5.4. Any $n \times n$ nonsingular Hermitian matrix over $\mathbb{F}_{q^2}$ is necessarily cogredient to the $n \times n$ identity matrix. And it is also cogredient to

$$\begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix},$$

when $n = 2\nu$ is even or $n = 2\nu + 1$ is odd, respectively. □

If H is an $n \times n$ Hermitian matrix of rank r, then $[r/2]$ is called the *index* of H. Clearly, the index of an Hermitian matrix is invariant under cogredience transformations. We denote

$$H_0 = \begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix} \tag{5.1}$$

and

$$H_1 = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix}, \tag{5.2}$$

where ν is the index of H_0 and H_1. In order to discuss these two cases simultaneously, we introduce the notation $n = 2\nu + \delta$ and H_δ, where $\delta = 0$ or 1.

Now let H be an $n \times n$ nonsingular Hermitian matrix over $\mathbb{F}_{q^2}$. An $n \times n$ matrix T over $\mathbb{F}_{q^2}$ is called a *unitary matrix* with respect to H if

$$TH\,{}^t\bar{T} = H.$$

Clearly, $n \times n$ unitary matrices with respect to a nonsingular Hermitian matrix H are nonsingular and they form a group with respect to matrix multiplication, called the *unitary group* of degree n with respect to H over $\mathbb{F}_{q^2}$ and denoted by $U_n(\mathbb{F}_{q^2}, H)$.

Let H_1 and H_2 be two nonsingular Hermitian matrices over $\mathbb{F}_{q^2}$. As in Section 3.1 we can prove that the unitary groups $U_n(\mathbb{F}_{q^2}, H_1)$ and $U_n(\mathbb{F}_{q^2}, H_2)$ are isomorphic. Therefore there is essentially only one unitary group of degree n over $\mathbb{F}_{q^2}$ within isomorphism. We usually regard this unitary group as the unitary group with respect to $I^{(n)}$, and denote it simply by $U_n(\mathbb{F}_{q^2})$. But when $n = 2\nu$ is even, it will sometimes also be regarded as the unitary group with respect to H_0 and denoted by $U_{2\nu}(\mathbb{F}_{q^2})$, and when $n = 2\nu + 1$ is odd, it will sometimes also be regarded as the unitary group with respect to H_1 and denoted by $U_{2\nu+1}(\mathbb{F}_{q^2})$.

There is an action of $U_n(\mathbb{F}_q, H)$ on $\mathbb{F}_{q^2}^{(n)}$ defined as follows:

$$\begin{aligned} \mathbb{F}_{q^2}^{(n)} \times U_n(\mathbb{F}_q, H) &\to \mathbb{F}_{q^2}^{(n)} \\ ((x_1, x_2, \ldots, x_n), T) &\mapsto (x_1, x_2, \ldots, x_n)T. \end{aligned}$$

The elements of $U_n(\mathbb{F}_{q^2}, H)$ are then called *unitary transformations*. The vector space $\mathbb{F}_{q^2}^{(n)}$, with the above action of the unitary group $U_n(\mathbb{F}_{q^2}, H)$, is called the n-dimensional *unitary space* over $\mathbb{F}_{q^2}$. According to the Erlangen Program the geometry of the unitary group $U_n(\mathbb{F}_{q^2}, H)$ or simply the *unitary geometry* over $\mathbb{F}_{q^2}$ aims at the study of properties of geometric figures of the unitary space $\mathbb{F}_{q^2}^{(n)}$ which are invariant under the unitary group $U_n(\mathbb{F}_{q^2}, H)$.

At first, we want to study how the vector subspaces of $\mathbb{F}_{q^2}^{(n)}$ are partitioned into orbits under the action of $U_n(\mathbb{F}_{q^2}, H)$. Parallel to Section 3.1 we shall give some definitions first.

Let H be an $n \times n$ nonsingular Hermitian matrix over $\mathbb{F}_{q^2}$. A vector $x \in \mathbb{F}_{q^2}^{(n)}$ is said to be *isotropic* or *non-isotropic* with respect to H, if $xH\,{}^t\bar{x} = 0$ or $xH\,{}^t\bar{x} \neq 0$, respectively. For example, if $H = I^{(n)}$, then all basis vectors $e_1, e_2, \ldots, e_n$ are non-isotropic. By Lemma 5.1 there is an element a of $\mathbb{F}_{q^2}$ such that $a\bar{a} = -1$, then the vector $e_1 + ae_2$ is isotropic with respect to $I^{(n)}$. However, for $H = H_0$, all basis vectors $e_1, e_2, \ldots, e_{2\nu}$ are isotropic. By Lemma 5.1, there is an element $b \in \mathbb{F}_{q^2}$ such that $b + \bar{b} = 1$, then the vector $e_1 + be_{\nu+1}$ is non-isotropic with respect to H_0.

Let P be an m-dimensional vector subspace of $\mathbb{F}_{q^2}^{(n)}$. As before we use the same letter P to denote a matrix representation of the vector subspace P, i.e., P is an $m \times n$ matrix of rank m whose rows form a basis of P. For an $n \times n$ nonsingular Hermitian matrix H, it is clear that $PH\,{}^t\bar{P}$ is Hermitian. If $PH\,{}^t\bar{P}$ is of rank r, we say that P is a *subspace of type* (m, r) with respect to H. Clearly $r \leq m \leq n$. In particular, subspaces of type $(m, 0)$ with respect to H are called m-dimensional *totally isotropic subspaces* with respect to H and subspaces of type (m, m) with respect to H are called m-dimensional *non-isotropic subspaces* with respect to H. It is clear that a subspace P is totally isotropic with respect to H if and only if $PH\,{}^t\bar{P} = 0$, and it is non-isotropic with respect to H if and only if $PH\,{}^t\bar{P}$ is nonsingular. Thus, a non-zero isotropic vector generates a 1-dimensional totally isotropic subspace and a non-isotropic vector generates a 1-dimensional non-isotropic subspace.

For example, if $s \leq \nu = [n/2]$,

$$\begin{array}{c} \overset{s}{} \quad \overset{\nu-s}{} \quad \overset{s}{} \quad \overset{\nu-s+\delta}{} \\ (\ I^{(s)} \quad 0 \quad 0 \quad 0 \quad) \end{array} \tag{5.3}$$

is a totally isotropic subspace of dimension s with respect to H_δ, and

$$\begin{array}{c} \overset{s}{} \quad \overset{s}{} \quad \overset{n-2s}{} \\ (\ I^{(s)} \quad aI^{(s)} \quad 0 \quad) \end{array}, \tag{5.4}$$

where $a\bar{a} = -1$, is a totally isotropic subspace of dimension s with respect to $I^{(n)}$. If $r \leq n$, then

$$(\ I^{(r)} \quad 0^{(r,n-r)}\ ) \tag{5.5}$$

is a non-isotropic subspace of dimension r with respect to $I^{(n)}$. If $0 \leq 2m - r \leq n$, then

$$\begin{array}{c} \overset{r}{} \quad \overset{m-r}{} \quad \overset{m-r}{} \quad \overset{n-2m+r}{} \\ \begin{pmatrix} I^{(r)} & 0 & 0 & 0 \\ 0 & I^{(m-r)} & aI^{(m-r)} & 0 \end{pmatrix} \end{array} \tag{5.6}$$

is a subspace of type (m, r) with respect to $I^{(n)}$.

Two vectors x and y of $\mathbb{F}_{q^2}^{(n)}$ are said to be *orthogonal* with respect to H, if $xH\,{}^t\bar{y} = 0$. Clearly, if $xH\,{}^t\bar{y} = 0$, then $yH\,{}^t\bar{x} = 0$. A vector which is orthogonal to itself is an isotropic vector, otherwise a non-isotropic vector.

Let P be an m-dimensional subspace of $\mathbb{F}_{q^2}^{(n)}$ and H be an $n \times n$ non-singular Hermitian matrix over $\mathbb{F}_{q^2}$. Denote by $P^\perp$ the set of vectors which are orthogonal to every vector of P with respect to H, i.e.,

$$P^\perp = \{y \in \mathbb{F}_{q^2}^{(n)} \mid yH\,{}^t\bar{x} = 0 \text{ for all } x \in P\}.$$

Obviously, $P^\perp$ is an $(n-m)$-dimensional subspace of $\mathbb{F}_{q^2}^{(n)}$, and is called the *dual subspace of* P with respect to H. For example, the dual subspace of (5.3) with respect to H_δ is

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s+\delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(\nu-s)} & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s+\delta)} \end{pmatrix} \end{array},$$

the dual subspace of (5.4) with respect to $I^{(n)}$ is

$$\begin{pmatrix} I^{(s)} & aI^{(s)} & 0 \\ 0 & 0 & I^{(n-2s)} \end{pmatrix},$$

and the dual subspace of (5.5) with respect to $I^{(n)}$ is

$$\begin{pmatrix} 0^{(n-r,r)} & I^{(n-r)} \end{pmatrix}.$$

More generally, the dual subspace of (5.6), which is of type (m, r) with respect to $I^{(n)}$, is

$$\begin{array}{c} \begin{matrix} r & m-r & m-r & n-2m+r \end{matrix} \\ \begin{pmatrix} 0 & I^{(m-r)} & aI^{(m-r)} & 0 \\ 0 & 0 & 0 & I^{(n-2m+r)} \end{pmatrix} \end{array},$$

which is of type $(n-m, n-2m+r)$.

The following series of Theorems, Lemmas, and Corollaries are parallel to the corresponding series in Section 3.1, i.e., Theorem 3.4 to Theorem 3.11.

Theorem 5.5. A subspace P is totally isotropic if and only if $P \subseteq P^\perp$; and P is non-isotropic if and only if $P \cap P^\perp = (0)$. □

The proof of the above theorem runs similarly to the proof of Theorem 3.4 and hence, is omitted.

Lemma 5.6. Let P be a subspace of type (m, r) with respect to the $n \times n$ nonsingular Hermitian matrix H. Then $2r \leq 2m \leq n + r$ and there is an $m \times m$ nonsingular matrix Q and an $(n - m) \times n$ matrix Z such that

$$\begin{pmatrix} QP \\ Z \end{pmatrix} H \, {}^t\overline{\begin{pmatrix} QP \\ Z \end{pmatrix}} = \begin{pmatrix} I^{(r)} & & & \\ & 0 & I^{(m-r)} & \\ & I^{(m-r)} & 0 & \\ & & & 0^{(\sigma)} \end{pmatrix},$$

where $\sigma = n + r - 2m$.

Proof. P is a subspace of type (m, r) and hence $PH\,{}^t\bar{P}$ is an $m \times m$ matrix of rank r. By Theorem 5.2 there is an $m \times m$ nonsingular matrix Q such that

$$Q(PH\,{}^t\bar{P})\,{}^t\bar{Q} = \begin{pmatrix} I^{(r)} & \\ & 0^{(m-r)} \end{pmatrix}.$$

Therefore $r \leq m$. There is an $(n - m) \times n$ matrix Z_1 such that

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix}$$

is nonsingular. We may write

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix} H \, {}^t\overline{\begin{pmatrix} QP \\ Z_1 \end{pmatrix}} = \begin{array}{c} \begin{matrix} r & m-r & n-m \end{matrix} \\ \begin{pmatrix} I & & H_{13} \\ & 0 & H_{23} \\ {}^t\bar{H}_{13} & {}^t\bar{H}_{23} & H_{33} \end{pmatrix} \begin{matrix} r \\ m-r \\ n-m \end{matrix} \end{array},$$

where H_{33} is an $(n - m) \times (n - m)$ Hermitian matrix. We have

$$\begin{pmatrix} I & & \\ & I & \\ -\,{}^t\bar{H}_{13} & & I \end{pmatrix} \begin{pmatrix} I & & H_{13} \\ & 0 & H_{23} \\ {}^t\bar{H}_{13} & {}^t\bar{H}_{23} & H_{33} \end{pmatrix} {}^t\overline{\begin{pmatrix} I & & \\ & I & \\ -\,{}^t\bar{H}_{13} & & I \end{pmatrix}}$$

$$= \begin{pmatrix} I & & \\ & 0 & H_{23} \\ & -\,{}^t\bar{H}_{23} & K_{33} \end{pmatrix},$$

where $K_{33} = H_{33} - {}^t\bar{H}_{13}H_{13}$. Since H is nonsingular, H_{23} is of rank $m-r$. Thus $m-r \le n-m$, which gives $2m \le n+r$, and there is an $(n-m)\times(n-m)$ nonsingular matrix B such that

$$H_{23}\,{}^t\bar{B} = (\ I^{(m-r)}\quad 0^{(m-r,\sigma)}\ ),$$

where $\sigma = n+r-2m$. Then

$$\begin{pmatrix} I & & \\ & I & \\ & & B \end{pmatrix}\begin{pmatrix} I & & \\ & 0 & H_{23} \\ & {}^t\bar{H}_{23} & K_{33} \end{pmatrix}\ {}^t\overline{\begin{pmatrix} I & & \\ & I & \\ & & B \end{pmatrix}}$$

$$= \begin{array}{c} \begin{matrix} r & m-r & m-r & \sigma \end{matrix} \\ \begin{pmatrix} I & & & \\ & 0 & I & 0 \\ & I & L_{33} & L_{34} \\ & 0 & {}^t\bar{L}_{34} & L_{44} \end{pmatrix} \end{array} \begin{matrix} r \\ m-r \\ m-r \\ \sigma \end{matrix},$$

where L_{33} and L_{44} are $(m-r)\times(m-r)$ and $\sigma\times\sigma$ Hermitian matrices, respectively. We can write

$$L_{33} = \begin{pmatrix} l_{11} & l_{12} & \dots & l_{1m-r} \\ \bar{l}_{12} & l_{22} & \dots & l_{2m-r} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{l}_{1m-r} & \bar{l}_{2m-r} & \dots & l_{m-r\,m-r} \end{pmatrix},$$

then $l_{11}, l_{22}, \dots, l_{m-r,m-r} \in \mathbb{F}_q$. By Lemma 5.1 there exist $l_1, l_2, \dots, l_{m-r} \in \mathbb{F}_{q^2}$ such that $l_{11} = l_1 + \bar{l}_1$, $l_{22} = l_2 + \bar{l}_2, \dots, l_{m-r\,m-r} = l_{m-r} + \bar{l}_{m-r}$. Let

$$L_3 = \begin{pmatrix} l_1 & l_{12} & \dots & l_{1m-r} \\ & l_2 & \dots & l_{2m-r} \\ & & \ddots & \vdots \\ & & & l_{m-r} \end{pmatrix},$$

then $L_{33} = L_3 + {}^t\bar{L}_3$. Thus

$$\begin{pmatrix} I & & \\ & I & \\ & -{}^t\bar{L}_3 & I \\ & -{}^t\bar{L}_{34} & & I \end{pmatrix}\begin{pmatrix} I & & & \\ & 0 & I & 0 \\ & I & L_{33} & L_{34} \\ & 0 & {}^t\bar{L}_{34} & L_{44} \end{pmatrix}\ {}^t\overline{\begin{pmatrix} I & & & \\ & I & & \\ & -{}^t\bar{L}_3 & I & \\ & -{}^t\bar{L}_{34} & & I \end{pmatrix}}$$

$$= \begin{pmatrix} I & & & \\ & 0 & I & \\ & I & 0 & \\ & & & L_{44} \end{pmatrix}.$$

Hence L_{44} is nonsingular. There is a $\sigma \times \sigma$ nonsingular matrix R such that $RL_{44}\,{}^t\bar{R} = I^{(\sigma)}$. Then

$$\begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & R \end{pmatrix} \begin{pmatrix} I & & & \\ & 0 & I & \\ & I & 0 & \\ & & & L_{44} \end{pmatrix} {}^t\overline{\begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & R \end{pmatrix}}$$

$$= \begin{pmatrix} I^{(r)} & & & \\ & 0 & I^{(m-r)} & \\ & I^{(m-r)} & 0 & \\ & & & I^{(\sigma)} \end{pmatrix}.$$

Clearly,

$$\begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & R \end{pmatrix} \begin{pmatrix} I & & & \\ & I & & \\ & -{}^t\bar{L}_3 & I & \\ & -{}^t\bar{L}_{34} & & I \end{pmatrix} \begin{pmatrix} I & & \\ & I & \\ & & B \end{pmatrix} \begin{pmatrix} I & & \\ & I & \\ -{}^t\bar{H}_{13} & & I \end{pmatrix}$$

is an $n \times n$ nonsingular matrix of the form

$$\begin{pmatrix} I^{(m)} & \\ L_1 & L_2 \end{pmatrix}.$$

Let

$$\begin{pmatrix} I^{(m)} & \\ L_1 & L_2 \end{pmatrix} \begin{pmatrix} QP \\ Z_1 \end{pmatrix} = \begin{pmatrix} QP \\ Z \end{pmatrix},$$

then the last assertion is achieved. □

Theorem 5.7. Subspaces of type (m, r) exist in the n-dimensional unitary space if and only if $2r \leq 2m \leq n + r$.

Proof. If there is a subspace of type (m, r) with respect to the $n \times n$ nonsingular Hermitian matrix H, then by Lemma 5.6 $2r \leq 2m \leq n + r$. Conversely, if $2r \leq 2m \leq n + r$, then (5.6) is a subspace of type (m, r) with respect to $I^{(n)}$. Let P be a subspace of type (m, r) with respect to $I^{(n)}$ and assume that $QH\,{}^t\bar{Q} = I^{(n)}$, then PQ is a subspace of type (m, r) with respect to H. □

Theorem 5.8. Let P_1 and P_2 be two m-dimensional subspaces of $\mathbb{F}_{q^2}^{(n)}$. Then there is a $T \in U_n(\mathbb{F}_{q^2}, H)$ such that $P_1 = AP_2T$, where A is an $m \times m$ nonsingular matrix over $\mathbb{F}_{q^2}$, if and only if P_1 and P_2 are of the same type with respect to H. In other words, $U_n(\mathbb{F}_{q^2}, H)$ acts transitively on each set of subspaces of the same type.

Proof. Same as the proof of Theorem 3.7. □

Corollary 5.9. Totally isotropic subspaces of $\mathbb{F}_{q^2}^{(n)}$ are of dimension $\leq \nu = [n/2]$, and there exist totally isotropic subspaces of dimension ν. Therefore maximal totally isotropic subspaces are of dimension ν. Moreover, any totally isotropic subspace is contained in a maximal totally isotropic subspace.

Proof. Similar to the proof of Corollary 3.8. □

Corollary 5.10. The dual subspace of a subspace of type (m, r) is of type $(n-m, n-2m+r)$.

Proof. By the computation before Theorem 5.5, we know that the dual subspace of (5.6), which is of type (m, r) with respect to $I^{(n)}$, is of type $(n-m,\ n-2m+r)$ with respect to $I^{(n)}$. Hence our corollary follows from Theorem 5.8. □

Theorem 5.11. The number of orbits of subspaces of $\mathbb{F}_{q^2}^{(n)}$ under the action of $U_n(\mathbb{F}_{q^2}, H)$ is equal to

$$([n/2]+1)^2, \qquad \text{if } n \text{ is even,}$$

or

$$([n/2]+1)([n/2]+2), \quad \text{if } n \text{ is odd.}$$

Proof. By Theorem 5.7, the number of orbits of subspaces of $\mathbb{F}_{q^2}^{(n)}$ under $U_n(\mathbb{F}_{q^2})$ is equal to the number of pairs of nonnegative integers m and r such that $2r \leq 2m \leq n+r$. But $2r \leq 2m \leq n+r$ is equivalent to $2m-n \leq r \leq m$. For a given m such that $0 \leq m \leq n$, if $2m \leq n$, then r can take $m+1$ values: $0, 1, \ldots, m$; and if $2m > n$, then r can take $n-m+1$ values: $2m-n,\ 2m-n+1, \ldots, m$. Therefore the number of orbits of subspaces of $\mathbb{F}_{q^2}^{(n)}$ under $U_n(\mathbb{F}_{q^2})$ is equal to

$$\sum_{m=0}^{[n/2]}(m+1) + \sum_{m=[n/2]+1}^{n}(n-m+1) = \begin{cases} ([n/2]+1)^2, & \text{if } n \text{ is even,} \\ ([n/2]+1)([n/2]+2), & \text{if } n \text{ is odd.} \end{cases}$$

□

Lemma 5.12. Let P_1 and P_2 be two $m \times n$ matrices of rank m. Then there exists an element $T \in U_n(\mathbb{F}_{q^2}, H)$ such that $P_1 = P_2 T$ if and only if $P_1 H\, {}^t\bar{P}_1 = P_2 H\, {}^t\bar{P}_2$.

Proof. Same as the proof of Lemma 3.11. □

Theorem 5.8 above tells us that the set of vector subspaces of $\mathbb{F}_{q^2}^{(n)}$ is partitioned into orbits under $U_n(\mathbb{F}_{q^2}, H)$ according to their types and Theorem 5.11 tells us their number. Our next goal is to compute the length of each orbit, which is the content of next section.

5.2 Anzahl Theorems in Unitary Geometry over Finite Fields

We follow the notation of the previous section.

Let H be an $n \times n$ nonsingular Hermitian matrix. Denote by $\mathcal{M}(m, r; n)$ the set of subspaces of $\mathbb{F}_{q^2}^{(n)}$ of type (m, r) with respect to H and put

$$N(m, r; n) = \mid \mathcal{M}(m, r; n) \mid .$$

We assert that $N(m, r; n)$ is independent of the particular choice of H. Suppose that H_1 and H_2 are two $n \times n$ nonsingular Hermitian matrices. By Corollary 5.3 there is an $n \times n$ nonsingular matrix Q such that $H_1 = Q H_2\, {}^t\bar{Q}$. It is clear that P is a subspace of type (m, r) with respect to H_1 if and only if PQ is a subspace of type (m, r) with respect to H_2. Thus there is a bijection from the set of subspaces of type (m, r) with respect to H_1 to the set of subspaces of type (m, r) with respect to H_2. Therefore $N(m, r; n)$ is independent of the particular choice of H.

The goal of the present section is to compute $N(m, r; n)$. If $r > m$ or $n + r - 2m < 0$, then by Theorem 5.7 we have $N(m, r; n) = 0$. Therefore in the following we always assume that

$$2r \leq 2m \leq n + r.$$

Let

$$M(m, r) = \begin{pmatrix} I^{(r)} & \\ & 0^{(m-r)} \end{pmatrix}. \tag{5.7}$$

If P is a subspace of type (m, r) with respect to H, then by Theorem 5.2 we can choose a matrix representation P of the subspace P such that

$$P H\, {}^t\bar{P} = M(m, r). \tag{5.8}$$

Denote by $n(m,r;n)$ the number of $m \times n$ matrices P of rank m which satisfy (5.8). In the same way as above we can prove that $n(m,r;n)$ is independent of the particular choice of H. Parallel to Lemma 3.12 we have

Lemma 5.13. Let $2r \leq 2m \leq n+r$. Then

$$n(m,r;n) = \mid U_r(\mathbb{F}_{q^2}) \mid\mid GL_{m-r}(\mathbb{F}_{q^2}) \mid q^{2r(m-r)} N(m,r;n).$$

□

Clearly $\mid U_r(\mathbb{F}_{q^2}) \mid = n(r,r;r)$ and we have already computed $\mid GL_{m-r}(\mathbb{F}_{q^2}) \mid$ in Theorem 1.6. Thus to compute $N(m,r;n)$ it is sufficient to compute $n(m,r;n)$.

Now let $2r \leq 2m \leq n+r$ and $0 \leq r_1, r_2, r_1+r_2 \leq m$. Let $n(r_1)$ be the number of $r_1 \times n$ matrices P of rank r_1 such that $P_1 H\, {}^t\bar{P}_1 = M(m,r)_{r_1}$, where $M(m,r)_{r_1}$ denotes the upper left corner $r_1 \times r_1$ submatrix of $M(m,r)$. For a fixed $r_1 \times n$ matrix P_1 of rank r_1 such that $P_1 H\, {}^t\bar{P}_1 = M(m,r)_{r_1}$, let $n(P_1, r_2)$ be the number of $r_2 \times n$ matrices P_2 such that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank $r_1 + r_2$ and $PH\, {}^t\bar{P} = M(m,r)_{r_1+r_2}$. Parallel to Lemma 3.13, we have

Lemma 5.14. Let $2r \leq 2m \leq n+r$ and $0 \leq r_1, r_2, r_1+r_2 \leq m$. Then $n(P_1, r_2)$ is independent of the choice of P_1, and

$$n(r_1+r_2) = n(r_1)n(P_1, r_2)$$

for any choice of P_1. □

This Lemma is a direct consequence of Lemma 5.12 and can be proved in the same way as Lemma 3.13; its proof is omitted here.

Now we are ready to compute $n(m,r;n)$. Let us start with a special case.

Lemma 5.15. Let $m \leq n$, then

$$n(m,m;n) = q^{mn-\frac{m(m+1)}{2}} \prod_{i=n-m+1}^{n} (q^i - (-1)^i).$$

Proof. For definiteness, we take $H = I^{(n)}$. Let P be an $m \times n$ matrix of rank m which satisfies $P\,{}^t\bar{P} = I^{(m)}$, then $n(m, m; n)$ is the number of such matrix P. Let

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix},$$

where $v_1, v_2, \ldots, v_m$ are the m row vectors of P; then

$$v_i\,{}^t\bar{v}_j = \delta_{ij},\ 1 \le i, j \le m.$$

Let $v_1 = (x_1, x_2, \ldots, x_n)$, then

$$x_1\bar{x}_1 + x_2\bar{x}_2 + \cdots + x_n\bar{x}_n = 1. \tag{5.9}$$

We assert that the number of vectors $(x_1, x_2, \ldots, x_n)$ in $\mathbb{F}_{q^2}^{(n)}$, which satisfy (5.9), is equal to $q^{n-1}(q^n - (-1)^n)$. For $n = 1$, our assertion follows from Lemma 5.1. Suppose that our assertion holds for $n-1$ and we are going to prove that it holds also for n. By induction hypothesis the number of vectors $(x_1, x_2 \ldots, x_{n-1}, 0)$ satisfying (5.9) is equal to $q^{n-2}(q^{n-1} - (-1)^{n-1})$. We then count the number of vectors $(x_1, x_2, \ldots, x_n)$ with $x_n \neq 0$, which satisfy (5.9). When $x_n \neq 0$, we have necessarily $x_1\bar{x}_1 + \cdots + x_{n-1}\bar{x}_{n-1} \neq 1$. There are $q^{2(n-1)} - q^{n-2}(q^{n-1} - (-1)^{n-1})$ vectors $(x_1, x_2, \ldots, x_{n-1})$ such that $x_1\bar{x}_1 + \cdots + x_{n-1}\bar{x}_{n-1} \neq 1$. Once $x_1, x_2, \ldots, x_{n-1}$ are chosen such that $x_1\bar{x}_1 + \cdots + x_{n-1}\bar{x}_{n-1} \neq 1$, by Lemma 5.1 there are $q + 1$ possible choices of $x_n \neq 0$ such that (5.9) holds. Therefore the number of vectors $(x_1, x_2, \ldots, x_n)$ in $\mathbb{F}_{q^2}^{(n)}$, which satisfy (5.9), is equal to

$$\begin{aligned} &q^{n-2}(q^{n-1} - (-1)^{n-1}) + (q^{2(n-1)} - q^{n-2}(q^{n-1} - (-1)^{n-1}))(q+1) \\ &\qquad = q^{n-1}(q^n - (-1)^n). \end{aligned}$$

This proves our assertion.

It follows from our assertion that there are $q^{n-1}(q^n - (-1)^n)$ possible choices of the first row v_1 of P. By Lemma 5.14, once the first row v_1 of P has been chosen, the number of possible choices of the second row of P is independent of the particular choice of v_1. Hence we may choose $v_1 = e_1$, then the second row $v_2 = (y_1, y_2, \ldots, y_n)$ of P satisfies

$$y_1 = 0,\ y_2\bar{y}_2 + \cdots + y_n\bar{y}_n = 1.$$

By the assertion proved above, there are $q^{n-2}(q^{n-1}-(-1)^{n-1})$ possible choices of v_2.

Proceeding in this way, by Lemma 5.14 once the first $m-1$ rows $v_1, v_2, \ldots, v_{m-1}$ of P have been chosen, the number of possible choices of the m-th row v_m of P is independent of the particular choice of $v_1, v_2, \ldots, v_{m-1}$. Hence we may choose $v_1 = e_1, v_2 = e_2, \ldots, v_{m-1} = e_{m-1}$, then the m-th row $v_m = (z_1, z_2, \ldots, z_n)$ of P satisfies

$$z_1 = z_2 = \cdots = z_{m-1} = 0, z_m\bar{z}_m + \cdots + z_n\bar{z}_n = 1.$$

By the assertion proved above, there are $q^{n-m}(q^{n-m+1}-(-1)^{n-m+1})$ possible choices of v_m.

Thus Lemma 5.15 follows immediately. □

As a special case of the above lemma, we have

Theorem 5.16.

$$|\, U_n(\mathbb{F}_{q^2})\,| = q^{\frac{n(n-1)}{2}} \prod_{i=1}^{n}(q^i-(-1)^i).$$

Proof. In fact, $|\, U_n(\mathbb{F}_{q^2})\,| = n(n,n;n)$. □

In order to compute $n(m,r;n)$, we compute another special case $n(m,0;n)$ first.

Lemma 5.17. Let $m \le [n/2]$, then

$$n(m,0;n) = q^{m(m-1)} \prod_{i=n-2m+1}^{n}(q^i-(-1)^i).$$

Proof. Let $n = 2\nu+\delta$ where $\delta = 0$ or 1, and we take $H = H_\delta$. Let P be an $m \times n$ matrix of rank m which satisfies $PH_\delta\,{}^t\bar{P} = 0^{(m)}$, then $n(m,0;n)$ is the number of such matrix P. Let

$$P = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix},$$

then

$$v_i H_\delta\,{}^t\bar{v}_j = 0,\ 1 \le i,j \le m.$$

Let $v_1 = (x_1, x_2, \ldots, x_{2\nu+\delta})$, then

$$\left.\begin{array}{ll} \sum_{i=1}^{\nu}(x_i\bar{x}_{\nu+i} + x_{\nu+i}\bar{x}_i) = 0, & \text{if } \delta = 0, \\ \sum_{i=1}^{\nu}(x_i\bar{x}_{\nu+i} + x_{\nu+i}\bar{x}_i) + x_{2\nu+1}\bar{x}_{2\nu+1} = 0, & \text{if } \delta = 1. \end{array}\right\} \tag{5.10}$$

Now we assert that the number of non-zero vectors $(x_1, x_2, \ldots, x_{2\nu+\delta})$ in $\mathbb{F}_{q^2}^{(2\nu+\delta)}$, which satisfy (5.10), is equal to $(q^n - (-1)^n)(q^{n-1} - (-1)^{n-1})$. When $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$, we have necessarily that $x_{2\nu+1} = 0$ if $\delta = 1$, and $x_{\nu+1}, x_{\nu+2}, \ldots, x_{2\nu}$ can be chosen arbitrarily in $\mathbb{F}_{q^2}$ but cannot be all zero. Hence the number of non-zero vectors $(0, \ldots, 0, x_{\nu+1}, \ldots, x_{2\nu+\delta})$ which satisfy (5.10) is equal to $q^{2\nu} - 1$. For any choice of $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$, suppose that $x_i \neq 0$, we can write (5.10) as

$$\left.\begin{array}{ll} x_i\bar{x}_{\nu+i} + x_{\nu+i}\bar{x}_i = -\sum_{j\neq i}(x_j\bar{x}_{\nu+j} + x_{\nu+j}\bar{x}_j), & \text{if } \delta = 0, \\ x_i\bar{x}_{\nu+i} + x_{\nu+i}\bar{x}_i = -\sum_{j\neq i}(x_j\bar{x}_{\nu+j} + x_{\nu+j}\bar{x}_j) - x_{2\nu+1}\bar{x}_{2\nu+1}, & \text{if } \delta = 1. \end{array}\right\}$$

$x_{\nu+j} (1 \leq j \leq \nu + \delta$ and $j \neq i)$ can be chosen arbitrarily in $\mathbb{F}_{q^2}$ and once they are chosen, by Lemma 5.1 there are q possible choices of $x_{\nu+i}$ such that (5.10) holds. There are $q^{2\nu} - 1$ possible choices of $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$, hence there are $(q^{2\nu}-1)q^{2(\nu+\delta-1)+1}$ vectors $(x_1, x_2, \ldots, x_{2\nu+\delta})$ with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$, which satisfy (5.10). Consequently, the number of non-zero vectors $(x_1, x_2, \ldots, x_{2\nu+\delta})$ which satisfy (5.10) is

$$(q^{2\nu} - 1) + (q^{2\nu} - 1)q^{2(\nu+\delta-1)+1} = (q^n - (-1)^n)(q^{n-1} - (-1)^{n-1}).$$

This proves our assertion.

It follows from our assertion that there are $(q^n - (-1)^n)(q^{n-1} - (-1)^{n-1})$ possible choices of the first row v_1 of P_1. By Lemma 5.14, once the first row v_1 of P has been chosen, the number of possible choices of the second row v_2 of P is independent of the particular choice of v_1. Hence we may choose $v_1 = e_1$ and then compute the number of possible choices of v_2. Let $v_2 = (y_1, y_2, \ldots, y_{2\nu+\delta})$, then $y_{\nu+1} = 0$ and

$$\left.\begin{array}{ll} \sum_{i=2}^{\nu}(y_i\bar{y}_{\nu+i} + y_{\nu+i}\bar{y}_i) = 0, & \text{if } \delta = 0, \\ \sum_{i=2}^{\nu}(y_i\bar{y}_{\nu+i} + y_{\nu+i}\bar{y}_i) + y_{2\nu+1}\bar{y}_{2\nu+1} = 0, & \text{if } \delta = 1. \end{array}\right\}$$

Thus y_1 can be chosen arbitrarily in $\mathbb{F}_{q^2}$. Since v_1 and v_2 are linearly independent, $y_2, \ldots, y_\nu, y_{\nu+2}, \ldots, y_{2\nu+\delta}$ cannot all equal zero. Hence by the assertion proved above there are $q^2(q^{n-2} - (-1)^{n-2})(q^{n-3} - (-1)^{n-3})$ possible choices of v_2.

Proceeding in this way, by Lemma 5.14 once the first $m-1$ rows $v_1, v_2, \ldots, v_{m-1}$ have been chosen, the number of possible choices of the m-th row

v_m of P is independent of the particular choice of $v_1, v_2, \ldots, v_{m-1}$. Hence we may choose $v_1 = e_1, v_2 = e_2, \ldots, v_{m-1} = e_{m-1}$ and then compute the number of possible choices of v_m. Let $v_m = (z_1, z_2, \ldots, z_{2\nu+\delta})$, then $z_{\nu+1} = \cdots = z_{\nu+m-1} = 0$ and

$$\left.\begin{array}{ll} \sum_{i=m}^{\nu}(z_i\bar{z}_{\nu+i} + z_{\nu+i}\bar{z}_i) = 0, & \text{if } \delta = 0, \\ \sum_{i=m}^{\nu}(z_i\bar{z}_{\nu+i} + z_{\nu+i}\bar{z}_i) + z_{2\nu+1}\bar{z}_{2\nu+1} = 0, & \text{if } \delta = 1. \end{array}\right\}$$

Thus $z_1, z_2, \ldots, z_{m-1}$ can be chosen arbitrarily in $\mathbb{F}_{q^2}$. Since $v_1, v_2, \ldots, v_m$ are linearly independent, $z_m, \ldots, z_\nu, z_{\nu+m}, \ldots, z_{2\nu+\delta}$ cannot all be zero. Therefore by the above assertion there are

$$q^{2(m-1)}(q^{n-2m+2} - (-1)^{n-2m+2})\,(q^{n-2m+1} - (-1)^{n-2m+1})$$

possible choices of v_m.

Hence Lemma 5.17 follows immediately. □

Lemma 5.18. Let $2r \le 2m \le n + r$, then

$$n(m, r; n) = q^{rn+(m-r)(m-r-1)-\frac{r(r+1)}{2}} \prod_{i=n+r-2m+1}^{n} (q^i - (-1)^i).$$

Proof. For definiteness we take $H = I^{(n)}$ and compute $n(m, r; n)$.

Let P be an $m \times n$ matrix of rank m which satisfies $P\,{}^t\bar{P} = M(m, r)$; then $n(m, r; n)$ is the number of such matrix P. Write P in block form

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{matrix} r \\ m-r \end{matrix},$$

then

$$P_1\,{}^t\bar{P}_1 = I^{(r)},$$
$$P_1\,{}^t\bar{P}_2 = 0^{(r,m-r)},$$

and

$$P_2\,{}^t\bar{P}_2 = 0^{(m-r)}.$$

Therefore there are $n(r, r; n)$ possible choices of P_1. By Lemma 5.14 once P_1 is chosen, the number of possible choices of P_2 is independent of the particular choice of P_1. Hence we may choose

$$P_1 = (\ I^{(r)} \quad 0^{(r,n-r)}\ )$$

and then compute the possible choices of P_2. From $P_1\,{}^t\bar{P}_2 = 0$, we deduce that P_2 is necessarily of the form

$$P_2 = (\ 0^{(m-r,r)}\quad P_{22}\ ),$$

where P_{22} is an $(m-r)\times(n-r)$ matrix. From $P_2\,{}^t\bar{P}_2 = 0$ we deduce

$$P_{22}\,{}^t\bar{P}_{22} = 0.$$

Therefore there are $n(m-r,0;n-r)$ possible choices of P_2. Thus

$$n(m,r;n) = n(r,r;n)\ n(m-r,0;n-r),$$

and our lemma follows from Lemmas 5.15 and 5.17. □

From Lemmas 5.13, 5.18 and Theorems 1.6, 5.16 follows the following Anzahl theorem in the geometry of unitary groups over finite fields.

Theorem 5.19. Let $2r \le 2m \le n+r$, then the number of subspaces of type (m,r) in the n-dimensional unitary space over $\mathbb{F}_{q^2}$ is given by

$$N(m,r;n) = q^{r(n+r-2m)}\frac{\displaystyle\prod_{i=n+r-2m+1}^{n}(q^i-(-1)^i)}{\displaystyle\prod_{i=1}^{r}(q^i-(-1)^i)\prod_{i=1}^{m-r}(q^{2i}-1)}.$$

□

Corollary 5.20. Let $0 \le m \le [n/2]$, then the number of m-dimensional totally isotropic subspaces in the n-dimensional unitary space over $\mathbb{F}_{q^2}$ is

$$N(m,0;n) = \frac{\displaystyle\prod_{i=n-2m+1}^{n}(q^i-(-1)^i)}{\displaystyle\prod_{i=1}^{m}(q^{2i}-1)}.$$

□

Corollary 5.21. Let $0 \le m \le n$, then the number of m-dimensional non-isotropic subspaces in the n-dimensional unitary space over $\mathbb{F}_{q^2}$ is

$$N(m,m;n) = q^{m(n-m)}\frac{\displaystyle\prod_{i=n-m+1}^{n}(q^i-(-1)^i)}{\displaystyle\prod_{i=1}^{m}(q^i-(-1)^i)}.$$

□

5.3 Singular Unitary Geometry over Finite Fields

We follow the notation of the previous two sections. Parallel to Chapter 3 in order to compute the number of subspaces of type (m_1, r_1) contained in a given subspace of type (m, r) in the n-dimensional unitary space over $\mathbb{F}_{q^2}$ we have to introduce the singular unitary spaces over $\mathbb{F}_{q^2}$.

Let

$$I_l = \begin{pmatrix} I^{(n)} & \\ & 0^{(l)} \end{pmatrix}.$$

The set of $(n+l) \times (n+l)$ nonsingular matrices T over $\mathbb{F}_{q^2}$ satisfying

$$T I_l {}^t\bar{T} = I_l$$

forms a group, called the *singular unitary group* of degree $n+l$ and rank n over $\mathbb{F}_{q^2}$ and denoted by $U_{n+l,n}(\mathbb{F}_{q^2})$. For $l = 0$, it reduces to the ordinary unitary group $U_{n,n}(\mathbb{F}_{q^2}) = U_n(\mathbb{F}_{q^2})$. It can be readily verified that $U_{n+l,n}(\mathbb{F}_{q^2})$ consists of all $(n+l) \times (n+l)$ nonsingular matrices of the form

$$T = \begin{array}{c} \begin{matrix} n & l \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} n \\ l \end{matrix} \end{array},$$

where $T_{11} {}^t\bar{T}_{11} = I^{(n)}$ and T_{22} is nonsingular. Parallel to Theorem 3.21 we have

Theorem 5.22.

$$| U_{n+l,n}(\mathbb{F}_{q^2}) | = q^{2nl+l(l-1)+\frac{n(n-1)}{2}} \prod_{i=1}^{n} (q^i - (-1)^i) \prod_{i=1}^{l} (q^{2i} - 1).$$

Proof. Similar to the proof of Theorem 3.21. □

Let $\mathbb{F}_{q^2}^{(n+l)}$ be the $(n+l)$-dimensional space of row vectors over $\mathbb{F}_{q^2}$. Clearly, $\mathbb{F}_{q^2}^{(n+l)}$ admits an action of $U_{n+l,n}(\mathbb{F}_{q^2})$ defined as follows:

$$\begin{aligned} \mathbb{F}_{q^2}^{(n+l)} \times U_{n+l,n}(\mathbb{F}_{q^2}) &\to \mathbb{F}_{q^2}^{(n+l)} \\ ((x_1, x_2, \ldots, x_{n+l}), T) &\mapsto (x_1, x_2, \ldots, x_{n+l})T. \end{aligned}$$

The vector space $\mathbb{F}_{q^2}^{(n+l)}$ together with this action of $U_{n+l,n}(\mathbb{F}_{q^2})$ is called the *singular unitary space* over $\mathbb{F}_{q^2}$. At first, we want to study how the

subspaces of $\mathbb{F}_{q^2}^{(n+l)}$ are subdivided into orbits under the singular unitary group $U_{n+l,n}(\mathbb{F}_{q^2})$.

An m-dimensional subspace P of $\mathbb{F}_{q^2}^{(n+l)}$ is said to be of type (m,r) if $PI_l\,{}^t\bar{P}$ is cogredient to $M(m,r)$ (cf. (5.7)). Denote the set of subspaces of type (m,r) in $\mathbb{F}_{q^2}^{(n+l)}$ by $\mathcal{M}(m,r;n+l,n)$ and let

$$N(m,r;n+l,n) = \mid \mathcal{M}(m,r;n+l,n) \mid .$$

For the ordinary unitary space, i.e., the case $l=0$, $\mathcal{M}(m,r;n+0,n)$ is simply $\mathcal{M}(m,r;n)$ which was studied previously in the last two sections.

Let E be the subspace of $\mathbb{F}_{q^2}^{(n+l)}$ generated by $e_{n+1}, e_{n+2}, \ldots, e_{n+l}$. An m-dimensional subspace P of $\mathbb{F}_{q^2}^{(n+l)}$ is called a *subspace of type* (m,r,k) if

(i) $PI_l\,{}^t\bar{P}$ is cogredient to $M(m,r)$, and

(ii) $\dim(P \cap E) = k$.

Denote the set of all subspaces of type (m,r,k) in $\mathbb{F}_{q^2}^{(n+l)}$ by $\mathcal{M}(m,r,k;n+l,n)$, and let

$$N(m,r,k;n+l,n) = \mid \mathcal{M}(m,r,k;n+l,n) \mid .$$

Parallel to Theorem 3.22 and Corollary 3.23 we have

Theorem 5.23. $\mathcal{M}(m,r,k;n+l,n)$ is non-empty if and only if

$$k \le l \qquad \text{and} \qquad 2r \le 2(m-k) \le n+r, \tag{5.11}$$

and if and only if

$$\max\{0, \frac{2m-n-r}{2}\} \le k \le \min\{l, m-r\}. \tag{5.12}$$

Moreover, if $\mathcal{M}(m,r,k;n+l,n)$ is non-empty, then it is an orbit of subspaces under $U_{n+l,n}(\mathbb{F}_{q^2})$. □

Corollary 5.24. $\mathcal{M}(m,r;n+l,n)$ is non-empty if and only if

$$\max\{0, \frac{2m-n-r}{2}\} \le \min\{l, m-r\}.$$

Moreover, $\mathcal{M}(m,r;n+l,n)$ is the disjoint union of $\mathcal{M}(m,r,k;n+l,n)$, where k satisfies (5.12). □

It follows from Theorem 5.23 that $N(m,r,k;n+l,n)=0$ whenever (5.12) does not hold. However, if (5.12) holds, then parallel to Theorem 3.24 we have

Theorem 5.25. Let

$$\max\{0,\frac{2m-n-r}{2}\}\leq k\leq \min\{l,m-r\}.$$

Then

$$N(m,r,k;n+l,n)=q^{r(n+r-2m+2k)+2(m-k)(l-k)}\times\frac{\prod_{i=n+r-2m+2k+1}^{n}(q^i-(-1)^i)\prod_{i=l-k+1}^{l}(q^{2i}-1)}{\prod_{i=1}^{r}(q^i-(-1)^i)\prod_{i=1}^{m-r-k}(q^{2i}-1)\prod_{i=1}^{k}(q^{2i}-1)}. \tag{5.13}$$

□

Corollary 5.26. Let

$$\max\{0,\frac{2m-n-r}{2}\}\leq \min\{l,m-r\}.$$

Then

$$N(m,r;n+l,n)=\sum_{k=\max\{0,[\frac{2m-n-r+1}{2}]\}}^{\min\{l,m-r\}}N(m,r,k;n+l,n),$$

where $N(m,r,k;n+l,n)$ is given by (5.13). □

We can also deduce an Anzahl theorem in the ordinary unitary geometry from the above corollary.

Let $\mathbb{F}_{q^2}^{(n)}$ be the n-dimensional row space on which the ordinary unitary group $U_n(\mathbb{F}_{q^2})$ acts. Let P be a fixed subspace of type (m,r) in $\mathbb{F}_{q^2}^{(n)}$. Denote by $\mathcal{M}(m_1,r_1;m,r;n)$ the set of subspaces of type (m_1,r_1) contained in P. Let

$$N(m_1,r_1;m,r;n)=\mid\mathcal{M}(m_1,r_1;m,r;n)\mid.$$

By Theorem 5.8, $N(m_1,r_1;m,r;n)$ is independent of the particular choice of the subspace P of type (m,r). Parallel to Theorems 3.26 and 3.27, we have

Theorem 5.27. $\mathcal{M}(m_1, r_1; m, r; n)$ is non-empty if and only if

$$\left.\begin{aligned} 2r \le 2m &\le n + r, \\ 2r_1 \le 2m_1 &\le n + r_1, \\ \text{and} \\ 0 \le r - r_1 &\le 2m - 2m_1, \end{aligned}\right\} \tag{5.14}$$

and these three conditions are equivalent to

$$\left.\begin{aligned} 2r \le 2m &\le n + r \\ \text{and} \\ \max\{0, \tfrac{2m_1 - r - r_1}{2}\} &\le \min\{m - r, m_1 - r_1\}. \end{aligned}\right\} \tag{5.15}$$

□

Theorem 5.28. Let

$$2r \le 2m \le n + r$$

and

$$\max\{0, \tfrac{2m_1 - r - r_1}{2}\} \le \min\{m - r, m_1 - r_1\}.$$

Then

$$\begin{aligned} &N(m_1, r_1; m, r; n) \\ &= \sum_{k=\max\{0, [\frac{2m_1 - r - r_1 + 1}{2}]\}}^{\min\{m-r, m_1 - r_1\}} q^{r_1(r + r_1 - 2m_1 + 2k) + 2(m_1 - k)(m - r - k)} \\ &\quad \times \frac{\prod_{i=r+r_1-2m_1+2k+1}^{r} (q^i - (-1)^i) \prod_{i=m-r-k+1}^{m-r} (q^{2i} - 1)}{\prod_{i=1}^{r_1} (q^i - (-1)^i) \prod_{i=1}^{m_1 - r_1 - k} (q^{2i} - 1) \prod_{i=1}^{k} (q^{2i} - 1)}. \end{aligned}$$

□

We can also study the similar problem in the singular unitary geometry. Now we consider again the $(n+l)$-dimensional row vector space $\mathbb{F}_q^{(n+l)}$ on which the singular unitary group $U_{n+l}(\mathbb{F}_{q^2})$ acts. Let P be a fixed subspace of type (m, r, k) in $\mathbb{F}_{q^2}^{(n+l)}$. Denote by $\mathcal{M}(m_1, r_1, k_1; m, r, k; n + l, n)$ the set of subspaces of type (m_1, r_1, k_1) contained in P. Let

$$N(m_1, r_1, k_1; m, r, k; n + l, n) = \mid \mathcal{M}(m_1, r_1, k_1; m, r, k; n + l, n) \mid,$$

which is independent of the particular choice of the subspace P of type (m, r, k). Then parallel to Theorems 3.28 and 3.29 we have the following two theorems.

Theorem 5.29. $\mathcal{M}(m_1, r_1, k_1; m, r, k; n+l, n)$ is non-empty if and only if

$$\left.\begin{array}{c} k_1 \le k \le l, \\ 2r \le 2(m-k) \le n+r, \\ 2r_1 \le 2(m_1-k_1) \le n_1+r_1, \end{array}\right\} \quad (5.16)$$

and

$$\left.\begin{array}{c} \\ \\ 0 \le r-r_1 \le 2(m-k)-2(m_1-k_1), \end{array}\right\}$$

and if and only if

$$\left.\begin{array}{c} k_1 \le k \le l, \\ 2r \le 2(m-k) \le n+r, \\ \\ \max\{0, \frac{2(m_1-k_1)-r-r_1}{2}\} \le \min\{m-k-r, m_1-k_1-r_1\}. \end{array}\right\} \quad (5.17)$$

and

□

Theorem 5.30. Assume that (5.17) holds. Then

$$\begin{aligned} &N(m_1, r_1, k_1; m, r, k; n+l, n) \\ &\quad = N(m_1-k_1, r_1; r+(m-k-r), r) N_2(k_1, k) q^{2(m_1-k_1)(k-k_1)}, \end{aligned}$$

where $N(m_1-k_1, r_1; r+(m-k-r), r)$ is given in Corollary 5.26 and $N_2(k_1, k)$ is the number of k_1-dimensional subspaces contained in a given k-dimensional vector space over $\mathbb{F}_{q^2}$, which can be obtained from $N(k_1, k)$ given in Theorem 1.7 by replacing q by q^2. □

The proofs of the above theorems are quite similar to the corresponding ones in singular symplectic geometry and will be omitted.

Finally, as another application of Theorem 5.22, let us now compute the number of Hermitian matrices of a given rank over a finite field.

Denote by $\mathcal{H}(n)$ the set of all $n \times n$ Hermitian matrices over $\mathbb{F}_{q^2}$. Define an action of $GL_n(\mathbb{F}_{q^2})$ on $\mathcal{H}(n)$ as follows:

$$\begin{array}{ccc} \mathcal{H}(n) \times GL_n(\mathbb{F}_{q^2}) & \to & \mathcal{H}(n) \\ (H, \quad T) & \mapsto & TH\,{}^t\bar{T}. \end{array}$$

Then $\mathcal{H}(n)$ is subdivided into orbits under the action of $GL_n(\mathbb{F}_{q^2})$. By Theorem 5.2 each orbit is characterized by the rank of the matrices contained in it. The orbit of all $n \times n$ Hermitian matrices of rank r $(0 \le r \le n)$ will be denoted by $\mathcal{H}(n, r)$. Then we have

Theorem 5.31. Let $0 \leq r \leq n$. Then

$$| \mathcal{H}(n,r) | = q^{\frac{r(r-1)}{2}} \frac{\prod_{i=n-r+1}^{n} (q^{2i}-1)}{\prod_{i=1}^{r}(q^i-(-1)^i)}.$$

Proof. Let

$$I_{n-r} = \begin{pmatrix} I^{(r)} & \\ & 0^{(n-r)} \end{pmatrix}.$$

Clearly the stabilizer of I_{n-r} in $GL_n(\mathbb{F}_{q^2})$ is the extended unitary group $U_{r+(n-r),r}(\mathbb{F}_{q^2})$ and

$$| \mathcal{H}(n,r) | = GL_n(\mathbb{F}_{q^2}) : U_{r+(n-r),r}(\mathbb{F}_{q^2}).$$

Hence our theorem follows from Theorems 1.6 and 5.22. □

Corollary 5.32. The number of $n \times n$ nonsingular Hermitian matrices over $\mathbb{F}_{q^2}$ is

$$q^{\frac{n(n-1)}{2}} \prod_{i=1}^{n}(q^i+(-1)^i).$$ □

5.4 Unitary Polarities and Polar Spaces Arising from Them

Now we change our language to the projective geometry language. Let us consider the $(n-1)$-dimensional projective space $PG(n-1,\mathbb{F}_{q^2})$ over $\mathbb{F}_{q^2}$ arising from the n-dimensional vector space $\mathbb{F}_{q^2}^{(n)}$ over $\mathbb{F}_{q^2}$. Let H be an $n \times n$ nonsingular Hermitian matrix over $\mathbb{F}_{q^2}$. Let P be an r-flat in $PG(n-1,\mathbb{F}_{q^2})$. Then

$$P^{\perp} = \{x \in \mathbb{F}_{q^2}^{(n)} \mid xH\,{}^t\bar{P} = 0\}$$

is an $(n-r-2)$-flat in $PG(n-1,\mathbb{F}_{q^2})$. Thus $\dim P + \dim P^{\perp} = n-2$. Moreover, $P \subset Q$ implies $P^{\perp} \supset Q^{\perp}$. Therefore the map

$$P \to P^{\perp}$$

from the set of flats in $PG(n-1,\mathbb{F}_{q^2})$ to itself is a correlation of $PG(n-1,\mathbb{F}_{q^2})$. Since

$$(P^{\perp})^{\perp} = P,$$

this correlation is a polarity, called the *unitary polarity* defined by the $n \times n$ nonsingular Hermitian matrix H, or simply a unitary polarity, or an *Hermitian polarity*. Denote by $\mathcal{P}(H)$ the set of absolutes of dimension $\leq [n/2]-1$ of the unitary polarity of $PG(n-1, \mathbb{F}_{q^2})$ defined by H and called it the *polar space* arising from the unitary polarity of $PG(n-1, \mathbb{F}_{q^2})$ defined by H.

An m-dimensional totally isotropic subspace of $\mathbb{F}_{q^2}^{(n)}$ with respect to H is called a *totally isotropic* $(m-1)$*-flat* in $PG(n-1, \mathbb{F}_{q^2})$ with respect to H. We deduce from Corollary 5.9 that maximal totally isotropic flats of $PG(n-1, \mathbb{F}_{q^2})$ are of dimension $[n/2]-1$. It follows from Theorem 5.5 that $\mathcal{P}(H)$ is the set of all totally isotropic flats with respect to H in $PG(n-1, \mathbb{F}_{q^2})$.

Let H_1 and H_2 be two $n \times n$ nonsingular Hermitian matrices over $\mathbb{F}_{q^2}$. By Corollary 5.3 there is an $n \times n$ nonsingular matrix A over $\mathbb{F}_{q^2}$ such that $H_1 = AH_2\,{}^t\bar{A}$. It can be readily verified that the map

$$\begin{aligned} \mathcal{P}(H_1) &\to \mathcal{P}(H_2) \\ P &\mapsto PA \end{aligned}$$

is a bijection such that $\dim PA = \dim P$, $(PA)^{\perp} = P^{\perp}A$ for any flat P in $\mathcal{P}(H_1)$, and for any two flats P and Q in $\mathcal{P}(H_1)$, $P \supset Q$ implies $PA \supset QA$. Therefore without loss of generality we can choose a particular $n \times n$ nonsingular Hermitian matrix H and study $\mathcal{P}(H)$. Now let $n = 2\nu + \delta$, where $\delta = 0$ or 1, we take H to be H_δ given by (5.1) and (5.2).

Theorem 5.33. The set $\mathcal{P}(H_\delta)$ of totally isotropic flats of $PG(n-1, \mathbb{F}_q)$ with respect to the $n \times n$ nonsingular Hermitian matrix H_δ forms a polar space of rank ν, where $\nu = [n/2]$. □

The proof of Theorem 5.33 runs similarly as that of Theorem 3.32 and will be omitted. Parallel to Corollary 3.35 we have

Corollary 5.34. The set of all totally isotropic points of $PG(3, \mathbb{F}_{q^2})$ or $PG(4, \mathbb{F}_{q^2})$ together with the set of all totally isotropic lines with respect to the non-singular Hermitian matrices

$$\begin{pmatrix} 0 & I^{(2)} \\ I^{(2)} & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & I^{(2)} & \\ I^{(2)} & 0 & \\ & & 1 \end{pmatrix}, \tag{5.18}$$

respectively, forms a generalized quadrangle. □

The generalized quadrangle formed by the totally isotropic points and lines with respect to the first matrix of (5.18) in $PG(3, \mathbb{F}_{q^2})$ is usually denoted by $H(3, q^2)$, and the one formed by the totally isotropic points and lines with respect to the second matrix of (5.18) in $PG(4, \mathbb{F}_q)$ is denoted by $H(4, q^2)$. It follows immediately from Theorem 5.8 that all lines of $H(d, q^2)$, where $d = 3$ or 4, contains the same number of points and all its points lie on the same number of lines. Moreover, we have

Corollary 5.35. The generalized quadrangles $H(d, q^2)$, where $d = 3$ or 4, have parameters

$$\begin{aligned} &s = q^2,\ t = q \quad \text{for} \quad H(3, q^2), \\ &s = q^2,\ t = q^3 \quad \text{for} \quad H(4, q^2). \end{aligned}$$

Proof. Clearly, every point of a totally isotropic line is totally isotropic, hence every line of $H(d, q^2)$ contains q^2+1 points and this gives $s = q^2$ for both $H(3, q^2)$ and $H(4, q^2)$. To compute t for $H(d, q^2)$, let us determine the number of lines of $H(d, q^2)$ passing through the point $(1, 0, 0, 0)$ or $(1, 0, 0, 0, 0)$ corresponding to the cases $d = 3$ and $d = 4$, respectively. We can assume that any line of $PG(d, \mathbb{F}_{q^2})$ containing $(1, 0, 0, 0)$ or $(1, 0, 0, 0, 0)$ is of the form

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x_2 & x_3 & x_4 \end{pmatrix} \qquad \text{or} \qquad \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & x_2 & x_3 & x_4 & x_5 \end{pmatrix},$$

where not all x_i are zero, respectively. If it is a line of $H(d, q^2)$, then it must be totally isotropic and we must have $x_3 = 0$ and $x_2\bar{x}_4 + \bar{x}_2 x_4 = 0$ or $x_2\bar{x}_4 + \bar{x}_2 x_4 + x_5\bar{x}_5 = 0$, respectively. Therefore it follows from Corollary 5.20 that $s = q$ or q^3 for $H(3, q^2)$ or $H(4, q^2)$, respectively.

Finally, parallel to the symplectic case, we can use the unitary polarity to derive a closed formula for the number of subspaces of a given type (m, r) in $\mathbb{F}_{q^2}^{(n)}$ containing a given subspace of type (m_1, r_1). Here we use the vector space language again.

Now let P_1 be a fixed subspace of type (m_1, r_1) in $\mathbb{F}_{q^2}^{(n)}$. Denote by $\mathcal{M}'(m_1, r_1;\ m, r; n)$ the set of subspaces of type (m, r) containing P_1. Let

$$N'(m_1, r_1; m, r; n) = \mid \mathcal{M}'(m_1, r_1; m, r; n) \mid .$$

By Theorem 5.8, $N'(m_1, r_1; m, r; n)$ is independent of the particular choice of the subspace P_1 of type (m_1, r_1). Parallel to Theorems 3.37 and 3.38, we have

Theorem 5.36. $\mathcal{M}'(m_1, r_1; m, r; n)$ is non-empty if and only if (5.14) holds. □

Theorem 5.37. Assume that (5.14) holds. Then

$$N'(m_1, r_1; m, r; n)$$
$$= \sum_{k=\max\{0,[\frac{2m_1-r-r_1+1}{2}]\}}^{\min\{m-r,m_1-r_1\}} q^{(n-2m+r)(r+r_1-2m_1+2k)+2(n-m-k)(m_1-r_1-k)}$$
$$\times \frac{\prod_{i=r+r_1-2m_1+2k+1}^{n-2m_1+r_1}(q^i-(-1)^i)\prod_{i=m_1-r_1-k+1}^{m_1-r_1}(q^{2i}-1)}{\prod_{i=1}^{n-2m+r}(q^i-(-1)^i)\prod_{i=1}^{m-r-k}(q^{2i}-1)\prod_{i=1}^{k}(q^{2i}-1)}.$$

□

5.5 Hermitian varieties in $PG(n, \mathbb{F}_{q^2})$ and $AG(n, \mathbb{F}_{q^2})$

The set of points $(x_0, x_1, \ldots, x_n)$ of $PG(n, \mathbb{F}_{q^2})$ which satisfy an equation of the form

$$\sum_{0\le i,k\le n} a_{ik}x_i\bar{x}_k = 0, \tag{5.19}$$

where a_{ik} $(0 \le i, k \le n)$ are elements of $\mathbb{F}_{q^2}$, $\bar{a}_{ik} = a_{ki}$ for all i, k, and not all a_{ik} are zero, is called an *Hermitian variety in* $PG(n, \mathbb{F}_{q^2})$ and the equation (5.19) is called its equation. An Hermitian variety and its equation are said to be *non-degenerate*, if its equation cannot be transformed into an equation with less than $n+1$ variables under projective transformations.

At first we would like to study how the equation of an Hermitian variety can be transformed into the simplest form under the projective group.

Theorem 5.38. Under projective transformations an Hermitian variety in $PG(n, \mathbb{F}_{q^2})$ can be carried into an Hermitian variety with one of the following equations as its equation

$$\sum_{i=0}^{r} x_i\bar{x}_i = 0\ (0 \le r \le n). \tag{5.20}$$

Moreover, any two of these equations are not projectively equivalent.

Proof. Introduce the coefficient matrix

$$A = (a_{ik}) \; 0 \leq i, k \leq n,$$

which is an $(n+1) \times (n+1)$ Hermitian matrix, then (5.19) can be written in the following matrix form

$$(x_0, x_1, \ldots, x_n) A \, {}^t\overline{(x_0, x_1, \ldots, x_n)} = 0. \tag{5.21}$$

Under a projective transformation

$$(x_0, x_1, \ldots, x_n) \mapsto (x_0, x_1, \ldots, x_n) T, \tag{5.22}$$

where $T \in GL_{n+1}(\mathbb{F}_q)$, (5.21) is carried into

$$(x_0, x_1, \ldots, x_n) T A \, {}^t\bar{T} \, {}^t\overline{(x_0, x_1, \ldots, x_n)} = 0.$$

Hence the coefficient matrix A is subjected to the cogredient transformation

$$A \mapsto T A \, {}^t\bar{T}.$$

Assume that A is of rank $r+1$, then by Theorem 5.2 there is an $(n+1) \times (n+1)$ nonsingular matrix T such that

$$T A \, {}^t\bar{T} = \begin{pmatrix} I^{(r+1)} & \\ & 0^{(n-r)} \end{pmatrix}.$$

Therefore under the projective transformation (5.22), (5.21) is carried into (5.20).

Since the rank of an Hermitian matrix is invariant under cogredient transformations, any two of the equations in (5.20) are not projectively equivalent. □

The following theorem is a special case of Corollary 5.20.

Theorem 5.39. The number of points lying on the non-degenerate Hermitian variety

$$\sum_{i=0}^{n} x_i^2 = 0$$

in $PG(n, \mathbb{F}_{q^2})$ is equal to

$$\frac{(q^{n+1} - (-1)^{n+1})(q^n - (-1)^n)}{q^2 - 1}.$$

□

Now let us study the Hermitian varieties in $AG(n, \mathbb{F}_{q^2})$.

The set of points $(x_1, x_2, \ldots, x_n)$ of $AG(n, \mathbb{F}_{q^2})$ which satisfy an equation of the form

$$\sum_{1 \le i,k \le n} a_{ik} x_i \bar{x}_k + \sum_{j=1}^{n} (b_j x_j + \bar{b}_j \bar{x}_j) + c = 0, \tag{5.23}$$

where a_{ik} $(1 \le i, k \le n)$ and b_j $(1 \le j \le n)$ are elements of $\mathbb{F}_{q^2}$, $\bar{a}_{ik} = a_{ki}$ for all i, k, not all a_{ik} are zero, and $c \in \mathbb{F}_q$, is called an *Hermitian variety in* $AG(n, \mathbb{F}_{q^2})$ and the equation (5.23) is called its equation. We define an Hermitian variety and its equation to be *non-degenerate* if its equation cannot be transformed into an equation with less than n variables under affine transformations. Parallel to Theorems 5.38 and 5.39 we have

Theorem 5.40. An Hermitian variety in $AG(n, \mathbb{F}_{q^2})$ can be carried under affine transformations into an Hermitian variety with one of the following equations as its equation

$$\sum_{i=1}^{r} x_i \bar{x}_i = 0 \, (1 \le r \le n)$$

$$\sum_{i=1}^{r} x_i \bar{x}_i + 1 = 0 \, (1 \le r \le n)$$

$$\sum_{i=1}^{r} x_i \bar{x}_i + x_{r+1} + \bar{x}_{r+1} = 0 \, (1 \le r \le n).$$

Moreover, any two of these equations are not affinely equivalent. □

Theorem 5.41. The numbers of points in $AG(n, \mathbb{F}_{q^2})$ lying on the non-degenerate Hermitian varieties are listed below.

equations	number of points
$\sum_{i=1}^{n} x_i \bar{x}_i = 0$	$(q^n - (-1)^n)(q^{n-1} - (-1)^{n-1}) + 1$
$\sum_{i=1}^{n} x_i \bar{x}_i + 1 = 0$	$q^{n-1}(q^n - (-1)^n)$
$\sum_{i=1}^{n-1} x_i \bar{x}_i + x_n + \bar{x}_n = 0$	q^{2n-3}

□

5.6 Comments

Unitary groups over finite fields were first studied by L. E. Dickson; in particular, Lemma 5.1 and Theorems 5.2 and 5.16 are due to him (cf.

Dickson, 1900). Unitary groups and unitary spaces over division rings with involutions were studied by J. Dieudonné and L. K. Hua independently; in particular, a generalization of Witt's theorem was obtained by them (cf. Dieudonné, 1952 and Hua, 1953). However, the present proof of Witt's theorem in the unitary space over finite fields (Theorem 5.8), which is based on Lemma 5.6, is due to the author. From Lemma 5.6 we deduce also Theorems 5.7, 5.11, and Lemma 5.12, which are also due to the author (cf. Wan, 1992a). Lemma 5.12 can be regarded as an improvement of Theorem 5.8 and is used in proving the Anzahl theorems in Section 5.2. Corollary 5.10 is due to B. Yang and W. Wei (cf. Yang and Wei, 1989).

All the results of Section 5.2, except Theorem 5.16, are due to Z. Wan and B. Yang (cf. Wan and Yang, 1964 and 1965). Corollary 5.20 was obtained independently by B. Segre and by R. C. Bose and I. M. Chakravati (cf. Segre 1965 and Bose and Chakravati, 1966).

All the results of Section 5.3, except Theorem 5.31, are due to the author (cf. Wan, 1993c and 1993d). Theorem 5.31 was obtained by L. Carlitz and J. Hodge when the characteristic is greater than 2 and by Y. Wang for the general case (cf. Carlitz and Hodge, 1955 and Wang, 1980).

Theorem 5.33 and Corollaries 5.34 and 5.35 in Section 5.4 are due to J. Tits (cf. Tits, 1959 and 1974). Theorem 5.37 is due to B. Yang and W. Wei (cf. Yang and Wei, 1989).

Theorems 5.38 and 5.40 in Section 5.5 are immediate consequences of Theorem 5.2 of L. E. Dickson.

5.7 Exercises

5.1 Give all the missing proofs of the theorems in this Chapter.

5.2 Let $2r \le 2m \le n + r$, $2r_1 \le 2m_1 \le n + r_1$, and P and Q are subspaces of types (m, r) and (m_1, r_1), respectively, in the n-dimensional unitary space over $\mathbb{F}_{q^2}$. Assume that $P \supseteq Q$. Prove that there is a matrix representation P of the subspace P such that the first m_1 rows of the matrix P span Q and

$$P\,{}^t\bar{P} = \begin{pmatrix} I^{(r_1)} & & & & & \\ & 0 & & I^{(s)} & & \\ & & 0^{(\sigma_1)} & & & \\ & I^{(s)} & & 0 & & \\ & & & & I^{(r_2)} & \\ & & & & & 0^{(\sigma_2)} \end{pmatrix},$$

where s and r_2 are nonnegative integers, $\sigma_1 = m_1 - r_1 - s \ge 0$, $\sigma_2 = m - m_1 - s - r_2 \ge 0$, $r = r_1 + 2s + r_2$, and $2m - 2m_1 \ge r - r_1 \ge 0$.

5.3 Let k be an integer ≥ 3. Denote by n the number of points in $PG(k-1, \mathbb{F}_{q^2})$ lying on the non-degenerate Hermitian variety

$$x_1\bar{x}_1 + x_2\bar{x}_2 + \cdots + x_k\bar{x}_k = 0.$$

For each one of these n points we choose a system of homogeneous coordinates and regard it as a k-dimensional column vector. Arrange these n column vectors in any order into a $k \times n$ matrix over $\mathbb{F}_{q^2}$, which will be denoted by G. Prove that G is of rank k. Then G can be regarded as a generator matrix of an (n, k) linear code over $\mathbb{F}_{q^2}$, which will be denoted by C. Prove that the weights of the non-zero code words of C take only two different values, which will be denoted by w_1 and w_2. Then compute w_1 and w_2 and the numbers of code words of weight w_1 and w_2, respectively.

Chapter 6
Geometry of Orthogonal Groups over Finite Fields of Odd Characteristic

6.1 Orthogonal Geometry over Finite Fields of Odd Characteristic

In this chapter, we assume that $\mathbb{F}_q$ is a finite field of odd characteristic. As in Section 1.3 we choose a fixed non-square element z of $\mathbb{F}_q^*$.

Let S be an $n \times n$ nonsingular symmetric matrix over $\mathbb{F}_q$. An $n \times n$ matrix T over $\mathbb{F}_q$ is said to be *orthogonal* with respect to S, if

$$TS\,{}^tT = S.$$

Clearly, $n \times n$ orthogonal matrices with respect to a nonsingular symmetric matrix S are nonsingular and they form a group with respect to matrix multiplication, called the *orthogonal group* of degree n with respect to S over $\mathbb{F}_q$ and denoted by $O_n(\mathbb{F}_q, S)$. The index of S is also called the *index* of the orthogonal group $O_n(\mathbb{F}_q, S)$. If S_1 and S_2 are two cogredient $n \times n$ nonsingular symmetric matrices over $\mathbb{F}_q$, then as in the symplectic case we can prove that $O_n(\mathbb{F}_q, S_1)$ and $O_n(\mathbb{F}_q, S_2)$ are isomorphic. Hence by Theorem 1.26 it is sufficient to discuss the four orthogonal groups with respect to the following four $n \times n$ nonsingular symmetric matrices

$$S_{2\nu} = \begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix}, \tag{6.1}$$

$$S_{2\nu+1,1} = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix}, \tag{6.2}$$

$$S_{2\nu+1,z} = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & z \end{pmatrix}, \tag{6.3}$$

$$S_{2\nu+2} = \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 1 & \\ & & & -z \end{pmatrix}, \tag{6.4}$$

where $n = 2\nu$, $2\nu+1$, $2\nu+1$, and $2\nu+2$, respectively. In order to cover these four cases, we introduce the notation $S_{2\nu+\delta,\Delta}$, where ν is its index and Δ denotes its definite part, i.e.,

$$\Delta = \begin{cases} \phi, & \text{if } \delta = 0, \\ (1) \text{ or } (z), & \text{if } \delta = 1, \\ \begin{pmatrix} 1 & \\ & -z \end{pmatrix}, & \text{if } \delta = 2. \end{cases}$$

The orthogonal group of degree $2\nu+\delta$ with respect to $S_{2\nu+\delta,\Delta}$ over $\mathbb{F}_q$ will be denoted by $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$. Clearly, $O_{2\nu+1}(\mathbb{F}_q, S_{2\nu+1,z}) = O_{2\nu+1}(\mathbb{F}_q, zS_{2\nu+1,z})$. Since $zS_{2\nu+1,z}$ and $S_{2\nu+1,1}$ are cogredient, the groups $O_{2\nu+1}(\mathbb{F}_q, zS_{2\nu+1,z})$ and $O_{2\nu+1}(\mathbb{F}_q, S_{2\nu+1,1})$ are isomorphic. It follows that $O_{2\nu+1,1}(\mathbb{F}_q)$ and $O_{2\nu+1,z}(\mathbb{F}_q)$ are isomorphic. Therefore actually only three types of orthogonal groups need to be considered.

For simplicity sometimes we write S for $S_{2\nu+\delta,\Delta}$, i.e., S is one of the four $(2\nu+\delta) \times (2\nu+\delta)$ nonsingular symmetric matrices (6.1) – (6.4).

There is an action of $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ on $\mathbb{F}_q^{(2\nu+\delta)}$ defined as follows:

$$\begin{aligned} &\mathbb{F}_q^{(2\nu+\delta)} \times O_{2\nu+\delta,\Delta}(\mathbb{F}_q) \to \mathbb{F}_q^{(2\nu+\delta)} \\ &((x_1, x_2, \ldots, x_{2\nu+\delta}), T) \mapsto (x_1, x_2, \ldots, x_{2\nu+\delta})T. \end{aligned}$$

The elements of $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ are then called *orthogonal transformations* with respect to S. The vector space $\mathbb{F}_q^{(2\nu+\delta)}$ together with the above group action of the orthogonal group $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ is called the $(2\nu+\delta)$-dimensional *orthogonal space* over $\mathbb{F}_q$ with respect to S. The *geometry of the orthogonal group* $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$, or simply the *orthogonal geometry* over $\mathbb{F}_q$, aims at the study of properties of geometric figures of the orthogonal space $\mathbb{F}_q^{(2\nu+\delta)}$ which are invariant under the orthogonal group $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$.

At first let us study how the vector subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ are partitioned into orbits under $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$.

Let P be an m-dimensional vector subspace of $\mathbb{F}_q^{(2\nu+\delta)}$. Then $PS\,{}^tP$ is an $m \times m$ symmetric matrix. By Theorem 1.26 $PS\,{}^tP$ is cogredient to one of the following four normal forms:

$$M(m, 2s, s) = \begin{pmatrix} 0 & I^{(s)} & \\ I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix},$$

$$M(m, 2s+1, s, 1) = \begin{pmatrix} 0 & I^{(s)} & & \\ I^{(s)} & 0 & & \\ & & 1 & \\ & & & 0^{(m-2s-1)} \end{pmatrix},$$

$$M(m, 2s+1, s, z) = \begin{pmatrix} 0 & I^{(s)} & & \\ I^{(s)} & 0 & & \\ & & z & \\ & & & 0^{(m-2s-1)} \end{pmatrix},$$

and

$$M(m, 2s+2, s) = \begin{pmatrix} 0 & I^{(s)} & & & \\ I^{(s)} & 0 & & & \\ & & 1 & & \\ & & & -z & \\ & & & & 0^{(m-2s-2)} \end{pmatrix}.$$

We use the symbol $M(m, 2s+\gamma, s, \Gamma)$ to represent any one of these four normal forms, where s is its *index*, $\gamma = 0, 1$, or 2, and Γ represents the definite part in these normal forms. In particular, when $\gamma = 1$, we have $\Gamma = (1)$ or (z); when $\gamma = 0$ or 2,

$$\Gamma = \phi \qquad \text{or} \qquad \begin{pmatrix} 1 & \\ & -z \end{pmatrix},$$

respectively, and may be omitted. If $PS\,{}^tP$ is cogredient to $M(m, 2s+\gamma, s, \Gamma)$, then P is called a *subspace of type* $(m, 2s+\gamma, s, \Gamma)$ with respect to S in $\mathbb{F}_q^{(2\nu+\delta)}$. Subspaces of type $(m, 2s, s)$, $(m, 2s+1, s, 1)$, $(m, 2s+1, s, z)$, and $(m, 2s+2, s)$ are also called *subspace of the hyperbolic type*, *the square type*, *the non-square type*, and *the elliptic type*, respectively. Moreover, subspaces of the hyperbolic type with $s = 0$ are called *totally isotropic subspaces* and subspaces of type $(m, 2s+\gamma, s, \Gamma)$ with $m = 2s+\gamma$ are called *non-isotropic subspaces*. It is clear that a subspace P is totally

isotropic if and only if $PS\,{}^tP = 0$, and it is non-isotropic if and only if $PS\,{}^tP$ is nonsingular.

For example, if $s \le \nu$, then

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.5}$$

is an s-dimensional totally isotropic subspace of $\mathbb{F}_q^{(2\nu+\delta)}$,

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(s)} & 0 & 0 \end{pmatrix} \end{array} \tag{6.6}$$

and

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(s)} & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(\delta)} \end{pmatrix} \end{array} \tag{6.7}$$

are non-isotropic subspaces of dimension $2s$ and $2s+\delta$, respectively. More generally, if $2s \le m \le \nu + s$ and we write $\nu + s - m = \sigma$, then

$$\begin{array}{c} \begin{matrix} s & m-2s & \sigma & s & m-2s & \sigma & \delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.8}$$

is a subspace of type $(m, 2s, s)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Let $\delta = 0$; if $2s+1 \le m \le \nu+s$ and we write $\nu + s - m = \sigma$, then

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \Gamma/2 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{6.9}$$

where $\Gamma = 1$ or z, is a subspace of type $(m, 2s+1, s, \Gamma)$; and if $2s+2 \le m \le \nu + s$ and we write $\sigma = \nu + s - m$, then

$$\begin{array}{c} \begin{matrix} s & 2 & m-2s-2 & \sigma & s & 2 & m-2s-2 & \sigma \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(2)} & 0 & 0 & 0 & \begin{pmatrix} 1/2 & \\ & -z/2 \end{pmatrix} & 0 & 0 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.10}$$

is a subspace of type $(m, 2s+2, s)$. Let $\delta = 1$; if $2s+1 \le m \le \nu+s+1$ and we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.11}$$

is a subspace of type $(m, 2s+1, s, \Delta)$; if $2s+1 \le m \le \nu+s$ and we write $\sigma = \nu + s - m$, then

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \Gamma/2 & 0 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.12}$$

is a subspace of type $(m, 2s+1, s, \Gamma)$, where $\Gamma \neq \Delta$; and if $2s+2 \le m \le \nu+s+1$ and we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{matrix} s & 1 & m-2s-2 & \sigma & s & 1 & m-2s-2 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & -z\Delta/2 & 0 & 0 & 0 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.13}$$

is a subspace of type $(m, 2s+2, s)$. Finally, let $\delta = 2$; if $2s+1 \le m \le \nu+s+1$ and we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & x & y \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{6.14}$$

where $x^2 - zy^2 = \Gamma = 1$ or z, is a subspace of type $(m, 2s+1, s, \Gamma)$; and if $2s+2 \le m \le \nu+s+2$ and we write $\sigma = \nu + s - m + 2$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-2 & \sigma & s & m-2s-2 & \sigma & 2 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I^{(2)} \\ 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{6.15}$$

is a subspace of type $(m, 2s+2, s)$.

Let v be a vector of $\mathbb{F}_q^{(2\nu+\delta)}$. v is called an *isotropic* or a *non-isotropic vector* if $vS^tv = 0$ or $vS^tv \neq 0$, respectively. Clearly, a non-zero vector v is isotropic or non-isotropic, if and only if the 1-dimensional subspace $\langle v \rangle$ generated by v is totally isotropic or non-isotropic, respectively. Non-isotropic vectors v are further divided into vectors of the *square type* and vectors of the *non-square type* according to $vS^tv \in \mathbb{F}_q^{*2}$ or not, respectively. Thus v is of the square type or the non-square type if and only if $\langle v \rangle$ is of the square type or the non-square type, respectively.

For example, all $e_1, e_2, \ldots, e_{2\nu}$ are isotropic vectors, $e_i + \frac{1}{2}e_{\nu+i} (1 \leq i \leq \nu)$ are non-isotropic vectors of the square type, and $e_i + \frac{z}{2}e_{\nu+i} (1 \leq i \leq \nu)$ are non-isotropic vectors of the non-square type.

Two vectors x and y of $\mathbb{F}_q^{(2\nu+\delta)}$ are said to be *orthogonal* with respect to S, if $xS^ty = 0$. Clearly, if $xS^ty = 0$, then $yS^tx = 0$, and isotropic vectors are vectors orthogonal to themselves.

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu+\delta)}$, we use the symbol $P^\perp$ to denote the set of vectors which are orthogonal to every vector of P with respect to S, i.e.,

$$P^\perp = \{y \in \mathbb{F}_q^{(2\nu+\delta)} \mid yS^tx = 0 \text{ for all } x \in P\}.$$

Clearly, $P^\perp$ is a $(2\nu+\delta-m)$-dimensional subspace of P and is called the *dual subspace of* P with respect to S.

For example, the dual subspace of (6.5) is

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & I^{(\nu-s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s)} & 0 \\ 0 & 0 & 0 & 0 & I^{(\delta)} \end{pmatrix} \end{array},$$

which is of type $(2\nu+\delta-s,\ 2(\nu-s)+\delta,\ \nu-s,\ \Delta)$, the dual subspace of (6.6) is

$$\begin{array}{c} \begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix} \\ \begin{pmatrix} 0 & I^{(\nu-s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s)} & 0 \\ 0 & 0 & 0 & 0 & I^{(\delta)} \end{pmatrix} \end{array},$$

which is of type $(2(\nu-s)+\delta,\ 2(\nu-s)+\delta,\ \nu-s,\ \Delta)$, and the dual subspace

of (6.7) is

$$\begin{matrix} s & \nu-s & s & \nu-s & \delta \end{matrix}$$
$$\begin{pmatrix} 0 & I^{(\nu-s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\nu-s)} & 0 \end{pmatrix},$$

which is of type $(2(\nu-s),\ 2(\nu-s),\ \nu-s)$. More generally, if $2s \le m \le \nu+s$, the dual subspace of (6.8) is

$$\begin{matrix} s & m-2s & \sigma & s & m-2s & \sigma & \delta \end{matrix}$$
$$\begin{pmatrix} 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I^{(\delta)} \end{pmatrix},$$

which is of type $(2\nu+\delta-m,\ 2\sigma+\delta,\ \sigma,\ \Delta)$, where $\sigma = \nu+s-m$. Let $\delta = 0$; if $2s+1 \le m \le \nu+s$, then the dual subspace of (6.9) is

$$\begin{matrix} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma \end{matrix}$$
$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & -\Gamma/2 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} \end{pmatrix},$$

which is of type $(2\nu-m,\ 2\sigma+1,\ \sigma,\ \Gamma_1)$, where $\sigma = \nu+s-m$ and

$$\Gamma_1 = \begin{cases} \Gamma, & \text{if } -1 \in \mathbb{F}_q^{*2}, \\ z \text{ or } 1, & \text{if } -1 \notin \mathbb{F}_q^{*2} \text{ and } \Gamma = 1 \text{ or } z, \text{ respectively;} \end{cases} \tag{6.16}$$

and if $2s+2 \le m \le \nu+s$, then the dual subspace of (6.10) is

$$\begin{matrix} s & 2 & m-2s-2 & \sigma & s & 2 & m-2s-2 & \sigma \end{matrix}$$
$$\begin{pmatrix} 0 & I^{(2)} & 0 & 0 & 0 & \begin{pmatrix} -1/2 & \\ & z/2 \end{pmatrix} & 0 & 0 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} \end{pmatrix},$$

which is of type $(2\nu-m,\ 2\sigma+2,\ \sigma)$, where $\sigma = \nu+s-m$. Let $\delta = 1$; if $2s+1 \le m \le \nu+s+1$, then the dual subspace of (6.11) is

$$\begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 \end{matrix}$$
$$\begin{pmatrix} 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 \end{pmatrix},$$

which is of type $(2\nu+1-m,\ 2\sigma,\ \sigma)$, where $\sigma=\nu+s-m+1$; if $2s+1\le m\le\nu+s$, then the dual subspace of (6.12) is

$$\begin{array}{cccccccccc} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma & 1 \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & -\Gamma/2 & 0 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array},$$

which is of type $(2\nu+1-m,\ 2\sigma+2,\ \sigma)$, where $\sigma=\nu+s-m$; and if $2s+2\le m\le\nu+s+1$, then the dual subspace of (6.13) is

$$\begin{array}{cccccccccc} s & 1 & m-2s-2 & \sigma & s & 1 & m-2s-2 & \sigma & 1 \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & z\Delta/2 & 0 & 0 & 0 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 \end{pmatrix} \end{array},$$

which is of type $(2\nu+1-m,\ 2\sigma+1,\ \sigma,\ \Gamma_2)$, where $\sigma=\nu+s-m+1$ and

$$\Gamma_2=\begin{cases} 1, & \text{if } \Delta=z, \\ z, & \text{if } \Delta=1. \end{cases} \tag{6.17}$$

Finally, let $\delta=2$; if $2s+1\le m\le\nu+s+1$, then the dual subspace of (6.14) is

$$\begin{array}{cccccccc} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 & 1 \\ \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & zy & x \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 & 0 \end{pmatrix} \end{array},$$

which is of type $(2\nu+2-m,\ 2\sigma+1,\ \sigma,\ \Gamma_3)$, where $\sigma=\nu+s-m+1$ and

$$\Gamma_3=\begin{cases} z \text{ or } 1, & \text{if } -1\in\mathbb{F}_q^{*2} \text{ and } \Gamma=1 \text{ or } z, \text{ respectively}, \\ \Gamma, & \text{if } -1\notin\mathbb{F}_q^{*2}; \end{cases} \tag{6.18}$$

and if $2s+2\le m\le\nu+s+2$, then the dual subspace of (6.15) is

$$\begin{array}{ccccccc} s & m-2s-2 & \sigma & s & m-2s-2 & \sigma & 2 \\ \begin{pmatrix} 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I^{(\sigma)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I^{(\sigma)} & 0 \end{pmatrix} \end{array},$$

which is of type $(2\nu + 2 - m,\ 2\sigma,\ \sigma)$, where $\sigma = \nu + s - m + 2$.

Let $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ denote the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ with respect to S in $\mathbb{F}_q^{(2\nu+\delta)}$. We shall prove that it is an orbit of subspaces under $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$. However, some preparatory results will be proved first. Parallel to Theorem 3.4 we have the following necessary and sufficient conditions for a subspace to be totally isotropic or non-isotropic.

Theorem 6.1. A subspace P is totally isotropic, if and only if $P \subseteq P^{\perp}$; and P is non-isotropic, if and only if $P \cap P^{\perp} = (0)$. □

Parallel to Lemma 3.5 we have

Lemma 6.2. Let P be a subspace of type $(m, 2s+\gamma, s, \Gamma)$ with respect to S in $\mathbb{F}_q^{(2\nu+\delta)}$. Then $2(\nu+s-m)+\delta+\gamma \geq 0$ and there exists an $m \times m$ nonsingular matrix Q and a $(2\nu+\delta-m) \times (2\nu+\delta)$ matrix Z such that

$$\begin{pmatrix} QP \\ Z \end{pmatrix} S \, {}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix} = \begin{pmatrix} \Lambda & & & \\ & 0 & I^{(m-2s-\gamma)} & \\ & I^{(m-2s-\gamma)} & 0 & \\ & & & \Sigma \end{pmatrix},$$

where $\Lambda = M(2s+\gamma, 2s+\gamma, s, \Gamma)$, Σ is a cogredience normal form of $\sigma \times \sigma$ nonsingular symmetric matrices and is uniquely determined by S and the type of P, and $\sigma = 2(\nu+s-m)+\delta+\gamma$.

Proof. Since P is a subspace of type $(m, 2s+\gamma, s, \Gamma)$, there is an $m \times m$ nonsingular matrix Q such that

$$QPS\,{}^tP\,{}^tQ = M(m, 2s+\gamma, s, \Gamma).$$

Let Z_1 be a $(2\nu+\delta-m) \times (2\nu+\delta)$ matrix such that

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix}$$

is nonsingular. We may write

$$\begin{pmatrix} QP \\ Z_1 \end{pmatrix} S \, {}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} = \begin{array}{c} \begin{matrix} 2s+\gamma & m-2s-\gamma & 2\nu+\delta-m \end{matrix} \\ \begin{pmatrix} \Lambda & 0 & A \\ 0 & 0 & B \\ {}^tA & {}^tB & C \end{pmatrix} \end{array} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ 2\nu+\delta-m \end{matrix},$$

where $\Lambda = M(2s+\gamma, 2s+\gamma, s, \Gamma)$ and C is a symmetric matrix. Let

$$R = \begin{pmatrix} I & & \\ & I & \\ -{}^tA\Lambda^{-1} & & I \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ 2\nu+\delta-m \end{matrix},$$

then

$$R \begin{pmatrix} QP \\ Z_1 \end{pmatrix} S \, {}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} {}^tR = \begin{pmatrix} \Lambda & 0 & 0 \\ 0 & 0 & B \\ 0 & {}^tB & D \end{pmatrix},$$

where D is symmetric. Since S is nonsingular, B is of rank $m-2s-\gamma$. Thus $m-2s-\gamma \le 2\nu+\delta-m$, which gives $\sigma = 2(\nu+s-m)+\delta+\gamma \ge 0$, and there exists a $(2\nu+\delta-m)\times(2\nu+\delta-m)$ nonsingular matrix L_1 such that

$$B\,{}^tL_1 = (I^{(m-2s-\gamma)} \;\; 0^{(m-2s-\gamma,\sigma)}).$$

Let

$$R_1 = \begin{pmatrix} I & \\ & L_1 \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix},$$

then

$$R_1 R \begin{pmatrix} QP \\ Z_1 \end{pmatrix} S \, {}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} {}^tR \, {}^tR_1 = \begin{pmatrix} \Lambda & & & \\ & 0 & I & 0 \\ & I & D_1 & D_2 \\ & 0 & {}^tD_2 & D_3 \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma \end{matrix}.$$

Since ${}^tD_1 = D_1$, we can assume that

$$D_1 = \begin{pmatrix} d_{11} & d_{12} & \dots & d_{1l} \\ d_{12} & d_{22} & \dots & d_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ d_{1l} & d_{2l} & \dots & d_{ll} \end{pmatrix},$$

where $l = m-2s-\gamma$. Let

$$L_2 = \begin{pmatrix} d_{11}/2 & d_{12} & \dots & d_{1l} \\ & d_{22}/2 & \dots & d_{2l} \\ & & \ddots & \vdots \\ & & & d_{ll}/2 \end{pmatrix},$$

then $D_1 = L_2 + {}^tL_2$. Let

$$R_2 = \begin{pmatrix} I & & & \\ & I & & \\ & -{}^tL_2 & I & \\ & -{}^tD_2 & & I \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma \end{matrix},$$

then

$$R_2R_1R\begin{pmatrix} QP \\ Z_1 \end{pmatrix} S\,{}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} {}^tR\,{}^tR_1\,{}^tR_2 = \begin{pmatrix} \Lambda & & & \\ & 0 & I & \\ & I & 0 & \\ & & & D_3 \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma \end{matrix}.$$

Let the cogredience normal form of D_3 be Σ; by Lemma 1.24 it is uniquely determined by S and the type of P. There is a $\sigma \times \sigma$ nonsingular matrix Q_3 such that

$$Q_3D_3\,{}^tQ_3 = \Sigma.$$

Let

$$R_3 = \begin{pmatrix} I & \\ & Q_3 \end{pmatrix} \begin{matrix} 2m-2s-\gamma \\ \sigma \end{matrix},$$

then

$$R_3R_2R_1R\begin{pmatrix} QP \\ Z_1 \end{pmatrix} S\,{}^t\!\begin{pmatrix} QP \\ Z_1 \end{pmatrix} {}^tR\,{}^tR_1\,{}^tR_2\,{}^tR_3 = \begin{pmatrix} \Lambda & & \\ & 0 & I \\ & I & 0 \\ & & & \Sigma \end{pmatrix}.$$

Notice that $R_3R_2R_1R$ is a matrix of the form

$$\begin{pmatrix} I & \\ L_3 & L_4 \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix},$$

hence we can assume that

$$R_3R_2R_1R\begin{pmatrix} QP \\ Z_1 \end{pmatrix} = \begin{pmatrix} QP \\ Z \end{pmatrix}$$

for some $(2\nu+\delta-m)\times(2\nu+\delta)$ matrix Z. Therefore

$$\begin{pmatrix} QP \\ Z \end{pmatrix} S \, {}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix} = \begin{pmatrix} \Lambda & & & \\ & 0 & I & \\ & I & 0 & \\ & & & \Sigma \end{pmatrix}.$$

□

From Lemma 6.2 we deduce immediately the condition for the existence of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$.

Theorem 6.3. Subspaces of type $(m, 2s+\gamma, s, \Gamma)$ exist in the $(2\nu+\delta)$-dimensional orthogonal space $\mathbb{F}_q^{(2\nu+\delta)}$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix (6.1), (6.2), (6.3), or (6.4) if and only if

$$2s+\gamma \le m \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, & \text{if } \gamma\neq\delta, \text{ or } \gamma=\delta \text{ and } \Gamma=\Delta, \\ \nu+s, & \text{if } \gamma=\delta=1 \text{ and } \Gamma\neq\Delta. \end{cases} \tag{6.19}$$

Proof. If (6.19) holds, then it follows from (6.8) – (6.15) that subspaces of type $(m, 2s+\gamma, s, \Gamma)$ exist in $\mathbb{F}_q^{(2\nu+\delta)}$. Conversely, suppose that there exists a subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$, then by Lemma 6.2 $S_{2\nu+\delta,\Delta}$ is cogredient to

$$\begin{pmatrix} \Lambda & & & \\ & 0 & I^{(m-2s-\gamma)} & \\ & I^{(m-2s-\gamma)} & 0 & \\ & & & \Sigma \end{pmatrix}, \tag{6.20}$$

where $\Lambda = M(2s+\gamma, 2s+\gamma, s, \Gamma)$, Σ is a cogredience normal form of $\sigma\times\sigma$ nonsingular symmetric matrices and is uniquely determined by the type of P, and $\sigma = 2(\nu+s-m)+\delta+\gamma \ge 0$. We discuss the cases $\delta = 0, 1$, and 2 in succession.

(a) $\delta = 0$. Then $S_{2\nu}$ is cogredient to (6.20). Thus the definite parts of Σ and Λ must be of the same size. But the definite part of Λ is a $\gamma\times\gamma$ matrix. Hence $\nu+s-m \ge 0$. Consequently

$$2s+\gamma \le m \le \nu+s. \tag{6.21}$$

(b) $\delta = 1$. Since $S_{2\nu+1,\Delta}$ is cogredient to (6.20), corresponding to the four cases: $\gamma = 0$, $\gamma = 1$ but $\Gamma = \Delta$, $\gamma = 1$ but $\Gamma \neq \Delta$, and $\gamma = 2$, the definite part of Σ is of order $1, 0, 2$, or 1, respectively. Thus

$$2s+\gamma \le m \le \begin{cases} \nu+s+\min\{1,\gamma\}, & \text{when } \gamma\neq 1, \text{ or } \gamma=1 \text{ and } \Gamma=\Delta, \\ \nu+s, & \text{when } \gamma=1 \text{ and } \Gamma\neq\Delta. \end{cases} \tag{6.22}$$

(c) $\delta = 2$. Since $S_{2\nu+2}$ is cogredient to (6.20), the definite part of Σ is of order $2-\gamma$. Hence

$$2s+\gamma \le m \le \nu+s+\gamma. \tag{6.23}$$

Assembling all these cases, we obtain (6.19). □

From Lemma 6.2 we deduce also the following celebrated Witt's transitivity theorem.

Theorem 6.4. Let P_1 and P_2 be two m-dimensional subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$, then there is a $T \in O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ such that $P_1 = AP_2T$, where A is an $m\times m$ nonsingular matrix, if and only if P_1 and P_2 are of the same type with respect to S. In other words, $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ acts transitively on each set of subspaces of the same type.

Proof. Same as the proof of Theorem 3.7. □

Parallel to Corollaries 3.8 and 3.9 we have

Corollary 6.5. Totally isotropic subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ are of dimension $\le \nu$, and there exist totally isotropic subspaces of dimension ν. Therefore maximal totally isotropic subspaces are of dimension ν. Moreover, any totally isotropic subspace is contained in a maximal totally isotropic subspace.

Proof. Similar to the proof of Theorem 3.8. □

Corollary 6.6. Denote the type of the dual subspace of a subspace of type $(m, 2s+\gamma, s, \Gamma)$ with respect to $S_{2\nu+\delta,\Delta}$ in $\mathbb{F}_q^{(2\nu+\delta)}$ by $(m, 2s+\gamma, s, \Gamma)^\perp$, where $(m, 2s+\gamma, s, \Gamma)$ satisfies (6.19). Then

$$(m, 2s, s)^\perp = (2\nu+\delta-m,\ 2(\nu+s-m)+\delta,\ \nu+s-m, \Delta),$$
$$\text{where } \delta = 0, 1, \text{ or } 2;$$

$$(m, 2s+1, s, \Gamma)^\perp = \begin{cases} (2\nu-m,\ 2(\nu+s-m)+1,\ \nu+s-m,\ \Gamma_1), \\ \quad \text{where } \Gamma_1 \text{ is defined by (6.16), when } \delta = 0, \\ (2\nu+1-m,\ 2(\nu+s-m+1),\ \nu+s-m+1), \\ \quad \text{when } \delta = 1 \text{ and } \Gamma = \Delta, \\ (2\nu+1-m,\ 2(\nu+s-m)+2,\ \nu+s-m), \\ \quad \text{when } \delta = 1 \text{ and } \Gamma \ne \Delta, \\ (2\nu+2-m,\ 2(\nu+s-m+1)+1, \\ \quad \nu+s-m+1,\ \Gamma_3), \\ \quad \text{where } \Gamma_3 \text{ is defined by (6.18), when } \delta = 2; \end{cases}$$

$$(m, 2s+2, s)^{\perp} = \begin{cases} (2\nu - m,\ 2(\nu+s-m)+2,\ \nu+s-m), \\ \quad \text{when } \delta = 0, \\ (2\nu+1-m,\ 2(\nu+s-m+1)+1, \\ \quad \nu+s-m+1,\ \Gamma_2), \\ \quad \text{where } \Gamma_2 \text{ is defined by (6.17), when } \delta = 1, \\ (2\nu+2-m,\ 2(\nu+s-m+2),\ \nu+s-m+2), \\ \quad \text{when } \delta = 2. \end{cases}$$

Proof. By the computation given before Theorem 6.1, we know that our corollary holds for subspaces (6.8) – (6.15), whence it follows from Theorem 6.4. □

From Theorems 6.3 and 6.4, we deduce

Theorem 6.7. The number of orbits of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ under the group $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ is equal to

$$2\nu(\nu+\delta+1) + 2^{\delta}.$$

Proof. By Witt's theorem the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ under $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ is equal to the number of subspace types $(m, 2s+\gamma, s, \Gamma)$ with respect to $S_{2\nu+\delta,\Delta}$ in $\mathbb{F}_q^{(2\nu+\delta)}$. By Theorem 6.3, the parameters m, s, and γ of the subspace type satisfy (6.19). Thus it is sufficient to enumerate the number of triples (m, s, γ) of non-negative integers which satisfy condition (6.19). We treat the three cases $\delta = 0, 1$, and 2 one by one.

(a) $\delta = 0$. In this case, (6.19) reduces to (6.21). If $\gamma = 0$, then (6.21) becomes $2s \le m \le \nu + s$, from which we deduce $0 \le m \le 2\nu$. Clearly, $2s \le m \le \nu + s$ is equivalent to $m - \nu \le s \le m/2$. Thus for $0 \le m \le \nu$, s can take $[m/2]+1$ values $0, 1, \ldots, [m/2]$, and for $\nu < m \le 2\nu$, s can take $[m/2] - (m-\nu) + 1$ values $m-\nu, m-\nu+1, \ldots, [m/2]$. Consequently, the number of subspace types $(m, 2s, s)$ in $\mathbb{F}_q^{(2\nu)}$ is equal to

$$\sum_{m=0}^{\nu} ([m/2]+1) + \sum_{m=\nu+1}^{2\nu} ([m/2] - (m-\nu) + 1) = \frac{1}{2}(\nu^2 + 3\nu + 2).$$

If $\gamma = 1$, (6.21) becomes $2s + 1 \le m \le \nu + s$, from which we deduce $1 \le m \le 2\nu - 1$. Clearly $2s + 1 \le m \le \nu + s$ is equivalent to $m - \nu \le s \le (m-1)/2$. Similar to the case $\gamma = 0$, the number of subspace types $(m, 2s+1, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu)}$ is equal to

$$\sum_{m=1}^{\nu} ([(m-1)/2]+1) + \sum_{m=\nu+1}^{2\nu-1} ([(m-1)/2] - (m-\nu) + 1) = \frac{1}{2}(\nu^2 + \nu).$$

Notice that when $\gamma = 1$, Γ can take two values 1 and z. Finally, let $\gamma = 2$, then (6.21) becomes $2s + 2 \leq m \leq \nu + s$, from which we deduce $2 \leq m \leq 2\nu - 2$. Clearly $2s+2 \leq m \leq \nu+s$ is equivalent to $m-\nu \leq s \leq (m-2)/2$. Similar to the case $\gamma = 0$, the number of subspace types $(m, 2s + 2, s)$ in $\mathbb{F}_q^{(2\nu)}$ is equal to

$$\sum_{m=2}^{\nu}([(m-2)/2]+1) + \sum_{m=\nu+1}^{2\nu-2}([(m-2)/2]-(m-\nu)+1) = \frac{1}{2}(\nu^2-\nu).$$

Summing up, we obtain

$$\frac{1}{2}(\nu^2+3\nu+2) + 2\cdot\frac{1}{2}(\nu^2+\nu) + \frac{1}{2}(\nu^2-\nu) = 2\nu^2+2\nu+1. \quad (6.24)$$

(b) $\delta = 1$. In this case, (6.19) reduces to (6.22). If $\gamma = 0$, then (6.22) becomes $2s \leq m \leq \nu + s$. As in case "$\delta = 0$ and $\gamma = 0$", the number of subspace types $(m, 2s, s)$ with respect to $S_{2\nu+1,\Delta}$ in $\mathbb{F}_q^{(2\nu+1)}$ is equal to

$$\frac{1}{2}(\nu^2+3\nu+2).$$

If $\gamma = 1$ and $\Gamma = \Delta$, then (6.22) becomes $2s + 1 \leq m \leq \nu + s + 1$, from which we deduce $1 \leq m \leq 2\nu + 1$. Clearly, $2s + 1 \leq m \leq \nu + s + 1$ is equivalent to $m - \nu - 1 \leq s \leq (m - 1)/2$. Therefore the number of subspace types $(m, 2s+1, s, \Delta)$ with respect to $S_{2\nu+1,\Delta}$ in $\mathbb{F}_q^{(2\nu+1)}$ is equal to

$$\sum_{m=1}^{\nu+1}([(m-1)/2]+1) + \sum_{m=\nu+2}^{2\nu+1}([(m-1)/2]-(m-\nu-1)+1)$$
$$= \frac{1}{2}(\nu^2+3\nu+2).$$

If $\gamma = 1$ and $\Gamma \neq \Delta$, then (6.22) becomes $2s + 1 \leq m \leq \nu + s$. As in case "$\delta = 0$ and $\gamma = 1$", the number of subspace types $(m, 2s + 1, s, \Gamma)$, where $\Gamma \neq \Delta$, with respect to $S_{2\nu+1,\Delta}$ in $\mathbb{F}_q^{(2\nu+1)}$ is equal to

$$\frac{1}{2}(\nu^2+\nu).$$

Finally, if $\gamma = 2$, then (6.22) becomes $2s + 2 \leq m \leq \nu + s + 1$, from which we deduce $2 \leq m \leq 2\nu$. Clearly, $2s + 2 \leq m \leq \nu + s + 1$ is equivalent to $m - \nu - 1 \leq s \leq (m - 2)/2$. Therefore the number of subspace types

$(m, 2s+2, s)$ with respect to $S_{2\nu+1,\Delta}$ in $\mathbb{F}_q^{(2\nu+1)}$ is equal to

$$\sum_{m=2}^{\nu+1}([(m-2)/2]+1)+\sum_{m=\nu+2}^{2\nu}([(m-2)/2]-(m-\nu-1)+1)$$
$$=\frac{1}{2}(\nu^2+\nu).$$

Summing up, we obtain

$$2\cdot\frac{1}{2}(\nu^2+3\nu+2)+2\cdot\frac{1}{2}(\nu^2+\nu)=2\nu^2+4\nu+2. \tag{6.25}$$

(c) $\delta=2$. In this case, (6.19) reduces to (6.23). If $\gamma=0$, (6.23) becomes $2s\le m\le\nu+s$. As in case "$\delta=0$ and $\gamma=0$", the number of subspace types $(m,2s,s)$ in $\mathbb{F}_q^{(2\nu+2)}$ is equal to

$$\frac{1}{2}(\nu^2+3\nu+2).$$

If $\gamma=1$, (6.23) becomes $2s+1\le m\le\nu+s+1$. As in case "$\delta=1,\gamma=1$, and $\Gamma=\Delta$", the number of subspace types $(m,2s+1,s,\Gamma)$ in $\mathbb{F}_q^{(2\nu+2)}$ is equal to

$$\frac{1}{2}(\nu^2+3\nu+2).$$

Notice that when $\gamma=1$, Γ can take two values 1 and z. Finally, let $\gamma=2$, then (6.23) becomes $2s+2\le m\le\nu+s+2$, from which we deduce $2\le m\le 2\nu+2$. Clearly, $2s+2\le m\le\nu+s+2$ is equivalent to $m-\nu-2\le s\le(m-2)/2$. Therefore the number of subspace types $(m,2s+2,s)$ in $\mathbb{F}_q^{(2\nu+2)}$ is equal to

$$\sum_{m=2}^{\nu+2}([(m-2)/2]+1)+\sum_{m=\nu+3}^{2\nu+2}([(m-2)/2]-(m-\nu-2)+1)$$
$$=\frac{1}{2}(\nu^2+3\nu+2).$$

Summing up, we obtain

$$4\cdot\frac{1}{2}(\nu^2+3\nu+2)=2(\nu^2+3\nu+2)=2\nu^2+6\nu+4. \tag{6.26}$$

Assembling (6.24), (6.25), and (6.26), we obtain Theorem 6.7.

In the next section we shall compute the length of each orbit $\mathcal{M}(m,2s+\gamma,s,\Gamma;2\nu+\delta,\Delta)$. For this purpose, the following result, which is parallel to Lemma 3.11, will be needed.

Lemma 6.8. Let P_1 and P_2 be two $m \times n$ matrices of rank m. Then there exists an element $T \in O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ such that $P_1 = P_2T$ if and only if $P_1 S\,{}^tP_1 = P_2 S\,{}^tP_2$.

Proof. Same as the proof of Lemma 3.11. □

6.2 Anzahl Theorems in Orthogonal Geometry over Finite Fields of Odd Characteristic

We follow the notation of the previous section. Let

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) = \mid \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) \mid .$$

The goal of the present section is to compute $N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$. If condition (6.19) fails to hold, then

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) = 0.$$

Hence in the following we always assume that (6.19) holds.

Let P be a subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$, then we can choose a matrix representation P of the subspace P such that

$$PS\,{}^tP = M(m, 2s+\gamma, s, \Gamma). \tag{6.27}$$

Denote by $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ the number of $m \times (2\nu+\delta)$ matrices P of rank m satisfying (6.27). Parallel to Lemma 3.12 we have

Lemma 6.9. Let $(m, 2s+\gamma, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that (6.19) holds. Then

$$\begin{aligned} n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) &= \mid O_{2s+\gamma,\Gamma}(\mathbb{F}_q) \mid\mid GL_{m-2s-\gamma}(\mathbb{F}_q) \mid \\ &\times q^{(2s+\gamma)(m-2s-\gamma)} N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta). \end{aligned}$$

□

Clearly,

$$\mid O_{2\nu+\delta,\Delta}(\mathbb{F}_q) \mid = n(2\nu+\delta,\ 2\nu+\delta, \nu, \Delta; 2\nu+\delta, \Delta).$$

Thus to compute $N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ it is sufficient to compute $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$.

Now let $0 \le r_1, r_2, r_1+r_2 \le m$. Let $n(r_1)$ be the number of $r_1 \times (2\nu+\delta)$ matrices P_1 of rank r_1 such that $P_1 S\,{}^tP_1 = M(m, 2s+\gamma, s, \Gamma)_{r_1}$. For a fixed

$r_1 \times (2\nu+\delta)$ matrix P_1 of rank r_1 such that $P_1 S\,{}^tP_1 = M(m, 2s+\gamma, s, \Gamma)_{r_1}$, let $n(P_1, r_2)$ be the number of $r_2 \times (2\nu+\delta)$ matrices P_2 such that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank $r_1 + r_2$ and $PS\,{}^tP = M(m, 2s+\gamma, s, \Gamma)_{r_1+r_2}$. Then parallel to Lemma 3.13, from Lemma 6.8 we deduce

Lemma 6.10. Let $(m, 2s+\gamma, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that (6.19) holds. Let $0 \le r_1, r_2, r_1+r_2 \le m$. Then $n(P_1, r_2)$ is independent of the choice of P_1 and

$$n(r_1 + r_2) = n(r_1)\, n(P_1, r_2)$$

for any choice of P_1. □

In computing $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ the following lemma is also needed.

Lemma 6.11. Let $(m, 2s+\gamma, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that (6.19) holds. Let $0 \le \gamma_1 \le \gamma$, Γ_1 be the upper left corner $\gamma_1 \times \gamma_1$ submatrix of Γ, and Γ_2 be the lower right corner $(\gamma-\gamma_1) \times (\gamma-\gamma_1)$ submatrix of Γ. Let

$$(2s+\gamma_1, 2s+\gamma_1, s, \Gamma_1)^{\perp} = (2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \Delta_1).$$

Then

$$\begin{aligned} n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) &= n(2s+\gamma_1, 2s+\gamma_1, s, \Gamma_1; 2\nu+\delta, \Delta) \\ &\times n(m-2s-\gamma_1, \gamma-\gamma_1, 0, \Gamma_2; 2\nu_1+\delta_1, \Delta_1). \end{aligned}$$

Proof. Let P be an $m \times (2\nu+\delta)$ matrix of rank m such that

$$PS\,{}^tP = M(m, 2s+\gamma, s, \Gamma).$$

Then $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ is the number of such matrices P. Write

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix},$$

where P_1 is the first $2s+\gamma_1$ rows of P_1 and P_2 is the last $m-2s-\gamma_1$ rows of P. Then

$$P_1 S\,{}^tP_1 = M(2s+\gamma_1, 2s+\gamma_1, s, \Gamma_1), \tag{6.28}$$

$$P_1 S\,{}^tP_2 = 0, \tag{6.29}$$

$$P_2 S\,{}^tP_2 = M(m-2s-\gamma_1, \gamma-\gamma_1, 0, \Gamma_2). \tag{6.30}$$

By Lemma 6.10,

$$n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) = n(2s+\gamma_1)\, n(P_1, m-2s-\gamma_1)$$

for any choice of P_1 satisfying (6.28). From (6.28), we deduce

$$n(2s+\gamma_1) = n(2s+\gamma_1, 2s+\gamma_1, s, \Gamma_1; 2\nu+\delta, \Delta).$$

For a fixed P_1 satisfying (6.28), by the hypothesis of the lemma, we can assume that

$$P_1^{\perp} S\, {}^t(P_1^{\perp}) = S_2 = M(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \Delta_1).$$

From (6.29) we deduce $P_2 \subseteq P_1^{\perp}$. Thus there is an $(m-2s-\gamma_1) \times (2\nu+\delta-2s-\gamma_1)$ matrix Q_2 such that

$$P_2 = Q_2 P_1^{\perp}.$$

Clearly Q_2 is of rank $m-2s-\gamma_1$ and is uniquely determined by P_2. From (6.30) we deduce

$$Q_2 S_2\, {}^t Q_2 = M(m-2s-\gamma_1, \gamma-\gamma_1, 0, \Gamma_2). \tag{6.31}$$

Since

$$P_2 \to Q_2$$

is a bijection, $n(P_1, m-2s-\gamma_1)$ is equal to the number of $(m-2s-\gamma_1) \times (2\nu+\delta-2s-\gamma_1)$ matrices Q_2 of rank $m-2s-\gamma_1$ satisfying (6.31). That is,

$$n(P_1, m-2s-\gamma_1) = n(m-2s-\gamma_1, \gamma-\gamma_1, 0, \Gamma_2; 2\nu_1+\delta_1, \Delta_1).$$

Hence our lemma is proved. □

After the foregoing preparations, we could now come to the computation of $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$. At first, we compute $n(m, 2s, s; 2\nu+\delta, \Delta)$, then $n(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$, and finally $n(m, 2s+2, s; 2\nu+\delta, \Delta)$.

Lemma 6.12. Let $s \le \nu$, then

$$n(s, 0, 0; 2\nu+\delta, \Delta) = q^{\frac{s(s-1)}{2}} \prod_{i=\nu-s+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1). \tag{6.32}$$

Proof. $n(s,0,0;2\nu+\delta,\Delta)$ is the number of $s\times(2\nu+\delta)$ matrices P of rank s satisfying the condition $PS\,{}^tP=0$. Write

$$P=\begin{pmatrix} v_1\\ v_2\\ \vdots\\ v_s\end{pmatrix},$$

where $v_1,v_2,\dots,v_s$ are the s row vectors of P. Then

$$v_iS\,{}^tv_j=0,\ 1\le i,j\le s.$$

The first row vector v_1 of P satisfies the condition $v_1S\,{}^tv_1=0$. Write $v_1=(x_1,x_2,\dots,x_{2\nu+\delta})$, then the condition $v_1S\,{}^tv_1=0$ is

$$\left.\begin{aligned} &\sum_{i=1}^{\nu}x_ix_{\nu+i}=0, && \text{if } \delta=0,\\ 2&\sum_{i=1}^{\nu}x_ix_{\nu+i}+\Delta x_{2\nu+1}^2=0, && \text{if } \delta=1,\\ 2&\sum_{i=1}^{\nu}x_ix_{\nu+i}+x_{2\nu+1}^2-zx_{2\nu+2}^2=0, && \text{if } \delta=2.\end{aligned}\right\}\tag{6.33}$$

Thus the number of possible choices of v_1 is the number of non-zero solutions of (6.33). But the number of solutions of (6.33) with $(x_1,x_2,\dots,x_\nu)\neq(0,0,\dots,0)$ is $(q^\nu-1)q^{\nu+\delta-1}$ and the number of non-zero solutions of (6.33) with $(x_1,x_2,\dots,x_\nu)=(0,0,\dots,0)$ is $(q^\nu-1)$. Therefore the number of possible choices of v_1 is

$$(q^\nu-1)(q^{\nu+\delta-1}+1).$$

By Lemma 6.10, once v_1 is chosen, the number of possible choices of v_2 is independent of the particular choice of v_1. Hence we may choose $v_1=e_1$ and then compute the number of possible choices of v_2. Write $v_2=(y_1,y_2,\dots,y_{2\nu+\delta})$, then from $v_1S\,{}^tv_2=0$ we deduce that $y_{\nu+1}=0$. Thus $v_2=(y_1,y_2,\dots,y_\nu,0,y_{\nu+2},\dots,y_{2\nu+\delta})$ and the condition $v_2S\,{}^tv_2=0$ is

$$\left.\begin{aligned} &\sum_{i=2}^{\nu}y_iy_{\nu+i}=0, && \text{if } \delta=0,\\ 2&\sum_{i=2}^{\nu}y_iy_{\nu+i}+\Delta y_{2\nu+1}^2=0, && \text{if } \delta=1,\\ 2&\sum_{i=2}^{\nu}y_iy_{\nu+i}+y_{2\nu+1}^2-zy_{2\nu+2}^2=0, && \text{if } \delta=2.\end{aligned}\right\}\tag{6.34}$$

Since v_1 and v_2 are linearly independent, $y_2, \ldots, y_\nu, y_{\nu+2}, \ldots, y_{2\nu+\delta}$ cannot all equal zero. Therefore y_1 can be chosen arbitrarily in $\mathbb{F}_q$ and $y_2, y_3, \ldots, y_\nu, y_{\nu+2}, y_{\nu+2}, \ldots, y_{2\nu+\delta}$ can be any non-zero solution of (6.34). By the same counting method of the last paragraph for v_1 we obtain that the number of possible choices of v_2 is

$$q(q^{\nu-1}-1)(q^{(\nu-1)+\delta-1}+1).$$

Proceeding in this way, once $v_1, v_2, \ldots, v_{s-1}$ are chosen, the number of possible choices of v_s is independent of the particular choice of the preceding $v_1, v_2, \ldots, v_{s-1}$. Hence we may choose $v_1 = e_1$, $v_2 = e_2, \ldots, v_{s-1} = e_{s-1}$ and then compute the number of possible choices of v_s. Write $v_s = (z_1, z_2, \ldots, z_{2\nu+\delta})$, then from $v_1 S^t v_s = v_2 S^t v_s = \cdots = v_{s-1} S^t v_s = 0$ we deduce that $z_{\nu+1} = z_{\nu+2} = \cdots = z_{\nu+s-1} = 0$. Thus $v_s = (z_1, z_2, \ldots, z_\nu, 0, \ldots, 0, z_{\nu+s}, \ldots, z_{2\nu+\delta})$ and the condition $v_s S^t v_s = 0$ is

$$\left.\begin{aligned} \sum_{i=s}^{\nu} z_i z_{\nu+i} &= 0, && \text{if } \delta = 0,\\ 2\sum_{i=s}^{\nu} z_i z_{\nu+i} + \Delta z_{2\nu+1}^2 &= 0, && \text{if } \delta = 1,\\ 2\sum_{i=s}^{\nu} z_i z_{\nu+i} + z_{2\nu+1}^2 - z z_{2\nu+2}^2 &= 0, && \text{if } \delta = 2. \end{aligned}\right\} \tag{6.35}$$

Since $v_1, v_2, \ldots, v_s$ are linearly independent, $z_s, z_{s+1}, \ldots, z_\nu, z_{\nu+s}, z_{\nu+s+1}, \ldots, z_{2\nu+\delta}$ cannot all equal zero. Therefore $z_1, z_2, \ldots, z_{s-1}$ can be chosen arbitrarily in $\mathbb{F}_q$ and $z_s, z_{s+1}, \ldots, z_\nu, z_{\nu+s}, z_{\nu+s+1}, \ldots, z_{2\nu+\delta}$ can be any non-zero solution of (6.35). By the same argument as before we know that the number of possible choices of v_s is

$$q^{s-1}(q^{\nu-s+1}-1)(q^{(\nu-s+1)+\delta-1}+1).$$

Then by repeated application of Lemma 6.10, (6.32) will be obtained. □

Lemma 6.13. Let $s \le \nu$, then

$$n(2s, 2s, s; 2\nu+\delta, \Delta) = q^{2s(\nu-s)+\delta s+s(s-1)} \prod_{i=\nu-s+1}^{\nu} (q^i - 1)(q^{i+\delta-1}+1). \tag{6.36}$$

Proof. $n(2s, 2s, s; 2\nu+\delta, \Delta)$ is the number of $2s \times (2\nu+\delta)$ matrices P of rank $2s$ which satisfy the condition

$$P S^t P = M(2s, 2s, s).$$

Write

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \begin{matrix} s \\ s \end{matrix},$$

then

$$\begin{aligned} P_1 S\,{}^t P_1 &= 0, \\ P_2 S\,{}^t P_2 &= 0, \\ P_1 S\,{}^t P_2 &= I^{(s)}. \end{aligned}$$

By Lemma 6.12 the number of possible choices of P_1 is

$$n(s, 0, 0; 2\nu + \delta, \Delta) = q^{\frac{s(s-1)}{2}} \prod_{i=\nu-s+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1).$$

By Lemma 6.10 once P_1 is chosen, the number of possible choices of P_2 is independent of the choice of P_1. Hence we may choose

$$P_1 = (I^{(s)},\ 0^{(s,2\nu+\delta-s)})$$

and then compute the number of possible choices of P_2.

Write

$$P_2 = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_s \end{pmatrix},$$

where $v_1, v_2, \ldots, v_s$ are s row vectors of P_2. At first, we compute the number of possible choices of v_1. From $e_1 S\,{}^t v_1 = 1$, $e_2 S\,{}^t v_1 = e_3 S\,{}^t v_1 = \cdots = e_s S\,{}^t v_1 = 0$ we deduce that v_1 is necessarily of the following form

$$v_1 = (x_1, x_2, \ldots, x_\nu, 1, 0, \ldots, 0, x_{\nu+s+1}, x_{\nu+s+2}, \ldots, x_{2\nu+\delta}).$$

And the condition $v_1 S\,{}^t v_1 = 0$ is

$$\left.\begin{aligned} x_1 + \sum_{i=s+1}^{\nu} x_i x_{\nu+i} &= 0, && \text{if } \delta = 0, \\ 2x_1 + 2\sum_{i=s+1}^{\nu} x_i x_{\nu+i} + \Delta x_{2\nu+1}^2 &= 0, && \text{if } \delta = 1, \\ 2x_1 + 2\sum_{i=s+1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - z x_{2\nu+2}^2 &= 0, && \text{if } \delta = 2. \end{aligned}\right\} \quad (6.37)$$

Thus $x_2, x_3, \ldots, x_s$ can be chosen arbitrarily in $\mathbb{F}_q$ and $x_1, x_{s+1}, \ldots, x_\nu$, $x_{\nu+s+1}, \ldots, x_{2\nu+\delta}$ should satisfy (6.37). Clearly, the number of solutions of (6.37) is

$$q^{\nu-s+\delta}q^{\nu-s} = q^{2(\nu-s)+\delta},$$

and hence the number of possible choices of v_1 is

$$q^{s-1}q^{2(\nu-s)+\delta} = q^{2(\nu-s)+\delta+(s-1)}.$$

Next we compute the number of possible choices v_2. Again by Lemma 6.10, once P_1 and v_1 are chosen, the number of possible choices of v_2 is independent of the choice of P_1 and v_1. Hence we may choose P_1 as above and $v_1 = e_{\nu+1}$, and then compute the number of possible choices of v_2. From $e_1 S^t v_2 = 0$, $e_2 S^t v_2 = 1$, $e_3 S^t v_2 = e_4 S^t v_2 = \cdots = e_s S^t v_2 = e_{\nu+1} S^t v_2 = 0$ we deduce that v_2 is necessarily of the following form

$$v_2 = (0, y_2, y_3, \ldots, y_\nu, 0, 1, 0, \ldots, 0, y_{\nu+s+1}, y_{\nu+s+2}, \ldots, y_{2\nu+\delta}),$$

and the condition $v_2 S^t v_2 = 0$ is

$$\left.\begin{array}{ll} y_2 + \sum_{i=s+1}^{\nu} y_i y_{\nu+i} = 0, & \text{if } \delta = 0, \\ 2y_2 + 2\sum_{i=s+1}^{\nu} y_i y_{\nu+i} + \Delta y_{2\nu+1}^2 = 0, & \text{if } \delta = 1, \\ 2y_2 + 2\sum_{i=s+1}^{\nu} y_i y_{\nu+i} + y_{2\nu+1}^2 - z y_{2\nu+2}^2 = 0, & \text{if } \delta = 2. \end{array}\right\} \quad (6.38)$$

Hence $y_3, y_4, \ldots, y_s$ can be chosen arbitrarily in $\mathbb{F}_q$ and $y_2, y_{s+1}, \ldots, y_\nu$, $y_{\nu+s+1}, \ldots, y_{2\nu+\delta}$ should satisfy (6.38). By the same argument of the above paragraph, we obtain that the number of possible choices of v_2 is

$$q^{2(\nu-s)+\delta+(s-2)}.$$

Proceeding in this way, finally, when $v_1, v_2, \ldots, v_{s-1}$ are chosen, by Lemma 6.10, the number of possible choices of v_s is independent of the choice of $P_1, v_1, v_2, \ldots, v_{s-1}$. Hence we may choose P_1 as above and $v_1 = e_{\nu+1}, v_2 = e_{\nu+2}, \ldots, v_{s-1} = e_{\nu+s-1}$, and then compute the number of possible choices of v_s. From $e_1 S^t v_s = e_2 S^t v_s = \cdots = e_{s-1} S^t v_s = 0$, $e_s S^t v_s = 1$, $e_{\nu+1} S^t v_s = e_{\nu+2} S^t v_s = \cdots = e_{\nu+s-1} S^t v_s = 0$ we deduce that v_s is necessarily of the following form

$$v_s = (0, \ldots, 0, z_s, \ldots, z_\nu, 0, \ldots, 0, 1, z_{\nu+s+1}, \ldots, z_{2\nu+\delta}),$$

and the condition $v_s S\,{}^t v_s = 0$ is

$$\left.\begin{aligned} z_s + \sum_{i=s+1}^{\nu} z_i z_{\nu+i} &= 0, && \text{if } \delta = 0,\\ 2z_s + 2\sum_{i=s+1}^{\nu} z_i z_{\nu+i} + \Delta z_{2\nu+1}^2 &= 0, && \text{if } \delta = 1,\\ 2z_2 + 2\sum_{i=s+1}^{\nu} z_i z_{\nu+i} + z_{2\nu+1}^2 - z z_{2\nu+2}^2 &= 0, && \text{if } \delta = 2. \end{aligned}\right\}$$

By the same argument as above, we know that the number of possible choices of v_s is

$$q^{2(\nu-s)+\delta}.$$

By repeated application of Lemma 6.10 we will obtain (6.36). □

Lemma 6.14. Let $2s \le m \le \nu + s$. Then

$$\begin{aligned} &n(m, 2s, s; 2\nu+\delta, \Delta)\\ &\quad = q^{2s(\nu-s)+\delta s+s(s-1)+\frac{(m-2s)(m-2s-1)}{2}} \prod_{i=\nu+s-m+1}^{\nu} (q^i - 1)(q^{i+\delta-1}+1). \end{aligned} \tag{6.39}$$

Proof. $n(m, 2s, s; 2\nu+\delta, \Delta)$ is the number of $m \times (2\nu+\delta)$ matrices P of rank m which satisfy the condition

$$P S\,{}^t P = M(m, 2s, s).$$

By Corollary 6.6

$$(2s, 2s, s)^{\perp} = (2(\nu-s)+\delta, 2(\nu-s)+\delta, \nu-s, \Delta).$$

Hence by Lemma 6.11 we have

$$\begin{aligned} &n(m, 2s, s; 2\nu+\delta, \Delta)\\ &\quad = n(2s, 2s, s; 2\nu+\delta, \Delta)\, n(m-2s, 0, 0; 2(\nu-s)+\delta, \Delta). \end{aligned}$$

Therefore our lemma follows from Lemmas 6.12 and 6.13. □

This finishes the computation of $n(m, 2s, s; 2\nu+\delta, \Delta)$.

Next we are going to compute $n(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$.

Lemma 6.15. Let

$$\nu \geq \begin{cases} 1, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ 0, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = \Delta. \end{cases}$$

Then

$$n(1,1,0,\Gamma;2\nu+\delta,\Delta) = \begin{cases} q^{\nu-1}(q^\nu-1), & \text{if } \delta = 0, \\ q^\nu(q^\nu-1), & \text{if } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ q^\nu(q^\nu+1), & \text{if } \delta = 1 \text{ and } \Gamma = \Delta, \\ q^\nu(q^{\nu+1}+1), & \text{if } \delta = 2. \end{cases} \tag{6.40}$$

Proof. By definition, $n(1,1,0,\Gamma;2\nu+\delta,\Delta)$ is the number of non-zero vectors $x = (x_1, x_2, \ldots, x_{2\nu+\delta})$ which satisfy the condition $xS^t x = \Gamma$, i.e., the number of non-zero solutions of the equations

$$\left.\begin{aligned} 2\sum_{i=1}^{\nu} x_i x_{\nu+i} &= \Gamma, && \text{if } \delta = 0, \\ 2\sum_{i=1}^{\nu} x_i x_{\nu+i} + \Delta x_{2\nu+1}^2 &= \Gamma, && \text{if } \delta = 1, \\ 2\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 - z x_{2\nu+2}^2 &= \Gamma, && \text{if } \delta = 2. \end{aligned}\right\} \tag{6.41}$$

The number of solutions of (6.41) with $(x_1, x_2, \ldots, x_\nu) \neq (0,0,\ldots,0)$ is

$$q^{\nu+\delta-1}(q^\nu-1),$$

and the number of non-zero solutions of (6.41) with $(x_1, x_2, \ldots, x_\nu) = (0,0,\ldots,0)$ is

$$\begin{cases} 0, & \text{if } \delta = 0, \\ 0, & \text{if } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ 2q^\nu, & \text{if } \delta = 1 \text{ and } \Gamma = \Delta, \\ q^\nu(q+1), & \text{if } \delta = 2. \end{cases}$$

□

Lemma 6.16. Let

$$s \leq \begin{cases} \nu-1, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ \nu, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = \Delta. \end{cases}$$

Then

$$\begin{aligned} &n(2s+1, 2s+1, s, \Gamma; 2\nu+\delta, \Delta) \\ &\quad = n(2s, 2s, s; 2\nu+\delta, \Delta) n(1, 1, 0, \Gamma; 2(\nu-s)+\delta, \Delta), \end{aligned} \tag{6.42}$$

where $n(2s, 2s, s; 2\nu+\delta, \Delta)$ is given by (6.36) and $n(1, 1, 0, \Gamma; 2(\nu-s)+\delta, \Delta)$ by (6.40).

Proof. $n(2s+1, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$ is the number of $(2s+1) \times (2\nu+\delta)$ matrices P of rank $2s+1$, satisfying

$$PS\,{}^tP = M(2s+1, 2s+1, s, \Gamma).$$

By Corollary 6.6

$$(2s, 2s, s)^{\perp} = (2(\nu-s)+\delta, 2(\nu-s)+\delta, \nu-s, \Delta).$$

Hence our lemma follows from Lemma 6.11. □

Lemma 6.17. Let

$$2s+1 \le m \le \begin{cases} \nu+s, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ \nu+s+1, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = \Delta. \end{cases}$$

Then

$$\begin{aligned} &n(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta) \\ &\quad = n(2s+1, 2s+1, s, \Gamma; 2\nu+\delta, \Delta) n_1, \end{aligned} \tag{6.43}$$

where $n(2s+1, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$ is given by (6.42) and

$$n_1 = \begin{cases} n(m-2s-1, 0, 0; 2(\nu-s-1)+1, \Gamma_1), \\ \quad \text{if } \delta = 0, \\ n(m-2s-1, 0, 0; 2(\nu-s-1)+2), \\ \quad \text{if } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ n(m-2s-1, 0, 0; 2(\nu-s)), \\ \quad \text{if } \delta = 1 \text{ and } \Gamma = \Delta, \\ n(m-2s-1, 0, 0; 2(\nu-s)+1, \Gamma_3), \\ \quad \text{if } \delta = 2, \end{cases} \tag{6.44}$$

in which Γ_1 and Γ_3 are defined by (6.16) and (6.18), respectively and those n appearing in the right hand side of (6.44) are given by (6.32).

Proof. $n(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$ is the number of $m \times (2\nu+\delta)$ matrices P of rank m which satisfy the condition

$$PS\,{}^tP = M(m, 2s+1, s, \Gamma).$$

By Corollary 6.6

$$(2s+1, 2s+1, s, \Gamma)^{\perp}$$
$$= \begin{cases} (2(\nu-s-1)+1, 2(\nu-s-1)+1, \nu-s-1, \Gamma_1), \\ \quad \text{if } \delta = 0, \\ (2(\nu-s), 2(\nu-s), \nu-s), \\ \quad \text{if } \delta = 1 \text{ and } \Gamma = \Delta, \\ (2(\nu-s-1)+2, 2(\nu-s-1)+2, \nu-s-1), \\ \quad \text{if } \delta = 1 \text{ and } \Gamma \neq \Delta, \\ (2(\nu-s)+1, 2(\nu-s)+1, \nu-s, \Gamma_3), \\ \quad \text{if } \delta = 2, \end{cases}$$

where Γ_1 and Γ_3 are defined by (6.16) and (6.18), respectively. Hence our lemma follows from Lemma 6.11. □

This completes the computation of $n(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$.

Finally let us compute $n(m, 2s+2, s; 2\nu+\delta, \Delta)$.

Lemma 6.18. Let

$$\nu \geq (2-\delta).$$

then

$$n(2,2,0; 2\nu+\delta, \Delta) = \begin{cases} q^{2\nu-2}(q^{\nu-1}-1)(q^{\nu}-1), & \text{if } \delta = 0, \\ q^{2\nu-1}(q^{\nu}-1)(q^{\nu}+1), & \text{if } \delta = 1, \\ q^{2\nu}(q^{\nu}+1)(q^{\nu+1}+1), & \text{if } \delta = 2. \end{cases} \tag{6.45}$$

Proof. $n(2,2,0; 2\nu+\delta, \Delta)$ is the number of $2 \times (2\nu+\delta)$ matrices P of rank 2 such that

$$PS\,{}^tP = \begin{pmatrix} 1 & 0 \\ 0 & -z \end{pmatrix}.$$

By Corollary 6.6

$$(1,1,0,1)^{\perp}$$
$$= \begin{cases} (2(\nu-1)+1, 2(\nu-1)+1, \nu-1, \Gamma_1), & \text{if } \delta = 0, \\ (2\nu, 2\nu, \nu), & \text{if } \delta = 1 \text{ and } \Delta = 1, \\ (2(\nu-1)+2, 2(\nu-1)+2, \nu-1), & \text{if } \delta = 1 \text{ and } \Delta = z, \\ (2\nu+1, 2\nu+1, \nu, \Gamma_3), & \text{if } \delta = 2, \end{cases}$$

where Γ_1 and Γ_3 are defined by (6.16) and (6.18), respectively, and in the present case

$$\Gamma_1 = 1 \text{ and } \Gamma_3 = z, \text{ if } -1 \in \mathbb{F}_q^{*2},$$
$$\Gamma_1 = z \text{ and } \Gamma_3 = 1, \text{ if } -1 \notin \mathbb{F}_q^{*2}.$$

Hence it follows from Lemma 6.11 that

$$n(2,2,0;2\nu+\delta,\Delta) = n(1,1,0,1;2\nu+\delta,\Delta)n_0,$$

where $n(1,1,0,1;2\nu+\delta,\Delta)$ is given by (6.40) and

$$n_0 = \begin{cases} n(1,1,0,-z;2(\nu-1)+1,\Gamma_1), & \text{if } \delta = 0, \\ n(1,1,0,-z;2\nu), & \text{if } \delta = 1 \text{ and } \Delta = 1, \\ n(1,1,0,-z;2(\nu-1)+2), & \text{if } \delta = 1 \text{ and } \Delta = z, \\ n(1,1,0,-z;2\nu+1,\Gamma_3), & \text{if } \delta = 2. \end{cases}$$

It can be easily seen that $-z$ and Γ_1 cannot both be square elements and that $-z$ and Γ_3 are either both square elements or both non-square elements. From Lemma 6.15 we deduce (6.45) immediately. □

Lemma 6.19. Let

$$\nu \geq s + (2-\delta).$$

Then

$$\begin{aligned} &n(2s+2,2s+2,s;2\nu+\delta,\Delta) \\ &\quad = n(2s,2s,s;2\nu+\delta,\Delta)n(2,2,0;2(\nu-s)+\delta,\Delta), \end{aligned} \tag{6.46}$$

where $n(2s,2s,s;2\nu+\delta,\Delta)$ and $n(2,2,0;2(\nu-s)+\delta,\Delta)$ are given by (6.36) and (6.45), respectively.

Proof. Proceed as in the proof of Lemma 6.16. □

Lemma 6.20. Let $2s+2 \leq m \leq \nu+s+\delta$. Then

$$n(m,2s+2,s;2\nu+\delta,\Delta) = n(2s+2,2s+2,s;2\nu+\delta,\Delta)n_2, \tag{6.47}$$

where $n(2s+2,2s+2,s;2\nu+\delta,\Delta)$ is given by (6.46) and

$$n_2 = \begin{cases} n(m-2s-2,0,0;2(\nu-s-2)+2), & \text{if } \delta = 0, \\ n(m-2s-2,0,0;2(\nu-s-1)+1,\Gamma_2), & \text{if } \delta = 1, \\ n(m-2s-2,0,0;2(\nu-s)), & \text{if } \delta = 2, \end{cases} \tag{6.48}$$

in which $\Gamma_2 = z$ or 1 if $\Delta = 1$ or z, respectively and those n appearing in the right hand side of (6.48) are given by (6.32).

Proof. Proceed as in the proof of Lemma 6.17. □

From Lemmas 6.13, 6.16, and 6.19 we deduce immediately

Theorem 6.21.

$$| O_{2\nu+\delta}(\mathbb{F}_q) | = q^{\nu(\nu+\delta-1)} \prod_{i=1}^{\nu} (q^i - 1) \prod_{i=0}^{\nu+\delta-1} (q^i + 1).$$

Proof. In fact

$$| O_{2\nu+\delta}(\mathbb{F}_q) | = n(2\nu+\delta, 2\nu+\delta, \nu, \Delta; 2\nu+\delta, \Delta).$$

□

From Lemmas 6.9, 6.14, 6.17, and 6.20 and Theorems 1.6 and 6.21, we deduce the following three Anzahl theorems.

Theorem 6.22. The number of subspaces of the hyperbolic type $(m, 2s, s)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where $2s \le m \le \nu + s$, is given by

$$N(m, 2s, s; 2\nu+\delta, \Delta) = q^{2s(\nu+s-m)+\delta s} \frac{\prod_{i=\nu+s-m+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1)}{\prod_{i=1}^{s} (q^i - 1) \prod_{i=0}^{s-1} (q^i + 1) \prod_{i=1}^{m-2s} (q^i - 1)}.$$

□

Corollary 6.23. The number of m-dimensional totally isotropic subspaces in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where $m \le \nu$, is given by

$$N(m, 0, 0; 2\nu+\delta, \Delta) = \frac{\prod_{i=\nu-m+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1)}{\prod_{i=1}^{m} (q^i - 1)}.$$

□

Theorem 6.24. The number of subspaces of type $(m, 2s+1, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+1 \le m \le \begin{cases} \nu+s, & \text{if } \delta=0, \text{ or } \delta=1 \text{ and } \Gamma \ne \Delta, \\ \nu+s+1, & \text{if } \delta=2, \text{ or } \delta=1 \text{ and } \Gamma=\Delta, \end{cases}$$

is given by

$$\begin{aligned} &N(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta) \\ &= q^{2s(\nu+s-m)+s(\delta+1)-(m-2s-1)} \frac{\displaystyle\prod_{i=\nu+s-m+2}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{\displaystyle\prod_{i=1}^{s}(q^i-1)\prod_{i=0}^{s}(q^i+1)\prod_{i=1}^{m-2s-1}(q^i-1)} \\ &\quad \times n_0(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta), \end{aligned}$$

where

$$\begin{aligned} &n_0(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta) \\ &= \begin{cases} q^{\nu-s-1}(q^{\nu+s-m+1}-1), & \text{if } \delta=0, \\ q^{\nu-s}(q^{\nu+s-m+1}-1), & \text{if } \delta=1 \text{ and } \Gamma \ne \Delta, \\ q^{\nu-s}(q^{\nu+s-m+1}+1), & \text{if } \delta=1 \text{ and } \Gamma=\Delta, \\ q^{\nu-s}(q^{\nu+s-m+2}+1), & \text{if } \delta=2. \end{cases} \end{aligned}$$

□

Theorem 6.25. The number of subspaces of the elliptic type $(m, 2s+2, s; 2\nu+\delta, \Delta)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+2 \le m \le \nu+s+\delta,$$

is given by

$$\begin{aligned} &N(m, 2s+2, s; 2\nu+\delta, \Delta) \\ &= q^{2s(\nu+s-m)+s(\delta+2)-2(m-2s-2)} \frac{\displaystyle\prod_{i=\nu+s-m+3}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{\displaystyle\prod_{i=1}^{s}(q^i-1)\prod_{i=0}^{s+1}(q^i+1)\prod_{i=1}^{m-2s-2}(q^i-1)} \\ &\quad \times n_0(m, 2s+2, s; 2\nu+\delta, \Delta), \end{aligned}$$

where

$$n_0(m, 2s+2, s; 2\nu+\delta, \Delta)$$
$$= \begin{cases} q^{2(\nu-s)-2}(q^{\nu+s-m+1}-1)(q^{\nu+s-m+2}-1), & \text{if } \delta = 0, \\ q^{2(\nu-s)-1}(q^{\nu+s-m+2}-1)(q^{\nu+s-m+2}+1), & \text{if } \delta = 1, \\ q^{2(\nu-s)}(q^{\nu+s-m+2}+1)(q^{\nu+s-m+3}+1), & \text{if } \delta = 2. \end{cases}$$

□

The above three Anzahl theorems can be combined into a single Anzahl theorem as follows.

Theorem 6.26. The number of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+\gamma \le m \le \begin{cases} v+s+\min\{\delta,\gamma\}, & \text{if } \gamma \ne \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta, \\ \nu+s, & \text{if } \gamma = \delta = 1, \text{ and } \Gamma \ne \Delta, \end{cases}$$

is given by

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$$
$$= q^{2s(\nu+s-m)+s(\delta+\gamma)-\gamma(m-2s-\gamma)} \frac{\displaystyle\prod_{i=\nu+s-m+\gamma+1}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{\displaystyle\prod_{i=1}^{s}(q^i-1) \prod_{i=0}^{s+\gamma-1}(q^i+1) \prod_{i=1}^{m-2s-\gamma}(q^i-1)}$$
$$\times n_0(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta), \tag{6.49}$$

where

$$n_0(m, 2s, s; 2\nu+\delta, \Delta) = 1;$$
$$n_0(m, 2s+1, s, \Gamma; 2\nu+\delta, \Delta)$$
$$= \begin{cases} q^{\nu-s-1}(q^{\nu+s-m+1}-1), & \text{if } \delta = 0, \\ q^{\nu-s}(q^{\nu+s-m+1}-1), & \text{if } \delta = 1 \text{ and } \Gamma \ne \Delta, \\ q^{\nu-s}(q^{\nu+s-m+1}+1), & \text{if } \delta = 1 \text{ and } \Gamma = \Delta, \\ q^{\nu-s}(q^{\nu+s-m+2}+1), & \text{if } \delta = 2; \end{cases}$$

$$n_0(m, 2s+2, s; 2\nu+\delta, \Delta)$$
$$= \begin{cases} q^{2(\nu-s)-2}(q^{\nu+s-m+1}-1)(q^{\nu+s-m+2}-1), & \text{if } \delta = 0, \\ q^{2(\nu-s)-1}(q^{\nu+s-m+2}-1)(q^{\nu+s-m+2}+1), & \text{if } \delta = 1, \\ q^{2(\nu-s)}(q^{\nu+s-m+2}+1)(q^{\nu+s-m+3}+1), & \text{if } \delta = 2. \end{cases}$$

□

6.3 Singular Orthogonal Geometry over Finite Fields of Odd Characteristic

Let

$$S_l = \begin{pmatrix} S & \\ & 0^{(l)} \end{pmatrix},$$

where $S = S_{2\nu+\delta,\Delta}$, $\delta = 0, 1,$ or 2, and

$$\Delta = \begin{cases} \phi, & \text{if } \delta = 0, \\ 1 \text{ or } z, & \text{if } \delta = 1, \\ \begin{pmatrix} 1 & \\ & -z \end{pmatrix}, & \text{if } \delta = 2. \end{cases}$$

The set of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices T over $\mathbb{F}_q$ satisfying

$$T S_l {}^t T = S_l$$

forms a group which will be called the *singular orthogonal group* of degree $2\nu+\delta+l$, rank $2\nu+\delta$, and with definite part Δ over $\mathbb{F}_q$ and denoted by $O_{2\nu+\delta+l,2\nu+\delta,\Delta}(\mathbb{F}_q)$, or simply $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$. Clearly, $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$ consists of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices of the form

$$T = \begin{array}{c} \begin{matrix} 2\nu+\delta & \quad l \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} 2\nu+\delta \\ l \end{matrix} \end{array},$$

where $T_{11} S {}^t T_{11} = S$ and T_{22} is nonsingular. Parallel to Theorem 3.21 we have

Theorem 6.27.

$$\begin{aligned} &| \, O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q) \, | \\ &\quad = q^{\nu(\nu+\delta-1)+l(2\nu+\delta)+\frac{l(l-1)}{2}} \prod_{i=1}^{\nu} (q^i - 1) \prod_{i=0}^{\nu+\delta-1} (q^i + 1) \prod_{i=1}^{l} (q^i - 1). \end{aligned}$$

□

Let $\mathbb{F}_q^{(2\nu+\delta+l)}$ be the $(2\nu+\delta+l)$-dimensional row vector space over $\mathbb{F}_q$. Clearly, we have an action of the group $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$ on the vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ defined as follows:

$$\begin{aligned} &\mathbb{F}_q^{(2\nu+\delta+l)} \times O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q) \to \mathbb{F}_q^{(2\nu+\delta+l)} \\ &\quad ((x_1, x_2, \ldots, x_{2\nu+\delta+l}), T) \mapsto (x_1, x_2, \ldots, x_{2\nu+\delta+l})T. \end{aligned}$$

The vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ together with this action is called the *singular orthogonal space* over $\mathbb{F}_q$. At first, we want to study how the subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$ are subdivided into orbits under $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$.

An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is said to be of *type* $(m, 2s+\gamma, s, \Gamma)$ if $PS_l\,{}^tP$ is cogredient to $M(m, 2s+\gamma, s, \Gamma)$ (cf. p. 253). Denote the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$ by $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \delta, \Delta)$, or simply by $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta)$ and let

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta) = |\ \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta)\ | .$$

For the ordinary orthogonal space, i.e., the case $l=0$, $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta)$ is simply $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ which was studied in the previous two sections.

Let E be the subspace of $\mathbb{F}_q^{(2\nu+\delta+l)}$ generated by $e_{2\nu+\delta+1}, e_{2\nu+\delta+2}, \ldots, e_{2\nu+\delta+l}$, then $\dim E = l$. An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is called a subspace of *type* $(m, 2s+\gamma, s, \Gamma, k)$, if

(i) $PS_l\,{}^tP$ is cogredient to $M(m, 2s+\gamma, s, \Gamma)$, and

(ii) $\dim(P \cap E) = k$.

Denote by $\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ the set of subspaces of type $(m, 2s+\gamma, s, \Gamma, k)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$ and let

$$N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta) = |\ \mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)\ | .$$

Parallel to Theorems 3.22, 3.24 and Corollaries 3.23, 3.25 we have

Theorem 6.28. *$\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ is non-empty if and only if*

$$\left.\begin{array}{l} k \le l \\ \text{and} \\ 2s+\gamma \le m-k \le \left\{\begin{array}{l} \nu+s+\min\{\delta,\gamma\}, \\ \quad \text{if } \gamma \ne \delta, \text{ or } \gamma=\delta \text{ and } \Gamma=\Delta, \\ \nu+s, \\ \quad \text{if } \gamma=\delta=1 \text{ and } \Gamma\ne\Delta, \end{array}\right. \end{array}\right\} \tag{6.50}$$

and if and only if

$$\min\{l, m-2s-\gamma\} \geq k \geq \begin{cases} \max\{0, m-\nu-s-\min\{\delta,\gamma\}\}, \\ \quad \text{if } \gamma \neq \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta, \\ \max\{0, m-\nu-s\}, \\ \quad \text{if } \gamma = \delta = 1 \text{ and } \Gamma \neq \Delta. \end{cases} \tag{6.51}$$

Moreover, if $\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ is non-empty, then it is an orbit of subspaces under $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$. □

Corollary 6.29. $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta)$ is non-empty if and only if

$$\min\{l, m-2s-\gamma\} \geq \begin{cases} \max\{0, m-\nu-s-\min\{\delta,\gamma\}\}, \\ \quad \text{if } \gamma \neq \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta, \\ \max\{0, m-\nu-s\}, \\ \quad \text{if } \gamma = \delta = 1 \text{ and } \Gamma \neq \Delta. \end{cases} \tag{6.52}$$

Moreover, $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta)$ is the disjoint union of all the orbits $\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ under $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$, where k satisfies (6.51). □

It follows from Theorem 6.28 that $N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta) = 0$ whenever (6.51) does not hold. However, if (6.51) holds, then parallel to Theorem 3.24 we have

Theorem 6.30. Suppose that (6.51) holds, then

$$\begin{aligned} &N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta) \\ &= N(m-k, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) q^{(m-k)(l-k)} \frac{\prod_{i=l-k+1}^{l}(q^i-1)}{\prod_{i=1}^{k}(q^i-1)}, \end{aligned} \tag{6.53}$$

where $N(m-k, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ is the number of subspaces of type $(m-k, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ and is given by (6.49). □

Corollary 6.31. Suppose that (6.52) holds, then

$$\begin{aligned} &N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, \Delta) \\ &\quad = \sum_k N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta), \end{aligned} \tag{6.54}$$

where the summation range of k is

$$[\max\{0, m-\nu-s-\min\{\delta,\gamma\}\},\ \min\{l, m-2s-\gamma\}],$$
$$\text{if } \gamma \neq \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta,$$

or

$$[\max\{0, m-\nu-s\},\ \min\{l, m-2s-\gamma\}],$$
$$\text{if } \gamma = \delta = 1 \text{ and } \Gamma \neq \Delta,$$

and $N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ is given by (6.53). □

We can also deduce an Anzahl theorem in ordinary orthogonal geometry from the above corollary.

Let $\mathbb{F}_q^{(2\nu+\delta)}$ be the $(2\nu+\delta)$-dimensional row vector space on which the ordinary orthogonal group $O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$ acts. Let P be a fixed subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Denote by $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ the set of subspaces of type $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)$ contained in P. Let

$$N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$$
$$= \mid \mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) \mid .$$

It is clear that $N(m_1, 2s_1+\gamma_1, s_1, \Gamma; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ is independent of the particular choice of the subspace P of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Parallel to Theorems 3.26 and 3.27 we have

Theorem 6.32. $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ is non-empty if and only if

$$\left.\begin{aligned}
&2s+\gamma \leq m \leq \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \quad \text{if } \gamma \neq \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta, \\ \nu+s, \\ \quad \text{if } \gamma = \delta = 1 \text{ and } \Gamma \neq \Delta, \end{cases} \\
&2s_1+\gamma_1 \leq m_1 \leq \begin{cases} \nu+s_1+\min\{\delta,\gamma_1\}, \\ \quad \text{if } \gamma_1 \neq \delta, \text{ or } \gamma_1 = \delta \text{ and } \Gamma_1 = \Delta, \\ \nu+s_1, \\ \quad \text{if } \gamma_1 = \delta = 1 \text{ and } \Gamma_1 \neq \Delta, \end{cases}
\end{aligned}\right\} \tag{6.55}$$

and

$$2m-2m_1 \geq \begin{cases} (2s+\gamma)-(2s_1+\gamma_1)+\mid \gamma-\gamma_1 \mid \geq 2 \mid \gamma-\gamma_1 \mid, \\ \text{if } \gamma_1 \neq \gamma, \text{ or } \gamma_1 = \gamma \text{ and } \Gamma_1 = \Gamma, \\ (2s+\gamma)-(2s_1+\gamma_1)+2 \geq 4, \\ \text{if } \gamma_1 = \gamma = 1 \text{ and } \Gamma_1 \neq \Gamma \end{cases}$$

and these conditions are equivalent to the following two conditions

$$2s+\gamma \le m \le \left\{ \begin{array}{l} \nu+s+\min\{\delta,\gamma\}, \\ \quad \text{if } \gamma \ne \delta, \text{ or } \gamma=\delta \text{ and } \Gamma=\Delta, \\ \nu+s, \\ \quad \text{if } \gamma=\delta=1 \text{ and } \Gamma\ne\Delta, \end{array} \right.$$

and

$$\begin{aligned} &\min\{m-2s-\gamma, m_1-2s_1-\gamma_1\} \\ &\quad \ge \left\{ \begin{array}{l} \max\{0, m_1-s-s_1-\min\{\gamma,\gamma_1\}\}, \\ \quad \text{if } \gamma_1\ne\gamma, \text{ or } \gamma_1=\gamma \text{ and } \Gamma_1=\Gamma, \\ \max\{0, m_1-s-s_1\}, \\ \quad \text{if } \gamma_1=\gamma=1 \text{ and } \Gamma_1\ne\Gamma. \end{array} \right. \end{aligned} \tag{6.56}$$

□

Theorem 6.33. Suppose that (6.55) holds. Then

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) \\ &\quad = N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; 2s+\gamma+(m-2s-\gamma), \Gamma), \end{aligned}$$

where $N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; 2s+\gamma+(m-2s-\gamma), \Gamma)$ is given by (6.54), or more precisely

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) \\ &\quad = \sum_k q^{2s_1(s+s_1-m_1+k)+s_1(\gamma+\gamma_1)-\gamma_1(m_1-2s_1-\gamma_1-k)+(m_1-k)(m-2s-\gamma-k)} \\ &\qquad \times n_0(m_1-k, 2s_1+\gamma_1, s_1, \Gamma_1; 2s+\gamma, \Gamma) \\ &\qquad \times \frac{\displaystyle\prod_{i=s+s_1-m_1+\gamma_1+k+1}^{s} (q^i-1)(q^{i+\gamma-1}+1) \prod_{i=m-2s-\gamma-k+1}^{m-2s-\gamma} (q^i-1)}{\displaystyle\prod_{i=1}^{s_1}(q^i-1) \prod_{i=0}^{s_1+\gamma_1-1}(q^i+1) \prod_{i=1}^{m_1-2s_1-\gamma_1-k}(q^i-1) \prod_{i=1}^{k}(q^i-1)}, \end{aligned} \tag{6.57}$$

where the summation range of k is

$$\begin{aligned} &\min\{m-2s-\gamma, m_1-2s_1-\gamma_1\} \ge k \\ &\quad \ge \left\{ \begin{array}{l} \max\{0, -s-s_1+m_1+\min\{\gamma_1,\gamma\}\}, \\ \quad \text{if } \gamma_1\ne\gamma, \text{ or } \gamma_1=\gamma \text{ and } \Gamma_1=\Gamma, \\ \max\{0, -s-s_1+m_1\}, \\ \quad \text{if } \gamma_1=\gamma=1 \text{ and } \Gamma_1\ne\Gamma; \end{array} \right. \end{aligned}$$

and

$$\begin{aligned} &n_0(m_1-k, 2s_1, s_1; 2s+\gamma, \Gamma) = 1; \\ &n_0(m_1-k, 2s_1+1, s_1, \Gamma; 2s+\gamma, \Gamma) \end{aligned}$$

$$= \begin{cases} q^{s-s_1-1}(q^{s+s_1-m_1+k+1}-1), & \text{if } \gamma = 0, \\ q^{s-s_1}(q^{s+s_1-m_1+k+1}+1), & \text{if } \gamma = 1 \text{ and } \Gamma_1 = \Gamma, \\ q^{s-s_1}(q^{s+s_1-m_1+k+1}-1), & \text{if } \gamma = 1 \text{ and } \Gamma_1 \neq \Gamma, \\ q^{s-s_1}(q^{s+s_1-m_1+k+2}-1), & \text{if } \gamma = 2; \end{cases}$$

$$n_0(m_1-k, 2s_1+2, s_1, \Gamma_1; 2s+\gamma, \Gamma)$$

$$= \begin{cases} q^{2(s-s_1)-2}(q^{s+s_1-m_1+k+1}-1)(q^{s+s_1-m_1+k+2}-1), & \text{if } \gamma = 0, \\ q^{2(s-s_1)-1}(q^{s+s_1-m_1+k+2}-1)(q^{s+s_1-m_1+k+2}+1), & \text{if } \gamma = 1, \\ q^{2(s-s_1)}(q^{s+s_1-m_1+k+2}+1)(q^{s+s_1-m_1+k+3}+1), & \text{if } \gamma = 2. \end{cases}$$

□

Now let us consider the similar problem in the singular orthogonal geometry. We consider again the $(2\nu+\delta+l)$-dimensional row vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ on which the singular orthogonal group $O_{2\nu+\delta+l,\Delta}(\mathbb{F}_q)$ acts. Let P be a fixed subspace of type $(m, 2s+\gamma, s, \Gamma, k)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$. Denote by $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ the set of subspaces of type $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1)$ contained in P. Let

$$N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$$
$$= \mid \mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta) \mid,$$

which is independent of the particular choice of the subspace P of type $(m, 2s+\gamma, s, \Gamma, k)$. Then parallel to Theorems 3.28 and 3.29 we have the following two theorems.

Theorem 6.34. $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta)$ is non-empty if and only if

$$\left.\begin{array}{l} k_1 \leq k \leq l, \\ 2s+\gamma \leq m-k \leq \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \quad \text{if } \gamma \neq \delta, \text{ or } \gamma = \delta \text{ and } \Gamma = \Delta, \\ \nu+s, \\ \quad \text{if } \gamma = \delta = 1 \text{ and } \Gamma \neq \Delta, \end{cases} \\ 2s_1+\gamma_1 \leq m_1-k_1 \leq \begin{cases} \nu+s_1+\min\{\delta,\gamma_1\}, \\ \quad \text{if } \gamma_1 \neq \delta, \text{ or } \gamma_1 = \delta \text{ and } \Gamma_1 = \Delta, \\ \nu+s_1, \\ \quad \text{if } \gamma_1 = \delta = 1 \text{ and } \Gamma_1 \neq \Delta, \end{cases} \end{array}\right\} \tag{6.58}$$

and

$$2(m-k)-2(m_1-k_1)$$
$$\geq \begin{cases} (2s+\gamma)-(2s_1+\gamma_1)+\mid \gamma-\gamma_1 \mid \geq 2 \mid \gamma-\gamma_1 \mid, \\ \text{if } \gamma_1 \neq \gamma, \text{ or } \gamma_1 = \gamma \text{ and } \Gamma_1 = \Gamma, \\ (2s+\gamma)-(2s_1+\gamma_1)+2 \geq 4, \\ \text{if } \gamma_1 = \gamma = 1 \text{ and } \Gamma_1 \neq \Gamma \end{cases}$$

and these four conditions are equivalent to the following three conditions

$$\left.\begin{array}{l} k_1 \le k \le l, \\ 2s+\gamma \le m-k \le \left\{\begin{array}{l} \nu+s+\min\{\delta,\gamma\}, \\ \quad \text{if } \gamma\neq\delta, \text{ or } \gamma=\delta \text{ and } \Gamma=\Delta, \\ \nu+s, \\ \quad \text{if } \gamma=\delta=1 \text{ and } \Gamma=\Delta, \end{array}\right. \\ \text{and} \\ \min\{m-k-2s-\gamma, m_1-k_1-2s_1-\gamma_1\} \\ \quad \ge \left\{\begin{array}{l} \max\{0, m_1-k_1-s-s_1-\min\{\gamma,\gamma_1\}\}, \\ \quad \text{if } \gamma_1\neq\gamma, \text{ or } \gamma_1=\gamma \text{ and } \Gamma_1=\Gamma, \\ \max\{0, m_1-k_1-s-s_1\}, \\ \quad \text{if } \gamma_1=\gamma=1 \text{ and } \Gamma_1\neq\Gamma. \end{array}\right. \end{array}\right\} \tag{6.59}$$

□

Theorem 6.35. Assume that (6.58) holds. Then

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l, \Delta) \\ &\quad = N(m_1-k_1, 2s_1+\gamma_1, s_1, \Gamma_1; 2s+\gamma+(m-k-2s-\gamma), \Gamma) \\ &\qquad \times N(k_1, k) q^{(m_1-k_1)(k-k_1)}, \end{aligned}$$

where $N(m_1-k_1, 2s_1+\gamma_1, s_1, \Gamma_1; 2s+\gamma+(m-k-2s-\gamma), \Gamma)$ and $N(k_1, k)$ are given in Corollary 6.31 and Theorem 1.7, respectively. □

As an application of Theorem 6.27, let us compute the number of symmetric matrices of a given rank over a finite field.

Denote by $\mathcal{S}(n)$ the set of all $n \times n$ symmetric matrices over $\mathbb{F}_q$. Define an action of $GL_n(\mathbb{F}_q)$ on $\mathcal{S}(n)$ as follows:

$$\begin{array}{ccc} \mathcal{S}(n) \times GL_n(\mathbb{F}_q) & \to & \mathcal{S}(n) \\ (S, \quad T) & \mapsto & TS^tT. \end{array}$$

Then $\mathcal{S}(n)$ is subdivided into orbits under the action of $GL_n(\mathbb{F}_q)$. By Theorem 1.26, these orbits are represented by all the cogredience normal forms

$$M(n, 2\nu+\delta, \nu, \Delta),$$

where $0 \le 2\nu+\delta \le n$, $\delta = 0, 1$, or 2,

$$\Delta = \left\{\begin{array}{ll} \phi, & \text{if } \delta = 0, \\ 1 \text{ or } z, & \text{if } \delta = 1, \\ \left(\begin{smallmatrix} 1 & \\ & -z \end{smallmatrix}\right), & \text{if } \delta = 2. \end{array}\right.$$

and z is a fixed non-square element of $\mathbb{F}_q^*$. Denote by $\mathcal{S}(n, 2\nu+\delta, \nu, \Delta)$ the orbit of $M(n, 2\nu+\delta, \nu, \Delta)$. Then we have

Theorem 6.36. Let $0 \le 2\nu+\delta \le n$, where $\delta = 0, 1$, or 2. Then

$$| \mathcal{S}(n, 2\nu+\delta, \nu, \Delta) | = q^{\nu(\nu+\delta)+\frac{1}{2}\delta(\delta-1)} \frac{\prod_{i=\nu+1}^{n} (q^i - 1)}{\prod_{i=0}^{\nu+\delta-1} (q^i + 1) \prod_{i=1}^{n-2\nu-\delta} (q^i - 1)}.$$

Proof. Clearly $| \mathcal{S}(n, 2\nu+\delta, \nu, \Delta) |$ is equal to the index of the stabilizer of $M(n, 2\nu+\delta, \nu, \Delta)$ in $GL_n(\mathbb{F}_q)$ and the stabilizer of $M(n, 2\nu+\delta, \nu, \Delta)$ is the singular orthogonal group $O_{2\nu+\delta+(n-2\nu-\delta),\Delta}(\mathbb{F}_q)$. That is,

$$| \mathcal{S}(n, 2\nu+\delta, \nu, \Delta) | = GL_n(\mathbb{F}_q) : O_{2\nu+\delta+(n-2\nu-\delta),\Delta}(\mathbb{F}_q).$$

Hence our theorem follows from Theorems 1.6 and 6.27. □

Denote the set of $n \times n$ symmetric matrices of rank r by $\mathcal{S}(n, r)$. Then we have

Corollary 6.37. Let $0 \le r \le n$. Then

$$| \mathcal{S}(n, r) | = q^{[r/2]([r/2]+1)} \frac{\prod_{i=[r/2]+1}^{n} (q^i - 1)}{\prod_{i=1}^{[r/2]} (q^i + 1) \prod_{i=1}^{n-r} (q^i - 1)}.$$

Proof. In fact

$$\mathcal{S}(n, r) = \begin{cases} \mathcal{S}(n, 2\nu, \nu) \cup \mathcal{S}(n, 2(\nu-1)+2, \nu-1), & \text{if } r = 2\nu \text{ is even,} \\ \mathcal{S}(n, 2\nu+1, \nu, 1) \cup \mathcal{S}(n, 2\nu+1, \nu, z), & \text{if } r = 2\nu+1 \text{ is odd.} \end{cases}$$

□

Moreover, we have

Corollary 6.38.

$$| \mathcal{S}(2\nu+\delta, 2\nu+\delta, \nu, \Delta) | = \begin{cases} \frac{1}{2} q^{\nu^2} (q^\nu + 1) \prod_{i=0}^{\nu-1} (q^{2i+1} - 1), & \text{if } \delta = 0, \\ \frac{1}{2} q^{\nu(\nu+1)} \prod_{i=0}^{\nu} (q^{2i+1} - 1), & \text{if } \delta = 1, \\ \frac{1}{2} q^{(\nu+1)^2} (q^{\nu+1} - 1) \prod_{i=0}^{\nu} (q^{2i+1} - 1), & \text{if } \delta = 2. \end{cases}$$

□

6.4 Orthogonal Polarities and Polar Spaces Arising from Them

In this section we change our language to the projective geometry language. Let us consider the $(2\nu+\delta-1)$-dimensional projective space $PG(2\nu+\delta-1,\mathbb{F}_q)$ over $\mathbb{F}_q$ arising from the $(2\nu+\delta)$-dimensional row vector space $\mathbb{F}_q^{(2\nu+\delta)}$. Let S be the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix (6.1), (6.2), (6.3), or (6.4). Let P be an r-flat in $PG(2\nu+\delta-1,\mathbb{F}_q)$, then

$$P^{\perp}=\{x\in\mathbb{F}_q^{(2\nu+\delta)}\mid xS\,{}^tP=0\}$$

is an $(2\nu+\delta-r-2)$-flat such that $P\subset Q$ implies $P^{\perp}\supset Q^{\perp}$. Thus the map

$$P\mapsto P^{\perp}$$

from the set of flats in $PG(2\nu+\delta-1,\mathbb{F}_q)$ to itself is a correlation of $PG(2\nu+\delta-1,\mathbb{F}_q)$. Moreover, since

$$(P^{\perp})^{\perp}=P,$$

this correlation is a polarity, called the *orthogonal polarity* defined by the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix S, or simply an orthogonal polarity.

An m-dimensional totally isotropic subspace of $\mathbb{F}_q^{(2\nu+\delta)}$ with respect to S is called a *totally isotropic* $(m-1)$*-flat* in $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to S. Denote by $\mathcal{P}(S)$, the set of all totally isotropic flats in $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to S. We deduce from Corollary 6.5 that maximal totally isotropic flats in $PG(2\nu+\delta-1,\mathbb{F}_q)$ are of dimension $\nu-1$. Therefore $\mathcal{P}(S)$ is the set of absolutes of dimension $\leq\nu-1$ of the orthogonal polarity defined by S.

Parallel to Theorem 3.33, we have

Theorem 6.39. The set $\mathcal{P}(S)$ of totally isotropic flats of $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix $S=S_{2\nu+\delta,\Delta}$ forms a polar space of rank ν. □

Parallel to Corollary 3.35, we have

Corollary 6.40. The set of totally isotropic points together with the set of totally isotropic lines of $PG(3,\mathbb{F}_q)$, $PG(4,\mathbb{F}_q)$, or $PG(5,\mathbb{F}_q)$ with respect

to the following nonsingular symmetric matrices

$$\begin{pmatrix} 0 & I^{(2)} \\ I^{(2)} & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(2)} & \\ I^{(2)} & 0 & \\ & & 1 \end{pmatrix}, \text{ or } \begin{pmatrix} 0 & I^{(2)} & & \\ I^{(2)} & 0 & & \\ & & 1 & \\ & & & -z \end{pmatrix}, \quad (6.60)$$

respectively, forms a generalized quadrangle. □

The generalized quadrangle formed by the totally isotropic points and the totally isotropic lines with respect to (6.60) in $PG(3, \mathbb{F}_q)$, $PG(4, \mathbb{F}_q)$, or $PG(5, \mathbb{F}_q)$ is usually denoted by $Q(3, q)$, $Q(4, q)$, or $Q(5, q)$, respectively. Parallel to Corollary 3.36, we have

Corollary 6.41. The generalized quadrangles $Q(d, q)$, where $d = 3, 4$, or 5, have parameters

$$\begin{aligned} &s = q, t = 1 \quad \text{for } Q(3, q), \\ &s = t = q \quad \text{for } Q(4, q), \\ &s = q, t = q^2 \quad \text{for } Q(5, q). \end{aligned}$$

□

The proof of this corollary is left as an exercise.

Finally, parallel to the symplectic case, we can use the orthogonal polarity to derive a closed formula for the number of subspace of a given type $(m, 2s + \gamma, s, \Gamma)$ with respect to $S_{2\nu+\delta,\Delta}$ in $\mathbb{F}_q^{(2\nu+\delta)}$ containing a given subspace of type $(m_1, 2s_1 + r_1, s_1, \Gamma_1)$.

Let P_1 be a fixed subspace of type $(m_1, 2s_1 + r_1, s_1, \Gamma_1)$ with respect to $S_{2\nu+\delta,\Delta}$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Denote by $\mathcal{M}'(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu + \delta, \Delta)$ the set of subspaces of type $(m, 2s + \gamma, s, \Gamma)$ containing P_1, and let

$$\begin{aligned} &N'(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1; m, 2s + \gamma, s, \Gamma; 2\nu + \delta, \Delta) \\ &\quad = \mid \mathcal{M}'(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1; m, 2s + \gamma, s, \Gamma; 2\nu + \delta, \Delta) \mid . \end{aligned}$$

Also by Theorem 6.5, $N'(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1; m, 2s + \gamma, s, \Gamma; 2\nu + \delta, \Delta)$ is independent of the particular choice of the subspace P_1 of type $(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1)$. Parallel to Theorems 3.37 and 3.38 we have

Theorem 6.42. $\mathcal{M}'(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ is nonempty if and only if (6.55) holds. □

Theorem 6.43. Assume that (6.55) holds. Then

$$N'(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) = N((m, 2s+\gamma, s, \Gamma)^{\perp}; (m_1, 2s_1+\gamma_1, s_1, \Gamma_1)^{\perp}; 2\nu+\delta, \Delta),$$

where $(m, 2s+\gamma, s, \Gamma)^{\perp}$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)^{\perp}$ are given in Corollary 6.6, and $N((m, 2s+\gamma, s, \Gamma)^{\perp}, (m_1, 2s_1+\gamma_1, s_1, \Gamma_1)^{\perp}; 2\nu+\delta, \Delta)$ is given in (6.57). □

6.5 Comments

Orthogonal groups over finite fields of odd characteristic were first studied by L.E. Dickson, in particular, Theorem 6.21 is due to him (cf. Dickson 1900). For the Witt's transitivity theorem over any field of characteristic $\neq 2$, see Witt (1937). The present proof of this theorem over finite fields (Theorem 6.4) is based on Lemma 6.2 which is due to the author. From Lemma 6.2 we deduce Theorems 6.3, 6.7, and 6.8 which are also due to the author (cf. Wan 1993a). Corollary 6.6 is due to B. Yang and W. Wei (cf. Yang and Wei 1990); however the special case when $(m, 2s+\gamma, s, \Gamma)$ is any non-isotropic subspace type is due to Z. Dai and X. Feng (cf. Dai and Feng 1965).

Most of the results of Section 6.2 are contained in the thesis of Z. Dai and X. Feng (cf. Dai and Feng 1965). However, Theorem 6.21 is due to L. E. Dickson and Corollary 6.23 is due to B. Segre (cf. Segre 1959 and also RayChaudhuri 1962).

All the results of Section 6.3, except Theorem 6.36, are due to the author (cf. Wan 1993c and 1993d). Theorem 6.36 was obtained by L. Carlitz (cf. Carlitz 1954a).

Theorem 6.39 and Corollaries 6.40, 6.41 are all due to J. Tits (cf. Tits 1959 and 1974). Theorem 6.43 is due to B. Yang and W. Wei (cf. Yang and Wei 1990).

6.6 Exercises

6.1 Prove Theorem 6.21 in detail.

6.2 Do the same thing for Theorems 6.22, 6.24, and 6.25.

6.3 Give the missing proofs of the theorems in Section 6.3.

6.4 Let $N = 2\nu+\delta+l-1$, where $\nu \geq 0$, $\delta = 0, 1$, or 2, and $l \geq 0$. Denote by Q the quadric in $PG(N, \mathbb{F}_q)$ defined by the quadratic equation

$$(x_0, x_1, \ldots, x_N) \begin{pmatrix} S_{2\nu+\delta,\Delta} & \\ & 0^{(l)} \end{pmatrix} {}^t(x_0, x_1, \ldots, x_N) = 0.$$

For any set S of points in $PG(N, \mathbb{F}_q)$ denote by $\mid S \mid$ the number of points in S and call $\mid S \mid$ the size of S.

(i) Prove that $\mid Q \mid = \mid H \mid$ for any hyperplane H in $PG(N, \mathbb{F}_q)$ if and only if $\delta = 1$.

(ii) Let $\delta = 1$ and $\Delta = 1$ or z, where z is a non-square element of $\mathbb{F}_q$. Assume that $\nu \geq 1$. Prove that the hyperplanes H of $PG(N, \mathbb{F}_q)$ are classified into three classes according to the size $\mid H \cap Q \mid$. Determine the three different values of $\mid H \cap Q \mid$ and compute the number of hyperplanes H having the same size $\mid H \cap Q \mid$ for $H \cap Q$.

6.5 Prove Corollary 6.41.

Chapter 7
Geometry of Orthogonal Groups over Finite Fields of Characteristic 2

7.1 Orthogonal Geometry over Finite Fields of Characteristic 2

In this chapter we assume that $\mathbb{F}_q$ is a finite field of characteristic 2. As in Section 1.4 we choose a fixed element α of $\mathbb{F}_q$ such that $\alpha \notin N = \{x^2 + x \mid x \in \mathbb{F}_q\}$.

Let G be an $n \times n$ regular matrix over $\mathbb{F}_q$. An $n \times n$ matrix T over $\mathbb{F}_q$ is said to be *orthogonal* with respect to G, if

$$TG\,{}^tT \equiv G.$$

It can be readily verified that $n \times n$ orthogonal matrices with respect to G are nonsingular and they form a group with respect to matrix multiplication, called the *orthogonal group* of degree n with respect to G over $\mathbb{F}_q$ and denoted by $O_n(\mathbb{F}_q, G)$. The index of G is also called the *index* of the orthogonal group $O_n(\mathbb{F}_q, G)$.

If G_1 and G_2 are two 'cogredient' $n \times n$ regular matrices over $\mathbb{F}_q$, then it is easy to prove that $O_n(\mathbb{F}_q, G_1)$ and $O_n(\mathbb{F}_q, G_2)$ are isomorphic. Hence by Theorem 1.35 it is sufficient to discuss the three orthogonal groups with respect to the following three $n \times n$ regular matrices

$$\begin{pmatrix} 0 & I^{(\nu)} \\ 0 & \end{pmatrix}, \begin{pmatrix} 0 & I^{(\nu)} & \\ 0 & & \\ & & 1 \end{pmatrix}, \text{and} \begin{pmatrix} 0 & I^{(\nu)} & & \\ 0 & & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}, \tag{7.1}$$

where $n = 2\nu$, $n = 2\nu + 1$, and $n = 2\nu + 2$, respectively.

From now on we assume that G is one of the three regular matrices in (7.1), where ν is the index of G. We use the symbol $O_{2\nu+\delta}(\mathbb{F}_q)$, where

$\delta = 0, 1$, or 2, to represent the three orthogonal groups with respect to the three $(2\nu+\delta)\times(2\nu+\delta)$ regular matrices in (7.1), respectively. The lower left corner $\delta\times\delta$ submatrix of G is called the *definite part* of G and is denoted by Δ. In particular, when $\delta = 0$, Δ disappears; when $\delta = 1$, $\Delta = (1)$; when $\delta = 2$, $\Delta = \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}$.

We have an action of $O_{2\nu+\delta}(\mathbb{F}_q)$ on $\mathbb{F}_q^{(2\nu+\delta)}$ defined as follows:

$$\begin{aligned} \mathbb{F}_q^{(2\nu+\delta)} \times O_{2\nu+\delta}(\mathbb{F}_q) &\to \mathbb{F}_q^{(2\nu+\delta)} \\ ((x_1, x_2, \ldots, x_{2\nu+\delta}), T) &\mapsto (x_1, x_2, \ldots, x_{2\nu+\delta})T. \end{aligned}$$

The elements of $O_{2\nu+\delta}(\mathbb{F}_q)$ are then called *orthogonal transformations* with respect to G. The vector space $\mathbb{F}_q^{(2\nu+\delta)}$, together with the above group action of the orthogonal group $O_{2\nu+\delta}(\mathbb{F}_q)$, is called the $(2\nu+\delta)$-dimensional *orthogonal space* over $\mathbb{F}_q$ with respect to G and *the geometry of the orthogonal group* $O_{2\nu+\delta}(\mathbb{F}_q)$, or simply the *orthogonal geometry* over $\mathbb{F}_q$, aims at the study of the properties of geometric figures of the orthogonal space $\mathbb{F}_q^{(2\nu+\delta)}$ which are invariant under the orthogonal group $O_{2\nu+\delta}(\mathbb{F}_q)$.

Now let us study how the vector subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ are partitioned into orbits under the action of $O_{2\nu+\delta}(\mathbb{F}_q)$.

By Theorem 1.35 there are three types of normal forms of $m\times m$ matrices over $\mathbb{F}_q$ under 'cogredience'. They are

$$M(m, 2s, s) = \begin{pmatrix} 0 & I^{(s)} & \\ & 0 & \\ & & 0^{(m-2s)} \end{pmatrix},$$

$$M(m, 2s+1, s) = \begin{pmatrix} 0 & I^{(s)} & & \\ & 0 & & \\ & & 0^{(m-2s-1)} & \\ & & & 1 \end{pmatrix},$$

and

$$M(m, 2s+2, s) = \begin{pmatrix} 0 & I^{(s)} & & & \\ & 0 & & & \\ & & \alpha & 1 & \\ & & & \alpha & \\ & & & & 0^{(m-2s-2)} \end{pmatrix}.$$

We use the symbol $M(m, 2s+\gamma, s)$ to cover these three types of normal forms, where s is its index and $\gamma = 0, 1$, or 2.

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2\nu+\delta)}$. If $PG\,{}^tP$ is cogredient to $M(m, 2s+\gamma, s)$, then P is called a *subspace of type* $(m, 2s+\gamma, s, \Gamma)$, with respect to G, where

$$\Gamma = \begin{cases} 1 \text{ or } 0, \text{ if } e_{2\nu+1} \in P \text{ or not, respectively, in case } \delta = \gamma = 1, \\ \phi \text{ and may be omitted, in all other cases.} \end{cases}$$

Subspaces of type $(m, 2s, s)$ and $(m, 2s+2, s)$ are also called *subspaces of the hyperbolic type* and *subspaces of the elliptic type*, respectively. Moreover, subspaces of the hyperbolic type with $s = 0$ are called *totally singular subspaces* and subspaces of type $(2s+\gamma, 2s+\gamma, s, \Gamma)$ are called *nonsingular subspaces*. Clearly, a subspace P is totally singular if and only if $PG\,{}^tP$ is alternate, and it is nonsingular if and only if $PG\,{}^tP$ is regular.

Let $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ denote the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ with respect to G in $\mathbb{F}_q^{(2\nu+\delta)}$. We shall prove that $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ is an orbit of subspaces under $O_{2\nu+\delta}(\mathbb{F}_q)$. We need some preparation.

Theorem 7.1. Elements of $O_{2\nu+1}(\mathbb{F}_q)$ are of the form

$$\begin{array}{c} \begin{matrix} \nu & \nu & 1 \end{matrix} \\ \begin{pmatrix} A & B & P \\ C & D & Q \\ & & 1 \end{pmatrix} \begin{matrix} \nu \\ \nu \\ 1 \end{matrix} \end{array},$$

where

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp_{2\nu}(\mathbb{F}_q);$$

and the map

$$\begin{aligned} O_{2\nu+1}(\mathbb{F}_q) &\to Sp_{2\nu}(\mathbb{F}_q) \\ \begin{pmatrix} A & B & P \\ C & D & Q \\ & & 1 \end{pmatrix} &\mapsto \begin{pmatrix} A & B \\ C & D \end{pmatrix} \end{aligned}$$

is an isomorphism of groups. Moreover, for any $T \in O_{2\nu+1}(\mathbb{F}_q)$, $e_{2\nu+1}T = e_{2\nu+1}$.

Proof. Let

$$T = \begin{pmatrix} A & B & P \\ C & D & Q \\ R & U & W \end{pmatrix}$$

be an arbitrary element of $O_{2\nu+1}(\mathbb{F}_q)$. From $TG\,{}^tT \equiv G$, where G is the middle one of the matrices in (7.1), we deduce

$$\begin{aligned} A\,{}^tB + P\,{}^tP &\equiv 0, \\ C\,{}^tD + Q\,{}^tQ &\equiv 0, \\ R\,{}^tU + W\,{}^tW &\equiv 1, \\ A\,{}^tD + B\,{}^tC &= I, \\ A\,{}^tU + B\,{}^tR &= 0, \\ C\,{}^tU + D\,{}^tR &= 0. \end{aligned}$$

From $A\,{}^tB + P\,{}^tP \equiv 0$ and $C\,{}^tD + Q\,{}^tQ \equiv 0$ we deduce that both $A\,{}^tB$ and $C\,{}^tD$ are congruent to diagonal matrices mod $\mathcal{K}_\nu$. Therefore

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} {}^tD & {}^tB \\ {}^tC & {}^tA \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix},$$

which implies that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

is nonsingular. The relations $A\,{}^tU + B\,{}^tR = 0$ and $C\,{}^tU + D\,{}^tR = 0$ can be written in the following matrix form

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} {}^tU \\ {}^tR \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

from which we deduce $U = 0$ and $R = 0$. Then from $R\,{}^tU + W\,{}^tW = 1$ we deduce $W = 1$. Therefore all elements of $O_{2\nu+1}(\mathbb{F}_q)$ are of the form

$$\begin{pmatrix} A & B & P \\ C & D & Q \\ & & 1 \end{pmatrix}.$$

Since $A\,{}^tB$ and $C\,{}^tD$ are 'cogredient' to diagonal matrices, we have

$$A\,{}^tB + B\,{}^tA = 0, \quad C\,{}^tD + D\,{}^tC = 0.$$

These two relations together with

$$A\,{}^tD + B\,{}^tC = I$$

imply that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in Sp_{2\nu}(\mathbb{F}_q).$$

Thus the first assertion of the theorem is proved.

From $A\,{}^tB + P\,{}^tP \equiv 0$ and $\mathbb{F}_q^2 = \mathbb{F}_q$ we deduce that P is uniquely determined by A and B. Similarly, Q is uniquely determined by C and D. Hence the second assertion of the theorem follows immediately.

Finally, it is quite obvious that for any $T \in O_{2\nu+1}(\mathbb{F}_q)$, $e_{2\nu+1}T = e_{2\nu+1}$.

□

Let v be a vector of $\mathbb{F}_q^{(2\nu+\delta)}$. v is called a *singular vector* or *nonsingular vector* with respect to G, if $vG\,{}^tv = 0$ or $vG\,{}^tv \neq 0$, respectively. Clearly, a non-zero vector v is singular if and only if the 1-dimensional subspace $\langle v \rangle$ generated by v is a totally singular subspace.

Lemma 7.2. Let G be an $n \times n$ indefinite regular matrix. Then there exist n linearly independent singular vectors with respect to G.

Proof. It is sufficient to prove our lemma for the three 'cogredience' normal forms of $n \times n$ regular matrices in (7.1), where $n = 2\nu + \delta$, $\nu \geq 1$, and $\delta = 0, 1$, or 2. We can choose $2\nu + \delta$ linearly independent singular vectors as follows:

For $\delta = 0$, we choose

$$v_i = e_i,\ i = 1, 2, \ldots, 2\nu.$$

For $\delta = 1$, we choose

$$\begin{aligned} &v_i = e_i,\ i = 1, 2, \ldots, 2\nu; \\ &v_{2\nu+1} = e_1 + e_{\nu+1} + e_{2\nu+1}. \end{aligned}$$

For $\delta = 2$, we choose

$$\begin{aligned} &v_i = e_i,\ i = 1, 2, \ldots, 2\nu; \\ &v_{2\nu+i} = e_1 + \alpha e_{\nu+1} + e_{2\nu+i},\ i = 1, 2. \end{aligned}$$

□

Lemma 7.3. Let P be a nonsingular subspace of type $(2s+\gamma, 2s+\gamma, s)$, where $\gamma = 0$ or 2, in $\mathbb{F}_q^{(2\nu+\delta)}$. Then there exists a subspace P^* of $\mathbb{F}_q^{(2\nu+\delta)}$ such that $V_{2\nu+\delta} = P \cup P^*$, $P \cap P^* = (0)$ and $P(G + {}^tG)\,{}^t(P^*) = 0$. Furthermore, the 'cogredience' norm form of $P^*G\,{}^t(P^*)$ is uniquely determined by G and the type of P (or the normal form of PG^tP) and is given in the following table:

$PG\,{}^tP\ P^*G\,{}^t(P^*)\ G$	$\begin{pmatrix} 0 & I^{(\nu)} \\ 0 & \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu)} & \\ 0 & & \\ & & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu)} & & \\ 0 & & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}$
$\begin{pmatrix} 0 & I^{(s)} \\ 0 & \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s)} \\ 0 & \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s)} & \\ 0 & & \\ & & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s)} & & \\ 0 & & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}$
$\begin{pmatrix} 0 & I^{(s)} & & \\ 0 & & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s-2)} & & \\ 0 & & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s-1)} & \\ 0 & & \\ & & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & I^{(\nu-s)} \\ 0 & \end{pmatrix}$

In particular, the definite part of $P^*G\,{}^t(P^*)$ is of degree $|\,\delta-\gamma\,|$.

Proof. If $\delta=0$ or 2, then the alternate matrix $G+{}^tG$ is nonsingular and we take the dual subspace $P^\perp$ of P with respect to $G+{}^tG$ as P^*. Clearly we have $\mathbb{F}_q^{(2\nu+\delta)}=P+P^*$, $P\cap P^*=(0)$, and $P(G+{}^tG)\,{}^t(P^*)=0$. For $\delta=1$, we write

$$P=\begin{matrix} & 2\nu & 1 \end{matrix}\ \begin{pmatrix} P_1 & P_2 \end{pmatrix}\ ,$$

then

$$PG\,{}^tP=P_1\begin{pmatrix} 0 & I^{(\nu)} \\ 0 & \end{pmatrix}{}^tP_1+P_2\,{}^tP_2.$$

We can assume that

$$PG\,{}^tP\equiv M(2s+\gamma,2s+\gamma,s).$$

It follows that

$$P_1\begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix}{}^tP_1=\begin{cases} \begin{pmatrix} 0 & I^{(s)} \\ I^{(s)} & 0 \end{pmatrix}, & \text{if } \gamma=0, \\ \begin{pmatrix} 0 & I^{(s)} & & \\ I^{(s)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 0 \end{pmatrix}, & \text{if } \gamma=2. \end{cases}$$

That is, P_1 is a non-isotropic subspace of type $(2s+\gamma,s+\gamma/2)$ in the 2ν-dimensional symplectic space with respect to

$$\begin{pmatrix} 0 & I^{(\nu)} \\ I^{(\nu)} & 0 \end{pmatrix}. \tag{7.2}$$

Let $P_1^{\perp}$ be the dual subspace of P_1 with respect to (7.2). Then $P_1^{\perp}$ is a non-isotropic subspace of dimension $2\nu-(2s+\gamma)$ in the 2ν-dimensional symplectic space. We may take

$$P^* = \begin{matrix} 2\nu & 1 \\ \end{matrix} \begin{pmatrix} 0 & 1 \\ P_1^{\perp} & 0 \end{pmatrix} \begin{matrix} 1 \\ 2\nu-(2s+\gamma) \end{matrix}.$$

The first statement of our lemma is completely proved.

Now let us prove the second statement. By the above choice of P^*, G is 'cogredient' to

$$\begin{pmatrix} P \\ P^* \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ P^* \end{pmatrix} \equiv \begin{pmatrix} PG\,{}^tP & \\ & P^*G\,{}^t(P^*) \end{pmatrix}.$$

Hence $P^*G\,{}^t(P^*)$ must be regular. We distinguish the cases $\delta=1$ and $\delta=0$ or 2.

If $\delta=1$ and $\gamma=0$, then $P^*G\,{}^t(P^*)$ is a $(2\nu-2s+1)\times(2\nu-2s+1)$ matrix and by Theorem 1.35 its 'cogredience' normal form must be

$$\begin{pmatrix} 0 & I^{(\nu-s)} & \\ & 0 & \\ & & 1 \end{pmatrix}.$$

If $\delta=1$ and $\gamma=2$, then $P^*G\,{}^t(P^*)$ is a $(2\nu-2s-1)\times(2\nu-2s-1)$ matrix and by Theorem 1.35 its 'cogredience' normal form must be

$$\begin{pmatrix} 0 & I^{(\nu-s-1)} & \\ & 0 & \\ & & 1 \end{pmatrix}.$$

Therefore the case $\delta=1$ is verified.

If $\delta=0$ or 2, then by Theorem 1.35 and Lemma 1.33, the second statement is also true. □

Lemma 7.4. Let P be a subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that

$$PG\,{}^tP \equiv M(m, 2s+\gamma, s).$$

Then

(i) For $\gamma=0$ or 2, there exists a $(2\nu+\delta-m)\times(2\nu+\delta)$ matrix Z such that

$$\begin{pmatrix} P \\ Z \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z \end{pmatrix} \equiv \begin{pmatrix} \Lambda & & \\ & 0 \quad I^{(m-2s-\gamma)} & \\ & 0 & \\ & & \Sigma_1 \end{pmatrix},$$

where $\Lambda = M(2s+\gamma, 2s+\gamma, s)$, Σ_1 is a 'cogredience' normal form of $\sigma_1 \times \sigma_1$ regular matrices with definite part of degree $\mid \delta - \gamma \mid$ and is uniquely determined by G and the type of P, and $\sigma_1 = 2(\nu + s - m) + \delta + \gamma \geq 0$;

(ii) For $\gamma = 1$ and $\delta = 0$ or 2, and for $\gamma = \delta = 1$ and $\Gamma = 0$, there exists a $(2\nu + \delta - m) \times (2\nu + \delta)$ matrix Z and an $m \times m$ matrix Q of the form

$$Q = \begin{pmatrix} I & & \\ & I & \\ & L & b \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \end{matrix}$$

such that

$$\begin{pmatrix} QP \\ Z \end{pmatrix} G \, {}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix}$$

$$\equiv \begin{cases} \begin{pmatrix} 0 & I^{(s)} & & & & \\ & 0 & & & & \\ & & 0 & & I^{(m-2s-1)} & \\ & & & \alpha & & 1 \\ & & & & 0 & \\ & & & & & \alpha \end{pmatrix}, & \text{if } \delta = 2 \text{ and } \nu + s - m + 1 = 0, \\ \begin{pmatrix} 0 & I^{(s)} & & & & \\ & 0 & & & & \\ & & 0 & & I^{(m-2s-1)} & \\ & & & 1 & & 1 \\ & & & & 0 & \\ & & & & & 0 \\ & & & & & & \Sigma_2 \end{pmatrix}, & \text{if } \delta \neq 2 \text{ or } \nu + s - m + 1 > 0, \end{cases}$$

where Σ_2 is a 'cogredience' normal form of $\sigma_2 \times \sigma_2$ regular matrices with definite part of degree δ and is uniquely determined by G and the type of P, and $\sigma_2 = 2(\nu + s - m) + \delta \geq 0$;

(iii) For $\gamma = \delta = 1$ and $\Gamma = 1$, there exists a $(2\nu + \delta - m) \times (2\nu + \delta)$ matrix Z and an $m \times m$ matrix Q of the form

$$Q = \begin{pmatrix} I & & \\ & I & \\ & L & 1 \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \end{matrix}$$

such that

$$\begin{pmatrix} QP \\ Z \end{pmatrix} G \, {}^t\!\begin{pmatrix} QP \\ Z \end{pmatrix} \equiv \begin{pmatrix} 0 & I^{(s)} & & & & \\ & 0 & & & & \\ & & 0 & & I^{(m-2s-1)} & \\ & & & 1 & & \\ & & & & 0 & \\ & & & & & \Sigma_3 \end{pmatrix},$$

where Σ_3 is a 'cogredience' normal form of $\sigma_3 \times \sigma_3$ regular matrices without definite part and is uniquely determined by G and the type of P, and $\sigma_3 = 2(\nu + s - m + 1) \geq 0$.

Proof. Choose a $(2\nu + \delta - m) \times (2\nu + \delta)$ matrix Z_1 such that

$$\begin{pmatrix} P \\ Z_1 \end{pmatrix}$$

is a $(2\nu + \delta) \times (2\nu + \delta)$ nonsingular matrix. Then we can write

$$\begin{pmatrix} P \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix}$$

$$\equiv \begin{cases} \begin{matrix} & 2s+\gamma & m-2s-\gamma & 2\nu+\delta-m \end{matrix} \\ \begin{pmatrix} \Lambda & & L_1 \\ & 0 & L_2 \\ & & B \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ 2\nu+\delta-m \end{matrix} \end{cases} , \quad \text{if } \gamma = 0 \text{ or } 2,$$

$$\equiv \begin{cases} \begin{matrix} & 2s & m-2s-1 & 1 & 2\nu+\delta-m \end{matrix} \\ \begin{pmatrix} \Lambda & & & L_1 \\ & 0 & & L_2 \\ & & 1 & L_3 \\ & & & B \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ 2\nu+\delta-m \end{matrix} \end{cases} , \quad \text{if } \gamma = 1.$$

We distinguish the following three cases.

(i) $\gamma = 0$ or 2. At first we assert that there is a $(2\nu + \delta) \times (2\nu + \delta)$ matrix R of the form

$$R = \begin{pmatrix} I & \\ A & I \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix}$$

such that

$$R \begin{pmatrix} P \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^tR$$

$$\equiv \begin{pmatrix} \begin{matrix} 2s+\gamma & m-2s-\gamma & 2\nu+\delta-m \end{matrix} \\ \begin{pmatrix} \Lambda & & \\ & 0 & L_2 \\ & & C \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ 2\nu+\delta-m \end{matrix} \end{pmatrix} . \tag{7.3}$$

In fact, if $\gamma = 0$, then we have

$$\Lambda = \begin{pmatrix} 0 & I^{(s)} \\ & 0 \end{pmatrix} .$$

Write

$$L_1 = \begin{pmatrix} L_{11} \\ L_{12} \end{pmatrix} \begin{matrix} s \\ s \end{matrix},$$

and we may take

$$R = \begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ {}^tL_{12} & {}^tL_{11} & & I \end{pmatrix} \begin{matrix} s \\ s \\ m-2s \\ 2\nu+\delta-m \end{matrix}.$$

If $\gamma = 2$, then we have

$$\Lambda = \begin{pmatrix} 0 & I^{(s)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}.$$

Write

$$L_1 = \begin{pmatrix} L_{11} \\ L_{12} \\ L_{13} \\ L_{14} \end{pmatrix} \begin{matrix} s \\ s \\ 1 \\ 1 \end{matrix},$$

and we may take

$$R = \begin{pmatrix} I & & & & & \\ & I & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & I & \\ {}^tL_{12} & {}^tL_{11} & {}^tL_{14} & {}^tL_{13} & & I \end{pmatrix} \begin{matrix} s \\ s \\ 1 \\ 1 \\ m-2s-2 \\ 2\nu+\delta-m \end{matrix}.$$

Since G is regular, the matrix in the right hand side of (7.3) is also regular. Hence L_2 is of rank $m-2s-\gamma$. Then $m-2s-\gamma \le 2\nu+\delta-m$ and there is a $(2\nu+\delta-m)\times(2\nu+\delta-m)$ nonsingular matrix A_1 such that

$$L_2\,{}^tA_1 = \left(I^{(m-2s-\gamma)}\ 0^{(m-2s-\gamma,\sigma_1)}\right),$$

where $\sigma_1 = 2(\nu+s-m)+\delta+\gamma \ge 0$. Let

$$R_1 = \begin{pmatrix} I & \\ & A_1 \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix},$$

then

$$R_1 R \begin{pmatrix} P \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^t\!R \, {}^t\!R_1$$

$$\equiv \begin{array}{c} \begin{matrix} 2s+\gamma & m-2s-\gamma & m-2s-\gamma & \sigma_1 \end{matrix} \\ \begin{pmatrix} \Lambda & & & \\ & 0 & I & 0 \\ & & D_1 & D_2 \\ & & & D_3 \end{pmatrix} \end{array} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma_1 \end{matrix} .$$

Let

$$R_2 = \begin{pmatrix} I & & & \\ & I & & \\ & D_1 & I & \\ & {}^t\!D_2 & & I \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma_1 \end{matrix} ,$$

then

$$R_2 R_1 R \begin{pmatrix} P \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^t\!R \, {}^t\!R_1 \, {}^t\!R_2 \equiv \begin{pmatrix} \Lambda & & & \\ & 0 & I & \\ & & 0 & \\ & & & D_3 \end{pmatrix} .$$

Let the 'cogredience' normal form of D_3 be Σ_1, then by Lemma 7.3 it is uniquely determined by G and the type of P, and its definite part is of degree $| \delta - \gamma |$. There is a $\sigma_1 \times \sigma_1$ nonsingular matrix Q_3 such that

$$Q_3 D_3 \, {}^t\!Q_3 \equiv \Sigma_1 .$$

Let

$$R_3 = \begin{pmatrix} I & & & \\ & I & & \\ & & I & \\ & & & Q_3 \end{pmatrix} \begin{matrix} 2s+\gamma \\ m-2s-\gamma \\ m-2s-\gamma \\ \sigma_1 \end{matrix} ,$$

then

$$R_3 R_2 R_1 R \begin{pmatrix} P \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^t\!R \, {}^t\!R_1 \, {}^t\!R_2 \, {}^t\!R_3 \equiv \begin{pmatrix} \Lambda & & & \\ & 0 & I & \\ & & 0 & \\ & & & \Sigma_1 \end{pmatrix} .$$

Notice that $R_3 R_2 R_1 R$ is a matrix of the following form

$$\begin{pmatrix} I & \\ B_1 & B_2 \end{pmatrix} \begin{matrix} m \\ 2\nu + \delta - m \end{matrix} ,$$

hence we can assume that

$$R_3R_2R_1R\begin{pmatrix} P \\ Z_1 \end{pmatrix} = \begin{pmatrix} P \\ Z \end{pmatrix}.$$

This finishes the proof of case (i).

(ii) $\gamma = 1$ and when $\delta = 1$ we assume further that $\Gamma = 0$. At first, as in the above case, there is a $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular matrix R of the form

$$R = \begin{pmatrix} I & \\ A & I \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix}$$

such that

$$R\begin{pmatrix} P \\ Z_1 \end{pmatrix} G\, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^tR$$

$$\equiv \begin{matrix} 2s & m-2s-1 & 1 & 2\nu+\delta-m \end{matrix}$$
$$\begin{pmatrix} \Lambda & & & \\ & 0 & & L_2 \\ & & 1 & L_3 \\ & & & C \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ 2\nu+\delta-m \end{matrix}.$$

Since G is regular, L_2 is of rank $m-2s-1$. Thus $m-2s-1 \le 2\nu+\delta-m$. As in case (i), there is a $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular matrix R_1 of the form

$$R_1 = \begin{pmatrix} I & \\ & A_1 \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix}$$

such that

$$R_1R\begin{pmatrix} P \\ Z_1 \end{pmatrix} G\, {}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix} {}^tR\, {}^tR_1$$

$$\equiv \begin{matrix} 2s & m-2s-1 & 1 & m-2s-1 & \sigma_2+1 \end{matrix}$$
$$\begin{pmatrix} \Lambda & & & & \\ & 0 & & I & 0 \\ & & 1 & L_4 & L_5 \\ & & & D_1 & D_2 \\ & & & & D_3 \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ \sigma_2+1 \end{matrix},$$

where $\sigma_2 = 2(\nu+s-m)+\delta$ and $\sigma_2+1 \ge 0$. Again as in case (i), there is a $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular matrix R_2 of the form

$$R_2 = \begin{pmatrix} I & & & \\ & I & & \\ & & 1 & \\ & D_1 & & I \\ & {}^tD_2 & & \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ \sigma_2+1 \end{matrix}$$

such that

$$R_2R_1R\begin{pmatrix} P \\ Z_1 \end{pmatrix} G\,{}^t\!\begin{pmatrix} P \\ Z_1 \end{pmatrix}\,{}^tR\,{}^tR_1\,{}^tR_2 \equiv \begin{pmatrix} \Lambda & & & & \\ & 0 & & I & \\ & & 1 & L_4 & L_5 \\ & & & 0 & \\ & & & & D_3 \end{pmatrix}.$$

Notice that R_2R_1R is a matrix of the form

$$\begin{pmatrix} I & \\ B_1 & B_2 \end{pmatrix} \begin{matrix} m \\ 2\nu+\delta-m \end{matrix},$$

hence we can assume that

$$R_2R_1R\begin{pmatrix} P \\ Z_1 \end{pmatrix} = \begin{pmatrix} P \\ Z_2 \end{pmatrix}.$$

Then

$$\begin{pmatrix} P \\ Z_2 \end{pmatrix} G\,{}^t\!\begin{pmatrix} P \\ Z_2 \end{pmatrix} \equiv \begin{pmatrix} \Lambda & & & & \\ & 0 & & I & \\ & & 1 & L_4 & L_5 \\ & & & 0 & \\ & & & & D_3 \end{pmatrix}.$$

Let

$$Q_1 = \begin{pmatrix} I & & \\ & I & \\ & L_4 & 1 \end{pmatrix} \begin{matrix} 2s \\ m-2s-1 \\ 1 \end{matrix}, \tag{7.4}$$

then

$$\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} G\,{}^t\!\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} \equiv \begin{pmatrix} \Lambda & & & & \\ & 0 & & I & \\ & & 1 & 0 & L_5 \\ & & & 0 & \\ & & & & D_3 \end{pmatrix}.$$

If $\sigma_2+1=0$, or $\sigma_2+1>0$ but $L_5=0$, let

$$v = e_m\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix},$$

then $v \neq 0,\ v \in P$ and

$$v(G+{}^tG)\,{}^t\!\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} = (0,0,\ldots,0),$$

hence

$$v(G + {}^tG) = (0, 0, \ldots, 0).$$

This implies that $\delta = 1$ and $\Gamma = 1$, which is not the case (ii) we are considering. Therefore in case (ii) we must have $\sigma_2 + 1 \geq 1$ and $L_5 \neq 0$.

Let

$$G_1 = \begin{pmatrix} 1 & L_5 \\ & D_3 \end{pmatrix},$$

then G_1 is a $(\sigma_2 + 2) \times (\sigma_2 + 2)$ regular matrix. We distinguish the cases when G_1 is definite or indefinite.

If G_1 is definite, we must have $\sigma_2 = 0$ and $D_3 \neq 0$. By the proof of Lemma 1.32, there exists a 2×2 nonsingular matrix A_3 of the form

$$A_3 = \begin{pmatrix} b & 0 \\ c & d \end{pmatrix}$$

such that

$$A_3 G_1 {}^tA_3 \equiv \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}.$$

Let

$$R_3 = \begin{array}{c} \begin{matrix} m-1 & 1 & m-2s-1 & 1 \end{matrix} \\ \begin{pmatrix} I & & & \\ & b & & 0 \\ & & I & \\ & c & & d \end{pmatrix} \end{array} \begin{matrix} m-1 \\ 1 \\ m-2s-1 \\ 1 \end{matrix},$$

then

$$R_3 \begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix} G \, {}^t\!\begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix} {}^tR_3$$

$$\equiv \begin{array}{c} \begin{matrix} 2s & m-2s-1 & 1 & m-2s-1 & 1 \end{matrix} \\ \begin{pmatrix} \Lambda & & & & \\ & 0 & & I & \\ & & \alpha & & 1 \\ & & & 0 & \\ & & & & \alpha \end{pmatrix} \end{array} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ 1 \end{matrix}.$$

Hence we must have $\delta = 2$. From $\sigma_2 = 0$ we deduce $\nu + s - m + 1 = 0$. Let

$$R_3 \begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix} = \begin{pmatrix} QP \\ Z \end{pmatrix}$$

then

$$Q = \begin{pmatrix} I^{(m-1)} & \\ & b \end{pmatrix} Q_1 = \begin{pmatrix} I^{(2s)} & & \\ & I^{(m-2s-1)} & \\ & bL_4 & b \end{pmatrix}$$

is an $m \times m$ matrix of the required form.

Now let G_1 be indefinite. Since $L_5 \neq 0$, $e_1(G_1 + {}^tG_1) \neq 0$. Let

$$\langle e_1 \rangle^\perp = \{x \in \mathbb{F}_q^{(\sigma_2+2)} \mid e_1(G_1 + {}^tG_1)\,{}^tx = 0\},$$

then $\dim \langle e_1 \rangle^\perp = \sigma_2 + 1$. By Lemma 7.2, there exist $\sigma_2 + 2$ linearly independent singular vectors with respect to the indefinite $(\sigma_2 + 2) \times (\sigma_2 + 2)$ matrix G_1. Hence there exists at least one non-zero singular vector $v_2 \notin \langle e_1 \rangle^\perp$, and we can assume that $e_1(G_1 + {}^tG_1)\,{}^tv_2 = 1$. Let Z_3 be a $\sigma_2 \times (\sigma_2 + 2)$ matrix such that

$$\begin{pmatrix} e_1 \\ v_2 \\ Z_3 \end{pmatrix}$$

is nonsingular, then

$$\begin{pmatrix} e_1 \\ v_2 \\ Z_3 \end{pmatrix} G_1 \, {}^t\!\begin{pmatrix} e_1 \\ v_2 \\ Z_3 \end{pmatrix} \equiv \begin{pmatrix} 1 & 1 & L_6 \\ & 0 & L_7 \\ & & D_4 \end{pmatrix}.$$

We also have

$$\begin{pmatrix} 1 & & \\ & 1 & \\ {}^tL_7 & {}^tL_6 & I \end{pmatrix} \begin{pmatrix} 1 & 1 & L_6 \\ & 0 & L_7 \\ & & D_4 \end{pmatrix} {}^t\!\begin{pmatrix} 1 & & \\ & 1 & \\ {}^tL_7 & {}^tL_6 & I \end{pmatrix} \equiv \begin{pmatrix} 1 & 1 & \\ & 0 & \\ & & D_5 \end{pmatrix}$$

for some $\sigma_2 \times \sigma_2$ matrix D_5. Write

$$\begin{pmatrix} e_1 \\ v_2 \\ Z_3 \end{pmatrix} = \begin{array}{c} \begin{matrix} 1 & 1 & \sigma_2 \end{matrix} \\ \begin{pmatrix} 1 & 0 & 0 \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{pmatrix} \begin{matrix} 1 \\ 1 \\ \sigma_2 \end{matrix} \end{array},$$

and let

$$R_4 = \begin{pmatrix} I^{(m-1)} & & & & \\ & 1 & & & \\ & & I^{(m-2s-1)} & & \\ & & & 1 & \\ & {}^tL_7 & & {}^tL_6 & I^{(\sigma_2)} \end{pmatrix} \begin{pmatrix} I^{(m-1)} & & & & \\ & 1 & & & \\ & & I^{(m-2s-1)} & & \\ & B_{21} & & B_{22} & B_{23} \\ & B_{31} & & B_{32} & B_{33} \end{pmatrix}$$

then

$$R_4 \begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} G \, {}^t\!\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} {}^tR_4$$

$$\equiv \begin{array}{c} \begin{matrix} 2s & m-2s-1 & 1 & m-2s-1 & 1 & \sigma_2 \end{matrix} \\ \begin{pmatrix} \Lambda & & & & & \\ & 0 & & I & & \\ & & 1 & & 1 & \\ & & & 0 & & \\ & & & & 0 & \\ & & & & & D_5 \end{pmatrix} \end{array} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ 1 \\ \sigma_2 \end{matrix} .$$

Let the 'cogredience' normal form of D_5 be Σ_2, by Lemma 7.3 it is uniquely determined by G and the type of P and its definite part is of degree δ. Let Q_5 be a $\sigma_2 \times \sigma_2$ nonsingular matrix such that

$$Q_5 D_5 \, {}^tQ_5 = \Sigma_2,$$

and let

$$R_5 = \begin{pmatrix} I^{(2m-2s)} & \\ & Q_5 \end{pmatrix},$$

then

$$R_5R_4 \begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} G \, {}^t\!\begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} {}^tR_4 \, {}^tR_5 \equiv \begin{pmatrix} \Lambda & & & & \\ & 0 & I & & \\ & 1 & & 1 & \\ & & 0 & & \\ & & & & \Sigma_2 \end{pmatrix}.$$

R_5R_4 is of the form

$$\begin{array}{c} \begin{matrix} m \end{matrix} \\ \begin{pmatrix} I & \\ B_3 & B_4 \end{pmatrix} \end{array} \begin{matrix} \\ 2\nu + \delta - m \end{matrix} .$$

If we put

$$R_5R_4 \begin{pmatrix} Q_1P \\ Z_2 \end{pmatrix} = \begin{pmatrix} QP \\ Z \end{pmatrix},$$

then $Q = Q_1$ is an $m \times m$ nonsingular matrix of the required form.

This finishes the proof of case (ii).

(iii) $\gamma = \delta = 1$ and $\Gamma = 1$. Proceeding as in case (ii) and adopting the same notation as case (ii), we can assume that there is an $m \times m$ matrix Q_1 of the form (7.4) and a $(2\nu + \delta - m) \times (2\nu + \delta)$ matrix Z_2 such that

$$\begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix}$$

is nonsingular and

$$\begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix} G \, {}^t\!\begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix}$$

$$\equiv \begin{array}{c} \begin{matrix} 2s & m-2s-1 & 1 & m-2s-1 & \sigma_3 \end{matrix} \\ \begin{pmatrix} \Lambda & & & & \\ & 0 & & I & \\ & & 1 & & L_5 \\ & & & 0 & \\ & & & & D_3 \end{pmatrix} \end{array} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ \sigma_3 \end{matrix},$$

where $\sigma_3 = 2(\nu + s - m + 1) \geq 0$. Write

$$\begin{pmatrix} Q_1 P \\ Z_2 \end{pmatrix} = \begin{pmatrix} P_1 \\ v_2 \\ Z_{21} \\ Z_{22} \end{pmatrix} \begin{matrix} m-1 \\ 1 \\ m-2s-1 \\ \sigma_3 \end{matrix},$$

then

$$\begin{pmatrix} v_2 \\ Z_{22} \end{pmatrix} G \, {}^t\!\begin{pmatrix} v_2 \\ Z_{22} \end{pmatrix} \equiv \begin{pmatrix} 1 & L_5 \\ & D_3 \end{pmatrix}.$$

By Lemma 7.3 there exists a $(\sigma_3 + 1) \times (\sigma_3 + 1)$ nonsingular matrix Q_3 such that

$$Q_3 \begin{pmatrix} v_2 \\ Z_{22} \end{pmatrix} G \, {}^t\!\begin{pmatrix} v_2 \\ Z_{22} \end{pmatrix} {}^t Q_3 \equiv \begin{array}{c} \begin{matrix} 1 & & \end{matrix} \\ \begin{pmatrix} 1 & & \\ & 0 & I \\ & & 0 \end{pmatrix} \end{array} \begin{matrix} 1 \\ \nu + s - m + 1 \\ \nu + s - m + 1 \end{matrix}.$$

Write

$$Q_3 \begin{pmatrix} v_2 \\ Z_{22} \end{pmatrix} = \begin{pmatrix} v_3 \\ Z_{23} \end{pmatrix},$$

then

$$\begin{pmatrix} P_1 \\ v_3 \\ Z_{21} \\ Z_{23} \end{pmatrix} G \, {}^t\!\begin{pmatrix} P_1 \\ v_3 \\ Z_{21} \\ Z_{23} \end{pmatrix}$$

$$\equiv \begin{array}{c} \begin{matrix} 2s & m-2s-1 & 1 & m-2s-1 & \nu+s-m+1 & \nu+s-m+1 \end{matrix} \\ \begin{pmatrix} \Lambda & & & & & \\ & 0 & & I & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & 0 & I \\ & & & & & 0 \end{pmatrix} \end{array} \begin{matrix} 2s \\ m-2s-1 \\ 1 \\ m-2s-1 \\ \nu+s-m+1 \\ \nu+s-m+1 \end{matrix} .$$

Therefore $v_3 G^t v_3 = 1$ and $v_3(G + {}^tG) = 0$. Hence we must have $v_3 = e_{2\nu+1}$. Since $\Gamma = 1$, $v_3 \in P$. Write

$$Q_3 = \begin{array}{c} \begin{matrix} 1 & \sigma_2 \end{matrix} \\ \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{pmatrix} \begin{matrix} 1 \\ \sigma_2 \end{matrix} \end{array} .$$

From $v_3 \in P$ we deduce $Q_{12} = 0$. And from $v_2 G^t v_2 = v_3 G^t v_3 = 1$ we deduce $Q_{11} = 1$. Thus we can assume that

$$\begin{pmatrix} P_1 \\ v_3 \\ Z_{21} \\ Z_{23} \end{pmatrix} = \begin{pmatrix} I^{(m)} & & & \\ & 1 & & \\ & & I^{(m-2s)} & \\ & Q_{21} & & Q_{22} \end{pmatrix} \begin{pmatrix} P_1 \\ v_2 \\ Z_{21} \\ Z_{22} \end{pmatrix} = \begin{pmatrix} Q_1 P \\ Z \end{pmatrix},$$

where Q_1 is an $m \times m$ nonsingular matrix of the required form. This finishes the proof of case (iii). □

From Lemma 7.4 we deduce the condition for the existence of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ immediately.

Theorem 7.5. Subspaces of type $(m, 2s+\gamma, s, \Gamma)$ exist in the $(2\nu+\delta)$-dimensional orthogonal space $\mathbb{F}_q^{(2\nu+\delta)}$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ regular matrix (7.1) if and only if

$$2s+\gamma \le m \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, & \text{if } \delta \ne 1, \text{ or } \gamma \ne 1, \text{ or } \gamma=\delta=1 \\ & \text{and } \Gamma = 1, \\ \nu+s, & \text{if } \gamma=\delta=1 \text{ and } \Gamma = 0. \end{cases} \tag{7.5}$$

Proof. At first, we prove the necessity of condition (7.5). The left inequality $2s+\gamma \le m$ in (7.5) is trivial, and it is enough to prove the right one. We follow the proof of Lemma 7.4 and consider the three cases separately.

(i) $\gamma = 0$ or 2. Since the definite part of Σ_1 is of degree $|\delta - \gamma|$,

$$\sigma_1 = 2(\nu+s-m)+\delta+\gamma \ge |\delta-\gamma| .$$

From the above inequality we deduce immediately

$$m \leq \nu + s + \min\{\delta, \gamma\}.$$

(ii) $\gamma = 1$ and $\delta = 0$ or 2, or $\gamma = \delta = 1$ and $\Gamma = 0$. As in the proof of Lemma 7.4 we distinguish the cases when G_1 is definite or not. If G_1 is definite, then we must have $\sigma_2 = 0$ and $\delta = 2$. But $\sigma_2 = 2(\nu+s-m)+\delta = 0$, hence $\nu + s - m + 1 = 0$. Therefore

$$m = \nu + s + 1 = \nu + s + \min\{\delta, \gamma\}.$$

If G_1 is indefinite, then the definite part of Σ_2 is of degree δ. Therefore

$$\sigma_2 = 2(\nu + s - m) + \delta \geq \delta.$$

Hence

$$m \leq \nu + s = \begin{cases} \nu + s + \min\{\delta, \gamma\}, & \text{if } \delta = 0, \\ \nu + s, & \text{if } \delta = 1. \end{cases}$$

(iii) $\gamma = \delta = 1$ and $\Gamma = 1$. In this case, we have

$$\sigma_3 = 2(\nu + s - m + 1) \geq 0.$$

Therefore

$$m \leq \nu + s + 1 = \nu + s + \min\{\delta, \gamma\}.$$

This completes the proof of the necessity of condition (7.5).

Now let us come to the proof of the sufficiency of condition (7.5).

If $\gamma = 0$, then (7.5) reduces to $2s \leq m \leq \nu + s$. If we write $\sigma = \nu + s - m$, then

$$\begin{array}{ccccccc} s & m-2s & \sigma & s & m-2s & \sigma & \delta \\ \left(\begin{array}{ccccccc} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & I^{(m-2s)} & 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array} \tag{7.6}$$

is a subspace of type $(m, 2s, s)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. This settles the case $\gamma = 0$.

Now assume that $\gamma \geq 1$. We distinguish the cases $\delta = 0, 1$, or 2.

Let $\delta = 0$, then (7.5) reduces to $2s + \gamma \leq m \leq \nu + s$. For $\gamma = 1$, we have $2s + 1 \leq m \leq \nu + s$; if we write $\sigma = \nu + s - m$, then

$$\begin{array}{cccccccc} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma \\ \left(\begin{array}{cccccccc} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{array}\right) \end{array} \tag{7.7}$$

is a subspace of type $(m, 2s+1, s)$ in $\mathbb{F}_q^{(2\nu)}$. For $\gamma = 2$, we have $2s+2 \le m \le \nu + s$; if we write $\sigma = \nu + s - m$, then

$$\begin{array}{c} \begin{matrix} s & 2 & m-2s-2 & \sigma & s & 2 & m-2s-2 & \sigma \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix} & 0 & 0 & 0 & \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} & 0 & 0 \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{7.8}$$

is a subspace of type $(m, 2s+2, s)$ in $\mathbb{F}_q^{(2\nu)}$.

Let $\delta = 2$, then (7.5) reduces to $2s + \gamma \le m \le \nu + s + \gamma$. For $\gamma = 1$, we have $2s+1 \le m \le \nu + s + 1$; if we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 2 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & (x\ y) \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}, \tag{7.9}$$

where $\alpha x^2 + xy + \alpha y^2 = 1$, is a subspace of type $(m, 2s+1, s)$ in $\mathbb{F}_q^{(2\nu+2)}$. For $\gamma = 2$, we have $2s+2 \le m \le \nu + s + 2$; if we write $\sigma = \nu + s - m + 2$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-2 & \sigma & s & m-2s-2 & \sigma & 2 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I^{(2)} \\ 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{7.10}$$

is a subspace of type $(m, 2s+2, s)$ in $\mathbb{F}_q^{(2\nu+2)}$.

Let $\delta = 1$, then (7.5) reduces to

$$2s + \gamma \le m \le \begin{cases} \nu + s + 1, & \text{if } \gamma = 2, \text{ or } \gamma = 1 \text{ and } \Gamma = 1, \\ \nu + s, & \text{if } \gamma = 1 \text{ but } \Gamma = 0. \end{cases}$$

For $\gamma = 1$ and $\Gamma = 1$, we have $2s+1 \le m \le \nu + s + 1$; if we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{matrix} s & m-2s-1 & \sigma & s & m-2s-1 & \sigma & 1 \end{matrix} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{7.11}$$

is a subspace of type $(m, 2s+1, s, 1)$ in $\mathbb{F}_q^{(2\nu+1)}$. For $\gamma = 1$ and $\Gamma = 0$, we have $2s+1 \le m \le \nu + s$; if we write $\sigma = \nu + s - m$, then

$$\begin{array}{c} \begin{array}{ccccccccc} s & 1 & m-2s-1 & \sigma & s & 1 & m-2s-1 & \sigma & 1 \end{array} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & I^{(m-2s-1)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{7.12}$$

is a subspace of type $(m, 2s+1, s, 0)$ in $\mathbb{F}_q^{(2\nu+1)}$. For $\gamma = 2$, we have $2s+2 \le m \le \nu + s + 1$; if we write $\sigma = \nu + s - m + 1$, then

$$\begin{array}{c} \begin{array}{ccccccccc} s & 1 & m-2s-2 & \sigma & s & 1 & m-2s-2 & \sigma & 1 \end{array} \\ \begin{pmatrix} I^{(s)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I^{(s)} & 0 & 0 & 0 & 0 \\ 0 & \alpha^{1/2} & 0 & 0 & 0 & \alpha^{1/2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \alpha^{-1/2} & 0 & 0 & \alpha^{1/2} \\ 0 & 0 & I^{(m-2s-2)} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array} \tag{7.13}$$

is a subspace of type $(m, 2s+2, s)$ in $\mathbb{F}_q^{(2\nu+1)}$. □

We deduce from Lemma 7.4 also

Theorem 7.6. Let P_1 and P_2 be two m-dimensional subspaces of the vector space $\mathbb{F}_q^{(2\nu+\delta)}$, then there is a $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ such that $P_1 = AP_2T$, where A is an $m \times m$ nonsingular matrix, if and only if P_1 and P_2 are of the same type. In other words, $O_{2\nu+\delta}(\mathbb{F}_q)$ acts transitively on each set of subspaces of the same type.

Proof. Suppose that P_1 and P_2 are of the same type, then parallel to the proof of Theorem 3.7 we can prove that there is a $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ and an $A \in GL_m(\mathbb{F}_q)$ such that $P_1 = AP_2T$.

Conversely, suppose that there is a $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ and an $A \in GL_m(\mathbb{F}_q)$ such that $P_1 = AP_2T$. Then

$$P_1 G\, {}^tP_1 \equiv AP_2TG\, {}^tT\, {}^tP_2\, {}^tA \equiv A(P_2G\, {}^tP_2)\, {}^tA.$$

It follows that if $\delta \neq 1$ or $\gamma \neq 1$, then P_1 and P_2 are of the same type. If $\delta = \gamma = 1$, then by Theorem 7.1 for any $T \in O_{2\nu+1}(\mathbb{F}_q)$, we have $e_{2\nu+1}T = e_{2\nu+1}$. It follows that $e_{2\nu+1} \in P_1$ if and only if $e_{2\nu+1} \in P_2$. Therefore in this case P_1 and P_2 are also of the same type. □

Corollary 7.7. Totally singular subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ are of dimension $\leq \nu$, and there exists totally singular subspaces of dimension ν. Therefore maximal totally singular subspaces are of dimension ν. Moreover, any totally singular subspace is contained in a maximal totally singular subspace. □

Corollary 7.8. In $(2\nu+1)$-dimensional orthogonal space over $\mathbb{F}_q$, both subspaces of type $(m, 2s, s)$ and of type $(m+2, 2s+2, s)$ do not contain the vector $e_{2\nu+1}$.

Proof. By Theorem 7.1 we know that for any $T \in O_{2\nu+1}(\mathbb{F}_q)$, $e_{2\nu+1}T = e_{2\nu+1}$. Thus by Theorem 7.6 it is sufficient to prove that there exists a subspace of type $(m, 2s, s)$ and also a subspace of type $(m, 2s+2, s)$ in $\mathbb{F}_q^{(2\nu+1)}$ such that both of them do not contain $e_{2\nu+1}$. It is clear that (7.6) with $\delta = 1$ is a subspace of type $(m, 2s, s)$ in $\mathbb{F}_q^{(2\nu+1)}$ and does not contain $e_{2\nu+1}$, and that (7.13) is a subspace of type $(m, 2s+2, s)$ in $\mathbb{F}_q^{(2\nu+1)}$ which does not contain $e_{2\nu+1}$. □

From Theorems 7.5 and 7.6 we deduce

Theorem 7.9. The number of orbits of subspaces $\mathbb{F}_q^{(2\nu+\delta)}$ under $O_{2\nu+\delta}(\mathbb{F}_q)$ is equal to

$$\frac{1}{2}[3\nu^2 + (\delta+1)(3\nu+2) + \delta(2-\delta)(\nu^2+2\nu)].$$

Proof. By Theorem 7.6 the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ under $O_{2\nu+\delta}(\mathbb{F}_q)$ is equal to the number of subspace types $(m, 2s+\gamma, s, \Gamma)$ with respect to $G_{2\nu+\delta}$ in $\mathbb{F}_q^{(2\nu+\delta)}$. By Theorem 7.5, the parameters m, s, and γ of the subspace type must satisfy (7.5). Thus it is sufficient to enumerate the number of triples (m, s, γ) of non-negative integers which satisfy condition (7.5). We treat the three cases $\delta = 0, 1$, and 2 one by one.

(a) $\delta = 0$. In this case, (7.5) reduces to

$$2s + \gamma \leq m \leq \nu + s.$$

Proceeding as in case (a) in the proof of Theorem 6.8 and noticing that there are only subspace types $(m, 2s, s)$, $(m, 2s+1, s)$, and $(m, 2s+2, s)$, we obtain that the number of orbits of subspaces of $\mathbb{F}_q^{(2\nu)}$ is equal to

$$\frac{1}{2}(\nu^2+3\nu+2) + \frac{1}{2}(\nu^2+\nu) + \frac{1}{2}(\nu^2-\nu) = \frac{1}{2}(3\nu^2+3\nu+2).$$

(b) $\delta = 1$. In this case, (7.5) reduces to

$$2s + \gamma \leq m \leq \begin{cases} \nu + s + \min\{\delta, \gamma\}, & \text{if } \gamma \neq 1, \text{ or } \gamma = 1 \text{ and } \Gamma = 1, \\ \nu + s, & \text{if } \gamma = 1 \text{ but } \Gamma = 0. \end{cases}$$

Proceeding as in case (b) in the proof of Theorem 6.8, we obtain that the number of orbits of subspaces in $\mathbb{F}_q^{(2\nu+1)}$ is equal to

$$2\nu^2 + 4\nu + 2.$$

(c) $\delta = 2$. In this case, (7.5) reduces to

$$2s + \gamma \leq m \leq \nu + s + \gamma.$$

Proceeding as in case (c) in the proof of Theorem 6.8, we obtain that the number of orbits of subspaces in $\mathbb{F}_q^{(2\nu+2)}$ is equal to

$$3 \cdot \frac{1}{2}(\nu^2 + 3\nu + 2) = \frac{3}{2}(\nu^2 + 3\nu + 2).$$

Theorem 7.9 is now completely proved. □

Finally, parallel to Lemma 3.11, we can also deduce

Lemma 7.10. Let P_1 and P_2 be two $m \times (2\nu+\delta)$ matrices of rank m such that both P_1 and P_2 represent subspaces of type $(m, 2s+\gamma, s)$, where $\gamma = 0$ or 2. Then there exists an element $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ such that $P_1 = P_2 T$ if and only if $P_1 G\, {}^tP_1 \equiv P_2 G\, {}^tP_2$. □

7.2 Anzahl Theorems in Orthogonal Geometry over Finite Fields of Characteristic 2

We follow the notation of the previous section. Let

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta) = \mid \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \mid .$$

The goal of the present section is to compute $N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$. If condition (7.5) fails to hold, then $N(m+\gamma, 2s+\gamma, s, \Gamma; 2\nu+\delta) = 0$. Hence in the following we always assume that condition (7.5) holds.

Let P be a subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$, then we can choose a matrix representation P of the subspace P such that

$$P G\, {}^tP \equiv M(m, 2s+\gamma, s). \tag{7.14}$$

Denote by $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ the number of $m \times (2\nu+\delta)$ matrices P of rank m which are matrix representations of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ and satisfy (7.14). Parallel to Lemma 3.12 we have

Lemma 7.11. Let $(m, 2s+\gamma, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that (7.5) holds. Then

$$n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$$
$$=\mid O_{2s+\gamma}(\mathbb{F}_q) \mid\mid GL_{m-2s-\gamma}(\mathbb{F}_q) \mid q^{(2s+\gamma)(m-2s-\gamma)} N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta).$$

Proof. Same as Lemma 3.12. □

Clearly,

$$\mid O_{2\nu+\delta}(\mathbb{F}_q) \mid = n(2\nu+\delta, 2\nu+\delta, \nu, \Gamma; 2\nu+\delta),$$

where $\Gamma = 1$ if $\delta = 1$. Thus to compute $N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ it is sufficient to compute $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$.

Now let $1 \le r_1, r_2, r_1+r_2 \le m$. Consider first the cases $\gamma \ne 1$ or $\delta \ne 1$. Let $n(r_1)$ be the number of $r_1 \times (2\nu+\delta)$ matrices P_1 of rank r_1 such that $P_1 G^t P_1 \equiv M(m, 2s+\gamma, s)_{r_1}$. For a fixed $r_1 \times (2\nu+\delta)$ matrix P_1 of rank r_1 such that $P_1 G^t P_1 \equiv M(m, 2s+\gamma, s)_{r_1}$, let $n(P_1, r_2)$ be the number of $r_2 \times (2\nu+\delta)$ matrices P_2 such that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank r_1+r_2 and $PG^tP \equiv M(m, 2s+\gamma, s)_{r_1+r_2}$. Then we have

Lemma 7.12. Let $(m, 2s+\gamma, s)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ with either $\gamma \ne 1$ or $\delta \ne 1$, and assume that (7.5) holds. Let $0 \le r_1, r_2, r_1+r_2 \le m$. Then $n(P_1, r_2)$ is independent of the choice of P_1 and

$$n(r_1+r_2) = n(r_1) n(P_1, r_2)$$

for any choice of P_1. □

Next consider the case $\gamma = \delta = 1$, but assume that $M(m, 2s+1, s)_{r_1}$ is without defect. Let $n_\Gamma(r_1+r_2)$, where $\Gamma = 1$ or 0, be the number of $(r_1+r_2) \times (2\nu+1)$ matrices P of rank r_1+r_2 such that the subspace P contains or does not contain the vector $e_{2\nu+1}$, respectively, and that $PG^tP \equiv M(m, 2s+1, s)_{r_1+r_2}$. For a fixed $r_1 \times (2\nu+1)$ matrix P_1 of rank r_1 such that $P_1 G^t P_1 \equiv M(m, 2s+1, s)_{r_1}$, let $n_\Gamma(P_1, r_2)$, where $\Gamma = 1$ or 0, be the number of $r_2 \times (2\nu+\delta)$ matrices P_2 such that the subspace P_2 contains or does not contain the vector $e_{2\nu+1}$, respectively, that

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$$

is of rank r_1+r_2, and that $PG^tP \equiv M(m, 2s+1, s)_{r_1+r_2}$. Then we have

Lemma 7.13. Let $(m, 2s+1, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+1)}$ and assume that (7.5) holds. Let $0 \le r_1, r_2, r_1 + r_2 \le n$ and assume that the matrix $M(m, 2s+1, s)_{r_1}$ is without defect. Then $n_\Gamma(P_1, r_2)$, where $\Gamma = 1$ or 0, is independent of the choice of P_1 and

$$n_\Gamma(r_1 + r_2) = n(r_1) n_\Gamma(P_1, r_2)$$

for any choice of P_1. □

Both Lemmas 7.12 and 7.13 can be proved in the same way as Lemma 3.13, and their proofs will be omitted.

Similar to Lemma 6.11 we have

Lemma 7.14. Let $(m, 2s+\gamma, s, \Gamma)$ be a given type of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and assume that (7.5) holds. Then

$$\begin{aligned} &n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \\ &\quad = n(2s, 2s, s; 2\nu+\delta)\, n(m-2s, \gamma, 0, \Gamma; 2(\nu-s)+\delta). \end{aligned} \tag{7.15}$$

If $\gamma = 2$, we also have

$$\begin{aligned} n(m, 2s+2, s; 2\nu+\delta) &= n(2s+2, 2s+2, s; 2\nu+\delta) \\ &\times n(m-2s-2, 0, 0; 2(\nu-s-2+\delta)+2-\delta). \end{aligned} \tag{7.16}$$

□

The proof of Lemma 7.14 relies on Lemma 7.3 and is the same as the proof of Lemma 6.11.

After the foregoing preparation we can now come to the computation of $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$. As in Section 6.2 we compute $n(m, 2s, s; 2\nu+\delta)$ first. Parallel to Lemmas 6.12, 6.13, and 6.14, we can prove

Lemma 7.15. Let $s \le \nu$, then

$$n(s, 0, 0; 2\nu+\delta) = q^{\frac{s(s-1)}{2}} \prod_{i=\nu-s+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1). \tag{7.17}$$

□

Lemma 7.16. Let $s \le \nu$, then

$$\begin{aligned} &n(2s, 2s, s; 2\nu+\delta) \\ &\quad = q^{2s(\nu-s)+\delta s+s(s-1)} \prod_{i=\nu-s+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1). \end{aligned} \tag{7.18}$$

□

Lemma 7.17. Let $2s \le m \le \nu + s$, then

$$n(m, 2s, s; 2\nu + \delta) = q^{2s(\nu-s)+\delta s+s(s-1)+\frac{(m-2s)(m-2s-1)}{2}} \prod_{i=\nu+s-m+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1). \quad (7.19)$$

□

The formulas and proofs of these three lemmas are the same as the formulas and proofs of the corresponding lemmas for the odd characteristic case.

Then we are going to compute $n(m, 2s+1, s, \Gamma; 2\nu + \delta)$.

Lemma 7.18. Let

$$\nu \ge \begin{cases} 1, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma = 0, \\ 0, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = 1. \end{cases}$$

Then

$$n(1, 1, 0, \Gamma; 2\nu + \delta) = \begin{cases} q^{\nu-1}(q^\nu - 1), & \text{if } \delta = 0, \\ q^{2\nu} - 1, & \text{if } \delta = 1 \text{ and } \Gamma = 0, \\ 1, & \text{if } \delta = 1 \text{ and } \Gamma = 1, \\ q^\nu(q^{\nu+1} + 1), & \text{if } \delta = 2. \end{cases} \quad (7.20)$$

Proof. The cases $\delta = 0$ or $\delta = 2$ can be proved in the same way as Lemma 6.15 and will not be repeated here. We consider only the case $\delta = 1$.

$n(1, 1, 0, 0; 2\nu + 1)$ is the number of vectors $x = (x_1, x_2, \ldots, x_{2\nu+1})$ with $(x_1, x_2, \ldots, x_{2\nu}) \ne (0, 0, \ldots, 0)$ such that $xG^t x = 1$, i.e., the number of solutions of the equation

$$\sum_{i=1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 = 1 \quad (7.21)$$

with $(x_1, x_2, \ldots, x_{2\nu}) \ne (0, 0, \ldots, 0)$. Clearly the number of solutions of (7.21) with $(x_1, x_2, \ldots, x_\nu) \ne (0, 0, \ldots, 0)$ is $q^\nu(q^\nu - 1)$, and the number of solutions of (7.21) with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$ but $(x_1, x_2, \ldots, x_{2\nu})$ $\ne (0, 0, \ldots, 0)$ is $q^\nu - 1$. Therefore

$$n(1, 1, 0, 0; 2\nu + 1) = q^\nu(q^\nu - 1) + (q^\nu - 1) = q^{2\nu} - 1.$$

Moreover, we have obviously

$$n(1,1,0,1;2\nu+1)=1.$$

□

Lemma 7.19. Let

$$m \le \begin{cases} \nu-1, & \text{if } \delta=0, \text{ or } \delta=1 \text{ and } \Gamma=0, \\ \nu, & \text{if } \delta=2, \text{ or } \delta=1 \text{ and } \Gamma=1. \end{cases}$$

Then

$$\begin{aligned} &n(m,1,0,\Gamma;2\nu+\delta) \\ &\quad = n(m-1,0,0;2\nu+\delta)\; n(1,1,0,\Gamma;2(\nu-m+1)+\delta)q^{m-1}, \end{aligned} \tag{7.22}$$

where $n(m-1,0,0;2\nu+\delta)$ and $n(1,1,0,\Gamma;2(\nu-m+1)+\delta)$ are given by (7.17) and (7.20), respectively.

Proof. $n(m,1,0,\Gamma;2\nu+\delta)$ is the number of $m\times(2\nu+\delta)$ matrices P of rank m such that

$$PG\,{}^tP \equiv M(m,1,0)$$

and when $\delta=1$, it is further required that P contains the vector $e_{2\nu+1}$ or not corresponding to the cases $\Gamma=1$ or 0, respectively. Write

$$P=\begin{pmatrix} P_1 \\ v \end{pmatrix} \begin{matrix} m-1 \\ 1 \end{matrix},$$

then

$$\begin{aligned} &P_1 G\,{}^tP_1 \equiv (m-1,0,0), \\ &P_1(G+{}^tG)\,{}^tv=0, \end{aligned}$$

and

$$vG\,{}^tv=1.$$

Therefore the number of possible choices of P_1 is $n(m-1,0,0;2\nu+\delta)$. By Lemmas 7.12 or 7.13, once P_1 is chosen, the number of possible choices of v is independent of the particular choice of P_1 and

$$n(m,1,0,\Gamma;2\nu+\delta)=n(m-1,0,0;2\nu+\delta)n_\Gamma(P_1,1), \tag{7.23}$$

where $\Gamma = 0$ or 1 if $\delta = 1$, and Γ disappears if $\delta \neq 1$. Thus we may choose

$$P_1 = (I^{(m-1)},\ O^{(m-1,2\nu+\delta-m-1)})$$

and then compute $n_\Gamma(P_1, 1)$, i.e., the number of possible choices of v. From $P_1(G + {}^tG)\,{}^tv = 0$ we deduce that v is of the form

$$v = (x_1, x_2, \ldots, x_\nu, \underbrace{0, \ldots, 0}_{m-1}, x_{\nu+m}, \ldots, x_{2\nu+\delta}),$$

and the condition $vG\,{}^tv = 1$ becomes

$$\begin{aligned}
&\sum_{i=m}^{\nu} x_i x_{\nu+i} = 1, && \text{if } \delta = 0,\\
&\sum_{i=m}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 = 1, && \text{if } \delta = 1,\\
&\sum_{i=m}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 = 1, && \text{if } \delta = 2.
\end{aligned}$$

Hence $x_1, x_2, \ldots, x_{m-1}$ can be chosen arbitrarily in $\mathbb{F}_q$ and $x_m, \ldots, x_\nu, x_{\nu+m}, \ldots, x_{2\nu+\delta}$ must satisfy the above equations. Therefore the number of possible choices of v is

$$q^{m-1} n(1, 1, 0, \Gamma; 2(\nu - m + 1) + \delta).$$

Substituting this number for $n_\Gamma(P_1, 1)$ in (7.23), we obtain (7.22). □

Lemma 7.20. Let

$$2s + 1 \leq m \leq \begin{cases} \nu + s, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma = 0,\\ \nu + s + 1, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = 1. \end{cases}$$

Then

$$\begin{aligned}
&n(m, 2s + 1, s, \Gamma; 2\nu + \delta)\\
&\quad = n(2s, 2s, s; 2\nu + \delta)\ n(m - 2s, 1, 0, \Gamma; 2(\nu - s) + \delta),
\end{aligned} \tag{7.24}$$

where $n(2s, 2s, s; 2\nu + \delta)$ and $n(m - 2s, 1, 0, \Gamma; 2(\nu - s) + \delta)$ are given by (7.18) and (7.22), respectively.

Proof. This is a special case of the first assertion of Lemma 7.14. Setting $\gamma = 1$ in (7.15), we get (7.24). □

Finally let us compute $n(m, 2s + 2, s; 2\nu + \delta)$.

Lemma 7.21. Let

$$\nu \geq 2 - \delta.$$

Then

$$n(2,2,0;2\nu+\delta) = \begin{cases} q^{2\nu-2}(q^{\nu-1}-1)(q^{\nu}-1), & \text{if } \delta = 0, \\ q^{2\nu-1}(q^{\nu}-1)(q^{\nu}+1), & \text{if } \delta = 1, \\ q^{2\nu}(q^{\nu}+1)(q^{\nu+1}+1), & \text{if } \delta = 2. \end{cases} \tag{7.25}$$

Proof. $n(2,2,0;2\nu+\delta)$ is the number of $2 \times (2\nu+\delta)$ matrices P of rank 2, satisfying

$$PG\,{}^tP \equiv \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}.$$

Write

$$P = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix},$$

where v_1 and v_2 are the two row vectors of P. Then

$$\begin{aligned} &v_1 G\,{}^t v_1 = \alpha, \\ &v_1 (G + {}^tG)\,{}^t v_2 = 1, \\ &v_2 G\,{}^t v_2 = \alpha. \end{aligned}$$

Since $v_1(G + {}^tG)\,{}^tv_2 = 1$, v_1 cannot be a multiple of $e_{2\nu+1}$ if $\delta = 1$. Let $\alpha^{1/2} \in \mathbb{F}_q$ be such that $(\alpha^{1/2})^2 = \alpha$. Then the number of possible choices of v_1 such that $v_1 G\,{}^t v_1 = \alpha$ and $v_1 \notin \langle e_{2\nu+1} \rangle$ if $\delta = 1$ is equal to the number of possible choices of $\alpha^{1/2} v_1$ such that $(\alpha^{1/2}v_1)G\,{}^t(\alpha^{1/2}v_1) = 1$ and $\alpha^{1/2}v_1 \notin \langle e_{2\nu+1} \rangle$; clearly the latter number is $n(1,1,0,0;2\nu+\delta)$.

Next we assert that once v_1 is chosen such that $v_1 G\,{}^t v_1 = \alpha$, the number of possible choices of v_2 such that $v_1(G + {}^tG)\,{}^tv_2 = 1$ and $v_2 G\,{}^t v_2 = \alpha$ is independent of the choice of v_1. In fact, let u_1 be another $(2\nu+\delta)$-dimensional vector such that $u_1 G\,{}^t u_1 = \alpha$ and $u_1 \notin \langle e_{2\nu+1} \rangle$ if $\delta = 1$. Then by case (ii) of Lemma 7.4 there exist two $(2\nu+\delta-1) \times (2\nu+\delta)$ matrices Z_1 and Z_2 such that

$$\begin{pmatrix} v_1 \\ Z_1 \end{pmatrix} G\,{}^t\begin{pmatrix} v_1 \\ Z_1 \end{pmatrix} \equiv \begin{pmatrix} u_1 \\ Z_2 \end{pmatrix} G\,{}^t\begin{pmatrix} u_1 \\ Z_2 \end{pmatrix} \equiv \begin{pmatrix} \alpha & 1 \\ & \alpha \end{pmatrix}, \text{ if } \delta = 2 \text{ and } \nu = 0,$$

or

$$\begin{pmatrix} \alpha^{1/2}v_1 \\ Z_1 \end{pmatrix} G \, {}^t\!\begin{pmatrix} \alpha^{1/2}v_1 \\ Z_1 \end{pmatrix} \equiv \begin{pmatrix} \alpha^{1/2}u_1 \\ Z_2 \end{pmatrix} G \, {}^t\!\begin{pmatrix} \alpha^{1/2}u_1 \\ Z_2 \end{pmatrix} \equiv \begin{pmatrix} 1 & 1 & \\ & 0 & \\ & & \Sigma_2 \end{pmatrix},$$

if $\delta \neq 2$ or $\nu > 0$.

Let

$$T = \begin{cases} \begin{pmatrix} u_1 \\ Z_2 \end{pmatrix}^{-1} \begin{pmatrix} v_1 \\ Z_1 \end{pmatrix}, & \text{if } \delta = 2 \text{ and } \nu = 0, \\ \begin{pmatrix} \alpha^{1/2}u_1 \\ Z_2 \end{pmatrix}^{-1} \begin{pmatrix} \alpha^{1/2}v_1 \\ Z_1 \end{pmatrix}, & \text{if } \delta \neq 2 \text{ or } \nu > 0, \end{cases}$$

then $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ and $v_1 = u_1 T$. Thus v_2 is a vector in $\mathbb{F}_q^{(2\nu+\delta)}$ such that $v_1(G + {}^tG)\,{}^tv_2 = 1$ and $v_2 G\,{}^tv_2 = \alpha$, if and only if v_2T^{-1} is a vector in $\mathbb{F}_q^{(2\nu+\delta)}$ such that $u_1(G + {}^tG)\,{}^t(v_2T^{-1}) = 1$ and $(v_2T^{-1})G\,{}^t(v_2T^{-1}) = \alpha$. This proves our assertion.

If $\delta = 0$ or 1, then we choose $v_1 = e_1 + \alpha e_{\nu+1}$; and if $\delta = 2$, then we choose $v_1 = e_{2\nu+1}$. We have to compute the number of possible choices of v_2. Let $v_2 = (x_1, x_2, \ldots, x_{2\nu+\delta})$. From $v_1(G + {}^tG)\,{}^tv_2 = 1$ we deduce that $x_{\nu+1} = 1 + \alpha x_1$ if $\delta = 0$ or 1, and $x_{2\nu+2} = 1$ if $\delta = 2$. From $v_2 G\,{}^tv_2 = \alpha$ we deduce

$$\left.\begin{array}{ll} \displaystyle\sum_{i=2}^{\nu} x_i x_{\nu+i} + \alpha x_1^2 + x_1 = \alpha, & \text{if } \delta = 0, \\ \displaystyle\sum_{i=2}^{\nu} x_i x_{\nu+i} + \alpha x_1^2 + x_1 + x_{2\nu+1}^2 = \alpha, & \text{if } \delta = 1, \\ \displaystyle\sum_{i=1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1} + \alpha = \alpha, & \text{if } \delta = 2. \end{array}\right\} \tag{7.26}$$

Thus the number of possible choices of v_2 is equal to the number of solutions of (7.26). But, the number of solutions of (7.26) with $(x_2, x_3, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$ if $\delta = 0$ or 1 and with $(x_1, x_2, \ldots, x_\nu) \neq (0, 0, \ldots, 0)$ if $\delta = 2$ is

$$\begin{array}{ll} q^{\nu-1}(q^{\nu-1} - 1), & \text{if } \delta = 0, \\ q^{\nu}(q^{\nu-1} - 1), & \text{if } \delta = 1, \\ q^{\nu}(q^{\nu} - 1), & \text{if } \delta = 2, \end{array}$$

respectively. And the number of solutions of (7.26) with $(x_2, x_3, \ldots, x_\nu) = (0, 0, \ldots, 0)$ if $\delta = 0$ or 1 and with $(x_1, x_2, \ldots, x_\nu) = (0, 0, \ldots, 0)$ if $\delta = 2$

is

$$\begin{array}{ll} 0, & \text{if } \delta = 0, \\ q^{\nu}, & \text{if } \delta = 1, \\ 2q^{\nu}, & \text{if } \delta = 2, \end{array}$$

respectively. Summing up, we obtain that once v_1 is chosen, the number of possible choices of v_2 is

$$\begin{array}{ll} q^{\nu-1}(q^{\nu-1}-1), & \text{if } \delta = 0, \\ q^{2\nu-1}, & \text{if } \delta = 1, \\ q^{\nu}(q^{\nu}+1), & \text{if } \delta = 2, \end{array}$$

respectively.

Therefore formula (7.25) follows from (7.20) and the above computation immediately. □

Lemma 7.22. Let

$$\nu \geq s + (2 - \delta).$$

Then

$$\begin{aligned} &n(2s+2, 2s+2, s; 2\nu+\delta) \\ &\quad = n(2s, 2s, s; 2\nu+\delta)\, n(2, 2, 0; 2(\nu-s)+\delta), \end{aligned} \tag{7.27}$$

where $n(2s, 2s, s; 2\nu+\delta)$ and $n(2, 2, 0; 2(\nu-s)+\delta)$ are given by (7.18) and (7.25), respectively.

Proof. This is a special case of the first assertion in Lemma 7.14. Setting $m = 2s+2$ and $\gamma = 2$ in (7.15), we obtain (7.27). □

We recall that by Lemma 7.14, if

$$2s + 2 \leq m \leq \nu + s + \delta,$$

then we have (7.16)

$$\begin{aligned} &n(m, 2s+2, s; 2\nu+\delta) = n(2s+2, 2s+2, s; 2\nu+\delta)\, n(m-2s-2, 0, 0; \\ &\quad 2(\nu-s-2+\delta)+(2-\delta)). \end{aligned}$$

We know that $n(2s+2, 2s+2, s; 2\nu+\delta)$ and $n(m-2s-2, 0, 0; 2(\nu-s-2+\delta)+(2-\delta))$ are given by (7.27) and (7.17), respectively.

Therefore the computation of $n(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ is now completed.

From Lemmas 7.16, 7.20, and 7.22 we deduce

Theorem 7.23.

$$| O_{2\nu+\delta}(\mathbb{F}_q) | = \begin{cases} q^{\nu(\nu+\delta-1)} \prod_{i=1}^{\nu}(q^i-1) \prod_{i=0}^{\nu+\delta-1}(q^i+1), & \text{if } \delta = 0 \text{ or } 2, \\ q^{\nu^2} \prod_{i=1}^{\nu}(q^{2i}-1), & \text{if } \delta = 1. \end{cases}$$

Proof. In fact,

$$| O_{2\nu+\delta}(\mathbb{F}_q) | = n(2\nu+\delta, 2\nu+\delta, \nu, \Gamma; 2\nu+\delta),$$

where $\Gamma = 1$ if $\delta = 1$. □

From Lemmas 7.11, 7.17, 7.20, 7.14, 7.22 and Theorems 1.5, 7.23, we deduce the following three Anzahl theorems.

Theorem 7.24. The number of subspaces of hyperbolic type $(m, 2s, s)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where $2s \le m \le \nu+s$, is given by

$$N(m, 2s, s; 2\nu+\delta) = q^{2s(\nu+s-m)+\delta s} \frac{\prod_{i=\nu+s-m+1}^{\nu}(q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{s}(q^i-1)\prod_{i=0}^{s-1}(q^i+1)\prod_{i=1}^{m-2s}(q^i-1)}.$$

□

Corollary 7.25. The number of m-dimensional totally isotropic subspaces in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where $m \le \nu$, is given by

$$N(m, 0, 0; 2\nu+\delta) = \frac{\prod_{i=\nu-m+1}^{\nu}(q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{m}(q^i-1)}.$$

□

Theorem 7.26. The number of subspaces of type $(m, 2s+1, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+1 \le m \le \begin{cases} \nu+s, & \text{if } \delta = 0, \text{ or } \delta = 1 \text{ and } \Gamma = 0, \\ \nu+s+1, & \text{if } \delta = 2, \text{ or } \delta = 1 \text{ and } \Gamma = 1 \end{cases}$$

is given by

$$\begin{aligned} &N(m, 2s+1, s, \Gamma; 2\nu+\delta) \\ &= q^{2s(\nu+s-m)+s(\delta+1)-(m-2s-1)} \frac{\displaystyle\prod_{i=\nu+s-m+2}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{2\displaystyle\prod_{i=1}^{s}(q^{2i}-1)\prod_{i=1}^{m-2s-1}(q^i-1)} \\ &\quad \times n_0(m, 2s+1, s, \Gamma; 2\nu+\delta), \end{aligned}$$

where

$$\begin{aligned} &n_0(m, 2s+1, s, \Gamma; 2\nu+\delta) \\ &= \begin{cases} 2q^{\nu-s-1}(q^{\nu+s-m+1}-1), & \text{if } \delta = 0, \\ 2q^{m-2s-1}(q^{2(\nu+s-m+1)}-1), & \text{if } \delta = 1,\ \Gamma = 0, \\ 2q^{m-2s-1}, & \text{if } \delta = 1,\ \Gamma = 1, \\ 2q^{\nu-s}(q^{\nu+s-m+2}+1), & \text{if } \delta = 2. \end{cases} \end{aligned}$$

□

Theorem 7.27. The number of subspaces of elliptic type $(m, 2s+2, s)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+2 \le m \le \nu+s+\delta,$$

is given by

$$\begin{aligned} &N(m, 2s+2, s; 2\nu+\delta) \\ &= q^{2s(\nu+s-m)+s(\delta+2)-2(m-2s-2)} \frac{\displaystyle\prod_{i=\nu+s-m+3}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{\displaystyle\prod_{i=1}^{s}(q^i-1)\prod_{i=0}^{s+1}(q^i+1)\prod_{i=1}^{m-2s-2}(q^i-1)} \\ &\quad \times n_0(m, 2s+2, s; 2\nu+\delta), \end{aligned}$$

where

$$n_0(m, 2s+2, s; 2\nu+\delta) = \begin{cases} q^{2(\nu-s)-2}(q^{\nu+s-m+1}-1)(q^{\nu+s-m+2}-1), & \text{if } \delta = 0, \\ q^{2(\nu-s)-1}(q^{\nu+s-m+2}-1)(q^{\nu+s-m+2}+1), & \text{if } \delta = 1, \\ q^{2(\nu-s)}(q^{\nu+s-m+2}+1)(q^{\nu+s-m+3}+1), & \text{if } \delta = 2. \end{cases}$$

□

The above three Anzahl theorems can be combined into a single one as follows.

Theorem 7.28. The number of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, where

$$2s+\gamma \le m \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, & \text{if } \delta \ne 1, \text{ or } \gamma \ne 1, \text{ or } \gamma = \delta = 1 \text{ and } \Gamma = 1, \\ \nu+s+\gamma-1, & \text{if } \delta = \gamma = 1 \text{ and } \Gamma = 0, \end{cases}$$

is given by

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta) = q^{2s(\nu+s-m)+s(\delta+\gamma)-\gamma(m-2s-\gamma)} \frac{\prod_{i=\nu+s-m+\gamma+1}^{\nu} (q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{s}(q^i-1) \prod_{i=0}^{s+\gamma-1}(q^i+1) \prod_{i=1}^{m-2s-\gamma}(q^i-1)} \times n_0(m, 2s+\gamma, s, \Gamma; 2\nu+\delta), \tag{7.28}$$

where

$$n_0(m, 2s, s; 2\nu+\delta) = 1;$$

$$n_0(m, 2s+1, s, \Gamma; 2\nu+\delta) = \begin{cases} 2q^{\nu-s-1}(q^{\nu+s-m+1}-1), & \text{if } \delta = 0, \\ 2q^{m-2s-1}(q^{2(\nu+s-m+1)}-1), & \text{if } \delta = 1,\ \Gamma = 0, \\ 2q^{m-2s-1}, & \text{if } \delta = 1,\ \Gamma = 1, \\ 2q^{\nu-s}(q^{\nu+s-m+2}+1), & \text{if } \delta = 2; \end{cases}$$

$$n_0(m, 2s+2, s; 2\nu+\delta) = \begin{cases} q^{2(\nu-s)-2}(q^{\nu+s-m+1}-1)(q^{\nu+s-m+2}-1), & \text{if } \delta = 0, \\ q^{2(\nu-s)-1}(q^{\nu+s-m+2}-1)(q^{\nu+s-m+2}+1), & \text{if } \delta = 1, \\ q^{2(\nu-s)}(q^{\nu+s-m+2}+1)(q^{\nu+s-m+3}+1), & \text{if } \delta = 2. \end{cases}$$

□

7.3 Singular Orthogonal Geometry over Finite Fields of Characteristic 2

We follow the notations of the previous two sections. Let

$$G_l = \begin{pmatrix} G & \\ & 0^{(l)} \end{pmatrix},$$

where

$$G = M(2\nu+\delta, 2\nu+\delta, \nu),$$

and $\delta = 0, 1$, or 2. The set of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices T over $\mathbb{F}_q$ satisfying

$$TG_l\,{}^tT \equiv G_l$$

forms a group, which will be called the *singular orthogonal group* of degree $2\nu+\delta+l$ and rank $2\nu+\delta$ over $\mathbb{F}_q$ and denoted by $O_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$, or simply $O_{2\nu+\delta+l}(\mathbb{F}_q)$. Clearly $O_{2\nu+\delta+l}(\mathbb{F}_q)$ consists of all $(2\nu+\delta+l) \times (2\nu+\delta+l)$ nonsingular matrices of the form

$$T = \begin{array}{c} \begin{matrix} 2\nu+\delta & \quad l \end{matrix} \\ \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix} \begin{matrix} 2\nu+\delta \\ l \end{matrix} \end{array},$$

where $T_{11}G\,{}^tT_{11} \equiv G$ and T_{22} is nonsingular. Parallel to Theorem 3.21 we have

Theorem 7.29.

$$\begin{aligned} &| O_{2\nu+\delta+l}(\mathbb{F}_q) | \\ &= \begin{cases} q^{\nu(\nu+\delta-1)+l(2\nu+\delta)+\frac{l(l-1)}{2}} \prod_{i=1}^{\nu}(q^i-1) \prod_{i=0}^{\nu+\delta-1}(q^i+1) \prod_{i=1}^{l}(q^i-1), & \\ \qquad\qquad \text{if } \delta = 0 \text{ or } 2, & \\ q^{\nu^2+l(2\nu+1)+\frac{l(l-1)}{2}} \prod_{i=1}^{\nu}(q^{2i}-1) \prod_{i=1}^{l}(q^i-1), \quad \text{if } \delta = 1. \end{cases} \end{aligned}$$

□

Let $\mathbb{F}_q^{(2\nu+\delta+l)}$ be the $(2\nu+\delta+l)$-dimensional row vector space over $\mathbb{F}_q$. We have an action of the group $O_{2\nu+\delta+l}(\mathbb{F}_q)$ on the vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ defined as follows:

$$\begin{aligned} &\mathbb{F}_q^{(2\nu+\delta+l)} \times O_{2\nu+\delta+l}(\mathbb{F}_q) \to \mathbb{F}_q^{(2\nu+\delta+l)} \\ &((x_1, x_2, \ldots, x_{2\nu+\delta+l}), T) \mapsto (x_1, x_2, \ldots, x_{2\nu+\delta+l})T. \end{aligned}$$

The vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ together with this action is called the *singular orthogonal space* over $\mathbb{F}_q$. At first, we want to study how the subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$ are subdivided into orbits under $O_{2\nu+\delta+l}(\mathbb{F}_q)$.

An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is said to be of *type* $(m, 2s+\gamma, s)$ if $PG_l\,{}^tP$ is 'cogredient' to $M(m, 2s+\gamma, s)$. In case $\delta = 1$, the subspaces of type $(m, 2s+1, s)$ are further subdivided into subspaces of *type* $(m, 2s+1, s, 0)$ and of *type* $(m, 2s+1, s, 1)$ corresponding to the cases when P does not contain or contains a vector of the form

$$(\underbrace{0, \ldots, 0}_{2\nu}, 1, \underbrace{* \ldots *}_{l}), \tag{7.29}$$

respectively. Thus the type of a subspace can be written as $(m, 2s+\gamma, s, \Gamma)$, where $\Gamma = 0$ or 1 in case $\delta = \gamma = 1$ and Γ disappears in all other cases. Denote the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$ by $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l, 2\nu+\delta)$, or simply $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l)$ and let

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l) = \mid \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l) \mid .$$

For the ordinary orthogonal case, i.e., the case $l = 0$, $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l)$ is simply $\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ which was studied in the previous two sections.

Let E be the subspace of $\mathbb{F}_q^{(2\nu+\delta+l)}$ generated by $e_{2\nu+\delta+1}, e_{2\nu+\delta+2}, \ldots$, and $e_{2\nu+\delta+l}$, then $\dim E = l$. An m-dimensional subspace P of $\mathbb{F}_q^{(2\nu+\delta+l)}$ is said to be of *type* $(m, 2s+\gamma, s, \Gamma, k)$, if

(i) P is of type $(m, 2s+\gamma, s, \Gamma)$, and

(ii) $\dim(P \cap E) = k$.

Denote by $\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$ the set of subspaces of type $(m, 2s+\gamma, s, \Gamma, k)$ and let

$$N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l) = \mid \mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l) \mid .$$

Parallel to Theorems 3.22, 3.24 and Corollaries 3.23, 3.25 we have

Theorem 7.30. $\mathcal{M}(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$ is non-empty if and only if

$$k \leq l$$

and

$$2s+\gamma \leq m-k$$

$$\leq \begin{cases} \nu + s + \min\{\delta, \gamma\}, \\ \qquad \text{if } \delta \neq 1, \text{ or } \gamma \neq 1, \text{ or } \gamma = \delta = 1 \text{ and } \Gamma = 1, \\ \nu + s, \\ \qquad \text{if } \gamma = \delta = 1 \text{ and } \Gamma = 0. \end{cases} \tag{7.30}$$

and if and only if

$$\min\{l, m - 2s - \gamma\} \geq k$$
$$\geq \begin{cases} \max\{0, m - \nu - s - \min\{\delta, \gamma\}\}, \\ \qquad \text{if } \delta \neq 1, \text{ or } \gamma \neq 1, \text{ or } \gamma = \delta = 1 \text{ and } \Gamma = 1, \\ \max\{0, m - \nu - s\}, \\ \qquad \text{if } \gamma = \delta = 1 \text{ and } \Gamma = 0. \end{cases} \tag{7.31}$$

Moreover, if $\mathcal{M}(m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l)$ is non-empty, then it is an orbit of subspaces under $O_{2\nu+\delta+l}(\mathbb{F}_q)$. □

Corollary 7.31. $\mathcal{M}(m, 2s + \gamma, s, \Gamma; 2\nu + \delta + l)$ is non-empty if and only if

$$\min\{l, m - 2s - \gamma\}$$
$$\geq \begin{cases} \max\{0, m - \nu - s - \min\{\delta, \gamma\}\}, \\ \qquad \text{if } \delta \neq 1, \text{ or } \gamma \neq 1, \text{ or } \gamma = \delta = 1 \text{ and } \Gamma = 1, \\ \max\{0, m - \nu - s\}, \\ \qquad \text{if } \gamma = \delta = 1 \text{ and } \Gamma = 0. \end{cases} \tag{7.32}$$

Moreover, $\mathcal{M}(m, 2s + \gamma, s, \Gamma; 2\nu + \delta + l)$ is the disjoint union of all the orbits $\mathcal{M}(m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l)$, where k satisfies (7.31). □

It follows from Theorem 7.30 that $N(m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l, \Delta) = 0$, whenever (7.31) does not hold. However, if (7.31) holds, then parallel to Theorem 3.24 we have

Theorem 7.32. Suppose that (7.31) holds. Then

$$N(m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l)$$
$$= N(m - k, 2s + \gamma, s, \Gamma; 2\nu + \delta) q^{(m-k)(l-k)} \frac{\prod_{i=l-k+1}^{l} (q^i - 1)}{\prod_{i=1}^{k} (q^i - 1)}, \tag{7.33}$$

where $N(m - k, 2s + \gamma, s, \Gamma; 2\nu + \delta)$ is the number of subspaces of type $(m - k, 2s + \gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$ and is given by (7.28). □

Corollary 7.33. Suppose that (7.32) holds. Then

$$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta+l) = \sum_{k} N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l), \tag{7.34}$$

where $N(m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$ is given by (7.33) and the summation range of k is

$$[\max\{0, m-\nu-s-\min\{\delta,\gamma\}\},\ \min\{l, m-2s-\gamma\}],$$
$$\text{if } \delta \neq 1, \text{ or } \gamma \neq 1, \text{ or } \gamma = \delta = 1 \text{ and } \Gamma = 1,$$

or

$$[\max\{0, m-\nu-s\},\ \min\{l, m-2s-\gamma\}],$$
$$\text{if } \gamma = \delta = 1 \text{ and } \Gamma = 0.$$

□

Now we shall compute the number of subspaces of a given type contained in a given subspace in ordinary orthogonal space over a finite field of characteristic 2. We need to modify the above results in singular orthogonal space for the case $\delta = 1$ as follows.

Let

$$\mathcal{M}(m, 2s+1, s, k; 2\nu+1+l) = \bigcup_{\Gamma=0,1} \mathcal{M}(m, 2s+1, s, \Gamma, k; 2\nu+1+l)$$

and

$$N(m, 2s+1, s, k; 2\nu+1+l) = |\ \mathcal{M}(m, 2s+1, s, k; 2\nu+1+l)\ |\ .$$

Then

$$N(m, 2s+1, s, k; 2\nu+1+l) = \sum_{\Gamma=0,1} N(m, 2s+1, s, \Gamma, k; 2\nu+1+l). \tag{7.35}$$

Let

$$\mathcal{M}(m, 2s+1, s; 2\nu+1+l) = \bigcup_{\Gamma=0,1} \mathcal{M}(m, 2s+1, s, \Gamma; 2\nu+1+l)$$

and

$$N(m, 2s+1, s; 2\nu+1+l) = \mathcal{M}(m, 2s+1, s; 2\nu+1+l).$$

Then

$$\begin{aligned} &N(m, 2s+1, s; 2\nu+1+l) \\ &\quad = \sum_{\Gamma=0,1} N(m, 2s+1, s, \Gamma; 2\nu+1+l). \end{aligned}$$

From Corollaries 7.31 and 7.33 we deduce immediately

Corollary 7.34. $\mathcal{M}(m, 2s+1, s; 2\nu+1+l)$ is non-empty if and only if

$$\min\{l, m-2s-1\} \geq \max\{0, m-\nu-s-1\}. \tag{7.36}$$

□

Corollary 7.35. Suppose that (7.36) holds. Then

$$\begin{aligned} &N(m, 2s+1, s; 2\nu+1+l) \\ &\quad = \sum_{k} N(m, 2s+1, s, k; 2\nu+1+l), \end{aligned} \tag{7.37}$$

where $N(m, 2s+1, s, k; 2\nu+1+l)$ is given by (7.35), and the summation range of k is

$$[\max\{0, m-\nu-s-1\},\ \min\{l, m-2s-\gamma\}].$$

□

Now let $\mathbb{F}_q^{(2\nu+\delta)}$ be the $(2\nu+\delta)$-dimensional row vector space on which the ordinary orthogonal group $O_{2\nu+\delta}(\mathbb{F}_q)$ acts. Let P be a fixed subspace of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$. Denote by $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ the set of subspaces of type $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)$ contained in P. Let

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \\ &\quad = |\ \mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta)\ |\ . \end{aligned}$$

It is clear that $N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ is independent of the particular choice of the subspace P of type $(m, 2s+\gamma, s, \Gamma)$ in $\mathbb{F}_q^{(2\nu+\delta)}$.

At first, consider the following two cases:

$$\begin{array}{ll} \text{(i)} & \delta = 1;\ \ \gamma \neq 1;\ \gamma_1 = 1 \text{ and } \Gamma_1 = 1; \\ \text{(ii)} & \delta = 1;\ \ \gamma = 1 \text{ and } \Gamma = 0;\ \ \gamma_1 = 1 \text{ and } \Gamma_1 = 1. \end{array}$$

We have

Theorem 7.36.

$$N(m_1, 2s_1+1, s_1, 1; m, 2s+\gamma, s; 2\nu+1) = 0, \text{ when } \gamma \neq 1.$$
$$N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 0; 2\nu+1) = 0.$$

Proof. The first equality follows from Corollary 7.8. The second equality follows from the definition of the types $(m, 2s+1, s, 0)$ and $(m, 2s+1, s, 1)$ of subspaces when $\delta = 1$. □

Next, consider the case

$$\text{(iii)}\quad \delta = 1;\ \ \gamma = 1 \text{ and } \Gamma = 1;\ \gamma_1 = 1 \text{ and } \Gamma_1 = 1.$$

We have

Theorem 7.37. $\mathcal{M}(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1)$ is non-empty if and only if

$$\left.\begin{array}{r} 2s+1 \leq m \leq \nu+s+1, \\ 2s_1+1 \leq m_1 \leq \nu+s_1+1, \\ \\ m-m_1 \geq s-s_1 \geq 0, \end{array}\right\} \quad \text{and} \tag{7.38}$$

and these three conditions are equivalent to the following two conditions

$$\left.\begin{array}{c} 2s+1 \leq m \leq \nu+s+1 \\ \text{and} \\ \min\{m-2s-1, m_1-2s_1-1\} \geq \max\{0, m_1-s-s_1-1\}. \end{array}\right\} \tag{7.39}$$

Moreover, if (7.39) holds, then

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)), \end{aligned} \tag{7.40}$$

which is given by (7.34).

Proof. It is easy to verify that (7.38) and (7.39) are equivalent. It is also clear that we can establish a bijection from $\mathcal{M}(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1)$ to $\mathcal{M}(m_1-1, 2s_1, s_1; m-1, 2s, s; 2\nu)$, from which we deduce that the former set is non-empty if and only if the latter is non-empty and that

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1-1, 2s_1, s_1; m-1, 2s, s; 2\nu). \end{aligned}$$

Similar to the proof of Theorem 3.26 we can show that $\mathcal{M}(m_1-1, 2s_1, s_1; m-1, 2s, s; 2\nu)$ is non-empty if and only if (7.39) holds, and similar to the proof of Theorem 3.27 we can show that if (7.39) holds, then

$$\begin{aligned} &N(m_1-1, 2s_1, s_1; m-1, 2s, s; 2\nu) \\ &\quad = N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)). \end{aligned}$$

□

Hence if (7.39) holds, we have (7.40).

Then consider the case

$$\text{(iv)} \quad \delta = 1; \quad \gamma = 1 \text{ and } \Gamma = 1; \ \gamma_1 = 1 \text{ and } \Gamma_1 = 0.$$

Let

$$\begin{aligned} &\mathcal{M}(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = \bigcup_{\Gamma_1=0,1} \mathcal{M}(m_1, 2s_1+1, s_1, \Gamma_1; m, 2s+1, s, 1; 2\nu+1) \end{aligned}$$

and

$$\begin{aligned} &N(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = \mathcal{M}(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1), \end{aligned}$$

then

$$\begin{aligned} &N(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = \sum_{\Gamma_1=0,1} N(m_1, 2s_1+1, s_1, \Gamma_1; m, 2s+1, s, 1; 2\nu+1). \end{aligned}$$

We have

Theorem 7.38. $\mathcal{M}(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1)$ is empty if (7.39) does not hold. Moreover, if (7.39) holds, then

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1, 2s_1+1, s_1; 2s+1+(m-2s-1)) \\ &\qquad -N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)). \end{aligned} \tag{7.41}$$

Proof. We have

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = \sum_{\Gamma=0,1} N(m_1, 2s_1+1, s_1, \Gamma_1; m, 2s+1, s, 1;\ 2\nu+1) \\ &\qquad -N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \\ &\qquad -N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1). \end{aligned}$$

From Theorem 7.37 we know that

$$N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \neq 0$$

if and only if (7.39) holds. Similar to the proof of Theorem 3.26 we can prove also that

$$N(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \neq 0$$

if and only if (7.39) holds. Therefore, if (7.39) does not hold, then

$$N(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1) = 0,$$

i.e., $\mathcal{M}(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1)$ is empty.

Now suppose that (7.39) holds. It follows from Theorem 7.37 that

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)). \end{aligned}$$

Similar to the proof of Theorem 3.27 we can prove that

$$\begin{aligned} &N(m_1, 2s_1+1, s_1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1, 2s_1+1, s_1; 2s+1+(m-2s-1)). \end{aligned}$$

Therefore, if (7.39) holds, we have (7.41). □

Finally, let us consider the remaining cases. Parallel to Theorems 3.26 and 3.27 we have the following two theorems.

Theorem 7.39. Let $(m, 2s+\gamma, s, \Gamma)$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)$ be two given types of subspaces in $\mathbb{F}_q^{(2\nu+\delta)}$, and suppose that cases (i), (ii), (iii), and (iv) do not occur. Then $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ is non-empty if and only if

$$\left.\begin{aligned} &2s+\gamma \le m \\ &\quad \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \qquad \text{if } \delta\neq 1, \text{ or } \gamma\neq 1, \text{ or } \delta=\gamma=1 \text{ and } \Gamma=1, \\ \nu+s, \\ \qquad \text{if } \delta=\gamma=1 \text{ and } \Gamma=0, \end{cases} \\ &2s_1+\gamma_1 \le m_1 \\ &\quad \le \begin{cases} \nu+s_1+\min\{\delta,\gamma_1\}, \\ \qquad \text{if } \delta\neq 1, \text{ or } \gamma_1\neq 1, \text{ or } \delta=\gamma_1=1 \text{ and } \Gamma_1=1, \\ \nu+s_1, \\ \qquad \text{if } \delta=\gamma_1=1 \text{ and } \Gamma_1=0, \end{cases} \\ &\text{and} \\ &2m-2m_1 \ge (2s+\gamma)-(2s_1+\gamma_1)+\mid \gamma-\gamma_1 \mid \ge 2\mid \gamma-\gamma_1 \mid, \end{aligned}\right\} \tag{7.42}$$

and these three conditions are equivalent to the following two conditions

$$\left.\begin{aligned} &2s+\gamma \le m \\ &\quad \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \qquad \text{if } \delta\neq 1, \text{ or } \gamma\neq 1, \text{ or } \delta=\gamma=1 \text{ and } \Gamma=1, \\ \nu+s, \\ \qquad \text{if } \delta=\gamma=1 \text{ and } \Gamma=0, \end{cases} \\ &\text{and} \\ &\min\{m-2s-\gamma, m_1-2s_1-\gamma_1\} \\ &\quad \ge \max\{0, m_1-s-s_1-\min\{\gamma,\gamma_1\}\}. \end{aligned}\right\} \tag{7.43}$$

□

Theorem 7.40. Let $(m, 2s+\gamma, s, \Gamma)$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)$ be two given types of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$ and suppose that cases (i), (ii), (iii), and (iv) do not occur and that (7.43) holds. Then

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \\ &\quad = N(m_1, 2s_1+\gamma_1, s_1; 2s+\gamma+(m-2s-\gamma)), \end{aligned} \tag{7.44}$$

which is given by (7.37) or (7.34) corresponding to the case $\gamma=\gamma_1=1$ or not. □

Notice that when $\delta = \gamma = \gamma_1 = 1$ and $\Gamma = 1$, (7.43) reduces to (7.39). We can combine Theorems 7.36, 7.37, 7.38, 7.39, and 7.40 into a single Anzahl theorem as follows.

Theorem 7.41. Let $(m, 2s+\gamma, s, \Gamma)$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1)$ be two given types of subspaces of $\mathbb{F}_q^{(2\nu+\delta)}$. Then

(a) When $\delta = 1, \gamma \neq 1, \gamma_1 = 1$, and $\Gamma_1 = 1$; or when $\delta = \gamma = \gamma_1 = 1, \Gamma = 0$, and $\Gamma_1 = 1$; or when (7.43) does not hold,

$$N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta) = 0.$$

(b) When $\delta = \gamma = \gamma_1 = 1, \Gamma = \Gamma_1 = 1$, and (7.43) holds,

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)). \end{aligned}$$

(c) When $\delta = \gamma = \gamma_1 = 1, \Gamma = 1, \Gamma_0 = 0$, and (7.43) holds,

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 0; m, 2s+1, s, 1; 2\nu+1) \\ &\quad = N(m_1, 2s_1+1, s_1; 2s+1+(m-2s-1)) \\ &\qquad -N(m_1-1, 2s_1, s_1; 2s+(m-2s-1)). \end{aligned}$$

(d) For any case which is not one of the cases (i), (ii), (iii), and (iv), assume that (7.43) holds, then

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1; m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \\ &\quad = N(m_1, 2s_1+\gamma_1, s_1; 2s+\gamma+(m-2s-\gamma)). \end{aligned}$$

□

Now let us consider the similar problem in the singular orthogonal geometry. Consider again the $(2\nu+\delta+l)$-dimensional row vector space $\mathbb{F}_q^{(2\nu+\delta+l)}$ over $\mathbb{F}_q$, on which the singular orthogonal group $O_{2\nu+\delta+l}(\mathbb{F}_q)$ acts. Let P be a fixed subspace of type $(m, 2s+\gamma, s, \Gamma, k)$ in $\mathbb{F}_q^{(2\nu+\delta+l)}$. Denote by $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$ the set of subspaces of type $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1)$ contained in P and let

$$\begin{aligned} &N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l) \\ &\quad = \mid \mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l) \mid, \end{aligned}$$

which is independent of the particular choice of P. We want to study the condition for the non-emptiness of $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+$

$\gamma, s, \Gamma, k; 2\nu+\delta+l)$ and compute $N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$.

Similar to Corollary 7.8 we have

Lemma 7.42. Let $\delta = 1$. Then any subspace of type $(m, 2s, s, \Gamma, k)$ or of type $(m, 2s+2, s, \Gamma, k)$ cannot contain a vector of the form (7.29). □

We start with the following two cases.

(i) $\delta = 1$; $\gamma \neq 1$; $\gamma_1 = 1$ and $\Gamma_1 = 1$;
(ii) $\delta = 1$; $\gamma = 1$ and $\Gamma = 0$; $\gamma_1 = 1$ and $\Gamma_1 = 1$.

Parallel to Theorem 7.36 we have

Theorem 7.43.

$$N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+\gamma, s, k; 2\nu+1+l) = 0, \text{if } \gamma \neq 1.$$
$$N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 0, k; 2\nu+1+l) = 0.$$

Proof. The first equality follows from Lemma 7.42. The second equality follows from the definition of the types $(m, 2s+1, s, 0, k)$ and $(m, 2s+1, s, 1, k)$ of subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$. □

Next, consider the case

(iii) $\delta = 1$; $\gamma = 1$ and $\Gamma = 1$; $\gamma_1 = 1$ and $\Gamma_1 = 1$.

Parallel to Theorem 7.37 we have

Theorem 7.44. $\mathcal{M}(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)$ is non-empty if and only if

$$\left.\begin{aligned} k_1 \leq k \leq l, \\ 2s+1 \leq m-k \leq \nu+s+1, \\ 2s_1+1 \leq m_1-k_1 \leq \nu+s_1+1, \end{aligned}\right\} \tag{7.45}$$

and

$$\left.\begin{aligned} (m-k)-(m_1-k_1) \geq s-s_1 \geq 0, \end{aligned}\right\}$$

and these four conditions are equivalent to the following three conditions

$$\left.\begin{aligned} k_1 \leq k \leq l, \\ 2s+1 \leq m-k \leq \nu+s+1, \end{aligned}\right\} \tag{7.46}$$

and

$$\left.\begin{aligned} \max\{0, m_1-k_1-s-s_1-1\} \\ \leq \min\{m-k-2s-1,\ m_1-k_1-2s_1-1\}. \end{aligned}\right\}$$

Moreover, if (7.46) holds, then

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+\delta+l) \\ &\quad = N(m_1-k_1-1, 2s_1, s_1; 2s+(m-k-2s-1)) \\ &\qquad \times N(k_1, k) q^{(m_1-k_1-1)(k-k_1)}, \end{aligned} \tag{7.47}$$

where $N(m_1-k_1-1, 2s_1, s_1; 2s+(m-k-2s-1))$ and $N(k_1, k)$ are given by Corollary 7.33 and Theorem 1.7, respectively.

Proof. It is easy to verify that (7.45) and (7.46) are equivalent. It is also clear that we can establish a bijection from $\mathcal{M}(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+1+\delta)$ to $\mathcal{M}(m_1-1, 2s_1, s_1, k_1; m-1, 2s, s, k; 2\nu+1+\delta)$. It follows that the first set is non-empty if and only if the second one is non-empty and

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) \\ &\quad = N(m_1-1, 2s_1, s_1, k_1; m-1, 2s, s, k; 2\nu+l). \end{aligned} \tag{7.48}$$

Similar to the proof of Theorem 3.28 we can show that $\mathcal{M}(m_1-1, 2s_1, s_1, k_1; m-1, 2s, s, k; 2\nu+\delta)$ is non-empty if and only if (7.46) holds. Similar to the proof of Theorem 3.29 we can show that

$$\begin{aligned} &N(m_1-1, 2s_1, s_1, k_1; m-1, 2s, s, k; 2\nu+l) \\ &\quad = N(m_1-k_1-1, 2s_1, s_1; 2s+(m-k-2s-1)) \\ &\qquad \times N(k_1, k) q^{(m_1-k_1-1)(k-k_1)}. \end{aligned} \tag{7.49}$$

From (7.48) and (7.49) we deduce (7.47). □

Then consider the case

$$\text{(iv)} \quad \delta = 1; \quad \gamma = 1 \text{ and } \Gamma = 1; \quad \gamma_1 = 1 \text{ and } \Gamma_1 = 0.$$

Clearly, we have

$$\begin{aligned} &N(m_1, 2s_1+1, s_1, 0, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) \\ &\quad = \sum_{\Gamma_1=0,1} N(m_1, 2s_1+1, s_1, \Gamma_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) \\ &\qquad - N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+l). \end{aligned}$$

Let

$$\begin{aligned} &\mathcal{M}(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) \\ &\quad = \bigcup_{\Gamma_1=0,1} \mathcal{M}(m_1, 2s_1+1, s_1, \Gamma_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l), \end{aligned}$$

and

$$\begin{aligned}&N(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\quad =\mid \mathcal{M}(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l).\end{aligned}$$

Then

$$\begin{aligned}&N(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\quad = \sum_{\Gamma_1=0,1} N(m_1, 2s_1+1, s_1, \Gamma_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l).\end{aligned}$$

Therefore

$$\begin{aligned}&N(m_1, 2s_1+1, s_1, 0, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\quad = N(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\qquad -N(m_1, 2s_1+1, s_1, 1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l).\end{aligned}$$

Similar to the proof of Theorem 3.28 we can prove that if (7.46) does not hold, then

$$N(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) = 0.$$

Similar to the proof of Theorem 3.29 we can prove that if (7.46) holds, then

$$\begin{aligned}&N(m_1, 2s_1+1, s_1, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\quad = N(m_1-k_1, 2s_1+1, s_1; 2s+1+(m-k-2s-1))\\ &\qquad \times N(k_1, k) q^{(m_1-k_1)(k-k_1)}.\end{aligned}$$

From the above discussion we conclude

Theorem 7.45. If (7.46) does not hold, then we have

$$N(m_1, 2s_1+1, s_1, 0, k_1; m, 2s+1, s, 1, k; 2\nu+1+l) = 0.$$

If (7.46) holds, then

$$\begin{aligned}&N(m_1, 2s_1+1, s_1, 0, k_1; m, 2s+1, s, 1, k; 2\nu+1+l)\\ &\quad = N(m_1-k_1, 2s_1+1, s_1; 2s+1+(m-k-2s-1))\\ &\qquad \times N(k_1, k) q^{(m_1-k_1)(k-k_1)}\\ &\qquad -N(m_1-k_1-1, 2s_1, s_1; 2s+(m-k-2s-1))\\ &\qquad\quad \times N(k_1, k) q^{(m_1-k_1-1)(k-k_1)},\end{aligned} \tag{7.50}$$

where $N(m_1-k_1, 2s_1+1, s_1; 2s+1+(m-k-2s-1))$, $N(m_1-k_1-1, 2s_1, s_1; 2s+(m-k-2s-1))$, and $N(k_1, k)$ are given in Corollaries 7.35, 7.33, and Theorem 1.7, respectively. □

Finally let us consider the remaining cases. Parallel to Theorems 3.28 and 3.29 we have the following two theorems.

Theorem 7.46. Let $(m, 2s+\gamma, s, \Gamma, k)$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1)$ be two given types of subspaces in $\mathbb{F}_q^{(2\nu+\delta+l)}$, and assume that cases (i), (ii), (iii), and (iv) do not occur. Then $\mathcal{M}(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$ is non-empty if and only if

$$\left.\begin{array}{l} k_1 \le k \le l, \\ 2s+\gamma \le m-k \\ \qquad \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \qquad \text{if } \delta \ne 1, \text{ or } \gamma \ne 1, \text{ or } \delta=\gamma=1 \text{ and } \Gamma=1, \\ \nu+s, \\ \qquad \text{if } \delta=\gamma=1 \text{ and } \Gamma=0, \end{cases} \\ 2s_1+\gamma_1 \le m_1-k_1 \\ \qquad \le \begin{cases} \nu+s_1+\min\{\delta,\gamma_1\}, \\ \qquad \text{if } \delta \ne 1, \text{ or } \gamma_1 \ne 1, \text{ or } \delta=\gamma_1=1 \text{ and } \Gamma_1=1, \\ \nu+s_1, \\ \qquad \text{if } \delta=\gamma_1=1 \text{ and } \Gamma_1=0, \end{cases} \\ \text{and} \\ \quad 2(m-k)-2(m_1-k_1) \\ \qquad \ge (2s+\gamma)-(2s_1+\gamma_1)+\mid \gamma-\gamma_1 \mid \ge 2 \mid \gamma-\gamma_1 \mid, \end{array}\right\} \tag{7.51}$$

and these four conditions are equivalent to the following three conditions

$$\left.\begin{array}{l} k_1 \le k \le l, \\ 2s+\gamma \le m-k \\ \qquad \le \begin{cases} \nu+s+\min\{\delta,\gamma\}, \\ \qquad \text{if } \delta \ne 1, \text{ or } \gamma \ne 1, \text{ or } \delta=\gamma=1 \text{ and } \Gamma=1, \\ \nu+s, \\ \qquad \text{if } \delta=\gamma=1 \text{ and } \Gamma=0, \end{cases} \\ \text{and} \\ \quad \max\{0, m_1-k_1-s-s_1-\min\{\gamma,\gamma_1\}\} \\ \qquad \le \min\{m-k-2s-\gamma, m_1-k_1-2s_1-\gamma_1\}. \end{array}\right\} \tag{7.52}$$

□

Theorem 7.47. Let $(m, 2s+\gamma, s, \Gamma, k)$ and $(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1)$ be two given types of subspaces of $\mathbb{F}_q^{(2\nu+\delta+l)}$, and assume that cases (i), (ii), (iii), and (iv) do not occur. Assume further that (7.52) holds. Then

$$N(m_1, 2s_1+\gamma_1, s_1, \Gamma_1, k_1; m, 2s+\gamma, s, \Gamma, k; 2\nu+\delta+l)$$

$$= N(m_1 - k_1, 2s_1 + \gamma_1, s_1, \Gamma_1; 2s + \gamma + (m - k - 2s - \gamma)) \times N(k_1, k) q^{(m_1 - k_1)(k - k_1)}, \tag{7.53}$$

where $N(m_1 - k_1, 2s_1 + \gamma_1, s_1, \Gamma_1; 2s + \gamma + (m - k - 2s - \gamma))$ and $N(k_1, k)$ are given in Corollary 7.33 and Theorem 1.7, respectively. □

Notice that (7.52) reduces to (7.46) when $\delta = \gamma = \gamma_1 = 1$ and $\Gamma = 1$.

We can combine Theorems 7.43, 7.44, 7.45, 7.46, and 7.47 into a single Anzahl theorem as follows.

Theorem 7.48. Let $(m, 2s + \gamma, s, \Gamma, k)$ and $(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1, k_1)$ be two given types of subspaces of $\mathbb{F}_q^{(2\nu + \delta + l)}$.

(a) When $\delta = 1$, $\gamma \neq 1$, $\gamma_1 = 1$, and $\Gamma_1 = 1$; or when $\delta = \gamma = \gamma_1 = 1$, $\Gamma = 0$, and $\Gamma_1 = 1$; or when (7.52) does not hold, then

$$N(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1, k_1; m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l) = 0. \tag{7.54}$$

(b) When $\delta = \gamma = \gamma_1 = 1$, $\Gamma = \Gamma_1 = 1$, and (7.52) holds, we have (7.49)

$$N(m_1, 2s_1 + 1, s_1, 1, k_1; m, 2s + 1, s, 1, k; 2\nu + \delta + l) = N(m_1 - k_1 - 1, 2s_1, s_1; 2s + (m - k - 2s - 1)) \times N(k_1, k) q^{(m_1 - k_1 - 1)(k - k_1)}.$$

(c) When $\delta = \gamma = \gamma_1 = 1$, $\Gamma = 1$, $\Gamma_0 = 0$, and (7.52) holds, we have (7.50)

$$\begin{aligned} &N(m_1, 2s_1 + 1, s_1, 0, k_1; m, 2s + 1, s_1, 1, k; 2\nu + \delta + l) \\ &\quad = N(m_1 - k_1, 2s_1 + 1, s_1; 2s + 1 + (m - k - 2s - 1)) \\ &\qquad \times N(k_1, k) q^{(m_1 - k_1)(k - k_1)} \\ &\qquad - N(m_1 - k_1 - 1, 2s_1, s_1; 2s + (m - k - 2s - 1)) \\ &\qquad\quad \times N(k_1, k) q^{(m_1 - k_1 - 1)(k - k_1)}. \end{aligned}$$

(d) For any case which is not one of the cases (i), (ii), (iii), and (iv), assume that (7.52) holds, then we have (7.53)

$$N(m_1, 2s_1 + \gamma_1, s_1, \Gamma_1, k_1; m, 2s + \gamma, s, \Gamma, k; 2\nu + \delta + l) = N(m_1 - k_1, 2s_1 + \gamma_1, s_1, \Gamma_1; 2s + \gamma + (m - k - 2s - \gamma)) \times N(k_1, k) q^{(m_1 - k_1)(k - k_1)}.$$

□

As an application of Theorem 7.29, let us compute the number of 'cogredient' classes of $n \times n$ matrices of a given 'rank' over a finite field of characteristic 2.

Denote by $\mathcal{C}(n)$ the set of all 'cogredient' classes of $n \times n$ matrices. Define an action of $GL_n(\mathbb{F}_q)$ on $\mathcal{C}(n)$ as follows:

$$\begin{array}{ccc} \mathcal{C}(n) \times GL_n(\mathbb{F}_q) & \to & \mathcal{C}(n) \\ (G, \quad T) & \mapsto & TG^tT, \end{array}$$

where G is an $n \times n$ matrix. It can be readily verified that the action is well-defined. Then $\mathcal{C}(n)$ is subdivided into orbits under the action of $GL_n(\mathbb{F}_q)$. By Theorem 1.35 these orbits are represented by all the 'cogredience' normal forms $M(n, 2\nu+\delta, \nu)$, where $0 \le 2\nu+\delta \le n$ and $\delta = 0, 1$, or 2. Denote the orbit of the cogredience class with $M(n, 2\nu+\delta, \nu)$ as a member by $\mathcal{C}(n, 2\nu+\delta, \nu)$. Then we have

Theorem 7.49. Let $0 \le 2\nu+\delta \le n$, where $\delta = 0, 1$, or 2. Then

$$| \mathcal{C}(n, 2\nu+\delta, \nu) | = \begin{cases} q^{\nu(\nu+\delta)+\frac{1}{2}\delta(\delta-1)} \dfrac{\prod_{i=\nu+1}^{n} (q^i - 1)}{\prod_{i=0}^{\nu+\delta-1} (q^i+1) \prod_{i=1}^{n-2\nu-\delta} (q^i-1)}, & \text{if } \delta = 0 \text{ or } 2, \\ q^{\nu(\nu+1)} \dfrac{\prod_{i=\nu+1}^{n} (q^i - 1)}{\prod_{i=1}^{\nu}(q^i+1) \prod_{i=1}^{n-2\nu-1} (q^i-1)}, & \text{if } \delta = 1. \end{cases}$$

Proof. Same as the proof of Theorem 6.36. □

Denote the set of 'cogredient' classes of $n \times n$ matrices of 'rank' r by $\mathcal{C}(n, r)$. Then we have

Corollary 7.50. Let $0 \le r \le n$. Then

$$| \mathcal{C}(n, r) | = q^{[r/2]([r/2]+1)} \frac{\prod_{i=[r/2]+1}^{n} (q^i - 1)}{\prod_{i=1}^{[r/2]} (q^i+1) \prod_{i=1}^{n-r} (q^i-1)}.$$

Proof. In fact,

$$\mathcal{C}(n,r) = \begin{cases} \mathcal{C}(n,2\nu,\nu) \cup \mathcal{C}(n,2(\nu-1)+2,\nu-1), & \text{if } r=2\nu \text{ is even,} \\ \mathcal{C}(n,2\nu+1,\nu), & \text{if } r=2\nu+1 \text{ is odd.} \end{cases}$$

□

Moreover, we have

Corollary 7.51.

$$|\ \mathcal{C}(2\nu+\delta,2\nu+\delta,\nu)\ | = \begin{cases} \frac{1}{2}q^{\nu^2}(q^\nu+1)\prod_{i=0}^{\nu-1}(q^{2i+1}-1), & \text{if } \delta=0, \\ q^{\nu(\nu+1)}\prod_{i=0}^{\nu}(q^{2i+1}-1), & \text{if } \delta=1, \\ \frac{1}{2}q^{(\nu+1)^2}(q^{\nu+1}-1)\prod_{i=0}^{\nu}q^{2i+1}-1), & \text{if } \delta=2. \end{cases}$$

□

7.4 Polar Spaces Arising from Orthogonal Geometry over Finite Fields of Characteristic 2

In this section we adopt the projective geometry language again. Let us consider the $(2\nu+\delta-1)$-dimensional projective space $PG(2\nu+\delta-1,\mathbb{F}_q)$ over $\mathbb{F}_q$ arising from the $(2\nu+\delta)$-dimensional row vector space $\mathbb{F}_q^{(2\nu+\delta)}$. Let G be the $(2\nu+\delta)\times(2\nu+\delta)$ regular matrix (7.1). An m-dimensional totally singular subspace of $\mathbb{F}_q^{(2\nu+\delta)}$ with respect to G will now be called a *totally singular* $(m-1)$*-flat* of $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to G. Denote by $\mathcal{P}(G)$ the set of all totally singular flats in $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to G.

Parallel to Theorem 3.33 and Corollary 3.34, we have

Theorem 7.52. The set $\mathcal{P}(G)$ of totally singular flats of $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ regular matrix G of the form (7.1) forms a polar space of rank ν. □

Corollary 7.53. The set of singular points of $PG(3,\mathbb{F}_q), PG(4,\mathbb{F}_q)$, or $PG(5,\mathbb{F}_q)$ together with the set of totally singular lines with respect to the following regular matrices

$$\begin{pmatrix} 0 & I^{(2)} \\ & 0 \end{pmatrix}, \begin{pmatrix} 0 & I^{(2)} & \\ & 0 & \\ & & 1 \end{pmatrix}, \text{or} \begin{pmatrix} 0 & I^{(2)} & & \\ & 0 & & \\ & & \alpha & 1 \\ & & & \alpha \end{pmatrix}, \tag{7.55}$$

respectively, forms a generalized quadrangle. □

The generalized quadrangle formed by the set of totally singular points and the set of totally singular lines with respect to (7.55) in $PG(3, \mathbb{F}_q)$, $PG(4, \mathbb{F}_q)$, or $PG(5, \mathbb{F}_q)$ is usually denoted by $Q(3, q)$, $Q(4, q)$, or $Q(5, q)$, respectively.

Parallel to Corollary 6.41 we have

Corollary 7.54. The generalized quadrangles $Q(d, q)$, where $d = 3, 4$, or 5, have parameters

$$\begin{array}{ll} s = q, t = 1 & \text{for } Q(3, q), \\ s = t = q & \text{for } Q(4, q), \\ s = q, t = q^2 & \text{for } Q(5, q). \end{array}$$

□

7.5 Comments

Orthogonal groups over finite fields of characteristic 2 were first studied by L.E. Dickson, in particular, Theorem 7.23 is due to him (cf. Dickson 1900). The transitivity theorem of orthogonal geometry over finite fields of characteristic 2 (Theorem 7.6) and Lemma 7.4 on which Theorem 7.6 is based are contained in the thesis of X. Feng and Z. Dai (cf. Feng and Dai 1965). Theorems 7.5 and 7.9 are due to the author (cf. Wan 1993b).

All the results of Section 7.2, except Theorem 7.23 and Corollary 7.25, are contained in Feng and Dai 1965, but their proofs are simplified in this book. Corollary 7.25 is due to B. Segre (cf. Segre 1959 and also RayChaudhuri 1962).

All the results of Section 7.3 are due to the author (cf. Wan 1993c and 1993d).

All the results of Section 7.4 are due to J. Tits (cf. Tits 1959 and 1974).

7.6 Exercises

7.1 Let G be any $n \times n$ regular matrix over a finite field $\mathbb{F}_q$ of characteristic 2. Prove that $n \times n$ orthogonal matrices over $\mathbb{F}_q$ with respect to G are nonsingular.

7.2 Give the missing proofs in Sections 7.2 and 7.3.

7.3 Let $N = 2\nu + \delta + l - 1$, where $\nu \geq 0$, $\delta = 0, 1$ or 2, and $l \geq 0$. Denote by Q the quadric in $PG(N, \mathbb{F}_q)$ defined by

$$(x_0, x_1, \ldots, x_N) \begin{pmatrix} G_{2\nu+\delta} & \\ & 0^{(l)} \end{pmatrix} {}^t(x_0, x_1, \ldots, x_N) = 0.$$

(i) Prove that $| Q | = | H |$ for any hyperplane H in $PG(N, \mathbb{F}_q)$ if and only if $\delta = 1$.

(ii) Assume that $\nu \geq 1$. Prove that the hyperplanes H of $PG(N, \mathbb{F}_q)$ are classified into three classes according to the size $| H \cap Q |$ of $H \cap Q$. Determine the three different values of $| H \cap Q |$ and compute the number of hyperplanes having the same size $| H \cap Q |$ of $H \cap Q$.

7.4 Prove Corollary 7.54.

Appendix A
Axiomatic Projective Geometry

A.1 Incidence Structures

Let P and L be two disjoint non-empty sets whose elements are called *points* and *lines*, respectively, and let $I \subset P \times L$ be a subset of $P \times L$, called an *incidence relation*. Then the triple (P, L, I) is called an *incidence structure*.

Example A.1. Take P to be the set of points of $PG(2, \mathbb{F}_q)$, L to be the set of lines of $PG(2, \mathbb{F}_q)$, and define

$$I = \{(p, l) \mid p \in P, l \in L \text{ such that } p \text{ is contained in } l\},$$

then (P, L, I) is an incidence structure. Similarly, $PG(n, \mathbb{F}_q)$, $AG(2, \mathbb{F}_q)$, and $AG(n, \mathbb{F}_q)$ also give rise to incidence structures. □

Let (P, L, I) be an incidence structure. If $(p, l) \in I$, we say that the point p and the line l are *incident* and also say that the point p *is contained in* (or *lies on*) the line l, or the line l *contains* (or *passes through*) the point p. Two or more points are said to be *collinear*, if they are contained in a common line, and two or more lines are said to be *concurrent*, if they contain a point in common. Two collinear points p and q are also said to be *adjacent*; if p and q are adjacent (or not *adjacent*), we write $p \sim q$ (or $p \not\sim q$, respectively); we agree that $p \sim p$ for any point p. Two distinct concurrent lines l_1 and l_2 are also called *intersecting*; if l_1 and l_2 intersect (or do not intersect), we write $l_1 \cap l_2 \neq \phi$ (or $l_1 \cap l_2 = \phi$, respectively).

In an incidence structure we define the *dual statement* of a statement by interchanging the words 'points' and 'lines', and 'is contained in' and 'contains'. For instance, in $PG(2, \mathbb{F}_q)$ the dual statement of "any two distinct points is contained in a unique line" is "any two distinct lines contain a unique point in common".

Two incidence structures (P, L, I) and (P', L', I') are said to be *isomorphic*, if there are two bijective maps

$$f_P : P \to P' \quad p \mapsto f_P(p) \qquad \text{and} \qquad f_L : L \to L' \quad l \mapsto f_L(l)$$

such that $(p, l) \in I$ if and only if $(f_P(p), f_L(l)) \in I'$. The pair of maps (f_P, f_L) is called an *isomorphism* from (P, L, I) to (P', L', I'). An isomorphism of an incidence structure to itself is called an *automorphism*.

Example A.2.

A projective transformation of $PG(2, \mathbb{F}_q)$ is a bijective point to point transformation. Since it carries lines into lines, it induces a bijective line to line transformation. This pair of transformations is an automorphism of the incidence structure of $PG(2, \mathbb{F}_q)$. □

Let (P, L, I) be an incidence structure. Now let $P' = L$, $L' = P$, and define $I' \subset P' \times L'$ so that $(l, p) \in I'$ if and only if $(p, l) \in I$. Then (P', L', I') is also an incidence structure, called the *dual* of (P, L, I). If the dual of an incidence structure (P, L, I) is isomorphic to itself, then we say that (P, L, I) is *self-dual*.

A.2 Axioms of Projective Planes and the Principle of Duality

Let $\Pi = (P, L, I)$ be an incidence structure. It is called a *projective plane*, if the following axioms are satisfied.

PP1 There is one and only one line containing two distinct points.

PP2 There is one and only one point contained in two distinct lines.

PP3 There exist four points, no three of which are collinear.

PP1, PP2, and PP3 are usually called the axioms of projective planes.

Let p and q be two distinct points of a projective plane. By PP1, there is a unique line containing both p and q, and we denote this line by pq.

Example A.3.

In Example A.1 we defined an incidence structure (P, L, I) of $PG(2, \mathbb{F}_q)$. Theorem 2.1 (ii) says that PP1 and PP2 hold in this incidence structure; PP3 also hold, for $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, and $(1, 1, 1)$ are four points in $PG(2, \mathbb{F}_q)$, no three of which are collinear. Therefore (P, L, I) is a projective plane in the new axiomatic sense. □

Theorem A.1. In any projective plane the dual statements of PP1, PP2, and PP3 also hold. Hence the *principle of duality* holds in any projective plane, i.e., if a statement holds in a projective plane, then its dual statement holds also.

Proof. Denote the dual of PPi by PPi'. Clearly, PP1$'$ is PP2 and PP2$'$ is PP1. Hence both PP1$'$ and PP2$'$ hold in any projective plane.

The dual of PP3 is

PP3$'$. There exist four lines, no three of which are concurrent.

Let us now prove PP3$'$. By PP3 in any projective plane there exist four points, no three of which are collinear. Let the four points be p, q, r, and s. Then pq, qr, rs, and sp are four lines, no three of which are concurrent.

□

Corollary A.2. The dual of a projective plane is also a projective plane.

□

Theorem A.3. In a projective plane every line contains at least three distinct points.

Proof. Let p, q, r, and s be four points in a projective plane, no three of which are collinear, and let l be any line of the plane. Then one and only one of the following three situations occurs: (i) l contains two of the four points; (ii) l goes through only one of the four points, and (iii) l does not go through any one of the four points. We consider these three cases separately.

Suppose that l contains two of the four points. Without loss of generality we can assume that l contains p and q. Then l coincides with the line pq. By PP2, l and the line rs meet at a point, call it t. Since p, r, and s are not collinear, $t \neq p$. Similarly $t \neq q$. Therefore p, q, and t are three distinct points contained in l.

Next, suppose that l goes through only one of the four points. Without loss of generality, we can assume that l goes through p. By PP2, l and the line qr meet at a point u, and l and the line rs meet at a point v. Then p, u, and v are three distinct points contained in l.

Finally, suppose that l does not go through any one of the four points. Let t, u, and v be the points of intersection of l with pq, pr, and ps, respectively. Then t, u, and v are three distinct points contained in l. □

The dual of Theorem A.3 is

Theorem A.4. In a projective plane every point is contained in at least three distinct lines. □

Let l be any line of a projective plane $\Pi = (P, L, I)$, and $[l]$ be the set of points contained in l. It follows from PP1 and Theorem A.3 that for any $l, l' \in L$, $[l] = [l']$ implies $l = l'$. Therefore we can interpret the line $l \in L$ as the point set $[l]$, or we can identify l and $[l]$. This justifies the terminologies 'contains' and 'is contained in'. Henceforth we shall use the notation $p \in l$ to denote that the point p is contained in the line l, and use other set theoretic notations. Moreover, we have

Theorem A.5. Let $\Pi = (P, L, I)$ and $\Pi' = (P', L', I')$ be two projective planes and let

$$f_P : P \to P'$$

be a bijective map. For any $l \in L$, define

$$f_L(l) = \{f_P(p) \mid p \in l\}.$$

If $f_L(l) \in L'$ for all $l \in L$, then the pair of maps (f_P, f_L) is an isomorphism from Π to Π'.

Proof. The injectiveness of f_P implies the injectiveness of f_L. For any $l' \in L'$ choose two distinct points p' and q' on l' and let $f_P^{-1}(p') = p$, $f_P^{-1}(q') = q$, then the line $l = pq$ will be mapped to a line containing p' and q', i.e., $f_L(l) = l'$, which proves the surjectiveness of f_L on L. That $(p, l) \in I$ if and only if $(f_P(p), f_L(l)) \in I'$ follows immediately from the definition of f_L. Therefore the pair of maps (f_P, f_L) is an isomorphism from Π to Π'. □

In virtue of Theorem A.5 a bijective map from the set of points of a projective plane Π to that of a projective plane Π', which carries lines of Π to that of Π', is also called an *isomorphism* from Π to Π'.

A.3 Finite Projective Planes

A projective plane $\Pi = (P, L, I)$ is called a *finite projective plane* if P is a finite set. We may deduce from (PP1) and (PP2) that in a projective plane $\Pi = (P, L, I)$, P is a finite set if and only if L is a finite set. Hence we can also define a projective plane Π to be a finite projective plane if L is a finite set.

Example A.4.

We know from Theorem 2.1 that $PG(2, \mathbb{F}_q)$ is a finite projective plane, it has q^2+q+1 points and also q^2+q+1 lines, each line of it contains $q+1$ points, and each point of it lies on $q+1$ lines. □

In general we have

Theorem A.6. For any finite projective plane there exists an integer $n \geq 2$ such that

(i) every line contains $n+1$ points;

(ii) every point is contained in $n+1$ lines;

(iii) the number of points in the plane is equal to the number of lines in the plane and is equal to n^2+n+1.

Proof. Let $\Pi = (P, L, I)$ be a finite projective plane, $p \in P, l \in L$, and p does not lie on l. Assume that l contains $n+1$ points and p is contained in $n'+1$ lines. Let $p_1, p_2, \ldots, p_{n+1}$ be the $n+1$ distinct points of l, then $pp_1, pp_2, \ldots, pp_{n+1}$ are $n+1$ distinct lines passing through p. Hence $n \leq n'$. Let $l_1, l_2, \ldots, l_{n'+1}$ be $n+1$ distinct lines passing through p, and let l and l_i meet at q_i $(1 \leq i \leq n'+1)$, then $q_1, q_2, \ldots, q_{n'+1}$ are $n'+1$ distinct points on l. Hence $n' \leq n$. Consequently $n = n'$. By Theorem A.3, $n \geq 2$.

Now let p' be any point of Π and $p' \neq p$. If p' does not lie on l, then by the above proof, p' is also contained in $n+1$ lines. Now suppose that p' lies on l. Let q be a point on l and $q \neq p'$. By Theorem A.4 there is a line l' passing through q such that l' is different from l and the line pq. Then p is not on l' and by the above proof l' contains $n+1$ points. But p' is not on l' and also by the above proof p' is contained in $n+1$ lines. Therefore every point of Π is contained in $n+1$ lines.

The dual of the argument of the above paragraph implies that every line of Π contains $n+1$ points. More precisely, let l' be any line of Π and $l' \neq l$. If l' does not contain p, then by the above proof, l' contains also $n+1$ points. Suppose that l' contains p. Let l'' be a line passing through p and $l'' \neq l'$. By Theorem A.3 there is a point q on l'' such that q is distinct from p and not on l. Since q is not contained in l, by the above proof q is contained in $n+1$ lines. But q is also not contained in l', also by the above proof l' contains $n+1$ points. Therefore every line of Π contains $n+1$ points.

There remains to prove (iii). Let l be a fixed line of Π. By (i) l contains $n+1$ points, let p be one of them. By (ii) p is contained in n lines distinct

from l and each of these n lines contains n points distinct from p. Thus we obtain n^2 point of Π not contained in l. Clearly, these n^2 points are different in pairs and they exhaust all the points of Π not contained in l. Therefore altogether Π has $n^2 + n + 1$ points.

The dual of the argument of the above paragraph yields that Π has $n^2 + n + 1$ lines, but the details are omitted. □

The natural number $n \geq 2$ in Theorem A.6 is called the *order* of the projective plane.

Now consider $PG(2, \mathbb{F}_q)$ again. Denote by P and L the set of points and the set of lines of $PG(2, \mathbb{F}_q)$, respectively, and define

$$I = \{(p, l) \mid p \in P,\ l \in L \text{ such that } p \in l\}.$$

We know that (P, L, I) is an incidence structure on $PG(2, \mathbb{F}_q)$. It was pointed out in Example A.3 that (P, L, I) is a projective plane. We write $PG(2, \mathbb{F}_q)$ for (P, L, I) for simplicity.

Theorem A.7. *$PG(2, \mathbb{F}_q)$ is a finite projective plane of order q and is self-dual.*

Proof. That $PG(2, \mathbb{F}_q)$ is a finite projective plane of order q follows from Theorem 2.1.

Now let us prove that $PG(2, \mathbb{F}_q)$ is self-dual. Let $P' = L$, $L' = P$ and define $I' \subset P' \times L'$ such that $(l, p) \in I'$ if and only if $(p, l) \in I$. Then (P', L', I') is the dual of (P, L, I). For $p \in P$, let (p_0, p_1, p_2) be the coordinates of the point p, and for $l \in P$ let $l_0x_0 + l_1x_1 + l_2x_2 = 0$ be its equation. Clearly the maps

$$P \to P'$$
$$(p_0, p_1, p_2) \mapsto p_0x_0 + p_1x_1 + p_2x_2 = 0$$

and

$$L \to L'$$
$$l_0x_0 + l_1x_1 + l_2x_2 = 0 \mapsto (l_0, l_1, l_2)$$

are well-defined bijections. Furthermore, $(p, l) \in I$ if and only if $l_0p_0 + l_1p_1 + l_2p_2 = 0$, hence if and only if $(l, p) \in I'$. Therefore (P, L, I) and (P', L', I') are isomorphic, i.e., $PG(2, \mathbb{F}_q)$ is self-dual. □

Corollary A.8. *For every prime power q, there exists a finite projective plane of order q.* □

Besides the prime powers, what positive integer can the order of a projective plane be? The Bruck-Ryser's Theorem asserts that if $n \equiv 1$ or $2 \pmod 4$ there cannot be a projective plane of order n unless n can be expressed as a sum of two integral squares (cf. Bruck and Ryser 1949). From this theorem it follows immediately that there does not exist any projective plane of order 6. Recently it was proved by a computer search that there does not exist any projective plane of order 10 (cf. Lam et al. 1989).

A.4 Collineations in Projective Planes

Let $\Pi = (P, L, I)$ be a projective plane. An automorphism of Π will now be called a *collineation* of Π. If we adopt the point of view that a line is identified with the set of points incident with it, then by Theorem A.5 an equivalent definition of a collineation is: A collineation of a projective plane $\Pi = (P, L, I)$ is a bijective map $f : P \to P$ such that $l \in L$ implies $f(l) \in L$. In the following we always understand the collineations of a projective plane according to the second definition.

Clearly, the composition of two collineations is a collineation and the inverse of a collineation is a collineation. Hence the set of all collineations of a projective plane forms a group under composition called the *collineation group* of the projective plane.

Example A.5.

Projective transformations of the projective plane $PG(2, \mathbb{F}_q)$ are examples of collineations. □

A collineation f of a projective plane $\Pi = (P, L, I)$ is called a *central collineation*, if there exists a point $c \in P$ such that $f(c) = c$ and for any line l passing through c, $f(l) = l$. An equivalent definition is: a collineation f is a central collineation if there exists a point $c \in P$ such that $f(c) = c$ and for any point $x \in P, x \neq c, f(x) \in cx$. The point c is called the *center* of central collineation.

The dual concept of a central collineation is an axial collineation. A collineation f of a projective plane $\Pi = (P, L, I)$ is called an *axial collineation*, if there exists a line $l \in L$ such that $f(l) = l$ and for any point p on l, $f(p) = p$. An equivalent definition is: a collineation f is an axial collineation, if there exists a line $l \in L$ such that f fixes every point of l. The line l is called the *axis* of the axial collineation.

Theorem A.9. Every non-identity central collineation has only one center.

Proof. Let f be a central collineation and suppose that f has two centers c and c'. For any point x not on cc', we have $f(cx) = cx$, $f(c'x) = c'x$. Since $x \in cx \cap c'x$, $f(x) \in f(cx) \cap f(c'x) = cx \cap c'x = x$. Therefore $f(x) = x$. Let y be any point on cc' and distinct from both c and c', and let l be a line passing through y and distinct from cc'. Then f fixes every point of l distinct from y, hence f also fixes y. Therefore f must be the identity map. □

The dual of Theorem A.9 is

Theorem A.10. Every non-identity axial collineation has only one axis. □

Theorem A.11. Every central collineation is also an axial collineation.

Proof. Let f be a central collineation with center c. By definition, any line passing through c is left fixed by f.

Suppose that there is a line l, not passing through c, which is left fixed by f. Let x be any point of l. Then $x = cx \cap l$, from which we deduce $f(x) = f(cx) \cap f(l) = cx \cap l = x$. Therefore f is an axial collineation with axis l.

Next suppose that all lines left fixed by f pass through c. Let l be any line not passing through c, then so is $f(l)$. Let $a = l \cap f(l)$. Then $a = l \cap ca$ and hence $f(a) = f(l) \cap ca = a$. We are going to prove that every point on ca is also fixed by f. Let $b \in ca \setminus \{c, a\}$, then $f(b) \in ca \setminus \{c, a\}$. Assume that $f(b) \neq b$. Let h be any line passing through b and distinct from ca, then $f(h)$ is a line passing through $f(b)$ and distinct from ca. Let $x = h \cap f(h)$. Since h does not pass through c, by the above argument $f(x) = x$. From $f(a) = a$ and $f(x) = x$ we deduce $f(ax) = ax$. By our assumption ax must pass through c. Then x is on ca. But b is on ca, hence $h = ca$, this is a contradiction. Therefore f is an axial collineation with axis ca. □

The dual of Theorem A.11 is

Theorem A.12. Every axial collineation is also a central collineation. □

Let f be a central collineation with center c, then by Theorem A.11 f has also an axis l. If f is not the identity map, then by Theorems A.9 and A.10 c and l are uniquely determined. We call f a (c, l)-*collineation*. If c lies on l, f is called an *elation*, and if c does not lie on l, f is called a *homology*.

Example A.6.

In $PG(2, \mathbb{F}_q)$, two projective transformations f_1 and f_2 are said to be projectively equivalent, if there exists a projective transformation f such that $f_1 = f \circ f_2 \circ f^{-1}$. By Theorem 2.10, if a projective transformation of $PG(2, \mathbb{F}_q)$ is an elation, it is projectively equivalent to a projective transformation of the form

$$(x_0, x_1, x_2) \mapsto (x_0, x_1, x_2) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & 0 & 1 \end{pmatrix},$$

where $\lambda \neq 0$, which is a (c, l)-collineation with $c = (1, 0, 0)$ and $l : x_2 = 0$. And if a projective transformation of $PG(2, \mathbb{F}_q)$ is a homology, it is projectively equivalent to a projective transformation of the form

$$(x_0, x_1, x_2) \mapsto (x_0, x_1, x_2) \begin{pmatrix} \lambda & & \\ & 1 & \\ & & 1 \end{pmatrix},$$

where $\lambda \neq 0$, which is a (c, l)-collineation with $c = (1, 0, 0)$ and $l : x_0 = 0$. □

Theorem A.13. A non-identity (c, l)-collineation fixes no point other than c and the points of l.

Proof. Let f be a non-identity (c, l)-collineation and assume that there is a point $p \notin \{c\} \cup l$ such that $f(p) = p$. Let l' be any line passing through p and let l and l' intersect at q. Then $f(p) = p$ and $f(q) = q$, hence $f(l') = l'$. Thus p is also a center of f, which contradicts to Theorem A.9. □

Let $\Pi = (P, L, I)$ be a projective plane, $c \in P$, and $l \in L$. Π is said to be (c, l)-transitive, if for any two points $x, x' \notin \{c\} \cup l$, such that c, x, x' are collinear, there exists a (c, l)-collineation carrying x into x'.

Theorem A.14. $PG(2, \mathbb{F}_q)$ is (c, l)-transitive for any $c \in P$ and $l \in L$.

Proof. For any $c \in P$ and $l \in L$, let x and x' be two points such that $x, x' \notin \{c\} \cup l$ and c, x, x' are collinear. We distinguish the following two cases:

(i) $c \in l$. By Theorem 2.10, without loss of generality we can assume that $c = (1, 0, 0)$, $l : x_2 = 0$, $x = (0, 0, 1)$, and $x' = (\lambda, 0, 1)$. Then the

(c, l)-collineation

$$(x_0, x_1, x_2) \mapsto (x_0, x_1, x_2) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \lambda & 0 & 1 \end{pmatrix},$$

which is an elation, carries x into x'.

(ii) $c \notin l$. By Theorem 2.10, without loss of generality, we can assume that $c = (1, 0, 0)$, $l : x_0 = 0$, $x = (1, 1, 1)$, and $x' = (\lambda, 1, 1)$. Then the (c, l)-collineation

$$(x_0, x_1, x_2) \mapsto (x_0, x_1, x_2) \begin{pmatrix} \lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is a homology, carries x into x'. □

A.5 Desargues Planes

Three non-collinear points x, y, z are said to form a *triangle* with vertices x, y, z and sides xy, yz, zx. Denote this triangle by xyz. Two triangles xyz and $x'y'z'$ are said to be *perspective from a point* c, if c, x, and x' are collinear, so are c, y, and y', and c, z, and z'; c is called the *center of perspectivity*. They are said to be *perspective from a line* l, if xy and $x'y'$ intersect at a point on l, so do yz and $y'z'$, and zx and $z'x'$; l is called the *axis of perspectivity*. A projective plane is called a *Desargues plane*, if for any two triangles xyz and $x'y'z'$, which are perspective from a point such that the six vertices are six distinct points, the lines xy and $x'y'$ are distinct, and so are the lines yz and $y'z'$, and the lines zx and $z'x'$, they are also perspective from a line.

Theorem A.15. If a projective plane $\Pi = (P, L, I)$ is (c, l)-transitive for any $c \in P$ and $l \in L$, then it is a Desargues plane.

Proof. Let xyz and $x'y'z'$ be two triangles which are perspective from a point c, and assume that x, y, z, x', y', z', and c are seven distinct points. Denote the point of intersection of xy and $x'y'$ by u, that of yz and $y'z'$ by v, and that of zx and $z'x'$ by w. By hypothesis, there is a (c, uv)-collineation f such that $f(y) = y'$. From $x = uy \cap cx$ we deduce $f(x) = uy' \cap cx = x'$. Similarly, $f(z) = z'$. From $w = zx \cap cw$ we deduce $f(w) = z'x' \cap cw = w$. By Theorem A.13 we have $w \in uv$ or $w = c$. If $w \in uv$, we are finished. If $w = c$, then $c = zz' \cap zx = z$, which is a contradiction. □

From Theorems A.14 and A.15 we deduce immediately

Theorem A.16. $PG(2, \mathbb{F}_q)$ is a finite Desargues plane. □

The converse of Theorem A.16 is also true.

Theorem A.17 (O. Veblen and J. W. Young). Any finite Desargues plane is isomorphic to a $PG(2, \mathbb{F}_q)$ for some prime power q. □

The proof of Theorem A.17 will not be included in this book, and interested readers may consult Veblen and Young (1916). Actually, Veblen and Young proved that any Desargue plane is isomorphic to a $PG(2, K)$ for some division ring K, and the finiteness assumption forces K to be a finite field.

A.6 Projective Spaces

Let $\mathcal{S} = (P, L, I)$ be an incidence structure. Assume that there are some distinguished subsets of P, called the *flats* of $\mathcal{S}$. Then $\mathcal{S}$ together with the flats, denoted by $\mathcal{S}$ again, is called a *projective space*, if the following axioms are satisfied:

PG1 There is one and only one line containing two distinct points.
(Let p and q be two distinct points. By PG1, there is a unique line containing both p and q, and we denote this line by pq.)

PG2 Any line contains at least three points.
(As in Section A.2, by PG1 and PG2 we can identify any line $l \in L$ with the point set $[l]$ consisting of all the points contained in l. Thus in the following lines are understood to be certain subsets of P.)

PG3 If two distinct lines l, l' have a common point p, if q, r are points $\neq p$ of l, and if s, t are points $\neq p$ of l', then the lines qs and rt have a common point.

PG4 Any flat F has the property that $p, q \in F$ implies $pq \subseteq F$, and any subset F of P having the property that $p, q \in F$ implies $pq \subseteq F$ is a flat.

PG1, PG2, PG3, and PG4 are called the axioms of projective spaces.

Example A.7. Clearly, projective planes defined in Section A.2 are projective spaces, and the flats of a projective plane $\Pi = (P, L, I)$ are the empty set ϕ, the single point sets $\{p\}$ where $p \in P$, the lines $l \in L$, and the whole point set P. □

Example A.8. The incidence structure arising from $PG(n, \mathbb{F}_q)$ together with the flats defined in Section 2.1 is a projective space. □

Moreover, we have

Theorem A.18. Let F be any flat of a projective space $\mathcal{S}$, which contains some lines of $\mathcal{S}$ as subsets. Define an incidence structure $\mathcal{F}$ whose set of points is F, whose set of lines is the set of lines of $\mathcal{S}$ contained in F, and whose incidence relation is induced by the incidence relation of $\mathcal{S}$. Then $\mathcal{F}$ together with the flats of $\mathcal{S}$ contained in F as its flats is also a projective space. □

The proof is immediate and will be omitted.

In a projective space $\mathcal{S}$, the empty set ϕ, the subsets $\{p\}$ consisting of a single point p, the lines $l \in L$, and the whole point set P are flats; but in general there are more flats as we can see from the projective space $PG(n, \mathbb{F}_q)$ when $n \geq 3$. Moreover, the intersection of any number of flats is also a flat. Thus, for any given set X of points, there is a minimal flat containing X, called the flat *spanned by* X and denoted by $\langle X \rangle$. A set X of points is called an *independent set of points*, if $p \notin \langle X \setminus \{p\} \rangle$ for every point $p \in X$. An independent set of points spanning the flat P is called a *basis* of $\mathcal{S}$.

A projective space $\mathcal{S}$ is called a *finite dimensional projective space*, if P has a finite independent set of points which spans P, i.e., $\mathcal{S}$ has a finite basis. In the following we shall confine ourselves to finite dimensional projective spaces.

Theorem A.19. Let $\mathcal{S} = (P, L, I)$ be a finite dimensional projective space. Then all independent sets spanning the flat P have the same cardinality. In other words, all bases have the same cardinality. □

We prove the following lemmas first.

Lemma A.20. Let F be a flat of a projective space and let p be a point not contained in F. Then the flat $\langle F \cup \{p\} \rangle$ consists of points on all lines pq for $q \in F$.

Proof. Let X be the set of points consisting of points on all lines pq for $q \in F$. We shall show that X is a flat, then it will be the smallest flat containing F and p and hence we have $X = \langle F \cup \{p\}\rangle$.

Let r and s be any two points of X. We must show that $rs \subseteq X$. If p, r, and s are collinear, then $rs = pr$; but r is on a line pq for $q \in F$, hence $pr = pq \subseteq X$, therefore $rs \subseteq X$. Suppose now that p, r, and s are not collinear. By definition of X, r is on a line pq for $q \in F$ and s is on a line pq' for $q' \in F$. Let t be any point of rs, and let tp and qq' intersect at q''. Since $q, q' \in F$, we have $q'' \in F$. But t is on $pt = pq''$, hence $t \in X$. Consequently, $rs \subseteq X$. □

Lemma A.21. Let x and y be two points of a projective space $\mathcal{S} = (P, L, I)$ and let X be a subset of P such that $x \notin \langle X\rangle$ and $x \in \langle X \cup \{y\}\rangle$, then $y \in \langle X \cup \{x\}\rangle$.

Proof. Without loss of generality, we can assume that $X = \langle X\rangle$ is a flat. By Lemma A.20, $x \in \langle X \cup \{y\}\rangle$ implies that x is on some line yq for $q \in X$. Since $x \notin X$, $q \neq x$. Thus y lies on the line xq and so $y \in \langle X \cup \{x\}\rangle$. □

The property stated in Lemma A.21 is called the *exchange property.*

Lemma A.22. Let $\{x_1, x_2, \ldots, x_n\}$ be an independent set of points in a projective space and let x be any point in the same projective space such that $x \notin \langle\{x_1, x_2, \ldots, x_n\}\rangle$. Then $\{x, x_1, x_2, \ldots, x_n\}$ is also an independent set of points.

Proof. We need to show that $x_i \notin \langle\{x, x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n\}\rangle$ for every i. Suppose that $x_i \in \langle\{x, x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n\}\rangle$ for some i. Since $\{x_1, x_2, \ldots, x_n\}$ is an independent set, $x_i \notin \langle\{x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n\}\rangle$. By the exchange property, $x \in \langle\{x_1, x_2, \ldots, x_n\}\rangle$, a contradiction. □

Let us now come to the proof of Theorem A.19.

Proof of Theorem A.19. Let $\{x_1, x_2, \ldots, x_n\}$ and $\{y_1, y_2, \ldots, y_m\}$ be two independent set of points spanning the flat P and $n \leq m$. Since $\mathcal{S}$ is a finite dimensional projective space, we can assume that n is finite. It cannot be the case that every $y_i \in \langle\{x_2, x_3, \ldots, x_n\}\rangle$, for otherwise, $x_1 \in P = \langle\{y_1, y_2, \ldots, y_m\}\rangle \subseteq \langle\{x_2, x_3, \ldots, x_n\}\rangle$, which contradicts to the assumption. Without loss of generality we can assume that $y_1 \notin \langle\{x_2, x_3, \ldots, x_n\}\rangle$. Since $y_1 \in \langle\{x_1, x_2, \ldots, x_n\}\rangle$, $x_1 \in \langle\{y_1, x_2, \ldots, x_n\}\rangle$ by the exchange property. Therefore $\langle\{y_1, x_2, \ldots, x_n\}\rangle = P$. By Lemma A.22 $\{y_1, x_2, \ldots, x_n\}$ is an independent set.

Similarly, if every $y_i \in \langle\{y_1, x_3, \ldots, x_n\}\rangle$, then $x_2 \in P \subseteq \langle\{y_1, x_3 \ldots, x_n\}\rangle$, which contradicts to the fact that $\{y_1, x_2, \ldots, x_n\}$ is an independent set. Without loss of generality we can assume that $y_2 \notin \langle\{y_1, x_3, \ldots, x_n\}\rangle$. But $y_2 \in \langle\{y_1, x_2, \ldots, x_n\}\rangle = P$, hence by the exchange property $x_2 \in \langle\{y_1, y_2, x_3, \ldots, x_n\}\rangle$. Therefore $\{y_1, y_2, x_3, \ldots, x_n\} = P$, and by Lemma A.22 $\{y_1, y_2, x_3, \ldots, x_n\}$ is an independent set.

Continuing in this way, we get eventually $\langle\{y_1, y_2, \ldots, y_n\}\rangle = P$, so that $n = m$. □

Let $\mathcal{S} = (P, L, I)$ be a finite dimensional projective space. The cardinal of any basis of $\mathcal{S}$ diminished by 1 is called the *dimension* of the projective space $\mathcal{S}$, and denoted by $\dim \mathcal{S}$. Clearly, $\dim \mathcal{S} \geq 1$ for any finite dimensional projective space $\mathcal{S}$ and the dimension of a projective plane is 2.

Corollary A.23. Any independent set of points of an n-dimensional projective space $\mathcal{S} = (P, L, I)$ can be supplemented to a base of $\mathcal{S}$. In particular, the cardinal of any independent set of points is $\leq n + 1$ and any independent set of $n + 1$ points is a base of $\mathcal{S}$.

Proof. Let $\{y_1, y_2, \ldots, y_d\}$ be an independent set of points. Since $\dim \mathcal{S} = n$, there is an independent set of $n + 1$ points $\{x_1, x_2, \ldots, x_{n+1}\}$ spanning P. Form the set $S = \{y_1, y_2, \ldots, y_d, x_1, x_2, \ldots, x_{n+1}\}$. For each i, if $x_i \in \langle\{y_1, y_2, \ldots, y_d, x_1, x_2, \ldots, x_{i-1}\}\rangle$, then delete it from S, and if $x_i \notin \langle\{y_1, y_2, \ldots, y_d, x_1, x_2, \ldots, x_{i-1}\}\rangle$, then keep it in S. Thus we obtain a set $S' = \{y_1, y_2, \ldots, y_d, x_{i_1}, \ldots, x_{i_e}\}$ which is a basis of $\mathcal{S}$. □

Corollary A.24. Let F be a flat of a finite dimensional projective space $\mathcal{S}$, which contains some lines of $\mathcal{S}$ as subsets and let $\mathcal{F}$ be the projective space with F as its set of points defined in Theorem A.18. Then $\mathcal{F}$ is also finite dimensional and $\dim \mathcal{F} \leq \dim \mathcal{S}$. □

Under the hypothesis of Corollary A.24 we also call $\dim \mathcal{F}$ the *dimension* of F and use the symbol $\dim F$ for $\dim \mathcal{F}$. Clearly $\dim F$ is the cardinal of any independent set of points of F diminished by 1.

If F is a flat of $\mathcal{S}$, which contains no lines of $\mathcal{S}$, then F must be a set consisting of a single point; in this case we define $\dim F = 0$. Moreover, if $F = \phi$, then we define $\dim F = -1$.

Corollary A.25. Let F_1 and F_2 be two flats of a finite dimensional projective space. Then $F_1 \subseteq F_2$ implies $\dim F_1 \leq \dim F_2$. Moreover, $F_1 \subseteq F_2$ and $\dim F_1 = \dim F_2$ implies $F_1 = F_2$. □

Corollary A.26. Let F_1 and F_2 be any two flats of a finite dimensional projective space. Then

$$\dim(F_1 \cap F_2) + \dim\langle F_1, F_2\rangle = \dim F_1 + \dim F_2.$$

□

The proofs of these two corollaries are left as exercises.

Theorem A.27. In any projective space, projective spaces with flats of dimension 2 as their sets of points are projective planes, i.e., they satisfy the axioms PP1, PP2, and PP3.

Proof. Let F be a 2-dimensional flat in a projective space $\mathcal{S}$. Denote the projective space with the flat F as its set of points again by F. PP1 is $PG1$ and hence holds in F.

Now let us prove PP2. Let l and l' be two distinct lines in F, and let p, q be two distinct points on l and r, s be two distinct points on l'. If either r or s lies on l, then l and l' will have a common point r or s, respectively. Now suppose that both r and s are not on l. Then p, q, and r are three non-collinear points and hence $\{p, q, r\}$ is an independent set of points of F. Since $\dim F = 2$, by Corollary A.23 $\langle\{p, q, r\}\rangle = F$. F can also be expressed as $F = \langle pq \cup \{r\}\rangle$. By Lemma A.20 the flat $\langle pq \cup \{r\}\rangle$ consists of points on all lines rt for $t \in pq$. In particular, the point s is on a line rt_0 where $t_0 \in pq$. Then the line $l' = rs = rt_0$ meets l at t_0. This proves that PP2 holds in F.

Then we prove PP3. Let p, q, and r be three independent points of F. Of course, they are non-collinear. On the line pq choose a third point s, and on the line pr choose a third point t. By $PG3$, the lines qr and st have a common point u. Then it is easy to verify that p, q, t, and u are four points of F, no three of which are collinear. □

Thus in any projective space, projective spaces with 2-dimensional flats as their sets of points are also called *planes*. Moreover, we have the following interesting results.

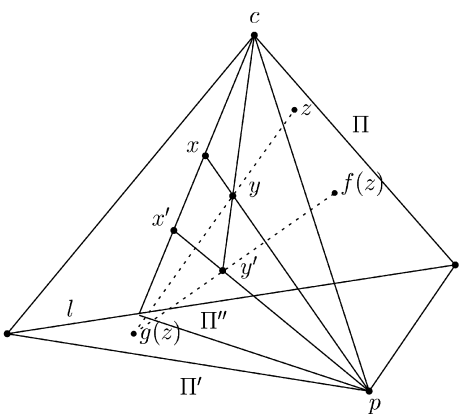
Theorem A.28. Any plane of a projective space of dimension $n \geq 3$ is a Desargues plane.

Proof. Let Π be a plane of a projective space $\mathcal{S} = (P, L, I)$ of $\dim n \geq 3$. Let c and l be any point and line of Π, respectively. Let x and x' be

distinct points of Π such that c, x, x' are collinear and $x, x' \notin \{c\} \cup l$. We shall construct a (c, l)-collineation f of Π such that $f(x) = x'$, which will prove that Π is (c, l)-transitive for any c and l. Then our theorem will follow from Theorem A.15.

Let $p \in P \setminus \Pi$. Let Π' be the plane containing p and l, and let Π'' be the plane containing p and the three collinear points c, x, x'. Choose $y \in xp \setminus \{x, p\}$ and set $y' = cy \cap px'$. Then y and y' are both in Π''.

Define a map g from Π to Π' by $g(z) = zy \cap \Pi'$ for every $z \in \Pi$, and define a map h from Π' to Π by $h(w) = wy' \cap \Pi$ for every $w \in \Pi'$. Then we define $f = h \circ g$. Clearly g and h are isomorphisms from Π to Π' and Π' to Π, respectively. Therefore f is an isomorphism of Π, in other words, f is a collineation of Π. Clearly, f fixes c and each point of l and maps x to x'. It remains to show that for any $z \in \Pi \setminus \{c\}$, the points $c, z, f(z)$ are collinear. $\langle\{z, y, y'\}\rangle$ is a plane distinct from Π and contains both c and $f(z)$. On the other hand, c, z, and $f(z)$ are all in Π. The intersection of the two planes $\langle\{z, y, y'\}\rangle$ and Π must be a line, and this line contains c, z, and $f(z)$. Hence c, z, and $f(z)$ are collinear. □



Two projective spaces $\mathcal{S}$ and $\mathcal{S}'$ are said to be *isomorphic*, if their underlying incidence structures are isomorphic. As the case of the projective plane, we also call a bijective map f from the set of points of $\mathcal{S}$ to that of $\mathcal{S}'$ an *isomorphism* from $\mathcal{S}$ to $\mathcal{S}'$, if it carries lines of $\mathcal{S}$ to that of $\mathcal{S}'$. In fact, it induces an isomorphism from $\mathcal{S}$ to $\mathcal{S}'$ in an obvious way as Theorem A.5 shows.

Then we have

Theorem A.29 (O. Veblen and J. W. Young). Any finite dimensional projective space of dimension $n \geq 3$ is isomorphic to a $PG(n, K)$ for some division ring K. □

The proof of this theorem will not be included in this book, and interested readers may consult Veblen and Young (1916).

Appendix B
Polar Spaces and Generalized Quadrangles

B.1 Polar Spaces

A set $\mathcal{P}$ of elements called points together with some distinguished subsets called subspaces is called a *polar space* of rank n, where n is an integer ≥ 1, if the following four axioms are satisfied.

PS1 A subspace together with the subspaces it contains forms an i-dimensional projective space with $-1 \leq i \leq n-1$.

PS2 The intersection of any two subspaces is a subspace.

PS3 Given a subspace P of dimension $n-1$ and a point $p \in \mathcal{P} \setminus P$ there exists a unique subspace Q containing p such that $\dim(P \cap Q) = n-2$; Q contains all those points of P such that each of them is contained in a 1-dimensional subspace together with p.

PS4 There exist two disjoint subspaces of dimension $n-1$.

The polar space whose set of points is $\mathcal{P}$ is also denoted by $\mathcal{P}$. PS1, PS2, PS3, and PS4 are called axioms of polar spaces.

Let $\mathcal{P}$ and $\mathcal{P}'$ be two polar spaces, and let f be a bijective map from the set of points of $\mathcal{P}$ to that of $\mathcal{P}'$. For any subspace P of $\mathcal{P}$, define

$$\bar{f}(P) = \{f(p) \mid p \in P\}.$$

Assume that $\bar{f}$ has the following properties.

(i) For any subspace P of $\mathcal{P}$, $\bar{f}(P)$ is a subspace of $\mathcal{P}'$.

(ii) $\bar{f}$ is a bijective map from the set of subspaces of $\mathcal{P}$ to that of $\mathcal{P}'$.

(iii) For subspaces P, Q of $\mathcal{P}$, $P \supset Q$ implies $\bar{f}(P) \supset \bar{f}(Q)$.

(iv) For any subspace P of $\mathcal{P}$, $\dim \bar{f}(P) = \dim P$.

Then f is called an *isomorphism* from the polar space $\mathcal{P}$ to the polar space $\mathcal{P}'$.

A polar space $\mathcal{P}$ is called a *finite polar space*, if the set $\mathcal{P}$ of points is a finite set.

Theorem B.1. The following are finite polar spaces:

(a) The polar space formed by the points of $PG(2\nu-1, \mathbb{F}_q)$, $\nu \geq 2$, together with the absolutes of dimension $\leq \nu-1$ of a symplectic polarity of $PG(2\nu-1, \mathbb{F}_q)$ as subspaces; this polar space has rank ν.

(b) The polar space formed by the points of a non-degenerate quadric in $PG(2\nu, \mathbb{F}_q)$, $\nu \geq 2$, together with the projective flats lying on the quadric as subspaces; this polar space has rank ν.

(c) The polar space formed by the points of a non-degenerate hyperbolic quadric in $PG(2\nu+1, \mathbb{F}_q)$, $\nu \geq 1$, together with the projective flats lying on the quadric as subspaces; this polar space has rank $\nu+1$.

(d) The polar space formed by the points of a non-degenerate elliptic quadric in $PG(2\nu+1, \mathbb{F}_q)$, $\nu \geq 2$, together with the projective flats lying on the quadric as subspaces; this polar space has rank ν.

(e) The polar space formed by the points of a non-degenerate Hermitian variety in $PG(n-1, \mathbb{F}_q)$, $n \geq 4$, together with the projective flats lying on the Hermitian variety as subspaces; this polar space has rank $[\frac{n}{2}]$.

Proof. (a) is actually Theorem 3.33. (b), (c), and (d) follows from Theorems 6.39 and 7.52. (e) is Theorem 5.33. □

In Section 4.5 we have also constructed some polar spaces from pseudo-symplectic polarities and proved that they are isomorphic to the polar space arising from a symplectic polarity. Thus they are not new.

It was pointed out in Section 3.5 that the definition of the polar space is due to J. Tits (cf. Tits, 1974). Moreover, a complete classification of the polar spaces of rank ≥ 3 has been obtained by him. Specializing his result to the finite case we have

Theorem B.2. Let $\mathcal{P}$ be a finite polar space of rank ≥ 3. Then $\mathcal{P}$ is isomorphic to one of (a), (b), (c), (d), (e) listed in Theorem B.1. □

For the proof of Theorem B.2, see Tits (1974).

However, for finite polar space of rank 2, a complete classification has not yet been achieved. Polar spaces of rank 2 are also called generalized quadrangles, which will be studied in next section.

B.2 Generalized Quadrangles

Let $Q = (P, L, I)$ be an incidence structure. It is called a *generalized quadrangle*, if the following four axioms are satisfied:

GQ1 Any line contains at least two distinct points;

GQ2 Any two distinct points are contained in at most one line;

GQ3 Given a line l and a point p not contained in l, there exists a unique line l' and a unique point p' such that l' contains p and p' is contained in both l and l';

GQ4 There exists a pair of non-concurrent lines.

GQ1, GQ2, GQ3, and GQ4 are called the axioms of generalized quadrangles.

As in Section A.2, by GQ1 and GQ2 we can identify any line $l \in L$ with the point set $[l]$ consisting of all the points contained in l. Thus the following lines may be understood to be certain subsets of P.

At first it should be remarked that in any generalized quadrangle there does not exist a triangle, i.e., the incidence structure consisting of three non-collinear points p, q, and r and three lines pq, qr, and rp. Here by pq we mean the line incident with both p and q. In fact, the existence of a triangle would contradict GQ3.

Example B.1. The simplest example of generalized quadrangles is the quadrangle, i.e., the incidence structure consisting of four points p_1, p_2, p_3, and p_4, no three of which are collinear, and four lines p_1p_2, p_2p_3, p_3p_4, and p_4p_1. More precisely, let $P = \{p_1, p_2, p_3, p_4\}, L = \{l_1, l_2, l_3, l_4\}$, and $I = \{(p_i, l_i), (p_{i+1}, l_i) \mid i = 1, 2, 3, 4 \text{ and } p_5 = p_1\}$, then (P, L, I) is called a quadrangle and is clearly a generalized quadrangle. □

Since generalized quadrangles are incidence structures, we can define the *dual statement* of a statement in a generalized quadrangle by interchanging the words "points" and "lines", and "is contained in" and "contains". The dual of GQi will be denoted by GQi', $i = 1, 2, 3, 4$. Then we have

Theorem B.3. In any generalized quadrangle the dual statements of GQ1, GQ2, GQ3, and GQ4 hold. Hence the *principle of duality* holds in any generalized quadrangle, i.e., if a statement holds in a generalized quadrangle, so is its dual statement.

Proof. The dual of GQ1 is

GQ1′. Any point is contained in at least two distinct lines.

It follows from GQ3 that any point lies on at least one line. Let p be a point lying on a line l. If there is a line l_1 not intersecting l, then by GQ3 there is a line l_2 containing p and $l_2 \cap l_1 \neq \phi$. Since $l \cap l_1 = \phi$, we have $l_2 \neq l$. Hence p lies on two distinct lines l and l_2. Now assume that all the lines $\neq l$ intersect l. If one of those lines $\neq l$ goes through p, we are finished. Suppose that all the lines $\neq l$ do not pass through p. By GQ4 there is a pair of non-concurrent lines l_1 and l_2. By assumption none of l_1 and l_2 is l, both l_1 and l_2 intersect l, and do not pass through p. By GQ1 and GQ2 there is a point q lying on l_1 but not lying on l. Then by GQ3 there is a unique line l_3 containing q and $l_3 \cap l_2 \neq \phi$. Since $q \notin l$, $l_3 \neq l$. By assumption $l_3 \cap l \neq \phi$. But $q \notin l$ and there are two lines l_1 and l_3, both of which contains q and intersect l; this contradicts GQ3. Therefore GQ1′ is proved.

The dual of GQ2 is

GQ2′. Any two distinct lines contains at most a point in common.

Clearly, GQ2′ is an immediate consequence of GQ2.

Clearly GQ3 is self-dual, i.e., the dual GQ3′ of GQ3 is GQ3 itself. Hence GQ3′ holds automatically.

The dual of GQ4 is

GQ4′. There exists a pair of non-adjacent points.

By GQ4 there exists a pair of non-concurrent lines l_1 and l_2. Let p be a point lying on l_1 and q_1, q_2 be two distinct points on l_2. By GQ3 either p and q_1 or p and q_2 are non-adjacent. □

Let $Q = (P, L, I)$ be a generalized quadrangle. By Theorem B.3, GQ1′, GQ2′, GQ3′, and GQ4′, hold in Q. Now let $P' = L, L' = P$, and define $I' \subset P' \times L'$ so that $(l, p) \in I'$ if and only if $(p, l) \in I$. Then in $Q' = (P', L', I')$, GQ1, GQ2, GQ3, and GQ4 hold. Thus Q' is also a generalized quadrangle, called *the dual of* Q. We have proved

Corollary B.4. The dual of a generalized quadrangle is a generalized

quadrangle. □

Two generalized quadrangles $Q = (P, L, I)$ and $Q' = (P', L', I')$ are said to be *isomorphic*, if their underlying incidence structures are isomorphic, i.e., if there are two bijective maps

$$\begin{array}{l} P \to P' \\ p \mapsto p' \end{array} \quad \text{and} \quad \begin{array}{l} L \to L' \\ l \mapsto l' \end{array}$$

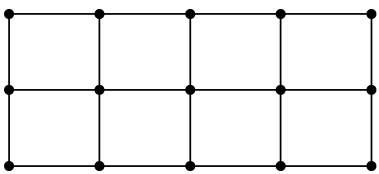
such that $(p, l) \in I$ if and only if $(p', l') \in I'$. As the case of projective planes, a bijective map from P to P' is also called an *isomorphism* from Q to Q', if it carries lines into lines. In fact, it induces an isomorphism from Q to Q' in an obvious way as Theorem A.5 shows.

A generalized quadrangle is called *self-dual*, if its incidence structure is self-dual, i.e., if it is isomorphic to its dual.

Example B.2. Clearly a quadrangle is self-dual and is the only generalized quadrangle having only four points, i.e., any generalized quadrangle having only four points is isomorphic to the quadrangle. □

A generalized quadrangle $Q = (P, L, I)$ is called a *finite generalized quadrangle*, if P is a finite set. If P is finite, we deduce immediately that L is also finite. From now on we restrict ourselves to finite generalized quadrangles and when we talk about generalized quadrangles we always mean that they are finite generalized quadrangles.

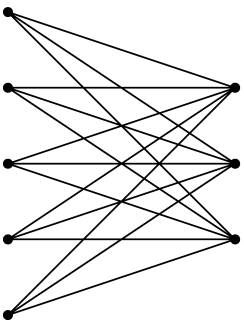
Example B.3. The following figure gives a generalized quadrangle with fifteen points and eight lines, in which every point lies on two lines and the lines are partitioned into two sets L_1 and L_2 such that each line intersects every line of the other set and no line of the set containing it, $\mid L_1 \mid = 5$, $\mid L_2 \mid = 3$, each line of L_1 has three points, and each line of L_2 has five points.



□

Example B.4. The dual of the generalized quadrangle in Example B.1 is the generalized quadrangle consisting of eight points and fifteen lines, in which the points are partitioned into two sets P_1 and P_2 such that each point is joined to every point of the other set and no point of the set

containing it, $\mid P_1 \mid = 5$, $\mid P_2 \mid = 3$, each point of P_1 lies on three lines, and each point of P_2 lies on five lines.



□

A generalized quadrangle in which every point lies on two lines is called a *grid*. The dual of a grid is called a *dual grid*. At first we want to determine the structure of the grid and the dual grid. The following lemma is useful.

Lemma B.5. In a generalized quadrangle any two non-adjacent points are on the same number of lines and any two non-concurrent lines contains the same number of points.

Proof. Let p and q be two non-adjacent points. Suppose that p lies on $t+1 \geq 2$ lines. Then by GQ3 q lies on at least $t+1$ lines. If q were on more than $t+1$ lines, then by GQ3 p would be on more than $t+1$ lines, which contradicts our assumption. Therefore q lies also on $t+1$ lines.

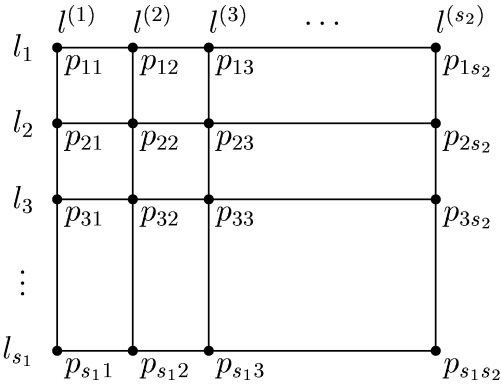
The dual of the above argument proves the second statement of the lemma.

□

Theorem B.6. Let $Q = (P, L, I)$ be a grid, i.e., every point of Q lies on two lines. Then the lines are partitioned into two sets L_1 and L_2 such that each line intersects every line of the other set and no line of the set containing it, and each line of L_1 (resp. L_2) contains $\mid L_2 \mid$ (resp. $\mid L_1 \mid$) points.

Proof. Let us pick up a line of Q, call it l_1 and suppose that l_1 contains s_2 points: $p_{11}, p_{12}, \ldots, p_{1s_2}$, say. By hypothesis, each $p_{1j} (1 \leq j \leq s_2)$ lies on two lines. Denote the line passing through p_{1j} and distinct from l_1 by $l^{(j)} (1 \leq j \leq s_2)$. By GQ3, no two of the $l^{(j)}$'s intersect. And by Lemma B.5, each $l^{(j)}$ contains the same number of points, say, s_1 points.

Let the points lying on $l^{(j)}$ be $p_{ij}(1 \le i \le s_1)$, where p_{1j} is the point of intersection of l_1 and $l^{(j)}$. Clearly, the points $p_{ij}(1 \le i \le s_1,\ 1 \le j \le s_2)$ are $s_1 s_2$ pairwise distinct points.



Through each $p_{i1}(1 \le i \le s_1)$ there passes two lines, one of them is $l^{(1)}$, let the other line be l_i. By GQ3, no two of the l_i's intersect. Since l_1 contains s_2 points, it follows from Lemma B.5 that each l_i contains also s_2 points. Since p_{1j} lies on l_1 and $l_1 \cap l_i = \phi$ for $i \ne 1$, by GQ3 $l^{(j)}$ must intersect l_i. After permuting the first indices of the points $p_{ij}(i = 1, 2, \ldots, s_1)$ we can assume that l_i and $l^{(j)}$ intersect at p_{ij}. Then $p_{i1}, p_{i2}, \ldots, p_{is_2}$ are all the s_2 points on $l_i(1 \le i \le s_1)$. Therefore we have the diagram of the preceding page.

It remains to prove that all the points $p_{ij}(i = 1, 2, \ldots, s_1;\ j = 1, 2, \ldots, s_2)$ and lines $l_i(i = 1, 2, \ldots, s_1)$, $l^{(j)}(j = 1, 2, \ldots, s_2)$ exhaust the points and lines of the grid Q, respectively. Let p be a point of Q. If p lies on l_1, we are finished. If p does not lie on l_1, by GQ3 there is a line passing through p and intersecting l_1, then this line must be one of $l^{(1)}, l^{(2)}, \ldots, l^{(s_2)}$, and we are also finished. Now let l be any line of Q. Pick up any point p on l. By the above proof p must be one of the points $p_{ij}(i = 1, 2, \ldots, s_1;\ j = 1, 2, \ldots, s_2)$. If $p = p_{ij}$, then l is either l_i or $l^{(j)}$.

Now let $L_1 = \{l_1, l_2, \ldots, l_{s_1}\}$ and $L_2 = \{l^{(1)}, l^{(2)}, \ldots, l^{(s_2)}\}$, then the partition $L = L_1 \cup L_2$ meets all the requirements of our theorem. □

The dual of Theorem B.6 is:

Theorem B.7. Let $Q = (P, L, I)$ be a dual grid. Then every line contains two points and the points are partitioned into two sets P_1 and P_2 such that each point is joined to every point of the other set and no point of the set containing it, and each point of P_1 (resp. P_2) lies on $\mid P_2 \mid$ (resp.

$|\ P_1\ |$) lines. □

Let p and q be two points of a generalized quadrangle. We recall that $p \sim q$ (resp. $p \not\sim q$) represents that p and q are adjacent (resp. non-adjacent), and we agree that $p \sim p$ for every $p \in P$.

Theorem B.8. Let $Q = (P, L, I)$ be a generalized quadrangle. Then one of the following is true:

(i) Q is a grid.

(ii) Q is a dual grid.

(iii) There are integers $s \geq 1$ and $t \geq 1$ such that each line contains $s+1$ points and each point lies on $t+1$ lines.

Proof. By GQ4′ there is a pair of points p and q of Q such that $p \not\sim q$. Let p lie on $t+1 \geq 2$ lines, then by Lemma B.5 so does q. We distinguish the following two cases:

Case 1. All points adjacent to p are also adjacent to q. In this case, by GQ3 every line passing through p contains only two points. Let l be any line not passing through p. We assert that l also contains only two points. By GQ3 there is a unique point r on l which is adjacent to p. If l contains more than two points, then l has at least two points r_1 and r_2 such that $r_1 \not\sim p$ and $r_2 \not\sim p$. By GQ1′, there is a line l' passing through p and $l' \neq pr$. Since l' passes through p, it has only two points. Denote the point of l', which is different from p, by s. By GQ3, s does not lie on l. Again by GQ3, $r_1 \sim s$ and $r_2 \sim s$, thus we have a triangle in Q, which is impossible. This proves our assertion. Since every line of Q contains only two points, by definition Q is a dual grid.

Case 2. There is a point r which is adjacent to p but not adjacent to q. Since q lies on $t+1$ lines and $r \not\sim q$, by Lemma B.5 r lies also on $t+1$ lines. By GQ3, any point not equal to p or r can be adjacent to at most one of p and r, hence by Lemma B.5 it lies also on $t+1$ lines. This proves that all points of Q lie on $t+1$ lines.

The dual of the above argument implies that either S is a grid or all lines of Q contains the same number $s+1 \geq 2$ points. □

The structure of the grids and dual grids are very transparent from Theorems B.6 and B.7, respectively. Hence from now on we restrict our attention to generalized quadrangles of type (iii).

If a generalized quadrangle Q is of type (iii), we shall say that it has parameters s and t. Thus by a generalized quadrangle with parameters

s and t we always mean a generalized quadrangle of type (iii). Clearly, if s and t are parameters of a generalized quadrangle, then t and s are parameters of the dual generalized quadrangle.

However, the three types of generalized quadrangles are not mutually exclusive. Let Q be a grid and $L = L_1 \cup L_2$ be the partition asserted in Theorem B.6, then Q falls into type (iii) if and only if $\mid L_1 \mid = \mid L_2 \mid$, and in this case Q has parameters $s = \frac{1}{2} \mid L \mid -1$ and $t = 1$. Similarly, let Q be a dual grid and $P = P_1 \cup P_2$ be the partition asserted in Theorem B.7, then Q falls into type (iii) if and only if $\mid P_1 \mid = \mid P_2 \mid$, and in this case Q has parameters $s = 1$ and $t = \frac{1}{2} \mid P \mid -1$.

In a generalized quadrangle the number of points is usually denoted by v and the number of lines is usually denoted by b. That is, $v = \mid P \mid$ and $b = \mid L \mid$.

Theorem B.9. In any generalized quadrangle with parameters s and t, $v = (s+1)(st+1)$ and $b = (t+1)(st+1)$.

Proof. Fix a line l. There are $v - (s+1)$ points not on l. By GQ3 each point not on l lies on a unique line meeting l. Hence an alternate way to count all the points not on l is to count all the points on the lines meeting l but not lying on l. There are $s+1$ points on l, through each of these $s+1$ points there passes t lines different from l, and on each of these $(s+1)t$ lines there are s points not lying on l. Therefore altogether there are $(s+1)ts$ points not on l. Hence $v - (s+1) = (s+1)ts$, or $v = (s+1)(st+1)$. Dually $b = (t+1)(st+1)$. □

Let us introduce the following notations. For any point p of a generalized quadrangle $Q = (P, L, I)$, define

$$p^{\perp} = \{x \in P \mid x \sim p\}.$$

For $p, q \in P$ and $p \neq q$, the trace of p and q is defined by

$$\text{tr}\,(p,q) = \{p,q\}^{\perp} = \{x \in P \mid x \sim p \text{ and } x \sim q\}.$$

Then we have

Lemma B.10. In a generalized quadrangle $Q = (P, L, I)$ with parameters s and t, we have

$$\mid p^{\perp} \mid = st + s + 1 \qquad \text{for any } p \in P,$$

$$\mid \text{tr}\,(p,q) \mid = \begin{cases} s+1 & \text{for any } p, q \in P \text{ such that } p \sim q \text{ and } p \neq q. \\ t+1 & \text{for any } p, q \in P \text{ such that } p \not\sim q. \end{cases}$$

Proof. Let $p \in P$. The number of lines passing through p is equal to $t+1$ and on each of these lines there are s points $\neq p$. Clearly this gives us precisely $(t+1)s$ points adjacent to p and $\neq p$. With the point p added, we obtain $\mid p^{\perp} \mid = st + s + 1$.

Now let $p, q \in P$. If $p \sim q$ and $p \neq q$, then tr (p, q) is the set of points on the line pq, and therefore $\mid$ tr $(p, q) \mid = s + 1$. Now suppose that $p \not\sim q$. There are $t + 1$ lines passing through p. By GQ3 there is one and only one point on each of these lines, which is adjacent to q. Clearly this gives precisely $t + 1$ points adjacent to both p and q, i.e., $\mid$ tr $(p, q) \mid = t + 1$. □

Example B.5.

Now let us construct an authentication code from a generalized quadrangle with parameters s and t as follows: Take an arbitrary point p. Let the source states be the $t+1$ lines which passes through p, the encoding rules be the points not adjacent to p, and the messages be the points of $p^{\perp} \setminus \{p\}$. Given a source state l, that is a line passing through p and an encoding rule $e \notin p^{\perp}$, by GQ3 there is a unique point m lying on l and adjacent to e. Then we say that the source state l is encoded by e into the message m. Thus we obtain an authentication code with parameters

$$\begin{aligned} \mid S \mid &= t + 1, \\ \mid E \mid &= (s+1)(st+1) - (st+s+1) = s^2 t \\ \mid M \mid &= (st+s+1) - 1 = s(t+1). \end{aligned}$$

Assume that the encoding rules are equally probable, i.e., each one has a probability of occurrence $1/s^2 t$. For an arbitrary message m, suppose that m lies on the line l through p. Then there are t lines passing through m and distinct from l and on each of these t lines there are s points distinct from m. These st points are all the encoding rules which will encode l into m. Therefore, if the opponent plays an impersonation attack, then

$$P_I = \frac{st}{s^2 t} = \frac{1}{s}.$$

Moreover, suppose that after the opponent observed a message m, he sends a message $m' \neq m$ trying to deceive the receiver. There are st possible choices of encoding rules to encode the source state, on which the message m lies on, into m. Clearly $m \not\sim m'$, and by Lemma B.10, $\mid$ tr $(m, m') \mid = t + 1$. But $p \in$ tr (m, m'). Hence there are t possible encoding rules which could encode two different source states into m and m', respectively. Therefore, if the opponent plays a substitution attack,

then

$$P_S = \frac{t}{st} = \frac{1}{s}.$$

□

The parameters s and t of a generalized quadrangle have some numerical restrictions, which will be given in the following two theorems.

Theorem B.11. In any generalized quadrangle with parameters s and t, $(s+t) \mid st(s+1)(t+1)$.

Proof. Let $Q = (P, L, I)$ be a generalized quadrangle with parameters s and t. Label the points of Q as $p_1, p_2, \ldots, p_v$. Define the adjacency matrix A of Q as follows:

$$A = (a_{ij})_{1 \le i,j \le v},$$

where

$$a_{ij} = \begin{cases} 1, & \text{if } i \neq j \text{ and } p_i \sim p_j, \\ 0, & \text{otherwise.} \end{cases}$$

Since through each point p_i there pass $t+1$ lines and on each of these $t+1$ lines lie s points distinct from p_j, $\sum_{i=1}^{v} a_{ij} = (t+1)s$ for all j. That is, the v-dimensional all 1 vector $(1, 1, \ldots, 1)$ is an eigenvector of A with eigenvalue $(t+1)s$. Let J be the $v \times v$ all 1 matrix

$$J = \begin{pmatrix} 1 & 1 & \ldots & 1 \\ 1 & 1 & \ldots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \ldots & 1 \end{pmatrix},$$

then $(1, 1, \ldots, 1)$ is also an eigenvector of J but with eigenvalue v. Let P be a $v \times v$ nonsingular matrix with the all 1 vector as its first row, then

$$PAP^{-1} = \begin{pmatrix} (t+1)s & 0 & \ldots & 0 \\ * & & & \\ \vdots & & A_1 & \\ * & & & \end{pmatrix}, PJP^{-1} = \begin{pmatrix} v & 0 & \ldots & 0 \\ * & & & \\ \vdots & & J_1 & \\ * & & & \end{pmatrix}.$$

Since J is of rank 1, we have necessarily $J_1 = 0$.

We want to find the other eigenvalues of A, i.e., those of A_1. We try to find some simple polynomials satisfied by A. For this purpose we compute A^2. Let

$$A^2 = (c_{ij})_{1 \le i,j \le v},$$

then it follows from Lemma B.10 that

$$c_{ij} = \begin{cases} (t+1)s, & \text{for } i = j, \\ t+1, & \text{for } i \neq j \text{ and } p_i \not\sim p_j, \\ s-1, & \text{for } i \neq j \text{ and } p_i \sim p_j. \end{cases}$$

Therefore

$$A^2 - (s-t-2)A - (t+1)(s-1)I^{(v)} = (t+1)J.$$

Consequently

$$A_1^2 - (s-t-2)A_1 - (t+1)(s-1)I^{(v-1)} = 0.$$

Thus the other eigenvalues of A are roots of the quadratic equation

$$\lambda^2 - (s-t-2)\lambda - (t+1)(s-1) = 0.$$

Solving, we obtain

$$\lambda_1 = s-1, \ \lambda_2 = -t-1.$$

Let the multiplicities of λ_1 and λ_2 be m_1 and m_2, respectively. Then

$$1 + m_1 + m_2 = v,$$

and

$$m_1(s-1) + m_2(-t-1) + (t+1)s = \text{ tr } A = 0.$$

Eliminating m_2 from the above two equalities, we obtain

$$m_1(s-1) + (v - m_1 - 1)(-t-1) + (t+1)s = 0.$$

Therefore

$$m_1 = \frac{st(s+1)(t+1)}{s+t}.$$

But m_1 must be an integer, hence $(s+t) \mid st(s+1)(t+1)$. □

Theorem B.12. In any generalized quadrangle with parameters s and t, if $s > 1$, then $t \leq s^2$. Dually, if $t > 1$, then $s \leq t^2$.

In order to prove this theorem, we prove the following lemma first.

Lemma B.13. Let $t_1, t_2, \ldots, t_d$ be d real numbers, where d is a positive integer. Then

$$d \sum_{i=1}^{d} t_i^2 \geq \left(\sum_{i=1}^{d} t_i \right)^2 .$$

Proof. Set $\overline{t} = (\sum_{i=1}^{d} t_i)/d$. Then

$$\begin{aligned} 0 \leq \sum_{i=1}^{d} (\overline{t} - t_i)^2 &= \sum_{i=1}^{d} (\overline{t}^2 - 2\overline{t}t_i + t_i^2) = d\overline{t}^2 - 2\overline{t} \sum_{i=1}^{d} t_i + \sum_{i=1}^{d} t_i^2 \\ &= \left(\sum_{i=1}^{d} t_i \right)^2 /d - 2 \left(\sum_{i=1}^{d} t_i \right)^2 /d + \sum_{i=1}^{d} t_i^2, \end{aligned}$$

so

$$d \sum_{i-1}^{d} t_i^2 \geq \left(\sum_{i=1}^{d} t_i \right)^2 . \qquad \square$$

Now let us come to the proof of Theorem B.12.

Proof of Theorem B.12. Let p and q be a pair of non-adjacent points of a generalized quadrangle $Q = \{P, L, I\}$ with parameters s and t. Let

$$X = \{x \in P \mid x \not\sim p \text{ and } x \not\sim q\},$$

then by Lemma B.10 and Theorem B.9

$$\begin{aligned} \mid X \mid &= v - 2 - 2(t+1)s + (t+1) \\ &= (s+1)(st+1) - 2 - (2s-1)(t+1) \\ &= s(st-1) - (s-1)t. \end{aligned}$$

Now assume that $s > 1$, then $\mid X \mid > 0$. Set $\mid X \mid = d$, and label the elements of X as $x_1, x_2, \ldots, x_d$. For each $x_i \in X$, let

$$X_i = \{y \in P \mid y \sim p, y \sim q, \text{ and } y \sim x_i\}^1, 1 \leq i \leq d,$$

and set $\mid X_i \mid = t_i$.

Count the number of ordered pairs (x, y) where $x \in X$ and $y \sim p, q$, and x in two different ways. Firstly, for a fixed $x_i \in X, y \sim p, q$, and x_i if and only if $y \in X_i$, and $\mid X_i \mid = t_i$; as x_i varies in X we get $\sum_{i=1}^{d} t_i$ ordered pairs (x, y) such that $x \in X$ and $y \sim p, q$, and x. Secondly, by Lemma B.10 there are $t+1$ points which are adjacent to both p and q, let y be any one of these $t+1$ points, then through y there are $t-1$ lines not passing through p or q and on each of these $t-1$ lines there are s points x different from y; all these $(t-1)s$ points are not adjacent to both p and q; this gives altogether $(t+1)(t-1)s$ pairs (x, y) such that $x \in X$ and $y \sim p, q, x$. Therefore

$$\sum_{i=1}^{d} t_i = (t+1)(t-1)s. \tag{B.1}$$

Now count the number of ordered triples (x, y, y'), where $x \in X$, both y and y' are adjacent to p, q, and x, and $y \neq y'$ in two different ways. On the one hand, for a fixed $x_i \in X$, $y, y' \sim p, q$, and x_i if and only if $y, y' \in X_i$. But $y \neq y'$ and $\mid X_i \mid = t_i$. Thus we get $t_i(t_i - 1)$ ordered triples (x_i, y, y') satisfying the required properties. On the other hand, there are $(t+1)t$ possible choices of ordered pairs (y, y') such that both y and y' are adjacent to p and q and $y \neq y'$. For each such ordered pair (y, y') we have necessarily $y \not\sim y'$ and hence by Lemma B.10 there are $t-1$ points adjacent to both y and y', and distinct from p and q. By GQ3, any of these $t-1$ points is not adjacent to p and not adjacent to q. This gives altogether $(t+1)t(t-1)$ ordered triples (x, y, y') satisfying the required properties. Therefore

$$\sum_{i=1}^{d} t_i(t_i - 1) = (t+1)t(t-1). \tag{B.2}$$

Combining (B.1) and (B.2) we obtain

$$\sum_{i=1}^{d} t_i^2 = (t+1)(t-1)(t+s).$$

[1] A triple of pairwise non-adjacent points in a generalized quadrangle is called a *triad*. Given a triad T, define $T^{\perp} = \cap_{x \in T} x^{\perp}$ and call the points in $T^{\perp}$ the *centers* of T. Thus for each $x_i \in X$, $\{p, q, x_i\}$ is a triad and X_i is the set of centers of the triad $\{p, q, x_i\}$.

By Lemma B.13 and (B.1) we obtain

$$d(t+1)(t-1)(t+s) \geq ((t+1)(t-1)s)^2$$

or

$$d(s+t) \geq (t^2-1)s^2.$$

Substituting $d = s(st-1) - (s-1)t$ into the above inequality, we obtain

$$d(s+t) - (t^2-1)s^2 = t(s-1)(s^2-t) \geq 0.$$

Since $s > 1$, $s^2 \geq t$. □

Corollary B.14. In any generalized quadrangle with parameters $s > 1$ and t, $s^2 = t$ if and only if every triad has a constant number of centers, in which case this constant number is $s+1$.

Proof. We follow the notation of the proofs of Theorem B.12 and Lemma B.13. By the proof of Theorem B.12 $s^2 = t$ if and only if $d\sum_{i=1}^{d} t_i^2 = (\sum_{i=1}^{d} t_i)^2$, hence by the proof of Lemma B.13 if and only if $\overline{t} = t_i$ for all $i = 1, 2, \ldots, d$. If $\overline{t} = t_i$ for all i, then (B.1) becomes

$$d\overline{t} = (t+1)(t-1)s.$$

Substituting $d = s(st-1) - (s-1)t$ and $t = s^2$ into the above equality, we obtain $\overline{t} = s+1$. □

Corollary B.15. In any generalized quadrangle with parameters s and t, if $s > 1$ and $t \neq s^2$, then $t \leq s^2 - s$.

Proof. From Theorem B.12 we can assume that $t = s^2 - x$, where x is a positive integer. From Theorem B.11 we deduce

$$s(s^2-x)(s+1)(s^2-x+1) \equiv 0 \pmod{s+s^2-x}.$$

Hence it follows that

$$x(x-2s) \equiv 0 \pmod{s+s^2-x}.$$

If $x \geq 2s$, then $t = s^2 - x \leq s^2 - 2s < s^2 - s$. On the other hand, if $x < 2s$, then $2s - x > 0$ and $x(2s-x) \geq s + s^2 - x$, which forces $x = s$ or $s+1$. Consequently, we have $t \leq s^2 - s$. □

As applications of Theorems B.11, B.12, and their corollaries, let us consider generalized quadrangles with parameters $s = 2$ or 3. When $s = 2$, by Theorem B.12 and Corollary B.15, the only possible values of t are 2 and 4. When $s = 3$, by these theorems and corollaries the only possible values of t are $3, 5, 6$, and 9. It is known that for $s = 2, t = 2$, or $s = 2, t = 4$, or $s = 3, t = 5$, or $s = 3, t = 9$, there is a unique generalized quadrangle, there exist no generalized quadrangles with parameters $s = 3, t = 6$, and there exist exactly two non-isomorphic generalized quadrangles with parameters $s = 3, t = 3$, which are dual to each other. The proofs of these results will not be included in this book, interested reader may consult Payne and Thas 1984.

In Chapters 3–7 we have already constructed three families of generalized quadrangles and computed their parameters. They are

I	$W(q)$	:	$s = t = q$.
II	$H(3, q^2)$	:	$s = q^2, t = q$;
	$H(4, q^2)$	:	$s = q^2, t = q^3$.
III	$Q(3, q)$	:	$s = q, t = 1$;
	$Q(4, q)$	:	$s = t = q$;
	$Q(5, q)$	:	$s = q, t = q^2$.

The q in I, II, and III denotes a prime power. Remember that we have used the same symbols $Q(d, q)$, where $d = 3, 4$, or 5, for the generalized quadrangles constructed in Chapters 6 and 7, and now we regard them as generalized quadrangles of the same family. The above three families are called the *classical generalized quadrangles*. Clearly $Q(3, q)$ is a grid. Moreover, we know that $W(q)$ is isomorphic to the dual of $Q(4, q)$, $Q(5, q)$ is isomorphic to the dual of $H(3, q^2)$, and $W(q)$ is self-dual if and only if q is even. The proofs of these results can be found in Payne and Thas 1984.

Besides the classical ones there are many non-classical constructions of generalized quadrangles. Interested readers may consult Payne and Thas (1984) and Payne (1990).

Finally, it should be pointed out that the definition of the generalized quadrangle and the construction of the classical generalized quadrangles are due to J. Tits (cf. Tits, 1959). Example B.5 is due to M. De Sorte (cf. De Sorte, 1988). Theorem B.11 is due to R. C. Bose (cf. Bose, 1972). Theorem B.12 is due to D. Higman, but the present proof is due to P. Cameron (cf. Higman 1971 and Cameron, 1974).

Appendix C
Critical Problems

C.1 A Critical Problem in Finite Vector Spaces

Critical problems are problems of current interest and have a vast literature. As applications of the Anzahl theorems in the geometry of classical groups over finite fields we restrict our study to critical problems in finite vector spaces on which some classical groups act. In what follows we need some concepts from matroid theory and the readers may consult Welsh (1976) or Oxley (1992).

Let us begin with a critical problem in a finite vector space on which the general linear group acts.

Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$, where n is a positive integer. A subspace P of $\mathbb{F}_q^{(n)}$ is said to *distinguish* S if $P \cap S = \phi$. The *critical exponent* of S, denoted by $c(S, \mathbb{F}_q^{(n)})$, is defined as the minimum non-negative integer s such that there exists a subspace of dimension $n - s$ distinguishing S. It can also be defined as the minimum non-negative integer s such that there exists an s-tuple of $(n-1)$-dimensional subspaces $(H_1, H_2, \ldots, H_s)$ distinguishing S, i.e., $(\cap_{i=1}^{s} H_i) \cap S = \phi$. By convention, we regard the intersection of 0 $(n-1)$-dimensional subspace to be $\mathbb{F}_q^{(n)}$. Since an $(n-s)$-dimensional subspace can be expressed as an intersection of s $(n-1)$-dimensional subspaces, the equivalence of these two definitions is clear.

A map f from $\mathbb{F}_q^{(n)}$ to $\mathbb{F}_q$ is called a *linear map* if

$$f(\lambda x + \mu y) = \lambda f(x) + \mu f(y) \text{ for all } x, y \in \mathbb{F}_q^{(n)} \text{ and } \lambda, \mu \in \mathbb{F}_q.$$

Denote the kernel of the linear map f by $\operatorname{Ker} f$. It is clear that if f is a non-zero linear map $\operatorname{Ker} f$ is an $(n-1)$-dimensional subspace of $\mathbb{F}_q^{(n)}$ and that if f is the zero map $\operatorname{Ker} f = \mathbb{F}_q^{(n)}$. The critical exponent of a set

of non-zero vectors S in $\mathbb{F}_q^{(n)}$ can also be defined as the minimum non-negative integer s such that there exists an s-tuple of linear maps from $\mathbb{F}_q^{(n)}$ to $\mathbb{F}_q$, $(f_1, f_2, \ldots, f_s)$, distinguishing S, i.e., $(\cap_{i=1}^{s} \operatorname{Ker} f_i) \cap S = \phi$. By convention we also regard the intersection of the kernel of 0-tuple of linear map as $\mathbb{F}_q^{(n)}$. The equivalence of the third definition with the preceding two is also clear.

Let X be a non-empty set of vectors of $\mathbb{F}_q^{(n)}$. Denote by $\langle X \rangle$ the subspace spanned by X. The *rank* of X, denoted by $r(X)$, is defined as $\dim \langle X \rangle$. If $X = \phi$, we agree that $\langle X \rangle = \phi$ and $r(X) = 0$.

Lemma C.1. Let X be a set of vectors in $\mathbb{F}_q^{(n)}$. Then the number of linear maps f from $\mathbb{F}_q^{(n)}$ to $\mathbb{F}_q$ such that $\operatorname{Ker} f \supseteq X$ is $q^{n-r(X)}$.

Proof. Let $v_1, \ldots, v_{r(X)}$ be a maximal set of linearly independent vectors in X. We can supplement them with vectors $v_{r(X)+1}, \ldots v_n$ in $\mathbb{F}_q^{(n)}$ such that $v_1, \ldots v_{r(X)}, v_{r(X)+1}, \ldots, v_n$ form a basis of $\mathbb{F}_q^{(n)}$. Let $f : \mathbb{F}_q^{(n)} \to \mathbb{F}_q$ be a linear map, then f is uniquely determined by its images on $v_1, \ldots, v_{r(X)}, v_{r(X)+1}, \ldots, v_n$. Clearly, $\operatorname{Ker} f \supseteq X$ if and only if $f(v_1) = \cdots = f(v_{r(X)}) = 0$ and $f(v_{r(X)+1}), \ldots f(v_n)$ are arbitrary elements in $\mathbb{F}_q$. The number of such f is $q^{n-r(X)}$. □

Theorem C.2. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$, $M(S)$ be the matroid on S defined by linear independence of vectors, $L(M(S))$ be the lattice of flats of $M(S)$, μ be the Möbius function on $L(M(S))$, and $\chi(L(M(S)), \lambda)$ be the characteristic polynomial of $L(M(S))$. Then the number of s-tuples of linear maps from $\mathbb{F}_q^{(n)}$ to $\mathbb{F}_q$ distinguishing S is

$$q^{s(n-r(S))} \chi(L(M(S)), q^s).$$

Proof. Consider first the case $s > 0$. Let X be a flat of $M(S)$. Denote by $g(s, X)$ the number of s-tuples of linear maps $(f_1, f_2, \ldots, f_s)$ such that $\cap_{i=1}^{s} \operatorname{Ker} f_i \supseteq X$. By Lemma C.1

$$g(s, X) = q^{s(n-r(X))}.$$

Denote by $f(s, X)$ the number of s-tuples of linear maps $(f_1, f_2, \ldots, f_s)$ such that $(\cap_{i=1}^{s} \operatorname{Ker} f_i) \cap S = X$. Since X is the intersection of the vector subspace $\cap_{i=1}^{s} \operatorname{Ker} f_i$ and S, it is a flat in $M(S)$. Then

$$g(s, X) = \sum_{Y \in L(M(S)) : Y \supseteq X} f(s, Y).$$

By Möbius inversion

$$f(s,Y) = \sum_{X \in L(M(S)): X \supseteq Y} \mu(Y,X) g(s,X).$$

Since S consists of non-zero vectors, the empty set ϕ is a flat in $M(S)$. In the above formula, let $Y = \phi$, then we have

$$\begin{aligned} f(s,\phi) &= \sum_{X \in L(M(S))} \mu(\phi,X) g(s,X) \\ &= \sum_{X \in L(M(S))} \mu(\phi,X) q^{s(n-r(X))} \\ &= q^{s(n-r(S))} \sum_{X \in L(M(S))} \mu(\phi,X) q^{s(r(S)-r(X))} \\ &= q^{s(n-r(S))} \chi(L(M(S)), q^s) \end{aligned}$$

and $f(s,\phi)$ is the number of s-tuples of linear maps $(f_1, f_2, \ldots, f_s)$ such that $(\cap_{i=1}^{s} \operatorname{Ker} f_i) \cap S = \phi$.

Then consider the case $s = 0$. From our convention we deduce that the number of 0-tuples of linear maps distinguishing S is 0 or 1 when $S \neq \phi$ or $S = \phi$, respectively. It is clear that $\chi(L(M(S)), 1) = 0$ or 1 when $S \neq \phi$ or $S = \phi$, respectively. Hence our theorem also holds for $s = 0$. □

Corollary C.3. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$ and $M(S)$, $L(M(S))$, $\chi(L(M(S)), \lambda)$ be the same as in Theorem C.2. Then

$$c(S, \mathbb{F}_q^{(n)}) = \min\{s \mid \chi(L(M(S)), q^s) \neq 0\}.$$

□

It follows from Corollary C.3 that the critical exponent of a set S of non-zero vectors in $\mathbb{F}_q^{(n)}$ depends only on the matroid structure of the set S.

Moreover, we have

Theorem C.4. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$. Then

$$c(S, \mathbb{F}_q^{(n)}) \leq r(S).$$

Proof. If $S = \phi$, then $\mathbb{F}_q^{(n)}$ distinguishes S. Hence $c(\phi, \mathbb{F}_q^{(n)}) = 0$. We also have $r(S) = r(\phi) = 0$ by convention. Therefore $c(\phi, \mathbb{F}_q^{(n)}) = r(\phi) = 0$.

Now let $S \neq \phi$ and let $r(S) = s$, then $\dim \langle S \rangle = s$. There is an $(n-s)$-dimensional subspace P such that $\mathbb{F}_q^{(n)} = P \oplus \langle S \rangle$ which implies $P \cap S = \phi$. Therefore $c(S, \mathbb{F}_q^{(n)}) \leq s = r(S)$. □

We can also use the first two definitions to characterize the critical exponent. First we have

Lemma C.5. Let X be a set of vectors of $\mathbb{F}_q^{(n)}$ and denote the rank of X by $r(X)$. Then the number of k-dimensional subspaces containing X is

$$\begin{bmatrix} n - r(X) \\ n - k \end{bmatrix}_q = \prod_{i=1}^{n-k} \frac{q^{k-r(X)+i} - 1}{q^i - 1}.$$

In particular, the number of $(n-1)$-dimensional subspaces containing X is

$$\begin{bmatrix} n - r(X) \\ 1 \end{bmatrix}_q = \frac{q^{n-r(X)} - 1}{q - 1}.$$

Proof. It is an immediate consequence of Corollary 1.9. □

Theorem C.6. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$, and $M(S)$, $L(M(S))$, $\chi(L(M(S)), \lambda)$ be the same as in Theorem C.2. Then the number of s-tuples of $(n-1)$-dimensional subspaces of $\mathbb{F}_q^{(n)}$ distinguishing S is

$$\frac{1}{(q-1)^s} \sum_{j=0}^{s} (-1)^{s-j} \binom{s}{j} q^{j(n-r(S))} \chi(L(M(S)), q^j).$$

Proof. We consider only the case $s > 0$. Let X be a flat of $M(S)$. Denote by $g(s, X)$ the number of s-tuples of $(n-1)$-dimensional subspaces of $\mathbb{F}_q^{(n)}$ containing X. By Lemma C.5

$$\begin{aligned} g(s, X) &= \left(\frac{q^{n-r(X)} - 1}{q - 1} \right)^s \\ &= \frac{1}{(q-1)^s} \sum_{j=0}^{s} (-1)^{s-j} \binom{s}{j} q^{j(n-r(X))}. \end{aligned}$$

By Möbius inversion, the number of s-tuples of $(n-1)$-dimensional subspaces of $\mathbb{F}_q^{(n)}$ distinguishing S is

$$\sum_{X \in L(M(S))} \mu(\phi, X) g(s, X)$$

$$= \frac{1}{(q-1)^s} \sum_{j=0}^{s} (-1)^{s-j} \binom{s}{j} q^{j(n-r(S))} \chi(L(M(S)), q^j).$$

□

Corollary C.7. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$ and $M(S)$, $L(M(S))$, $X(L(M(S)), \lambda)$ be the same as in Theorem C.2. Then

$$c(S, \mathbb{F}_q^{(n)}) = \min\{s \mid \sum_{j=0}^{s} (-1)^{s-j} \binom{s}{j} q^{j(n-r(S))} \chi(L(M(S)), q^j) \neq 0\}.$$

□

Theorem C.8. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$, and $M(S)$, $L(M(S))$, $\chi(L(M(S)), \lambda)$ be the same as in Theorem C.2. Then the number of $(n-s)$-dimensional subspaces of $\mathbb{F}_q^{(n)}$ distinguishing S is

$$\frac{1}{\prod_{j=0}^{s-1}(q^s - q^j)} \sum_{j=0}^{s} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} s \\ j \end{bmatrix}_q q^{(s-j)(n-r(S))} \chi(L(M(S)), q^{s-j}).$$

Proof. Let X be a flat of $M(S)$. Denote by $g(s, X)$ the number of $(n-s)$-dimensional subspaces containing X. By Lemma C.5

$$\begin{aligned} g(s, X) &= \prod_{i=1}^{s} \frac{q^{n-s-r(X)+i} - 1}{q^i - 1} \\ &= \prod_{j=0}^{s-1} \frac{q^{n-r(X)} - q^j}{q^s - q^j}. \end{aligned} \tag{C.1}$$

By mathematical induction on s, we can prove

$$\sum_{j=0}^{s} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} s \\ j \end{bmatrix}_q \lambda^{s-j} = (\lambda - 1)(\lambda - q) \ldots (\lambda - q^{s-1}). \tag{C.2}$$

Substituting (C.2) with $\lambda = q^{n-r(X)}$ into (C.1), we have

$$g(s, X) = \frac{1}{\prod_{j=0}^{s-1}(q^s - q^j)} \sum_{j=0}^{s} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} s \\ j \end{bmatrix} q^{(s-j)(n-r(X))}.$$

By Möbius inversion, the number of $(n-s)$-dimensional subspaces of $\mathbb{F}_q^{(n)}$ distinguishing S is

$$\sum_{X \in L(M(S))} \mu(\phi, X) g(s, X)$$
$$= \frac{1}{\prod_{j=0}^{s-1}(q^s - q^j)} \sum_{j=0}^{s} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} s \\ j \end{bmatrix}_q q^{(s-j)(n-r(S))} \chi(L(M(S)), q^{s-j}).$$

□

Corollary C.9. Let S be a set of non-zero vectors in $\mathbb{F}_q^{(n)}$ and $M(S)$, $L(M(S))$, $\chi(L(M(S)), \lambda)$ be the same as in Theorem C.2. Then

$$c(S, \mathbb{F}_q^{(n)})$$
$$= \min\{s \mid \sum_{j=0}^{s} (-1)^j q^{\binom{j}{2}} \begin{bmatrix} s \\ j \end{bmatrix} q^{(s-j)(n-r(S))} \chi(L(M(S)), q^{s-j}) \neq 0\}.$$

□

C.2 Critical Problems in Finite Unitary Spaces

Let n be a positive integer and $\mathbb{F}_{q^2}^{(n)}$ be the n-dimensional unitary space, i.e., the n-dimensional vector space over $\mathbb{F}_{q^2}$ on which the unitary group $U_n(\mathbb{F}_{q^2})$ of degree n over $\mathbb{F}_{q^2}$ acts.

Let S be a set of non-zero vectors in $\mathbb{F}_{q^2}^{(n)}$. The *unitary critical exponent* of S, denoted by $c_{\text{uni}}(S, \mathbb{F}_{q^2}^{(n)})$, is defined as the minimum non-negative integer s such that there exists an $(n-s)$-dimensional non-isotropic subspace P distinguishing S, i.e., $P \cap S = \phi$. Since an $(n-s)$-dimensional non-isotropic subspace can be expressed as an intersection of s $(n-1)$-dimensional non-isotropic subspaces, the unitary critical exponent of S can also be defined as the minimum non-negative integer s such that there exists an s-tuple of $(n-1)$-dimensional non-isotropic subspaces $(H_1, H_2, \ldots, H_s)$ distinguishing S, i.e., $(\cap_{i=1}^{s} H_i) \cap S = \phi$.

Let X be a non-empty set of vectors in $\mathbb{F}_{q^2}^{(n)}$. If the subspace $U = \langle X \rangle$ spanned by X is of type (m, r), where m is the dimension of U and r is the rank of $U\,{}^t\overline{U}$, then X is also said to be of type (m, r). By Theorem 5.7, $2r \leq 2m \leq n + r$. We also write $m = m(X)$ and $r = r(X)$ for simplicity. If $X = \phi$, we agree that $\langle X \rangle = \phi$ and $m(X) = r(X) = 0$.

Lemma C.10. Let X be a set of vectors of type (m, r) in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$, where $2r \leq 2m \leq n + r$, and let k be a positive integer $\leq n$. Then the number of k-dimensional non-isotropic subspaces containing X equals

$$q^{(n-k)(k-r)} \prod_{i=1}^{n-k} \frac{q^{k+r-2m+i} - (-1)^{k+r-2m+i}}{q^i - (-1)^i}.$$

Proof. When $X = \phi$, our lemma follows from Corollary 5.21. In fact, the number of k-dimensional non-isotropic subspaces containing $X = \phi$ is the number $N(k, k; n)$ of k-dimensional non-isotropic subspaces, which is equal to

$$\begin{aligned} N(k, k; n) &= q^{k(n-k)} \prod_{i=1}^{k} \frac{q^{n-k+i} - (-1)^{n-k+i}}{q^i - (-1)^i} \\ &= q^{k(n-k)} \prod_{i=1}^{n-k} \frac{q^{k+i} - (-1)^{k+i}}{q^i - (-1)^i}. \end{aligned}$$

Now we assume that $X \neq \phi$. Let P be a k-dimensional non-isotropic subspace in the unitary space $\mathbb{F}_{q^2}^{(n)}$. Then $P \supseteq X$ if and only if $P \supseteq < X >$. If $k < 2m - r$, by Theorem 5.36 the number of k-dimensional non-isotropic subspaces containing X is 0 and the formula in Lemma C.10 also gives 0. If $k \geq 2m - r$, our lemma follows from Theorem 5.37. □

Corollary C.11. Let X be a set of vectors of type (m, r) in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$, where $2r \leq 2m \leq n+r$. Then the number of $(n-1)$-dimensional non-isotropic subspaces containing X equals

$$\frac{q^{n-r-1}(q^{n+r-2m} - (-1)^{n+r-2m})}{q+1}.$$

□

Theorem C.12. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$, $M(S)$ be the matroid on S defined by linear independence of vectors, $L(M(S))$ be the lattice of flats of the matroid $M(S)$, and μ be the Möbius function on $L(M(S))$. Then the number of $(n-s)$-dimensional non-isotropic subspaces distinguishing S equals

$$\sum_{X \in L(M(S))} \mu(\phi, X) q^{s(n-s-r(X))} \prod_{i=1}^{s} \frac{q^{n-s+r(X)-2m(X)+i} - (-1)^{n-s+r(X)-2m(X)+i}}{q^i - (-1)^i}.$$

Proof. Let X be a flat of $M(S)$ and assume that X is of type $(m(X), r(X))$. Denote by $g(s, X)$ the number of $(n-s)$-dimensional non-isotropic subspaces containing X. By Lemma C.10

$$g(s,X) = q^{s(n-s-r(X))} \prod_{i=1}^{s} \frac{q^{n-s+r(X)-2m(X)+i} - (-1)^{n-s+r(X)-2m(X)+i}}{q^i - (-1)^i}.$$

Denote by $f(s, X)$ the number of $(n-s)$-dimensional non-isotropic subspace P such that $P \cap S = X$. Then

$$g(s,X) = \sum_{Y \in L(M(S)): Y \supseteq X} f(s,Y).$$

By Möbius inversion,

$$f(s,Y) = \sum_{X \in L(M(S)): X \supseteq Y} \mu(Y,X) g(s,X).$$

For $Y = \phi$, $f(s, \phi)$ is the number of $(n-s)$-dimensional non-isotropic subspaces distinguishing S. The theorem is proved. □

Corollary C.13. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$, and $M(S)$, $L(M(S))$, μ be as in Theorem C.12. Then

$$c_{\mathrm{uni}}(S, \mathbb{F}_{q^2}^{(n)}) = \min \left\{ s \mid \sum_{X \in L(M(S))} \mu(\phi, X) q^{s(n-s-r(X))} \times \prod_{i=1}^{s} \frac{q^{n-s+r(X)-2m(X)+i} - (-1)^{n-s+r(X)-2m(X)+i}}{q^i - (-1)^i} \neq 0 \right\}.$$

□

In a similar way from Corollary C.11 we deduce

Theorem C.14. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$. Then the number of s-tuples of $(n-1)$-dimensional non-isotropic subspaces distinguishing S is

$$\sum_{X \in L(M(S))} \mu(\phi, X) \left(\frac{q^{n-r(X)-1}(q^{n+r(X)-2m(X)} - (-1)^{n+r(X)-2m(X)})}{q+1} \right)^s.$$

□

Corollary C.15. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$. Then

$$c_{\mathrm{uni}}(S, \mathbb{F}_{q^2}^{(n)}) = \min \left\{ s \mid \sum_{X \in L(M(S))} \mu(\phi, X) \times \left(\frac{q^{n-r(X)-1}(q^{n+r(X)-2m(X)} - (-1)^{n+r(X)-2m(X)})}{q+1} \right)^s \neq 0 \right\}.$$

□

It follows from Corollaries C.13 or C.15 that the unitary critical exponent of a set S of non-zero vectors in the unitary space $\mathbb{F}_{q^2}^{(n)}$ depends only on the matroid structure of S.

The orbit of $(n-1)$-dimensional subspaces in $\mathbb{F}_{q^2}^{(n)}$ under the general linear group $GL_n(\mathbb{F}_{q^2})$ is partitioned into two orbits under the unitary group $U_n(\mathbb{F}_{q^2})$, one of which consists of all $(n-1)$-dimensional non-isotropic subspaces and the other one consists of all $(n-1)$-dimensional isotropic subspaces or subspaces of type $(n-1, n-2)$. The critical exponent of a set S of non-zero vectors in $\mathbb{F}_{q^2}^{(n)}$ is defined as the minimum non-negative integer s such that there exists an s-tuple of $(n-1)$-dimensional subspaces distinguishing S, and the unitary critical exponent of S is defined as the minimum non-negative integer s such that there exists an s-tuple of $(n-1)$-dimensional non-isotropic subspaces distinguishing S. Now we call the latter the *unitary critical exponent of S defined by $(n-1)$-dimensional non-isotropic subspaces* and denote it by $c_{\mathrm{uni}}^{(n)}(S, \mathbb{F}_{q^2}^{(n)})$. In a similar way we define the *unitary critical exponent of S defined by $(n-1)$-dimensional isotropic subspaces* as the minimum non-negative integer s such that there exists an s-tuple of $(n-1)$-dimensional isotropic subspaces distinguishing S and denote it by $c_{\mathrm{uni}}^{(i)}(S, \mathbb{F}_{q^2}^{(n)})$.

Lemma C.16. Let X be a set of vectors of type (m, r) in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$, where $2r \leq 2m \leq n+r$. Then the number of $(n-1)$-dimensional isotropic subspaces containing X is equal to

$$N'(m, r; n-1, n-2; n) = \frac{q^{2(m-r)} \prod_{j=0}^{1} (q^{n-2m+r-j} - (-1)^{n-2m+r-j}) + (q^{2(m-r)} - 1)}{q^2 - 1}.$$

Proof. When $X = \phi$, the number of $(n-1)$-dimensional isotropic subspaces containing X is equal to the number of $(n-1)$-dimensional isotropic subspace and by Theorem 5.19 the latter is equal to

$$N(n-1, n-2; n) = \frac{(q^n - (-1)^n)(q^{n-1} - (-1)^{n-1})}{q^2 - 1}.$$

By convention $m = r = 0$ and then the formula of $N'(m, r; n-1, n-2; n)$ gives

$$N'(0, 0; n-1, n-2; n) = \frac{(q^n - (-1)^n)(q^{n-1} - (-1)^{n-1})}{q^2 - 1}.$$

Therefore our lemma is true for the case $X = \phi$.

Now assume that $X \neq \phi$. Consider first the case $m = r$. By Theorem 5.36 when $m = n$ or $n-1$, the number of $(n-1)$-dimensional isotropic subspaces containing X is zero, and the formula of $N'(m, r; n-1, n-2; n)$ gives also

$$N'(m, r; n-1, n-2; n) = 0.$$

When $m \leq n-2$, by Theorem 5.37 we have the formula of $N'(m, m; n-1, n-2; n)$. Then consider the case $m > r$. The formula of $N'(m, r; n-1, n-2; n)$ is a special case of the formula in Theorem 5.37. □

Theorem C.17. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$. Then the number of s-tuples of $(n-1)$-dimensional isotropic subspaces distinguishing S is

$$\sum_{X \in L(M(S))} \mu(\phi, X) N'(m(X), r(X); n-1, n-2; n)^s.$$

Proof. Similar to the proof of Theorem C.12. □

Corollary C.18. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$. Then

$$c_{\text{uni}}^{(i)}(S, \mathbb{F}_{q^2}^{(n)}) = \min \left\{ s \mid \sum_{X \in L(M(S))} \mu(\phi, X) N'(m(X), r(X); n-1, n-2; n)^s \neq 0 \right\}.$$

□

Finally, we have

Theorem C.19. Let S be a set of non-zero vectors in the n-dimensional unitary space $\mathbb{F}_{q^2}^{(n)}$. Then

$$c_{\text{uni}}^{(n)}(S, \mathbb{F}_{q^2}^{(n)}) \geq c(S, \mathbb{F}_{q^2}^{(n)})$$

and

$$c_{\text{uni}}^{(i)}(S, \mathbb{F}_{q^2}^{(n)}) \geq c(S, \mathbb{F}_{q^2}^{(n)}). \qquad \square$$

The above study of the critical problems in the finite unitary space can be carried over to the finite pseudo-symplectic and orthogonal spaces in a parallel way, but the details will not be repeated. However, for the symplectic space of dimension 2ν over $\mathbb{F}_q$ the set of $(2\nu-1)$-dimensional subspaces forms also an orbit under the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$, thus the symplectic critical exponent will be defined in a different way in the next section.

C.3 A Critical Problem in Finite Symplectic Spaces

As in Section 3.1 let ν be a positive integer,

$$K = \begin{pmatrix} 0 & I^{(\nu)} \\ -I^{(\nu)} & 0 \end{pmatrix},$$

$Sp_{2\nu}(\mathbb{F}_q)$ be the symplectic group of degree 2ν with respect to K over $\mathbb{F}_q$, and $\mathbb{F}_q^{(2\nu)}$ be the 2ν-dimensional symplectic space over $\mathbb{F}_q$, i.e., the 2ν-dimensional vector space $\mathbb{F}_q^{(2\nu)}$ on which the symplectic group $Sp_{2\nu}(\mathbb{F}_q)$ acts.

A set of vectors X in $\mathbb{F}_q^{(2\nu)}$ is called an *isotropic set of vectors* if $uK\,{}^t v = 0$ for all $u, v \in X$. Denote by $\langle X \rangle$ the subspace spanned by X. Clearly, $\langle X \rangle$ is a totally isotropic subspace if X is an isotropic set of vectors. The rank of a set of vectors X is defined as the dimension of $\langle X \rangle$, which is denoted by $r(X)$, and X is called a rank-r set. When $X = \phi$, then $\langle X \rangle = \phi$ and we agree that $r(X) = 0$.

Let S be a set of nonzero vectors in the 2ν-dimensional symplectic space over $\mathbb{F}_q$. A totally isotropic subspace P is said to distinguish S if $P \cap S = \phi$. The symplectic critical exponent of S, denoted by $c_{\text{symp}}(S, \mathbb{F}_q^{(2\nu)})$, is defined as the minimum positive integer $s \leq \nu+1$ such that there exists an $(\nu+1-s)$-dimensional totally isotropic subspace distinguishing S. Since any $(\nu+1-s)$-dimensional totally isotropic subspace is an intersection of

s maximal totally isotropic subspaces, the symplectic critical exponent of S can also be defined as the minimum positive integer s such that there exist an s-tuple of maximal totally isotropic subspaces $(P_1, P_2, \ldots, P_s)$ such that $\cap_{i=1}^{s} P_i$ distinguishes S.

Let S be a set of non-zero vectors in $\mathbb{F}_q^{(2\nu)}$, $M(S)$ be the matroid on S defined by linear independence of vectors, and $L(M(S))$ be the lattice of flats of G. An *isotropic flat* is a flat which is also an isotropic set of vectors. Clearly, subsets of isotropic set of vectors are also isotropic. It follows that the collection of isotropic flats forms an ideal in the lattice $L(M(S))$ and this ideal will be denoted by $L_I(M(S))$.

Lemma C.20. Let X be a rank-r isotropic set of vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$ and k be an integer satisfying $\nu \geq k \geq r$. Then the number of k-dimensional totally isotropic subspaces in $\mathbb{F}_q^{(2\nu)}$ containing X is

$$\prod_{i=0}^{k-r-1} \frac{q^{2\nu-2r-2i}-1}{q^{k-r-i}-1}.$$

Proof. When $r = 0$ our lemma follows from Corollary 3.19. Now assume that $r > 0$. Let P be a k-dimensional totally isotropic subspace. Then $P \supset X$ if and only if $P \supset \langle X \rangle$. Therefore our lemma is a special case of Theorem 3.38. □

Corollary C.21. The number of maximal totally isotropic subspaces in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$ containing a given rank-r isotropic set of vectors is

$$\prod_{i=0}^{\nu-r-1} (q^{\nu-r-i}+1).$$

□

Theorem C.22. Let S be a set of non-zero vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$, $M(S)$ be the matroid on S defined by linear independence of vectors, $L(M(S))$ be the lattice of flats of the matroid G, $L_I(M(S))$ be the ideal of isotropic flats in the lattice $L(M(S))$, and μ be the Möbius function on $L(M(S))$. Then for any positive integer $s \leq \nu + 1$

the number of $(\nu+1-s)$-dimensional totally isotropic subspaces distinguishing S is equal to

$$\sum_{X\in L_I(M(S)):r(X)\le\nu+1-s}\mu(\phi,X)\prod_{i=0}^{\nu-s-r(X)}\frac{q^{2\nu-2r(X)-2i}-1}{q^{\nu+1-s-r(X)-i}-1}.$$

Proof. Let X be a flat of $M(S)$. Denote by $g(s,X)$ the number of $(\nu+1-s)$-dimensional totally isotropic subspaces containing X. If X is isotropic and $r(X)\le\nu+1-s$, then by Lemma C.20

$$g(s,X)=\prod_{i=0}^{\nu-s-r(X)}\frac{q^{2\nu-2r(X)-2i}-1}{q^{\nu+1-s-r(X)-i}-1}.$$

If X is not isotropic or X is isotropic but $r(X)>\nu+1-s$, then

$$g(s,X)=0.$$

Denote by $f(s,X)$ the number of $(\nu+1-s)$-dimensional totally isotropic subspaces P such that $P\cap S=X$. Then

$$g(s,X)=\sum_{Y\in L_I(M(S)):Y\supseteq X}f(s,Y).$$

By Möbius inversion

$$\begin{aligned}f(s,Y)&=\sum_{X\in L_I(M(S)):X\supseteq Y}\mu(Y,X)g(s,X)\\&=\sum_{\substack{X\in L_I(M(S)):X\supseteq Y\\ \text{and } r(X)\le\nu+1-s}}\mu(Y,X)\prod_{i=0}^{\nu-s-r(X)}\frac{q^{2\nu-2r(X)-2i}-1}{q^{\nu+1-s-r(X)-i}-1}.\end{aligned}$$

For $Y=\phi$, $f(s,\phi)$ is the number of $(\nu+1-s)$-dimensional totally isotropic subspaces distinguishing S. The theorem is proved. □

Corollary C.23. Let S be a set of non-zero vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$ and $M(S), L(M(S)), L_I(M(S)), \mu$ be as in Theorem C.22. Then

$$\begin{aligned}&c_{\text{symp}}(S,\mathbb{F}_q^{(2\nu)})\\&\quad=\min\left\{s\mid\sum_{\substack{X\in L_I(M(S))\\ \text{with } r(X)\le\nu+1-s}}\mu(Y,X)\prod_{i=0}^{\nu-s-r(X)}\frac{q^{2\nu-2r(X)-2i}-1}{q^{\nu+1-s-r(X)-i}-1}\neq 0\right\}.\end{aligned}$$

□

In a similar way from Corollary C.21 we deduce

Theorem C.24. Let S be a set of non-zero vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$. Then the number of s-tuples of maximal totally isotropic subspaces distinguishing S is

$$\sum_{X\in L_I(M(S))} \mu(\phi, X) \prod_{i=0}^{\nu-r(X)-1} (q^{\nu-r(X)-i}+1)^s.$$

□

Corollary C.25. Let S be a set of non-zero vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$. Then

$$c_{\mathrm{symp}}(S, \mathbb{F}_q^{(2\nu)}) = \min\left\{ s \mid \sum_{X\in L_I(M(S))} \mu(\phi, X) \prod_{i=0}^{\nu-r(X)-1} q^{\nu-r(X)-i}+1)^s \neq 0 \right\}.$$

□

Moreover, we have

Theorem C.26. Let S be a set of nonzero vectors in the 2ν-dimensional symplectic space $\mathbb{F}_q^{(2\nu)}$. Then

$$c_{\mathrm{symp}}(S, \mathbb{F}_q^{(2\nu)}) \leq \lfloor \tfrac{1}{2} c(S, \mathbb{F}_q^{(2\nu)}) \rfloor + 1,$$

where $\lfloor \frac{1}{2} c(S, \mathbb{F}_q^{(2\nu)}) \rfloor$ is the largest integer $\leq \frac{1}{2} c(S, \mathbb{F}_q^{(2\nu)})$.

Proof. Let $s = c(S, \mathbb{F}_q^{(2\nu)})$ and P be a $(2\nu - s)$-dimensional subspace distinguishing S. Clearly any totally isotropic subspace contained in P is of dimension

$$\geq \begin{cases} \nu - \frac{1}{2}s, & \text{if } s \text{ is even,} \\ \nu - \frac{1}{2}(s-1), & \text{if } s \text{ is odd.} \end{cases}$$

Hence,

$$c_{\mathrm{symp}}(S, \mathbb{F}_q^{(2\nu)}) \leq \begin{cases} \nu + 1 - (\nu - \frac{1}{2}s) = \frac{1}{2}s + 1, & \text{if } s \text{ is even,} \\ \nu + 1 - (\nu - \frac{1}{2}(s-1)) = \frac{1}{2}(s-1) + 1, & \text{if } s \text{ is odd.} \end{cases}$$

That is,

$$c_{\mathrm{symp}}(S, \mathbb{F}_q^{(2\nu)}) = \lfloor \tfrac{1}{2} c(S, \mathbb{F}_q^{(2\nu)}) \rfloor + 1.$$

□

The above discussion of the critical problem in the finite symplectic space can be carried over to the finite pseudo-symplectic, unitary, and orthogonal spaces in a parallel way. The details will not be repeated but we remark that in the finite orthogonal space over a field of characteristic 2, we should use singular sets of vectors and totally singular subspaces to replace the isotropic sets of vectors and totally isotropic subspaces.

C.4 Comments

The study of the critical problem in finite vector spaces was started by H. H. Crapo and G.-C. Rota and Theorems C.1–C.4 were obtained by them (cf. Crapo and Rota, 1970). Theorems C.6–C.9 are due to T. A. Dowling (cf. Dowling, 1971).

Section C.2 is due to the author (cf. Wan, 2001).

Section C.3 is due to J. P. S. Kung (cf. Kung, 1997), but there are corrections in the formulas in Theorem C.22 and Corollary C.23 (cf. Wan, 2001).

Appendix D
Moor-Penrose Generalized Inverses of Matrices over Finite Fields

D.1 Definition and Properties

Let $\mathbb{F}_q$ be a finite field and $\mathcal{M}(m \times n)$ the set of $m \times n$ matrices over $\mathbb{F}_q$. In general we assume m is not necessarily equal to n. As a generalization of the Moor-Penrose generalized inverse of an $m \times n$ matrix over the real or complex field, the Moor-Penrose *generalized inverse*, or in short, the M-P *inverse* of $A \in \mathcal{M}(m \times n)$ is defined in the following way. An $n \times m$ matrix X over $\mathbb{F}_q$ is called an M-P inverse of A if it satisfies the following conditions:

$$\left.\begin{array}{rcl} AXA & = & A, \\ XAX & = & X, \\ {}^t(XA) & = & XA, \\ {}^t(AX) & = & AX. \end{array}\right\} \tag{D.1}$$

The matrix A is called M-P *invertible* if A has an M-P inverse. From the definition we see if X is an M-P inverse of A, then A is also an M-P inverse of X, and ${}^tX$ is an M-P inverse of ${}^tA$.

Let A be an $m \times n$ matrix over $\mathbb{F}_q$. Denote by $L(A)$ the vector subspace spanned by all the rows of A in $\mathbb{F}_q^{(n)}$. Let R be an r-dimensional subspace of $\mathbb{F}_q^{(n)}$ and we fix a matrix representation of the subspace R, which will also be denoted by R. Similarly, let C be an r-dimensional subspace of $\mathbb{F}_q^{(m)}$ and we fix a matrix representation C of the subspace C. Denote by $\mathcal{M}(m \times n, r)$ the set of $m \times n$ matrices of rank r over $\mathbb{F}_q$ and by $\mathcal{M}(m \times n, r, C, R)$ the set of matrices A in $\mathcal{M}(m \times n, r)$ with $L({}^tA) = C$ and $L(A) = R$. Denote by $\mathcal{M}(r, n)$ the set of r-dimensional subspaces of

$\mathbb{F}_q^{(n)}$. Then the following disjoint union is clear:

$$\mathcal{M}(m \times n, r) = \dot{\bigcup_{(C,R) \in \mathcal{M}(r,m) \times \mathcal{M}(r,n)}} \mathcal{M}(m \times n, r, C, R).$$

Thus to study $\mathcal{M}(m \times n, r)$ it is sufficient to study $\mathcal{M}(m \times n, r, C, R)$ for given $C \in \mathcal{M}(r, m)$ and $R \in \mathcal{M}(r, n)$.

Lemma D.1. Let $A \in \mathcal{M}(m \times n, r)$ and $C \in \mathcal{M}(r, m)$, $R \in \mathcal{M}(r, n)$. Then $A \in \mathcal{M}(m \times n, r, C, R)$ if and only if $A = {}^tCaR$ for some $a \in GL_r(\mathbb{F}_q)$. Moreover, the map $a \mapsto {}^tCaR$ is a bijection from $GL_r(\mathbb{F}_q)$ to $\mathcal{M}(m \times n, r, C, R)$.

Proof. If $A = {}^tCaR$ for some $a \in GL_r(R)$, clearly we have $L(A) = R$ and $L({}^tA) = C$. Therefore $A \in \mathcal{M}(m \times n, r, C, R)$. Conversely, assume that $A \in \mathcal{M}(m \times n, r, C, R)$, then rank $A = r$ $L({}^tA) = C$, and $L(A) = R$. There exist $P \in GL_m(\mathbb{F}_q)$ and $Q \in GL_n(\mathbb{F}_q)$ such that

$$A = P \begin{pmatrix} I^{(r)} & 0 \\ 0 & 0 \end{pmatrix} Q. \tag{D.2}$$

Thus

$$R = L(A) = L((I^{(r)} \ \ 0)Q),$$

hence $(I^{(r)} \ \ 0)Q = bR$ for some $b \in GL_r(\mathbb{F}_q)$. Similarly,

$$C = L({}^tA) = L((I^{(r)} \ \ 0)\,{}^tP),$$

hence $(I^{(r)} \ \ 0)\,{}^tP = dC$ for some $d \in GL_r(\mathbb{F}_q)$. Therefore

$$A = P \begin{pmatrix} I^{(r)} & 0 \\ 0 & 0 \end{pmatrix} Q = P \begin{pmatrix} I^{(r)} \\ 0 \end{pmatrix} (I^{(r)} \ \ 0)Q = {}^t(dC)bR = {}^tCaR,$$

where $a = {}^t d\, b \in GL_r(\mathbb{F}_q)$.

Assume that ${}^tCaR = {}^tCbR$, where $a, b \in GL_r(\mathbb{F}_q)$. Then ${}^tC(a-b)R = 0$. Since R is an $r \times n$ matrix of rank r, ${}^tC(a-b) = 0$. Since C is an $r \times m$ matrix of rank r, $a - b = 0$. This proves the injectiveness. □

Lemma D.2. Let $C \in \mathcal{M}(r, m)$, $R = \mathcal{M}(r, n)$, and $A \in \mathcal{M}(m \times n, r, C, R)$. Assume that A is M-P invertible and let X be an M-P inverse of A. Then $X \in \mathcal{M}(n \times m, r, R, C)$.

Proof. We have

$$L(A) = L(AXA) \subseteq L(XA) = L({}^tA{}^tX) \subseteq L({}^tX)$$

and

$$L(X) = L(XAX) \subseteq L(AX) = L({}^tX{}^tA) \subseteq L({}^tA).$$

Thus

$$\dim L(A) \leq \dim L({}^tX) = \dim L(X) \leq \dim L({}^tA) = \dim L(A).$$

Therefore

$$\dim L(A) = \dim L({}^tX) \quad \text{and} \quad \dim L({}^tA) = \dim L(X).$$

Then $R = L(A) = L({}^tX)$ and $C = L({}^tA) = L(X)$. Hence $X \in \mathcal{M}(n \times m, r, R, C)$. □

Lemma D.3. Let $C \in \mathcal{M}(r, m)$, $R \in \mathcal{M}(r, n)$, and $A \in \mathcal{M}(m \times n, r, C, R)$. Then the following hold.

(i) $\operatorname{rank}(A\,{}^tA) = \operatorname{rank}(R\,{}^tR)$,

(ii) $\operatorname{rank}({}^tAA) = \operatorname{rank}({}^tCC)$,

(iii) $\dim(L(A) \cap L(A)^{\perp}) + \operatorname{rank}(R\,{}^tR) = r$,

(iv) $\dim(L({}^tA) \cap L({}^tA)^{\perp}) + \operatorname{rank}(C\,{}^tC) = r$.

Proof. (i) By Lemma D.1 we assume that $A = {}^tCaR$, where $a \in GL_r(\mathbb{F}_q)$. There are $P \in GL_m(\mathbb{F}_q)$ and $Q \in GL_n(\mathbb{F}_q)$ such that (D.2) holds. By the proof of Lemma D.1, $(I^{(r)}\ 0)Q = bR$ and $(I^{(r)}\ 0)\,{}^tP = dC$, where $b, d \in GL_r(\mathbb{F}_q)$. Then

$$\begin{aligned} A\,{}^tA &= ({}^tCaR)\,{}^t({}^tCaR) = {}^tCaR\,{}^tR\,{}^taC \\ &= P \begin{pmatrix} I^{(r)} \\ 0 \end{pmatrix} {}^td^{-1}aR\,{}^tR\,{}^tad^{-1}(I^{(r)}\ 0)\,{}^tP = P \begin{pmatrix} {}^td^{-1}aR\,{}^tR\,{}^tad^{-1} & 0 \\ 0 & 0 \end{pmatrix} {}^tP \end{aligned}$$

hence

$$\operatorname{rank}(A\,{}^tA) = \operatorname{rank}({}^td^{-1}aR\,{}^tR\,{}^tad^{-1}) = \operatorname{rank}(R\,{}^tR).$$

(ii) Let $B = {}^tA$, then $B = {}^tR\,{}^taC$, thus $B \in \mathcal{M}(n \times m, r, R, C)$. Therefore $\operatorname{rank}({}^tAA) = \operatorname{rank}(B\,{}^tB) = \operatorname{rank}(C\,{}^tC)$ by (i).

(iii) We have $L(A) = R$. Thus

$$\begin{aligned} L(A) \cap L(A)^{\perp} &= \{x \in \mathbb{F}_q^{(n)} \mid x = yR \text{ for some } y \in \mathbb{F}_q^{(r)} \text{ and } R\,{}^t x = 0\} \\ &= \{x \in \mathbb{F}_q^{(n)} \mid x = yR \text{ for some } y \in \mathbb{F}_q^{(r)} \text{ and } R\,{}^t R\,{}^t y = 0\}. \end{aligned}$$

Therefore

$$\dim(L(A) \cap L(A)^{\perp}) = r - \operatorname{rank}(R\,{}^t R).$$

(iv) Let $B = {}^t A$, then $B \in \mathcal{M}(n \times m, r, R, C)$. We have

$$\dim(L({}^t A) \cap L({}^t A)^{\perp}) = \dim(L(B) \cap L(B)^{\perp}) = r - \operatorname{rank}(C\,{}^t C),$$

by (iii). □

The $m \times n$ matrix A over $\mathbb{F}_q$ defines a linear map from $\mathbb{F}_q^{(m)}$ to $\mathbb{F}_q^{(n)}$ in the following way

$$\begin{aligned} \mathbb{F}_q^{(m)} &\to \mathbb{F}_q^{(n)} \\ x &\mapsto xA, \end{aligned}$$

which will also be denoted by A.

Theorem D.4 (Criteria of M-P Invertibility). Let $C \in \mathcal{M}(r, m)$, $R \in \mathcal{M}(r, n)$ and $A \in \mathcal{M}(m \times n, r, C, R)$. Then the following are equivalent:

(i) A is M-P invertible,

(ii) $\operatorname{rank}(C\,{}^t C) = \operatorname{rank}(R\,{}^t R) = r$,

(iii) $\operatorname{rank}(A\,{}^t A) = \operatorname{rank}({}^t A A) = r$,

(iv) $L(A) \cap L(A)^{\perp} = (0)$ and $L({}^t A) \cap L({}^t A)^{\perp} = (0)$,

(v) $\mathbb{F}_q^{(n)} = \operatorname{Im} A \oplus (\operatorname{Im} A)^{\perp}$ and $\mathbb{F}_q^{(m)} = \operatorname{Ker} A \oplus (\operatorname{Ker} A)^{\perp}$, where $\oplus$ denotes the direct sum.

Proof. By Lemma D.1 we can assume that $A = {}^t CaR$ for some $a \in GL_r(\mathbb{F}_q)$.

(i) $\Rightarrow$ (ii). Assume that (i) holds. Let X be an M-P inverse of A. By Lemma D.2 $X \in \mathcal{M}(n \times m, r, R, C)$. Again by Lemma D.1 $X = {}^t RbC$ for some $b \in GL_r(\mathbb{F}_q)$. Then

$${}^t CaR = A = AXA = {}^t CaR\,{}^t RbC\,{}^t CaR.$$

By the second statement of Lemma D.1, $a = aR\,{}^tRbC\,{}^tCa$. Thus $a^{-1} = R\,{}^tRbC\,{}^tC$, which shows that both $R\,{}^tR$ and $C\,{}^tC$ are nonsingular, hence $\text{rank}\,(R\,{}^tR) = \text{rank}\,(C\,{}^tC) = r$.

(ii) $\Rightarrow$ (i). Assume that (ii) holds. Let $b = (R\,{}^tR)^{-1}a^{-1}(C\,{}^tC)^{-1}$, then $b \in GL_r(\mathbb{F}_q)$. Let $X = {}^tRbC$, then it is easy to verify (D.1) for A and X; for instance,

$$\begin{aligned} AXA &= ({}^tCaR)\,{}^tR(R\,{}^tR)^{-1}a^{-1}(C\,{}^tC)^{-1}C({}^tCaR) \\ &= {}^tCaR = A. \end{aligned}$$

Thus X is an M-P inverse of A.

(ii) $\Leftrightarrow$ (iii). It follows from Lemma D.3 (i) and (ii).

(ii) $\Leftrightarrow$ (iv). It follows from Lemma D.3 (iii) and (iv).

(iv) $\Leftrightarrow$ (v). Note that $L(A) = \text{Im}\,A$ and $L({}^tA)^{\perp} = \text{Ker}\,A$. Thus $L(A)^{\perp} = (\text{Im}\,A)^{\perp}$ and $L({}^tA) = (\text{Ker}\,A)^{\perp}$. Then

$$\begin{aligned} L(A) \cap L(A)^{\perp} = (0) &\Leftrightarrow \text{Im}\,A \cap (\text{Im}\,A)^{\perp} = (0) \\ &\Leftrightarrow \mathbb{F}_q^{(n)} = \text{Im}\,A \oplus (\text{Im}\,A)^{\perp} \end{aligned}$$

and

$$\begin{aligned} L({}^tA) \cap L({}^tA)^{\perp} = (0) &\Leftrightarrow (\text{Ker}\,A)^{\perp} \cap \text{Ker}\,A = (0) \\ &\Leftrightarrow \mathbb{F}_q^{(m)} = (\text{Ker}\,A)^{\perp} \oplus \text{Ker}\,A. \end{aligned}$$

□

Theorem D.5 (Uniqueness and Formula of M-P Inverse). Let $C \in \mathcal{M}(r, m)$, $R \in \mathcal{M}(r, n)$, and $A \in \mathcal{M}(m \times n, r, C, R)$, and assume that A is M-P invertible. Then the M-P inverse of A is unique and is equal to ${}^tR(CA\,{}^tR)^{-1}C$.

Proof. By Lemma D.1 we can assume that $A = {}^tCaR$ for some $a \in GL_r(\mathbb{F}_q)$. Let X be an M-P inverse of A, then by Lemma D.2 $X \in \mathcal{M}(n \times m, r, R, C)$ and by Lemma D.1 $X = {}^tRbC$ for some $b \in GL_r(\mathbb{F}_q)$. By the proof of (i) $\Rightarrow$ (ii) of Theorem D.4, $a^{-1} = R\,{}^tRbC\,{}^tC$. Therefore

$$b = (R\,{}^tR)^{-1}a^{-1}(C\,{}^tC)^{-1} = ((C\,{}^tC)a(R\,{}^tR))^{-1} = (CA\,{}^tR)^{-1}.$$

Hence the M-P inverse X of A is unique and $X = {}^tR(CA\,{}^tR)^{-1}C$. □

In the sequel, we will denote by A^+ the unique M-P inverse of A. Then $A^+ = {}^tR(CA\,{}^tR)^{-1}C$.

Corollary D.6. Let A be an $m \times n$ matrix over $\mathbb{F}_q$ of rank r.

(i) Assume that $A\,{}^tA$ is nonsingular. Then A is M-P invertible and $A^+ = {}^tA(A\,{}^tA)^{-1}$.

(ii) Assume both $C\,{}^tC$ and $R\,{}^tR$ to be nonsingular. Then for any $A = {}^tCaR$, where $a \in GL_r(\mathbb{F}_q)$, we have $A^+ = R^+a^{-1}\,{}^t(C^+)$.

Proof. (i) Since $A\,{}^tA$ is $m \times m$ and nonsingular, $r = \text{rank } A = m$. Take $C = a = I^{(r)}$ and $R = A$, then $A = {}^tCaR \in \mathcal{M}(m \times n, r, C, R)$. Since both $C\,{}^tC$ and $R\,{}^tR$ are nonsingular, by Theorem D.4, (i) $\Leftrightarrow$ (ii), A is M-P invertible. By Theorem D.5 $A^+ = {}^tR(CA\,{}^tR)^{-1}C = {}^tA(A\,{}^tA)^{-1}$.

(ii) By Theorem D.4, (i) $\Leftrightarrow$ (ii), A is M-P invertible. Then by Theorem D.5 $A^+ = {}^tR(CA\,{}^tR)^{-1}C$, and by (i) above $R^+ = {}^tR(R\,{}^tR)^{-1}$ and $C^+ = {}^tC(C\,{}^tC)^{-1}$. But $C\,{}^tC$ is a symmetric matrix, so is $(C\,{}^tC)^{-1}$ and ${}^t(C^+) = (C\,{}^tC)^{-1}C$. Then

$$\begin{aligned} A^+ &= {}^tR(CA\,{}^tR)^{-1}C = {}^tR(C\,{}^tCaR\,{}^tR)^{-1}C \\ &= {}^tR(R\,{}^tR)^{-1}a^{-1}(C\,{}^tC)^{-1}C = R^+a^{-1}\,{}^t(C^+). \end{aligned}$$

Denote by $\mathcal{M}(m \times n, r)^*$ the set of M-P invertible $m \times n$ matrices of rank r over $\mathbb{F}_q$ and by $\mathcal{M}(r, n)^*$ the set of r-dimensional subspaces R such that $R\,{}^tR$ is nonsingular.

Theorem D.7. $\mathcal{M}(m \times n, r)^*$ can be expressed as a disjoint union

$$\mathcal{M}(m \times n, r)^* = \dot{\bigcup_{(C,R) \in \mathcal{M}(r,m)^* \times \mathcal{M}(r,n)^*}} \{\, {}^tCaR \mid a \in GL_r(\mathbb{F}_q)\},$$

and

$$\mathcal{M}(m \times n, r)^* = \mid \mathcal{M}(r, m)^* \mid\mid GL_r(\mathbb{F}_q) \mid\mid \mathcal{M}(r, n)^* \mid .$$

Proof. By Lemma D.1 and Theorem D.4. □

D.2 Construction of Enumeration of M-P Invertible Matrices

By Theorem D.7 the problem of constructing and enumerating M-P invertible $m \times n$ matrices of rank r is reduced to that of constructing and enumerating the elements in $GL_r(\mathbb{F}_q)$ and that of constructing and enumerating the elements in $\mathcal{M}(m, r)^*$ and the elements in $\mathcal{M}(n, r)^*$. The first problem can be solved by the method of the proof of Lemma 1.5 and

Theorem 1.6. For the second problem, we distinguish the cases when q is odd or even.

First, let q be odd. By Theorem 1.26 there is an $n \times n$ nonsingular matrix T such that

$$TI^{(n)}\,{}^tT = S_{2\nu+\delta,\Delta} = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & \Delta \end{pmatrix},$$

where $n = 2\nu + \delta$, $\delta = 0, 1$, or 2,

$$\Delta = \begin{cases} \phi, & \text{when } \delta = 0, \\ (1) \text{ or } (z), & \text{when } \delta = 1, \\ \binom{1}{-z}, & \text{when } \delta = 2, \end{cases}$$

and z is a fixed non-square element of $\mathbb{F}_q$. More precisely, for even n, we have

$$\delta = \begin{cases} 0, & \text{when } n \equiv 0 \pmod 4 \text{ or } q \equiv 1 \pmod 4, \\ 2, & \text{otherwise,} \end{cases}$$

and for odd even, we have $\delta = 1$ and

$$\Delta = \begin{cases} (1), & \text{when } n \equiv 1 \pmod 4 \text{ or } q \equiv 1 \pmod 4, \\ (z), & \text{otherwise.} \end{cases}$$

For any $R \in \mathcal{M}(r, n)^*$, i.e., R is an r-dimensional subspace of $\mathbb{F}_q^{(n)}$ such that $R\,{}^tR$ is nonsingular, let $R_1 = RT^{-1}$, then

$$R\,{}^tR = RT^{-1}S_{2\nu+\delta,\Delta}\,{}^tT^{-1}\,{}^tR = R_1 S_{2\nu+\delta,\Delta}\,{}^tR_1.$$

Since $R\,{}^tR$ is nonsingular, R_1 is non-isotropic with respect to $S_{2\nu+\delta,\Delta}$. But r-dimensional non-isotropic subspaces with respect to $S_{2\nu+\delta,\Delta}$ are partitioned into orbits of subspaces of types $(r, 2s+\gamma, s, \Gamma)$ with $r = 2s+\gamma$, where

$$\Gamma = \begin{cases} \phi, & \text{when } \gamma = 0, \\ (1) \text{ or } (z), & \text{when } \gamma = 1, \\ \binom{1}{-z}, & \text{when } \gamma = 2 \end{cases}$$

(cf. Section 6.1). Let R_1 be an r-dimensional subspace of type $(r, 2s + \gamma, s, \Gamma)$ with $\gamma = 2s + \gamma$, then there is a matrix representation R_1 of the

subspace R_1 such that

$$R_1 S_{2\nu+\delta,\Delta}\, {}^tR_1 = \begin{pmatrix} 0 & I^{(s)} & \\ I^{(s)} & 0 & \\ & & \Gamma \end{pmatrix}.$$

A method for constructing such subspaces R_1 can be found in Section 6.2. To each such R_1 we apply the transformation $T \in GL_n(\mathbb{F}_q)$, then we obtain an $R = R_1 T \in \mathcal{M}(r,n)^*$. Clearly, the cardinality of $\mathcal{M}(r,n)^*$ follows from the Anzahl theorems in orthogonal geometry over $\mathbb{F}_q$ of characteristic $\neq 2$ (i.e., Theorem 6.26), and we have

Theorem D.8 (Enumeration of $\mathcal{M}(r,n)^*$). Let q be odd. Then

$$|\,\mathcal{M}(r,n)^*\,| = \begin{cases} \displaystyle\sum_{2t=2s+\gamma,\gamma=0 \text{ or } 2} N(2t, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta), & \text{if } r = 2t, \\ \displaystyle\sum_{\Gamma=(1) \text{ or } (z)} N(2t+1, 2t+1, s, \Gamma; 2\nu+\delta, \Delta), & \text{if } r = 2t+1, \end{cases}$$

where $N(r, 2s+\gamma, s, \Gamma;\ 2\nu+\delta, \Delta)$ is given in (6.49). □

Next let q be even. By Corollary 4.2 there is an $n \times n$ nonsingular matrix T such that

$$TI^{(n)}\,{}^tT = S_\delta = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & \Delta \end{pmatrix},$$

where $n = 2\nu + \delta$, $\delta = 1$ or 2, and

$$\Delta = \begin{cases} (1), & \text{when } \delta = 1, \\ \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right), & \text{when } \delta = 2. \end{cases}$$

For any $R \in \mathcal{M}(r,n)^*$, let $R_1 = RT^{-1}$, then

$$R\,{}^tR = RT^{-1} S_\delta\, {}^tT^{-1}\, {}^tR = R_1 S_\delta\, {}^tR_1.$$

Since $R\,{}^tR$ is nonsingular, R_1 is non-isotropic with respect to S_δ. But r-dimensional non-isotropic subspaces with respect to S_δ are partitioned into orbits of subspaces of type $(r, 2s+\tau, s, \epsilon)$ with $r = 2s+\tau$, where $\tau = 0, 1,$ or 2 and $\epsilon = 0$ or 1 (cf. Section 4.2). Let R_1 be an r-dimensional

subspace of type $(r, 2s+\tau, s, \epsilon)$ with $r = 2s+\tau$, then there is a matrix representation R_1 of the subspace R_1 such that

$$R_1 S_\delta {}^t R_1 = \begin{pmatrix} 0 & I^{(s)} & \\ I^{(s)} & 0 & \\ & & \Gamma \end{pmatrix},$$

where

$$\Gamma = \begin{cases} 0, & \text{when } \tau = 0, \\ 1, & \text{when } \tau = 1, \\ \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right), & \text{when } \tau = 2, \end{cases}$$

and, moreover,

$$e_{2\nu+1} \notin R_1, \text{ when } \epsilon = 0; \quad e_{2\nu+1} \in R_1, \text{ when } \epsilon = 1.$$

A method for constructing such subspaces R_1 can be found in Section 4.2. To each such R_1 we apply the transformation $T \in GL_n(\mathbb{F}_q)$, then we obtain an $R = R_1 T \in \mathcal{M}(r,n)^*$. Clearly, the cardinality of $\mathcal{M}(r,n)^*$ follows from the Anzahl theorem in pseudo-symplectic geometry over $\mathbb{F}_q$ of characteristic 2 (i.e., Theorem 4.14), and we have

Theorem D.9 (Enumeration of $\mathcal{M}(r,n)^*$). Let q be even. Then

$$|\mathcal{M}(r,n)^*| = \begin{cases} \displaystyle\sum_{\epsilon=0,1} N(2t+1, 2t+1, t, \epsilon; 2\nu+\delta), & \text{if } r = 2t+1, \\ \displaystyle\sum_{\epsilon=0,1} \sum_{2t=2s+\tau, \tau=0,2} N(2t, 2s+\tau, s, \epsilon; 2\nu+\delta), & \text{if } r = 2t, \end{cases}$$

where $N(r, 2s+\tau, s, \epsilon;\ 2\nu+\delta)$ is given in (4.19). □

D.3 Generalization to Matrices over $\mathbb{F}_{q^2}$

As an analogy to the Moor-Penrose generalized inverse of a complex matrix we can formulate the definition of Moor-Penrose generalized inverse of a matrix over $\mathbb{F}_{q^2}$ as follows.

Let A be an $m \times n$ matrix over $\mathbb{F}_{q^2}$. A Moor-Penrose generalized inverse (or, simply, an M-P inverse) over $\mathbb{F}_{q^2}$ of A is an $n \times m$ matrix X such that the following conditions are satisfied.

$$\left.\begin{array}{rcl} AXA &=& A, \\ XAX &=& X, \\ {}^t\overline{(XA)} &=& XA, \\ {}^t\overline{(AX)} &=& AX. \end{array}\right\} \tag{D.3}$$

If A has an M-P inverse over $\mathbb{F}_{q^2}$, then A is said to be M-P invertible over $\mathbb{F}_{q^2}$.

From the above definition it is clear that if A has an M-P inverse X over $\mathbb{F}_{q^2}$ then A is also an M-P inverse of X over $\mathbb{F}_{q^2}$ and ${}^t\overline{X}$ is an M-P inverse of ${}^t\overline{A}$ over $\mathbb{F}_{q^2}$.

Now let $L_1(A)$ be the vector subspace spanned by the rows of A over $\mathbb{F}_{q^2}$, then $L_1(A) \subseteq \mathbb{F}_{q^2}^{(n)}$. Denote by $\mathcal{M}_1(m \times n, r)$ the set of $m \times n$ matrices of rank r over $\mathbb{F}_{q^2}$, and by $\mathcal{M}_1(r, n)$ the set of r-dimensional subspaces of $\mathbb{F}_{q^2}^{(n)}$. Let $C \in \mathcal{M}_1(r, m)$ and $R \in \mathcal{M}_1(r, n)$, then denote by $\mathcal{M}_1(m \times n, r, C, R)$ the set of matrices A in $\mathcal{M}_1(m \times n, r)$ with $L_1({}^t\overline{A}) = C$ and $L_1(A) = R$. The following disjoint union is clear:

$$\mathcal{M}_1(m \times n, r) = \dot{\bigcup_{(C,R) \in \mathcal{M}_1(r,m) \times \mathcal{M}_1(r,n)}} \mathcal{M}_1(m \times n, r, C, R).$$

Thus to study $\mathcal{M}_1(m \times n, r)$ it is sufficient to study $\mathcal{M}_1(m \times n, r, C, R)$ for given $C \in \mathcal{M}_1(r, m)$ and $R \in \mathcal{M}_1(r, n)$.

Parallel to Lemmas D.1, D.2, and D.3 we have

Lemma D.10. Let $A \in \mathcal{M}_1(m \times n, r)$ and $C \in \mathcal{M}_1(r, m)$, $R \in \mathcal{M}_1(r, n)$. Then $A \in \mathcal{M}_1(m \times n, r, C, R)$ if and only if $A = {}^t\overline{C}aR$ for some $a \in GL_r(\mathbb{F}_{q^2})$. Moreover, the map $a \mapsto {}^t\overline{C}aR$ is a bijection from $GL_r(\mathbb{F}_{q^2})$ to $\mathcal{M}_1(m \times n, r, C, R)$. □

Lemma D.11. Let $C \in \mathcal{M}_1(r, m)$, $R \in \mathcal{M}_1(r, n)$, and $A \in \mathcal{M}_1(m \times n, r, C, R)$. Assume that A is M-P invertible and let X be an M-P inverse of A. Then $X \in \mathcal{M}_1(n \times m, r, R, C)$. □

Lemma D.12. Let $C \in \mathcal{M}_1(r, m)$, $R \in \mathcal{M}_1(r, n)$ and $A \in \mathcal{M}_1(m \times n, r, C, R)$. Then the following holds.

(i) $\operatorname{rank}(A\,{}^t\overline{A}) = \operatorname{rank}(R\,{}^t\overline{R})$,

(ii) $\operatorname{rank}({}^t\overline{A}A) = \operatorname{rank}({}^t\overline{C}C)$,

(iii) $\dim(L_1(A) \cap L_1(A)^{\perp}) + \operatorname{rank}(R\,{}^t\overline{R}) = r$,

(iv) $\dim(L_1({}^t\overline{A}) \cap L_1({}^t\overline{A})^{\perp}) + \operatorname{rank}({}^t\overline{C}C) = r$. □

The $m \times n$ matrix A over $\mathbb{F}_{q^2}$ defines a linear map from $\mathbb{F}_{q^2}^{(m)}$ to $\mathbb{F}_{q^2}^{(n)}$ in the following way

$$\mathbb{F}_{q^2}^{(m)} \to \mathbb{F}_{q^2}^{(n)}$$

$$x \mapsto xA,$$

which will also be denoted by A.

Parallel to Theorems D.4 and D.5 we have

Theorem D.13 (Criterion of M-P Invertibility over $\mathbb{F}_{q^2}$). Let $C \in \mathcal{M}_1(r,m)$, $R \in \mathcal{M}_1(r,n)$, and $A \in \mathcal{M}_1(m \times n, r, C, R)$. Then the following are equivalent:

(i) A is M-P invertible over $\mathbb{F}_{q^2}$,

(ii) $\operatorname{rank}(C\,{}^t\overline{C}) = \operatorname{rank}(R\,{}^t\overline{R}) = r$,

(iii) $\operatorname{rank}(A\,{}^t\overline{A}) = \operatorname{rank}({}^t\overline{A}A) = r$,

(iv) $L_1(A) \cap L_1(A)^\perp = (0)$ and $L_1({}^t\overline{A}) \cap L_1({}^t\overline{A})^\perp = (0)$,

(v) $\mathbb{F}_{q^2}^{(n)} = \operatorname{Im} A \oplus (\operatorname{Im} A)^\perp$ and $\mathbb{F}_{q^2}^{(m)} = \operatorname{Ker} A \oplus (\operatorname{Ker} A)^\perp$.

Theorem D.14 (Uniqueness and Formula of M-P Inverse). Let $C \in \mathcal{M}_1(r,m)$, $R \in \mathcal{M}_1(r,n)$, and $A \in \mathcal{M}_1(m \times n, r, C, R)$, and assume that A is M-P invertible over $\mathbb{F}_{q^2}$. Then the M-P inverse of A over $\mathbb{F}_{q^2}$ is unique and is equal to ${}^t\overline{R}(CA\,{}^t\overline{R})^{-1}C$. □

In the sequel we will denote by A^+ the unique M-P inverse of A over $\mathbb{F}_{q^2}$. Then $A^+ = {}^t\overline{R}(CA\,{}^t\overline{R})^{-1}C$.

Corollary D.15. Let A be an $m \times n$ matrix over $\mathbb{F}_{q^2}$ of rank r.

(i) Assume that $A\,{}^t\overline{A}$ is nonsingular. Then A is M-P invertible over $\mathbb{F}_{q^2}$ and $A^+ = {}^t\overline{A}(A\,{}^t\overline{A})^{-1}$.

(ii) Assume both $C\,{}^t\overline{C}$ and $R\,{}^t\overline{R}$ to be nonsingular. Then for any $A = {}^t\overline{C}aR$, where $a \in GL_r(\mathbb{F}_{q^2})$, we have $A^+ = R^+a^{-1}\,{}^t(C^+)$. □

Denote by $\mathcal{M}_1(m \times n, r)^*$ the set of M-P invertible $m \times n$ matrices over $\mathbb{F}_{q^2}$ of rank r and by $\mathcal{M}_1(n,r)^*$ the set of r-dimensional subspaces R of $\mathbb{F}_{q^2}^{(n)}$ such that $R\,{}^t\overline{R}$ is nonsingular. Then parallel to Theorem D.7 we have

Theorem D.16. $\mathcal{M}_1(m \times n, r)^*$ can be expressed as a disjoint union

$$\mathcal{M}_1(m \times n, r)^* = \dot{\bigcup_{(C,R) \in \mathcal{M}_1(r,m)^* \times \mathcal{M}_1(r,n)^*}} \{{}^t\overline{C}aR \mid a \in GL_r(\mathbb{F}_{q^2})\},$$

and

$$|\,\mathcal{M}_1(m \times n, r)^*\,| = |\,\mathcal{M}_1(r,m)^*\,|\,|\,GL_r(\mathbb{F}_{q^2})\,|\,|\,\mathcal{M}_1(r,n)^*\,|\,. \qquad \square$$

$| GL_r(\mathbb{F}_{q^2}) |$ is given in Theorem 1.6. For $| \mathcal{M}_1(n,r)^* |$ we have

Theorem D.17 (Enumeration of $\mathcal{M}_1(n,r)^*$).

$$| \mathcal{M}_1(n,r)^* | = N(r,r,n),$$

where $N(r,r;n)$ is given in Corollary 5.21.

Proof. Parallel to the proof of Theorem D.8 or D.9. □

D.4 Comments

The definition of Moor-Penrose generalized inverse of a matrix over the real or complex field is generalized to that over $\mathbb{F}_q$ by E. Dawson and C. K. Wuan, who give a key arrangement scheme using this definition (cf. Dawson and Wuan, 1997). A necessary and sufficient condition for a matrix over $\mathbb{F}_q$ to be M-P invertible (i.e., (i) $\Leftrightarrow$ (v) in Theorem D.4) is given by C. K. Wu and E. Dawson (cf. Wu and Dawson, 1998). All the other equivalent conditions in Theorem D.4 and the other results of Sections D.1 and D.2 are obtained by Z. Dai and Y. Zhang (cf. Dai and Zhang, 2001). Section D.3 is due to Z. Dai and Z. Wan (cf. Dai and Wan, 2001).

Appendix E
Representations of Forms by Forms in a Finite Field

Another beautiful application of the Anzahl theorems of the geometry of classical groups is the representation of forms by forms of the same type in a finite field. We begin with the case of bilinear forms.

E.1 Representations of Bilinear Forms

Let A be an $m \times n$ matrix over $\mathbb{F}_q$ and B be a $t \times u$ matrix over $\mathbb{F}_q$. Denote by $n^{(b)}_{t\times m,u\times n}(A,B)$ the number of pairs of $t \times m$ and $u \times n$ matrices (X, Y) over $\mathbb{F}_q$ such that

$$XA\,{}^tY = B. \tag{E.1}$$

Stated geometrically, $n^{(b)}_{t\times m,u\times n}(A,B)$ is the number of representations of the bilinear form

$$(x_1, x_2, \ldots, x_t)B\,{}^t(y_1, y_2, \ldots, y_u)$$

by the bilinear form

$$(z_1, z_2, \ldots, z_m)A\,{}^t(w_1, w_2, \ldots, w_n).$$

Let us deduce formulas for $n^{(b)}_{t\times m,u\times n}(A,B)$. Clearly there is no loss of generality in assuming that one or both of A and B are in their normal forms under equivalence transformations. We recall that an $m \times n$ matrix A is equivalent to one of the normal forms

$$\begin{pmatrix} I^{(r)} & \\ & 0^{(m-r,n-r)} \end{pmatrix}, \quad r = 0, 1, 2, \ldots, \min\{m, n\}, \tag{E.2}$$

where r is the rank of A.

Theorem E.1. Let A be an $m \times n$ matrix of rank r over $\mathbb{F}_q$, where $0 \le r \le \min\{m, n\}$, and B be a $t \times u$ matrix over $\mathbb{F}_q$. Then

$$n^{(b)}_{t\times m,u\times n}(A, B) = q^{t(m-r)+u(n-r)} n^{(b)}_{t\times r,u\times r}(I^{(r)}, B). \tag{E.3}$$

Proof. We can assume that A is of the form (E.2). Let X be a $t \times m$ matrix and Y be a $u \times n$ matrix satisfying (E.1). Write X and Y in the following block forms:

$$X = \begin{array}{c} \begin{array}{cc} r & m-r \end{array} \\ \begin{pmatrix} X_1 & X_2 \end{pmatrix} \end{array} \begin{array}{c} \\ t \end{array}$$

and

$$Y = \begin{array}{c} \begin{array}{cc} r & n-r \end{array} \\ \begin{pmatrix} Y_1 & Y_2 \end{pmatrix} \end{array} \begin{array}{c} \\ u \end{array};$$

then

$$XA\,{}^tY = X_1 I^{(r)}\,{}^tY_1 = B.$$

Therefore X_2 and Y_2 can be arbitrary, and X_1 and Y_1 satisfy $X_1 I^{(r)}\,{}^tY_1 = B$. Hence (E.3) follows. □

Thus our problem is reduced to compute $n^{(b)}_{t\times r,u\times r}(I^{(r)}, B)$.

Theorem E.2. Let B be a $t \times u$ matrix of rank s over $\mathbb{F}_q$, where $0 \le s \le \min\{t, u\}$. Then

$$n^{(b)}_{t\times r,u\times r}(I^{(r)}, B) = \begin{cases} 0 & \text{if } r < s, \\ q^{s(r-s)+s(s-1)/2} \displaystyle\prod_{i=r-s+1}^{r} (q^i - 1)\, n^{(b)}_{(t-s)\times(r-s),(u-s)\times(r-s)}(I^{(r-s)}, 0^{(t-s,u-s)}) & \\ & \text{if } r \ge s. \end{cases} \tag{E.4}$$

Proof. We can assume that

$$B = \begin{pmatrix} I^{(s)} & \\ & 0^{(t-s,u-s)} \end{pmatrix}.$$

Let X be a $t \times r$ matrix and Y be a $u \times r$ matrix such that

$$XI^{(r)}\,{}^tY = B. \tag{E.5}$$

Clearly, rank $XI^{(r)}\,{}^tY \le r$. Thus, if $r < s$, then $n^{(b)}_{t\times r,u\times r}(I^{(r)}, B) = 0$.

Now consider the case $r \ge s$. Write X and Y in the following block forms

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \begin{matrix} s \\ t-s \end{matrix}$$

and

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \begin{matrix} s \\ u-s \end{matrix},$$

then (E.5) can be written as

$$X_1\,{}^tY_1 = I^{(s)}, \tag{E.6}$$

$$X_1\,{}^tY_2 = 0, \tag{E.7}$$

$$X_2\,{}^tY_1 = 0, \tag{E.8}$$

and

$$X_2\,{}^tY_2 = 0. \tag{E.9}$$

From (E.6) we deduce that both X_1 and Y_1 are of rank s. We assert that for any $s \times r$ matrix X_1 of rank s, the number of $(t-s)\times r$ matrices X_2, $s \times r$ matrices Y_1, and $(u-s)\times r$ matrices Y_2 for which (E.6)–(E.9) hold is independent of the particular choice of X_1. Let X_1 and X_1^* be two $s \times r$ matrices of rank s, then there is an $r \times r$ nonsingular matrix P such that

$$X_1^* = X_1P.$$

Assume that X_2, Y_1, and Y_2 are $(t-s)\times r$, $s \times r$, and $(u-s)\times r$ matrices, respectively, for which (E.6)–(E.9) hold. Let

$$X_2^* = X_2P, \quad Y_1^* = Y_1\,{}^tP^{-1}, \quad Y_2^* = Y_2\,{}^tP^{-1};$$

then X_1^*, X_2^*, Y_1^*, and Y_2^* satisfy also (E.6)–(E.9). This proves our assertion.

By Lemma 1.5 the number of $s \times r$ matrices of rank s over $\mathbb{F}_q$ is equal to

$$q^{s(s-1)/2} \prod_{i=r-s+1}^{r} (q^i - 1). \tag{E.10}$$

Given any $s \times r$ matrix X_1 of rank s, we have to compute the number of $(t-s)\times r$ matrices X_2, $s\times r$ matrices Y_1, and $(u-s)\times r$ matrices Y_2 such that (E.6)–(E.9) hold. We may choose

$$X_1 = \begin{pmatrix} I^{(s)} & 0^{(s,r-s)} \end{pmatrix}.$$

Then from (E.6) and (E.7) we deduce

$$Y_1 = \begin{array}{c} \begin{matrix} s & r-s \end{matrix} \\ (I^{(s)} \quad Y_{12}) \end{array}$$

and

$$Y_2 = \begin{array}{c} \begin{matrix} s & r-s \end{matrix} \\ (0 \quad Y_{22}) \end{array} \begin{matrix} \\ u-s \end{matrix}.$$

Write

$$X_2 = \begin{array}{c} \begin{matrix} s & r-s \end{matrix} \\ (X_{21} \quad X_{22}) \end{array} \begin{matrix} \\ t-s \end{matrix};$$

then from (E.8) and (E.9) we deduce

$$X_{21} + X_{22}\,{}^tY_{12} = 0 \tag{E.11}$$

and

$$X_{22}\,{}^tY_{22} = 0, \tag{E.12}$$

respectively. The number of pairs of $(t-s)\times(r-s)$ and $(u-s)\times(r-s)$ matrices (X_{22}, Y_{22}) satisfying (E.12) is denoted by

$$n^{(b)}_{(t-s)\times(r-s),(u-s)\times(r-s)}(I^{(r-s)},\ 0^{(t-s,u-s)}).$$

Given any pair of $(t-s)\times(r-s)$ and $(u-s)\times(t-s)$ matrices (X_{22}, Y_{22}) satisfying (E.12) and any $s\times(r-s)$ matrix Y_{12}, there is a uniquely determined $(t-s)\times s$ matrix X_{21} satisfying (E.11). Therefore given any $s\times r$ matrix X_1 of rank s, the number of $(t-s)\times r$ matrices X_2, $s\times r$ matrices Y_1, and $(u-s)\times r$ matrices Y_2 for which (E.6)–(E.9) hold is equal to

$$q^{s(r-s)}n^{(b)}_{(t-s)\times(r-s),(u-s)\times(r-s)}(I^{(r-s)},\ 0^{(t-s,u-s)}). \tag{E.13}$$

Multiplying (E.13) by (E.10), we obtain the second formula of (E.4). □

Corollary E.3. Let $r \geq s$. Then

$$n^{(b)}_{s\times r,s\times r}(I^{(r)}, I^{(s)}) = q^{s(r-s)+s(s-1)/2}\prod_{i=r-s+1}^{r}(q^i-1).$$

□

Theorem E.4.

$$n^{(b)}_{t\times m,u\times m}(I^{(m)},0^{(t,u)})$$
$$=\sum_{k=0}^{\min\{m,u\}} q^{t(m-k)+k(k-1)/2}\frac{\prod_{i=m-k+1}^{m}(q^i-1)\prod_{i=u-k+1}^{u}(q^i-1)}{\prod_{i=1}^{k}(q^i-1)}. \quad \text{(E.14)}$$

Proof. $n^{(b)}_{t\times m,u\times m}(I^{(m)},0^{(t,u)})$ is the number of pairs of $t\times m$ and $u\times m$ matrices (X,Y) such that

$$X\,{}^tY=0^{(t,u)}. \quad \text{(E.15)}$$

Let Y be a $u\times m$ matrix of rank k; then $k\le\min\{u,m\}$. By Theorem 1.10 the number of $u\times m$ matrices of rank k is equal to

$$q^{k(k-1)/2}\begin{bmatrix}u\\k\end{bmatrix}_q\prod_{i=m-k+1}^{m}(q^i-1),$$

where

$$\begin{bmatrix}u\\k\end{bmatrix}_q=\frac{\prod_{i=u-k+1}^{u}(q^i-1)}{\prod_{i=1}^{k}(q^i-1)}$$

is the number of k-dimensional subspaces of the u-dimensional row vector space $\mathbb{F}_q^{(u)}$. Given any $u\times m$ matrix Y of rank k, the number of $t\times m$ matrices X satisfying (E.15) is equal to $q^{t(m-k)}$. Hence (E.14) follows. □

Corollary E.5.

$$\sum_{k=0}^{\min\{m,u\}} q^{t(m-k)+k(k-1)/2}\frac{\prod_{i=m-k+1}^{m}(q^i-1)\prod_{i=u-k+1}^{u}(q^i-1)}{\prod_{i=1}^{k}(q^i-1)}$$
$$=\sum_{k=0}^{\min\{m,t\}} q^{u(m-k)+k(k-1)/2}\frac{\prod_{i=m-k+1}^{m}(q^i-1)\prod_{i=t-k+1}^{t}(q^i-1)}{\prod_{i=1}^{k}(q^i-1)}.$$

Proof. The number of pairs of $t \times m$ and $u \times m$ matrices (X, Y) such that $X\,{}^tY = 0^{(t,u)}$ is equal to the number of pairs of $u \times m$ and $t \times m$ matrices (Y, X) such that $Y\,{}^tX = 0^{(u,t)}$. □

From Theorems E.1, E.2, and E.4 it is not difficult to deduce a closed formula for $n^{(b)}_{t\times m,u\times n}(A, B)$, where A is an $m \times n$ matrix and B is a $t \times u$ matrix.

Theorem E.6. Let A be an $m \times n$ matrix of rank r over $\mathbb{F}_q$, where $0 \le r \le \min\{m, n\}$, and B be a $t \times u$ matrix of rank s over $\mathbb{F}_q$. Then

$$n^{(b)}_{t\times m,u\times n}(A, B)$$

$$= \begin{cases} 0 & \text{if } r < s, \\ q^{t(m-r)+u(n-r)+s(r-s)+s(s-1)/2} \displaystyle\prod_{i=r-s+1}^{r} (q^i - 1) & \end{cases}$$

$$\times \sum_{k=0}^{\min\{r-s,u-s\}} q^{(t-s)(r-s-k)+k(k-1)/2} \frac{\displaystyle\prod_{i=r-s-k+1}^{r-s} (q^i - 1) \prod_{i=u-s-k+1}^{u-s} (q^i - 1)}{\displaystyle\prod_{i=1}^{k} (q^i - 1)}$$

if $r \ge s$.

□

E.2 Representations of Alternate Forms

Now let us compute the number of representations of an alternate form by another alternate form over $\mathbb{F}_q$. In matrix language, let A be an $m \times m$ alternate matrix over $\mathbb{F}_q$ and B be a $t \times t$ alternate matrix over $\mathbb{F}_q$. Denote by $n^{(a)}_{t\times m}(A, B)$ the number of $t \times m$ matrices X over $\mathbb{F}_q$ such that

$$X A\,{}^tX = B. \tag{E.16}$$

Our purpose is to deduce formulas for $n^{(a)}_{t\times m}(A, B)$.

Clearly, there is no loss of generality in assuming that one or both of A and B are in their normal forms under cogredience transformations. By Theorem 3.1 an $m \times m$ alternate matrix A is cogredient to one of the

normal forms

$$\begin{pmatrix} 0 & I^{(\nu)} & \\ -I^{(\nu)} & 0 & \\ & & 0^{(m-2\nu)} \end{pmatrix}, \quad \nu = 0, 1, 2, \cdots, \left[\frac{m}{2}\right], \tag{E.17}$$

where ν is the *index* of A. Let

$$K_{2\nu} = \begin{pmatrix} 0 & I^{(\nu)} \\ -I^{(\nu)} & 0 \end{pmatrix}.$$

Then we have

Theorem E.7. Let A be an $m \times m$ alternate matrix of index ν over $\mathbb{F}_q$, where $2\nu \le m$, and B be a $t \times t$ alternate matrix over $\mathbb{F}_q$. Then

$$n^{(a)}_{t\times m}(A, B) = q^{t(m-2\nu)} n^{(a)}_{t\times 2\nu}(K_{2\nu}, B). \tag{E.18}$$

Proof. We can assume that A is of the form (E.17). Let X be an $t \times m$ matrix over $\mathbb{F}_q$ satisfying (E.16). Write X in the block form

$$\begin{matrix} & 2\nu & m-2\nu & \\ X = & (X_1 & X_2 &) \ t; \end{matrix}$$

then

$$X A\, {}^tX = X_1 K_{2\nu}\, {}^tX_1 = B.$$

Therefore X_2 can be any $t \times (m - 2\nu)$ matrix and X_1 is a $t \times 2\nu$ matrix satisfying $X_1 K_{2\nu}\, {}^tX_1 = B$. Hence (E.18) follows. □

Theorem E.8. Let B be a $t \times t$ alternate matrix of index ν_1 over $\mathbb{F}_q$, where $2\nu_1 \le t$. Then

$$\begin{aligned} & n^{(a)}_{t\times 2\nu}(K_{2\nu}, B) \\ & = \begin{cases} 0 & \text{if } \nu_1 > \nu, \\ n^{(a)}_{2\nu_1\times 2\nu}(K_{2\nu}, K_{2\nu_1}) n^{(a)}_{(t-2\nu_1)\times 2(\nu-\nu_1)}(K_{2(\nu-\nu_1)}, 0^{(t-2\nu_1)}) & \\ \quad \text{if } \nu_1 \le \nu. \end{cases} \end{aligned} \tag{E.19}$$

Proof. We can assume that

$$B = \begin{pmatrix} K_{2\nu_1} & \\ & 0^{(t-2\nu_1)} \end{pmatrix}.$$

Let X be a $t \times 2\nu$ matrix over $\mathbb{F}_q$ satisfying

$$X K_{2\nu}\, {}^tX = B. \tag{E.20}$$

Write X in the block form

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \begin{matrix} 2\nu_1 \\ t-2\nu_1 \end{matrix}. \tag{E.21}$$

From (E.20) we deduce

$$\begin{aligned} &X_1 K_{2\nu}\, {}^tX_1 = K_{2\nu_1}, \\ &X_1 K_{2\nu}\, {}^tX_2 = 0^{(2\nu_1, t-2\nu_1)}, \end{aligned} \tag{E.22}$$

and

$$X_2 K_{2\nu}\, {}^tX_2 = 0^{(t-2\nu_1)}.$$

If $\nu_1 > \nu$, then there is no X_1 satisfying (E.22) and, hence, $n^{(a)}_{t\times 2\nu}(K_{2\nu}, B) = 0$.

Now consider the case $\nu_1 \le \nu$. For any $2\nu_1 \times 2\nu$ matrix X_1 satisfying (E.22), clearly, we have rank $X_1 = 2\nu_1$. The $2\nu_1$ rows of X_1 span a subspace of dimension $2\nu_1$, which will also be denoted by X_1. Define

$$X_1^{\perp} = \{v \in \mathbb{F}_q{}^{(2\nu)} \mid X_1 K_{2\nu}\, {}^tv = 0\}.$$

Then $X_1^{\perp}$ is a subspace of dimension $2(\nu - \nu_1)$. Clearly, $X_1 K\, {}^t(X_1^{\perp}) = 0$. By (E.22) $X_1 \cap X_1^{\perp} = \{0\}$. Denote any $2(\nu - \nu_1) \times 2\nu$ matrix over $\mathbb{F}_q$ whose rows form a basis of $X_1^{\perp}$ also by $X_1^{\perp}$, then

$$\begin{pmatrix} X_1 \\ X_1^{\perp} \end{pmatrix}$$

is a $2\nu \times 2\nu$ nonsingular matrix and

$$\begin{pmatrix} X_1 \\ X_1^{\perp} \end{pmatrix} K_{2\nu}\, {}^t\begin{pmatrix} X_1 \\ X_1^{\perp} \end{pmatrix} = \begin{pmatrix} K_{2\nu_1} & \\ & X_1^{\perp} K_{2\nu}\, {}^t(X_1^{\perp}) \end{pmatrix}.$$

Therefore $X_1^{\perp} K_{2\nu}\, {}^t(X_1^{\perp})$ is a $2(\nu - \nu_1) \times 2(\nu - \nu_1)$ nonsingular alternate matrix. We can assume that

$$X_1^{\perp} K_{2\nu}\, {}^t(X_1^{\perp}) = K_{2(\nu-\nu_1)}.$$

It is clear that for any $(t - 2\nu_1) \times 2(\nu - \nu_1)$ matrix T satisfying

$$T K_{2(\nu-\nu_1)}\, {}^tT = 0, \tag{E.23}$$

the $t \times 2\nu$ matrix

$$\begin{pmatrix} X_1 \\ TX_1^{\perp} \end{pmatrix}$$

satisfies (E.20). Conversely, for any $t \times 2\nu$ matrix (E.21) satisfying (E.20), there is a uniquely determined $(t - 2\nu_1) \times 2(\nu - \nu_1)$ matrix T such that $X_2 = TX_1^{\perp}$ and T satisfies (E.23). Hence the second formula of (E.19) follows. □

Lemma 3.15 can be restated as

Theorem E.9. Let $\nu_1 \leq \nu$. Then

$$n^{(a)}_{2\nu_1 \times 2\nu}(K_{2\nu}, K_{2\nu_1}) = q^{\nu_1(2\nu-\nu_1)} \prod_{i=\nu-\nu_1+1}^{\nu} (q^{2i} - 1). \tag{E.24}$$

It remains to compute $n^{(a)}_{(t-2\nu_1)\times 2(\nu-\nu_1)}(K_{2(\nu-\nu_1)}, 0^{(t-2\nu_1)})$.

Theorem E.10.

$$n^{(a)}_{t\times 2\nu}(K_{2\nu}, 0^{(t)}) = \sum_{k=0}^{\min\{t,\nu\}} q^{\frac{1}{2}k(k-1)} \frac{\prod_{i=t-k+1}^{t} (q^i - 1) \prod_{i=\nu-k+1}^{\nu} (q^{2i} - 1)}{\prod_{i=1}^{k} (q^i - 1)}. \tag{E.25}$$

Proof. Let X be a $t \times 2\nu$ matrix over $\mathbb{F}_q$ such that

$$XK_{2\nu}\,{}^tX = 0. \tag{E.26}$$

Assume that the rows of X span a subspace U of dimension k. Then $k \leq t$ and U is a k-dimensional totally isotropic subspace with respect to $K_{2\nu}$. Hence $k \leq \nu$. Let $u_1, u_2, \cdots, u_k$ be a basis of U. Then there is a uniquely determined $t \times k$ matrix T of rank k such that

$$X = T \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_k \end{pmatrix}. \tag{E.27}$$

Conversely, for any $t \times k$ matrix T of rank k, define X by (E.27), then X satisfies (E.26) and the rows of X span U. By Corollary 3.19 the number

of k-dimensional totally isotropic subspaces with respect to $K_{2\nu}$ is equal to

$$\frac{\prod_{i=\nu-k+1}^{\nu}(q^{2i}-1)}{\prod_{i=1}^{k}(q^i-1)}.$$

By Lemma 1.5 the number of $t \times k$ matrices of rank k is equal to

$$q^{\frac{1}{2}k(k-1)} \prod_{i=t-k+1}^{t}(q^i-1).$$

Hence we have (E.25). □

From Theorems E.7–E.10 we deduce

Theorem E.11. Let A be an $m \times m$ alternate matrix of index ν over $\mathbb{F}_q$, where $2\nu \le m$, and B be a $t \times t$ alternate matrix of index ν_1 over $\mathbb{F}_q$, where $2\nu_1 \le t$. Then

$$n^{(a)}_{t\times m}(A,B) = \begin{cases} 0 & \text{if } \nu_1 > \nu, \\ q^{t(m-2\nu)+\nu_1(2\nu-\nu_1)} \prod_{i=\nu-\nu_1+1}^{\nu}(q^{2i}-1) \\ \quad \times \sum_{k=0}^{\min\{t-2\nu,\nu-\nu_1\}} q^{(1/2)k(k-1)} \dfrac{\prod_{i=t-2\nu_1-k+1}^{t-2\nu_1}(q^i-1)\prod_{i=\nu-\nu_1-k+1}^{\nu-\nu_1}(q^{2i}-1)}{\prod_{i=1}^{k}(q^i-1)}, \\ \qquad \text{if } \nu_1 \le \nu. \end{cases}$$

□

E.3 Representations of Hermitian Forms

Now let us study the number of representations of an Hermitian form over $\mathbb{F}_{q^2}$ by another Hermitian form. Same as the preceding sections we use matrix language.

Now let A be an $m \times m$ Hermitian matrix over $\mathbb{F}_q$ and B be a $t \times t$ Hermitian matrix over $\mathbb{F}_q$. Denote by $n^{(h)}_{t\times m}(A, B)$ the number of $t \times m$ matrices X over $\mathbb{F}_q$ such that

$$XA\,{}^t\overline{X} = B.$$

Now we shall deduce formulas for $n^{(h)}_{t\times m}(A, B)$. There is no loss of generality in assuming that one or both of A and B are in their normal forms under cogredience transformations. By Theorem 5.2 we can assume that A is cogredient to one of normal forms

$$\begin{pmatrix} I^{(r)} & \\ & 0^{(m-r)} \end{pmatrix}, \quad r = 0, 1, 2, \ldots, m,$$

where r is the rank of A. Parallel to Theorems E.7–E.11 of Section E.2 we have the following theorems whose proofs are the same as the corresponding ones in Section E.2 and will be omitted.

Theorem E.12. Let A be an $m \times m$ Hermitian matrix of rank r over $\mathbb{F}_{q^2}$, where $r \le m$, and B be a $t \times t$ Hermitian matrix over $\mathbb{F}_{q^2}$. Then

$$n^{(h)}_{t\times m}(A, B) = q^{(1/2)t(m-r)} n^{(h)}_{t\times r}(I^{(r)}, B). \qquad \square$$

Theorem E.13. Let B be a $t \times t$ Hermitian matrix of rank r_1 over $\mathbb{F}_{q^2}$, where $r_1 \le t$. Then

$$n^{(h)}_{t\times r}(I^{(r)}, B) = \begin{cases} 0, & \text{if } r_1 > r, \\ n^{(h)}_{r_1\times r}(I^{(r)}, I^{(r_1)}) n^{(h)}_{(t-r_1)\times(r-r_1)}(I^{(r-r_1)}, 0^{(t-r_1)}), & \text{if } r_1 \le r. \end{cases} \qquad \square$$

Lemma 5.15 can be restated as follows.

Theorem E.14. Let $r_1 \le r$. Then

$$n^{(h)}_{r_1\times r}(I^{(r)}, I^{(r_1)}) = q^{r_1 r - \frac{1}{2}r_1(r_1+1)} \prod_{i=r-r_1+1}^{r} (q^i - (-1)^i). \qquad \square$$

Theorem E.15.

$$n^{(h)}_{t\times r}(I^{(r)}, 0^{(t)}) = \sum_{k=0}^{\min\{t,[r/2]\}} q^{k(k-1)} \frac{\displaystyle\prod_{i=t-k+1}^{t} (q^{2i}-1) \prod_{i=r-2k+1}^{r} (q^i - (-1)^i)}{\displaystyle\prod_{i=1}^{k} (q^{2i}-1)}.$$

$\square$

Theorem E.16. Let A be an $m \times m$ Hermitian matrix of rank r over $\mathbb{F}_{q^2}$, where $r \le m$, and B be a $t \times t$ Hermitian matrix of rank r_1 over $\mathbb{F}_{q^2}$, where $r_1 \le t$. Then

$$n^{(h)}_{t\times m}(A,B) = \begin{cases} 0, & \text{if } r_1 > r, \\ q^{(1/2)t(m-r)+r_1 r-(1/2)r_1(r_1+1)} \displaystyle\prod_{i=r-r_1+1}^{r} (q^i-(-1)^i) \end{cases}$$

$$\times \sum_{k=0}^{\min\{t,[r/2]\}} q^{k(k-1)} \frac{\displaystyle\prod_{i=t-k+1}^{t}(q^{2i}-1) \prod_{i=r-2k+1}^{r}(q^i-(-1)^i)}{\displaystyle\prod_{i=1}^{k}(q^{2i}-1)}, \quad \text{if } r_1 \le r.$$

□

E.4 Representations of Quadratic Forms (Odd Characteristic Case)

Let $\mathbb{F}_q$ be a finite field of odd characteristic. We will compute the number of representations of a quadratic form over $\mathbb{F}_q$ by another quadratic form. We use the matrix language again.

Let A be an $m \times m$ symmetric matrix over $\mathbb{F}_q$ and B be a $t \times t$ symmetric matrix over $\mathbb{F}_q$. Denote by $n^{(q)}_{t\times m}(A,B)$ the number of $t \times m$ matrices X over $\mathbb{F}_q$ such that

$$XA\,{}^tX = B.$$

There is no loss of generality in assuming that one or both of A and B are in their normal forms under cogredience transformations. By Theorem 1.26 we can assume that

$$A = \begin{pmatrix} S_{2\nu+\delta,\Delta} & \\ & 0^{(m-2\nu-\delta)} \end{pmatrix},$$

where $0 \le \nu \le \left[\frac{m-\delta}{2}\right], \delta = 0, 1,$ or 2,

$$S_{2\nu+\delta,\Delta} = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & \Delta \end{pmatrix},$$

and Δ denotes the definite part, i.e.,

$$\Delta = \begin{cases} \phi, & \text{if } \delta = 0, \\ (1) \text{ or } (z), & \text{if } \delta = 1, \\ \left(\begin{smallmatrix} 1 & \\ & -z \end{smallmatrix}\right), & \text{if } \delta = 2. \end{cases}$$

ν is the index of A and $2\nu+\delta$ is the rank of A.

Parallel to Theorems E.7–E.10 of Section E.2 we have the following theorems whose proofs are omitted.

Theorem E.17. Let A be an $m\times m$ symmetric matrix and B be a $t\times t$ symmetric matrix, both over $\mathbb{F}_q$. Assume that A is of index ν, rank $2\nu+\delta$, where $\delta=0$, 1, or 2 and $2\nu+\delta\le m$, and definite part Δ. Then

$$n^{(q)}_{t\times m}(A,B)=q^{t(m-2\nu-\delta)}n^{(q)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta},B). \qquad \square$$

Theorem E.18. Let B be a $t\times t$ symmetric matrix over $\mathbb{F}_q$ and assume that B is of index ν_1, rank $2\nu_1+\delta_1$, where $\delta_1=0$, 1, or 2 and $2\nu_1+\delta_1\le t$, and definite part Δ_1, where

$$\Delta_1=\begin{cases}\phi, & \text{when } \delta_1=0,\\ (1) \text{ or } (z), & \text{when } \delta_1=1,\\ \left(\begin{smallmatrix}1&\\&-z\end{smallmatrix}\right), & \text{when } \delta_1=2.\end{cases}$$

(i) If the condition

$$\begin{aligned}&2\nu_1+\delta_1\le t\\ &\quad\le\begin{cases}\nu+\nu_1+\min\{\delta,\delta_1\}, & \text{when } \delta_1\ne\delta, \text{ or } \delta_1=\delta \text{ and } \Delta_1=\Delta,\\ \nu+\nu_1, & \text{when } \delta_1=\delta=1 \text{ and } \Delta_1\ne\Delta,\end{cases}\end{aligned} \tag{E.28}$$

does not hold, then

$$n^{(q)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta},B)=0.$$

(ii) If condition (E.28) holds, then there exist uniquely determined ν_2,δ_2, and Δ_2, where $\delta_2=0$, 1, or 2 and

$$\Delta_2=\begin{cases}\phi, & \text{when } \delta_2=0,\\ (1) \text{ or } (z), & \text{when } \delta_2=1,\\ \left(\begin{smallmatrix}1&\\&-z\end{smallmatrix}\right), & \text{when } \delta_2=2,\end{cases}$$

such that

$$\begin{pmatrix} S_{2\nu_1+\delta_1,\Delta_1} & \\ & S_{2\nu_2+\delta_2,\Delta_2}\end{pmatrix}$$

is cogredient to $S_{2\nu+\delta,\Delta}$ and

$$n^{(q)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, B) = n^{(q)}_{(2\nu_1+\delta_1)\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1+\delta_1,\Delta_1}) \times n^{(q)}_{(t-2\nu_1-\delta_1)\times(2\nu_2+\delta_2)}(S_{2\nu_2+\delta_2,\Delta_2}, 0). \qquad \square$$

Theorem E.19. (i) If $\nu \geq \nu_1$, then

$$n^{(q)}_{2\nu_1\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1}) = q^{2\nu_1(\nu-\nu_1)+\delta\nu_1+\nu_1(\nu_1-1)} \prod_{i=\nu-\nu_1+1}^{\nu} (q^i-1)(q^{i+\delta-1}+1).$$

(ii) If

$$\nu \geq \begin{cases} \nu_1+1, & \text{when } \delta=0, \text{ or } \delta=1 \text{ and } \Delta_1 \neq \Delta, \\ \nu_1, & \text{when } \delta=2, \text{ or } \delta=1 \text{ and } \Delta_1=\Delta, \end{cases}$$

then

$$n^{(q)}_{(2\nu_1+1)\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1+1,\Delta_1}) = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1}) n^{(q)}_{1\times(2(\nu-\nu_1)+\delta)}(S_{2(\nu-\nu_1)+\delta,\Delta}, \Delta_1),$$

where $n^{(q)}_{2\nu_1\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1})$ is given by (i) and

$$n^{(q)}_{1\times(2(\nu-\nu_1)+\delta)}(S_{2(\nu-\nu_1)+\delta,\Delta}, \Delta_1) = \begin{cases} q^{\nu-\nu_1-1}(q^{\nu-\nu_1}-1), & \text{when } \delta=0, \\ q^{\nu-\nu_1}(q^{\nu-\nu_1}-1), & \text{when } \delta=1 \text{ and } \Delta_1\neq\Delta, \\ q^{\nu-\nu_1}(q^{\nu-\nu_1}+1), & \text{when } \delta=1 \text{ and } \Delta_1=\Delta, \\ q^{\nu-\nu_1}(q^{\nu-\nu_1+1}+1), & \text{when } \delta=2. \end{cases}$$

(iii) If $\nu \geq \nu_1 + (2-\delta)$, then

$$n^{(q)}_{(2\nu_1+2)\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1+2,\Delta_1}) = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1}) n^{(q)}_{2\times(2(\nu-\nu_1)+\delta)}\left(S_{2(\nu-\nu_1)+\delta,\Delta}, \left(\begin{smallmatrix}1\\ &-z\end{smallmatrix}\right)\right),$$

where $n^{(q)}_{2\nu_1\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, S_{2\nu_1})$ is given in (i) and

$$n^{(q)}_{2\times(2(\nu-\nu_1)+\delta)}(S_{2(\nu-\nu_1)+\delta,\Delta}, \left(\begin{smallmatrix}1\\ &-z\end{smallmatrix}\right)) = \begin{cases} q^{2(\nu-\nu_1)-2}(q^{\nu-\nu_1-1}-1)(q^{\nu-\nu_1}-1), & \text{when } \delta=0, \\ q^{2(\nu-\nu_1)-1}(q^{\nu-\nu_1}-1)(q^{\nu-\nu_1}+1), & \text{when } \delta=1, \\ q^{2(\nu-\nu_1)}(q^{\nu-\nu_1}+1)(q^{\nu-\nu_1+1}+1), & \text{when } \delta=2. \end{cases} \qquad \square$$

Theorem E.20.

$$n^{(q)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta,\Delta}, 0^{(t)})$$
$$= \sum_{k=0}^{\min\{t,\nu\}} q^{\frac{1}{2}k(k-1)} \frac{\prod_{i=t-k+1}^{t}(q^i-1)\prod_{i=\nu-k+1}^{\nu}(q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{k}(q^i-1)}. \qquad \square$$

Now it is routine to write down a closed formula for $n^{(q)}_{t\times m}(A,B)$, where A is an $m\times m$ symmetric matrix of index ν and definite part Δ, and B is a $t\times t$ symmetric matrix of index ν, and definite part Δ_1.

E.5 Representations of Quadratic Forms (Even Characteristic Case)

Let $\mathbb{F}_q$ be a finite field of even characteristic. We will study the number of representations of a quadratic form over $\mathbb{F}_q$ by another quadratic form. We will also use the matrix language.

Let A be an $m\times m$ matrix over $\mathbb{F}_q$ and B be a $t\times t$ matrix over $\mathbb{F}_q$. Denote by $n^{(q)}_{t\times m}(A,B)$ the number of $t\times m$ matrices X over $\mathbb{F}_q$ such that

$$X\,A\,{}^tX \equiv B.$$

We recall that for any two $t\times t$ matrices B and B_1 over $\mathbb{F}_q$ $B\equiv B_1$ means that $B-B_1$ is alternate.

Our problem now is to deduce formulas of $n^{(q)}_{t\times m}(A,B)$. There is no loss of generality in assuming that one or both of A and B are in their normal forms under "cogredience" transformations. By Theorem 1.35 we can assume that

$$A = \begin{pmatrix} G_{2\nu+\delta} & \\ & 0^{(m-2\nu-\delta)} \end{pmatrix},$$

where $0\le\nu\le\left[\frac{m-\delta}{2}\right]$, $\delta=0,1,$ or 2,

$$G_{2\nu+\delta} = \begin{pmatrix} 0 & I^{(\nu)} & \\ & 0 & \\ & & \Delta \end{pmatrix},$$

Δ denotes the definite part, i.e.,

$$\Delta = \begin{cases} \phi, & \text{if } \delta = 0, \\ (1), & \text{if } \delta = 1, \\ \left(\begin{smallmatrix} \alpha & 1 \\ & \alpha \end{smallmatrix}\right), & \text{if } \delta = 2, \end{cases}$$

and $\alpha \in \mathbb{F}_q \setminus N$. ν is the *index* of A and $2\nu + \delta$ is the *"rank"* of A.

Parallel to Theorems E.7 and E.17 we have

Theorem E.21. Let A be an $m \times m$ matrix over $\mathbb{F}_q$ and B be a $t \times t$ matrix over $\mathbb{F}_q$. Assume that A is of index ν and "rank" $2\nu + \delta$, where $\delta = 0$, 1, or 2 and $2\nu + \delta \leq m$. Then

$$n^{(q)}_{t\times m}(A, B) = q^{t(m-2\nu-\delta)} n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B). \qquad \square$$

Thus our problem reduces to the finding of formulas for $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B)$.

Parallel to Theorems E.8 and E.18, and by using Lemma 7.3 we deduce

Theorem E.22. Let B be a $t \times t$ matrix of index ν_1 and "rank" $2\nu_1 + \delta_1$.

(i) If the condition

$$2\nu_1 + \delta_1 \leq t \leq \nu + \nu_1 + \min\{\delta, \delta_1\} \tag{E.29}$$

does not hold, then

$$n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B) = 0.$$

(ii) If $\delta_1 = 0$ and condition (E.29) holds, then

$$\begin{aligned} &n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B) \\ &\quad = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1}) n^{(q)}_{(t-2\nu_1)\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, 0^{(t-2\nu_1)}). \end{aligned}$$

(iii) If $\delta_1 = 1$ and condition (E.29) holds, then

$$\begin{aligned} &n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B) \\ &\quad = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1}) n^{(q)}_{(t-2\nu_1)\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, E_{11}), \end{aligned}$$

where E_{11} is the $(t - 2\nu_1) \times (t - 2\nu_1)$ matrix with a 1 in its (1,1) position and 0's elsewhere.

(iv) If $\delta_1 = 2$ and condition (E.29) holds, then

$$n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, B) = n^{(q)}_{(2\nu_1+2)\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1+2}) \times n^{(q)}_{(t-2\nu_1-2)\times(2(\nu-\nu_1)+\delta-2)}(G_{2(\nu-\nu_1-2+\delta)+2-\delta}, 0^{(t-2\nu_1-2)}).$$

□

Thus our problem reduces to the computing of $n^{(q)}_{(2\nu_1+\delta_1)\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1+\delta_1})$, $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, 0^{(t)})$, and $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, E_{11})$, where E_{11} is a $t\times t$ matrix with a 1 in its $(1,1)$ position and 0's elsewhere.

From Lemmas 7.16, 7.18, 7.20, and 7.22 we deduce

Theorem E.23. Let q be even.

(i) If $\nu \geq \nu_1$, then

$$n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1}) = q^{2\nu_1(\nu-\nu_1)+\delta\nu_1+\nu_1(\nu_1-1)} \prod_{i=\nu-\nu_1+1}^{\nu} (q^i-1)(q^{i+\delta-1}+1).$$

(ii) If

$$\nu \geq \begin{cases} \nu_1+1, & \text{when } \delta = 0, \\ \nu_1, & \text{when } \delta = 1 \text{ or } 2, \end{cases}$$

then

$$n^{(q)}_{(2\nu_1+1)\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1+1}) = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1}) n^{(q)}_{1\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, G_1),$$

where $n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1})$ is given in (i) and

$$n^{(q)}_{1\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, G_1) = \begin{cases} q^{\nu-\nu_1-1}(q^{\nu-\nu_1}-1), & \text{if } \delta = 0, \\ q^{2(\nu-\nu_1)}, & \text{if } \delta = 1, \\ q^{\nu-\nu_1}(q^{\nu-\nu_1+1}+1), & \text{if } \delta = 2. \end{cases}$$

(iii) If $\nu \geq \nu_1 + (2-\delta)$, then

$$n^{(q)}_{(2\nu_1+2)\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1+2}) = n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1}) n^{(q)}_{2\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, G_2),$$

where $n^{(q)}_{2\nu_1\times(2\nu+\delta)}(G_{2\nu+\delta}, G_{2\nu_1})$ is given in (i) and

$$n^{(q)}_{2\times(2(\nu-\nu_1)+\delta)}(G_{2(\nu-\nu_1)+\delta}, G_2) = \begin{cases} q^{2(\nu-\nu_1)-2}(q^{\nu-\nu_1-1}-1)(q^{\nu-\nu_1}-1), & \text{if } \delta=0,\\ q^{2(\nu-\nu_1)-1}(q^{\nu-\nu_1}-1)(q^{\nu-\nu_1}+1), & \text{if } \delta=1,\\ q^{2(\nu-\nu_1)}(q^{\nu-\nu_1}+1)(q^{\nu-\nu_1+1}+1), & \text{if } \delta=2. \end{cases}$$

□

Parallel to Theorems E.10 and E.20 we derive in a similar way

Theorem E.24.

$$n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, 0^{(t)}) = \sum_{k=0}^{\min\{t,\nu\}} q^{\frac{1}{2}k(k-1)} \frac{\prod_{i=t-k+1}^{t}(q^i-1)\prod_{i=\nu-k+1}^{\nu}(q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{k}(q^i-1)}.$$

□

Now let us come to the computation of $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, E_{11})$.

Theorem E.25. Let q be even. Then

$$n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, E_{11}) = \sum_{k=0}^{\min\{\nu,t-1\}} q^{\frac{1}{2}k(k+1)} \frac{\prod_{i=t-k}^{t-1}(q^i-1)\prod_{i=\nu-k+1}^{\nu}(q^i-1)(q^{i+\delta-1}+1)}{\prod_{i=1}^{k}(q^i-1)} \times n^{(q)}_{(2(\nu-k)+\delta)\times 1}(G_{2(\nu-k)+\delta}, G_1),$$

where $n^{(q)}_{1\times(2(\nu-k)+\delta)}(G_{2(\nu-k)+\delta}, G_1)$ is given in Theorem E.23(ii).

Proof. It is the same thing to compute $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, E_{tt})$, where E_{tt} is the $t\times t$ matrix with a 1 in its (t,t) position and 0's elsewhere. We

know that $n^{(q)}_{t\times(2\nu+\delta)}(G_{2\nu+\delta}, E_{tt})$ is the number of $t \times (2\nu+\delta)$ matrix P such that

$$PG_{2\nu+\delta}\,{}^tP \equiv \begin{pmatrix} 0^{(t-1)} & \\ & 1 \end{pmatrix}.$$

Write

$$P = \begin{matrix} \\ t-1 \\ 1 \end{matrix} \overset{2\nu+\delta}{\begin{pmatrix} P_1 \\ v \end{pmatrix}};$$

then

$$P_1 G_{2\nu+\delta}\,{}^tP_1 \equiv 0, \tag{E.30}$$

$$P_1(G_{2\nu+\delta} + {}^tG_{2\nu+\delta})\,{}^tv = 0, \tag{E.31}$$

and

$$vG_{2\nu+\delta}\,{}^tv = 1. \tag{E.32}$$

Suppose that the rows of P_1 generate a subspace U of dimension k. Then U is totally singular and $k \le \min\{\nu, t-1\}$. Let U be a matrix representation of the subspace U, then

$$P_1 = T_{P_1}U,$$

where T_{P_1} is a $(t-1)\times k$ matrix of rank k and is uniquely determined by P_1 for a fixed choice of the matrix representation U. Since T_{P_1} is of rank k, (E.30) and (E.31) are equivalent to

$$UG_{2\nu+\delta}\,{}^tU \equiv 0$$

and

$$U(G_{2\nu+\delta} + {}^tG_{2\nu+\delta})\,{}^tv = 0,$$

respectively. Therefore for any $(t-1)\times(2\nu+\delta)$ matrix P_1, whose rows generate a k-dimensional totally singular subspace U, the number of $(2\nu+\delta)$-dimensional row vectors v such that $\binom{P_1}{v}$ satisfies (E.30)–(E.32) depends only on U.

Now let U and V be two k-dimensional totally singular subspaces, and P_1 and Q_1 be two $(t-1)\times(2\nu+\delta)$ matrices whose rows generate U and V, respectively. We assert that the number of $(2\nu+\delta)$-dimensional row vectors v such that $\binom{P_1}{v}$ satisfy (E.30)–(E.32) is equal to the number of row vectors u such that $\binom{Q_1}{u}$ satisfy (E.30)–(E.32). Let

$$P_1 = T_{P_1}U$$

and

$$Q_1 = T_{Q_1} V,$$

where T_{P_1} and T_{Q_1} are $(t-1) \times k$ matrices of rank k. By the discussion of the above paragraph we can assume that $T_{P_1} = T_{Q_1}$. We have

$$UG\,{}^tU \equiv\ VG\,{}^tV \equiv 0,$$

then by Lemma 7.10 there is a $T \in O_{2\nu+\delta}(\mathbb{F}_q)$ such that

$$V = UT.$$

T_{P_1} is a $(t-1) \times k$ matrix of rank k, from which it follows that for a row vector v $\binom{P_1}{v}$ satisfies (E.30)–(E.32) if and only if $\binom{Q_1}{vT}$ satisfies (E.30)–(E.32). This proves our assertion.

By Corollary 7.25 the number of k-dimensional totally singular subspaces is equal to

$$\frac{\displaystyle\prod_{i=\nu-k+1}^{\nu} (q^i - 1)(q^{i+\delta-1} + 1)}{\displaystyle\prod_{i=1}^{k} (q^i - 1)}$$

and the number of $(t-1) \times k$ matrices of rank k can be obtained from Lemma 1.5. Thus it remains to compute the number of possible choices of column vectors v such that $\binom{P_1}{v}$ satisfies (E.30)–(E.32), once P_1 is chosen. By the discussion of the above paragraph, we may choose

$$P_1 = \begin{pmatrix} I^{(k)} & \\ & 0^{(t-1-k,2\nu+\delta-k)} \end{pmatrix}$$

and then compute the number of possible choices of v. From $P_1(G_{2\nu+\delta} + {}^tG_{2\nu+\delta})\,{}^tv = 0$ we deduce that v is of the form

$$v = (x_1, x_2, \ldots, x_\nu, \underbrace{0, 0, \ldots, 0}_{k}, x_{\nu+k+1}, \ldots, x_{2\nu+\delta})$$

and the condition $vG_{2\nu+\delta}\,{}^tv = 1$ becomes

$$\begin{aligned}
&\sum_{i=k+1}^{\nu} x_i x_{\nu+i} = 1, && \text{if } \delta = 0,\\
&\sum_{i=k+1}^{\nu} x_i x_{\nu+i} + x_{2\nu+1}^2 = 1, && \text{if } \delta = 1,\\
&\sum_{i=k+1}^{\nu} x_i x_{\nu+i} + \alpha x_{2\nu+1}^2 + x_{2\nu+1}x_{2\nu+2} + \alpha x_{2\nu+2}^2 = 1, && \text{if } \delta = 2.
\end{aligned}$$

Hence $x_1, x_2, \ldots, x_k$ can be chosen arbitrarily in $\mathbb{F}_q$ and the number of possible choices of $x_{k+1}, \ldots, x_\nu, x_{\nu+k+1}, \ldots, x_{2\nu+\delta}$ is $n^{(q)}_{1\times(2(\nu-k)+\delta)}(G_{2(\nu-k)+\delta}, G_1)$, which is given in Theorem E.23(iv). Therefore once P_1 is chosen, the number of possible choices of v is

$$q^k n^{(q)}_{1\times(2(\nu-k)+\delta)}(G_{2(\nu-k)+\delta}, G_1).$$

Theorem E.25 is now proved. □

Now it is routine to write down a formula for $n^{(q)}_{t\times m}(A, B)$.

E.6 Representations of Symmetric Bilinear Forms over Finite Fields of Even Characteristic

In this section we also assume that $\mathbb{F}_q$ is of even characteristic. Let us compute the number of representations of a symmetric bilinear form over $\mathbb{F}_q$ by another symmetric bilinear form. We use the matrix language. Let A be an $m \times m$ symmetric matrix over $\mathbb{F}_q$ and B be a $t \times t$ symmetric matrix over $\mathbb{F}_q$. Denote by $n^{(s)}_{t\times m}(A, B)$ the number of $t \times m$ matrices X such that

$$XA\,{}^tX = B.$$

Our problem is to deduce formulas for $n^{(s)}_{t\times m}(A, B)$. Since q is even, symmetric matrices are either alternate or nonalternate. If A is alternate and B is nonalternate, then, clearly, $n^{(s)}_{t\times m}(A, B) = 0$. If both A and B are alternate, then this is the case we studied in Section E.2. Therefore we need to consider only the case when A is nonalternate.

There is no loss of generality in assuming that one or both of A and B are in their normal forms under cogredience transformations. By Theorem 4.1 we can assume that

$$A = \begin{pmatrix} S_{2\nu+\delta} & \\ & 0^{(m-2\nu-\delta)} \end{pmatrix},$$

where $0 \le \nu \le [\frac{m-\delta}{2}], \delta = 0, 1$ or 2,

$$S_{2\nu+1} = \begin{pmatrix} 0 & I^{(\nu)} & \\ I^{(\nu)} & 0 & \\ & & 1 \end{pmatrix},$$

$$S_{2\nu+2} = \begin{pmatrix} 0 & I^{(\nu)} & & \\ I^{(\nu)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 1 \end{pmatrix}.$$

ν is the *index* of A.

Parallel to Theorems E.7–E.11 of Section E.2 we have Theorems E.26–E.30 in the following

Theorem E.26. Let A be an $m \times m$ nonalternate symmetric matrix of index ν and rank $2\nu + \delta$, where $\delta = 1$ or 2 and $2\nu + \delta \leq m$, and let B be a $t \times t$ symmetric matrix, both over $\mathbb{F}_q$. Then

$$n^{(s)}_{t\times m}(A, B) = q^{t(m-2\nu-\delta)} n^{(s)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta}, B).$$

Theorem E.27. Let $\nu \geq 0$, $\delta = 1$ or 2, and B be a $t \times t$ symmetric matrix over $\mathbb{F}_q$. Assume that B is of index ν_1 and rank $2\nu_1 + \delta_1$, where $\delta_1 = 0$, 1, or 2 and $2\nu_1 + \delta_1 \leq t$.

(i) If $\nu_1 > \nu$, or $\nu_1 = \nu$ and $\delta_1 > \delta$, then

$$n^{(q)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta}, B) = 0.$$

(ii) If $\nu_1 = \nu$ and $\delta_1 \leq \delta$, or $\nu_1 < \nu$, then there exist ν_2 and δ_2, where $\delta_2 = 0$, 1, or 2, such that

$$\begin{pmatrix} S_{2\nu_1+\delta_1} & \\ & S_{2\nu_2+\delta_2} \end{pmatrix}$$

is cogredient to $S_{2\nu+\delta}$ and

$$\begin{aligned} n^{(s)}_{t\times(2\nu+\delta)}(S_{2\nu+\delta}, B) &= n^{(s)}_{(2\nu_1+\delta_1)\times(2\nu+\delta)}(S_{2\nu+\delta}, S_{2\nu_1+\delta_1}) \\ &\quad \times n^{(s)}_{(t-2\nu_1-\delta_1)\times(2\nu_2+\delta_2)}(S_{2\nu_2+\delta_2}, 0^{(t-2\nu_1-\delta_1)}). \end{aligned}$$

□

The proofs of these two theorems are left as an exercise (Exercise E.5).

Theorem E.28. Let $\delta = 1$ or 2 and $\delta_1 = 0$, 1, or 2. Assume that $\nu = \nu_1$ and $\delta_1 \leq \delta$, or $\nu_1 < \nu$. Then

$$\begin{aligned} &n^{(s)}_{(2\nu_1+\delta_1)\times(2\nu+\delta)}(S_{2\nu+\delta}, S_{2\nu_1+\delta_1}) \\ &\quad = |Ps_{2\nu_1+\delta_1}(\mathbb{F}_q)| \sum_{\varepsilon_1=0}^{1} N(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \varepsilon_1; 2\nu+\delta), \end{aligned}$$

where $|Ps_{2\nu_1+\delta_1}(\mathbb{F}_q)|$ is given by Theorem 4.7 and $N(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \varepsilon_1; 2\nu+\delta)$ is given by Theorem 4.14.

Proof. Let X be a $(2\nu_1+\delta_1)\times(2\nu+\delta)$ matrix over $\mathbb{F}_q$ satisfying

$$XS_{2\nu+\delta}{}^tX = S_{2\nu_1+\delta_1}. \tag{E.33}$$

Then rank $X = 2\nu_1+\delta_1$ and the row vectors of X span a $(2\nu_1+\delta_1)$-dimensional subspace of $\mathbb{F}_q{}^{(2\nu+\delta)}$. This subspace is either of type $(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, 0)$ or of type $(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, 1)$. Conversely, let U be a subspace of type $(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \varepsilon_1)$, where $\varepsilon_1 = 0$ or 1, then we can find a $(2\nu_1+\delta_1)\times(2\nu+\delta)$ matrix X such that the row vectors form a basis of U and (E.33) holds. Let X and X_1 be two $(2\nu_1+\delta_1)\times(2\nu+\delta)$ matrices such that both of them satisfy (E.33), then the rows of X and the rows of X_1 span the same subspace if and only if there is a $T \in Ps_{2\nu_1+\delta_1}(\mathbb{F}_q)$ such that $XT = X_1$. Hence Theorem E.28 follows immediately. □

Theorem E.29.

$$n^{(s)}_{(2\nu+\delta)\times t}(S_{2\nu+\delta}, 0^{(t)})$$
$$= \begin{cases} \displaystyle\sum_{k=0}^{\min\{t,\nu\}} q^{\frac{1}{2}k(k-1)} \prod_{i=t-k+1}^{t} (q^i-1) N(k,0,0,0;2\nu+1), & \text{if } \delta = 1, \\ \displaystyle\sum_{k=0}^{\min\{t,\nu+1\}} q^{\frac{1}{2}k(k-1)} \prod_{i=t-k+1}^{t} (q^i-1) \sum_{\varepsilon_1=0}^{1} N(k,0,0,\varepsilon_1;2\nu+2), & \text{if } \delta = 2, \end{cases}$$

where $N(k,0,0,0;2\nu+1)$ and $N(k,0,0,\varepsilon_1;2\nu+2)$ are given by Theorem 4.14. □

The proof of Theorem E.29 is left as an exercise (Exercise E.5).

From Theorems E.26–E.29 we deduce

Theorem E.30. Let A be an $m\times m$ nonalternate symmetric matrix of index ν and rank $2\nu+\delta$, where $\delta = 1$ or 2 and $2\nu+\delta \le m$, and let B be a $t\times t$ symmetric matrix of index ν_1 and rank $2\nu_1+\delta_1$, $\delta_1 = 0, 1$, or 2, and $2\nu_1+\delta_1 \le t$, both over $\mathbb{F}_q$.

If $\nu_1 > \nu$ or $\nu_1 = \nu$ and $\delta_1 > \delta$,

$$n^{(s)}_{t\times m}(A,B) = 0.$$

If $\nu_1 = \nu$ and $\delta_1 \le \delta$,

$n^{(s)}_{t\times m}(A,B)$

$$= q^{t(m-2\nu-\delta)}|Ps_{2\nu_1+\delta_1}(\mathbb{F}_q)| \sum_{\varepsilon_1=0}^{1} N(2\nu_1+\delta_1, 2\nu_1+\delta_1, \nu_1, \varepsilon_1; 2\nu+\delta)$$

$$\times \begin{cases} \sum_{k=0}^{\min\{t,\nu\}} q^{\frac{1}{2}k(k-1)} \prod_{i=t-k+1}^{t} (q^i-1) N(k,0,0,0;2\nu+1), & \text{when } \delta=1, \\ \sum_{k=0}^{\min\{t,\nu+1\}} q^{\frac{1}{2}k(k-1)} \prod_{i=t-k+1}^{t} (q^i-1) \sum_{\varepsilon_1=0}^{1} N(k,0,0,\varepsilon_1;2\nu+2), & \text{when } \delta=2, \end{cases}$$

where $|Ps_{2\nu_1+\delta_1}(\mathbb{F}_q)|$ is given by Theorem 4.7 and N's are given by Theorem 4.14. □

E.7 Comments

The representation of a form by another one of the same type in a finite field of odd characteristic was studied by L. Carlitz in 1954 (cf. Carlitz, 1954a and 1954b). He obtained formulas for the number of representations of a quadratic (or skew-symmetric) form by another quadratic (or skew-symmetric) form in a finite field of odd characteristic and, together with J. H. Hodges, formulas for the number of representations of an hermitian form by another hermitian form also in a finite field of odd characteristic (cf. Carlitz and Hodge, 1955). They used exponential sums. Using the Anzahl theorems in the geometry of various classical groups the problem was studied by the author in its full generality (cf. Wan, 1995). On the one hand, the restriction that the finite field is of odd characteristic is removed. On the other hand, in addition to the quadratic, skew-symmetric, and hermitian forms studied by Carlitz, the bilinear, alternate bilinear, and symmetric bilinear forms are also studied. This appendix follows closely Wan (1995).

E.8 Exercises

5.1 Write down the detailed proofs of Theorems E.12–E.16.

5.2 Write down the detailed proofs of Theorems E.17–E.20.

5.3 Write down a closed formula for $n^{(q)}_{t\times m}(A,B)$, where A is an $m\times m$ symmetric matrix of index ν and definite part Δ and B is a $\times t$ symmetric matrix of index ν_1 and definite part Δ_1, both over a finite field of odd characteristic.

5.4 Write down the detailed proofs of Theorems E.21–E.24.

5.5 Prove Theorems E.26 and E.27.

5.6 Prove Theorem E.29.

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Notation

Groups

$GL_n(\mathbb{F}_q)$	the general linear group of degree n over $\mathbb{F}_q$, 2
$PGL_n(\mathbb{F}_q)$	the projective general linear group of degree $n+1$ over $\mathbb{F}_q$, 76
$AGL_n(\mathbb{F}_q)$	the affine group of the n-dimensional affine space over $\mathbb{F}_q$, 20
$Sp_{2\nu}(\mathbb{F}_q)$	the symplectic group of degree 2ν over $\mathbb{F}_q$, 113
$Sp_{2\nu+l,\nu}(\mathbb{F}_q)$	the singular symplectic group of degree $2\nu+l$ and index ν over $\mathbb{F}_q$, 131
$Ps_{2\nu+\delta}(\mathbb{F}_q)$	the pseudo-symplectic group of degree $2\nu+\delta$, where $\delta = 1$ or 2, over $\mathbb{F}_q$ (of characteristic 2), 164
$P_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$	the singular pseudo-symplectic group of degree $2\nu+\delta+l$ and rank $2\nu+\delta$ over $\mathbb{F}_q$ (of characteristic 2), 168
$U_n(\mathbb{F}_{q^2})$	the unitary group of degree n over $\mathbb{F}_{q^2}$, 229
$U_{n+l,n}(\mathbb{F}_q)$	the singular unitary group of degree $n+l$ and rank n over $\mathbb{F}_{q^2}$, 243
$O_{2\nu+\delta,\Delta}(\mathbb{F}_q)$	the orthogonal group of degree $2\nu+\delta$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix $S_{2\nu+\delta,\Delta}$ over $\mathbb{F}_q$ (of characteristic $\neq 2$), 258
$O_{2\nu+\delta+l,2\nu+\delta,\Delta}(\mathbb{F}_q)$	the singular orthogonal group of degree $2\nu+\delta+l$, rank $2\nu+\delta$, and with definite part Δ over $\mathbb{F}_q$ (of characteristic $\neq 2$), 288

$O_{2\nu+\delta}(\mathbb{F}_q)$ the orthogonal group of degree $2\nu+\delta$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ nonsingular symmetric matrix $G_{2\nu+\delta}$ over $\mathbb{F}_q$ (of characteristic 2), 302

$O_{2\nu+\delta+l,2\nu+\delta}(\mathbb{F}_q)$ the singular orthogonal group of degree $2\nu+\delta+l$ and rank $2\nu+\delta$ over $\mathbb{F}_q$ (of characteristic 2), 336

Spaces – general

$\mathbb{F}_q^{(n)}$ the n-dimensional row vector space over $\mathbb{F}_q$, 1

$\mathcal{M}(m,n)$ the set of m-dimensional subspaces of $\mathbb{F}_q^{(n)}$, 408

$N(m,n)$ the number of m-dimensional subspaces of $\mathbb{F}_q^{(n)}$, 4

$n(m,n)$ the number of $m\times n$ matrices of rank m over $\mathbb{F}_q$, 4

$N(k,m,n)$ the number of k-dimensional subspaces contained in a given m-dimensional subspace of $\mathbb{F}_q^{(n)}$, 6

$N'(k,m,n)$ the number of m-dimensional subspaces containing a given k-dimensional subspace of $\mathbb{F}_q^{(n)}$, 7

$\begin{bmatrix} n \\ m \end{bmatrix}_q$ the Gaussian coefficient, 6

$AG(2,\mathbb{F}_q)$ the affine plane over $\mathbb{F}_q$, 9

$AG(n,\mathbb{F}_q)$ the n-dimensional affine space over $\mathbb{F}_q$, 8

$PG(2,\mathbb{F}_q)$ the projective plane over $\mathbb{F}_q$, 64

$PG(n,\mathbb{F}_q)$ the n-dimensional projective space over $\mathbb{F}_q$, 63

$PG(2,K)$ the projective plane over a division ring K, 365

$PG(n,K)$ the n-dimensional projective space over a division ring K, 371

$\dim(U+u)$ the dimension of the affine flat $U+u$, 19

$\dim F$ the dimension of the flat F, 72, 369

$\dim\mathcal{S}$ the dimension of the projective space $\mathcal{S}$, 369

$\langle X\rangle$ the flat spanned by the set X of points, 367

$P^\perp$ the dual subspace (or flat) of the subspace (or flat) P, 115, 149, 218, 220, 231, 249, 262, 296

(P,L,I) an incidence structure, 355

$p\sim q$ the points p and q are collinear or $p=q$, 355

$p\not\sim q$ the points p and q are not collinear, 355

Spaces – symplectic

(m, s)	a type of subspace in the 2ν-dimensional symplectic space over $\mathbb{F}_q$, 114
$\mathcal{M}(m, s; 2\nu)$	the set of subspaces of type (m, s) in the 2ν-dimensional symplectic space over $\mathbb{F}_q$, 121
$N(m, s; 2\nu)$	$\mid \mathcal{M}(m, s; 2\nu) \mid$, 121
$N(m_1, s_1; m, s; 2\nu)$	the number of subspaces of type (m_1, s_1) contained in a given subspace of type (m, s) in the 2ν-dimensional symplectic space over $\mathbb{F}_q$, 137
$N'(m_1, s_1; m, s; 2\nu)$	the number of subspaces of type (m, s) containing a given subspace of type (m_1, s_1) in the 2ν-dimensional symplectic space over $\mathbb{F}_q$, 154

Spaces – pseudo-symplectic

$(m, 2s + \tau, s, \varepsilon)$	a type of subspace in $(2\nu + \delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$ (of characteristic 2), 170
$\mathcal{M}(m, 2s + \tau, s, \varepsilon; 2\nu + \delta)$	the set of subspaces of type $(m, 2s + \tau, s, \varepsilon)$ in the $(2\nu + \delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$ (of characteristic 2), 179
$N(m, 2s + \tau, s, \varepsilon; 2\nu + \delta)$	$\mid \mathcal{M}(m, 2s + \tau, s, \varepsilon; 2\nu + \delta) \mid$, 187
$(m, 2s + \tau, s, \varepsilon)^{\perp}$	the type of subspaces which are duals of subspaces of type $(m, 2s + \tau, s, \varepsilon)$ in the $(2\nu + \delta)$-dimensional pseudo-symplectic space over $\mathbb{F}_q$ (of characteristic 2), 220

Spaces – unitary

(m, r)	a type of subspace in n-dimensional unitary space over $\mathbb{F}_{q^2}$, 230
$\mathcal{M}(m, r; n)$	the set of subspaces of type (m, r) in the n-dimensional unitary space over $\mathbb{F}_{q^2}$, 236
$N(m, r; n)$	$\mid \mathcal{M}(m, r; n) \mid$, 236
$N(m_1, r_1; m, r; n)$	the number of subspaces of type (m_1, r_1) contained in a given subspace of type (m, r) in the n-dimensional unitary space over $\mathbb{F}_{q^2}$, 246

$N'(m_1, r_1; m, r; n)$ — the number of subspaces of type (m, r) containing a given subspace of type (m_1, r_1) in the n-dimensional unitary space over $\mathbb{F}_{q^2}$, 251

Spaces – orthogonal

$(m, 2s+\gamma, s, \Gamma)$ — a type of subspace in $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$, 260, 303

$(m, 2s+\gamma, s, \Gamma)^{\perp}$ — the type of the dual subspaces of the subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$ (of characteristic $\neq 2$), 269

$\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ — the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$ (of characteristic 2), 303

$\mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ — the set of subspaces of type $(m, 2s+\gamma, s, \Gamma)$ in the $(2\nu+\delta)$-dimensional orthogonal space over $\mathbb{F}_q$ (of characteristic $\neq 2$), 265

$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta)$ — $\mid \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta) \mid$, 324

$N(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta)$ — $\mid \mathcal{M}(m, 2s+\gamma, s, \Gamma; 2\nu+\delta, \Delta) \mid$, 273

Polar spaces

$\mathcal{P}(K)$ — the polar space arising from the symplectic polarity of $PG(2\nu-1, \mathbb{F}_q)$ defined by the $2\nu \times 2\nu$ nonsingular alternate matrix K, 149

$\mathcal{P}(S_1)$ — the polar space consisting of all totally isotropic flats with respect to S_1 in $PG(2\nu, \mathbb{F}_q)$, 219

$\mathcal{P}(S_2)$ — the polar space consisting of all totally isotropic flats with respect to S_2 and containing $e_{2\nu+1}$ in $PG(2\nu+1, \mathbb{F}_q)$, 219

$\mathcal{P}(H)$ — the polar space arising from the unitary polarity of $PG(n-1, \mathbb{F}_{q^2})$ defined by the $n \times n$ nonsingular Hermitian matrix H, 249

$\mathcal{P}(S)$ — the polar space arising from the orthogonal polarity of $PG(n-1, \mathbb{F}_q)$ defined by the $n \times n$ nonsingular symmetric matrix S, 297

$\mathcal{P}(G)$ the polar space consisting of totally singular flats of $PG(2\nu+\delta-1,\mathbb{F}_q)$ with respect to the $(2\nu+\delta)\times(2\nu+\delta)$ regular matrix G, 353

Generalized quadrangles

$W(q)$ the generalized quadrangle with the points of $PG(3,\mathbb{F}_q)$ as its points and totally isotropic lines with respect to a 4×4 nonsingular alternate matrix as its lines, 153

$H(3,q^2)$ the generalized quadrangle formed by the isotropic points and totally isotropic lines of $PG(3,\mathbb{F}_{q^2})$ with respect to a 4×4 nonsingular Hermitian matrix, 250

$H(4,q^2)$ the generalized quadrangle formed by the isotropic points and totally isotropic lines of $PG(4,\mathbb{F}_{q^2})$ with respect to a 5×5 nonsingular Hermitian matrix, 250

$Q(3,q)$ the generalized quadrangle formed by the isotropic (or singular) points and totally isotropic (or singular) lines of $PG(3,\mathbb{F}_q)$ with respect to a nondegenerate quadric of the hyperbolic type in $PG(3,\mathbb{F}_q)$, 297, 353

$Q(4,q)$ the generalized quadrangle formed by the isotropic (or singular) points and totally isotropic (or singular) lines of $PG(4,\mathbb{F}_q)$ with respect to a nondegenerate quadric in $PG(4,\mathbb{F}_q)$, 297, 353

$Q(5,q)$ the generalized quadrangle formed by the isotropic (or singular) points and totally isotropic (or singular) lines of $PG(5,\mathbb{F}_q)$ with respect to a nondegenerate quadric of the elliptic type in $PG(5,\mathbb{F}_q)$, 297, 353

$p^\perp$ $\{x\in P \mid x\sim p\}$, where P is the set of points of a generalized quadrangle and $p\in P$, 382

tr (p,q) $\{x\in P \mid x\sim p \text{ and } x\sim q\}$, where P is the set of points of a generalized quadrangle, $p,q\in P$, and $p\neq q$, 382

$T^\perp$ the centers of a triad T in a generalized quadrangle, 386

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